Trishna's

IIT JEE Super Course in Chemistry

INORGANIC CHEMISTRY



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PEARSON

Super Course in Chemistry

INORGANIC CHEMISTRY

for the IIT-JEE

Volume 2

Trishna Knowledge Systems

A division of
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Preface

The IIT-JEE, the most challenging amongst national level engineering entrance examinations, remains on the top of the priority list of several lakhs of students every year. The brand value of the IITs attracts more and more students every year, but the challenge posed by the IIT-JEE ensures that only the *best* of the aspirants get into the IITs. Students require thorough understanding of the fundamental concepts, reasoning skills, ability to comprehend the presented situation and exceptional problem-solving skills to come on top in this highly demanding entrance examination.

The pattern of the IIT-JEE has been changing over the years. Hence an aspiring student requires a step-by-step study plan to master the fundamentals and to get adequate practice in the various types of questions that have appeared in the IIT-JEE over the last several years. Irrespective of the branch of engineering study the student chooses later, it is important to have a sound conceptual grounding in Mathematics, Physics and Chemistry. A lack of proper understanding of these subjects limits the capacity of students to solve complex problems thereby lessening his/her chances of making it to the top-notch institutes which provide quality training.

This series of books serves as a source of learning that goes beyond the school curriculum of Class XI and Class XII and is intended to form the backbone of the preparation of an aspiring student. These books have been designed with the objective of guiding an aspirant to his/her goal in a clearly defined step-by-step approach.

- **Master the Concepts and Concept Strands!**
  This series covers all the concepts in the latest IIT-JEE syllabus by segregating them into appropriate units. The theories are explained in detail and are illustrated using solved examples detailing the different applications of the concepts.
- **Let us First Solve the Examples—Concept Connectors!**
  At the end of the theory content in each unit, a good number of "Solved Examples" are provided and they are designed to give the aspirant a comprehensive exposure to the application of the concepts at the problem-solving level.
- **Do Your Exercise—Daily!**
  Over 200 unsolved problems are presented for practice at the end of every chapter. Hints and solutions for the same are also provided. These problems are designed to sharpen the aspirant's problem-solving skills in a step-by-step manner.
- **Remember, Practice Makes You Perfect!**
  We recommend you work out ALL the problems on your own – both solved and unsolved – to enhance the effectiveness of your preparation.

A distinct feature of this series is that unlike most other reference books in the market, this is not authored by an individual. It is put together by a team of highly qualified faculty members that includes IITians, PhDs etc from some of the best institutes in India and abroad. This team of academic experts has vast experience in teaching the fundamentals and their application and in developing high quality study material for IIT-JEE at T.I.M.E. (Triumphant Institute of Management Education Pvt. Ltd), the number 1 coaching institute in India. The essence of the combined knowledge of such an experienced team is what is presented in this self-preparatory series. While the contents of these books have been organized keeping in mind the specific requirements of IIT-JEE, we are sure that you will find these useful in your preparation for various other engineering entrance exams also.

We wish you the very best!

CHAPTER

1

PERIODIC TABLE, PERIODIC PROPERTIES, S-BLOCK ELEMENTS AND PRINCIPLES OF INORGANIC QUALITATIVE ANALYSIS

CHAPTER OUTLINE

Preview

STUDY MATERIAL

- Periodic Table and Periodic Properties
- Mendeleev's Periodic Table
- Modern Periodic Table
- General Structure of Long form Periodic Table
- Periodic Table and Electronic Configuration
- s, p, d, f-Blocks of Elements
- Periodic Properties
- Group-I
- s-Block Elements
  - Concept Strands (1-5)
- Group II-Alkaline Earth metals
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- Principles of Inorganic qualitative analysis
- Reactions of anions
- Analysis of cations

TOPIC GRIP

- Subjective Questions (10)
- Straight Objective Type Questions (5)
- Assertion–Reason Type Questions (5)
- Linked Comprehension Type Questions (6)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

IIT ASSIGNMENT EXERCISE

- Straight Objective Type Questions (80)
- Assertion–Reason Type Questions (3)
- Linked Comprehension Type Questions (3)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

ADDITIONAL PRACTICE EXERCISE

- Subjective Questions (10)
- Straight Objective Type Questions (40)
- Assertion–Reason Type Questions (10)
- Linked Comprehension Type Questions (9)
- Multiple Correct Objective Type Questions (8)
- Matrix-Match Type Questions (3)

PERIODIC TABLE AND PERIODIC PROPERTIES

MENDELEEV'S PERIODIC TABLE

The first successful attempt to classify elements was made by Dimitri Ivanovich Mendeleev, a Russian scientist, in 1869. He put forward the periodic law which states that the physical and chemical properties of elements are a periodic function of their atomic masses.

Mendeleev arranged the elements in the increasing order of their atomic masses in horizontal rows called periods and vertical columns called groups. Only 63 elements were known in his days. He left vacant spaces for undiscovered elements and corrected the atomic masses in the case of some elements so that they fitted into correct spaces in the periodic table. He even predicted the properties of these undiscovered elements and their compounds as an average of the properties of neighbouring elements of the same group. In some cases he disregarded their atomic masses to place them in the correct positions in periodic table. The greatness of his achievement has no parallel in chemistry when we remember that the electronic configuration and atomic number were not known in those days.

MODERN PERIODIC TABLE

In 1912, Moseley found that the square root of frequency $\left(\sqrt{\upsilon}\right)$ of characteristic X-rays emitted by an element was proportional to its atomic number (Z)

$$\sqrt{\upsilon} = a(Z - b)$$

where, a and b are constants. On the basis of this observation Moseley suggested that atomic number is a more fundamental property of elements than atomic masses. He proposed the modern periodic law which states that the physical and chemical properties of elements are a periodic function of atomic number. The periodic table based on atomic number is called the modern periodic table. The most widely used form of modern periodic table is the Long form periodic table. Modern periodic table could resolve some of the defects of Mendeleev's periodic table, viz., position of isotopes, relative positions of Argon and potassium, Tellurium and iodine, cobalt and Nickel, Thorium and protactinium etc.

GENERAL STRUCTURE OF LONG FORM PERIODIC TABLE

The modern long form periodic table consists of seven horizontal rows called periods. The first is a very short period of two elements. Second and third are short periods of 8 elements each. Fourth and fifth are long periods of 18 elements each. Sixth is a very long period of 32 elements. Seventh is an incomplete period of about 28 elements.

There are eighteen vertical columns called groups. Groups are numbered from 1 to 18 according to IUPAC. Formerly the groups were numbered from 0 to 8. 1 to 7 groups were subdivided into A and B subgroups. Group 8 had three vertical columns containing the ferrous metals and platinum metals. Each vertical column is a family of closely related elements.

PERIODIC TABLE AND ELECTRONIC CONFIGURATION

The periodicity of properties is due to recurrence of similar outer electronic configuration after regular intervals of atomic number. Elements of a particular group have the same outer electronic configuration which imparts similar physical and chemical properties on the element. Thus all the halogens of group 17 have seven electrons in the outermost shell. This makes them highly electronegative and reactive elements with strong oxidizing property.

The general structures of the periodic table reflects the electronic configuration and the order of filling of various subshells according to aufbau principle.

Periods	Subshell				No. of elements in a period
1	1s				2
2	2s			2p	8
3	3s			3p	8
4	4s		3d	4p	18
5	5s		4d	5p	18
6	6s	4f	5d	6p	32
7	7s	5f	6d	7p	32 (if complete)

s, p, d, f-BLOCKS OF ELEMENTS

One advantage of the long form periodic table is that it is divided into four blocks called s, p, d and f-blocks. This classification is based on the subshell into which the distinguishing electron enters. The general characteristics of the four blocks are discussed below.

(i) s-block

s-block consists of elements in which the s-subshell is being filled and includes groups 1 and 2. They have the following characteristics.

1. They have the general electronic configuration ns^{1-2}
2. They are highly electropositive metals.
3. They are highly reactive and in reactions they lose their outermost electrons to form cations in the +1 or +2 oxidation state.
4. They never exhibit variable valency.

(ii) p-block

p-block consists of elements in which the p-subshell is being filled and includes groups 13 to 18.

1. The general electronic configuration is ns^2np^{1-6} where n is the principal quantum number of the outermost shell. He with configuration $1s^2$ is an exception.
2. p-block consists of metals, all the metalloids and all the nonmetals.
3. Together with s-block elements they constitute the representative elements or main group elements.
4. The elements of group 18 are called noble gases which are chemically very unreactive.
5. They exhibit variable valency to a limited extent. The valencies of these elements differ by units of two. For example, sulphur exhibits valencies 2, 4 and 6. Halogens exhibit valencies 1, 3, 5 and 7.

(iii) d-block

d-block consists of elements in which the d-orbitals are being filled and includes groups 3 to 12. They are called transition elements because they form a smooth transition between the highly electropositive s-block elements and the not so electropositive p-block elements.

1. All the elements in d-block are metals. Therefore, they are also called transition metals.
2. The general outer electronic configuration is $(n-1)d^{1-10}ns^{0-2}$. They have two outershells incomplete.
3. They exhibit variable valency. Their valencies differ by units of one. Thus Manganese exhibits the valencies 2, 4, 5, 6 and 7.
4. They form mostly coloured compounds.
5. They form large number of complex compounds.
6. The elements and compounds of d-block are mostly paramagnetic due to unpaired electrons in the d-subshell.

(iv) f-block

f-block consists of elements in which the f-orbitals in the antipenultimate shell are being filled. They are the fourteen elements following the lanthanum (lanthanoids) and fourteen elements following Actinium (Actinoids). All are in the group 3 along with La and Ac. Since there is nospace to write all the thirty elements in group 3 the twentyeight lanthanoids and actinoids are placed below the main structure of the periodic table as a separate block. They are also called inner transition elements because they occur inside the transition elements series.

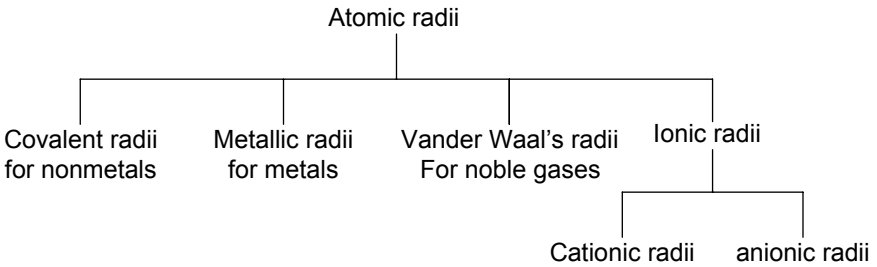
1. The outer electronic configuration is $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$.
2. They are all metals resembling transition metals.
3. They exhibit variable valency.
4. They form coloured compounds and many complex compounds.
5. The elements and their compounds are mostly paramagnetic.

PERIODIC PROPERTIES

Periodic properties are properties which show a gradation in one direction along a period and show gradation in the opposite direction down a group. They show repetition of similar properties after regular intervals when we go form left to right in the periodic table. The reason for the periodicity of properties lies in the repetition of similar electronic configuration after regular intervals. The important periodic properties are atomic radii, ionization energy electron gain enthalpy, oxidation states, electropositivity, oxidizing and reducing power etc.

(i) Atomic radius

Atomic radius is the distance from the nucleus of an atom to the outermost shell. Since electron clouds do not have a definite boundary it is impossible to measure the atomic radius accurately. To overcome this we define different types of atomic radii which can be experimentally determined.

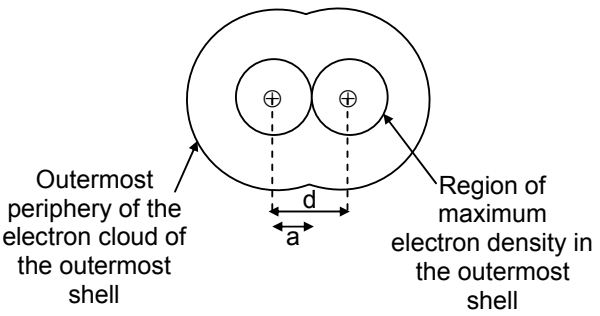


(a) Covalent radius

Covalent radius is half the distance between the nuclei of two covalently bonded atoms of the same element in a molecule. In the diagram "d" is the internuclear distance and "a" is the covalent radius. It is actually the distance from the nucleus to the region of maximum electron density in the outermost shell.

Covalent radii decrease from left to right in a period because of increasing nuclear charge and incomplete shielding by electrons of the same shell or in other words because of increasing effective nuclear charge.

Element	Li	Be	B	C	N	O	F	Ne
Covalent radii (pm)	123	89	80	77	76	74	72	160



For Ne the atomic radius is very large because it is the vander Waal's radius. Down a group the covalent radii increases because more and more shells are added as we go down a group. As shown ahead.

Element	Covalent radius (pm)	Element	Covalent radius (pm)
H	30	F	72
Li	123	Cl	99
Na	157	Br	114
K	203	I	133
Rb	216	At	140
Cs	235		

Since bond lengths decrease in the order
Single bond > double bond > triple bond
Atomic radii also decrease as bond order of the atom increases.

Element	C	N	O	S
Single bond radius (pm)	77	74	66	104
Double bond radius (pm)	67	65	57	95
Triple bond radius (pm)	60	55	-	-

(b) Metallic radius

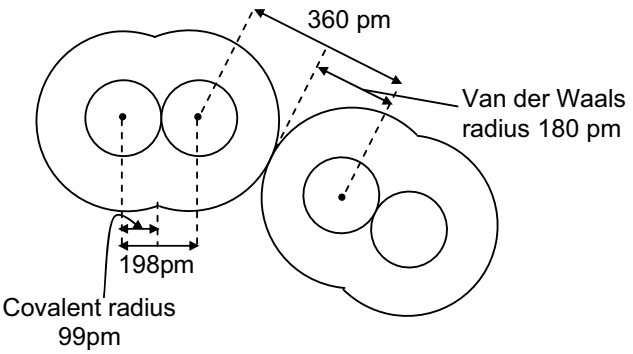
Metallic radius is half the distance between the nuclei of two adjacent atoms in a metallic crystal. It decreases from left to right in a period and increases down a group.

Li – 152 pm Na – 186 pm
Na – 186 pm Mg – 160 pm
K – 231 pm Al – 143 pm
Rb – 244 pm Cs – 262 pm

(c) Vander Waals radii

Vander Waals radii of an atom is half the distance between the nuclei of two nonbonded atoms at their closest approach

Usually covalent radius is taken as the atomic radius of non metals, metallic radius the atomic radius of metals and vander Waals radius the atomic radius of noble gases.

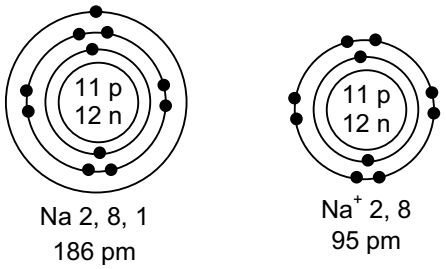


Covalent and van der Walls radii in chlorine molecule

(d) Ionic Radius

Ionic radius is the distance from the nucleus of an ion to the (outermost periphery of its) outermost shell. The following generalisations about the ionic radii are worth mentioning.

1. A cation is smaller than the atom from which it is formed. This is because when a cation is formed, usually the outermost shell is lost completely and also because the effective nuclear charge (ENC) has increased.



2. An anion is larger than the atom from which it is formed. This is because the ENC has decreased.
   Cl – 99 F – 72 pm
   Cl^- – 184 F^- – 136 pm
3. As more and more electrons are removed from an atom ionic radii decreases further and further.
   Fe – 117 pm
   Fe^{2+} – 78 pm
   Fe^{3+} – 64.5 pm
4. Size of isoelectronic speices decreases as atomic number increases
   e.g., $\underset{140\,pm}{O^{2-}} > \underset{133\,pm}{F^-} > \underset{95\,pm}{Na^+} > \underset{72\,pm}{Mg^{2+}}$
5. Ionic radii increases down a group and decreases in a period from left to right

(ii) Ionization enthalpy

Ionization Enthalpy, also known as the ionization potential or ionization energy is the minimum amount of energy required to remove the outermost (most losely bound) electron from an isolated gaseous atom (x) in its ground state.

$$X_{(g)} + \text{Ionization enthalpy} \rightarrow X_{(g)}^{+} + e^{-}$$

$\Delta_i H$ for the above reaction is the ionization energy of the atom, x. The unit of ionization enthalpy is k J mol^{-1}. Formerly ev and kcal mol^{-1} were used. The enthalpy change for the removal of the first electron is called the first ionization enthalpy I, and that for the 2nd electron is the second ionization enthalpy I_2. I_2 is always larger than I, because in the electron is removed from an already positive ion. I_3 is still larger.

$$I, < I_2 < I_3 \text{..... etc.}$$

Factors affecting Ionization Enthalpy

1. Size of the atom

As atomic size increases the attraction of the nucleus on the outermost electron decreases and ionization enthalpy decreases.

2. Nuclear charge

As nuclear charge increases ionization enthalpy increases.

3. Shielding of outer electron by inner electrons

Inner electrons neutralize part of the nuclear charge experienced by the outermost electron. This is called screening or shielding. Screening effect depends on the nature of the inner orbital. Screening decreases in the order s > p > d > f. The actual nuclear charge experienced by the outer electrons is called effective nuclear charge (E N C).

$$ENC = Z - \sigma$$

Where, Z is the nuclear charge or atomic number and σ is the screening constant.

4. Nature of the orbital

When all other factors are the same ionization energy decreases in the order s > p > d > f. This is because s electron cloud has maximum density near the nucleus. p, d and f do not penetrate near to the nuclear like s electrons.

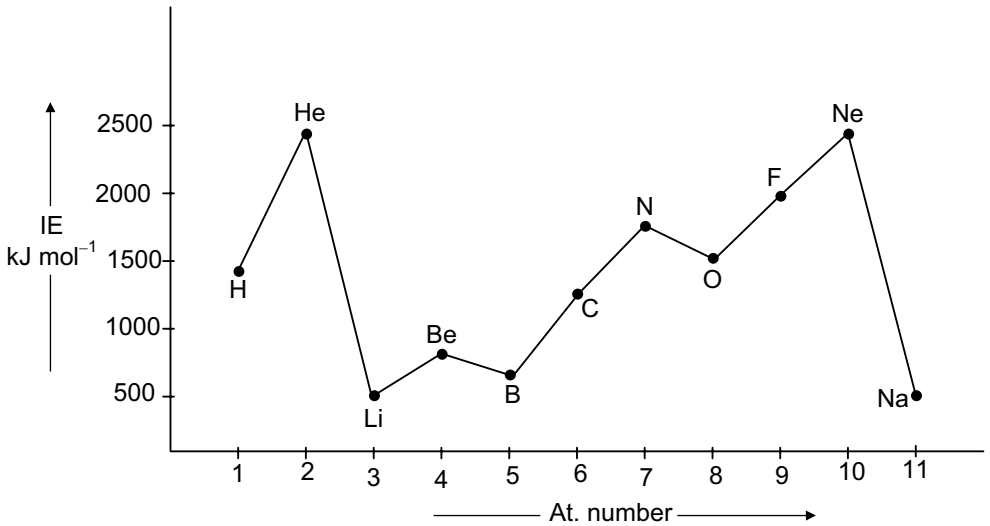
5. Exceptional stability of completely filled and half filled subshells

Half filled and completely filled subshells are much more stable than the subshells which are otherwise filled. The reason is the greater exchange energy possessed by these arrangements.

The energy required to remove an electron from such a subshell is comparatively large.

Variation of ionization energy along a period and down a group

I E increases along a period from left to right as the effective nuclear charge (ENC) increases. Along a period the electrons are added to the same shell. Electrons of the same shell do not effectively screen the nuclear charge from one another. Therefore, ENC increases along a period and IE increases. But the increase is not smooth.



In the graph maxima occur at noble gases because they have a stable closed shell electronic structure.

Minima occur at alkaline metals because of their large size, small ENC and an electronic configuration which is one electron more than a stable octet structure. Be and N have higher IE than their immediate neighbours on the left and right because they have completely filled s–sub shell and a half filled p–sub shell respectively. Moreover, B and O have lower I E than Be and N because when they lose one electron they attain stable half filled electronic configurations.

Ionization energy decreases down a group as the size of the atoms increases and the ENC remains almost the same. IE of some of the elements are given below (in k J mol^{-1})

H 1312							He 2372
Li 520	Be 899	B 801	C 1086	N 1402	O 1314	F 1681	Ne 2080
Na 496						Cl 1256	Ar 1520
K 419						Br 1142	Kr 1381
Rb 403						I 1008	Xe 1170
Cs 374							Rn 1037

(iii) Electron gain Enthalpy ($\Delta_{eg}H$)

Electron gain enthalpy is the enthalpy change taking place when an electron is added to a neutral gaseous atom (X) in its ground state

$$X_{(g)} + e^- \rightarrow X^-_{(g)}$$

For most of the elements energy is released and Therefore, $\Delta_{eg}H$ is a negative quantity. In some cases it is a positive quantity indicating absorption energy e.g., noble gases.

Factors affecting the Nature of Electron gain enthalpy

1. Size of the atom

As size increases magnitude of $\Delta_{eg}H$ decreases e.g., down the group 1 and group 17, the magnitude of $\Delta_{eg}H$ decreases.

2. Effective nuclear charge – ENC

As ENC increases $\Delta_{eg}H$ decreases. E.g., along second period.

3. Special stability of half filled and completely filled sub shells

Atoms with half filled and completely filled sub shells have very low $\Delta_{eg}H$.

e.g., noble gases, Element of group 15.

Along a period electron gain Enthalpy increases and down a group it decreases. Electron gain enthalpy was formerly called electron affinity which is the energy released when an e^- is added to an atom. $\Delta_{eg}H$ and EA are numerically equal but opposite in sign.

Elements of group 18 have positive $\Delta_{eg}H$ because they have already a stable octet structure.

H −73							He +48
Li −60	Be +66	B −83	C −122	N +31	O −141	F −328	Ne +116
Na −53	Mg +67	Al −50	Si −119	P −74	S −200	Cl −349	Ar +96
K −48						Br −325	Kr +96
Rb −47						I −295	Xe +77
Cs −46						Af −270	Rm +68

In group 17 $\Delta_{eg}H$ increases as expected but F has an abnormally low value. This is because F has seven electrons and its size is very small. Therefore, the electrons are already crowded and the eighth electron is not welcomed by the atom. Therefore, the energy released is not as great as expected. Thus chlorine the element with the highest electron affinity i.e., highest, negative electron gain enthalpy.

(iv) Electronegativity (X)

Electronegativity is the power of an atom in a molecule to attract shared electrons to itself. Electronegativity increases when size decreases and EAN increases . Electronegativity therefore, increases in a period and decreases in a group.

Linus pauling calculated the electronegativity of elements by measuring the polarity of bonds or more correctly the bond energies.

$$X_B - X_A = 0.1017\left[E_{A-B} - \frac{1}{2}\left(E_{A-A} + E_{B-B}\right)\right]^{1/2}$$

Where, X_A and X_B are electronegativities of elements A and B, E_{A-B}, E_{A-A} and E_{B-B} are bond energies of the molecules AB, A_2 and B_2, respectively in kJ mol^{-1}. He arbitrarily fixed the electronegativity of fluorine as 4 and calculated the electronegativity of other elements by comparison.

According to Mulliken electronegativity of an element is the average of ionization energy and electron affinity in eV.

$$X^M = \frac{IE + EA}{2}$$

So that Mulliken electronegativity values agree with Paulings values he had to divide his values with a constant

$$X^m \text{ on Pauling scale} = \frac{IE + EA}{540}$$

Where, IE and EA are in k J mol^{-1}

Because it is a combination of IE and EA which are determined by atomic radii, atomic number and screening by electrons, electronegativity is the sum total of all other properties of an atom. Therefore, electronegativity is the single most important property of atoms which determines the chemical properties of elements.

Factors which affect Electronegativity

1. Size of the atom

When the size decreases electronegativity increases.

2. ENC

When the effective nuclear charge increases, electronegativity increases.

3. Oxidation State

As oxidation state (positive) increases electronegativity increases.

4. Hybridization

Greater the s-character, greater will be the electronegativity.

Electronegativity increases along a period and decreases down a group. Therefore, the most electronegative element is Fluorine and the least electronegative (most electropositive) element is caesium (or Francium).

The nature of chemical bond formed between two atoms is determined by the elctronegativity difference between them. If the electronegativity difference is large, the bond formed is ionic. If the difference is small, the bond is covalent. It has been calculated that when these differences is 1.7 the bond will be 50% ionic and 50% covalent. Therefore, if the differences is less than 1.7 the bond is predominantly covalent and if the difference is more than 1.7 the bond is predominantly ionic.

The approximate percentage of ionic character in a covalent molecule is calculated using the Hanney–Smith equation. According to which the percentage of ionic character = $16 (X_B - X_A) + 3.5 (X_B - X_A)^2$ where X_A and X_B are the electronegativities of bonded atoms.

Table of electronegativities (Pauling's Scale)

H						
2.1						
Li	Be	B	C	N	O	F
1.0	1.5	2.0	2.5	3.0	3.5	4.0
Na					S	Cl
0.9						3.0
K						Br
0.8						2.8
Rb						I
0.8						2.5
Cs						At
0.7						2.2

Senderson scale and Allred-Rochow scale are two other electronegativity scales Senderson scale is based on stability ratio of atom.

(v) Electropositivity (Metallic Character)

Electropositivity is the tendency of an atom to lose electrons to form a positive ion

$$M \rightarrow M^+ + e^-$$

Electropositivity decreases along a period and increases down a group. The most electropositive element is Caesium (or Francium) and the least electropositive element is Fluorine.

(vi) Valency and Oxidation States

Valency towards hydrogen increases from 1 to 4 in groups 1, 2, 13 and 14 among representative elements and then decreases from 4 to 1 in groups 14 to 17. Valency towards Oxygen and other electronegative element increases from 1 to 7 in groups 1, 2, 13..... 17.

Group	1	2	13	14	15	16	17
Element	Na	Mg	Al	Si	P	S	Cl
Hydride	NaH	MgH_2	AlH_3	SiH_4	PH_3	H_2S	HCl
Valency	1	2	3	4	3	2	1
Oxide	Na_2O	MgO	Al_2O_3	SiO_2	P_2O_5	SO_3	Cl_2O_7
Valency	1	2	3	4	5	6	7

Valency of an atom is either equal to the number of valency electrons or equal to (8 – numbers of valency electrons).

(vii) Oxidizing power and reducing power

Reducing agent is a supplier of electrons and an oxidizing agent is an acceptor of electrons. Elements with low IE are good reducing agents e.g., elements of groups 1 and 2. Elements with large electron affinity are oxidizing agents. e.g., elements of groups 16 and 17. Oxidizing power increases along a period and decreases down a group.

S-BLOCK ELEMENTS

GROUP-I

s-block consists of the elements, lithium (Li), sodium (Na), potassium (K), rubidium (Rb), caesium (Cs) and francium (Fr). They readily dissolve in water to form strongly alkaline hydroxide, so they are called alkali metals. Francium (Fr) is radioactive. Its longest-lived isotope $_{87}Fr^{223}$ has a half-life of only 21 minutes.

$$^{223}_{87}\text{Fr} \rightarrow {}^{223}_{88}\text{Ra} + {}^{0}_{-1}\text{e}$$

Their general electronic configuration is [noble gas] ns^1. The elements are highly reactive metals and good conductors of heat and electricity. They are soft, malleable and ductile with low melting and boiling points.

Occurrence and abundance

Li is a relatively less (the 35th most) abundant element by weight and is mainly found as silicate minerals, *spodumene* $LiAl(SiO_3)_2$ and *lepidolite* $Li_2(F/OH)_2Al_2(SiO_3)_3$.

Sodium and potassium are the 7th and 8th most abundant elements, respectively, by weight in the earth's crust. NaCl and KCl occur in large amounts in seawater. The largest source of Na is the rock salt (NaCl). Seawater contains nearly 3% NaCl.

$Na_2B_4O_7.10H_2O$ (*borax*), $Na_2CO_3.NaHCO_3.2H_2O$ (*trona*), $NaNO_3$ (*Chile salt peter*) Na_2SO_4 (*mirabilite*) etc. are natural deposits. Potassium occurs mainly as KCl (*sylvine*), a mixture of KCl and NaCl (*sylvinite*) and the double salt $KCl.MgCl_2.6H_2O$ (*carnallite*). There is no convenient source for Rb and Cs and these elements are obtained as byproducts from lithium processing. Francium does not occur appreciably in nature due to its nuclear instability.

General characteristics of alkali metals

Atomic properties

(i) Electronic configuration: All the elements have one valence electron (ns^1) over the electronic configuration of inert gases. This electron is very well screened from the nuclear charge in these elements. It is easily lost to form M^+ ions with inert gas configuration. Therefore, alkali metals are uniformly univalent.

(ii) Atomic and ionic size: They are the largest in each period in the periodic table. Atomic size gradually increases down the group with increase in atomic number as more and more shells are added to the atom. M^+ ions are smaller than the parent atoms.

(iii) Ionization enthalpy: Ionisation energies for the atoms in this group are considerably low and decrease down the group from Li to Cs due to increase in size and high screening effect.

Physical properties

(i) Density: They are silvery white, soft and light elements. They have low density due to large atomic size. Density increases down the group. However, K is lighter than Na. Li is the lightest metal known.

(ii) Melting point and boiling point: All the alkali metals are soft and have low melting and boiling points due to low metallic bond energy of the atoms which in turn is due to the fact that the metallic bonding is due to one electron per metal atom. m.p and b.p decrease down the group.

(iii) Photoelectric effect: Higher alkali metals eject electrons when irradiated with visible light due to their low ionization potential. K and Cs are used in photoelectric cells.

(iv) Flame colouration: Alkali metals and their salts impart characteristic colours to the nonluminous Bunsen flame. The characteristic colours (and wavelengths) are

Li - crimson red (670.8 nm)
Na - golden yellow (589.2 nm)
K - lilac (766.5 nm red and 404 nm – violet)
Rb - red violet (purple) (780.0 nm)
Cs - blue (455.5 nm)

(v) Mobility and electrical conductivity of ions in aqueous solution: They exist as hydrated ions $M^+(H_2O)_x$ in the aqueous solution. Extent of hydration decreases with increase in ionic size from Li^+ to Cs^+. The mobility of ion and electrical conductivity in aqueous medium decreases as the hydrated radius increases in the order, $Cs^+ > Rb^+ > K^+ > Na^+ > Li^+$

(vi) Reducing character (in gaseous state): It depends on ionization energy of atoms and hence increases in the order Li < Na < K < Rb < Cs.

(vii) Reducing character (in solution): Depends on standard electrode potential (E^0) which is related to its sublimation, ionization and hydration enthalpies. Hence the increasing order is Na < K < Rb < Cs < Li. The abnormally large reducing power of lithium is due to its large hydration energy.

(viii) Reaction with liquid NH_3: Alkali metals dissolve in liquid NH_3 giving coloured solutions and are stable. The dilute solutions are intense blue but the colour changes to copper or bronze with increasing concentration above 3 M. The blue colour is due to solvated (ammoniated) electrons.

$$M + (x + y)NH_3 \rightarrow [M(NH_3)_x]^+ + [e(NH_3)_y]^-$$

ammoniated electron.

Blue solution on evaporation leaves metal as residue.

If the blue solution is allowed to stand, the colour slowly fades and finally disappears due to the formation of metal amide and H_2. The formation of amides is catalysed by transition metal ions like Fe^{3+}, even in traces.

$$M^+_{(am)} + e^-_{(am)} + NH_{3(l)} \rightarrow MNH_{2(am)} + \frac{1}{2}H_{2_{(g)}}$$

('am' denotes solution in ammonia)

At concentrations above 3 M, solutions are copper or bronze coloured and have metallic luster because metal ion clusters with metallic bonding are formed. These solutions are highly conducting and the conductivity is as high as that of pure metals. The blue solution is paramagnetic whereas bronze solution is diamagnetic. These solutions act as powerful reducing agents. They may even reduce an aromatic ring.

Abnormal behaviour of Li

The anomalous property of lithium is mainly due to the exceptionally small size of its atom and ion, (Li^+). Its ionic potential is the greatest of all alkali metal ions.

$$\text{Ionic potential} = \frac{\text{charge}}{\text{radius}}$$

Large ionic potential imparts high polarising power on an ion (see Fajan's rules). Increased covalent nature of its compounds is due to high polarizing power of Li^+, which in turn is responsible for their solubility in non-polar organic solvents.

E.g., LiCl is soluble in alcohol.

Li shows diagonal relationship with magnesium.

The main points of difference between Li and the rest of alkali metals and resemblance to magnesium—diagonal relationship

Li is much harder, its melting point and boiling point are higher than those of other alkali metals.

Li is the least reactive alkali metal but the strongest of all reducing agents in aqueous solution.

When burnt in air, it forms the monoxide (Li_2O) and the nitride Li_3N, while other elements form peroxides (M_2O_2) and super oxides (MO_2) and no nitride. Li_3N is ionic and reacts with water giving NH_3

$$Li_3N + 3H_2O \rightarrow 3LiOH + NH_3\uparrow$$

Li_2CO_3 is thermally unstable unlike other alkali metal carbonates and hence it decomposes on heating to oxide and carbondioxide.

$$Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2\uparrow$$

$LiHCO_3$ exists only in solution and does not exist as a solid while other alkali metal bicarbonates exist in solution as well as solid.

LiCl is deliquescent and crystallizes as a hydrate $LiCl.2H_2O$. Other alkali metal chlorides do not form such hydrates. Lithium halides are partially covalent and Therefore, soluble in alcohol.

Unlike other alkali metals, Li does not form the acetylide on reacting with acetylene.

Li shows diagonal relationship with Mg. Li resembles Mg more than the elements of its own group because Li and Mg have almost same ionic potentials.

$LiNO_3$ on heating decomposes to give the oxide, NO_2 and O_2, while nitrates of other alkali metals form nitrites and O_2 under similar condition.

$$4LiNO_3 \xrightarrow{\Delta} 2Li_2O + 4NO_2 + O_2$$

$$2NaNO_3 \xrightarrow{\Delta} 2NaNO_2 + O_2$$

Unlike other alkali metal hydroxides, LiOH has a low solubility in water and loses water on heating.

$$2LiOH \xrightarrow{\Delta} Li_2O + H_2O$$

Fluoride, oxalate, carbonate, phosphate, hydroxide and oxide of lithium are sparingly soluble in water while those of other alkali metals are readily soluble.

Li_2SO_4 does not form alums.

Li has higher hydration enthalpy (due to the smallest size of its ion) which accounts for its highest negative value of E^0 and hence highest reducing power.

CONCEPT STRAND

Concept Strand 1

A colourless solid (A) on heating liberates a colourless gas (C), leaving behind a colourless residue (B). (B) gives the following reactions.

(i) With dil.H_2SO_4 it gives a colourless residue (D) along with a reddish - brown gas.
(ii) With metallic sodium, it gives a white greyish amorphous solid (F) and a colourless gas (E).
(iii) With NH_4Cl, it gives a colourless solid (G) along with colourless gas (E) and water.
(iv) With dil.H_2SO_4 and urea, it gives two gases in addition to water and (D) and one of the gases is (E) itself.
(v) (C) forms a white powder with strongly heated Mg, which on hydrolysis gives $Mg(OH)_2$.
(vi) (A) and (D) impart yellow colour to a non – luminous flame.

Identify (A) – (G) and give proper equations.

Solution

The given information is as follows:

$$\underset{\text{colourless solid}}{(A)} \xrightarrow{\text{heat}} \underset{\text{solid}}{(B)} + \underset{\text{colourless gas}}{(C)}$$

(i) $(B) + H_2SO_4 \rightarrow (D) +$ reddish brown fumes

(ii) $(B) + Na \rightarrow \underset{\text{colourless gas}}{(E)} + \underset{\text{white amporphous solid}}{(F)}$

(iii) $(B) + NH_4Cl \rightarrow \underset{\text{colourless gas}}{(E)} + \underset{\text{colourless solid}}{(G)} + H_2O$

(iv) $(B) + H_2SO_4 + CO(NH_2)_2 \rightarrow (D) + (E) + H_2O +$ colourless gas

(v) $(C) + Mg \xrightarrow{\text{heat}} \text{white powder} \xrightarrow{H_2O} Mg(OH)_2$.

Gas (C) in equation which forms white powder with Mg should be O_2. Hence (C) should be O_2.

(A) and (D) impart yellow colour to flame. Hence (A) and (D) should be sodium salts. (B) gives off reddish brown gas of NO_2 with H_2SO_4. Hence (A) should be $NaNO_3$.

Equations:

$$\underset{(A)}{2NaNO_3} \xrightarrow{\text{heat}} \underset{(B)}{2NaNO_2} + \underset{(C)}{O_2}$$

(i) $\underset{(B)}{2NaNO_2} + \underset{(\text{dilute})}{H_2SO_4} \rightarrow \underset{(D)}{Na_2SO_4} + 2HNO_2$

$$3HNO_2 \rightarrow HNO_3 + H_2O + 2NO$$

$$2NO + O_2 \longrightarrow \underset{\text{reddish brown}}{2NO_2}$$

(ii) $\underset{(B)}{2NaNO_2} + 6Na \rightarrow \underset{(F)}{4Na_2O} + \underset{(E)}{N_2}$

(iii) $\underset{(B)}{NaNO_2} + NH_4Cl \rightarrow \underset{(G)}{NaCl} + \underset{(E)}{N_2} + 2H_2O$

(iv) $\underset{(B)}{2NaNO_2} + H_2SO_4 \rightarrow \underset{(D)}{Na_2SO_4} + 2HNO_2$

$$2HNO_2 + CO(NH_2)_2 \rightarrow CO_2 + 2\underset{(E)}{N_2} + 3H_2O$$

(v) $2Mg + O_2 \rightarrow 2MgO$

Solubility and hydration

Li^+ being very small is heavily hydrated in aqueous solution while Cs^+ is larger and least hydrated. Hence in aqueous solutions, hydrated Li^+ ion is larger than hydrated Cs^+ ion. The hydration number is the average number of water molecules associated with the metal ion. Water molecules which touch the metal ion directly constitute the primary shell of water. Thus Li^+ is tetrahedrally surrounded by four water molecules, Rb^+ and Cs^+ are octahedrally surrounded by six water molecules, etc. A secondary layer of water molecules, further hydrates the ions which are held by weak ion-dipole attractive forces. The secondary hydration decreases from lithium to cesium and accounts for the larger hydrated radius of Li^+.

Compounds with carbon

Lithium forms an ionic carbide Li_2C_2 when heated with carbon, while other metals of group I form similar carbides when heated with ethyne. The carbide ion in these compounds exists as $(C \equiv C)^{2-}$. These carbides upon reaction with water form acetylene. Hence they are called acetylides.

$$2Li + 2C \rightarrow Li_2C_2$$

$$Na + C_2H_2 \rightarrow NaHC_2 \xrightarrow{Na} Na_2C_2$$

$$Na_2C_2 + 2H_2O \rightarrow 2NaOH + C_2H_2$$

Structure of Group I metals

At normal temperature, all the group I metals adopt a body-centred cubic type of lattice with a co-ordination number of 8. At very low temperatures, lithium forms hexagonal close-packed structure with a co-ordination number of 12.

Important compounds of Group I metals

Oxides

Group I metals burn in air to form oxides, but the product varies depending on the metal. Lithium forms the monoxide Li_2O (and some peroxide Li_2O_2). Sodium forms the peroxide Na_2O_2 (and some monoxide Na_2O) while the others form superoxides of the type MO_2.

The monoxides are ionic. Li_2O and Na_2O are pure white solids but surprisingly K_2O is pale yellow and Cs_2O is orange. Li_2O, Na_2O, K_2O and Rb_2O have anti-fluorite structures. Cs_2O has anti-$CdCl_2$ layer structures.

Preparation

Na_2O is prepared by burning sodium in a limited supply of air.

$$4Na + O_2 \rightarrow 2Na_2O$$

The oxide of sodium can be prepared in a pure form by heating sodium nitrite with sodium metal.

$$2NaNO_2 + 6Na \rightarrow 4Na_2O + N_2$$

The oxide of potassium can also be prepared by heating caustic potash with potassium.

$$2KOH + 2K \rightarrow 2K_2O + H_2$$

Properties

Metal oxides are strongly basic. The monoxide, M_2O, dissolve in water to give the corresponding hydroxides.

$$Na_2O + H_2O \rightarrow 2NaOH$$

$$K_2O + H_2O \rightarrow 2KOH$$

Peroxides and superoxides

Preparation

When sodium metal is heated in an excess of oxygen or air, sodium peroxide is obtained.

$$2Na + O_2 \rightarrow Na_2O_2$$

Other alkali metals react with O_2 to form superoxide of the type MO_2.

$$M + O_2 \rightarrow MO_2 \ (M = K, Rb, Cs)$$

Sodium superoxide is prepared by reacting Na_2O_2 with O_2 at 450°C and 300 atm pressure.

$$Na_2O_2 + O_2 \rightarrow 2NaO_2.$$

Properties

The peroxides are diamagnetic and strong oxidizing agents.

Na_2O_2 is pale yellow due to the presence of traces of superoxide NaO_2. It is used industrially for bleaching wool, pulp, paper and fabric. It is used to purify the air in

submarines since it removes CO_2 and produce O_2. (It is also called 'oxone').

$$2Na_2O_2 + 2CO_2 \rightarrow 2Na_2CO_3 + O_2$$

It reacts with water or acid producing H_2O_2 and NaOH.

$$Na_2O_2 + 2H_2O \rightarrow 2NaOH + H_2O_2$$

The superoxide (MO_2) contains O_2^- ion, which has an unpaired electron. Hence they are paramagnetic and are coloured. LiO_2 and NaO_2 are yellow, KO_2 is orange, RbO_2 is brown and CsO_2 is orange.

Peroxides are powerful oxidizing agents.

$$Na_2O_2 + Cr^{3+} \rightarrow CrO_4^{2-}$$

$$Na_2O_2 + CO \rightarrow Na_2CO_3$$

$$2Na_2O_2 + 2CO_2 \rightarrow 2Na_2CO_3 + O_2$$

Superoxides are even stronger oxidizing agents as compared to peroxides and give both H_2O_2 and O_2 with either water or acids.

$$KO_2 + 2H_2O \rightarrow KOH + H_2O_2 + \frac{1}{2}O_2$$

KO_2 is used in submarines and breathing masks because it produces O_2 and removes CO_2.

$$4KO_2 + 2CO_2 \rightarrow 2K_2CO_3 + 3O_2$$

The stability of peroxides and superoxides increases with increase in metal ion size.

$$LiO_2 < NaO_2 < KO_2 < RbO_2 < CsO_2$$

$$Li_2O_2 < Na_2O_2 < K_2O_2 < Rb_2O_2 < Cs_2O_2$$

Hydroxides

The important hydroxides are sodium hydroxide (caustic soda) and potassium hydroxide (caustic potash).

Preparation

(i) By Gossage method

$$Na_2CO_3 + Ca(OH)_2 \rightarrow 2NaOH + CaCO_3\downarrow$$

(ii) By electrolysis of aqueous NaCl during which H_2 is liberated at iron cathode and Cl_2 is evolved at carbon anode and NaOH remains in the solution near cathode.

$$2NaCl + 2H_2O \rightarrow 2NaOH + H_2 + Cl_2$$

(iii) By Castner Kellner process

Aqueous solution of NaCl is electrolyzed with carbon, mercury and iron electrodes to form chlorine, NaOH and H_2.

$$2NaCl + 2H_2O \rightarrow 2NaOH + H_2 + Cl_2$$

(iv) Reaction of metal with H_2O

Alkali metals liberate hydrogen when treated with H_2O. The reaction becomes increasingly violent on descending the group.

$$2M + 2H_2O \rightarrow 2MOH + H_2\uparrow$$

Na and K always catch fire. Reaction of Cs is explosive in nature.

$Li \rightarrow Li^+ + \bar{e}$ has largest negative value of ΔG. So Li liberates more energy than other metals when it reacts with water. But it is surprising that Li reacts with water less violently as compared to Na and K. The explanation lies in the kinetics, that is the rate at which the reaction proceeds, rather than in the thermodynamics, that is the total amount of heat liberated.

(v) Reaction of metal oxides or superoxide with H_2O.

$$KO_2 + 2H_2O \rightarrow KOH + H_2O_2 + \frac{1}{2}O_2$$

$$K_2O + H_2O \rightarrow 2KOH$$

Basic character and solubility of hydroxides

Both basic character and solubility of hydroxides of group I metals increase down the group .

$$LiOH < NaOH < KOH < RbOH < CsOH.$$

This is because, on moving down the group, the ionic size of metal cation increases and M–O bond strength decreases, i.e., lattice energy decreases.

Properties

(i) $NaOH + SO_2 \rightarrow NaHSO_3$

(ii) $Fe_2(SO_4)_3 + 6NaOH \rightarrow 2Fe(OH)_3\downarrow + 3Na_2SO_4$

(iii) $NaOH + CO \xrightarrow[6-10\ atm]{180^\circ C} HCOONa$

(iv) $NaOH + NH_4Cl \rightarrow NaCl + H_2O + NH_3$

(v) $Zn(OH)_2 + 2NaOH \rightarrow Na_2[ZnO_2] + 2H_2O$

(vi) $Al(OH)_3 + NaOH \rightarrow Na[AlO_2] + 2H_2O$

(vii) $2NaOH\ (cold) + Cl_2 \rightarrow NaCl + NaOCl + 2H_2O$

(viii) $6NaOH\ (hot) + 3Cl_2 \rightarrow NaClO_3 + 5NaCl + 3H_2O$

(ix) $3NaOH + P_4 + 3H_2O \rightarrow 3NaH_2PO_2 + PH_3$

Carbonates

The carbonates of sodium exist in different forms, viz., Na_2CO_3 (*soda ash*), $Na_2CO_3.H_2O$ (*crystal carbonate*), $Na_2CO_3.7H_2O$, $Na_2CO_3.10H_2O$ (*washing soda or salt-soda*).

The carbonate of potassium K_2CO_3 is called *pearl ash.*

Preparation

(i) $2NaOH + CO_2 \rightarrow Na_2CO_3 + H_2O$

(ii) Leblanc process

The reactions involved are given below.

(a) $NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$

(b) $NaHSO_4 + NaCl \xrightarrow{\Delta} Na_2SO_4 + HCl$

(c) $Na_2SO_4 + CaCO_3 + 4C \xrightarrow{\Delta} Na_2CO_3 + CaS + 4CO.$

(iii) Solvay process

Brine (con. aq. NaCl) is ammoniated and then carbonated when $NaHCO_3$ precipitates out. It is calcined to get Na_2CO_3.

(a) $NH_3 + CO_2 + H_2O \rightarrow NH_4HCO_3$

(b) $NaCl + NH_4HCO_3 \rightarrow NaHCO_3\downarrow + NH_4Cl$

(c) $2NaHCO_3 \xrightarrow[\Delta]{410K} Na_2CO_3 + H_2O + CO_2$

CO_2 is obtained by the calcination of limestone

$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$

CaO is heated with the solution of NH_4Cl to regenerate NH_3

$2NH_4Cl + CaO \rightarrow 2NH_3 + CaCl_2 + H_2O$

Thus the only by – product is $CaCl_2$.

(iv) Precht's process for the preparation of K_2CO_3

The important reactions are as follows.

$2KCl + 3MgCO_3 + CO_2 + 9H_2O \rightarrow 2[MgCO_3.KHCO_3.4H_2O] + MgCl_2$

$2[MgCO_3.KHCO_3.4H_2O] \xrightarrow[\Delta]{410K} 2MgCO_3 + K_2CO_3 + 9H_2O + CO_2$

Properties

(i) An aqueous solution of Na_2CO_3 is alkaline due to partial hydrolysis.

$Na_2CO_3 + 2H_2O \rightleftharpoons 2NaOH + H_2CO_3$

(ii) $Na_2CO_3 + SO_2 \rightarrow Na_2SO_3 + CO_2$

$Na_2CO_3 + \underset{\text{excess}}{2SO_2} + H_2O \rightarrow 2NaHSO_3 + CO_2$

(iii) The melting point of K_2CO_3 decreases when mixed with sodium carbonate. This mixture is called the fusion mixture. It is used to dissolve many insoluble inorganic substances.

(iv) $K_2CO_3 + \underset{\text{steam}}{H_2O} \xrightarrow{\text{red heat}} 2KOH + CO_2$

(v) Thermal stability and solubility of alkali metal carbonates in water increases down the group.

$Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3 < Cs_2CO_3$

CONCEPT STRAND

Concept Strand 2

A sample of fusion mixture weighing 2.5 g is dissolved in 250 mL water and 25 mL of this solution is completely neutralized by 15.5 mL of 0.25 M HCl. Calculate the percentage of sodium carbonate in the fusion mixture?

Solution

Let 'x' be the weight of Na_2CO_3 and 'y' that of K_2CO_3

$x + y = 2.5$ — (1)

$\frac{x}{53} + \frac{y}{69} = 15.5 \times 0.25 \times 0.01$ — (2)

Solving (1) and (2), x = 0.575 g and y = 1.925 g.

$\therefore$ Percentage of Na_2CO_3 = 23%.

Bicarbonates

Sodium hydrogen carbonate or sodium bicarbonate ($NaHCO_3$) is also known as baking soda because on heating it decomposes to give bubbles of CO_2 which cause holes in cakes, pastries etc., thereby making them light and fluffy.

Preparation

(i) $Na_2CO_3 + H_2O + CO_2 \rightarrow 2NaHCO_3$

(ii) It is also obtained as an intermediate product in the manufacture of Na_2CO_3 by Solvay process.

Properties

$$2NaHCO_3 \xrightarrow{373K} Na_2CO_3 + CO_2 + H_2O$$

$$NaHCO_3 + H_2O \rightleftharpoons NaOH + H_2CO_3$$

Thermal stability and solubility of alkali metal bicarbonates in water increases in the order,

$$NaHCO_3 < KHCO_3 < RbHCO_3 < CsHCO_3$$

$NaHCO_3$ is not very soluble in water and Therefore, it makes a basis for the Solvay process of manufacture of sodium carbonate. K_2CO_3 cannot be prepared by Solvay process because $KHCO_3$ is fairly soluble in water, as compared to $NaHCO_3$. All bicarbonates form carbonates on heating.

$$2\ NaHCO_3 \rightarrow Na_2CO_3 + CO_2 + H_2O$$

Chlorides

Sodium chloride is called rock salt while potassium chloride is called sylvine.

Preparation

$$2KBr + Cl_2 \rightarrow 2KCl + Br_2$$

$$HCl + NaOH \rightarrow NaCl + H_2O$$

$$HCl + KOH \rightarrow KCl + H_2O$$

Manufacture

By the evaporation of sea water under the heat of sun. It is also obtained from salt mines as rock salt.

Properties

Electrolysis of molten NaCl produces sodium at the cathode and chlorine at the anode, while aqueous solution of NaCl gives H_2 and NaOH at cathode and chlorine at anode. NaCl is used in the manufacture of Na_2CO_3 by Leblanc and Solvay process.

For the same alkali metal ion, the melting point and boiling point always follow the order: fluorides > chlorides > bromides > iodides as the covalent character increases .

All halides are soluble in water. LiF is sparingly soluble in water due to its high lattice energy. Low solubility of CsI is due to small hydration energy of its ions.

CONCEPT STRAND

Concept Strand 3

Explain why KCl is more ionic in nature than NaCl.

Solution

According to Fajans' rule, larger the size of the cation, lesser will be its polarizing power and greater will be the ionic character.

Sulphates

The anhydrous Na_2SO_4 is known as salt cake, while decahydrate $Na_2SO_4.10H_2O$ is called *Glauber's salt*. K_2SO_4 occurs as *kainite* ($K_2SO_4.MgSO_4.MgCl_2.6H_2O$) and *Schoenite* ($K_2SO_4.MgSO_4.6H_2O$).

Properties

Unlike sodium sulphate, crystals of potassium sulphate do not contain water of crystallization.

(i) $Na_2SO_4 + 4C \rightarrow Na_2S + 4CO$

(ii) $K_2SO_4 + 4C \rightarrow K_2S + 4CO$

The increasing order of the thermal stability of alkali metal sulphates is

$$Li_2SO_4 < Na_2SO_4 < K_2SO_4 < Rb_2SO_4 < Cs_2SO_4$$

CONCEPT STRANDS

Concept Strand 4

Why the solubility of Na_2SO_4 in water increases with the increase of temperature up to 32°C and thereafter decreases?

Solution

The initial solubility is due to the hydrated salt. Below 32°C the decahydrate salt ($Na_2SO_4.10H_2O$) crystallizes out. Above 32°C, the anhydrous salt (Na_2SO_4) separates out, which has a negative temperature coefficient of solubility.

Concept Strand 5

(i) I_2O_5 is used to estimate a polluting poisonous gas. Which is the gas and what are the reactions involved in the estimation?

(ii) Discuss the structure and bonding in I_3^-. Why is NaI_3 unstable while RbI_3 is stable in the solid state?

Solution

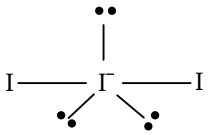
(i) CO

$I_2O_5 + 5CO \rightarrow 5CO_2 + I_2$

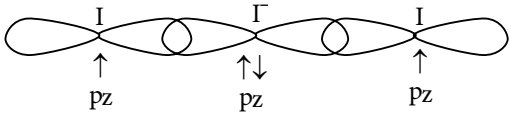
$2Na_2S_2O_3 + I_2 \rightarrow 2NaI + Na_2S_4O_6$

(ii) Structure is linear.

Note that I^- is isoelectronic with Xe. I_3^- is isoelectronic and isostructural with XeF_2. Central I carries negative charge and is sp^3d hybridized.



It can also be explained by molecular theory by a 3–centre bond.



The doubly filled p_z orbital of I^- overlaps simultaneously with p orbitals of two iodine atoms to form a 4–electron–3–centre bond.

Large anions are generally stabilized by large cations. Therefore, I_3^- is stabilized by K^+ and Rb^+ and not by Na^+.

GROUP II-ALKALINE EARTH METALS

Group ll comprises of the elements beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), barium (Ba) and radium (Ra). These are called alkaline earth metals. Beryllium shows diagonal relationship with aluminium. Radium is radioactive. It decays by α-emission

$$^{226}_{88}Ra \rightarrow {}^{222}_{86}Rn + {}^{4}_{2}He\ .$$

General Characteristics of Alkaline earth metals

Atomic properties

(i) Electronic configuration: Their outermost electronic configuration is ns^2, outside a noble gas core.

(ii) Atomic and ionic sizes are smaller than those of alkali metals in the corresponding periods due to increased nuclear charge. The outermost s-electrons in these metals (of Group I and II) are held very weakly by the nucleus. So the elements in Group I and II have larger atomic radii than the elements, which follow them in their respective periods.

(iii) Ionization enthalpies: The first ionization enthalpies of the alkaline earth metals are higher than those of Group I. The second ionization enthalpies of the alkaline earth metals are smaller than those of Group I.(why?)

Physical properties

(i) Melting and boiling points

Metallic bonding in group II metals is weak because of their large atomic size and because they have only two electrons per metal atom for metallic bonding. Hence Group I and II elements have lower melting and boiling points than other metals. Alkaline earth metals have fairly higher melting and boiling points than the alkali metals.

(ii) Electrical and thermal conductance

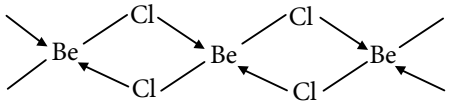
They are good conductors of heat and electricity. The conductivity of alkaline earth metals is more than that of alkali metals because they have two electrons per metal atom for metallic bonding.

(iii) Flame colour

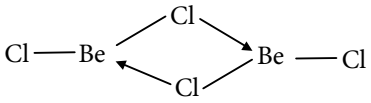
Be and Mg do not impart any colour to the flame due to higher ionization energy. Ca imparts brick red colour, Sr imparts crimson colour, Ba imparts apple green colour and Ra imparts crimson red colour.

Structure of $BeCl_2$

In solid state, $BeCl_2$ exists as a polymer. Be is tetrahedrally surrounded by four Cl atoms (sp^3 hybridization).



In vapour state below 900°C, $BeCl_2$ exists as a dimer (Be is in sp^2 hybridization).



In vapour state above 900°C, $BeCl_2$ exists as a linear monomeric molecule (Be is in sp hybridization).

$$Cl—Be—Cl$$

Important compounds of Group II metals

Oxides (MO)

Preparation

(i) Mg burns in air to form a mixture of magnesium oxide and magnesium nitride.
$Mg + air \rightarrow MgO + Mg_3N_2$

(ii) Thermal decomposition of carbonates.
$CaCO_3 \xrightarrow[\Delta]{1000-1200K} CaO + CO_2$

(iii) $MgCl_2 + Ca(OH)_2 \rightarrow Mg(OH)_2 + CaCl_2$
$Mg(OH)_2 \xrightarrow[\Delta]{} MgO + H_2O$

Properties

Monoxides are colourless ionic solids.

MgO is not very reactive, especially if it has been ignited at high temperature, hence it is used as a refractory. MgO is also known as magnesia. These oxides react exothermically with water to form hydroxides.

$$SrO + H_2O \rightarrow Sr(OH)_2$$

$$CaO + H_2O \rightarrow Ca(OH)_2$$

Peroxides (MO_2)

Preparation

Peroxides are formed with increasing ease and increasing stability as the metal ions become larger. The peroxides of group II are prepared by the action of H_2O_2 on the respective hydroxides.

$$MgSO_4 \xrightarrow{NaOH} Mg(OH)_2 \xrightarrow{H_2O_2} MgO_2 + 2H_2O$$

$$Ca(OH)_2 + H_2O_2 \rightarrow CaO_2.2H_2O \xrightarrow{\Delta} CaO_2$$

Barium peroxide by formed by passing air over BaO at 500°C.

$$BaO \xrightarrow{O_2} BaO_2.$$

They liberate H_2O_2 when treated with dilute acids.

$$BaO_2 + H_2SO_4 \rightarrow BaSO_4\downarrow + H_2O_2$$

Hydroxides

Preparation

(i) $MgCl_2 + 2KOH \rightarrow Mg(OH)_2 + 2KCl$
(ii) $CaO + H_2O \rightarrow Ca(OH)_2$

Properties

(i) When CO_2 is passed through lime water ($Ca(OH)_2$) a precipitate of $CaCO_3$ is formed. On passing more of CO_2 it dissolves forming soluble $Ca(HCO_3)_2$.

$$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O \xrightarrow{\text{excess } CO_2} Ca(HCO_3)_2$$

(ii) A mixture of CaO and NaOH is known as sodalime.
(iii) $Ca(OH)_2 + 2HCl \rightarrow CaCl_2 + H_2O$
(iv) $Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2.H_2O$
dry slaked lime bleaching powder

The hydroxides of Group I (except LiOH) are stable to heat but those of Group II decompose on heating.

$$Mg(OH)_{2\,(s)} \rightarrow MgO + H_2O_{(g)}$$

The solubility and basic character of hydroxides of alkaline earth metals increase in the order.

$$Be(OH)_2 < Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$$

CONCEPT STRAND

Concept Strand 6

(i) A sample of bleaching powder weighing 0.852 g on reaction with dil.H_2SO_4 liberates Cl_2, which is then passed into KI solution and titrated against 0.3 M thiosulphate solution. If the volume of thiosulphate consumed is 20 mL, find the percentage of available chlorine in the sample.
(ii) Al_2O_3 is not attacked by conc.H_2SO_4, but on fusion with $NaHSO_4$ it reacts. Give the balanced equation for this reaction.

Solution

(i) $Cl_2 + 2KI \rightarrow 2KCl + I_2$

$2S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$

No. of millimoles of thio = $20 \times 0.3 = 6$

6 millimoles of thio ≡ 3 millimoles of Cl_2

$$\therefore \text{ Percentage of } Cl_2 \text{ liberated} = \frac{3 \times 10^{-3} \times 71}{0.852} \times 100 = 25\%$$

(ii) $Al_2O_3 + 6NaHSO_4 \rightarrow Al_2(SO_4)_3 + 3Na_2SO_4 + 3H_2O.$

Carbonates

$MgCO_3$ occurs as magnesite. A mixture of carbonates of Mg and Ca occurs as dolomite [$CaCO_3.MgCO_3$]. $CaCO_3$ occurs as limestone, marble and calcite crystals.

Preparation

(i) $Mg(OH)_2 + CO_2 \rightarrow MgCO_3 + H_2O$
(ii) $Ca(OH)_2 + CO_2 \rightarrow CaCO_3\downarrow + H_2O$
(iii) $CaCl_2 + Na_2CO_3 \rightarrow CaCO_3\downarrow + 2NaCl$
(iv) $CaC_2O_4 \xrightarrow{\Delta} CaCO_3 + CO$

Properties

(i) $MgCO_3 \xrightarrow{\Delta} MgO + CO_2$
(ii) $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$
(iii) $CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2$

As the size of the cation increases its power to polarize the CO_3^{2-} ion decreases and Therefore, the thermal stability of carbonates increases down a group.

$$BeCO_3 < MgCO_3 < CaCO_3 < SrCO_3 < BaCO_3$$

$BeCO_3$ is so unstable that it decomposes into oxide and CO_2 even at room temperature. It is stored in an atmosphere of CO_2 under pressure. Carbonate decompose to oxides because of increase in stability which in turn is due to increased lattice energy as large CO_3^{2-} ion is replaced by smaller O^{2-} ion.

CONCEPT STRANDS

Concept Strand 7

A mixture of carbonates of calcium and magnesium weighing 1.5 g is heated till there is no weight loss. If the weight of the final residue is 0.75 g, what is the percentage composition of the mixture?

Solution

Let the weight of $CaCO_3$ be 'x' and $MgCO_3$ be 'y'

$x + y = 1.5$ — (1)

$\underset{100\ g}{CaCO_{3(s)}} \rightarrow \underset{56\ g}{CaO_{(s)}} + CO_2\uparrow$

$\underset{84\ g}{MgCO_{3(s)}} \rightarrow \underset{40\ g}{MgO_{(s)}} + CO_2\uparrow$

Weight of CaO obtainable from 'x' g $CaCO_3 = \frac{56x}{100}$ g

Weight of MgO obtainable from 'y' g $MgCO_3 = \frac{40y}{84}$ g

$\therefore$ $0.56\,x + 0.476\,y = 0.75$ —(2)

From (1) and (2), x = 0.43 g and y = 1.07 g

Percentage of $CaCO_3$ = 28.67%

Concept Strand 8

A white compound (A), which is only sparingly soluble in water exists in two different crystalline forms. (A) on strong heating liberates a colourless gas turning lime water milky along with a white residue (B). (B) on reaction with water gives a water soluble product (C). (C) is then treated with H_2O_2, followed by dehydration gives (D). (D) on reaction with dil.HCl regenerates H_2O_2 along with another compound (E). (E) is widely used for removing ice on roads and is better than NaCl for the same. (E) is also used to make concrete set more quickly and to improve its strength. (E) on treatment with dil.H_2SO_4 gives a sparingly soluble compound (F). (F) on strong heating gives (B) and a gas (G) as the products. A concentrated solution of (C) on treatment with Cl_2 gives a mixture of compounds and the mixture (H) is used for bleaching purpose. Identify, (A) – (H) and give the related reactions.

Solution

(A) is $CaCO_3$, which exists in the crystalline forms as calcite and aragonite.

$\underset{(A)}{CaCO_3} \rightarrow \underset{(B)}{CaO} + CO_2\uparrow$

$CaO + H_2O \rightarrow \underset{(C)}{Ca(OH)_2}$

$Ca(OH)_2 + H_2O_2 \rightarrow CaO_2.2H_2O \xrightarrow{\Delta} \underset{(D)}{CaO_2}$

$CaO_2 + 2HCl \rightarrow \underset{(E)}{CaCl_2} + H_2O_2$

$CaCl_2$ + ice freezes at –55°C compared to NaCl + H_2O, which freezes at –18°C.

$CaCl_2 + H_2SO_4 \rightarrow \underset{(F)}{CaSO_4} + 2HCl$

$CaSO_4 \xrightarrow{1100°C} \underset{(B)}{CaO} + \underset{(G)}{SO_3}\uparrow$

$3Ca(OH)_2 + 2Cl_2 \rightarrow \underset{\text{Bleaching powder (H)}}{Ca(OCl)_2.Ca(OH)_2.CaCl_2.2H_2O}$

Bicarbonates

Preparation

(i) $MgCO_3 + CO_2 + H_2O \rightarrow Mg(HCO_3)_2$

(ii) $CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2$

Properties

(i) $Mg(HCO_3)_2 \xrightarrow{\Delta} MgCO_3 + CO_2 + H_2O$

(ii) $Ca(HCO_3)_2 \xrightarrow{\Delta} CaCO_3 + CO_2 + H_2O$

The dissolved bicarbonates of Mg and Ca cause temporary hardness of water.

CONCEPT STRAND

Concept Strand 9

A sample of hard water contains 244 ppm HCO_3^-, 144 ppm SO_4^{2-} and 71 ppm Cl^- ions as their calcium salts. Find the weight of CaO of 80% purity required to remove HCO_3^- from 20,000 litres of this water. If the Ca^{2+} ions in the treated water are completely exchanged with hydrogen ions, what would be its pH? (Assume $CaCO_3$ to be completely insoluble in water and the dissolved CO_2 is negligible).

Solution

No. of moles of $HCO_3^- = \dfrac{244\times10^{-3}\times20{,}000}{61} = 80$

No. of moles of HCO_3^- as $Ca(HCO_3)_2 = 40$

$Ca(HCO_3)_2 + CaO \rightarrow 2CaCO_3 + H_2O$

No. of moles of CaO required = 40

Since the purity of CaO is only 80%, the actual amount of CaO required = 50 moles = 2.8 kg

No. of moles of $SO_4^{2-} = \dfrac{144\times10^{-3}\times20{,}000}{96} = 30$

No. of moles of $Cl^- = \dfrac{71\times10^{-3}\times20{,}000}{35.5} = 40$

After the removal of $Ca(HCO_3)_2$ as $CaCO_3$, the water contains 30 moles of $CaSO_4$ and 20 moles of $CaCl_2$

$\therefore$ Total no. of moles of Ca^{2+} in solution = 30 + 20 = 50 moles

When this is exchanged with H^+, total no. of moles of $H^+ = 2 \times 50 = 100$

$\therefore$ Molar concentration of $H^+ = \dfrac{100}{20000} = 5\times10^{-3} \Rightarrow$ pH = 2.3

Chlorides

Magnesium chloride exists as $MgCl_2.6H_2O$. When the mineral carnallite ($KCl.MgCl_2.6H_2O$) is powdered, boiled with water and cooled, KCl crystallizes out while $MgCl_2$ remains in the solution. When this solution is concentrated, $MgCl_2.6H_2O$ separates out. Anhydrous $MgCl_2$ is obtained by passing dry Cl_2 over heated metal or over a heated mixture of MgO + coke.

$$Mg + Cl_2 \rightarrow MgCl_2$$

$$MgO + C + Cl_2 \rightarrow MgCl_2 + CO$$

Properties

(i) $MgCl_2.6H_2O \xrightarrow[\Delta]{} MgO + 5H_2O + 2HCl$

(ii) $MgCl_2.6H_2O \xrightarrow[\text{heat}]{\text{dry HCl gas}} MgCl_2 + 6H_2O$ (this is also preparation of anhydrous $MgCl_2$)

(iii) $\underset{\text{saturated solution}}{MgCl_2.6H_2O} + 5MgO + H_2O \rightarrow \underset{\text{Sorel cement}}{MgCl_2.5MgO.xH_2O} + (6\text{-}x)H_2O$

$CaCl_2$ is deliquescent and is used in desiccators. $CaCl_2$ is not used for drying NH_3 because $CaCl_2$ forms an addition product with NH_3 viz., $CaCl_2.8NH_3$.

CONCEPT STRAND

Concept Strand 10

How is anhydrous $MgCl_2$ prepared from $MgCl_2.6H_2O$?

Solution

By heating $MgCl_2.6H_2O$ in a current of HCl gas. The presence of HCl gas checks up the hydrolysis of magnesium chloride by its own water of crystallization.

Sulphates

Magnesium sulphate occurs as *kieserite* ($MgSO_4.H_2O$), *epsom salt* ($MgSO_4.7H_2O$) and *kainite* ($KCl.MgSO_4.3H_2O$). Calcium sulphate occurs in nature as anhydrite ($CaSO_4$) and *gypsum* ($CaSO_4.2H_2O$). Gypsum on heating at 325K form a hemihydrate called *plaster of Paris*, $CaSO_4.\frac{1}{2}H_2O$. When

made into a paste, plaster of paris sets to a hard mass with expansion in volume. The final product is $CaSO_4.2H_2O$.

Preparation

(i) $Ca(OH)_2 + H_2SO_4 \rightarrow CaSO_4 + 2H_2O$

(ii) $CaCl_2 + Na_2SO_4 \rightarrow CaSO_4 + 2NaCl$

(iii) $\underset{\text{gypsom}}{CaSO_4.2H_2O} \xrightarrow{150°C}$.

$$\underset{\text{plaster of paris}}{CaSO_4.\frac{1}{2}H_2O} \xrightarrow{200°C} \underset{\text{Dead burnt plaster}}{CaSO_4}$$

(iv) $MgCO_3 + H_2SO_4 \rightarrow MgSO_4 + CO_2 + H_2O$

(v) $MgSO_4.7H_2O \xrightarrow{150°C} MgSO_4.H_2O \xrightarrow{227°C} MgSO_4$

Properties

(i) $MgSO_4 \xrightarrow[\Delta]{900°C} MgO + SO_3$

(ii) $CaSO_4 \xrightarrow[\Delta]{1100°C} CaO + SO_3$

(iii) $CaSO_4 + 4C \xrightarrow[\Delta]{} CaS + 4CO$

The increasing order of the thermal stability of alkaline earth metal sulphates is $BeSO_4 < MgSO_4 < CaSO_4 < SrSO_4 < BaSO_4$.

Solubility of sulphates decreases down the group, $BeSO_4$ and $MgSO_4$ are soluble in water while sulphates of Ca, Sr and Ba are insoluble . The reason is the decrease of hydration energy as the size of cation increases.

CONCEPT STRAND

Concept Strand 11

How will you separate $CaSO_4$ from a solid mixture of $CaSO_4$ and $MgSO_4$?

Solution

$MgSO_4$ is water soluble, where as $CaSO_4$ is insoluble. They are separated by extraction with water.

Additional points

Lithium cannot be stored in kerosene oil because it is very light and floats over surface. Hence it is kept wrapped in paraffin wax. Other alkali metals can be stored in kerosene oil.

Microcosmic salt, $Na(NH_4)HPO_4.4H_2O$ is sometimes used in place of borax for performing bead test for coloured basic radicals.

All alkali metals except Li dissolve in Hg forming amalgams with the evolution of heat. These amalgams are used as reducing agents in organic chemistry.

Lithium stearate $C_{17}H_{35}COOLi$ (lithium soap) is used in making automobile grease.

Alkali metal halides react with halogens and inter halogens forming ionic polyhalides.

$$KI + I_2 \rightarrow K[I_3]$$

$$KBr + ICl \rightarrow K[BrICl]$$

$$KF + BrF_3 \rightarrow K[BrF_4]$$

Li is alloyed with Pb to make white metal, which is used to make bearings for motor engines.

Sodium-lead (Na-Pb) alloy was used in the preparation of antiknock compound $Pb(C_2H_5)_4$.

Armour plates are made from Li-Mg alloy.

Liquid Na and sodium-potassium alloy are used as coolants in nuclear reactors.

Large amounts of Na_2SO_4 are used to make soaps, detergents, paper, textiles and glass.

Li metal is used as a scavenger in the metallurgical operations to remove oxygen and nitrogen because Li reacts with O_2 and N_2.

NaOCl is used as a bleach and a disinfectant. $NaHCO_3$ is called baking soda and used in baking powder.

A granular material prepared by mixing quick lime and NaOH solution, followed by heating to dryness, is known as soda lime. It is used as an absorbent for CO_2 and hence used in organic chemistry for decarboxylation.

KOH is used to make potassium phosphates and also soft soaps (potassium stearate), both of which are used in liquid detergent.

KNO_3 is used in most of the explosives. 95% of potassium salts are used as fertilizers for plants.

KBr finds extensive use in photography to manufacture AgBr.

Some complexes of alkali metals with crown ethers are known. The term cryptand comes from the fact that metal ion remains hidden in the structure.

Sodium vapour is used to fill street lamps.

Cs is used in vapour form in the atomic clock.

High thermal conductivity of Na makes it a very effective heat transfer medium in atomic reactors.

The stability of hydrides decreases in the order, LiH > NaH > KH > RbH > CsH

Sodium peroxide is used as a source of oxygen under the name 'oxone'.

A mixture of Na_2CO_3 and K_2CO_3 is called 'fusion mixture'.

Pure NaCl is not hygroscopic. It shows hygroscopic nature due to traces of impurities of $MgCl_2$ and $CaCl_2$.

A mixture of Na_2O_2 and dil.HCl is called 'soda bleach'.

A mixture of Na_2CO_3 and CaS is called 'black ash'.

'Magnesia cement' or 'Sorel's cement' is $MgCl_2.5MgO.xH_2O$. It is prepared by adding water to an intimate mixture of MgO and $MgCl_2$.

A mixture of calcium cyanamide and carbon is known 'nitrolim' and is used as nitrogenous fertilizer.

Magnesium perchlorate $Mg(ClO_4)_2$ is used as a drying agent under the name 'anhydrone'.

Ca^{2+} and Mg^{2+} ions in a solution and hardness of water can be estimated volumetrically by EDTA.

The main component of tooth enamel is hydroxyapatite, $Ca_5(PO_4)_3OH$ or $3Ca_3(PO_4)_2.Ca(OH)_2$.

In Castner-Kellner cell for the manufacture of NaOH, iron and mercury acts as cathode, graphite block acts as anode. Since hydrogen has high over voltage at mercury cathode, sodium is preferentially discharged forming amalgam with mercury.

CaH_2 is known as hydrolith.

$BaSO_4$ is used for X-ray imaging of digestive system.

Chlorophyll contains Mg.

A solution of $Ba(OH)_2$ is known as baryta and is used to detect CO_2 in place of lime water.

$Be(OH)_2$ is amphoteric where as hydroxides of other elements of Group II are basic.

Ca is the most abundant alkaline earth element.

BeO when heated with C at about 2000°C Be_2C is formed. The metals Mg, Ca, Sr and Ba form carbides of the formula MC_2.

BeH_2 is covalent and has polymeric structure involving three-centre H-bridges.

CONCEPT STRANDS

Concept Strand 12

When 5 g of a colourless solid (A) is heated, it gives 1.36 g of a colourless compound (B), (which is slightly soluble in water) along with 3.1 g of a brown gas (C) and a colourless paramagnetic gas (D). The metal salt does not give any colour in the flame test. Identify (A) – (D).

Solution

Compound (A) on heating gives a brown gas (C) which should be NO_2 and a colourless paramagnetic gas (D) which should be oxygen. Hence, (A) should be a nitrate compound, which upon heating gives metal oxide, NO_2 and O_2. Since the metal nitrate, on heating gives two gases, NO_2 and O_2, the metal ion cannot be a monovalent ion except lithium. It should be a multivalent ion. Since the metal salt does not give any colour to the flame, it cannot be lithium. The metal oxide obtained is slightly soluble in water. So it should be a divalent metal.

$\therefore \quad \underset{(A)}{2M(NO_3)_2} \rightarrow \underset{(B)}{2MO} + \underset{(C)}{4NO_2} + \underset{(D)}{O_2}$

In the above reaction 0.54 g of O_2 is liberated by 5 g of metal nitrate.

$\therefore$ 32 g of O_2 is liberated by $\left(\frac{5}{0.54}\times 32\right)$ = 296.3 g of metal nitrate

$\therefore$ Molecular weight of $M(NO_3)_2$ is $\frac{296.3}{2}$ = 148.14 g

$\therefore$ Atomic weight of metal = 148.15 – 28 – 96 = 24.15 g

Hence, the metal is Mg and the metal salt is magnesium nitrate.

$2Mg(NO_3)_2 \rightarrow 2MgO + 4NO_2 + O_2$

So, A is $Mg(NO_3)_2$, B is MgO, C is NO_2 and D is O_2

Concept Strand 13

Complete and balance the following equations.

(i) $Cl^- + K_2Cr_2O_7 + H_2SO_4 \rightarrow$ ___ + ___ + $K_2SO_4 + H_2O$

(ii) $NaOH +$ ___ + $NH_3 \rightarrow$ ___ + KI + NaI + H_2O.

(iii) ___ + ___ + $H_2SO_4 \rightarrow K_2SO_4 + CrO_2Cl_2 + BaSO_4$ + ___

Solution

(i) $4Cl^- + K_2Cr_2O_7 + 3H_2SO_4 \rightarrow 2CrO_2Cl_2 + 2SO_4^{2-} + K_2SO_4 + 3H_2O$

(ii) $3NaOH + 2K_2HgI_4 + NH_3 \rightarrow HgO.Hg(NH_2)I + 4KI + 3NaI + 2H_2O$

(iii) $2BaCl_2 + K_2Cr_2O_7 + 3H_2SO_4 \rightarrow K_2SO_4 + 2BaSO_4 + 2CrO_2Cl_2 + 3H_2O$

Concept Strand 14

A certain amount of anhydrous $CuSO_4$ is dissolved in water and made upto 500 mL. From the made up solution, 25 mL is added to excess potassium iodide and the liberated iodine is titrated against 0.5 M $Na_2S_2O_3$. It consumes 10 mL of thio sulphate. Calculate the amount of $CuSO_4$ dissolved in 500 mL.

Solution

$2Cu^{2+} + 4I^- \rightarrow 2CuI + I_2$

$2S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$

5 millimoles of thio ≡ 5 millimoles of Cu^{2+} (or) 5 millimoles of $CuSO_4$

∴ No. of millimoles in 500 mL = 5 × 20

∴ Weight of $CuSO_4$ = 100 × 0.1595 g = 15.95 g.

Concept Strand 15

In an industrial operation, 25000 kg of sodium bisulphate is prepared from 46.8% (m / v) brine and 5 M sulphuric acid. Calculate the amount of brine and sulphuric acid required.

Solution

$NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$

No. of moles of $NaHSO_4 = \frac{25\times10^6}{120} = 2.08\times10^5$

2.08×10^5 moles of $NaHSO_4 \equiv 2.08 \times 10^5$ moles of NaCl ≡ 2.08×10^5 moles of H_2SO_4

No. of moles of NaCl in 1 L brine $= \frac{468}{58.5} = 8$

∴ The volume of brine required = 26000 L

The volume of 5 M H_2SO_4 required = 41600 L

Concept Strand 16

A white compound (A) on strong heating gives a basic oxide (B) and a colourless gas (C). When (C) is passed into ammoniated brine solution, it gives a white precipitate (D). (A) is soluble in dil.HCl and this solution on treatment with $Ca(OH)_2$ gives a precipitate (E) and a clear solution (F). (F) is made alkaline and $(NH_4)_2C_2O_4$ solution is added to get a precipitate (G). (G) dissolved in dil.H_2SO_4 and also decolourises $KMnO_4$ solution. (E) on strong heating gives (B). The solution of (A) in dil.HCl when mixed with NH_4Cl and Na_3PO_4 solution gives a white precipitate (H). Identify (A) – (H) and give all the reactions involved.

Solution

$\underset{(A)}{MgCO_3} \xrightarrow{\Delta} \underset{(B)}{MgO} + \underset{(C)}{CO_2}$

$NH_3 + NaCl + \underset{(C)}{CO_2} + H_2O \rightarrow$

$\underset{(D)}{NaHCO_3} + NH_4Cl$

In presence of NaCl, $NaHCO_3$ is insoluble in water, due to common - ion effect

$\underset{(A)}{MgCO_3} + 2HCl \rightarrow MgCl_2 + CO_2 + H_2O$

$MgCl_2 + Ca(OH)_2 \rightarrow \underset{(E)}{Mg(OH)_2} + \underset{(F)}{CaCl_2}$

$\underset{(F)}{CaCl_2} + (NH)_2C_2O_4 \rightarrow \underset{(G)}{CaC_2O_4} + 2NH_4Cl$

$\underset{(G)}{CaC_2O_4} + H_2SO_4 \rightarrow CaSO_4 + H_2C_2O_4$

$5C_2O_4^{2-} + 2MnO_4^- + 16H^+ \rightarrow \underset{\text{colourless}}{2Mn^{2+}} + 5CO_2 + 8H_2O$

$\underset{(E)}{Mg(OH)_2} \xrightarrow{\Delta} \underset{(B)}{MgO} + H_2O$

$MgCl_2 + NH_4Cl + Na_3PO_4 \rightarrow \underset{(H)}{Mg(NH_4)PO_4} + 3NaCl.$

PRINCIPLES OF INORGANIC QUALITATIVE ANALYSIS

Introduction

An inorganic salt is made up of two ions—a positively charged ion called cation or basic radical and a negatively charged ion called anion or acidic radical.

The cations and anions in a simple salt or a mixture of salts can be identified by a process called qualitative analysis.

REACTIONS OF ANIONS

The anions discussed in this unit are sulphide (S^{2-}), chloride(Cl^-), bromide (Br^-), Iodide (I^-), Nitrate (NO_3^-), Sulphate (SO_4^{2-}). The anions are divided into three classes.

(a) Class A

These are anions which produce a gas when treated with dilute acid (HCl or H_2SO_4)

E.g., CO_3^{2-}, S^{2-}, SO_3^{2-}, NO_2^-, $S_2O_3^{2-}$ etc.

(b) Class B

These are anions which produce a gas or vapour when treated with concentrated H_2SO_4

E.g., F^-, Cl^-, Br^-, I^-, NO_3^-, CH_3COO^- etc.

(c) Class C

These are anions which produce a characteristic precipitate with a reagent.

E.g., SO_4^{2-}, PO_4^{3-}, CrO_4^{2-} etc.

Typical reactions of the various anions are given below.

Reactions of sulphide

(i) Solubility

All sulphides are insoluble in water except those of alkali metals.

(ii) Action of dil. HCl

Hydrogen sulphide gas is evolved when a sulphide is treated with dil.HCl or dil.H_2SO_4 H_2S is a colourless gas with smell of rotten eggs. It turns lead acetate paper (filter paper strip dipped in lead acetate) black.

$$FeS + 2HCl \rightarrow FeCl_2 + H_2S$$

$$H_2S + Pb(OCOCH_3)_2 \rightarrow PbS\downarrow + 2CH_3COOH$$

Black

(iii) Action of sodium nitroprusside

Water soluble sulphides produce a purple (violet) colour with sodium nitroprusside solution in presence of alkali.

$$Na_2S + Na_2[Fe(CN)_5NO] \rightarrow Na_4[Fe(CN)_5NOS]$$

purple

Similarly, H_2S gas turns a filter paper dipped in sodium nitroprusside solution made alkaline with NaOH violet.

Reaction of Chloride

(i) Solubility in water

Chlorides are generally soluble in water except those of, silver, lead, mercurous and cuprous ($PbCl_2$ is soluble in hot water).

(ii) Action of conc. H_2SO_4

Chlorides produces a colourless fuming gas in the cold or when slightly warmed. The gas is hydrogen chloride.

$$CaCl_2 + H_2SO_4 \rightarrow CaSO_4 + 2HCl\uparrow$$

Fumes become thick when blown across the mouth of the test tube due to moisture in exhaled air. It also produces thick white fumes (NH_4Cl crystals) with a glass rod dipped in NH_4OH.

$$HCl + NH_3 \rightarrow NH_4Cl$$

(iii) Action of conc. H_2SO_4 + MnO_2

When a chloride is warmed with conc. H_2SO_4 and manganese dioxide chlorine is evolved as a greenish yellow gas.

$$H_2SO_4 + Cl^- \rightarrow HCl + HSO_4^-$$

$$2HCl + MnO_2 + H_2SO_4 \rightarrow Cl_2\uparrow + 2H_2O + MnSO_4$$

(iv) Action of silver nitrate

A solution of chloride in water or dil.HNO_3 produces a curdy white ppt of AgCl.

$$BaCl_2 + 2AgNO_3 \rightarrow 2AgCl\downarrow + Ba(NO_3)_2$$

AgCl ppt is insoluble in dil. HNO_3 but readily soluble in NH_4OH

$$AgCl + 2NH_4OH \rightarrow \underset{\text{Silver ammine complex (soluble)}}{[Ag(NH_3)_2]Cl} + 2H_2O$$

AgCl is reprecipitated if the above solution is acidified with dil HNO_3.

$$[Ag(NH_3)_2]Cl + 2HNO_3 \rightarrow AgCl\downarrow + 2NH_4NO_3$$

(v) Chromyl chloride test

When a solid chloride is heated with conc. H_2SO_4 and potassium dichromate, deep red vapours of chromyl chloride CrO_2Cl_2 are evolved. When these vapours are passed into NaOH solution a yellow solution of Na_2CrO_4 is formed when this solution is acidified with acetic acid and lead acetate is added, a yellow precipitate of $PbCrO_4$ is formed.

$$2MgCl_2 + K_2Cr_2O_7 + 3H_2SO_4 \xrightarrow{\Delta} K_2SO_4 + 2MgSO_4 + 3H_2O + \underset{\text{red vapours}}{2CrO_2Cl_2}$$

$$CrO_2Cl_2 + 4NaOH \rightarrow 2NaCl + \underset{\text{(yellow solution)}}{Na_2CrO_4} + 2H_2O$$

$$Na_2CrO_4 + (CH_3COO)_2Pb \rightarrow \underset{\text{(yellow ppt.)}}{PbCrO_4\downarrow} + 2CH_3COONa$$

Reactions of Bromide

(i) Solubility in water

All bromides are soluble in water except those of Hg_2^{2+}, Cu^+, Ag^+. $PbBr_2$ is insoluble in cold water but soluble in hot water.

(ii) Action of con. H_2SO_4

Bromides produce hydrogen bromide a colourless gas fuming in moist air when treated with conc. H_2SO_4 in the cold. Slowly the gas turns yellowish and then reddish brown due to Br_2 produced by the oxidation of HBr by H_2SO_4 and the solution turns red due to dissolved Br_2.

$$ZnBr_2 + H_2SO_4 \rightarrow ZnSO_4 + \underset{\text{white fumes}}{2HBr\uparrow}$$

$$2HBr + H_2SO_4 \rightarrow 2H_2O + \underset{\text{red fumes}}{SO_2} + Br_2$$

These reactions are accelerated by warming.

(iii) Con. H_2SO_4 and MnO_2

When warmed with conc. H_2SO_4 and MnO_2 reddish brown vapours of bromine are evolved profusely.

$$ZnBr_2 + MnO_2 + 2H_2SO_4 \rightarrow Br_2 + ZnSO_4 + MnSO_4 + 2H_2O$$

(iv) Action of silver nitrate

A solution of a bromide in water or dil.HNO_3 produce a pale yellow ppt with silver nitrate solution. The precipitate is insoluble in dil.HNO_3 and soluble in a large excess of NH_4OH.

$$NaBr + AgNO_3 \rightarrow \underset{\text{(Pale yellow)}}{AgBr} + NaNO_3$$

$$AgBr + \underset{\text{excess}}{2NH_4OH} \rightarrow [Ag(NH_3)_2]Br + 2H_2O$$

(v) Action of Cl_2 water

When chlorine water is added drop wise to a solution of bromide, the solution turns orange red due to the displacement of Br_2 by Cl_2. If carbon tetrachloride or chloroform is added to this solution and shaken, the solvent layer turns reddish brown due to extraction of Br_2 to solvent.

$$2NaBr + Cl_2 \rightarrow 2NaCl + Br_2$$

If excess of chlorine water is added solution turns pale yellow due to formation of bromine monochloride.

$$Br_2 + Cl_2 \rightarrow 2BrCl$$

Reactions of Iodide

(i) Solubility

All iodides are soluble in water except those of Ag^+, Hg_2^{2+}, Hg^{2+}, Cu^+ and Pb^{2+}.

(ii) Action of con. H_2SO_4

When an iodide is treated with conc.H_2SO_4 in the cold or when warmed, white fumes of HI along with violet fumes of iodine are evolved. Iodine condenses on the cooler upper sides of test tube as a dark brown deposit. Solution turns brown due to dissolved iodine.

$$2NaI + H_2SO_4 \rightarrow Na_2SO_4 + 2HI$$

$$2HI + H_2SO_4 \rightarrow 2H_2O + SO_2 + I_2$$

(iii) Con. H_2SO_4 and MnO_2

When warmed with conc.H_2SO_4 and MnO_2 there is profuse evolution of violet vapours of I_2.

$$2KI + MnO_2 + 2H_2SO_4 \rightarrow K_2SO_4 + MnSO_4 + 2H_2O + I_2$$

(iv) Action of $AgNO_3$

A yellow ppt of AgI is formed. It is insoluble in dil.HNO_3 and NH_4OH.

$$KI + AgNO_3 \rightarrow \underset{\text{Yellow}}{AgI\downarrow} + KNO_3$$

(v) Action of Chlorine water

When Cl_2 water is added to an aqueous solution of an iodide in drops, solution turns brown due to liberated iodine. When this solution is shaken with CCl_4 or $CHCl_3$ the organic layer turns violet due to extraction of I_2.

$$2KI + Cl_2 \rightarrow 2KCl + I_2\uparrow$$

Reaction of nitrate (NO_3^-)

(i) Solubility

All nitrates are soluble in water.

(ii) Action of heat

Alkali metal nitrates evolve oxygen and form solid nitrites. Ammonium nitrate decompose to N_2O and H_2O. Noble metal (Ag, Au, Hg etc.,) nitrates decompose to metal, $NO_2 + O_2$. Other nitrates decompose to O_2, NO_2 and metal oxide.

$$NaNO_3 \rightarrow 2NaNO_2 + O_2$$

$$NH_4NO_3 \rightarrow N_2O\uparrow + 2H_2O$$

$$2AgNO_3 \rightarrow 2AgNO_2 + O_2$$
$$\downarrow$$
$$2Ag + 2NO_2$$

$$Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2\uparrow + O_2\uparrow$$

(iii) Action of hot conc.H_2SO_4

When a nitrate is heated with conc. H_2SO_4 first white fumes of HNO_3 are formed. Slowly reddish brown vapours of NO_2 appear. Solution remains colourless (distinction from Br^-)

$$Pb(NO_3)_2 + H_2SO_4 \rightarrow PbSO_4 + 2HNO_3\uparrow$$

$$4HNO_3 \xrightarrow{\Delta} 2H_2O + 4NO_2 + O_2$$

(iv) Action of Copper turning and conc. H_2SO_4

Reddish brown fumes of NO_2 are evolved profusely when a nitrate is heated with conc.H_2SO_4 and copper turnings. (or paper balls). The solution turns blue due to $Cu(NO_3)_2$.

$$Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$$

(v) Brown ring test

A solution of the nitrate in water or dil.H_2SO_4 is mixed with an equal volume of freshly prepared ferrous sulphate solution. Conc.H_2SO_4 is added slowly down the sides of the test tube to form a bottom layer. A brown ring is formed at the junction of the two liquids due to the formation of $FeSO_4$. NO or more correctly $[Fe(H_2O)_5NO]SO_4$.

$$KNO_3 + H_2SO_4 \rightarrow HNO_3 + KHSO_4$$

$$6FeSO_4 + 3H_2SO_4 + 2HNO_3 \rightarrow 3Fe_2(SO_4)_3 + 2NO + 4H_2O$$

$$FeSO_4 + NO \rightarrow FeSO_4.NO \text{ (brown complex)}$$

(vi) Action of Al powder and NaOH

When a nitrate is warmed with Al powder or zinc dust and sodium hydroxide solution ammonia is evolved which is detected by smell and which turns moist red litmus paper blue.

$$Zn + 2NaOH \rightarrow Na_2ZnO_2 + 2[H]$$

$$NaNO_3 + 8[H] \rightarrow NaOH + 2H_2O + NH_3$$

Reactions of Sulphate (SO_4^{2-})

(i) Solubility

All sulphates are soluble in water except those of Ca^{2+}, Sr^{2+}, Ba^{2+} and Hg^{2+}. Sulphates of Ca^{2+} and Hg^{2+} are slightly soluble.

(ii) Action of $BaCl_2$

A white crystalline precipitate of $BaSO_4$ is formed when a sulphate is mixed with $BaCl_2$ solution. The ppt is insoluble in dil. HCl and dil. HNO_3.

(iii) Action of lead acetate solution

When lead acetate is added to a solution of sulphate white ppt of $PbSO_4$ is formed. The ppt is soluble in NaOH, ammonium acetate and hot conc.H_2SO_4.

$$MgSO_4 + Pb(CH_3COO)_2 \rightarrow Mg(CH_3COO)_2 + PbSO_4\downarrow$$

Preparation of Sodium carbonate extract of anions

The anion of a salt can be brought into aqueous solution as the sodium salt, by extracting the salt with a concentrated solution of Na_2CO_3. Moreover, this eliminates interfering cations of the salt.

Sodium carbonate extract is prepared by boiling a little (1 g) of the salt with thrice its weight (3 g) of sodium carbonate in water (15 – 20 ml) for about 10 minutes. For example, if the salt is $CaSO_4$ a double decomposition reaction takes place producing Na_2SO_4 and $CaCO_3$.

$$CaSO_4 + Na_2CO_3 \rightarrow Na_2SO_4 + CaCO_3$$

The principle behind the process is that all sodium salts are water soluble and carbonates of all metals other than alkali metals are the most insoluble salts of these metals. The digested solution is filtered and the filtrates is called sodium carbonate extract.

Tests for anion in the sodium carbonates extract

1. Test for halides with $AgNO_3$

1 ml of the extract is acidified with dil. HNO_3 till effervescence (CO_2) ceases and boiled to remove dissolved CO_2. $AgNO_3$ is added.

(i) Curdy white ppt. readily soluble in NH_4OH confirms chloride.
(ii) Pale yellow ppt. soluble in a large excess of NH_4OH confirms bromide.
(iii) Yellow ppt. insoluble in NH_4OH confirms iodide.

2. Test for nitrate (Brown ring test)

A little of the extract is acidified with dil. H_2SO_4, mixed with freshly prepared ferrous sulphate solution and conc. H_2SO_4 is added slowly to form a bottom layer. A brown ring between the junction of the two liquids confirms nitrate.

3. Test for sulphate

A little of the extract is acidified with dilute HCl and $BaCl_2$ is added. White ppt. confirms sulphate.

ANALYSIS OF CATIONS

The cations discussed in this unit are silver(Ag^+), lead (Pb^{2+}), Mercurous (Hg_2^{2+}), cupric (Cu^{2+}), Bismuth (Bi^{3+}), Ferric (Fe^{3+}), chromium (Cr^{3+}) Aluminium (Al^{3+}), Calcium (Ca^{2+}), Barium (Ba^{2+}), Zinc (Zn^{2+}), Manganese (Mn^{2+}), and Magnesium (Mg^{2+}).

Dry reactions

1. Flame test

Volatile salts impart characteristic colour to the non luminous flame of Bunsen burner. A paste of a sample is made with conc.HCl and taken at the tip of a preheated platinum wire. When the platinum wire with the sample paste in the tip is introduced into the non luminous part of the flame, there is a change in the colour of the flame characteristic of the cation.

Flame colour	Cation
Brick red	Calcium
Apple green	Barium
Green or blue	Copper
Golden yellow	Sodium
Lilac(light violet)	Potassium
Crimson red	Strontium

2. Borax bead test

When a small quantity of borax ($Na_2B_4O_7.10H_2O$) is heated at the end of a platinum wire, it loses water of crystallization and forms a transparent glassy solid known as borax bead.

$$Na_2B_4O_7.10H_2O \xrightarrow[\text{strongly}]{\text{heat}}$$

$$Na_2B_4O_7 \xrightarrow[\text{heating}]{\text{further}} \underbrace{2NaBO_2 + B_2O_3}_{\text{Borax bead}}$$

On heating a small crystal of a salt with the glassy bead, a coloured metaborate is formed in the oxidising flame and a different colour is produced in the reducing flame. For example

$$CuO + B_2O_3 \rightarrow$$

$$\xrightarrow[\text{flame}]{\text{oxidising}} \underset{\text{blue}}{Cu(BO_2)_2} \xrightarrow[\text{flame}]{\text{reducing}} \underset{\text{colourless}}{CuBO_2} \longrightarrow \underset{\text{red}}{\text{Cu}}$$

$$NiO + B_2O_3 \rightarrow \underset{\text{Brown}}{Ni(BO_2)_2} \longrightarrow \underset{\text{grey}}{\text{Ni}}$$

Metal	Colour of the bead	
	Oxidizing Flame	Reducing flame
Copper	Blue or green $[Cu(BO_2)_2]$	Brown red (Cu)
Iron	Yellowish brown $[Fe(BO_2)_3]$	Green $[Fe(BO_2)_2]$
Chromium	Green $[Cr(BO_2)_3]$	Green $[Cr(BO_2)_3]$
Managanese	Violet $[Mn(BO_2)_3]$	Colourless $[Mn(BO_2)_2]$
Nickel	Violet or reddish brown $[Ni(BO_2)_2]$	Grey (Ni)

3. Charcoal cavity test

The given salt is heated with fusion mixture (a mixture of Na_2CO_3 and K_2CO_3) in a charcoal cavity by a reducing flame. The salts decompose to form the corresponding metal carbonates. The metal carbonates undergo decomposition to give the corresponding metal oxides. The metal oxides formed get further reduced to the metal by means of carbon to give a metallic bead.

$$CuCl_2 + Na_2CO_3 \rightarrow CuCO_3 + 2NaCl$$

$$CuCO_3 \rightarrow CuO + CO_2$$

$$CuO + C \rightarrow Cu + CO$$

Red bead of copper is formed.

$$PbCl_2 + Na_2CO_3 \rightarrow PbCO_3 + 2NaCl$$

$$PbCO_3 \xrightarrow{\Delta} PbO + CO$$

$$PbO + C \rightarrow Pb + CO$$

PbO is brown when hot and yellow when cold. Finally Pb is obtained as a grey metallic bead.

Zn^{2+} forms ZnO which is yellow while hot and white when cold. No metallic bead is formed.

Red metallic bead is formed in the case of Cu^{2+}, shining white bead is formed in the case of Ag^+.

1. Reactions of mercurous ion (Hg_2^{2+})

(i) Action of dil.HCl

White ppt of Hg_2Cl_2.

$$Hg_2(NO_3)_2 + 2HCl \rightarrow Hg_2Cl_2 + 2HNO_3$$

Hg_2Cl_2 is insoluble in dilute acids.

(ii) Action of NH_4OH on Hg_2Cl_2

Hg_2Cl_2 undergoes disproportionation to form a grey ppt of mercury and mercuric amidochloride.

$$Hg_2Cl_2 + 2NH_4OH \rightarrow \underbrace{Hg(NH_2)Cl + Hg}_{(\text{grey ppt})} + 2H_2O + NH_4Cl$$

(iii) Action of aquaregia on above ppt.

The grey ppt dissolves to form $HgCl_2$ which produces silky white precipitate of Hg_2Cl_2 with stannous chloride (aquaregia is a mixture of conc. HNO_3 and con. HCl in the ratio 1 : 3).

$$3HCl + HNO_3 \rightarrow NOCl + 2H_2O + Cl_2$$

$$Hg + Cl_2 \rightarrow HgCl_2$$

$$Hg_2Cl_2 + Cl_2 \rightarrow 2HgCl_2$$

$$2HgCl_2 + SnCl_2 \rightarrow \underset{\text{(silky white ppt)}}{Hg_2CI_2 \downarrow} + SnCl_4$$

With excess of $SnCl_2$ the white ppt turns grey due to Hg

$$Hg_2Cl_2 + SnCl_2 \rightarrow \underset{\text{(Grey of black ppt)}}{2Hg \downarrow} + SnCl_4$$

2. Reactions of silver (Ag^+)

(i) Action of dil.HCl

The white precipitate of AgCl is formed.

$$AgNO_3 + HCl \rightarrow AgCl\downarrow + HNO_3$$

AgCl is insoluble in dil.HNO_3 but soluble in ammonia.

(ii) Action of NH_4OH on AgCl

The AgCl ppt dissolves as diammine silver complex.

$$AgCl + 2NH_4OH \rightarrow [Ag(NH_3)_2]Cl + H_2O$$

(iii) Action of KI

Silver salts produce a yellow ppt of AgI which is insoluble in dil. HNO_3 and NH_4OH

$$AgNO_3 + KI \rightarrow \underset{(yellow)}{AgI\downarrow} + KNO_3$$

(iv) Action of K_2CrO_4

Silver salts produce a brick red ppt of Ag_2CrO_4 which is soluble in dilute acids and NH_4OH. It is insoluble in water and acetic acid.

$$2AgNO_3 + K_2CrO_4 \rightarrow Ag_2CrO_4 + 2KNO_3$$

3. Reaction of lead (Pb^{2+})

(i) Action of dil. HCl

White precipitate of $PbCl_2$ is produced.

$$Pb(NO_3)_2 + 2HCl \rightarrow PbCl_2\downarrow + 2HNO_3$$

The ppt. of $PbCl_2$ is soluble in hot water but sparingly soluble in cold water.

(ii) Action of dil. H_2SO_4

White ppt of $PbSO_4$ is formed.

$$Pb(NO_3)_2 + H_2SO_4 \rightarrow PbSO_4\downarrow + 2HNO_3$$

The ppt. is soluble in hot conc. H_2SO_4 as hydrogen sulphate.

$$PbSO_4 + H_2SO_4 \rightarrow \underset{\text{Soluble}}{Pb(HSO_4)_2}$$

$PbSO_4$ is also soluble in concentrated (6M) ammonium acetate as an acetate complex.

$$PbSO_4 + 4CH_3COONH_4 \rightarrow \underset{\text{Soluble}}{(NH_4)_2[Pb(CH_3COO)_4]} + (NH_4)_2SO_4$$

(iii) Action of Potassium chromate solution

Yellow precipitate of $PbCrO_4$ is insoluble in acetic acid but partially soluble in NaOH, due to the formation of plumbite complex.

To a small part of the solution containing Pb^{2+} ions potassium chromate solution is added.

$$Pb(NO_3)_2 + K_2CrO_4 \rightarrow \underset{(yellow)}{PbCrO_4\downarrow} + 2KNO_3$$

(iv) Action of KI solution

A yellow precipitate of PbI_2 is formed.

$$Pb(NO_3)_2 + 2KI \rightarrow PbI_2 + 2KNO_3$$

PbI_2 dissolve in hot water to form a colourless solution, and on cooling separates as golden spangles.

(v) Action of dil HCl and H_2S

Black ppt of PbS is formed when acid concentration is < 2M.

$$Pb(NO_3)_2 + H_2S \rightarrow \underset{(black)}{PbS\downarrow} + 2HNO_3$$

PbS is soluble in hot dilute or concentrated HNO_3 or conc. HCl.

$$3PbS + 8HNO_3 \rightarrow 3Pb(NO_3) + 3S + 2NO + 4H_2O$$

4. Reactions of mercuric ion (Hg^{2+})

(i) Action of dil HCl and H_2S

Black ppt of HgS is formed.

$$HgCl_2 + H_2S \rightarrow HgS\downarrow + 2HCl$$

HgS is insoluble in dil. HNO_3

(ii) Action of aquaregia on HgS

HgS dissolves to form $HgCl_2$.

$$3HgS + 2HNO_3 + 6HCl \rightarrow 3HgCl_2 + 2NO + 4H_2O + 3S$$

(iii) Action of $SnCl_2$ on the above solution of $HgCl_2$

A silky white precipitate of Hg_2Cl_2 is formed which turns grey and black with excess of $SnCl_2$.

$$2HgCl_2 + SnCl_2 \rightarrow \underset{\text{silky white}}{Hg_2CI_2} + SnCl_4$$

$$Hg_2Cl_2 + SnCl_2 \rightarrow \underset{\text{black}}{2Hg} + SnCl_2$$

5. Reaction of copper ion (Cu^{2+})

(i) Action of dil HCl and H_2S

Black precipitate of CuS is formed

$$CuSO_4 + H_2S \rightarrow CuS\downarrow + H_2SO_4$$

(ii) Action of dil. HNO_3 on CuS

CuS dissolves in hot dilute HNO_3 to form a blue solution of $Cu(NO_3)_2$

$$3CuS + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 3S + 2NO + 4H_2O$$

(iii) Action of NH_4OH on $Cu(NO_3)_2$

When NH_4OH is added to $Cu(NO_3)_2$ drop wise a pale blue ppt of $Cu(OH)_2$ is formed which dissolves in excess NH_4OH to form an intense blue solution containing tetraammine copper (II) ion

$$Cu(NO_3)_2 + 2NH_4OH \rightarrow \underset{\text{pale blue ppt}}{Cu(OH_2)\downarrow} + 2NH_4NO_3$$

$$Cu(OH)_2 + 4NH_4OH \rightarrow \underset{\text{intense blue solution}}{[Cu(NH_3)_4](OH)_2} + 4H_2O$$

(iv) Action of potassium ferrocyanide

Copper salts produce a chocolate brown precipitate of Copper(II) ferrocyanide which is insoluble in acetic acid. The tetra amine complex may be acidified with acetic acid and treated with pot. Ferrocyanide.

$$[Cu(NH_3)_4]^{2+} + 4CH_3COOH \rightarrow Cu^{2+} + 4CH_3COONH_4.$$

$$Cu^{2+} + K_4Fe(CN)_6 \rightarrow Cu_2[Fe(CN)_6]\downarrow + 4K^+$$

6. Reactions of Bismuth ion (Bi^{3+})

(i) Action of dil HCl and H_2S

Black precipitate of Bi_2S_3

$$2Bi(NO_3)_3 + 3H_2S \rightarrow Bi_2S_3\downarrow + 6HNO_3$$

Bi_2S_3 is soluble in dil.HNO_3 and dil H_2SO_4

(ii) Action of NH_4OH

$Bi(OH)_2$ a white ppt is formed which is insoluble in excess.

$$Bi(NO_3)_3 + 3NH_4OH \rightarrow Bi(OH)_3 + 3NH_4NO_3$$

(iii) Action of excess water

When a solution of $Bi(OH)_3$ in dil HCl is added to excess of water in a beaker the $BiCl_3$ is hydrolysed to a white ppt of bismuth oxychloride BiOCl which produces an opalescence

$$BiCl_3 + H_2O \rightarrow BiOCl\downarrow + 2HCl$$

(iv) Action of alkaline Sodium stannite in $BiCl_3$

A black ppt of Bi is formed.

$$BiCl_3 + 3NaOH \rightarrow Bi(OH)_3 + 3NaCl$$

$$2Bi(OH)_3 + 3Na_2SnO_2 \rightarrow 2Bi\downarrow + 3H_2O + 3Na_2SnO_3$$

7. Reactions of Ferrous ion (Fe^{2+})

Ferrous salts are pale green

(i) Action of NH_4Cl and NH_4OH

Dirty green $Fe(OH)_2$ is incompletely precipitated.

$$FeSO_4 + 2NH_4OH \rightleftharpoons Fe(OH)_2\downarrow + (NH_4)_2SO_4$$

(ii) Action of potassium ferrocyanide

White precipitates of potassium ferrous ferrocyanide

$$FeSO_4 + K_4[Fe(CN)_6] \rightarrow K_2Fe[Fe(CN)_6]\downarrow + K_2SO_4$$

usually the ppt has a blue tinge due to Fe^{3+} in Fe^{2+}

(iii) Action of potassium ferricynade

A dark blue precipitate (Turnbull's blue) is formed. First, ferrocynide ions oxidize Ferrous to ferric ions. Then ferric ferrocyanide is formed.

$$Fe^{2+} + Fe(CN)_6^{3-} \rightarrow Fe^{3+} + Fe(CN)_6^{4-}$$

$$4Fe^{3+} + 3Fe(CN)_6^{4-} \rightarrow Fe_4[Fe(CN)_6]_3$$

[Earlier it was formulated as ferrous ferricyanide $Fe_3[Fe(CN)_6]_2$ wrongly and was named turnbull's blue.]

8. Reactions of Ferric ion (Fe^{3+})

Ferric salts are yellow or brown.

(i) Action of NH4Cl and NH4OH

Reddish brown precipitate of $Fe(OH)_3$ is formed. The ppt is insoluble in excess of reagent, NaOH or sodium peroxide and water $FeCl_3 + 3NH_4OH \rightarrow Fe(OH)_3\downarrow + 3NH_4Cl$

The precipitate dissolves in dil.HCl to form $FeCl_3$.

(ii) Action potassium ferrocyanide

A deep blue precipitate of ferric ferrocyanide (prussian blue) is obtained.

$$4FeCl_3 + 3K_4[Fe(CN)_6] \rightarrow Fe_4[Fe(CN)_6]_3 + 12\ KCl$$

(iii) Action of potassium ferricyanide

A reddish brown solution due to soluble ferric ferricyanide is formed.

$$Fe^{3+} + Fe(CN)_6^{3-} \rightarrow Fe[Fe(CN)_6]$$

(iv) Action of Ammonium thiocynate

A blood red colouration due to ferric thiocyanate.

$$FeCl_3 + 3NH_4\ CNS \rightarrow Fe(CNS)_3 + 3NH_4Cl$$

9. Reaction of Aluminium ion (Al^{3+})

(i) Action of NH4Cl and NH4OH

White gelatinous precipitate of $Al(OH)_3$

$$AlCl_3 + 3NH_4OH \rightarrow Al(OH)_3 + 3NH_4Cl$$

$Al(OH)_3$ is soluble in all strong acids (HCl, H_2SO_4) and strong alkali (NaOH, KOH)

$$Al(OH)_3 + 3HCl \rightarrow AlCl_3 + 3H_2O$$

$$Al(OH)_3 + NaOH \rightarrow Na[Al(OH)_4]$$

$Na[Al(OH)_4]$ is usually formulated as $NaAlO_2 + 2H_2O$

When a solution of $NaAl(OH)_4$ is acidified the $Al(OH)_3$ is precipitated, and a large excess of acid causes the $Al(OH)_3$ to dissolve.

$$Al(OH)^-_4 + H^+ \rightarrow Al(OH)_3\downarrow + H_2O$$

$$Al(OH)_3 + 3H^+ \rightarrow Al^{3+} + 3H_2O$$

Moreover, $Al(OH)_3$ can be precipitated from $NaAl(OH)_4$ by boiling with NH_4Cl

$$NaAl(OH)_4 + NH_4Cl \rightarrow Al(OH)_3\downarrow + NaCl + NH_3\uparrow + H_2O$$

(ii) Cobalt nitrate ash test

The ppt of $Al(OH)_3$ is dissolved in a few drops of conc. HNO_3 and one drop of cobalt nitrate. A filter paper strip dipped in this solution is ignited. A blue ash (Thenard's blue) is produced due to the production of the mixed oxide $CoO.Al_2O_3$ or $CoAl_2O_4$.

(iii) Test with alizarin dye

Add a drop of alizarin solution to a solution of Aluminium salt in the HCl. Add excess of ammonia. Aluminium hydroxide is precipitated which absorbs alizarin to form a red lake. (Lake is a coloured precipitate in a slightly coloured back ground)

(iv) Test with Aluminon reagent

A drop of Aluminon reagent is added to a precipitates of Aluminium hydroxide. [obtained in test (i)]

The dye is absorbed on $Al(OH)_3$ to form a bright red lake.

10. Reactions of Chromium Cr^{3+}

Chromium salts are green or violet.

(i) Action of NH_4Cl and NH_4OH.

A bluish green precipitate of chromic hydroxide is formed.

$$CrCl_3 + 3NH_4OH \rightarrow Cr(OH)_3\downarrow + 3NH_4Cl$$

(ii) Action of sodium peroxide and water on $Cr(OH)_3$

When the precipitated $Cr(OH)_3$ is boiled with water and sodium peroxide it dissolves as yellow Sodium chromate.

$$Na_2O_2 + 2H_2O \rightarrow 2NaOH + H_2O_2$$

$$2Cr(OH)_3 + 3H_2O_2 + 4NaOH \rightarrow 2Na_2CrO_4 + 8H_2O$$

(iii) Action of lead acetate on chromate

When the above solution is acidified with acetic acid and lead acetate is added a yellow ppt of $PbCrO_4$ is formed.

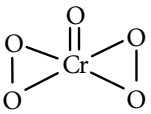
$$Na_2CrO_4\downarrow + Pb(CH_3COO)_2 \rightarrow PbCrO_4\downarrow + 2NaCH_3COO$$

(iv) Action of H_2O_2 in acid medium on Sodium chromate

When the filtrate from reaction (ii) is acidified with dil H_2SO_4, mixed with 2ml ether or amyl alcohol and 1ml H_2O_2, a blue colouration is formed which is extracted into the organic layer by gentle shaking.

$$Na_2CrO_4 + H_2SO_4 + 2H_2O_2 \rightarrow CrO_5 + Na_2SO_4 + 3H_2O$$

The blue colour is due to perchromic acid or chromium peroxide, CrO_5, with the structure,



11. Reactions of Manganese (Mn^{2+})

Manganous salts are pale pink or brown.

(i) Action of NH_4Cl and NH_4OH

Initially no precipitate is formed. But slowly Mn^{2+} is oxidized in alkaline medium by air to Mn^{4+} and dirty white ppt of MnO(OH) is formed

$$2MnSO_4 + 4NH_4OH + O_2 \rightarrow 2MnO(OH)_2 + (NH_4)_2SO_4$$

(ii) Action of NH_4Cl, NH_4OH and H_2S

Flesh coloured (pink) precipitates of MnS

$$MnSO_4 + H_2S \rightarrow MnS\downarrow + H_2SO_4$$

MnS is soluble in dilute HCl

(iii) Action of sodium hydroxide

$Mn(OH)_2$ is precipitated as a white ppt which turns brown due to slow oxidation by air to $MnO(OH)_2$.

$$MnSO_4 + 2NaOH \rightarrow Mn(OH)_2\downarrow + Na_2SO_4$$

$$2Mn(OH)_2 + O_2 \rightarrow 2MnO(OH)_2\downarrow$$

$Mn(OH)_2$ is rapidly converted to brown $MnO(OH)_2$ by treating with bromine water or H_2O_2.

(iv) Action of con. HNO_3 and lead peroxide (permanganic acid test).

When the ppt of $Mn(OH)_2$ or $MnO(OH)_2$ or the original salt (take only a very small amount) is boiled with con. HNO_3 and lead dioxide, diluted with water and let stand supernatant liquid turns purple (violet- red) due to the formation of permanganic acid

$$PbO_2 + 2HNO_3 \rightarrow Pb(NO_3)_2 + H_2O + [O]$$

$$2MnO(OH)_2 + 3[O] \rightarrow \underset{\text{purple}}{2HMnO_4} + H_2O$$

OR

$$2Mn(NO_3)_2 + 3H_2O + 5[O] \rightarrow \underset{\text{purple}}{2HMnO_4} + 4\,HNO_3$$

Oxidation of Mn^{2+} or Mn^{+4} to permanganic acid may also be effected by other oxidizing agents

(a) Potassium peroxidisulphate, $K_2S_2O_8$ with Ag^+ ions as catalyst
(b) Sodium bismuthate, $NaBiO_3$

12. Reactions of Zinc (Zn^{2+}) ion

(i) Action of NH_4Cl, NH_4OH and H_2S

white ppt of ZnS which is soluble dil HCl

$$ZnSO_4 + H_2S \rightarrow ZnS + H_2SO_4$$

(ii) Action of NaOH

A white ppt of $Zn(OH)_2$ soluble in excess NaOH to form Sodium zincate Na_2ZnO_2 or $Na_2[Zn(OH)_4]$

$$ZnSO_4 + 2NaOH \rightarrow Zn(OH)_2 + Na_2SO_4$$

$$Zn(OH)_2 + 2NaOH \rightarrow Na_2[Zn(OH)_4]$$

(iii) Action of H_2S on Sodium Zincate

White ppt of ZnS is formed $Na_2Zn(OH)_4 \rightarrow H_2S \rightarrow ZnS\downarrow$ $2NaOH + 2H_2O$

(iv) Action of $K_4Fe(CN)_6$

When pot. ferrocyanide is added to a zinc salt or the sodium zincate after acidification with acetic acid, a white ppt of zinc ferrocynide is formed.

$$Na_2ZnO_2 + 4HOAc \rightarrow 2NaOAc + Zn(OAc)_2 + 2H_2O$$

$$2Zn(OAc)_2 + K_4Fe(CN)_6 \rightarrow Zn_2Fe(CN)_6 + 4KOAc$$

(v) Ash test

A little of the ppt of ZnS or $Zn(OH)_2$ was heated with con. HNO_3 and a drop of cobalt nitrate. Filter paper dipped in this solution is ignited. Green ash of ZnO.CoO

(Rinmann's green) is produced.

13. Reactions of Barium ion (Ba^{2+})

Barium salts are colourless.

(i) Action of NH_4Cl NH_4OH and $(NH_4)_2CO_3$

White crystalline precipitate of $BaCO_3$ soluble in acetic acid is formed.

$$BaCl_2 + (NH_4)_2CO_3 \rightarrow BaCO_3\downarrow + 2NH_4Cl$$

(ii) Action of K_2CrO_4

The above precipitate is dissolved in hot acetic acid and K_2CrO_4 is added to one portion. Lemon yellow ppt of $BaCrO_4$ insoluble in acetic acid and soluble in mineral acids is formed.

$$(CH_3COO)_2Ba + K_2CrO_4 \rightarrow BaCrO_4\downarrow + 2CH_3COOK$$

(iii) Action of dil H_2SO_4

Dil.H_2SO_4 is added to a solution of $BaCrO_4$ in con.HCl. White crystalline ppt of $BaSO_4$ is formed. It is insoluble in all acids.

$$BaCl_2 + H_2SO_4 \rightarrow BaSO_4\downarrow + 2HCl$$

(iv) Action of ammonium oxalate

White ppt of BaC_2O_4.

$$BaCl_2 + (NH_4)_2C_2O_4 \rightarrow BaC_2O_4\downarrow + 2NH_4Cl$$

14. Reactions of Calcium ion (Ca^{2+})

Calcium salts are colourless.

(i) Action of NH_4Cl, NH_4OH and $(NH_4)_2CO_3$

White crystalline ppt of $CaCO_3$ soluble in acetic acid

$$CaCl_2 + (NH_4)_2CO_3 \rightarrow CaCO_3 + 2NH_4Cl$$

(ii) Action of K_2CrO_4 and acetic acid

The above ppt is dissolved in acetic acid and K_2CrO_4 is added. No ppt is formed since $CaCrO_4$ is soluble in acetic acid. (Distinction from Ba^{2+})

(iii) Action of NH_4OH and ammonium oxalate

White crystalline ppt of calcium oxalate

$$CaCl_2 + (NH_4)_2C_2O_4 \rightarrow 2NH_4Cl + CaC_2O_4\downarrow$$

15. Reactions of magnesium (Mg^{2+})

Magnessium salts are colourless

(i) Action of NH4Cl, NH4OH and disodium hydrogen phosphates

White crystalline ppt of NH_4MgPO_4 is formed slowly.

$Mg(NH_4)PO_4$ has a tendency to form supersaturated solution. Therefore, usually precipitation is effected by scratching the inside of the test tube or by shading

$$MgCl_2 + Na_2HPO_4 + NH_4OH \rightarrow Mg(NH_4)PO_4 + 2NaCl + H_2O$$

(ii) Action of magneson reagent

To salt solution add magneson reagent followed by NaOH till alkaline.

White $Mg(OH)_2$ is precipitated as blue ppt due to absorbed magneson.

(iii) Ash test

When a magnesium salt is boiled with con.HNO_3 and a drop of $Co(NO_3)_2$ and ignited on a filter paper pink ash due to MgO.CoO is obtained

Classification of cations into groups

A mixture of cations in solution may be separated by precipitating one or more cations using a suitable reagent. Depending on the precipitating agent cations (basic radicals) are classified into six groups.

Table 1

Group	Cations	Group reagent
1	Hg_2^{2+}, Ag^+, Pb^{2+}	dil. HCl
2	Pb^{2+}, Cu^{2+}, Bi^{3+}	dil. HCl + H_2S
3	Fe^{3+}, Al^{3+}, Cr^{3+}, Mn^{2+}	NH_4Cl + NH_4OH
4	Mn^{2+}, Zn^{2+}	NH_4Cl + NH_4OH + H_2S
5	Ba^{2+}, Ca^{2+}	NH_4Cl + NH_4OH + $(NH_4)_2CO_3$
6	Mg^{2+}	NH_4Cl + NH_4OH + Na_2HPO_4

Preparation of the solution of the salt or mixture

The following solvents are tried in the order given.

(1) water
(2) dilute HCl
(3) conc. HCl
(4) dilute HNO_3
(5) conc. HNO_3
(6) aquaregia

The best solvent is dil.HCl. If group I is suspected solution is made in dil.HNO_3.

Intergroup separation

Table 2

Dilute HCl is added to the solution of the mixture.
Filter

Residue:
Group I cations as chlorides. Hg_2Cl_2, $PbCl_2$, AgCl

Filtrate : H_2S passed. Filter

Residue:
Group II cations as sulphides PbS, CuS, Bi_2S_3, HgS

Filtrate: H_2S is boiled off. If Fe^{2+} is present boil with a few drops of conc. HNO_3. Add NH_4Cl and excess of NH_4OH. Filter

Residue:
Group III cations as hydroxides $Fe(OH)_3$, $Al(OH)_3$, $Cr(OH)_3$, $MnO(OH)_2$

Filterate: Add NH_4Cl excess NH_4OH and pass H_2S.Filter

Residue:
Group IV cations as sulphides MnS, ZnS

Filtrate : Boil off H_2S. Add NH_4Cl, excess NH_4OH and $(NH_4)_2CO_3$. Filter

Residue:
Group V cations as carbonates $BaCO_3$ $CaCO_3$

Filtrate: Concentrated to a small volume. Add NH_4Cl, NH_4OH and Na_2HPO_4, Mg is precipitated as NH_4MgPO_4

Table 3
Analysis of group I cations

The precipitate obtained in group I ($PbCl_2$), AgCl, Hg_2Cl_2) is boiled with water and filtered.

Residue:
AgCl and Hg_2Cl_2. Excess of NH_4OH is added. Filtered.

Residue:
Hg + $Hg(NH_2)$ Cl
Evaporate to dryness with aquaregia. Extracted with water and add $SnCl_2$ silky white ppt confirms Hg_2^{2+}

Filtrate:
$[Ag(NH_3)_2]Cl$
1. Acidify with dil. HNO_3 white ppt
2. Add KI yellow ppt Ag confirmed

Filtrate:
$PbCl_2$
1. Add K_2CrO_4
Bright yellow ppt.
Add KI yellow ppt
Dissolves in hot water and reprecipitates as golden spangles. Pb^{2+} confirmed.

Analysis of group II cations

H_2S is passed through the filtrate from groupI. CuS, Bi_2S_3, HgS and PbS are precipitated

Table 4 Analysis of group II cations

The precipitate obtained in the second group (PbS, HgS, CuS, Bi_2S_3, all are black) is boiled with dilute HNO_3 and filtered.

Residue:	**Filtrate :** $Pb(NO_3)_2$, $Cu(NO_3)_2$, $Bi(NO_3)_3$.		
(HgS)	dil. H_2SO_4 is added and Filttered		
The residue is evaporated nearly to dryness with aqua regia (1volume. HNO_3 + 3Vol. con.HCl).	**Residue:**	**Filtrate:**	
Residue is extracted with water and added to stannous chloride – silky white ppt shows Hg^{2+}	White $PbSO_4$.	Blue: Copper present	
	The residue is dissolved in conc.Ammonium acetate	Colourless: Copper absent	
	(1) To one portion add KI. yellow ppt soluble in hot water and on cooling separate as golden spangles – Pb^{2+} confirmed	Add NH_4OH is excess. Filter	
	(2) To another portion add K_2CrO_4. Bright yellow ppt – confirms Pb^{2+}	**Residue:**	**Filtrate:**
		Colourless, $Bi(OH)_3$	Intense blue
		Dissolve in dil HCl	$[Cu(NH_3)_4]SO_4$
		(1). One portion is added to excess water. White opalescence confirms Bi^{3+}	(1). Through one portion acidified with HCl pass H_2S Black ppt confirms Cu^{2+}
		(2) Alkaline sodium stannite is added to another portion Black ppt confirms Br^{3+}	(2) Another portion is acidified with acetic acid and potassium ferrocyanide is added.
			Brown ppt confirms copper

Filtrate from group II is boiled to expel H_2S. Boil with a few drops of con. HNO_3 to convert any Fe^{2+} present, to Fe^{3+}. Add NH_4Cl and excess of NH_4OH filter. Residue may contain $Fe(OH)_3$ (reddish brown), $Al(OH)_3$ (white), $Cr(OH)_3$ (bluish green) and $MnO(OH)_2$ (dirty white).

Table 5 Analysis of group III cations

The precipitate is boiled with water and Sodium peroxide until evolution of oxygen ceases. Filter

Residue:	**Filtrate:**
$Fe(OH)_3$, $MnO(OH)_2$	Yellow: Cr^{3+} present
1. One part is dissolved in dil. HCl and tested as follows	(i) One portion of filtrate is acidified with acetic acid and lead acetate added–yellow ppt confirms Cr^{3+}
(i) One part is treated with Amonium thiocyanate–Blood red colouration confirms Fe^{3+}	(ii) Another portion is acidified with dil HCl and NH_4OH is added in excess–white gelatinous ppt confirms Al^{3+}
(ii) Another part treated with ppt ferrocynide – Intense blue ppt confirms Fe^{3+}	(iii) The above ppt is boiled with conc.HNO_3 and a drop of $Co(NO_3)_2$
2. Another part of ppt is boiled with conc. HNO_3 and PbO_2. Dilute with water. Let stand–superendant liquid coloured purple–Mn^{2+} confirmed	Ignite on a filter paper – blue ash confirms Al^{3+}

Table 6 Analysis of group IV cations

Filtrate from group III is made ammoniacal and H_2S is passed. Filter Residue may contain ZnS (white), and MnS (flesh coloured)	
The precipitates in the group IV is boiled with dil. HCl. Boil off H_2S completely Add NaOH in excess, filter	
Residue	**Filtrate**
White turning brown $Mn(OH)_2$. Boil the ppt in conc. HNO_3 and lead dioxide, dilute with water and allowed to stand. *Supernatant* liquid is coloured purple.Mn^{2+} is confirmed	Contains Na_2ZnO_2 (i) H_2S is passed through one portion. White ppt of ZnS confirms Zn^{2+} (ii) Another portion is acidified with acetic acid and pot. ferrocyanide is added white ppt confirms Zn^{2+} (iii) A little of the ZnS ppt is dissolved in conc.HNO_3 and a drop of $Co(NO_3)_2$. Filter paper dipped in this is ignited green ash confirmes Zn^{2+}

Filtrate from group IV is boiled to expel $H_2S.NH_4OH$ and $(NH_4)_2CO_3$ are added. Filter. The residue may contain $BaCO_3$ and $CaCO_3$.

Table 7 Analysis of group V cations

The Precipitate in group V is dissolved in minimum amount of hot dilute acetic acid. Potassium chromate solution is added in slight excess and filtered.

	Residue:		**Filtrate:**
(i)	Yellow ppt of ($BaCrO_4$) - Barium present. The ppt is dissolved cone.HCl	(i)	$Ca(CH_3COO)_2$ Add NH_4OH and ammonium oxalate. white crystalline ppt of a CaC_2O_4. Ca^{2+} confirmed
(ii)	To one portion add dil H_2SO_4 white ppt. Ba^{2+}confirmed	(ii)	The above ppt is dissolved in con. HCl and flame test is conducted. Brick red flame. Ca^{2+} confirmed
(iii)	Another portion is evaporated to a paste and flame test is conducted –Apple green flame. Ba^{2+} confirmed		

Table 8 Analysis of group VI cations

Filtrate from group V is concentrated to a small volume

(i) To one portion add NH_4Cl, NH_4OH and Na_2HPO_4 – White crystalline ppt – Mg^{2+}confirmed

(ii) To another portion add magneson reagent and NaOH, – Blue precipitate – Mg^{2+} is confirmed.

(iii) The precipitates from, test (i) is dissolved in con. HNO_3 and a drop of $Co(NO_3)_2$. Filter paper dipped in this solution is ignited—pink ash confirms Mg^{2+}

Physico–Chemical Principles involved in the separation of cations

Group I

Dilute HCl precipitates the insoluble chlorides of Hg_2^{2+}, Ag^+ and Pb^{2+}. $PbCl_2$ is separated from the rest by making advantage of the fact that $PbCl_2$ is soluble in hot water. $HgCl_2$ and AgCl are separated on the basis of the high solubility of AgCl in NH_4OH forming a soluble complex $[Ag(NH_3)_2]$

Cl while Hg_2Cl_2 undergoes a disproportionation reaction producing an insoluble grey mixture of $Hg(NH_2)Cl$ (white) and Hg(black).

Group II

The role of dil.HCl during the passage of H_2S is to suppress the dissociation of the weak electrolyte H_2S decreasing the concentration of S^{2-} ions by common ion effect. When the con. of S^{2-} is low the solubility products of only HgS, PbS, CuS and Bi_2S_3 are exceeded. If there is no HCl sulphides of group IV would also be precipitated in groupII itself. Remember that a sparingly soluble salt is precipitated when its ionic product (IP) exceeds solubility product (K_{sp}). HgS is separated from the rest based on the fact that CuS and Bi_2S_3 dissolve is dilute HNO_3 while HgS remain insoluble. When excess of NH_4OH is added to a mixture of $Cu(NO_3)_2$ and $Bi(NO_3)_3$ bismuth is precipitated as $Bi(OH)_3$ while Copper is precipitated as $Cu(OH)_2$ but dissolves in excess of ammonia as the complex $[Cu(NH_3)_4](OH)_2$.

Group III

Here NH_4OH is added in presence of NH_4Cl. The function of NH_4Cl is to suppress the dissociation of NH_4OH by common ion effect thereby decreasing the concentration of OH^- ions. At the decreased concentration of OH^-, K_{sp} of only $Fe(OH)_3$, $Al(OH)_3$ and $Cr(OH)_3$ are exceeded by their IP in solution. In the alkaline medium existing in group III Mn^{2+} is slowly oxidized to Mn^{4+} and is precipitated as hydrated manganese dioxide $MnO(OH)_2$ or $MnO_2.xH_2O$. To a large extent the precipitation of $MnO(OH)_2$ can be prevented by having a high concentration of NH_4Cl in group III.

When the precipitates are boiled with water and Na_2O_2 (or a mixture of NaOH solution and H_2O_2) $Al(OH)_3$ dissolves in NaOH as sodium aluminate ($NaAlO_2$) and $Cr(OH)_3$ is oxidized by H_2O_2 and NaOH to Na_2CrO_4 and dissolves.

$Fe(OH)_3$ and $MnO(OH)_2$ remain insoluble and are separated by filtration.

Group IV

The function of NH_4OH in group IV is to increases the dissociation of H_2S by combining with the H^+

$$H_2S \rightleftharpoons 2H^+ + S^{2-}$$

$$NH_4OH + H^+ \rightarrow NH_4^+ + H_2O$$

The MnS and ZnS are dissolved in dilute HCl

$$ZnS + 2HCl \rightarrow ZnCl_2 + H_2S$$

The H_2S is to be completely removed by boiling and NaOH is added. Zn^{2+} forms $Zn(OH)_2$ precipitate which dissolves in NaOH to form Na_2ZnO_2.$Mn(OH)_2$ is precipitated and is insoluble in excess of NaOH, If H_2S is not boiled off completely ZnS is precipitated on adding NaOH and the former is not soluble in excess of NaOH making the separation of Zn^{2+} and Mn^{2+} impossible.

Group V

Ba^{2+} and Ca^{2+} are precipitated as carbonates by adding $(NH_4)_2CO_3$ in presence of NH_4OH and NH_4Cl. In the absence of NH_4Cl and NH_4OH, Mg^{2+} also will be precipitated in group V as $MgCO_3$.

The carbonates are dissolved in minimum hot dilute acetic acid and K_2CrO_4 is added. $BaCrO_4$ is insoluble in acetic acid and is precipitated. Ca^{2+} remain in solution since $CaCrO_4$ is soluble in acetic acid.

Group VI

Mg^{2+} is precipitated as $Mg(NH_4)PO_4$. Other metal ions must be absent because they also form a precipitate with Na_2HPO_4.

Detection of cations using dyes

Metal ion	Reagent	Observation
Al^{3+}	Aluminon	Red lake
	Alizarin	Red lake
	Alizarin– s	Red lake
Mg^{2+}	Magneson I	Blue lake
	Magneson II	Blue lake

"Lake" is an intensely coloured precipitate in a lightly coloured solution. The red precipitate in the red lake produced by aluminium is the $Al(OH)_3$ precipitate with aluminon or alizarin absorbed on it. The bright red colour of the solution is due to the dissolved dye in solution.

SUMMARY

Periodic Table and Periodic Properties	
Mendeleev's Periodic table	Mendeleev's Periodic law–Atomic mass considered to be fundamental–groups–Periods–anomalous pairs Prediction of undiscovered elements.
Modern Periodic table	Moseley's X–ray study – $\sqrt{\upsilon}$ = a(Z–b) –importance of atomic number–Modern Periodic law.
General structure of modern long form of Periodic table	Modern periodic table known as Bohr table–Designed by Rang and Warner 7 Periods and 18 groups–Arrangement of elements according to electron filling.
Classification of elements	s, p, d, and f block elements
Periodic properties	Atomic radius, co-valent radius, metallic radius van der Waals radius, ionic radius. Ionization enthalpy–electron gain enthalpy–electronegativity, and factors affecting them. Electropositivity, valency and oxidation state, oxidizing power–reducing power.
Group I – Alkali metals	
General	Occurrence and abundance of alkali metals.
General Characteristics	Electronic configuration–atomic and ionic size–ionization enthalpy – Density–melting point–boiling point–photoelectric effect–mobility and conductivity of ions in aqueous solution – reaction with NH_3.
Abnormal behavior of Lithium	Ionic potential–diagonal relation to Magnesium
Oxides to alkali metals	Oxides, peroxides and superoxides
Hydroxides	Caustic soda–preparation and reaction Basic character and solubility of hydroxides
Carbonates	Soda ash–Na_2CO_3 Pearl ash–K_2CO_3 Solvay Process–Leblanc process–Precht's process
Other compound	NaCl, Glauber's salt $Na_2SO_4.10H_2O$
Group –II – Alkaline earth metals	
General characteristics	Atomic properties and Physical properties–structure of $BeCl_2$
Oxides of alkaline earth metals	Magnesium Oxide and Magnesium peroxide
Hydroxides	$Mg(OH)_2$ and $Ca(OH)_2$–Basic nature and solubility of hydroxides.

Carbonates	$MgCO_3$ and $CaCO_3$–solubility and thermal stability of carbonates
Sulphates	Gypsum, plaster of paris, epsom salt Solubility of sulphates
Additional Point	Storage of Lithium–Micro cosmic salt–Uses of alkali metals and alkaline earth metals and their compounds
Principles of Inorganic qualitative analysis	
Reactions of anions	Reaction of CO_3^{2-}, NO_2^-, SO_3^{2-}, S^{2-}, $S_2O_3^{2-}$, F^-, Cl^-, Br^-, I^-, NO_3^-, CH_3COO^-, SO_4^{2-}, PO_4^{3-} and CrO_4^{2-}
Test for anions using Na_2CO_3 extract	Test for halides with $AgNO_3$ – Brown ring test–test for sulphate.
Analysis of cation	Action of heat–Borax bead test–Charcoal Cavity test–preparation of Solution of salt
Inter group separation of cations	Scheme of analysis
Group I	Group reagent–dil HCl ions present–Hg_2^{2+}, Ag^+ and Pb^{2+} identification and confirmation.
Group II	Group reagent–H_2S in presence of dil HCl ions identified–Pb^{2+}, Hg^{2+}, Cu^2, Bi^{3+}, Cd^{2+} identification and confirmation
Group III	Group reagent–NH_4Cl and NH_4OH ions identified–Al^{3+}, Cr^{3+} Fe^{2+} and Fe^{3+} identification and confirmation
Group IV	Group reagent H_2S in presence of NH_4Cl and NH_4OH ions identified Zn^{2+}, Mn^{2+}, Co^{2+} and Ni^{2+}
Group V	Group reagent–NH_4Cl, NH_4OH and $(NH_4)_2CO_3$ ions identified–Ba^{2+}, Sr^{2+} and Ca^{2+}
Group VI	Group reagent–NH_4Cl, NH_4OH and Na_2HPO_4 ion identified–Mg^{2+}
Physico Chemical Principles involved in the separation of Cations	Group–I, Group II, Group–III, Group IV, Group–V, and Group VI
Detection of Cations using dyes	Al^{3+} Aluminon, Alizarin and Alizarin–s Mg^{2+} Magneson–I, Magneson–II

TOPIC GRIP

Subjective Questions

1. Among Na^+ and Ag^+, which is a good lewis acid? Why?
2. A chloride of iron is contaminated with the chloride of aluminum. Identify a reagent that can remove the aluminium from iron (through reaction).
3. Due to high reactivity of alkali metals towards air, they are stored in kerosene. Lithium is not stored in kerosene. Explain.
4. Comparing K_2O and CaO, which do you think is more basic in nature?
5. How do the hydrides of Beryllium and Calcium differ from each other?
6. Give the reaction of calcium with
   (i) Carbon at 1100°C
   (ii) water at room temperature
   (iii) phosphorus at high temperature
7. Magnesium oxide is used as a refractory material. What are the characteristic features that make it so?
8. An effervescence is observed on treating carbonates with dilute acid. But on treating lead carbonate with dilute sulphuric acid, the effervescence stops after some time. Explain.
9. When chlorine gas is passed through an aqueous solution of a potassium bromide containing chloroform a brown colour is imparted to the organic layer. What would happen if excess of chlorine gas is passed?
10. The white precipitate of AgCl dissolves in excess NH_3 due to formation of a complex. Give any two reagents that can precipitate out silver salt from this solution.

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

11. Lead iodide dissolves in a conc. solution of potassium iodide due to the formation of
    (a) $K_2[PbI_4]$ (b) $H_2[PbI_4]$
    (c) $[PbI_2]I_2$ (d) $K[PbI_3]$
12. Among the alkali metals, the one with the highest heat of atomization is
    (a) Li (b) Cs (c) Rb (d) K
13. When sodium is exposed to air for a long time, it is converted to
    (a) Na_2O (b) Na_2O_2
    (c) NaOH (d) Na_2CO_3
14. Epsom salt ($MgSO_4.7H_2O$) on strong heating gives
    (a) $MgSO_4.H_2O$ (b) $MgO + SO_2 + O_2$
    (c) Anhydrous $MgSO_4$ (d) $MgO + SO_3$

15. Alkali metals have lower density than alkaline earth metals because
(a) alkaline earth metals have stronger covalent bonds
(b) Atomic size decreases across the period
(c) alkaline earth metals have two electrons in the valence shell
(d) alkali metals have lower ionization energy



Assertion–Reason Type Questions

Directions: **Each question contains Statement -1 and Statement -2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**
(a) Statement -1 is True, Statement -2 is True; Statement -2 is a correct explanation for Statement -1
(b) Statement -1 is True, Statement -2 is True; Statement -2 is NOT a correct explanation for Statement -1
(c) Statement -1 is True, Statement -2 is False
(d) Statement -1 is False, Statement -2 is True

16. Statement 1
In solution, lithium ion moves slowly under the effect of electric current.

and

Statement 2
The size of the ion directly determines the electrical conductivity of the alkali salt solution.

17. Statement 1
Lithium iodide has a lower dipole moment than expected.

and

Statement 2
By Fajans rule Li^+ has greater Polarizability and I^- has greater polarizing power.

18. Statement 1
$CaCO_3$ is insoluble in water but dissolves in soda water.

and

Statement 2
Soda water is acidic in nature.

19. Statement 1
Chlorides of both silver and mercury are soluble in NH_4OH.

and

Statement 2
Formation of complex leads to solubility in many cases.

20. Statement 1
On heating nitrates with zinc in aqueous NaOH, a pungent smelling gas is formed.

and

Statement 2
Nitrates are readily reduced by nascent hydrogen.

Linked Comprehension Type Questions

Directions: **This section contains 2 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

Hardness of water is caused by soluble salts of mainly calcium and magnesium. Hard water does not produce a ready and permanent lather with soap. Hard water produces precipitate with anions like carbonate. Hardness can be removed by precipitating Ca^{2+} and Mg^{2+} as their insoluble salts like carbonates or by absorbing them on ion exchanger or by sequestration by complexing agents.

21. Temporary hardness is caused by bicarbonates of Ca^{2+} and Mg^{2+}. It can be removed by
(a) boiling.
(b) treating with calogen
(c) treating with ion exchange resins.
(d) All the above

22. A substance added to detergents to remove hardness is
(a) sodium bicarbonate
(b) sodium stearate
(c) synthetic zeolites
(d) All the above

23. Which of the following causes hardness when present in water?
(a) $MgCl_2$
(b) $Ca(OH)_2$
(c) $FeSO_4$
(d) All the above

Passage II

Hydroxides of group 1 metals are prepared by action of water on the metal.

$$2M + 2H_2O \rightarrow 2MOH + H_2$$

MOH are strong bases. Sodium reacts rapidly with evolution of heat while potassium reacts so violently that H_2 produced bursts instantly into flame. NaOH and KOH are useful reagents in the laboratory and are called caustic soda and caustic potash respectively.

24. When NaOH reacts with sulphur, two products are formed apart from H_2O. The one reagent that can be used to identify the compounds when each of them are given in pure form is
(a) $AgNO_3$
(b) alk $KMnO_4$
(c) lead acetate
(d) Iodine solution

25. The fusion of caustic potash is usually carried out in vessels made of iron because
(a) Iron withstands high temperature
(b) Iron acts as poison for decomposition of KOH
(c) Iron is unaffected by KOH
(d) Iron does not allow reaction of KOH with atmospheric air

26. In organic chemistry, the reagent alcoholic potash is widely used, but alcoholic soda is unheard of because
(a) potassium is less dense than sodium
(b) potassium hydroxide is more reactive than NaOH
(c) NaOH reacts with ethanol
(d) KOH is more soluble in ethanol than NaOH

Multiple Correct Objective Type

Directions: **Each question in this section has four suggested answers of which ONE OR MORE answers will be correct.**

27. Calcium nitride is dissolved in water to give B and a gas A escapes. Through the solution of B, CO_2 gas is passed, it turns turbid and finally clears up. The true statement is
(a) The final solution will turn turbid on heating
(b) Solution B if allowed to stand will give a white layer on the surface
(c) Gas A will turn acidified dichromate paper green
(d) From the final solution the crystals of solute can be separated easily

28. While testing for nitrates in the presence of a nitrite salt, which of the following should be added
(a) HNO_3 (b) NH_4Cl
(c) NH_2CONH_2 with dil H_2SO_4 (d) NH_4OH

29. Beryllium being extremely small in size has a unique chemical behavior. Which of the following are unique only to beryllium?
(a) Beryllium does not combine directly with hydrogen to form BeH_2
(b) Beryllium does not dissolve in liquid ammonia
(c) beryllium halides, except flouride are water insoluble
(d) Beryllium forms beryllate with caustic alkali

Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

30.

Column I	**Column II**
(a) Washing soda	(p) Water soluble
(b) Glaubers salt	(q) loses weight on exposure to air
(c) Magnesia	(r) Basic
(d) Partially dehydrated Gypsum	(s) with water sets to hard mass

IIT ASSIGNMENT EXERCISE

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

31. Fire due to sodium in a laboratory should not be extinguished using water because
(a) sodium is highly electropositive
(b) Sodium floats over water
(c) Sodium is generally stored in kerosene
(d) Sodium reacts with atmospheric air

32. The metal ion that dissolves completely in sodium hydroxide (excess) is
(a) Ni^{2+} (b) Cu^{2+} (c) Zn^{2+} (d) Cd^{2+}

33. Sodium formate can be prepared by passing a gas X through aqueous NaOH. The gas X is
(a) CO_2
(b) CO
(c) water gas
(d) mixture of CO and CO_2

34. Lithium fluoride is insoluble in water due to
(a) small size of both Li^+ and F^-
(b) covalent character of LiF
(c) low density of Li F
(d) high electronegativity of fluorine

35. With respect to magnetic property the true statement is
(a) Both alkali metals and their ions are paramagnetic
(b) Both alkali metals and their ions are diamagnetic
(c) Alkali metals are paramagnetic while ions are diamagnetic
(d) Alkali metals are diamagnetic while ions are paramagnetic

36. LiCl has a lower melting point than NaCl because
(a) Li^+ is a smaller ion than Na^+
(b) Li has high ionization potential
(c) NaCl has more crystal defects
(d) NaCl combines with more halides to from polyhalides

37. The compound NH_4Cl can be distinguished from KCl by
(a) passing CO_2 gas
(b) treating with NaOH
(c) heating the salt with conc. H_2SO_4
(d) treating with $K_4[Fe(CN)_6]$

38. The salt that does not impart colour to bunsen burner flame is
(a) LiCl (b) $MgCl_2$ (c) CsCl (d) $SrCl_2$

39. The blue colour of solution of potassium in liquid ammonia, is due to the presence of
(a) solvated electron (b) impurity in K atoms (c) potassium amide (d) oxide

40. Sodium is extracted by electrolysis of fused sodium chloride, mixed with $CaCl_2$ and NaF. The gas liberated at the graphite anode is
(a) Cl_2 (b) F_2 (c) Na (d) Cl_2 and F_2

41. Black ash produced during Leblanc Process, is a mixture of
(a) $Na_2CO_3 + Na_2S$ (b) $Na_2S + CaCO_3$ (c) $Ca(OH)_2 + Na_2CO_3$ (d) $CaS + Na_2CO_3$

42. The element with which sodium hydroxide reacts is
(a) Carbon (b) Nitrogen (c) Phosphorus (d) hydrogen

43. In the Liebig's method of carbon estimation CO_2 is absorbed in KOH while NaOH is not used. The preference to KOH is because
(a) KOH is less deliquescent than NaOH
(b) KOH has higher molecular weight than NaOH
(c) KOH is a stronger base than NaOH
(d) K_2CO_3 is more water soluble than Na_2CO_3

44. Among the halides of sodium, the one with highest melting point is
(a) NaCl (b) NaI (c) NaF (d) NaBr

45. Metallic Bonds in alkaline earth metal is stronger than alkali metals because
(a) alkaline earth metals have bigger kernels
(b) alkaline earth metals have two electrons in the outermost shell
(c) alkaline earth metals have higher mass
(d) alkaline earth metals have higher ionization potential

46. The alkaline earth metal which is not attacked by water even at high temperatures,
(a) does not form hydrides
(b) forms polymeric hydrides
(c) gives crimson red colour to flame
(d) forms the most unstable hydride

47. Among the following, the oxide insoluble in water is
(a) Li_2O (b) MgO (c) BaO (d) Cs_2O

48. The gas released when Beryllium is dissolved in alkali is
(a) O_2 (b) N_2 gas evolution (c) H_2 (d) O_2 and H_2

49. The gas evolved when magnesium is heated with hot concentrated sulphuric acid is
(a) SO_2 (b) SO_3 (c) H_2 (d) O_2

50. $MgCl_2.6H_2O$ is strongly heated in air, the product formed is
(a) $MgCl_2.5H_2O$ (b) $MgCl_2.H_2O$ (c) $MgCl_2$ (anhydrous) (d) Mg(OH)Cl

51. The hydroxide among the following which have least K_{sp} is
(a) LiOH (b) $Be(OH)_2$ (c) $Ba(OH)_2$ (d) CsOH

52. Chlorides of alkali metals are more soluble than chlorides of alkaline earth metals,. because alkaline earth metal chlorides
(a) cannot be hydrolyzed
(b) have low hydration energy
(c) have high lattice energy
(d) have higher density

53. Among the given oxides, the odd one out is
(a) Li_2O (b) Cs_2O (c) BeO (d) BaO

54. Fire caused due to presence of magnesium cannot be extinguished by carbondioxide because
(a) Mg reacts with CO_2
(b) fires due to Mg cannot be extinguished
(c) Mg has high density
(d) Mg reacts with air much faster in the presence of CO_2

55. Beryllium forms coordination compound. The maximum coordination number that Be can show is
(a) 2 (b) 4 (c) 6 (d) 3

56. Crystalline compounds of alkaline earth metals have water of crystallization e.g., $MgCl_2.6H_2O$. This is not seen in NaCl. This is because alkaline earth metals
(a) have small size and high charge
(b) have higher ionization energy
(c) hybridise differently than alkali metals
(d) are less electropositive

57. The compound that may be used to produce green colour in fireworks may be
(a) $Mg(NO_3)_2$ (b) $Sr(OH)_2$ (c) $Sr(NO_3)_2$ (d) $Ba(NO_3)_2$

58. The correct order of ease of formation of nitride among the following ions is
(a) $Ba^{2+} > Mg^{2+} > Be^{2+}$ (b) $Ba^{2+} > Be^{2+} > Mg^{2+}$ (c) $Be^{2+} \geq Mg^{2+} > Ba^{2+}$ (d) $Be^{2+} > Mg^{2+} \simeq Ba^{2+}$

59. The element with highest standard reduction potential is
(a) Mg (b) Be (c) Sr (d) Ba

60. In the analysis of cation, if the group III reagent NH_4Cl, is replaced by $(NH_4)_2SO_4$.
(a) the test can be continued as $(NH_4)_2SO_4$ gives common ion NH_4^+
(b) there will not be sufficient OH^- as 2 moles of NH_4^+ is released
(c) Both Cl^- and SO_4^{2-} are conjugate base of strong acid, hence test can be continued.
(d) The anion SO_4^{2-} may precipitate cations of subsequent groups in group III.

61. During the confirmatory test of Pb^{2+} using KI, if a large excess of KI is added, then
(a) The golden spangles attain small particle size (b) The golden spangles disappear
(c) The golden spangles are seen with violet vapours of I_2 (d) The golden spangles attain a more orange colour

62. A metal ion gives a white precipitate on a adding KCl which disappears on heating . The metal ion is
(a) Ag^+ (b) Hg_2^{2+} (c) Al^{3+} (d) Pb^{2+}

63. A gas turns orange acidified dichromate paper green. If the gas is passed through water saturated with H_2S gas, the solution turns
(a) black (b) into a thick brown mass
(c) into a white turbid solution (d) into a gelatinous mass

64. A certain compound on heating with dil.HCl gives out a gas that turns lead acetate paper black. The compound may be
(a) PbS (b) Na_2S (c) $PbSO_4$ (d) Na_2CO_3

65. Carbonates and sulphites on reaction with dil.H_2SO_4 gives out gases which turn lime water milky. The gases can be distinguished using
(a) dichromate paper (b) lead acetate paper (c) starch iodide paper (d) litmus paper

66. The formula of the brown coloured complex formed on passing NO gas through dilute $FeSO_4$ solution is
(a) $[Fe(H_2O)NO].SO_4$ (b) $Fe(NO)\ SO_4$
(c) $[Fe(H_2O)_5SO_4]\ NO.\ 2H_2O$ (d) $[Fe(H_2O)_5NO]\ SO_4$

67. In order to confirm the presence of acetate ions in a salt, the reagent that should be used is
(a) $FeCl_2$ (b) $Fe(CNS)_3$ (c) neutral $FeCl_3$ (d) NH_4OH

68. The ions given below, give out fumes in air on adding con.H_2SO_4. The one that will intensify on adding a pinch of MnO_2 is
(a) NO_3^- (b) NO_2^- (c) $C_2O_4^{2-}$ (d) I^-

69. To identify sulphate, $BaCl_2$ is added. If the salt is silver sulphate, the reagent used to check the sulphate should be
(a) $Ba(NO_3)_2$, because all nitrates are soluble in water
(b) $BaCl_2$ because $BaSO_4$ formed is insoluble in Con. HNO_3
(c) $CaCl_2$, because $CaSO_4$ is also water insoluble
(d) $CaCl_2$, because $CaSO_4$ is dissolves in con. HNO_3

70. To the sodium carbonate extract, CCl_4 and chlorine water are added. If I^- is present, the chloroform layer turns violet. The compound that can be added instead of Cl_2 water is
(a) dil.HCl (b) dil.H_2SO_4 (c) $CHCl_3$ (d) oxalic acid

71. For identification of anions, sodium carbonate extract is prepared because
(a) sodium carbonate is basic in nature
(b) all sodium salts are water soluble
(c) bulkier ions like CO_3^{2-}, SO_4^{2-}, $C_2O_4^{2-}$ can be separated from smaller ions like F^-, OH^- etc
(d) sodium carbonate does not decompose on heating

72. Which of the following salts will not decolourise permanganate solution?
(a) $FeSO_4$ (b) Na_2SO_4 (c) $Na_2C_2O_4$ (d) K_2SO_3

73. Which of the following cations cannot be precipitated as their sulphates?
(a) Pb^{2+} (b) Sr^{2+} (c) Zn^{2+} (d) Ba^{2+}

74. A gelatinous white precipitate formed in the Group III of basic radical analysis dissolves
(a) in both NH_4OH and NaOH (b) in NaOH and not in NH_4OH
(C) in NH_4OH and not in NaOH (d) neither in NH_4OH nor in NaOH

75. A 1st group precipitate, does not dissolve in hot water nor in ammonia. If copper turnings are added to its solution,
(a) a white precipitate is obtained (b) solution turns green
(c) a grey deposit is seen on copper (d) a white precipitate is formed which turns black

76. Sodium stannite solution used for testing Bismuth is
(a) $SnCl_2$ in excess NH_4OH (b) $Sn(OH)_2$ in excess dil.HCl
(c) $SnCl_2$ with ammonia added drop wise (d) $SnCl_2$ in excess NaOH

77. Before adding the group III reagent, solution should be boiled sufficiently to remove H_2S. This is done so that
(a) group III does not precipitate as sulphide (b) group IV radicals remain in the filtrate
(c) Group V radicals do not precipitate out (d) Group III radicals do not get reduced by H_2S

78. The metal sulphides soluble in dil.HCl are
(a) NiS and CoS (b) NiS and PbS (c) MnS and ZnS (d) ZnS and PbS

79. Which of the following compounds can be used in the lab in order to generate H_2S gas?
(a) thioacetamide (b) thiourea (c) NH_4SCN (d) NH_4CNS

80. Which of the following ions would give a precipitate on reaction with excess of KCN?
(a) Pb^{2+} (b) Ag^+ (c) Cu^{2+} (d) Cd^{2+}

81. The correct order of reactivity of alkali metals with hydrogen is
(a) Cs < Rb < K (b) Li < Na < K (c) K > Na > Cs (d) Li > K > Na

82. Consider the reaction $MI + I_2 \rightarrow MI_3$ when M is an alkali metal. M cannot be
(a) Na (b) K (c) Rb (d) Cs

83. The gas evolved when the blue solution of alkali metal in ammonia is allowed to stand is
(a) NH_3 (b) H_2
(c) NO_2 (d) mixture of oxides of nitrogen

84. The compound that liberates two different gases on heating is
(a) $NaNO_3$ (b) $Li(NO_3)$ (c) $MgCO_3$ (d) $Mg(OH)_2$

85. Sodium sulphide can be used for the volumetric estimation of iodine because
(a) its aqueous solution is alkaline (b) it is readily oxidized by air
(c) it is used to make organic dyestuff (d) it hydrolyzes appreciably in water

86. On adding $AgNO_3$ to aqueous KI, a yellow precipitate is formed. This will dissolve in
(a) hot water (b) dil. HCl (c) NH_4OH (d) excess KCN

87. Sodium is miscible with K, Rb, Cs in all proportion while lithium is immiscible. This is because
(a) Lithium has very small size
(b) Lithium is less electropositive
(c) Lithium forms covalent bonds
(d) Lithium has a high melting point

88. Dilute solution of alkali metals in liquid ammonia is dark blue in colour. On standing, the colour
(a) turns brown
(b) turns bronze coloured with metallic lustre
(c) intensifies
(d) slowly fades and disappears

89. The hardest metal among the following is
(a) Li (b) Cs (c) K (d) Fr

90. The lithium compound, that is water soluble is
(a) Li_2CO_3 (b) $LiNO_3$ (c) LiF (d) $Li_3(PO_4)$

91. Alkaline earth metals form hydrated halides. On moving down the group from magnesium onwards water of hydration per molecule
(a) increases
(b) decreases
(c) remains same
(d) does not show any trend

92. Beryllium dissolves readily in dil aqueous HNO_3 but not in cold concentrated HNO_3 because
(a) Be reacts only on heating
(b) Be becomes passive
(c) presence of water favours reaction
(d) Be is amphoteric in nature

93. The metal that would liberate hydrogen gas on reaction with KOH is
(a) Ba (b) Sr (c) Mg (d) Be

94. Among the carbonates of alkali metals, only lithium carbonate is unstable because
(a) small cation stabilizes small anion
(b) CO_3^{2-} ion is resonance stabilized
(c) CO_3^{2-} reacts with Li^+
(d) CO_3^{2-} is oxidizing in nature

95. Sodium sulphate is soluble in water because
(a) sulphate is a large anion
(b) sodium sulphate has greater hydration energy
(c) solution formed is neutral
(d) Na_2SO_4 is salt of strong base and weak acid

96. The main constituent of sea shell
(a) is a salt of weak acid and weak base
(b) is also found in burnt plaster
(c) is soluble in hot water
(d) is soluble in water saturated with CO_2

97. Acetylene is used as a ripening agent for fruits. The compound used to produce acetylene is
(a) Mg_3C (b) Be_2C (c) CaC_2 (d) CaC_2O_4

98. The oxide of an s-block metal M forms a soluble complex with aq.NaOH. The hydroxide of the metal M
(a) is water insoluble
(b) is insoluble in dil. HCl
(c) turns milky on passing CO_2 gas
(d) is used as primary standard in titration

99. The correct order of 1st ionization energy of the following is
(a) Be > Li > Mg > Na (b) Be > Mg > Na > Li (c) Be > Mg > Li > Na (d) Li > Be > Mg > Na

100. An alkaline earth metal on heating in nitrogen gives a product X. X on hydrolysis gives a basic insoluble compound and a gas that turns $CuSO_4$ solution dark blue. The metal is
(a) Be (b) Ca (c) Mg (d) Ba

101. Among the following, the electron deficient compound that forms three centre two electron bonds is
(a) BeH_2 (b) CaH_2 (c) $MgCl_2$ (d) $BaCl_2$

102. Among the following reagents, the one that is not used to test Pb^{2+} is
(a) KCN (b) $K_4[Fe(CN)_6]$ (c) KI (d) K_2CrO_4

103. A colourless gas that turns cadmium acetate paper yellow is
(a) H_2S (b) SO_2 (c) NO_2 (d) H_2O

104. The formula of the waxy deposit obtained on a wet glass rod when NaF is heated with sulphuric acid is
(a) $Na_2[SiF_4]$ (b) H_4SiO_4 (c) $H_2[SiF_6]$ (d) $Na_2[SiF_6]$

105. The precipitate formed by adding silver nitrate to calcium chloride is insoluble in
(a) NaCN (b) KCN (c) $Na_2S_2O_3$ (d) dil.HNO_3

106. A certain compound on heating with con.H_2SO_4 gives out gas which turns lime water milky and burns with a blue edged flame. The compound may be
(a) Na_2CO_3 (b) Na_2SO_3 (c) Na_2SO_4 (d) $Na_2C_2O_4$

107. Which of the following soluble chloride produces a turbidity on addition of water?
(a) $ZnCl_2$ (b) $BiCl_3$ (c) $AlCl_3$ (d) $CdCl_2$

108. Even though the original salt given is Fe^{3+}, while moving to the third group, it might become Fe^{2+}, because of the action of
(a) dil HCl (b) H_2S (c) NH_4OH (d) dil HNO_3

109. A hot solution of metal chloride gives a precipitate on cooling. The metal ion gives yellow precipitate with K_2CrO_4. The metal chloride is
(a) AgCl (b) $BaCl_2$ (c) $SrCl_2$ (d) $PbCl_2$

110. Nessler's reagent is
(a) HgI_2 dissolved in excess KI
(b) alkaline solution of HgI_2 dissolved in excess KI
(c) Hg_2I_2 dissolved in excess KI
(d) solution of HgI_2 and Hg_2I_2 in excess KI

Assertion–Reason Type Questions

Directions: **Each question contains Statement -1 and Statement -2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement -1 is True, Statement -2 is True; Statement -2 is a correct explanation for Statement -1
(b) Statement -1 is True, Statement -2 is True; Statement -2 is NOT a correct explanation for Statement -1
(c) Statement -1 is True, Statement -2 is False
(d) Statement -1 is False, Statement -2 is True

111. **Statement 1**
Potassium is less dense than sodium.

and

Statement 2
Due to increase in atomic volume, density decreases down the group.

112. **Statement 1**
Beryllium chloride in water gives an acidic solution.

and

Statement 2
Solid $BeCl_2$ is a polymer containing halogen bridges.

113. **Statement 1**
Potassium chromate can be used to distinguish between sulphides and sulphites in acid solution.

and

Statement 2
The gases liberated by sulphide and sulphite on reaction with acid are different.

Linked Comprehension Type Questions

Directions: **This section contains 1 paragraph. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

When a chloride salt is heated with $K_2Cr_2O_7$ and a few drops of con.H_2SO_4. The formation of red vapours of chromyl chloride confirms the presence of chloride ions.

$$4NaCl + K_2Cr_2O_7 + 3H_2SO_4 \rightarrow K_2SO_4 + \underset{\text{Orange yellow vapours}}{2CrO_2Cl_2} + 2Na_2SO_4 + 3H_2O$$

When the red brown vapours are passed through NaOH, a yellow solution is formed

$$CrO_2Cl_2 + 4NaOH \rightarrow NaCl + \underset{\text{yellow solution}}{Na_2CrO_4} + 2H_2O$$

The resulting solution is acidified with acetic acid and lead acetate is added.

114. During the above test, with an unknown salt, formation of a yellow precipitate with $Pb(CH_3COO)_2$ confirms
(a) chloride only (b) Cl^-, Br^-, I^- (c) chloride or bromide (d) chloride or chromate

115. The solution is neutralized with acetic acid before adding lead acetate. This is because
(a) lead acetate will ionize better (b) a white precipitate may be formed
(c) sodium acetate formed is water soluble (d) lead acetate reacts only in acidic medium

116. It is very important to perform the experiment in a clean dry test tube because
(a) in the presence of moisture red vapours cannot be observed
(b) presence of moisture leads to explosive reaction
(c) Cl^- present in tap water may interfere with analysis
(d) presence of water causes turbidity

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers of which ONE OR MORE answers will be correct.**

117. Ionic compounds of alkaline earth metals are less soluble in water compared to alkali metals because alkaline earth metals have
(a) small size (b) higher charge
(c) higher ionization energy (d) higher density

118. A salt which is analysed contains I^- as anion. Which of following reagents will give a positive test?
(a) acidified $KMnO_4$ (b) lead acetate (c) $CuSO_4$ (d) $K_4[Fe(CN)_6]$

119. Beryllium hydride differs from hydrides of other alkaline earth metals in

(a) difficulty in preparation by direct combination
(b) reducing nature
(c) being a covalent compound
(d) being a polymeric, electron deficient compound



Matrix-Match Type Questions

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

120.

Column I	**Column II**
(a) Li_2O	(p) low solubility in water
(b) MgO	(q) gives H_2O_2 on hydrolysis
(c) Na_2O_2	(r) coloured
(d) KO_2	(s) paramagnetic

ADDITIONAL PRACTICE EXERCISE

Subjective Questions

121. Comment on the water solubility of an alkali metal hydroxide, which forms an oxide on heating.

122. Among PbO, Na_2O_2 HgO and ZnO, identify the oxide, which satisfies the following conditions and formulate the reactions. "The oxide is yellow coloured and on exposure to moist air it forms two different compounds of the metal, which are colourless and water soluble".

123. Ammonia gas is burnt in air and the products are cooled to 20°C and passed over a heated s block metal (M) to get a white solid which on hydrolysis gives a water insoluble product. Carbide of (M) on hydrolysis gives propyne. Identify (M).

124. Mg_3N_2 and $MgCl_2$ kept in two separate closed containers are dissolved in water at room temperature and a small amount of HCl gas is introduced into these containers. Indicate the possible reactions in the whole process.

125. Solubility of the sulphates of alkaline earth metals decreases down the group while that of the hydroxides increases. Account for the difference in behaviour.

126. Why an excess concentration of chloride ion must be avoided in group I analysis?

127. When a solution containing Pb^{2+} is treated with KI, an yellow precipitate soluble in excess KI is obtained. Represent the reactions involved.

128. An alkali metal (A) burns in air to form two oxides (B) and (C). (B) is a pale yellow coloured monoxide and it reacts violently with water to give (D). (C) is deep yellow coloured as well as paramagnetic. (A) on controlled oxidation in air at 300°C gives another oxide (E) which is orange coloured and diamagnetic. (D) reacts with O_3 to form the compounds (F) and (G) in two separate reactions. Water and oxygen are the other products of the reaction. (F) is a chrome yellow powder which on strong heating gives (B) with the liberation of O_2. (G) is an orange coloured solid as well as paramagnetic. Identify (A) to (G)

129. Identify (A) – (D).

(i) When a metal ion solution is made alkaline and H_2S gas is passed through it, a precipitate soluble in dil.HCl is obtained H_2S is boiled off from the solution and on further treatment with an excess of a reagent (A) gives a brown precipitate.

(ii) A colourless salt (B) is soluble in water and alcohol. (B) on strong heating gives a brown gas and leaves a metallic residue. Its solution gives a brick red precipitate with K_2CrO_4 solution. A solution of (B) gives a white precipitate with thiosulphate and is soluble in excess thiosulphate.

(iii) A gas (C) when passed through acidified $K_2Cr_2O_7$ solution, changes its orange colour to green. A white turbidity is obtained when H_2S gas is bubbled through a solution of (C) in water.

(iv) When excess water is added to a solution of $Bi(OH)_3$ dissolved in HCl, a white precipitate (D) is obtained.

130. A mixture contains nitrate and bromide. How will you conduct the brown ring test for nitrate without interference from bromide?

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

131. Among the alkali metals, a metal, whose oxide can be formed by heating the hydroxide

(a) form a nitride
(b) form highly soluble hydroxide
(c) form soluble carbonates
(d) is very soft

132. Sodium peroxide on exposure to moist air gives
(a) NaOH, NaO_2 (b) NaOH, Na_2CO_3 (c) NaOH only (d) NaO_2, Na_2CO_3

133. Among alkali metal chlorides only lithium forms hydrates and exists as $LiCl.2H_2O$ because
(a) Lithium shows diagonal relationship with magnesium (b) Lithium has small size compared to others
(c) Lithium has the highest ionization potential (d) hydrated lithium ion has larger size

134. You are provided with dil HCl, dil.H_2SO_4 and NaCl solutions. The pair of reagents that you would choose to identify each would be
(a) Litmus paper and Na_2CO_3 (b) Litmus paper and $BaCl_2$
(c) $BaCl_2$ and $AgNO_3$ (d) $CaCl_2$ and $AgNO_3$

135. In the fifth group analysis of metal cations, if reagent $(NH_4)_2CO_3$ is replaced by Na_2CO_3,
(a) routine analysis can be continued (b) precipitate formed may contain Magnesium
(c) sodium added will interfere in the sixth group (d) $[CO_3^{2-}]$ cannot help exceed K_{sp}

136. When 4.86 g of a mixture of Na_2CO_3 and $NaHCO_3$ is heated to 200°C, 0.224 L of CO_2 at STP is liberated. The mole percentage of Na_2CO_3 in the mixture is
(a) 30 (b) 40 (c) 50 (d) 60

137. When $CoCl_2$ is treated with acetic acid and potassium nitrite, an yellow precipitate is obtained. The composition of this yellow precipitate is
(a) $Co(NO_2)_2$ (b) $Co(NO_2)_3$ (c) $K_3[Co(NO_2)_6]$ (d) $K_2Co(NO_2)_4$

138. Potassium ferrocyanide is a good reagent to test for both
(a) Mn^{2+} and Cu^{2+} (b) Pb^{2+} and Fe^{3+} (c) Cu^{2+} and Zn^{2+} (d) Fe^{3+} and Ba^{2+}

139. When a colourless aqueous solution of a metal chloride (A) and a metal hydroxide (B) are mixed together, a white precipitate (C) is obtained. The filtrate on treatment with an alkali gives (B). Therefore, (A), (B) and (C) are respectively
(a) NaCl, $Ca(OH)_2$ and Na_2O (b) $BiCl_3$, $Ca(OH)_2$ and PbO_2
(c) $BiCl_3$, $Mg(OH)_2$ and Bi_2O_3 (d) $MgCl_2$, $Ca(OH)_2$ and $Mg(OH)_2$

140. In the preparation of gun powder, potassium nitrate is used and not sodium nitrate. This is because
(a) $NaNO_3$ is deliquescent (b) $NaNO_3$ reacts with S
(c) KNO_3 is hygroscopic (d) KNO_3 itself is an explosive

141. A metal (M) has an electronic configuration $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,4s^1$. Formula of its sulphate is
(a) MSO_4 (b) $M(SO_4)_2$ (c) $M_2(SO_4)_2$ (d) M_2SO_4

142. Limestone is a mineral containing $CaCO_3$. Such mineral deposits of Na or K are rare because
(a) Na_2CO_3 and K_2CO_3 are unstable in nature (b) $CaCO_3$ dissolves in acidic rainwater
(c) Na_2CO_3 and K_2CO_3 are not stable to heat (d) Na_2CO_3 and K_2CO_3 are soluble in water

143. Sodium hydroxide stored in glass bottle should not be stoppered with a glass stopper because
(a) Sodium hydroxide reacts with silica in glass (b) NaOH decomposes on standing liberating O_2
(c) Reaction of NaOH with water is exothermic (d) NaOH reacts with air in the bottle and might explode

144. Among the s- block elements, the metals that do not react with cold water is/are
(a) Li, Mg, Be (b) Li, Be (c) Mg, Be (d) Be only

145. A 1.0g mixture of anhydrous Na_2CO_3 and K_2CO_3 are completely neutralized by 30 ml of 0.5N HCl. The amount of Na_2CO_3 in ;the sample is approximately
(a) 0.12g (b) 0.53g (c) 0.88g (d) 0.44g

146. Among the following hydroxides, the most water soluble hydroxide is
(a) $Mg(OH)_2$ (b) LiOH (c) NaOH (d) CsOH

147. Which of the following compounds exist only in solution?
(a) Li_3PO_4 (b) $NaHCO_3$ (c) $[Li(NH_3)_4]I$ (d) $LiHCO_3$

148. Identify the correct statement
(a) Lithium forms covalent carbides
(b) Only lithium among alkali metal react with carbon
(c) Only Be, Mg and Ca among alkaline earth metal form carbides
(d) Alkali metals generally form ionic carbides

149. The reason for high solubility of lithium perchlorate in acetone is
(a) polar nature of acetone (b) covalent bonds in perchlorate anion
(c) small size of lithium ion (d) difference in size between the anion and cation

150. s-block elements are always produced by
(a) thermal decomposition (b) self reduction
(c) hydrometallurgy (d) electrolytic reduction

151. Which of the following reagents have a different reaction with Be as compared to other Group 2 metals?
(a) dil.HCl (b) CO_2 gas (c) dil.HNO_3 (d) NaOH

152. Alkaline earth metals readily generate hydrogen with
(a) NH_3 (b) dil.HCl (c) H_2O (d) NaOH

153. The value of ionic radius of Be^{2+} ion is only an approximate value because
(a) Be always forms covalent bonds (b) Free Be^{2+} ions do not exist
(c) Be and its compounds are extremely toxic (d) Be^{2+} has noble gas configuration.

154. During extraction of magnesium from sea water, the reaction

$$Ca(OH)_2 + Mg^{2+} \rightarrow Mg(OH)_2 + Ca^{2+} \text{ is used.}$$

The reaction proceed because of
(a) low basic nature of $Mg(OH)_2$ compared to $Ca(OH)_2$ (b) low solubility of the Ca^{2+} salt formed
(c) low K_{sp} value of $Mg(OH)_2$ salt (d) low reactivity of $Mg(OH)_2$ salt

155. Hardness of water, caused by salts of calcium and magnesium, can be removed by exchanging Ca^{2+} and Mg^{2+} for Na^+, because
(a) sodium is a monovalent element
(b) sodium ion has smaller size than magnesium (ion)
(c) sodium salts are generally water soluble
(d) compounds of sodium are less stable than that of Ca or Mg

156. Which of the following properties is not related to small size of alkaline earth metals?
(a) harder than alkali metal (b) form divalent ion rather than monovalent
(c) have higher density than alkali metal (d) have higher melting point than alkali metal

157. The compound, formed on heating MgO and carbon in a current of dry chlorine, is
(a) water insoluble (b) is used in making Sorel's cement
(c) highly efflorescent (d) used in preparation of hydrocabons

158. Magnesium sulphate may be prepared by dissolving dolomite in boiling dilute H_2SO_4. The byproduct can be separated from $MgSO_4$ by
(a) filtration
(b) distillation
(c) solvent extraction
(d) byproduct formed are gaseous, so automatically removed

159. Among the carbonates of alkaline earth metals, $BeCO_3$ is the least stable because
(a) Be^{2+} has the lowest charge/radius
(b) Be^{2+} has low nuclear charge
(c) Be^{2+} has low mass compared to CO_3^{2-}
(d) Be^{2+} is the smallest ion

160. Alkaline earth metal halides
(a) form polyhalides like alkali metal halides
(b) exist in the form of hydrates and are hygroscopic
(c) fume in air due to hydrolysis
(d) exist as monomer only in vapour phase

161. Among the following ions, the one that will not decolourise $KMnO_4$ solution is
(a) SO_3^{2-} (b) CO_3^{2-} (c) I^- (d) NO_2^-

162. Sodium hydroxide is not suitable to separate
(a) Al^{3+} and Fe^{3+} (b) Al^{3+} and Sn^{2+} (c) Al^{3+} and Fe^{2+} (d) Al^{3+} and Cr^{3+}

163. In the IV group cation analysis, H_2S gas is passed through alkaline solution. If a neutral solution is used instead, for Zn^{2+}
(a) complete precipitation will occur
(b) partial precipitation will occur as Zn^{2+} exists as its chloride.
(c) No precipitation occurs.
(d) precipitation occurs only if solution is diluted enough

164. A solution contains Al^{3+}, Zn^{2+} and Ca^{2+}. To separate Al^{3+} from this mixture, the reagent used should be
(a) NaOH
(b) HCl
(c) NH_4OH
(d) $NH_4Cl + NH_4OH + (NH_4)_2CO_3$

165. Permanganic acid test for Mn^{2+} is conducted by heating with PbO_2 and conc. HNO_3. What happens if HNO_3 is replaced by HCl?
(a) Intense pink colour is produced
(b) Only a pale pink colour is obtained
(c) No pink colour is produced
(d) Green colour of MnO_4^{2-} is obtained.

166. Sodium carbonate on reaction with acid liberates CO_2 gas. Which of the following will not liberate CO_2?
(a) dil.HCl (b) dil.H_3PO_4 (c) dil.H_3BO_3 (d) dil.H_2SO_4

167. Silver nitrate cannot be used to test for
(a) CO_3^{2-} (b) SO_3^{2-} (c) $S_2O_3^{2-}$ (d) F^-

168. Barium nitrate is a better reagent than $BaCl_2$ to confirm presence of SO_4^{2-} in the given salt because
(a) Barium sulphate is insoluble in con.HCl
(b) Most metal nitrates are water soluble
(c) $BaCl_2$ is more water soluble than $Ba(NO_3)_2$
(d) Some metal sulphates dissolve in dil.HCl

169. The reagent that can be used to distinguish between sodium nitrate and sodium bromide is
(a) alkaline $FeSO_4$ (b) $BaCl_2$ (c) heating with H_2SO_4 (d) $AgNO_3$

170. Hydrated $BeCl_2$ in solution, has Be^{2+} surrounded by four molecules of water act as
(a) Lewis base (b) Arrhenius acid (c) Arrhenius base (d) Lewis acid



Assertion–Reason Type Questions

Directions: **Each question contains Statement -1 and Statement -2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement -1 is True, Statement -2 is True; Statement -2 is a correct explanation for Statement -1
(b) Statement -1 is True, Statement -2 is True; Statement -2 is NOT a correct explanation for Statement -1

(c) Statement -1 is True, Statement -2 is False
(d) Statement -1 is False, Statement -2 is True

171. Statement 1
Among alkali metals, Caesium has the lowest work function for photoelectric effect.

and

Statement 2
Ionization Energy decreases down the group.

172. Statement 1
Ca^{2+} ions cannot be precipitated as its sulphate using ammonium sulphate.

and

Statement 2
Calcium sulphate is sparingly soluble in water but appreciably soluble in conc. solution of $(NH_4)_2SO_4$.

173. Statement 1
The violet colour of iodine gas disappears in the presence of sodium hydroxide.

and

Statement 2
In an auto redox reaction, the iodine is both reduced and oxidized.

174. Statement 1
There is a preliminary test that can be used to distinguish between NaCl and KCl.

and

Statement 2
Hydrated potassium ion is larger than hydrated sodium ion.

175. Statement 1
Oxalate ions cannot be confirmed in the presence of fluoride ions using $CaCl_2$.

and

Statement 2
$CaCl_2$ has high affinity for water which is not seen in the CaF_2.

176. Statement 1
It is important to keep solution free from any oxidising agent before passing H_2S gas in Group II cation analysis.

and

Statement 2
H_2S has sulphur in negative oxidation state.

177. Statement 1
Reactivity of alkali metals with hydrogen increases from Lithium to cesium.

and

Statement 2
Order of reactivity in general increases down the group due to decrease in ionization energy.

178. Statement 1
There is a compound of sodium, which on hydrolysis gives a compound that can decolourise PbS.

and

Statement 2
Solution of alkali metal in liquid ammonia are reducing in nature.

179. Statement 1
The amount of KI that should be added to confirm Pb^{2+} as golden spangles in a salt analysis should be large.

and

Statement 2
PbI_2 is less soluble in water compared to $PbCl_2$.

180. Statement 1
The solubility of chlorides of alkali metals always decreases on going down the group.

and

Statement 2
The solubility of halides of alkali metals always decreases on going down the group.

Linked Comprehension Type Questions

Directions: **This section contains 3 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

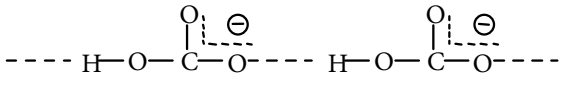
Group I metals are highly electropositive and thus form strong bases and have quite stable oxosalts. The carbonates are remarkably stable and will melt before they eventually decompose into oxides at temperatures above 1000°c. Li_2CO_3 is considerably less stable and decomposes more readily.

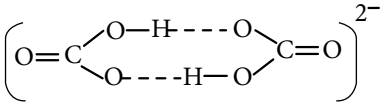
Group I metals form stable bicarbonates as they are strongly basic. No other metals form solid bicarbonates.

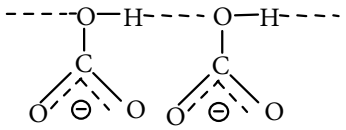
181. The only other bicarbonate known in solid state other than alkali metal bicarbonates is
(a) $NH_4(HCO_3)$ (b) $Ba(HCO_3)_2$ (c) $AgHCO_3$ (d) $Mg(HCO_3)_2$

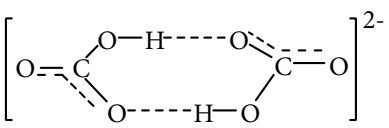
182. The process that cannot be used for preparation of a potassium salt because of difference in solubility between another potassium and sodium salt is
(a) Solvay process (b) Leblanc process (c) Precht's process (d) Castner-Kellnerprocess

183. Identify the structure of $KHCO_3$ in the crystalline state, from the following

(a) 

(b) 

(c) 

(d) 

Passage II

Halides of alkaline earth metals MX_2 can be made by heating the metals with the halogen or by the action of halogen acid on either the metal or the carbonate. The beryllium halides are covalent, hygroscopic and fume in air. They do not conduct electricity.

184 $BeCl_2$ fumes in air because of
(a) hydration (b) hydrolysis
(c) low melting point (d) reaction with O_2 in air

185. BeF_2 is prepared and finally crystallized from its aqueous solution. This salt is heated. The product formed is
(a) $Be(OH)_2$ (b) anhydrous BeF_2 (c) $[Be(H_2O)_2]F_2$ (d) $BeF_2.2H_2O$

186. $BeCl_2$ differs from BeH_2 with respect to
(a) polymeric structure (b) hydrolysis to give amphoteric hydroxide
(c) covalent character (d) three center two electron bond

Passage III

All nitrates are water soluble. The nitrates of mercury and bismuth yield basic salts on treatment with water. These are soluble in dilute nitric acid.

Nitrates on heating with con.H_2SO_4 gives out reddish brown fumes of NO_2 and also a pungent acid vapour of nitric acid which fumes in the air .The fumes intensify on addition of copper turnings. Nitrates are confirmed by formation of a brown ring with $FeSO_4$ and con.H_2SO_4

187. In the reaction of nitrates with con.H_2SO_4, nitrate with nitrogen in +5 state is converted to NO_2 (+4state). So NO_3^- is reduced. The element oxidized is
(a) Sulphur (b) Oxygen (c) SO_4^{2-} (d) Sulphur and Oxygen

188. On adding copper turnings, the fumes intensifies. The change noticed in the reaction mixture is
(a) No change in the reaction mixture (b) reaction mixture also looks brown
(c) reaction mixture turns blue (d) changes to a black colloidal solution

189. If bromides and iodides are present, the brown ring test is a failure. This is because
(a) the halogen is liberated
(b) Br^- is more reactive than the nitrates
(c) under the reaction conditions NO_2 reacts with Br_2 giving NOBr
(d) Ferrous sulphate is reduced by I^-

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers of which ONE OR MORE answers will be correct.**

190. The types of bonds present in hydrated sodium peroxide is
(a) ionic bon (b) hydrogen bond (c) covalent bond (d) van der Waals forces

191. Beryllium nitride differs from the other alkaline earth metal nitrides in
(a) being volatile (b) being covalent (c) not showing hydrolysis (d) water of hydration

192. The formation of the yellow precipitate of potassium hexanitrito cobaltate (III) is a good test to confirm
(a) Nickel (b) cobalt (c) Nitrates (d) nitrites

193. Exceptionally lithium carbonate is least unstable to heat compared to other alkali metals because
(a) Li^+ is strongly polarizing
(b) of electro positive nature of Li^+
(c) higher lattice energy of oxide compared to carbonate
(d) of alkaline nature of the carbonate

194. Which of the following processes will give $CaCO_3$ as the product?
(a) heating calcium sulphate with carbon
(b) heating calcium bicarbonate
(c) saturating aqueous calcium sulphate with ammonia and CO_2
(d) heating aqueous calcium cyanamide

195. Compared to the alkali metal compounds, alkaline earth metal compounds are less soluble in water because
(a) they are less electro positive than alkali metals
(b) they are less basic in nature
(c) they are divalent ions
(d) they are smaller than alkali metals

196. Which of the following will not give a metal oxide on heating?
(a) $NaNO_3$ (b) CsOH (c) LiOH (d) $SrCO_3$

197. The curdy white precipitate formed while performing the confirmatory test for chlorides is soluble in
(a) NaOH (b) KCN (c) $Na_2S_2O_3$ (d) con.HNO_3

Matrix-Match Type Questions

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

198.

Column I	**Column II**
(a) SO_2	(p) decolourises $KMnO_4$
(b) NH_3	(q) turns Baryta water turbid
(c) NO_2	(r) Absorbed by KOH
(d) CO_2	(s) turns red litmus blue

199.

Column I	**Column II**
(a) Sodium	(p) Amphoteric hydroxide
(b) Magnesium	(q) halides are hygroscopic
(c) Lithium	(r) Reacts slowly with hot water
(d) Beryllium	(s) water soluble hydroxide

200.

Column I	**Column II**
(a) $Na_2S_2O_3$	(p) On heating gives out gas that turns lime water milky.
(b) Na_2CO_3	(q) Decomposes on heating
(c) $NaNO_3$	(r) Aqueous solution is basic
(d) $NaHCO_3$	(s) Exists as hydrates

SOLUTIONS

ANSWERS KEYS

Topic Grip

1. Ag^+
2. NaOH
3. Lithium is lighter (Less dense than kerosene)
4. K_2O
5. Ionic and polymeric
6. CaC_2, $Ca(OH)_2$ and Ca_3P_2
7. Heat conductor, electrical insulator, inertness and high M.P
8. $PbSO_4$ coating is formed on the surface of $PbCO_3$
9. BrCl and HBrO
10. HNO_3, KI
11. (a) **12.** (a) **13.** (d)
14. (b) **15.** (b) **16.** (c)
17. (c) **18.** (b) **19.** (d)
20. (a) **21.** (a) **22.** (c)
23. (d) **24.** (a) **25.** (c)
26. (d)
27. (a), (b)
28. (b), (c)
29. (a), (c), (d)
30. (a) → (p), (q), (r)
(b) → (p), (q)
(c) → (r), (s)
(d) → (s)

IIT Assignment Exercise

31. (a) **32.** (c) **33.** (b)
34. (a) **35.** (c) **36.** (a)
37. (b) **38.** (b) **39.** (a)
40. (a) **41.** (d) **42.** (c)
43. (d) **44.** (c) **45.** (b)
46. (b) **47.** (b) **48.** (c)
49. (a) **50.** (d) **51.** (b)
52. (c) **53.** (c) **54.** (a)
55. (b) **56.** (a) **57.** (d)
58. (d) **59.** (b) **60.** (d)
61. (b) **62.** (d) **63.** (c)
64. (b) **65.** (a) **66.** (d)
67. (c) **68.** (d) **69.** (a)
70. (b) **71.** (b) **72.** (b)
73. (c) **74.** (b) **75.** (c)
76. (d) **77.** (b) **78.** (c)
79. (a) **80.** (a) **81.** (a)
82. (a) **83.** (b) **84.** (b)
85. (b) **86.** (d) **87.** (a)
88. (d) **89.** (a) **90.** (b)
91. (b) **92.** (b) **93.** (d)
94. (a) **95.** (b) **96.** (d)
97. (c) **98.** (a) **99.** (c)
100. (c) **101.** (a) **102.** (b)
103. (a) **104.** (b) **105.** (d)
106. (d) **107.** (b) **108.** (b)
109. (d) **110.** (b) **111.** (c)
112. (b) **113.** (d) **114.** (a)
115. (c)
116. (a)
117. (a), (b)
118. (a), (b), (c)
119. (a), (d)
120. (a) → (p)
(b) → (p)
(c) → (q), (r)
(d) → (q), (r), (s)

Additional Practice Exercise

121. Low solubility
122. Na_2O_2
123. Magnesium
124. White fumes with container containing Mg_3N_2 due to formation of NH_4Cl
125. For sulphates hydration energy decreases more rapidly than lattic enthalpy while for hydroxides lattice Enthalpy decreases more rapidly than hydration Enthalpy hence the difference in solubility pattern.
126. Excess concentration of Cl^- results in the formation of soluble complex like $[PbCl_4]^{2-}$
127. $Pb^{2+} + 2I^- \rightarrow PbI_2$
$PbI_2 + 2I^- \rightarrow [PbI_4]^{2-}$
128. A – K B – K_2O
C – KO_2 D – KOH
E – K_2O_2 F – K_2O_4
G – KO_3
129. A – NaOH B – $AgNO_3$
C – SO_2 D – BiOCl
130. Br^- in solution is precipitated as AgBr by adding Ag_2SO_4 and then Brown ring test is conducted.
131. (a) **132.** (b) **133.** (b)
134. (b) **135.** (b) **136.** (d)
137. (c) **138.** (c) **139.** (d)
140. (a) **141.** (d) **142.** (d)
143. (a) **144.** (c) **145.** (a)
146. (d) **147.** (d) **148.** (b)
149. (c) **150.** (d) **151.** (d)
152. (b) **153.** (b) **154.** (c)
155. (c) **156.** (b) **157.** (b)
158. (a) **159.** (d) **160.** (b)
161. (b) **162.** (b) **163.** (b)
164. (c) **165.** (c) **166.** (c)
167. (d) **168.** (b) **169.** (d)
170. (d) **171.** (a) **172.** (d)
173. (a) **174.** (c) **175.** (a)
176. (a) **177.** (d) **178.** (b)
179. (d) **180.** (c) **181.** (a)
182. (a) **183.** (d) **184.** (b)
185. (a) **186.** (d) **187.** (b)
188. (c)
189. (a)
190. (a), (b), (c)
191. (a), (b)
192. (b), (d)
193. (a), (c)
194. (b), (c), (d)
195. (c), (d)

196. (a), (b)
197. (b), (c)
198. (a) → (p), (q), (r)
(b) → (s)
(c) → (p), (r)
(d) → (q), (r)
199. (a) → (s)
(b) → (q), (r)
(c) → (q), (r), (s)
(d) → (p), (q)
200. (a) → (q), (s)
(b) → (r), (s)
(c) → (q)
(d) → (p), (q), (r)

HINTS AND EXPLANATIONS

Topic Grip

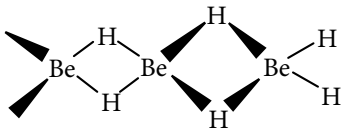
1. On comparing Na^+ and Ag^+, Ag^+ is a stronger lewis acid because both Na^+ and Ag^+ have almost the same ionic size, Ag^+ has higher nuclear charge and has a pseudo noble gas configuration. Therefore, it has greater polarising power, hence a better lewis acid.

2. NaOH, Both the chlorides of iron and aluminum react with NaOH to form the hydroxide. But aluminum hydroxide is soluble in excess NaOH while $Fe(OH)_3$ remains insoluble.

3. Lithium is also reactive
But since lithium is less dense than kerosene, it floats on kerosene. So it is wrapped in paraffin wax.

4. Alkali metal oxides are more basic than alkaline earth metal oxides because of
(i) lower ionization energy
(ii) larger cation size
(iii) smaller charge on alkli metal ion there is stronger attraction between alkaline earth metal ion M^{2+} or Ca^{2+} and O^{2-}
Therefore, K_2O is more basic than CaO.

5. Beryllium hydrides are difficult to prepare and are less stable than others. While CaH_2 is prepared by direct combination.
CaH_2 is ionic while BeH_2 is polymeric and exists as



forms an electron deficient compound.

6. (i) $Ca + 2C \xrightarrow{1100°C} CaC_2$
(ii) $Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$
Calcium reacts with cold water slowly.
(iii) $3Ca + 2P \xrightarrow{\Delta} \underset{\text{Calcium phosphide}}{Ca_3P_2}$

7. Magnesium oxide is a good refractory material because of
(a) High melting point (2800°C approximately)
(b) Chemical inertness even at high temperature
(c) Very good conductor of heat
(d) electrical insulator

8. On reaction of $PbCO_3$ with dil. H_2SO_4, initially CO_2 to formed, but insoluble $PbSO_4$ is also formed. This forms a protective layer on the remaining salt and does not allow further reaction. Therefore, effervescence stops after some time.

9. The initial reaction is Br^- is converted to Br_2 or chlorine replaces bromine from the salt.

$$2KBr + Cl_2 \rightarrow 2KCl + Br_2$$

Br_2 dissolves in chloroform to give the brown colour
When excess chlorine is passed
(i) Chlorine may react with liberated bromine to form bromine mono chloride which is pale yellow coloured.
$Br_2 + Cl_2 \rightarrow 2BrCl.$
(or)
(ii) Due to oxidizing nature of Cl_2 gas. Br_2 is oxidized to hypobromous acid

$$Br_2 + 2H_2O + Cl_2 \rightarrow \underset{\text{Hypobromous acid}}{2HBrO} + 2HCl$$

10. $AgCl + 2NH_4OH \rightarrow [Ag(NH_3)_2]\,Cl + 2H_2O$
Regents that can precipitate out silver are
(a) HNO_3 (b) KI
(a) $[Ag(NH_3)_2]Cl + 2HNO_3$

$$\rightarrow \underset{\text{White ppt}}{AgCl} + 2NH_4NO_3$$

(b) $[Ag(NH_3)_2]Cl + KI \rightarrow \underset{\substack{\text{yellow}\\\text{precipitate}}}{AgI} + KCl + 2NH_3$

11. PbI_2 dissolves in excess conc. KI due to formation of the complex $K_2[PbI_4]$.

12. Metallic bond strength decreases down the group. As the metallic bond strength decreases, it is easier to atomise the metal. so the first element will have the highest heat of atomization.

13. A piece of alkali metal when exposed to air finally forms the metal carbonate.

$$2Na + \frac{1}{2}O_2 \longrightarrow Na_2O$$

$$\xrightarrow{H_2O} 2NaOH \xrightarrow{CO_2} Na_2CO_3 + H_2O$$

14. $MgSO_4.7H_2O \xrightarrow[-6H_2O]{150^\circ C} MgSO_4.H_2O$

$\xrightarrow{200^\circ C} \underset{\text{Anhydrous}}{MgSO_4} + H_2O$

$MgSO_4 \xrightarrow{\Delta} MgO + SO_3$

15. Density = $\frac{\text{mass}}{\text{volume}}$

Atomic size decreases across the period and atomic mass increases across the period so density increases.

16. Lithium being the smallest ion, is heavily hydrated hence has a larger hydrated radius. Therefore, it has the least ionic mobility.

It is the size of hydrated ion that determines the electrical conductivity.

17. Statement II should be

Since Li^+ is a small cation, it has greater polarizing power and as I^- is a large anion, it is easily polarizable, hence the bond has greater covalent character. So the dipole moment is less than expected.

18. Soda water is CO_2 in water, hence acidic. $CaCO_3$ dissolves in Soda water due to the formation of $Ca(HCO_3)_2$

$CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2$

19. Statement II is true

Only AgCl is soluble in ammonia

$HgCl_2$ and Hg_2Cl_2 give a black precipitate

$AgCl + 2NH_4OH \rightarrow [Ag(NH_3)_2]Cl + 2H_2O$

$Hg_2Cl_2 + 2NH_4OH \rightarrow H_2N - Hg - Cl + Hg \downarrow$

$+NH_4Cl + 2H_2O$

20. $Zn + 2NaOH \rightarrow Na_2ZnO_2 + 2H$

$NaNO_3 + 8H \rightarrow NaOH + 2H_2O + \underset{\text{pungent gas}}{NH_3}\uparrow$

21. Boiling converts bicarbonates into insoluble carbonates.

$Ca(HCO_3)_2 \xrightarrow{\Delta} CaCO_3 + H_2O + CO_2.$

22. $NaHCO_3$ and sodium stearate cannot remove hardness. Synthetic zeolites can remove hardness by ion exchange.

23. $MgCl_2$, $Ca(OH)_2$ and $FeSO_4$ introduce Mg^{2+}, Ca^{2+} and Fe^{2+} ions and cause hardness to water.

24. $NaOH + 4S \xrightarrow{\Delta} Na_2S_2O_3 + 2Na_2S + 3H_2O.$

$AgNO_3$ forms black Ag_2S precipitate with $Na_2S.AgNO_3$ forms white $Ag_2S_2O_3$ turning into black Ag_2S with excess $Na_2S_2O_3$. alk. $KMnO_4$ and lead acetate can confirm Na_2S only. Iodine can be used to identify $Na_2S_2O_3$ only.

25. Caustic potash is highly reactive. It attacks many metals. Iron, Nickel, and Silver remain unattacked under these conditions.

26. Due to higher solubility of KOH in ethanol, alcoholic potash is commonly used.

27. $Ca_3N_2 + 3H_2O \rightarrow \underset{\text{SolutionB}}{3Ca(OH)_2} + \underset{\text{Gas A}}{2NH_3}$

$Ca(OH)_2 + CO_2 \rightarrow \underset{\text{turbidity}}{CaCO_3} + H_2O$

$CaCO_3 + CO_2 + H_2O \rightarrow \underset{\text{Clear solution}}{Ca(HCO_3)_2}$

$Ca(OH)_2$ On standing forms a white layer on top due to formation of $CaCO_3$ on the surface by action of atmospheric CO_2.

$Ca(HCO_3)_2$ exists only in solution.

28. While testing for nitrates, if nitrites are present they would interfere with the test;

Addition of NH_4Cl or urea ensures removal of nitrites as N_2

$NaNO_2 + NH_4Cl \rightarrow NaCl + NH_4NO_3 \xrightarrow{}$

$2H_2O + N_2\uparrow$

$2NaNO_2 + H_2SO_4 \rightarrow Na_2SO_4 + 2HNO_2$

$2HNO_2 + CO(NH_2)_2 \rightarrow CO_2 + N_2 + 3H_2O$

29. (a) Be does not directly combine with H_2

$2BeCl_2 + LiAlH_4 \rightarrow 2BeH_2 + LiCl + AlCl_3$

(b) All alkaline earth metals dissolve in ammonia.

(c) Be halides being covalent are water insoluble except BeF_2 which is ionic.

(d) Be is amphoteric while others are basic.

30. (a) $\rightarrow$ (p), (q), (r)

(b) $\rightarrow$ (p), (q)

(c) $\rightarrow$ (r), (s)

(d) $\rightarrow$ (s)

Washing soda – $Na_2CO_3.10H_2O$. undergoes efflorescence on exposure to air.

Water soluble and basic salt.

Glauber's Salt – $Na_2SO_4.10H_2O$ show efflorescence water soluble neutral.

Magnesia – MgO water insoluble, basic, sets to hard mass when mixed with $MgCl_2$ solution and used as magnesia cement.

Gypsum – $CaSO_4.2H_2O$

loses water only on heating and sets to a hard mass.

IIT Assingment Exercise

31. Due to high electropositive nature of sodium, it reacts with water vigorously and the reaction is highly exothermic. Hydrogen evolved during the reaction will also catch fire leading to spread of fire.

32. $ZnCl_2 + 2NaOH \rightarrow Zn(OH)_2 + 2NaCl$

$\xrightarrow{NaOH} \underset{\substack{\text{Soluble Sodium}\\ \text{zincate}}}{Na_2ZnO_2}$

33. $NaOH + CO \xrightarrow[5-10atm]{200°C} \underset{\text{Sodiumformate}}{HCOONa}$.

34. Since Lattice energy $\alpha \frac{1}{d^2}$

due to small size of Li^+ and F^-, LiF has greater lattice energy, hence low solubility.

35. Alkali metals are ns^1 configuration hence paramagnetic. The metal ions are M^+ ie ns^0 configuration hence diamagnetic.

36. Due to small size of Li^+, LiCl is covalent while NaCl is ionic. Hence LiCl has a lower melting point

37. On reaction with NaOH, NH_4Cl alone evolves ammonia

$NH_4Cl + NaOH \rightarrow NH_3\uparrow + NaCl + H_2O$.

38. $MgCl_2$

Magnesium has very high ionization energy.

39. Due to the presence of solvated electrons or ammoniated electrons, the solution is coloured.

40. $CaCl_2$ and NaF are added to decrease the melting point. They do not get electrolyzed under the voltage conditions.

41. Na_2SO_4 formed is heated with coal dust

$Na_2SO_4 + 4C \rightarrow Na_2S + 4CO\uparrow$

$Na_2S + CaCO_3 \rightarrow \underbrace{CaS + Na_2CO_3}_{\text{Black ash}}$

Sodium carbonate is extracted from black ash by extracting with water.

42. a, b, d show no reaction

$P_4 + 3NaOH + 3H_2O \rightarrow \underset{\substack{\text{Sodium hydrogen}\\ \text{phosphite}}}{3NaH_2PO_2} + \underset{\text{Phosphine}}{PH_3}$

43. K_2CO_3 is more water soluble than Na_2CO_3, it does not separate out when CO_2 is passed into KOH.

$$2KOH + CO_2 \rightarrow K_2CO_3 + H_2O$$

44. In NaF, F has small size, Therefore, the compound is more ionic. Hence higher melting point.

45. Due to the presence of two valence electrons as compared to one in alkali metals, the bond formation is stronger hence stronger metallic bonds are formed in alkaline earth metals.

46. Beryllium is not attacked by water because of a protective oxide coating.

Beryllium forms polymeric hydrides

47. (b)

48. $Be + 2NaOH + 2H_2O \rightarrow \underset{\text{Sidiumberyllate}}{Na_2BeO_2.2H_2O} + H_2$

49. $Mg + \underset{\text{dil.}}{H_2SO_4} \rightarrow MgSO_4 + H_2$

$Mg + \underset{\text{conc}}{2H_2SO_4} \rightarrow MgSO_4 + SO_2 + H_2O$

50. $MgCl_2.6H_2O$ on dehydration undergoes hydrolyzis by its by its water of hydration.

$MgCl_2 . 6H_2O \xrightarrow{\Delta} Mg(OH)Cl + HCl + 5H_2O$

$Mg(OH)Cl \xrightarrow{\Delta} MgO + HCl$

It should be dehydrated in a stream of dry HCl gas.

51. Only $Be(OH)_2$ is water insoluble

so it has the least K_{sp}

52. due to small size, lattice energy is higher hence solubility decreases.

53. Only BeO is amphoteric, all others are basic.

54. Magnesium being more electropositive reacts with CO_2 keeping the reaction going.

$2Mg + CO_2 \rightarrow 2MgO + C$

55. Be is $Is^2\ 2s^2$ and does not have d orbitals.

Due to small size and high charge, they can hold more molecules of water of crystallization.

56. Due to small size and high charge, alkaline earth metals can be easily hydrated.

57. Barium imparts apple green colour to the flame in flame test.

58. As the size of cation increases, the ease of formation nitride decreases.

59. Because Be has the highest. Ionization energy, and highest enthalpy of atomization it has the highest standard reduction potential.
i.e., the least negative nature.

60. Presence of SO_4^{2-} may precipitate out Ba^{2+} and Sr^{2+} as their sulphates. Hence $(NH_4)_2SO_4$ cannot be used.

61. An excess of a more concentrated (6M) Solution of KI dissolves the ppt.

$$PbCl_2 + 2KI \rightarrow \underset{\text{golden spangles}}{PbI_2\downarrow} + 2KCl$$

$$PbI_2 + 2KI \rightarrow \underset{\text{Soluble}}{K_2[PbI_4]}$$

62. $PbCl_2$ is insoluble in cold and soluble in hot solution.

63. The gas is SO_2.reduces Cr^{6+} to Cr^{3+}

$$2H_2S + SO_2 \rightarrow \underset{\substack{\text{colloidal}\\ \text{sulphur}}}{3S} + 2H_2O$$

colloidal sulphur gives the white turbidity

64. $Na_2S + 2HCl \rightarrow 2NaCl + H_2S$

$$(CH_3COO)_2Pb + H_2S \rightarrow \underset{\text{black}}{PbS} + 2CH_3COOH$$

(Pb, Cd, Ni, Co) sulphides do not decomposes with dil acid and require con.HCl

65. The gases are CO_2 and SO_2. Both give white precipitate of $CaCO_3$ with time water.
SO_2 turns acidified dichromate paper green

$$\underset{\text{Orange}}{Cr^{6+}} \xrightarrow{SO_2} \underset{\text{green}}{Cr^{3+}}$$

66. $FeSO_4.7H_2O + NO \rightarrow [Fe(H_2O)_5NO]SO_4 + 2H_2O$

67. Acetates are tested using neutral ferric chloride

$$FeCl_3 + 3CH_3COONa \rightarrow \underset{\text{wine red colour}}{(CH_3COO)_3Fe} + 3NaCl$$

68. $2KI + 2H_2SO_4 \rightarrow 2KHSO_4 + 2HI$
$MnO_2 + H_2SO_4 \rightarrow MnSO_4 + H_2O + (O)$
$2HI + [O] \rightarrow H_2O + I_2$.

69. $Ag_2SO_4 + BaCl_2 \rightarrow 2AgCl\downarrow + BaSO_4\downarrow$
Both are precipitates hence test for sulphate will be unsatisfactory.
In order to form a water soluble salt of
Ag^+ $Ba(NO_3)_2$ is used
$Ag_2SO_4 + Ba(NO_3)_2 \rightarrow 2AgNO_3 + BaSO_4\downarrow$.

70. Cl_2 water is an oxidizing agent which oxidizes I^- to I_2 (−1 to 0 oxidation state)
So an oxidizing agent is required
Among the four Con.HNO_3 is an oxidizing agent.

71. As all sodium salts are water soluble, the given salt on heating with sodium carbonate, undergo double decomposition reaction and convert the anions present into soluble sodium salt. So all anions in the mixture are brought into solution.

72. Both Na^+ and SO_4^{2-} cannot be oxidized.

73. Among the given options only $ZnSO_4$ is water soluble.

74. The precipitate is $Al(OH)_3$.
$Al(OH)_3$ formed in group III dissolves in NaOH and not in NH_4OH.

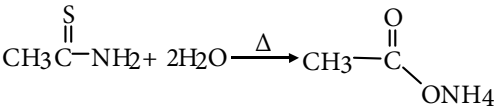
75. Group I ppt contain chlorides of Pb^{2+}, Ag^+, Hg_2^{2+}
$PbCl_2$ is soluble in hot water.
AgCl dissolves in excess ammonia.
So the salt is Hg_2Cl_2.
When copper is added it displace Hg^{2+} from solution. So grey deposits can be seen on copper.

76. $$(SnCl_2 + 2NaOH \rightarrow \underset{\text{ppt}}{Sn(OH)_2} + 2NaCl$$

$$Sn(OH)_2 + 2NaOH \rightarrow \underset{\text{Sodiums tan nite (soluble)}}{Na_2SnO_2} + 2H_2O$$

77. If H_2S gas is present in an alkaline medium provided by Group III reagents, S^{2-} concentration increases. Hence high K_{sp} of group IV sulphides is reached hence Group IV sulphides precipitate out.

78. Both MnS and ZnS form the soluble chlorides.

79. 

80. Ag^+, Cu^{2+}, Cd^{2+} on reaction with KCN gives precipitate which is soluble in excess KCN. forming complexes $[Ag(CN)_2]^-$, $[Cu(CN)_4]^{3-}$, $[Cd(CN)_4]^{2-}$ respectively.
Pb forms $Pb(CN)_2$ which is insoluble in excess KCN
$Pb^{2+} + 2CN^- \rightarrow Pb(CN)_2\downarrow$.

81. Cs < Rb < K. As electro positive nature decreases or electronegative nature increases reaction with hydrogen increases.

82. Halides of K, Rb, Cs combine with extra halogen atoms to form the polyhalides. This is because large cation stabilizes large anion.

83. $2M + 2NH_3 \rightarrow 2MNH_2 + H_2$

84. $NaNO_3 \rightarrow NaNO_2 + \frac{1}{2}O_2$

$4LiNO_3 \rightarrow 2Li_2O + 4NO_2 + O_2$

$MgCO_3 \rightarrow MgO + CO_2$

$Mg(OH)_2 \rightarrow MgO + H_2O$

85. It is easily oxidized by air to thiosulphate which can be used for iodine titrations.

$2Na_2S + 2O_2 + H_2O \rightarrow Na_2S_2O_3 + 2NaOH$

$2Na_2S_2O_3 + I_2 \rightarrow 2NaI + Na_2S_4O_6$

86. $AgNO_3 + KI \rightarrow \underset{\text{yellow ppt}}{AgI} + KNO_3$

$AgI + 2KCN \rightarrow K(Ag(CN)_2] + KI$

87. Due to small size of lithium. As it is much smaller than other atoms, it remains immiscible.

88. On standing, the colour fades away due to the formation of metal amide. It turns bronze coloured on increasing the concentration.

89. Lithium is the hardest alkali metal. Due to small size lithium forms the strongest metallic bonds. Hardness decreases on moving down the group.

90. For salts of strong acids, the lithium salt is usually the most soluble in water of the alkali metal salts, due to high solvation energy. Whereas, for weak acids the Li salts are usually less soluble than those of other alkali metals due to high lattice energy.

91. On moving down the group, cation size increases hence water of hydration decreases.

92. Conc. HNO_3 passivates Be, due to its oxidizing nature. A passive layer of BeO is formed on the surface.

93. Only Be is amphoteric. All others are basic

$Be + NaOH \rightarrow Na_2[Be(OH)_4] + H_2$

94. Li^+ is a small ion, while CO_3^{2-} is a large ion. Hence Li_2CO_3 is unstable.

95. In Na_2SO_4 hydration energy is greater than the lattice energy.

96. Main constituent – $CaCO_3$

$CaCO_3 + H_2O + CO_2 \rightarrow Ca(HCO_3)_2$

$CaCO_3$ will dissolve water saturated with CO_2

97. Calcium carbide

CaC_2 on hydrolysis liberates acetylene

$CaC_2 + 2H_2O \rightarrow Ca(OH)_2 + \underset{\text{acetylene}}{C_2H_2}$

98. The only oxide that dissolves in NaOH among s block element is BeO. BeO is amphoteric, all others are basic $Be(OH)_2$ is water insoluble.

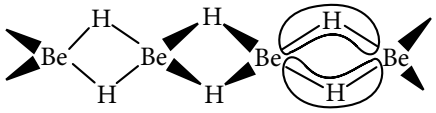
99. Ionisation energy decreases down the group and increases across the period.

100. $3Mg + N_2 \rightarrow Mg_3N_2 \xrightarrow{H_2O}$

$\underset{\text{Insolu ble, basic}}{Mg(OH)_2} + NH_3$

$Be(OH)_2$ is insoluble and amphoteric all others are basic and soluble. NH_3 forms a complex $[Cu(NH_3)_4]SO_4$ with $CuSO_4$.

101. BeH_2



3c-2e-Bond

102. with KCN

$Pb^{2+} + 2CN^- \rightarrow \underset{\text{white ppt}}{Pb(CN)_2}\downarrow$

$Pb^{2+} + 2I^- \rightarrow \underset{\substack{\text{yellow ppt}\\ \text{golden yellow spangles}}}{PbI_2}\downarrow$

$Pb^{2+} + CrO_4^{2+} \rightarrow \underset{\text{yellow ppt}}{PbCrO_4}\downarrow$

103. $Cd^{2+} + S^{2-} \rightarrow \underset{\text{yellow ppt}}{CdS}$

104. $NaF + H_2SO_4 \rightarrow NaHSO_4 + HF$

$SiO_2 + 4HF \rightarrow SiF_4\uparrow + H_2O$

$2SiF_4 + 4H_2O \rightarrow 2H_2[SiF_6] + \underset{\text{Waxy deposit}}{H_4SiO_4}$

105. AgCl dissolves in NaCN, KCN, $Na_2S_2O_3$ due to formation of complex and is insoluble in dil HNO_3.

106. a, b, c give out CO_2, SO_2 gases respectively which turn lime water milky but no blue edged flame. $Na_2C_2O_4$ when heated with conc. H_2SO_4 produces a mixture of CO and CO_2

$Na_2C_2O_4 + H_2SO_4 \rightarrow H_2C_2O_4 + Na_2SO_4$

$H_2C_2O_4 \xrightarrow{\Delta} CO + CO_2 + H_2O$

CO_2 turns lime water milky

CO burns with blue edged flame

107. $BiCl_3$

$BiCl_3 + H_2O \rightarrow \underset{\text{milkiness}}{BiOCl} + 2HCl$

108. Fe^{3+} may be partly reduced to Fe^{2+} by H_2S which is a strong reducing agent.

109. $PbCl_2$ is soluble in hot water and insoluble in cold water.

110. $HgCl_2 + 2KI \rightarrow 2KCl + \underset{\text{scarlet red ppt}}{HgI_2}$

$HgI_2 + 2KI \rightarrow \underset{\substack{\text{water soluble}\\ \text{colourless}}}{K_2[HgI_4]}$

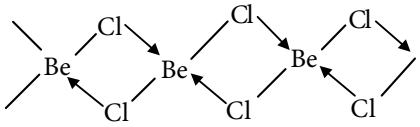
Alkaline solution of $K_2[HgI_4]$ is nessler's reagent.

111. Statement 2 is false. Even though there is an increase in volume it is more than compensated by increase in mass. So density increases down the group. In the case of Na and K, due to large volume of K, density decreases only on moving from Na to K.

112. $BeCl_2$ is acidic due to hydrolysis to form $Be(OH)_2$ and HCl.

$BeCl_2 + 2H_2O \rightarrow Be(OH)_2 + HCl$

Solid $BeCl_2$ exists as



113. On reaction with acid, sulphides give H_2S gas and sulphites give SO_2 gas. Even through both gases are different, both are reducing so both will reduce Cr^{6+} to green Cr^{3+}. Hence K_2CrO_4 cannot be used to distinguish the two.

$Na_2S + 2HCl \rightarrow 2NaCl + H_2S\uparrow$

$Na_2SO_3 + 2HCl \rightarrow 2NaCl + H_2O + SO_2\uparrow$

Reaction with K_2CrO_4

$2CrO_4^{2-} + 3H_2S + 10H^+ \rightarrow 2Cr^{3+} + 3S\downarrow + 8H_2O$

$2CrO_4^{2-} + 3SO_2 + 4H^+ \rightarrow$

$2Cr^{3+} + 3SO_4^{2-} + 2H_2O$

114. Chromyl chloride test is applicable only for chloride salts. (Chromates may be tested with NaCl instead of $K_2Cr_2O_7$ in the above test).

115. If sodium hydroxide is not neutralized with acetic acid, NaOH reacts with $Pb(CH_3COO)_2$. to form a white precipitate of $Pb(OH)_2$.

116. The chromyl chloride vapours formed get hydrolysed by moisture in the test tube to give chromic acid, so the vapours do not come out of the test tube.

$CrO_2Cl_2 + 2H_2O \rightarrow H_2CrO_4 + 2HCl$

117. Due to small size and higher charge, the lattice energy increases

$$L.E \propto \frac{q_1 q_2}{d^2}$$

When L.E > hydration Energy, compound remains insoluble.

118. $2MnO_4^- + 10I^- + 16H^+ \rightarrow 2Mn^{2+} + 8H_2O + 5I_2$

$2I^- + Pb(CH_3COO^-) \rightarrow \underset{\text{yellow ppt}}{PbI_2\downarrow} + CH_3COOH$

$CuSO_4 + 2I^- \rightarrow CuI_2 + SO_4^{2-}$

$2CuI_2 \rightarrow \underset{\text{white ppt}}{CuI_2} + I_2$

119. Beryllium hydride can be prepared by heating $BeCl_2$ with lithium borohydride. All hydrides are reducing in nature. Both BeH_2 and MgH_2 are covalent in nature BeH_2 exists as a polymer and forms electron deficient 3 centre 2 electron bonds.

120. (a) $\rightarrow$ (p)

(b) $\rightarrow$ (p)

(c) $\rightarrow$ (q), (r)

(d) $\rightarrow$ (q) (r), (s)

Li_2O and MgO have low water solubility. Na_2O_2 being a peroxide gives H_2O_2 on hydrolysis Na_2O_2 is coloured due to the presence of some superoxide.

KO_2 is a superoxide. O_2^- an odd electron molecule is coloured and paramagnetic due to presence of odd electron liberates H_2O_2 and O_2 on hydrolysis, hence highly oxidizing.

Additional Practice Exercise

121. Among the hydroxides of alkali metals, only LiOH gives Li_2O on heating.

$2LiOH \xrightarrow{\Delta} Li_2O + H_2O$

It has the lowest solubility in water among the hydroxides of the alkali metal.

122. The oxide is Na_2O_2

$Na_2O_2 + 2H_2O \rightarrow 2NaOH + H_2O_2$

$2NaOH + CO_2 \rightarrow Na_2CO_3 + H_2O$

123. $4NH_3 + 3O_2 \xrightarrow{\Delta} 2N_2\uparrow + 6H_2O$

At 20°C H_2O becomes liquid

Ca or Mg + $N_2 \rightarrow Mg_3N_2$ or Ca_3N_2

(Na and K do not react with N_2)

$$\left.\begin{matrix}Ca_3N_2\\ \text{or } Mg_3N_2\end{matrix}\right\} + 6H_2O \rightarrow \left.\begin{matrix}3Mg(OH)_2\\ 3Ca(OH)_2\end{matrix}\right\} + 2NH_3$$

$Ca(OH)_2$ is sparingly soluble whereas $Mg(OH)_2$ is insoluble in water. Mg_2C_3 produce propyne on hydrolysis while CaC_2 gives acetylene.

$\therefore$ (M) is Mg

124. $Mg_3N_2 + 6H_2O \rightarrow 3Mg(OH)_2 + 2NH_3$

Ammonia gas gives white fumes of NH_4Cl with HCl. But, $MgCl_2$ forms hexahydrate at room temperature (it gives MgO and HCl only on heating).

N^{3-} has a higher charge density (small size and high charge) and hence hydrolysis takes place readily. But, Cl^- being a weak conjugate base, does not undergo hydrolysis.

125. Lattice energy of salts decreases and hydration energy also decreases down the group. The former increases and the latter decreases solubility. For hydroxides and Fluorides lattice energy decreases more rapidly than hydration energy increasing the solubility. For other salts hydration energy decreases more rapidly than lattice energy thus decreasing the solubility.

126. A large excess of chloride ion must be avoided because AgCl and $PbCl_2$ tend to react with chloride ions forming soluble salts of complex ions.

$$AgCl + Cl^- \rightleftharpoons [AgCl_2]^-$$

$$PbCl_2 + 2Cl^- \rightleftharpoons [PbCl_4]^{2-}$$

127. $Pb^{2+} + 2I^- \rightarrow PbI_2$ (yellow ppt.)

$$PbI_2 + 2I^- \rightarrow \underset{\text{soluble}}{[PbI_4]^{2-}}$$

128. Since (A) forms only its oxides with air, it may be Na, K, Rb or Cs. (Li forms a mixture of the oxide and nitride). Since the monoxide (B) formed from (A) is pale yellow coloured, (A) should be potassium as Na_2O is white, Rb_2O s bright yellow and Cs_2O is orange coloured.

$$4K + O_2 \rightarrow \underset{(B)}{2K_2O}$$

The deep yellow coloured as well as the paramagnetic oxide of K is KO_2 (C).

(B) reacts violently with water to give (D)

$$K_2O + H_2O \rightarrow \underset{(D)}{2KOH}$$

Diamagnetic oxide should be O_2^{2-} $\therefore$ (E) is K_2O_2

KOH reacts with ozone to give KO_3 and K_2O_4, but only K_2O_4 on strong heating gives K_2O (B)

$$2KOH + O_3 \rightarrow K_2O_4 + H_2O$$

Tetroxide (F) is a chrome yellow powder

$$2K_2O_4 \xrightarrow{\Delta} 2K_2O + 3O_2 \uparrow$$

$$2KOH + 5O_3 \rightarrow \underset{(G)}{2KO_3} + H_2O + O_2$$

(G) is paramagnetic orange coloured compound.

(A) – Potassium

(B) – Potassium mono oxide, K_2O

(C) – Potassium super oxide, KO_2

(D) – Potassium hydroxide, KOH

(E) – Potassium peroxide, K_2O_2

(F) – Potassium tetroxide, K_2O_4

(G) – Potassium ozonide, KO_3

129. (i) When H_2S is passed in alkaline medium, sulphides of group II and group IV metals will precipitate out. Sulphides soluble in dilute. HCl are ZnS and MnS. When $ZnCl_2$ and $MnCl_2$ are treated with excess of (A), $ZnCl_2$ gives $Zn(OH)_2$ which is soluble in excess NaOH, whereas $MnCl_2$ gives a brown precipitate of $Mn(OH)_2$.

$\therefore$ (A) should be NaOH

$$ZnCl_2 + NaOH \rightarrow Zn(OH)_2 \xrightarrow{NaOH} \underset{\substack{\text{Sodium zincate}\\ \text{(Soluble)}}}{Na_2ZnO_2}$$

$$MnCl_2 + NaOH \rightarrow \underset{\text{Brown ppt}}{Mn(OH)_2}$$

(ii) (B) is $AgNO_3$

$$2AgNO_3 \xrightarrow{\Delta} 2Ag + \underset{\text{Brown gas}}{2NO_2} + O_2$$

$$2AgNO_3 + K_2CrO_4 \rightarrow \underset{\text{Red ppt}}{Ag_2CrO_4} + 2KNO_3$$

$$2AgNO_3 + Na_2S_2O_3 \rightarrow Ag_2S_2O_3 + 2NaNO_3$$

$$Ag_2S_2O_3 + 3Na_2S_2O_3 \rightarrow \underset{\text{(soluble)}}{2Na_3[Ag(S_2O_3)_2]}$$

(iii) $\underset{\text{(orange)}}{K_2Cr_2O_7} + 3SO_2 + H_2SO_4$

$$\rightarrow K_2SO_4 + \underset{\text{(green)}}{Cr_2(SO_4)_3} + H_2O$$

$$2H_2S + SO_2 \rightarrow 3S + 2H_2O$$

The white turbidity is due to colloidal sulphur. (C) is SO_2 gas

(iv) $Bi(OH)_3 + 3HCl \rightarrow BiCl_3 + 3H_2O$

$$BiCl_3 + H_2O \rightleftharpoons \underset{\text{(White ppt)}}{BiOCl} + 2HCl$$

(D) is BiOCl, Bismuth oxychloride

130. A solution of silver sulphate is added slowly till all Br^- is precipitated as AgBr. AgBr is filtered off Brown ring test is conducted in the filtrate as usual by adding fresh $FeSO_4$ solution and conc.H_2SO_4 slowly through the sides of the test tube. Brown ring is produced.

131. The metal may be Lithium or any of the alkaline earth metals.

132. $Na_2O_2 + 2H_2O \rightarrow 2NaOH + H_2O_2$
$2NaOH + CO_2 \rightarrow Na_2CO_3 + H_2O.$

133. Due to small size of lithium, it has greater hydration energy compared to others, hence it exists as hydrated salt.

134. Litmus paper can be used to separate out: NaCl. Among dil.HCl, and dil.H_2SO_4, $BaCl_2$ can be used to identify dil.H_2SO_4

135. Group V reagent are NH_4OH and $(NH_4)_2CO_3$.
NH_4^+ reduces CO_3^{2-} concentration by the reaction.

$$NH_4^+ + CO_3^{2-} \rightleftharpoons NH_3 + HCO_3^-$$

Hence if Na_2CO_3 is added, it results in a large concentration of CO_3^{2-} causing $MgCO_3$ to precipitate.

136. Mg^{2+} is precipitated as basic carbonate
$MgCO_3 . Mg(OH)_2$
On heating, only $NaHCO_3$ decomposes and not Na_2CO_3

$$\underset{\text{0.02 moles 1.68 g}}{2NaHCO_3} \xrightarrow{\Delta} Na_2CO_3 + H_2O + \underset{\substack{\text{0.224 L at STP}\\ \text{0.01 moles}}}{CO_2}$$

0.02 moles 0.224 L at STP
1.68 g 0.01 moles

$\therefore$ $Na_2CO_3 = 3.18\ g = 0.03$ moles

Mole percentage of $Na_2CO_3 = \dfrac{0.03}{0.05} = 60$

137. $Co^{2+} + 7NO_2^- + 2H^+ + 3K^+ \rightarrow K_3[Co(NO_2)_6]\downarrow + NO + H_2O$

138. $2Cu^{2+} + [Fe(CN)_6]^{4-} \rightarrow Cu_2[Fe(CN)_6]$
Chocolate brown ppt.
$3Zn^{2+} + 2k^+ + 2Fe(CN)_6^{4-} \rightarrow K_2Zn_3[Fe(CN)_6]_2$
White ppt.

139. $\underset{(A)}{MgCl_2} + \underset{(B)}{Ca(OH)_2} \rightarrow CaCl_2 + \underset{(C)}{Mg(OH)_2}\downarrow$

$$CaCl_2 + 2NaOH \rightarrow \underset{(B)}{Ca(OH)_2} + 2NaCl$$

140. $NaNO_3$ is a deliquescent substance and changes its physical state by absorbing moisture.

141. M_2SO_4, since the valency of the metal is one.

142. $CaCO_3$ is insoluble in water while K_2CO_3 and Na_2CO_3 are readily soluble in water. So even if minerals containing Na_2CO_3 were formed, they would dissolve in rain and water.

143. NaOH reacts with $SiO_2 \rightarrow Na_2SiO_3 + H_2O$
This may jam the stopper. Hence cork stopper should be used.

144. Mg reacts with steam or hot water
Be does react even at high temperature.

145. Let weight of Na_2CO_3 and K_2CO_3 be xg and
$(1 - x)$ g

$\therefore$ No.of equivalents $= \dfrac{x}{53} + \dfrac{(1-x)}{69}$

$\therefore$ $\dfrac{x}{53} + \dfrac{(1-x)}{69} = 0.5 \times 30 \times 10^{-3}$

$69x + 53 - 53x = 53 \times 69 \times 0.5 \times 30 \times 10^{-3}$

$16x = 54.8 - 53$

$x = \dfrac{1.8}{16} = 0.116 \cong 0.12g$

146. Solubility of hydroxide increases down the group.

147. Bicarbonates of lithium and alkaline earth metals exists only in solution.

148. Only lithium forms carbides among alkali metals. It forms ionic carbide.
Group 2 metals ,– all react with carbon to form carbides.

149. Since anion is very large and cation is very small there is anion-anion repulsion and consequently lattice energy is small. Therefore, it is soluble in water and acetone.

150. s-block elements are strong reducing agents so they are always reduced electrolytically.

151. Be is amphoteric hence it reacts with NaOH liberating H_2 gas
While other Group 2 elements are basic and do not give this reaction.

152. Be and Mg do not react with water
Only Be reacts with NaOH with NH_3, H_2 is not liberated readily.

153. Be^{2+} does not exist freely but only in the polymeric form,
The ion always exists as $[Be(H_2O)_4]^{2+}$.

154. $Mg(OH)_2$ is less soluble in water compared to $Ca(OH)_2$. $Mg(OH)_2$ precipitates out.

155. Soap is sodium stearate which is water soluble. If Ca^{2+} and Mg^{2+} are present, they form calcium or magnesium stearate which are insoluble and form scum. If Ca^{2+} and Mg^{2+} are replaced by Na^+, there will be no scum formation.

156. a, c, d are related to small size of alkaline earth metal as compared to the alkali metal.

157. $MgO + C + Cl_2 \rightarrow MgCl_2 + CO$
$MgCl_2$ is the compound formed. It is highly water soluble, deliquescent.
Saturated solution of $MgCl_2$ mixed with MgO forms a hard marble like mass called sorrel's cement.

158. $\underset{\text{Dolomite}}{MgCO_3.CaCO_3} + 2H_2SO_4 \rightarrow MgSO_4 + CaSO_4\downarrow + 2H_2O + CO_2\uparrow$
On going down the group solubility of sulphates decreases so $MgSO_4$ is water soluble while $CaSO_4$ is water insoluble hence filtered.

159. Small cation stabilizes a small anion Be^{2+} is a small cation while CO_3^{2-} is a large anion hence $BeCO_3$ is unstable

160. They do not form polyhalides
(c) and (d) are true only for Beryllium

161. CO_3^{2-} is not reducing
$SO_3^{2-} + 2H^+ \rightarrow SO_2 + H_2O$
$5SO_2 + 2MnO_4^- + 2H_2O \rightarrow 2Mn^{2+} + 5SO_4^{2-} + 4H^+$
$2MnO_4^- + 10I^+ + 16H^+ \rightarrow 2Mn^{2+} + 8H_2O + 5I_2$
$2MnO_4^- + 5NO_2^- + 6H^+ \rightarrow 2Mn^{2+} + 5NO_3^- + 3H_2O$

162. Al^{3+} is soluble in excess NaOH.
In (a), (c), (d), Fe^{3+}, Fe^{2+} and Cr^{3+} form precipitate with NaOH that are insoluble in excess.
But Sn^{2+} forms soluble sodium stannite with NaOH, hence cannot be separated from Al^{3+} using NaOH.

163. Initially Zn^{2+} is dissolved in HCl (Group I) so Zn^{2+} exists as $ZnCl_2$.
$ZnCl_2 + 2H_2O \rightarrow Zn(OH)_2 + 2HCl$
Since H^+ ions are produced in solution, dissociation of H_2S will be controlled. Hence after some initial precipitation stops when $[H^+]$ reaches 0.3 M

164. Zn^{2+} an d Ca^{2+} are soluble in ammonia due to formation of complex.
Al^{3+} dissolves in NaOH;' but forms gelatinous white precipitate of $Al(OH)_3$ with NH_4OH. $(NH_4)_2CO_3$ will precipitate all the three.

165. The test fails in the presence of HCl as.HCl is oxidized by PbO_2

166. H_3BO_3 is weaker than H_2CO_3
hence it will not liberate CO_2 from the carbonate.

167. $AgNO_3$gives precipitates with a, b, c
If cannot be used to test F^- because AgF is water soluble.

168. $Ba(NO_3)_2 + MSO_4 \rightarrow BaSO_4\downarrow + M(NO_3)_2$
$BaCl_2 + MSO_4 \rightarrow BaSO_4 + MCl_2$.
$M(NO_3)_2$ formed is always water soluble. But all metal chlorides are not water soluble.
e.g., AgCl, $PbCl_2$.

169. Only NaBr will give a pale yellow precipitate

170. Due to small size of Be^{2+}, in aqueous
$BeCl_2$ exists as $[Be(H_2O)_4]^{2+}$
The acts as a electron pair acceptor
Hence a Lewis acid.

171. Statement 1 is true
Statement 2 is true
As the ionization energy decreases, the energy required to eject the outermost election ie work function decreases. So among alkali metals cesium has the least work function.

172. Statement 1 is false. $CaSO_4$ is partially precipitated.
Statement 2 is true.
$CaSO_4$ formed by reaction of Ca^{2+} with $(NH_4)_2SO_4$, dissolves partially in excess $(NH_4)_2SO_4$ due to formation of complex.
$CaSO_4 + (NH_4)_2SO_4 \rightarrow (NH_4)_2[Ca(SO_4)$
soluble complex.

173. $I_2 + 2NaOH \rightarrow NaI + NaIO + H_2O$
disproportionation reaction or auto redox reaction

(0) (−1) (+2)

$I_2 \rightarrow I^- + OI^-$

Since I_2 reacts in presence of NaOH, the violet colour disappears as both I^- and OI^- are colourless.

174. Hydrated K^+ is smaller than hydrated Na^+. Larger the ion, lesser the hydration. As K^+ is a larger ion than Na^+, it is less hydrated.

NaCl and KCl can be distinguished using flame test. As their ionization energies are different they impact different colours to the flame.

175. $CaCl_2 + Na_2C_2O_4 \rightarrow CaC_2O_4 \downarrow 2NaCl$

$CaCl_2 + 2NaF \rightarrow CaF_2 \downarrow + 2NaCl.$

In the presence of fluorides, both oxalate and fluoride of Ca^{2+} form white precipitates. Hence oxalate cannot be confirmed.

The second statement is true and is the right reason because it indicates $CaCl_2$ is water soluble while the fluoride is not.

176. S in H_2S is in — 2 0xidation state.

hence it is a good reducing agent.

Therefore, it is important to keep the solution free from oxidizing agent, as H_2S can get oxidized to colloidal sulphur.

177. As ionization energy decreases, reactivity increases. But when it comes to hydrogen they form ionic hydrides of the type M^+H^-. Ease of formation of hydrides decreases down the group, as size of cation increases and lattice energy decreases.

178. $Na_2O_2 + 2H_2O \rightarrow 2NaOH + H_2O_2$

H_2O_2 is oxidizing agent. If oxidizes Pbs (Black) into $PbSO_4$ (White).

Solution of alkali metal in ammonia gives out ammoniated electrons which make the solution reducing.

179. I^- is a larger anion than Cl^-. So PbI_2 is more covalent, hence less water soluble.

The amount of KI added should be just enough to precipitate PbI_2. Excess KI may lead to formation of complex $K_2[PbI_4]$ which is water soluble.

$2KI + PbCl_2 \rightarrow 2KCl + PbI_2$

$2KI + PbI_2 \rightarrow K_2[PbI_4]$ soluble

180. On going down the group, ionic size increases.

As size increase both lattice energy and hydration energy decreases but decrease in lattice is very less .compared to decrease in hydration energy. This explains solubility of chlorides bromides and iodides.

In the case of fluorides, due to the small size of the fluoride ion, increase in size of alkali metals is significant, hence lattice energy decrease is significant so solubility increases down the group.

181. Solid ammonium bicarbonate ($NH_4(HCO_2)$)

182. In solvay process, the bicarbonate is formed as an intermediate due to high solubility of $KHCO_3$ compared to $NaHCO_3$, K_2CO_3 cannot be prepared by solvay's process.

183. In the crystalline state $KHCO_3$ exists as a dimer

Crystal structure shows that there is hydrogen bonding in $KHCO_3$ and the anion exists as a dimer.

184. $BeCl_2 + 2H_2O \rightarrow Be(OH)_2 + 2HCl.\uparrow$

it fumes due to formation of HCl.

185. BeF_2 or $BeCl_2$ made from aqueous solution exists as the hydrate. On heating it undergoes hydrolysis due to the water of hydration.

$$[Be(H_2O)_4]F_2 \xrightarrow{\Delta} Be(OH)_2 + 2HF + 2H_2O$$

186. Both $BeCl_2$ and BeH_2 exist as polymers in the solid state. Though the structure is similar, and both show clustering, the hydride has a three centre two election bond while the halides have halogen bridges in which a halogen atom bonded to one beryllium atom uses a lone pair of elections to form a coordinate bond to another beryllium atom.

187. $4NO_3^- + 2H_2SO_4 \longrightarrow 4NO_2\uparrow + O_2\uparrow + 2SO_4^{2-} + 2H_2O$

(+5) $\xrightarrow{+4e^-}$ (+4)

(−2) → (0)

188. $2NO_3^- + 4H_2SO_4 + 3Cu \rightarrow 3Cu^{2+} + 2NO\uparrow$

$4SO_4^{2-} + 4H_2O$

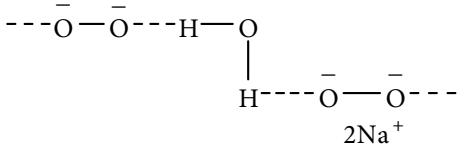
$2NO\uparrow + O_2 \rightarrow 2NO_2\uparrow$

Brown fumes

Due to formation of Cu^{2+}, solution turns blue

189. On addition of acid bromide would liberate Br_2 (Bromine) and iodide will liberate I_2 (violet) due to the liberation of these halogens, the brown ring test is not reliable.

190. (a), (b), (c)



191. Only Be_3N_2 is volatile and covalent while the other alkaline earth metal nitrides are non volatile and have ionic nature.

192. $Co^{2+} + 7NO_2^- + 2H^+ + 3K^+ \rightarrow \underset{\text{yellow ppt}}{K_3[Co(NO_2)_6]} + NO\uparrow + H_2O$

can be used to test for Co^{2+}, K^+ and NO_2^-

193. Li^+ polarizes the CO_3^{2-}.

So CO_3^{2-} is decomposed to $O^{2-} + CO_2$

Moreover due to small size of O^{2-}, Li_2O has higher lattice energy, hence more stable and reaction is favourable.

$Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2$

194. a – $CaSO_4 + 4C \rightarrow CaS + 4CO\uparrow$

b – $Ca(HCO_3)_2 \xrightarrow{\Delta} CaCO_3\downarrow H_2O + CO_2$

c – $CaSO_4 + 2NH_3 + CO_2 + H_2O \rightarrow (NH_4)_2SO_4 + CaCO_3$

d – $CaCN_2 + 5H_2O \rightarrow CaCO_3 + 2NH_4OH$

195. Due to high charge and small size, they have high lattice energy; Therefore, solubility decreases.

196. CsOH is stable, $NaNO_3$ gives $NaNO_2$ on heating

197. $Cl^- + AgNO_3 \rightarrow \underset{\text{curdy white precipitate}}{AgCl\downarrow} + NO_3^-$

This is soluble in NH_4OH, KCN, NaCN, $Na_2S_2O_3$ due to formation of complex.

198. (a) $\rightarrow$ (p), (q), (r)
(b) $\rightarrow$ (s)
(c) $\rightarrow$ (p), (r)
(d) $\rightarrow$ (q), (r)

SO_2 and NO_2 are reducing in nature, so they are easily oxidized by $KMnO_4$.

Baryta water is $Ba(OH)_2$. Both CO_2 and SO_2 turn it milky due to formation of $BaCO_3$ and $BaSO_3$ respectively.

SO_2, NO_2 and CO_2 are acidic gases. So they are easily absorbed by the base KOH.

NH_3 is a base so it turns red litmus blue.

199. (a) $\rightarrow$ (s)
(b) $\rightarrow$ (q), (r)
(c) $\rightarrow$ (q), (r), (s)
(d) $\rightarrow$ (p), (q)

Sodium reacts vigorously with water and forms water soluble basic hydroxide.

Due to small size Mg, Be and Li form hydrated halides

Magnesium reacts slowly with hot water. If forms basic insoluble hydroxide.

Lithium reacts slowly with water and forms basic water soluble hydroxide. Solubility is low.

Beryllium does not react with water and forms amphoteric water insoluble hydroxides.

200. (a) $\rightarrow$ (q), (s)
(b) $\rightarrow$ (r), (s)
(c) $\rightarrow$ (q)
(d) $\rightarrow$ (p), (q), (r)

$Na_2S_2O_3$ and Na_2CO_3 give out SO_2 and CO_2 respectively only on reaction with dil. acid. So(P) matches only with D.

$4Na_2S_2O_3 \xrightarrow{\Delta} 3Na_2SO_4 + Na_2S_5$

$2NaNO_3 \xrightarrow{\Delta} 2NaNO_3 + O_2$

$2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + H_2O + CO_2$

Na_2CO_3 and $NaHCO_3$ are salts of weak acid and strong base.

$Na_2S_2O_3$ exists as $Na_2S_2O_3.5H_2O$

Na_2CO_3 exists as $Na_2CO_3.10H_2O$

CHAPTER

2

The p-BLOCK ELEMENTS

CHAPTER OUTLINE

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Alumina, Aluminium Chloride and Alums

Carbon (C)

Oxides And Oxy-Acids of Carbon

Oxyacids of Carbon

Silicon (Si)

Nitrogen (N)
- Concept Strands (2-3)

Phosphorus (P)
- Concept Strands (4-7)

Oxygen (O)
- Concept Strands (8-9)

Sulphur (S)

Halogens (F, Cl, Br, I, At)
- Concept Strands (10-11)

Xenon Fluorides (XeF_2, XeF_4, XeF_6)
- Concept Strand 12

Commercially Available Fertilizers
- Concept Strand 13

TOPIC GRIP
- Subjective Questions (10)
- Straight Objective Type Questions (5)
- Assertion–Reason Type Questions (5)
- Linked Comprehension Type Questions (6)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

IIT ASSIGNMENT EXERCISE
- Straight Objective Type Questions (80)
- Assertion–Reason Type Questions (3)
- Linked Comprehension Type Questions (3)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

ADDITIONAL PRACTICE EXERCISE
- Subjective Questions (10)
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- Matrix-Match Type Questions (3)

THE p-BLOCK ELEMENTS

These are elements in which the last electron enter the outermost p-orbitals. Except Helium all the p-block elements have the general outer electronic configuration $ns^2\ np^{1-6}$, but the inner core of electronic configuration is not the same for all. They are placed in groups 13 to 18 of the periodic table. The major factors that decide the properties of p-block elements are, atomic size ionisation energy, electron gain enthalpy and electro negativity. The properties of second period p-block elements are significantly influenced by the absence of 'd' orbitals; where as in the case of heavier p-block elements it is the presence of 'd' and or 'f' orbitals that are the influeneing factors. p-block elements include metals, metalloids and non metals. They have the ability to react with, s, d and f-block elements and also with elements of p-block itself.

BORON (B)

Introduction

Boron (B) belongs to the category of p-block elements and is a member of the group 13 of the periodic table, the other members of the group being aluminium (Al), gallium (Ga), indium (In) and thallium (Tl). The metallic character (or electropositive nature) of the elements in this group increases as we go down the group from boron to thallium. Boron is a non-metal and it mainly forms covalent compounds. It resembles silicon (group 14) in its properties more closely than any other member of its own group (diagonal relationship). Some of important properties of boron are listed below:

Atomic number	Atomic weight	Electronic configuration	Atomic radius (A°)	Ionic radius (A°) B^{3+} ion	1st ionization energy kJ mol^{-1}	Oxidation state
5	10.82	$1s^2 2s^2 2p^1$	0.8	0.3	800	+3

The first three ionization energies of boron are considerably higher than the corresponding values for the other elements of the groups. Therefore, ionic compounds containing B^{3+} ions are not formed but only covalent compounds (involving electron sharing) are predominant and this is a dominant feature of boron chemistry. Another important aspect of boron chemistry is that it has only three valence electrons for covalent bond formation utilizing the four orbitals s, p_x, p_y, p_z. Therefore, electron deficiency arises in boron compounds, which makes them act as good Lewis acids.

Occurrence and extraction (isolation) of boron

Concentrated deposits of boron occur in the form of orthoboric acid H_3BO_3, $Na_2B_4O_7.10H_2O$ (borax) and $Na_2B_4O_7.4H_2O$ (kernite).

It was first isolated by Davy in 1807 by electrolysis of moist boric acid.

Boron is generally prepared by the reduction of boric oxide, B_2O_3 with magnesium or sodium at high temperature.

$$B_2O_3 + 3Mg \rightarrow 3MgO + 2B$$

$$B_2O_3 + 6Na \rightarrow 2B + 3Na_2O$$

The product is boiled with dilute HCl to remove metallic impurities. The purity of the amorphous form so obtained is 95-98% and it is brown in colour.

Pure crystalline form of boron is prepared by two methods: (i) by pyrolysis of boron iodide (ii) by thermal decomposition of diborane, B_2H_6.

$$2Bl_3 \xrightarrow[1200°C]{\text{red hot tungsten filament}} 2B + 3I_2$$ (known as Van Arkel's method)

$$B_2H_6 \xrightarrow{\text{heat}} 2B + 3H_2$$

Properties

Crystalline boron is black in colour and is extremely hard. It has a metallic lustre but possess low electrical conductivity. It is chemically inert and is not attacked by either conc. HNO_3 or conc.H_2SO_4. Boiling HCl has no effect on it.

On the other hand, amorphous boron is quite reactive and some of its reactions are given below.

(i) Reaction with air or oxygen

$$4B + 2O_2 \xrightarrow[700^\circ C]{\text{high temp.}} 2B_2O_3$$

A similar reaction occurs with sulphur to give B_2S_3

(ii) Reaction with N_2

$$2B + N_2 \xrightarrow[\text{temp}]{\text{high}} 2BN \text{ (Boron nitride)}$$

Boron nitride has a layered structure like graphite. When heated under pressure it is converted into a hard crystalline form resembling diamond. This adamantine form is harder than diamond.

(iii) Reaction with halogen

$$2B + 3Cl_2 \xrightarrow[\text{temp}]{\text{high}} 2BCl_3$$

A similar reaction occurs with other halogens. A molecule like BF_3 is trigonal planar with a bond angle of 120°. The shape is in accordance with VSEPR theory and sp^2 hybridization. Boron halides are covalent in nature as shown by their solubility in organic solvents. They are easily hydrolysed giving rise to acidic solutions.

$$BCl_3 + 3H_2O \rightarrow H_3BO_3 + 3HCl$$

(iv) Boron is oxidised by hot conc.HNO_3 to boric acid

$$B + 3HNO_3 \rightarrow H_3BO_3 + 3NO_2$$

(v) Reaction with fused alkali

It reacts with fused alkali liberating hydrogen.

$$2B + 6NaOH \rightarrow 2Na_3BO_3 + 3H_2$$

(vi) Reaction with NH_3

$$2B + 2NH_3 \xrightarrow[\text{temp}]{\text{High}} 2BN + 3H_2$$

CONCEPT STRAND

Concept Strand 1

Boron nitride exists in two different crystalline forms. Explain the differences in their structures.

Solution

Soft slippery form has a layer structure similar to graphite. When the soft form is heated to 3000°C at 70000 atmosphere pressure, a hard form called borazon is formed, which has diamond like structure (and is harder than diamond).

Uses of boron

Boron is used

(i) for making boron steel.
(ii) for making boron carbide (B_4C_3), which is used as control rods for capturing neutrons in nuclear reactors. It is also used as an abrasive.

It is used for making borosilicate glass which has a much lower thermal expansion compared to soda glass.

Oxides of boron

Boron sesquioxide, B_2O_3

It is prepared by the dehydration of boric acid.

$$\underset{\text{orthoboric acid}}{2H_3BO_3} \xrightarrow[-2H_2O]{373\text{ K}} \underset{\text{metaboric acid}}{2HBO_2} \xrightarrow[-H_2O]{\text{red heat}} \underset{\text{boric oxide}}{B_2O_3}$$

It is an acidic oxide and dissolves in alkali to give borates. It also reacts with basic metallic oxides forming salts known as borates or meta borates. However it is possible to force B_2O_3 to behave as a basic oxide by reacting with very strongly acidic compounds like P_2O_5 or As_2O_5 to form boron phosphate and boron arsenate, respectively.

$$B_2O_3 + P_2O_5 \rightarrow 2BPO_4$$

Boron hydrides

Boron forms a large number of covalent compounds with hydrogen known as boranes. Two series of boranes with the general formulae $B_nH_{(n+4)}$ and $B_nH_{(n+6)}$ generally known as nido boranes and arachno boranes, respectively, are the important ones. Among the former class, the foremost one is diborane (B_2H_6) followed by B_5H_9, B_6H_{10} etc. Among the latter class, the first member is B_4H_{10} (tetraborane) and it is followed by B_5H_{11}, B_6H_{12} etc.

The interest in the study of boranes arises from their unusual stoichiometry and structure. In recent years, they have gained importance because of their potential value as high energy fuels. A detailed account of diborane is undertaken below.

Diborane, B_2H_6

This is the simplest and widely studied of all hydrides. It is an important reagent in synthetic organic chemistry.

Preparation

(i) On a large scale, it is prepared by reduction of boron trifluoride gas with sodium hydride at 180°C. The reaction is

$$2BF_{3(g)} + 6NaH \xrightarrow{180^\circ C} B_2H_{6(g)} + 6NaF$$

(ii) In the laboratory, it is prepared by the oxidation of sodium borohydride with iodine in the polyether solvent, diglyme $(CH_3OCH_2CH_2)_2O$.

$$2Na[BH_4] + I_2 \rightarrow B_2H_6 + 2NaI + H_2$$

(iii) Another method of preparation involves the reduction of BF_3 – diethyl ether complex with sodium borohydride in diglyme. The reaction is

$$4[(C_2H_5)_2O.BF_3] + 3NaBH_4 \rightarrow 3Na[BF_4] + 4(C_2H_5)_2O + 2B_2H_6$$

(iv) B_2H_6 is also prepared by reduction of BCl_3 with lithium aluminimum hydride in anhydrous ether according to the reaction

$$3LiAlH_4 + 4BCl_3 \rightarrow 3LiCl + 3AlCl_3 + 2B_2H_6$$

Properties

Diborane is a highly reactive, colourless spontaneously inflammable gas. It is stable only at low temperatures. When heated in the temperature range, 100-250°C, it gives a number of higher boranes. The process involves decomposition as well as polymerization.

Reactions

(i) It readily catches fire in air and reacts explosively with O_2.

$$B_2H_6 + 3O_2 \rightarrow B_2O_3 + 3H_2O \quad \Delta H = -2165 \text{ kJ mol}^{-1}$$

(ii) It is instantly hydrolysed by water to give boric acid and H_2

$$B_2H_6 + 6H_2O \rightarrow 2H_3BO_3 + 3H_2$$

(iii) Diborane dissolves in strong alkali giving metaborates and hydrogen.

$$B_2H_6 + 2KOH + 2H_2O \rightarrow 2KBO_2 + 6H_2$$

(iv) It reacts with halogens under different conditions to form different products.

$$B_2H_6 + 6Cl_2 \xrightarrow{\text{room temperature}} 2BCl_3 + 6HCl$$

(vigorous reaction)

$$B_2H_6 + Br_2 \xrightarrow{100^\circ C} B_2H_5Br + HBr \text{ (slow reaction)}$$

(v) Diborane reacts with hydrogen halides to give the halodiborane and hydrogen.

$$B_2H_6 + HCl \rightarrow B_2H_5Cl + H_2$$

(vi) Hydrides of alkali metals react with diborane to give the borohydride of the metal.

$$B_2H_6 + 2NaH \rightarrow 2NaBH_4$$

(vii) Diborane reacts with alkenes and alkynes to form alkylborane products which can be converted to hydrocarbon, alcohols and ketones by treating them with suitable reagents. These reactions are known as hydroboration reactions.

$$6RCH = CH_2 + B_2H_6 \rightarrow 2B(CH_2CH_2R)_3$$ where –R is alkyl group

$$(RCH_2CH_2)_3B + 3H_2O_2 \xrightarrow{KOH} RCH_2CH_2OH + B(OH)_3$$

(viii) Boranes can accept electron pairs and thus act as Lewis acids. They react with tertiary amines to give simple adducts through symmetrical cleavage of diborane.

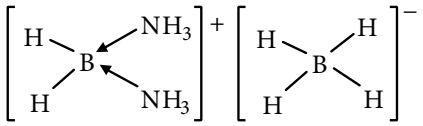
$$B_2H_6 + 2Me_3N \rightarrow 2[Me_3N.BH_3]$$

With excess of NH_3, primary and secondary amines at low temperature, diborane reacts to give an ionic adduct through unsymmetrical cleavage of diborane.

$$B_2H_6 + 2NH_3 \rightarrow [B_2H_6.2NH_3]$$

diborane diammoniate

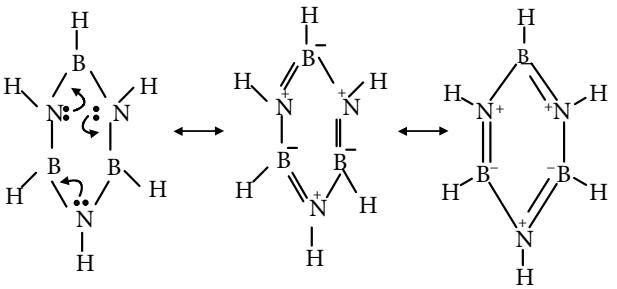
The adduct has an ionic structure



At higher temperatures, under the molar ratio of diborane to NH_3 as 1:2, a new compound known as borazine or borazole with the molecular formula $B_3N_3H_6$ is formed.

$$3B_2H_6.2NH_3 \xrightarrow{\Delta} 2B_3N_3H_6 + 12H_2$$

Borazine is aromatic in nature and has a ring structure similar to benzene and is some times referred as 'inorganic benzene'. The structure of borazine is



Borazine can also be made by reacting BCl_3 with NH_4Cl at 140°C and treating the resulting product with $NaBH_4$.

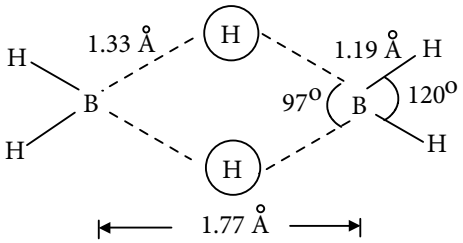
$$3BCl_3 + 3NH_4Cl \xrightarrow[140°C]{-9HCl} B_3N_3Cl_3H_3$$

$$\xrightarrow[NaBH_4]{-3HCl} B_3N_3H_6$$

Structure of diborane

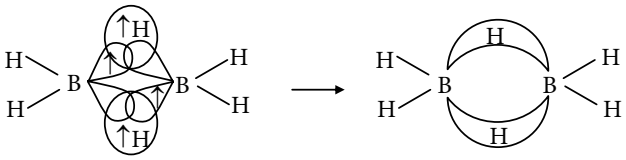
Boranes are interesting compounds from the point of view of their structure and bonding. There are not enough valence electrons to form the conventional two electron bonds between all the adjacent pairs of atoms. This can be seen from the fact that in diborane there are twelve valence electrons (three from each of two B atoms and six from six H atoms) and such compounds are known as electron deficient compounds.

The structure of diborane may be depicted as



Structure of diborane

In this structure, the two bridging H atoms (circled) are in a plane perpendicular to the rest of the molecule and lie above and below the plane. Boron atoms are in sp^3 hybridization. The terminal four H atoms are bonded to the two boron atoms (connected by full lines) by normal covalent bonds (with two electrons shared between two atoms) and may be described as two centre two electron (2c – 2e) bonds. The electron deficiency is overcome with the formation of two 3-centre bridge bonds.



The two bridges involve one electron from each B atom and one electron from each H atom. Thus there are four electrons forming two bridges. These bridging links may be described as three centre two electron bonds (3c – 2e). Because of their shape they are also called banana bonds.

Borohydrides

Boron forms a number of borohydride anions, the most important of them being the tetrahedral $[BH_4]^-$ ion. Sodium borohydride is prepared by treating trimethyl borate with sodium hydride or by the action of diborane on NaH.

$$4B(OMe)_3 + 4NaH \xrightarrow[250°C\ (in\ THF\ solvent)]{Hight\ presure} NaBH_4 + 3Na[B(OMe)_4]$$

$$B_2H_6 + 2NaH \xrightarrow{ether} 2NaBH_4$$

Other metal borohydrides are made by treating $NaBH_4$ with the required metal chloride.

The alkali metal borohydrides are useful reducing agents in organic and physical chemistry.

Boric acids

Boron forms a number of acids, which may be referred to as boric acids. In solid state, boric acid exists in three different forms known as

(i) Orthoboric acid, H_3BO_3
(ii) Metaboric acid, HBO_2
(iii) Tetraboric acid, $H_2B_4O_7$

In aqueous solutions, meta and tetraboric acid are converted into orthoboric acid. However, salts from all the three acids are known. A brief account of the most important acid of all viz, orthoboric acid will be presented.

Orthoboric acid, H_3BO_3

Preparation

There are two methods of preparation of boric acid.

(i) The mineral, colemanite, which has the composition $Ca_2B_6O_{11}.5H_2O$ is suspended in boiling water and treated with SO_2 gas, when orthoboric acid is formed

$$Ca_2B_6O_{11} + 4SO_2 + 11H_2O \rightarrow 2Ca(HSO_3)_2 + 6H_3BO_3\downarrow$$

(ii) A second method involves treatment of a hot concentrated solution of borax with the required quantity of conc.H_2SO_4. On cooling after the reaction, crystals of boric acid separate out.

$$Na_2B_4O_7 + H_2SO_4 + 5H_2O \rightarrow Na_2SO_4 + 4H_3BO_3$$

Properties

It crystallizes in the form of white flakes. It is sparingly soluble in cold water but the solubility increases very much in hot water. It is volatile in steam and gives a green colour to the non-luminous bunsen flame. It behaves like a weak acid ($K_a = 6 \times 10^{-10}$). It does not donate protons like other Bronsted acids but accepts OH^- ions. It is to be considered as a Lewis acid with the formula $B(OH)_3$ and the reaction in aqueous solution may be represented as

$$B(OH)_3 + 2H_2O \rightleftharpoons H_3O^+ + B(OH)_4^-$$

Reactions

(i) Boric acid decomposes on heating at 100°C to metaboric acid. When heated to 160°C, it gives tetraboric acid and at red heat forms a glassy mass of B_2O_3.

$$H_3BO_3 \xrightarrow{100^0\,C} HBO_2 \xrightarrow{160^0\,C}$$
$$H_2B_4O_7 \xrightarrow{\text{red heat}} B_2O_3$$

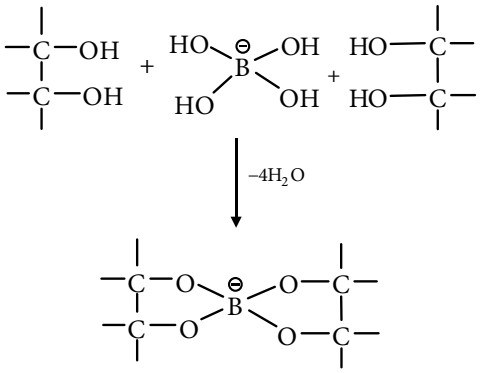
(ii) On reaction with strong NaOH, it forms sodium metaborate

$$B(OH)_3 + NaOH \rightarrow NaBO_2 + 2H_2O$$

(iii) On treatment with sodium peroxide, it forms sodium metaborate peroxohydrate $NaBO_2.H_2O_2.3H_2O$.

(iv) Although boric acid behaves like a weak acid and cannot be titrated with a base like NaOH, in the presence of polyhydroxy compounds like glycerol, it can be treated as a strong acid and can be titrated with a base using phenolphthalein as indicator. However, the polyhydroxy compound must be one with –OH groups on adjacent carbon atoms, as such compounds form strong complexes with $B(OH)_4^-$ and drive the reaction to the right. The neutralization reaction may be given as

$$B(OH)_3 + NaOH \rightleftharpoons Na[B(OH)_4]$$



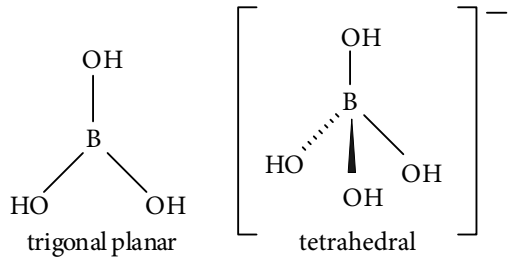
(v) H_3BO_3 dissolves in aqueous HF to give fluoroboric acid according to the reaction

$$H_3BO_3 + 4HF \rightarrow H^+[BF_4]^- + 3H_2O$$

Fluoroboric acid is a strong acid and $[BF_4^-]$ ion is tetrahedral.

Structure of boric acid and borate ion

The structure of $B(OH)_3$ and $B(OH)_4^-$ may be represented as



Borax, $Na_2B_4O_7.10H_2O$

Preparation

(i) Borax is prepared by boiling boric acid with soda ash according to the reaction

$$4H_3BO_3 + Na_2CO_3 \rightarrow Na_2B_4O_7 + 6H_2O + CO_2$$

(ii) It is also prepared from colemanite by digestion with hot Na_2CO_3 solution. The precipitated $CaCO_3$ is filtered off. The filtrate is cooled and large crystals of borax separate out.

$$2Ca_2B_6O_{11} + 2Na_2CO_3 + H_2O \rightarrow 3Na_2B_4O_7 + Ca(OH)_2 + 3CaCO_3$$

A current of CO_2 is passed through the mother liquor to convert the metaborate also into borax.

$$4NaBO_2 + CO_2 \rightarrow Na_2B_4O_7 + Na_2CO_3$$

Properties

Borax exists in three forms (i) Prismatic borax (decahydrate), $Na_2B_4O_7.10H_2O$, which is obtained when a solution of the salt is crystallized at room temperature (below 60°C). (ii) Jeweller's borax (pentahydrate), $Na_2B_4O_7.5H_2O$ which is obtained when the solution is crystallized above 60°C (iii) Borax glass (anhydrous), $Na_2B_4O_7$, which is obtained by heating $Na_2B_4O_7.10H_2O$ above its melting point until all the water of crystallization is lost.

(i) It is soluble in water and its aqueous solution is alkaline due to hydrolysis which may be written as $Na_2B_4O_7 + 7H_2O \rightarrow 2NaOH + 4H_3BO_3$

(ii) A solution of borax can be titrated with a mineral acid like HCl using methyl orange as indicator.

$$Na_2B_4O_7 + 2HCl + 5H_2O \rightarrow 2NaCl + 4H_3BO_3$$

Uses

Borax finds application as a primary standard in acid-base titrimetry. It is used for standardizing acids. Borax is used for testing metal ions in qualitative analysis by employing the borax bead.

Borax, when heated on a platinum loop swells up on losing water. Subsequently, it melts and decomposes to sodium metaborate and boron trioxide, which form a transparent glassy bead.

$$Na_2B_4O_7 \rightarrow \underbrace{2NaBO_2 + B_2O_3}_{\text{glassy bead}}$$

This transparent mass reacts with a number of metallic oxides yielding characteristic coloured beads. For example, with copper it forms a blue bead due to the formation of cupric metaborate ($Cu(BO_2)_2$). This on heating in the reducing bunsen flame turns brown-red due to reduction to metallic copper.

Borax also finds application in the preparation of buffers. The other uses of borax are

(i) as a preservative
(ii) as a mild antiseptic
(iii) as a flux in silver soldering
(iv) in flame retardant for wood and fabrics
(v) in pottery industry for glazing earthenware, china and sanitary ware
(vi) in the manufacture of borosilicate glass

ALUMINA, ALUMINIUM CHLORIDE AND ALUMS

Alumina, Al_2O_3

Preparation

The oxide of aluminium, Al_2O_3, is commonly known as alumina. It occurs in nature as bauxite, corundum (also called $\alpha - Al_2O_3$), ruby, sapphire and other precious gemstones. There is another form, known as $\gamma - Al_2O_3$ (activated alumina). Alumina can be prepared in the laboratory by heating aluminium in oxygen or by dehydrating aluminium hydroxide. It can also be made by igniting its nitrate or ammonium alum.

$$4Al + 3O_2 \rightarrow 2Al_2O_3$$

$$2Al(OH)_3 \xrightarrow{\Delta} Al_2O_3 + 3H_2O$$

$$2Al(NO_3)_3 \xrightarrow{\Delta} Al_2O_3 + 6NO_2 + 1\frac{1}{2}O_2$$

Corundum (or $\alpha - Al_2O_3$) actually occurs as a mineral and it can be made by heating $Al(OH)_3$ or γ - Al_2O_3 above 1000°C. A number of coloured forms of corundum like ruby, sapphire are prepared by strongly heating Al_2O_3 with small quantities of oxides like Cr_2O_3. γ -Al_2O_3 (activated alumina) is made by dehydrating $Al(OH)_3$ below 450°C.

Properties

Aluminium oxide is a white amorphous powder which becomes insoluble in acids when heated above 850°C. It is an amphoteric oxide and dissolves in acids and alkalies.

Corundum is a very hard material and is used to polish glass. Emery, which is used to make emery paper, is an impure form of corundum containing iron oxide and silica as impurities. Corundum has a high melting point above 2000°C and is used as refractory material. Ruby which is a mixed oxide containing alumina is very hard, and is used in jewellery and as bearing in watches.

γ -Al_2O_3 (activated alumina) is used as an adsorbent in chromatography.

Aluminium chloride, $AlCl_3$

Preparation

(i) Anhydrous aluminium chloride is prepared by passing dry chlorine over heated metal.

$$2Al + 3Cl_{2(g)} \rightarrow 2AlCl_{3(s)}$$

(ii) It can also be prepared by heating aluminium in a stream of dry HCl gas.

$$2Al + 6HCl_{(g)} \rightarrow 2AlCl_3 + 3H_2$$

(iii) The anhydrous salt is commonly prepared by heating a mixture of aluminium oxide and carbon at 870°C in a stream of dry chlorine.

$$Al_2O_3 + 3C + 3Cl_2 \rightarrow 2AlCl_3 + 3CO\uparrow$$ (Mac Affe process)

(iv) Aluminium chloride hexahydrate crystals separate out when a solution of the metal or its oxides or its hydroxide taken in hydrochloric acid is carefully evaporated. The hexahydrate decomposes to alumina on heating. Therefore, anhydrous $AlCl_3$ cannot be obtained by heating hydrated $AlCl_3.6H_2O$.

$$2AlCl_3.6H_2O \xrightarrow{\Delta} Al_2O_3 + 6HCl + 9H_2O$$

Properties

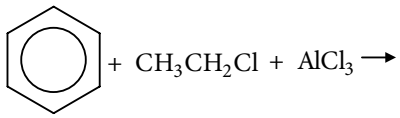
The anhydrous salt is white and crystalline but hygroscopic in nature. It sublimes at 183°C. It strongly fumes in moist air due to formation of HCl gas.

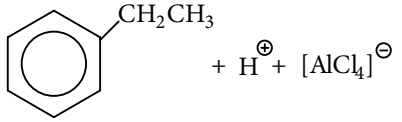
$$2AlCl_3 + 6H_2O \rightarrow 2Al(OH)_3 + 6HCl$$

The hydrated salt is colourless, crystalline and deliquescent in nature.

Uses

$AlCl_3$ is an important industrial chemical. It is used as a catalyst in a number of Friedel Craft's type of reactions involving acylation or alkylation. Ethyl benzene and styrene are two compounds produced by Friedel Craft's reaction using anhydrous $AlCl_3$ as catalyst. The formation of ethyl benzene may be represented as:



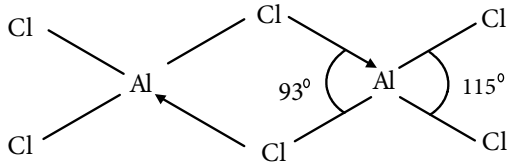


Other important applications of aluminium chloride are:

(i) in petroleum industry for isomerisation of hydrocarbons.
(ii) in dyestuff industry for making the raw material, anthraquinone etc.

Structure

From vapour density measurements, it is shown that $AlCl_3$ is a monomer above 800°C and it is predominantly covalent. It has a trigonal planar structure and the aluminium atom uses three sp^2 hybrid orbitals.



Structure of $AlCl_3$ dimer

Below 400°C in the vapour phase it exists as a dimer Al_2Cl_6 and retains this structure when dissolved in non-polar solvents like benzene.

In this molecule, each aluminium atom utilizes four sp^3 hybrid orbitals to accommodate eight electrons.

In aqueous solution, due to high enthalpy of hydration of Al^{3+}, the covalent bonds in the dimer are broken to give complex of aluminium and chloride ions according to

$$Al_2Cl_6 + 12H_2O \rightarrow 2[Al(H_2O)_6]^{3+} + 6Cl^-$$

Alums

Alums are double sulphates of a trivalent metal (Al^{3+}, Fe^{3+}, Cr^{3+}) and a univalent metal (Na^+, K^+, NH_4^+) with 24 water molecules of crystallisation $M'_2SO_4.M'''_2(SO_4)_3.24H_2O$ or $M'M'''(SO_4)_2.12H_2O$. Where M' is univalent metal and M''' is trivalent metal.

Aluminium ions crystallize from aqueous solutions containing other salts like K_2SO_4, $(NH_4)_2SO_4$ to form double salts. They are alums and have the general formula $MAl(SO_4)_2.12H_2O$ in which M is univalent cation (K^+, NH_4^+ or Na^+). The crystals are octahedral and are obtained in pure form. An example of such an alum is potash alum, $KAl(SO_4)_2.12H_2O$ or $K_2SO_4.Al_2(SO_4)_3.24H_2O$ which finds application as a mordant in dyeing . When this is applied to cloth, Al^{3+} ions get precipitated on it as $Al(OH)_3$ which binds the dye to the cloth in the form of alumina crystals.

The double salts dissociate in aqueous medium into their constituent ions $[M(H_2O)_6]^+$, $[Al(H_2O)_6]^{3+}$ and two SO_4^{2-} ions. Other M^{3+} ions forming alums similar to those of Al^{3+} are Cr^{3+}, Fe^{3+} etc. For example, chrome alum has the formula $KCr(SO_4)_2.12H_2O$ and ammonium ferric alum is represented as $NH_4Fe(SO_4)_2.12H_2O$. Alums are used as:

(i) coagulating agents for colloidal particles in water.
(ii) mordant in dyeing and calico printing
(iii) in tanning of leather
(iv) styptic to arrest bleeding

CARBON (C)

Introduction

Carbon (C) belongs to group 14 of the periodic table, wherein the other members are silicon (Si), germanium (Ge), tin (Sn) and lead (Pb). There is a striking transition from non-metallic to metallic character in these elements with increasing atomic number. The first member of the group, carbon, is a typical non metal while the end members tin and lead are typical metals. In this group, the formation of simple M^{4+} ions in gaseous state is not known because of the high ionization energies. These elements attain inert gas configuration by forming four covalent bonds using sp^3 hybrid orbitals. Hence these compounds have tetrahedral shape.

Allotropes of carbon

There are several allotropic modifications of carbon, which are both crystalline like diamond and graphite and amorphous like coal charcoal, lampblack. The most important of them are diamond and graphite. Another variety of carbon consisting of discrete molecules such as C_{60} are clusters of carbon atoms, known as fullerenes. However, a detailed treatment of the crystalline forms, diamond and graphite only will be presented here.

Diamond

It is the purest form of carbon. Diamonds occur naturally i.e., they are mined in some parts of the world. They are also made synthetically i.e., as artificial diamonds. Moissan first prepared artificial diamonds in 1893 using an electric furnace. In recent years, American and Swedish scientists produced artificial diamonds from graphite employing high pressure of about 70,000 atmospheres and prolonged heating at 3000°C. Diamond is a highly unreactive material.

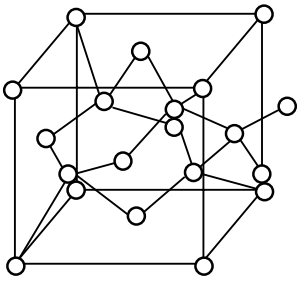
Properties

Diamond crystallizes in the cubic system. It is the hardest natural substance known and has a density of 3.51 g cm^{-3}. It has a high refractive index of 2.4. It is transparent, and a bad conductor of electricity. It has a high melting point of about 3930°C.

The action of chemical reagents on diamond is slow. For example, a mixture of $K_2Cr_2O_7$ and conc.H_2SO_4 oxidizes diamond slowly to CO_2 at 200°C. Diamond burns in air at 900°C or above leaving only a trace of ash. Diamond reacts with fluorine at 700°C to form carbon tetrafluoride. Diamond can be transformed into graphite by heating in vacuum at a temperature of 1800°C.

Structure of diamond

Each carbon atom is linked tetrahedrally (sp^3 hybridization) to four other carbon atoms using strong covalent bonds in diamond, the C–C bond distance being 1.54 A° and bond angle is 109.5°. The tetrahedra are linked together in a three dimensional giant polymeric network. Its unit cell is cubic. Strong covalent bonds extend in all directions. This is responsible for the high melting point, hardness and chemical inertness. Since all the electrons are used for σ bonding there are no mobile electrons or delocalized π bonds. That explains the electrical insulating property.



Unit cell of diamond

Uses

Large diamonds are cut into gemstones. Industrially, it is used for making drills, and as abrasive powder for cutting and polishing diamonds.

Graphite

Graphite occurs in native state. Graphite is also produced artificially (in USA) according to Acheson process, where a mixture of sand and petroleum coke is strongly heated in an electric furnace for over twenty four hours. An alternating current is then passed through the system via two large carbon electrodes connected by a set of carbon rods. At first, silicon carbide is formed, which then decomposes to silicon and graphite. Silicon volatilizes off at this temperature.

$$SiO_2 + 3C \rightarrow SiC + 2CO$$

$$SiC \rightarrow Si + C$$

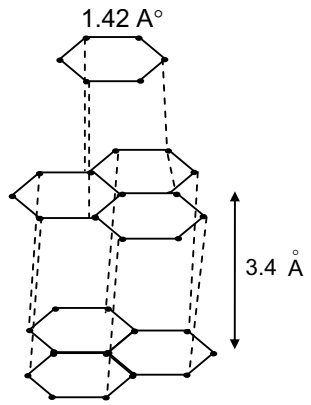
Properties

Graphite is a dark grey crystalline substance with a slight metallic lustre. It is a soft material having a density of 2.22 g cm^{-3}, and is a good conductor of heat and electricity. It burns in air at 700°C forming CO_2. It reacts with fluorine at 500°C It reacts with a mixture of conc.HNO_3 at 600°C in the presence of a little potassium chlorate to give a yellow insoluble solid commonly known as graphite oxide. Chromic acid oxidizes it to CO_2.

Structure of graphite

The difference in crystal structures of graphite and diamond explains their difference in properties. In graphite, each carbon atom is bound to three neighboring carbon atoms using sp^2 hybrid orbitals, which results in a hexagonal planar structure. Thus only three electrons of each carbon

atom are involved in σ-covalent bond formation, the other free electron in each carbon atom being available for electrical conduction through a π bond. The hexagonal layers (each layer is 3.4 A° apart from another) are held together by weak van der Waals forces. It is thus easy to slide one layer over another. This explains the excellent lubricating properties of graphite. The structure of graphite is shown in the following diagram. Bond order in graphite is 1.33. Bond length is 1.42 A°.



Structure of graphite

Graphite occurs in two forms as α – graphite and β – graphite. In α – graphite, the arrangement of the layer is of the ABAB type with the third layer exactly above the first layer while in β – graphite the layers are ABCABC type. The two forms are interconvertible.

Uses

Graphite finds application
(i) in making lead pencils
(ii) in making refractory crucibles
(iii) as electrodes for electric furnaces
(iv) as a moderator in nuclear reactors
(v) as a lubricant in machinery

OXIDES AND OXY-ACIDS OF CARBON

Oxides of carbon

Carbon forms more oxides than other elements and their oxides differ from those of other elements becaue they contain pπ - pπ multiple bonds between carbon and oxygen. Two of the oxides, which are most stable and important, are carbon monoxide and carbon dioxide. There are three less stable sub-oxides viz., C_3O_2, C_5O_2 and $C_{12}O_9$. It also form even less stable graphite oxides C_2O and C_2O_3.

A detailed account of the two oxides, carbon monoxide and carbon dioxide is given here.

Carbon monoxide, CO

Preparation

Carbon monoxide is formed from the incomplete combustion of carbon, charcoal or coke in a limited supply of air or O_2.

(i) A convenient method of preparation of CO in the laboratory is by the dehydration of formic acid or sodium formate with conc. H_2SO_4 according to the reaction,

$$HCOOH \xrightarrow{\text{conc. } H_2SO_4} CO + H_2O$$

$$HCOONa + H_2SO_4 \rightarrow NaHSO_4 + H_2O + CO$$

The evolved gas is washed with NaOH solution to remove traces of CO_2 and SO_2 and collected over water or mercury.

(ii) Another method of preparation involves the dehydration of oxalic acid or sodium oxalate by heating with conc.H_2SO_4. according to

$$\begin{matrix} COOH \\ | \\ COOH \end{matrix} \longrightarrow CO_2 + CO + H_2O$$

$$Na_2C_2O_4 + H_2SO_4 \rightarrow Na_2SO_4 + H_2C_2O_4 \rightarrow Na_2SO_4 + H_2O + CO_2 + CO$$

The liberated CO_2 is removed by passing the mixture of gases through potassium hydroxide solution.

(iii) Carbon dioxide can be reduced with red hot carbon to give CO.

$$CO_2 + C \rightleftharpoons 2CO$$

(iv) Pure CO can be obtained by thermal decomposition of nickel carbonyl.

$$Ni(CO)_4 \overset{180°C}{\rightleftharpoons} Ni + 4CO$$

Properties

It is a colourless, odourless gas. It is highly poisonous. Its poisonous nature is due to the fact that it coordinates with

haemoglobin of blood to form carboxy haemoglobin. It has a very high bond energy of 1070 kJ mol^{-1} as compared to other diatomic molecules.

The presence of CO can be detected by the fact that it burns with a blue flame.

It is slightly soluble in water but is readily absorbed by an ammoniacal solution of cuprous chloride giving a compound of the composition $CuCl.CO.2H_2O$.

Reactions of CO

(i) It reacts with NaOH solution at 160°C under pressure (6 – 10 atm) to give sodium formate.

$CO + NaOH \rightarrow HCOONa$

(ii) It readily combines with oxygen and sulphur to give CO_2 and COS respectively.

$$CO + \frac{1}{2}O_2 \rightarrow CO_2$$

$$8CO + S_8 \rightarrow \underset{\text{(carbonyl sulphide)}}{8COS}$$

(iii) It reacts with halogens, like Cl_2 (in presence of sunlight) to give phosgene, which is a highly toxic gas.

$$CO + Cl_2 \rightarrow \underset{\text{(Phosgene)}}{COCl_2}$$

(iv) CO acts as a strong reducing agent. It reduces many metal oxides like CuO, Fe_2O_3, PbO at red heat to the metallic state.

$$CuO + CO \rightarrow Cu + CO_2$$

$$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$$

$$PbO + CO \rightarrow Pb + CO_2$$

Ammoniacal $AgNO_3$ is reduced to metallic silver.

(v) CO can act as a ligand by donating a pair of electrons to many transition metals forming metal carbonyls. It forms strong bonds with those metals in which both σ and π bonding occur. For example, metals like Ni, Fe, Cr form carbonyls like $Ni(CO)_4$, $Fe(CO)_5$, $Cr(CO)_6$. The reactions of Fe and Ni with CO may be represented as

$$Fe + 5CO \xrightarrow[\text{under high pressure}]{200^0\,C} Fe(CO)_5$$

$$Ni + 4CO \xrightarrow{\text{room temperature}} Ni(CO)_4$$

The reaction involving Ni is the basis of Mond's process for the purification of nickel.

(vi) CO reduces palladium chloride to a green or black precipiate of the metal (test for CO).

$$PdCl_2 + CO + H_2O \rightarrow Pd + CO_2 + 2HCl$$

(vii) It liberates iodine when passed through a solution of iodine pentoxide

$$5CO + I_2O_5 \rightarrow 5CO_2 + I_2$$

This reaction is used to detect and estimate CO as the liberated iodine can be titrated with $Na_2S_2O_3$.

CO and its mixtures with other gases as fuel

Carbon monoxide is a valuable fuel and its combustion to form CO_2 is an exothermic process, with the evolution of heat of about 565 kJ mol^{-1}.

Carbon monoxide is a constituent of three industrial fuels.

(i) Water gas, an equimolar mixture of CO and H_2
(ii) Producer gas, a mixture of CO and N_2
(iii) Coal gas, a mixture of CO, H_2, CH_4 and CO_2.

Water gas is prepared by passing steam through red hot coke. It is a highly endothermic reaction. Water gas is a valuable fuel and it has high calorific value.

$$C + H_2O \rightarrow CO + H_2 \qquad \Delta H° = +131 \text{ kJ mol}^{-1}$$

Producer gas is made by passing air through red hot coke.

$$2C_{(s)} + O_{2(g)} \rightarrow 2CO_{(g)}\ \Delta H° = -221 \text{ kJ mol}^{-1}$$

Producer gas has a lower calorific value and is not an efficient fuel.

Coal gas is produced by the destructive distillation of coal.

Structure of CO

The structure of CO may be represented as

$$:\overset{-}{C}:::\overset{+}{O}: \text{ or } :\overset{-}{C}\equiv\overset{+}{O}: \text{ or } :C\overset{\leftarrow}{=}O:$$

It is a linear molecule and assumed to be "sp" hybridized.

From the point of molecular orbital theory, the arrangement of electrons in the various orbitals may be depicted as

$$\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2 \begin{cases} \pi 2p_x^2 \\ \pi 2p_y^2 \end{cases} \sigma 2p_z^2$$ with a bond order of 3.

Carbon dioxide, CO_2

Carbon dioxide occurs in the atmosphere to the extent of about 0.03 volume percent. There are several methods of obtaining carbon dioxide.

(i) In an industrial scale, it is obtained as a byproduct in the manufacture of hydrogen.

$$CO + H_2O \rightarrow CO_2 + H_2$$

It is also recovered as a byproduct in the fermentation of molasses.

$$C_6H_{12}O_6 \xrightarrow{\text{yeast}} 2C_2H_5OH + 2CO_2.$$

Calcination of limestone in lime kilns yields CO_2.

$$CaCO_3 \xrightarrow[\text{heat}]{\text{strong}} CaO + CO_2$$

(ii) On a laboratory scale, CO_2 is prepared by the decomposition of carbonates with dilute acids.

$$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$$

By the decomposition of $NaHCO_3$ on heating, CO_2 is formed.

$$2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$$

(iii) By burning carbon in excess of air, CO_2 can be obtained.

$$C + O_2 \rightarrow CO_2$$

Properties of CO_2

It is a colourless, odourless gas. It is heavier than air, its specific gravity being 1.53 relative to air. It is not combustible and it is not a supporter of combustion. It is fairly soluble in cold water and forms the weak acid, carbonic acid. Thus it is an acidic oxide.

The gas can be liquified under pressure between -57°C and 31°C. Solid CO_2 is produced as snow white solid by expanding the gas from cylinder. Solid CO_2 sublimes at -78°C under atmospheric pressure. It is called dry ice or cardice.

Uses of CO_2

CO_2 is used in the manufacture of urea, a nitrogenous fertilizer.

$$CO_2 + 2NH_3 \xrightarrow[\text{at high pressure}]{180°C} \underset{\substack{\text{Ammonium}\\\text{Carbamate}}}{NH_2CO_2NH_4}$$

$$\longrightarrow \underset{\text{Urea}}{NH_2CONH_2} + H_2O$$

Carbon dioxide is also used to neutralize alkalis because of its acidic nature. It also finds application as a fire extinguisher. It is used as a refrigerant. Dry ice is used as a coolant for fish and meat.

Detection of CO_2

The presence of CO_2 can be detected by its action on lime water or baryta water (i.e., $Ca(OH)_2$ or $Ba(OH)_2$), where insoluble $CaCO_3$ or $BaCO_3$ is formed. So it turns lime water milky, but the passage of excess CO_2 into lime water results in the disappearance of cloudiness due to the formation of soluble calcium bicarbonate.

$$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$$

$$CaCO_3 + H_2O + CO_2 \rightarrow Ca(HCO_3)_2$$

Photosynthesis and respiration

Carbon dioxide is an important constituent in photosynthesis as green plants synthesize glucose from it in sunlight.

$$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$$

The reverse process associated with the above reaction involves oxidation of glucose and is known as respiration. It is associated with a release of energy.

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy}$$

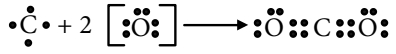
Greenhouse effect due to CO_2

The presence of carbon dioxide in atmosphere decreases the amount of heat loss from the earth by radiation since CO_2 absorbs strongly in the infrared region. This causes global warming and is known as 'greenhouse effect'.

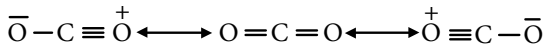
Other gases like water vapour, oxides of nitrogen, freons, aerosols also contribute to the green house effect.

Structure of CO_2

The Lewis structure for CO_2 may be written as

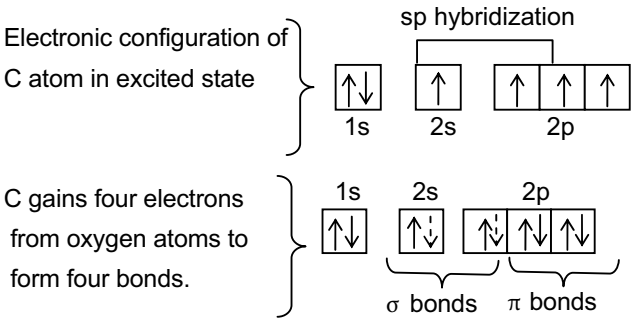


Carbon dioxide may be considered as resonance hybrid of the following structures.



The bonding in CO_2 may be described as follows:

Two σ bonds are formed from two "sp" hybrid orbitals. They point in opposite directions. These orbitals overlap with the p-orbitals from oxygen atoms forming a linear molecule. The remaining p_y and p_z orbitals on carbon atom form π bonds by overlapping with p orbitals on the oxygen atoms. These π bonds are in planes at right angles to each



other. Carbon dioxide has zero dipole moment and hence has a linear shape.

Carbon suboxide, C_3O_2

It is a relatively uncommon substance that is prepared by the dehydration of malonic acid with P_2O_5 in vacuum at 140 - 150°C. It is a foul smelling gas.

$$HOOC - CH_2 - COOH \xrightarrow[\text{in vacuum}]{\text{heat, } 150^\circ C} O = C = C = C = O + 2H_2O$$

The molecule is linear.

Properties

The gas polymerizes to a yellow solid at room temperature. At higher temperatures, it gives red and purple solids on polymerization.

The gas reacts with HCl to give an acid chloride and with water it forms malonic acid.

$$C_3O_2 + 2HCl \rightarrow CH_2(COCl)_2$$

$$C_3O_2 + 2H_2O \rightarrow CH_2(COOH)_2$$

OXYACIDS OF CARBON

Carbonic acid:

Carbon dioxide dissolves in water to give carbonic acid according to the reaction,

$$CO_2 + H_2O \rightleftharpoons H_2CO_3$$

The acid has never been isolated and actually it exists as a loose hydrate in water and may be formulated as $(HO)_2CO$. It acts like a weak dibasic acid which ionizes in two steps.

$$H_2CO_3 \rightleftharpoons H^+ + HCO_3^- \quad K_1 = 3.5 \times 10^{-7}$$

$$HCO_3^- \rightleftharpoons H^+ + CO_3^{2-} \quad K_2 = 3.7 \times 10^{-11}$$

On neutralization with alkali, it forms two series of salts, the bicarbonate and normal carbonate salts.

$$H_2CO_3 + NaOH \rightarrow NaHCO_3 + H_2O$$

$$NaHCO_3 + NaOH \rightarrow Na_2CO_3 + H_2O$$

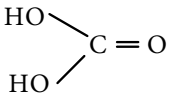
Alkali metal carbonates are soluble in water. Aqueous solutions of carbonates are alkaline due to hydrolysis reaction, which may be represented as

$$CO_3^{2-} + H_2O \rightleftharpoons HCO_3^- + OH^-$$

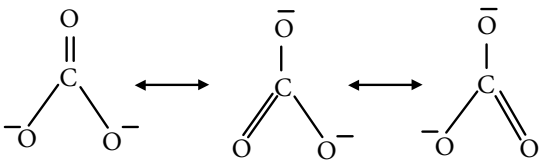
A solution of Na_2CO_3 can be titrated in two stages. First, it can be half neutralized to give the bicarbonate using phenolphthalein as indicator. A complete neutralization requiring two equivalents of acid can be followed by using methyl orange as indicator. Thus, by this approach, it is possible to estimate the individual components of a mixture of bicarbonate and carbonate.

Structure of carbonic acid and carbonates

The structure of carbonic acid may be given as



The carbonate ion has a trigonal planar shape. In this ion, the carbon atom forms three sp^2 hybrid orbitals and forms three σ bonds, one each with oxygen atoms. In addition, a π bond is formed between the electrons in p_z orbital of carbon with the electron in the p_z orbital of one of the oxygen atoms. The carbonate ion may be considered to be a resonance hybrid of the following structures.



SILICON (SI)

Introduction

Silicon has the properties of a non-metal and some of its important atomic properties are presented below.

Atomic number	Atomic weight	Electronic configuration	Covalent radius (A°)	Ionization energy ($I_1 + I_2 + I_3 + I_4$) kJ mol^{-1}	Electronegativity (Pauling scale)	Oxidation states
14	28.06	[Ne] $3s^2 3p^2$	1.18	4354	1.8	+4 and +2 (+2 is not common)

All the elements of group 14 exhibit valencies of 2 and 4 except carbon, which shows a valency of four. The divalent state becomes more stable in the lower end of the group due to inert pair effect. Silicon and other higher members of the group show a covalency greater than four in some of their complexes (similar to $[SiF_6]^{2-}$) due to the presence of vacant 'd' orbitals. The tendency to form bonds like C-C (as in carbon compounds) decreases as one goes down the group because of the decreasing bond energies of Si-Si, Ge-Ge etc. Formation of bonds between atoms of the same element forming chains is called catentation. Tendency for catenation decreases down the group, C > Si > Ge ≈ Sn > Pb.

Occurrence

After oxygen, silicon is the most abundant element and comprises approximately 25.7% of the earth's crust. Many minerals, rocks and soils are made up of silica and silicates.

Preparation of silicon

Silicon exists in two forms, amorphous and crystalline varieties. There are many methods for the preparation of elemental silicon, which are discussed below.

(i) Elemental silicon (in crystalline form) is obtained by reducing silicon dioxide with highly pure coke in an electric furnace. To prevent the formation of silicon carbide, excess of silicon dioxide must be present.

$$SiO_2 + 2C \rightarrow Si + 2CO\uparrow$$

It is also possible to use calcium carbide for this purpose in which case the reaction is

$$SiO_2 + CaC_2 \rightarrow Si + Ca + 2CO\uparrow$$

(ii) An amorphous form of silicon is prepared by mixing well powdered quartz with magnesium and heating the mixture in fire clay crucible. The magnesium oxide formed along with silicon is removed by dissolution with dilute hydrochloric acid.

$$SiO_2 + 2Mg \xrightarrow{\Delta} Si + 2MgO$$

(iii) High purity silicon required in electronics industry is prepared by converting silicon of ordinary grade to silicon tetrachloride. $SiCl_4$ is then purified by distillation and finally reduced by Mg or Zn.

$$Si + 2Cl_2 \rightarrow SiCl_4$$

$$SiCl_4 + 2Mg \rightarrow Si + 2MgCl_2$$

(iv) High purity silicon is also prepared by the pyrolysis of SiH_4.

$$SiH_4 \xrightarrow{\Delta} Si + 2H_2$$

Ultra pure silicon can be obtained from this variety by a process known as zone refining.

Properties

Silicon has a metallic lustre and blue grey colour. Amorphous silicon is a dark brown powder. Silicon is a semi conductor.

Reactions of silicon

(i) Reaction with oxygen: Silicon does not react with oxygen even on strong heating.

(ii) Reaction with chlorine: Silicon burns when heated in chlorine to form $SiCl_4$

$$Si + 2Cl_2 \rightarrow SiCl_4$$

Reaction with fluorine: It combines with fluorine spontaneously in cold to give SiF_4.

$$Si + 2F_2 \rightarrow SiF_4$$

(iii) Reaction with Na_2CO_3. Fusion with Na_2CO_3 results in the formation of sodium silicate.

$$Si + Na_2CO_3 \rightarrow Na_2SiO_3 + C$$

(iv) Reaction with concentrated alkali: A concentrated solution of alkali reacts with silicon and liberates hydrogen.

$$Si + 2KOH + H_2O \rightarrow K_2SiO_3 + 2H_2$$

(v) Reaction with water: Silicon reacts with boiling water to liberate hydrogen

$$Si + 2H_2O \xrightarrow[\text{boiling conditions}]{\text{under}} SiO_2 + 2H_2$$

(vi) Reaction with magnesium: Silicon reacts with magnesium at the high temperature of electric furnace to give magnesium silicide, Mg_2Si.

$$Si + 2Mg \rightarrow Mg_2Si$$

Uses

Silicon is widely used in the manufacture of silicone polymers. Highly pure silicon is used in microelectronics industry in the manufacture of transistors and computer chips.

Silicon also finds use in the manufacture of alloys like silicon steel, which is resistant to the corrosive action of acids. Other alloys like magnesium silicon bronze are known.

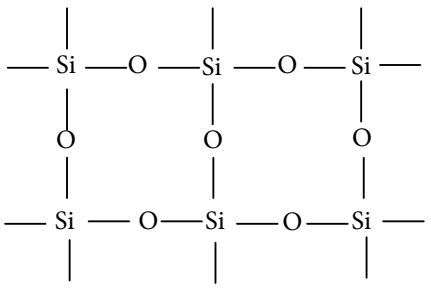
In the manufacture of steel castings, silicon is used as a deoxidizer. It is also used in the manufacture of ferrosilicon.

Compounds of silicon

Oxides of silicon

Silicon dioxide, SiO_2 (or silica)

Silicon dioxide is the most commonly known oxide and is found mostly in the form of sand and quartz. SiO_2 follows a large three-dimensional structure, which may be represented as



Si — O network in SiO_2

In this large molecule, each silicon atom is linked tetrahedrally to four oxygen atoms.

Properties of silica

Silica has a high melting point but softens below about 1600°C. It is inactive and colourless. Traces of metal ions impart colour to it resulting in some precious gemstones like amethysts (violet), rose quartz (pink), smoky quartz (brown) etc. Also, flint contains SiO_2 as a major constituent.

Silica is an acidic oxide and does not react with acids. However, it reacts with HF forming SiF_4 and this reaction is used in qualitative analysis to detect silicates and fluorides.

(i) Reaction with alkali: Silica slowly dissolves in aqueous alkali but the dissolution is more rapid in fused alkali or carbonate.

$$nSiO_2 + 2nNaOH \rightarrow (Na_2SiO_3)_n + nH_2O$$

$$SiO_2 + Na_2CO_3 \rightarrow Na_2SiO_3 + CO_2$$

This reaction is responsible for the ground glass stoppers sticking hard in reagent bottles containing alkali.

(ii) Reaction with halogens: Among halogen, only fluorine attacks silica in cold to form SiF_4.

$$SiO_2 + 2F_2 \rightarrow SiF_4 + O_2$$

SiF_4 hydrolyses when it comes in contact with a drop of water to give silicic acid and hexafluorosilicic acid. Silicic acid forms a water deposit.

$$SiF_4 + 4H_2O \rightarrow Si(OH)_4\downarrow + 4HF$$

$$SiF_4 + 2HF \rightarrow H_2SiF_6$$

Other forms of silica

Two other forms of silica that must be mentioned are (i) silica gel (ii) kieselguhr.

Silica gel is amorphous and quite porous. It is made by dehydrating silicic acid and contains about four percent water. Another method of preparation consists of treating a solution of sodium silicate with ammonium sulphate. It is a white translucent gel. It finds application as a catalyst and as drying agent and in chromatography. Kieselguhr is a fine powder. It is used as an abrasive and as an inert filler. Geliginite is a mixture of the explosive nitrobenzene (liquid) and inert kiesulguhr (solid).

Uses of silica

Silica in the form of quartz is an important piezoelectric material. It finds application in gas lighters and also for making crystal oscillators used in radios.

A vitreous form of silica is resistant to shock. It is used for making laboratory glassware, prisms, lenses etc.

Silica in the form of sand is used for making porcelain glass, silica bricks, which are used as lining material for furnaces.

Silicon hydrides or silanes

Silicon forms hydrides with the general formula Si_nH_{2n+2}. They exist as straight chain or branched chain compounds containing up to eight silicon atoms. SiH_4 (monosilane or silicomethane) will be briefly discussed.

Preparation of mono silane

(i) When magnesium silicide Mg_2Si is treated with H_2SO_4 or H_3PO_4, a mixture of silanes SiH_4 to Si_4H_{10} are formed in different proportion, the major components being SiH_4 and Si_2H_6.

(ii) SiH_4 is prepared by the reduction of $SiCl_4$ with $LiAlH_4$ or LiH at low temperature (0°C) in diethyl ether.

$$SiCl_4 + LiAlH_4 \rightarrow SiH_4 + AlCl_3 + LiCl$$

Properties

Silanes are more reactive than alkanes. Silanes are colourless gases at room temperature, are poisonous and have unpleasant smell.

(i) Reaction in air: They ignite spontaneously in air to form silicon dioxide.

$$SiH_4 + 2O_2 \rightarrow SiO_2 + 2H_2O$$

(ii) Action of water: Silanes are readily hydrolysed by water containing traces of alkali.

$$SiH_4 + 2H_2O \xrightarrow{OH^-} SiO_2 + 4H_2$$

(iii) Monosilane reacts with chlorine violently to give $SiCl_4$.

$$SiH_4 + 4Cl_2 \rightarrow SiCl_4 + 4HCl$$

(iv) Monosilane decomposes at red heat to give silicon and hydrogen.

$$SiH_4 \xrightarrow[\text{heat}]{\text{red}} Si + 2H_2$$

This method is used for the preparation of ultra pure silicon in semiconductor industry.

(v) Silanes hydrolyse readily in alkali to form silicates and H_2

$$SiH_4 + 2KOH + H_2O \rightarrow K_2SiO_3 + 4H_2$$

(vi) Silver nitrate is reduced to metallic silver by SiH_4.

$$SiH_4 + 4AgNO_3 \rightarrow Si + 4Ag + 4HNO_3$$

Silicon halides

Silicon forms tetrahalides, which are covalent and tetrahedral. Two of the important halides are $SiCl_4$ and SiF_4.

Preparation

(i) $SiCl_4$: $SiCl_4$ is prepared by the action of dry chlorine on heated silicon.

$$Si + 2Cl_2 \rightarrow SiCl_4$$

A second method involves heating a mixture of silica and charcoal in a current of chlorine.

$$SiO_2 + 2C + 2Cl_2 \rightarrow SiCl_4 + 2CO$$

Silicon tetrachloride is a colourless, fuming liquid having boiling point, 58°C.

(ii) SiF_4: SiF_4 is prepared by heating a mixture of CaF_2 and sand with excess conc.H_2SO_4.

$$2CaF_2 + SiO_2 + 2H_2SO_4 \rightarrow 2CaSO_4 + SiF_4\uparrow + 2H_2O$$

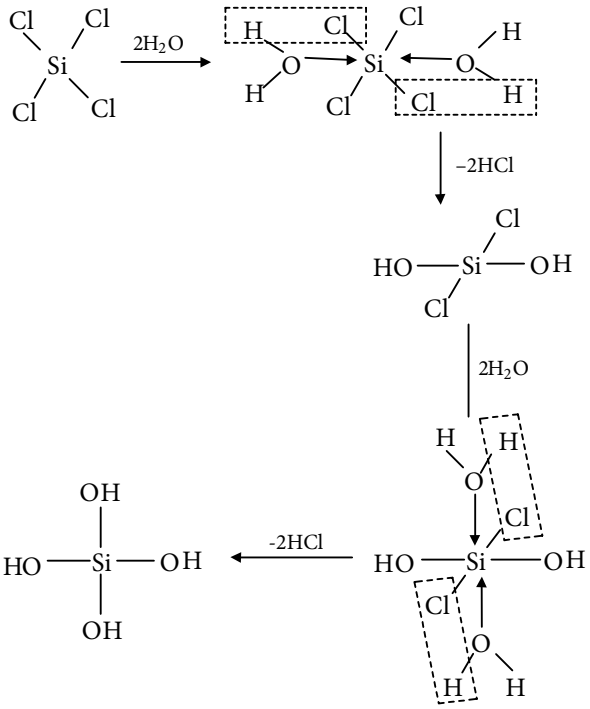
It is also prepared by the action of fluorine on silicon.

$$Si + 2F_2 \rightarrow SiF_4\uparrow$$

SiF_4 is a fuming gas with a pungent odour.

Hydrolysis

(i) $SiCl_4$: $SiCl_4$ rapidly hydrolyses on contact with water to form silicic acid.



$$SiCl_4 + 4H_2O \rightarrow Si(OH)_4 + 4HCl$$

$$Si(OH)_4 \rightarrow H_2SiO_3 + H_2O$$

The hydrolysis reaction may be depicted in the following way.

In the above scheme, the co-ordination of water molecules to the central silicon atom is assumed to take place through the vacant 'd' orbitals of silicon and silicon passes through a maximum covalency of six. This also exlains why CCl_4 is not hydrolysied by water.

(ii) SiF_4 : SiF_4 hydrolyses rapidly to give silicic acid a gelatinous white precipitate.

$$SiF_4 + 4H_2O \rightarrow Si(OH)_4 + 4HF$$

The HF formed reacts with unchanged SiF_4 to form hexafluorosilicic acid which is soluble in water.

$$SiF_4 + 2HF \rightarrow H_2SiF_6$$

Sodium silicate (water glass), Na_2SiO_3

It is prepared by heating sodium carbonate with sand

$$SiO_2 + Na_2CO_3 \rightarrow Na_2SiO_3 + CO_2$$

The fused mass is broken into smaller pieces and heated with water under pressure. The resulting liquid is concentrated to a viscous mass and is known as water glass. Its approximate composition is $Na_2SiO_3.SiO_2.6H_2O$.

It is used for fire proofing of wood and also as an adhesive. It also finds application as a filler in soaps.

Silicon carbide, SiC

It is also known as carborundum and is prepared by heating sand or quartz with excess of coke in an electric furnace at a temperature of 2000 – 2500°C.

$$SiO_2 + 3C \rightarrow SiC + 2CO$$

Silicon carbide is very hard, infusible and chemically inert. Pure SiC is pale yellow to colourless.

Properties

It is resistant to attack by most reagents. However, it is attacked by fused alkali in presence of air

$$SiC + 4NaOH + 2O_2 \rightarrow Na_2SiO_3 + Na_2CO_3 + 2H_2O$$

It reacts with chlorine at 100°C according to the reaction

$$SiC + 2Cl_2 \rightarrow SiCl_4 + C$$

Structure of SiC

It has a three dimensional structure of Si and carbon atoms. The lattice is the same as that of diamond, with each atom of one kind tetrahedrally surrounded by four atoms of the other kind.

Uses of SiC

It is used as an abrasive, instead of emery. It is used for making grinding stones, knife sharpeners. It is also used for making heating elements in furnaces. Because of its refractory nature and good heat conductivity, it is used for making crucibles, which are employed for melting metals.

Silicates

Introduction

Earth's crust is made of silicate minerals, aluminosilicate clays or silica. Silicates are present in various forms such as rocks, sand, as building materials in the form of granite, bricks and ceramics.

From a structural point of view, silicates may be described as a complex network (of solids) in which silicon is coordinated tetrahedrally to four oxygen atoms. Silicates are thus based on $(SiO_4)^{4-}$ tetrahedral units. The electronegativity difference between oxygen and silica being 1.7, is indicative of the Si – O bond having an appreciable ionic character (~ 50%).

The silicate units are formed by the sp^3 hybridization of silicon orbitals. Thus, Si forms four bonds with the oxygen atoms by utilizing its sp^3 hybrid orbitals and thus give rise to SiO_4^{4-} units. The SiO_4^{4-} tetrahedra may exist as discrete units or as large polymer units by sharing oxygen atoms. The oxygen atoms are often close packed.

Classification of silicates

The classification is based on the manner in which the $(SiO_4)^{4-}$ tetrahedral units are linked to each other.

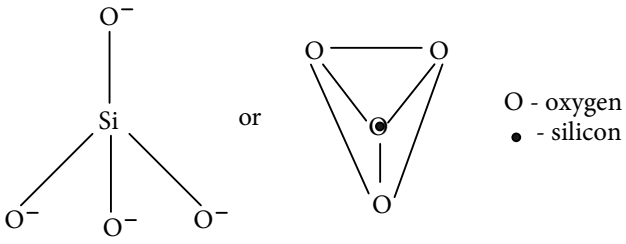
The different classes of silicates are:

(i) orthosilicates
(ii) pyrosilicates
(iii) cyclic silicates
(iv) chain silicates
(v) two dimensional sheet silicates
(vi) three dimensional network silicates

A brief description of the various classes of silicates is given below:

Orthosilicates

In this class, the SiO_4^{4-} tetrahedra are discrete and there is no sharing of corner oxygen atoms.

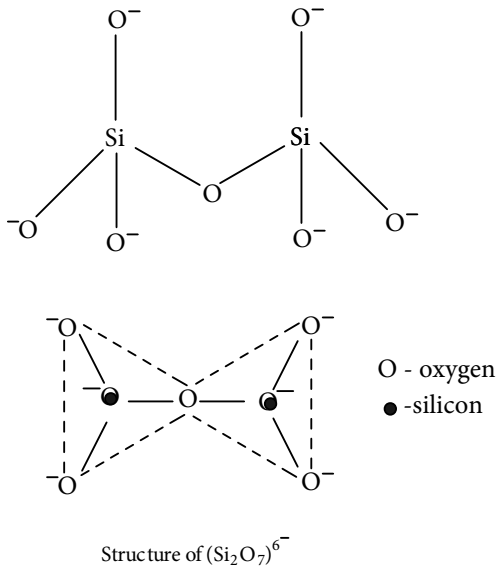


Structutre of $(SiO_4)^{4-}$

Mg_2SiO_4 (olivine) Be_2SiO_4 (phenacite) $ZrSiO_4$ (zircon) and Zn_2SiO_4 (willemite) are examples of this category.

Pyrosilicates

In this case, two tetrahedral units are joined by sharing the oxygen atom at one corner, giving the unit $[Si_2O_7]^{6-}$.

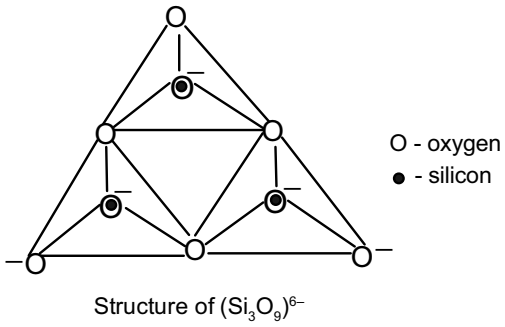


Structure of $(Si_2O_7)^{6-}$

Examples of this category are hemimorphite, $Zn_3(Si_2O_7).Zn(OH)_2.H_2O$ and thortveitite, $Sc_2(Si_2O_7)$.

Cyclic silicates

These have ring structures with the general formula $\left(SiO_3^{2-}\right)_n$ in which two oxygen atoms of each tetrahedron are shared. A simple structure of this class of silicates, where n is 3 may be represented as:

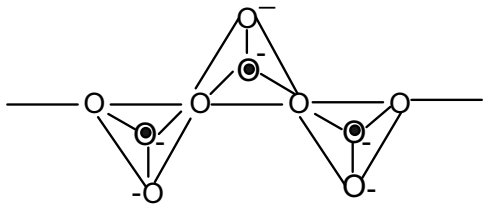


Structure of $(Si_3O_9)^{6-}$

This structure occurs in wollastonite, $Ca_3(Si_3O_9)$, benitoite $BaTi(Si_3O_9)$ catapleite, $Na_2Zr(Si_3O_9).2H_2O$ and beryl, $Be_2Al_2(Si_6O_{18})$.

Chain silicates

Simple chain silicates or pyroxenes are formed by the sharing of two oxygen atoms on each tetrahedron. The general formula of the ion is the same as in the case of cyclic silicates $\left(SiO_3^{2-}\right)_n$. A typical structure of a chain silicate is shown below.



Structure of a chain silicate

An example of such a silicate is spodumene, $LiAl(SiO_3)_2$ (a mineral of lithium). Asbestos belongs to this class but its structure is more complex. In asbestos two simple chains are joined together by shared oxygen atoms and it belongs to the class of silicates called amphibole, which are double stranded chain silicate. E.g., $Ca_2Mg_5(Si_4O_{11})_2(OH)_2$.

Two dimensional sheet silicates

It is possible to build two dimensional networks of SiO_4 tetrahedra, where the SiO_4 units share three corners with other tetrahedra forming a two-dimensional sheet having an empirical formula $\left(Si_2O_5^{2-}\right)_n$. There are strong bonds within the Si – O sheet but much weaker forces hold each sheet to the next one.

Examples of such structures are clay minerals like kaolinite, mica, montmorillonites (like Bentonite, Fullers earth).

Three dimensional network silicates

In this case, the three-dimensional networks, all the four corners of SiO_4 tetrahedra are shared. Aluminosilicates, zeolites, feldspar are examples of this type. Lapis lazuli is a semiprecious stone, which contains ultramarine ($Na_8[AlSiO_4)_6]S_2$). When there are no metal ions, the three dimensional network of SiO_4 tetrahedra will be just silica SiO_2.

Importance of silicates

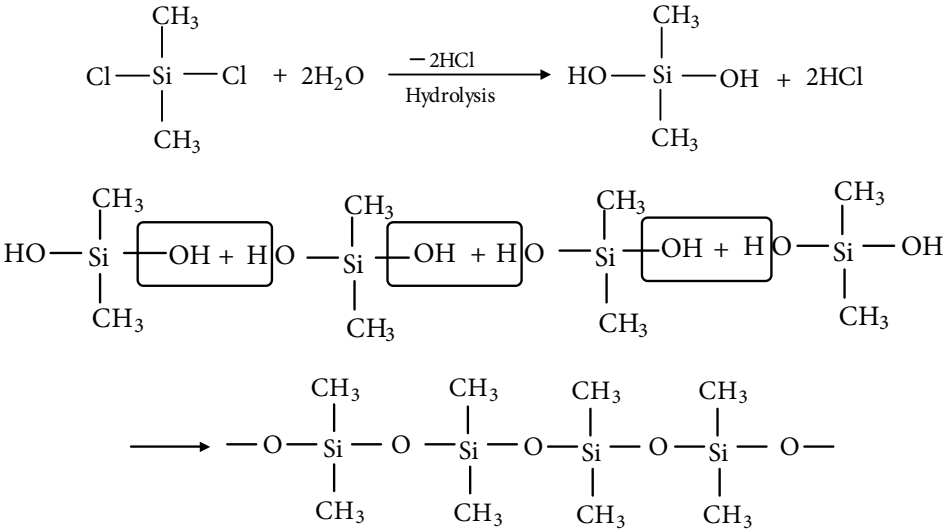
Silicates in the form of clay minerals are used for absorbing chemicals. Mica is used as an electrical insulator. Asbestos finds application as thermal insulating material. Silicates are important in cement, glass and ceramic industries.

Synthetic zeolites made in the laboratory are used as ion-exchangers and molecular sieves. They have honeycomb like structures with empty channels. They have Ca^{2+}, Na^+ ions in addition to Al^{3+} in their structures and have typical formulae as $Na_2(Al_2Si_3O_{10}).2H_2O$ and $Ca(Al_2Si_7O_{18}).6H_2O$.

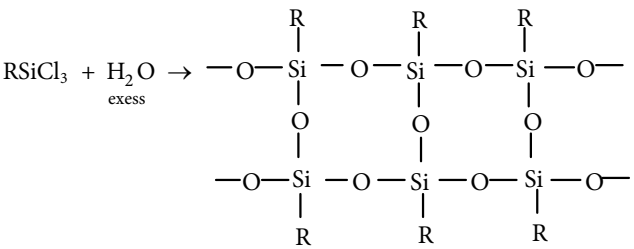
Silicones (Siloxanes)

(i) Silicones are a group of synthetic organosilicon polymers. They contain repeating $R_2SiO–Si(R_2)O–$ units ($R = CH_3$ or C_6H_5). Since their formula is similar to that of a ketone (R_2CO), they are referred to as silicones.

(ii) Manufacture of silicones:
The starting materials for the manufacture of silicones are alkyl or aryl substituted chlorosilanes. For example, hydrolysis of dimethyldichlorosilane ($(CH_3)_2SiCl_2$) form straight chain polymers leaving an active –OH group at the end. Thus the polymerization continues and the chain length grows.
The relevant reactions are shown below.



The hydrolysis of monoalkyl trichlorosilane, $RSiCl_3$ (where, $R = CH_3$) gives a complex cross linked polymer, which can be represented as



cross linked silicone polymer

Uses of silicones

Silicones can be obtained in the form of oils, viscous liquids, resins or rubber like materials depending on the chain length and the nature of groups attached to silicon atoms. They are chemically inert, resistant to heat and also have good electrical insulating capacity. They also have lubricating and water repellant properties. Because of these properties, they find a variety of applications such as:

(i) in high temperature oil baths

(ii) in high vacuum pumps (because of their low vapour pressure)

(iii) as lubricants in low temperature work
(iv) in the manufacture of water proof cloth
(v) in paints and enamels to make them resistant to higher temperature, light and chemicals.
(vi) in the manufacture of silicone rubber.

NITROGEN (N)

Introduction

Nitrogen (N) belongs to group 15 of the periodic table, the other elements of the group being phosphorus (P), arsenic (As), antimony (Sb) and bismuth (Bi). They have the valence shell electronic configuration of ns^2 np^3. Elements of this group show a significantly wider range of chemical behaviour than other groups of representative elements. There is a transition from non-metallic to metallic character as we go down the group. While nitrogen and phosphorus are essentially non-metals (electronegative), arsenic and antimony are metalloids (semi metals) and bismuth shows metallic character (weakly electropositive). The ionzation potential values decrease gradually down the group although nitrogen has a high first ionization potential. The larger values of ionisation energies (compared to elements of group 14) are attributed to the greater nuclear charge and smaller size of the elements.

Valence states

The valence shell of the elements contains two-paired electrons and three unpaired 'p' electrons. It is Therefore, possible for them to form three covalent bonds and thereby attain noble gas configuration using all their p-electrons. Since elements other than nitrogen have vacant 'd' orbitals also, one of the 'ns' electrons can be excited to 'nd' orbital so that there are five unpaired electrons in the valence shell. They can be used to form five-covalent bonds and P, As and Sb thus exhibit five covalent state which is not possible with nitrogen.

Oxidation states

Positive oxidation states of +3 and +5 are possible depending on whether only three p-electrons or all the five (two 's' and three 'p') electrons are involved in bond formation.

Because of the inert pair effect (tendency of 's' electrons to remain inert), the stability of +3 oxidation state increases and that of +5 state decreases as we go down the group. It is observed that nitrogen and phosphorus compounds are mainly covalent. But As, Sb, and Bi show increasing tendency to form cations. Elements in the +3 oxidation state exhibit electron donor properties because of the presence of lone pairs.

Except bismuth, all other elements show −3 oxidation state (like in NH_3 and AsH_3). The tendency to exhibit this oxidation state decreases as we go down the group. Thus the stability of the hydrides decreases from NH_3 to BiH_3.

The +5 oxidation state in higher members (like in Sb and Bi) is realized through covalent bonding.

Nature of the bond in compounds containing M^{3-} ions

The atoms of elements in this group may accept three electrons from highly electropositive metals like Li, Na etc., to form M^{3-} ions. However, the tendency to form M^{3-} ions decreases as one goes down the group (or increasing atomic size). Nitrogen, being smallest in size and most electronegative in the group forms N^{3-} ions (in compounds like Na_3N, Mg_3N_2).

Occurrence

Nitrogen is an important constituent of air making upto 78% by volume of the atmosphere. It is an essential constituent of all plant and animal matter in which it occurs in combination with carbon, hydrogen and oxygen as proteins.

It is also an essential component of the amino acids. It does not exist in large amounts in earth's crust but some nitrate salts are present as deposits. For example, as Chile salt petre ($NaNO_3$) in northern Chile.

Some important atomic properties of nitrogen are given below:

Atomic number	Atomic weight	Electronic configuration	Atomic radius (A°)	Ionization energy (1st) kJ mol^{-1}	Electro negativity	Oxidation states
7	14.01	[He] $2s^2 2p^3$	0.75	1403	3.0	−3, −2, −1, 0, +1, +2, +3, +4 and +5 (-3, +3 and +5 are the most common oxidation states)

Difference between nitrogen and other members of the group

Nitrogen differs from the rest of the members of the group due to

(i) small size
(ii) high electronegativity
(iii) absence of d orbitals in the outermost shell.

A striking difference in the physical state between nitrogen and the other members of the groups is that nitrogen is a gas while others are solids. Nitrogen exhibits a great tendency to form pπ–pπ multiple bonds with itself and also with other elements like carbon and oxygen. But the strength of the multiple bonds (relative to single bonds) decreases markedly as we go to higher members of the group. Elemental nitrogen is a stable, chemically inert diatomic molecule with a triple bond (N ≡ N, bond length of 1.09 A°) between the atoms. The other elements, phosphorus, arsenic and antimony do not form diatomic molecules in their elemental state.

The following differences in chemical reactivity between nitrogen and other members of the group are noteworthy.

(i) Nitrogen reacts with oxygen only at high temperature but phosphorus combines with oxygen at normal temperature.
(ii) Nitrogen exhibits a large number of oxidation states, -3 to +5 whereas this is not the case with others. NH_3 (-3), N_2H_4(-2), NH_2OH(-1), N_2(0), N_2O(-1), NO(2), N_2O_3(3), NO_2(4), N_2O_5(5).
(iii) Nitrogen halides (except NF_3) are generally unstable while the halides of other elements are stable.
(iv) The tri and pentoxides of nitrogen are monomeric while those of other elements are dimeric e.g., N_2O_3, P_4O_6, N_2O_5, P_4O_{10}, As_4O_6, Sb_4O_6.
(v) Nitrogen shows very little tendency for catenation (like –C–C–) due to weakness of N – N single bond, whereas phosphorus shows marked tendency for catenation.
(vi) Nitrogen forms a maximum of four covalent bonds For example, in compounds containing NH_4^+ or NR_4^+ where, R is CH_3 or C_2H_5). This is due to the absence of d-orbitals in nitrogen while higher members of the group can expand their valence shell to form penta and hexa coordinated derivatives like PCl_5, AsF_5, PF_6^- etc.
(vii) Nitrogen forms stable pπ-pπ bonds easily whereas this is not possible in the case of the higher members. Multiple bonding involving dπ-pπ interactions occurs readily in the case of higher members.

Commercial preparation

On a commercial scale, nitrogen is obtained by the fractional distillation of liquid air. Nitrogen distills out first due to its lower boiling point (-195°C) followed by oxygen (boiling point = –183°C).

Laboratory preparation

(i) Nitrogen is obtained by warming ammonium nitrite, which is done by heating by reacting an equimolar mixture of ammonium chloride and sodium nitrite in aqueous solution.

$$NH_4Cl + NaNO_2 \rightarrow NH_4NO_2 + NaCl$$
$$NH_4NO_2 \rightarrow N_2 + 2H_2O$$

(ii) Ammonium dichromate, on heating, decomposes to form nitrogen.

$$(NH_4)_2Cr_2O_7 \rightarrow Cr_2O_3 + 4H_2O + N_2$$

(iii) By heating sodium azide at 300°C

$$2NaN_3 \rightarrow 2Na + 3N_2\uparrow$$

Properties and uses of N_2

Nitrogen is a colourless and odourless gas. It is slightly soluble in water. It is not combustible and does not support combustion.

Nitrogen finds use in several ways.

(i) To provide an inert atmosphere in iron, steel and other metallurgical industries.
(ii) In the manufacture of NH_3 and calcium cyanamide.
(iii) Liquid N_2 is used in cryogenics.

Reactions

(i) Nitrogen, on heating with metals like magnesium, lithium and alkaline earth metals like Ca or Ba forms corresponding nitrides. These compounds are ionic and nitrogen is present as nitride ion N^{3-}

$$3Mg + N_2 \rightarrow Mg_3N_2$$
$$6Li + N_2 \rightarrow 2Li_3N$$

Ionic nitrides liberate ammonia when treated with water

$$Mg_3N_2 + 6H_2O \rightarrow 3Mg(OH)_2 + 2NH_3$$

(ii) Non metals like boron, phosphorus and silicon also give nitrides under red heat. All these nitrides are covalent in nature.

$$2B + N_2 \rightarrow 2BN$$
$$3Si + 2N_2 \rightarrow Si_3N_4$$

(iii) Nitrogen, on heating with calcium carbide, gives calcium cyanamide.

$$CaC_2 + N_2 \rightarrow CaCN_2 + C$$

(iv) Nitrogen combines with oxygen when passed through an electric arc at 3000°C to form NO.(Yield is only 1%)

$$N_2 + O_2 \rightarrow 2NO$$

(v) N_2 and H_2 combine to form ammonia when passed over finely divided iron at 725K and 200 – 1000 atm pressure (Haber process).

$$N_2 + 3H_2 \rightleftharpoons 2NH_3$$

Nitrogen cycle

Nitrogen is involved in a continuous exchange between atmosphere, soil, sea and living organisms. Nitrates, nitrites and ammonium compounds present in the soil are absorbed by the plants. Plants are eaten by animals, whose excreta return nitrogen to the soil in the form of urea and uric acid. A second way of loss of nitrogen from soil is through leaching of nitrates into sea where the marine plant life is supported. A third way of nitrogen loss from soil is through denitrifying bacteria, which convert nitrates in soil into ammonia or nitrogen gas and this is released into atmosphere. As against this, there is nitrogen fixation into the soil by nitrifying bacteria which fix nitrogen as nitrates or nitrites and these are released to the soil. Thus a continuous circulation of nitrogen exists in nature and it is maintained so from atmosphere to plants, animals and soil and back to atmosphere. This is referred as nitrogen cycle.

Some important compounds of nitrogen

Hydrides

Hydrazoic acid or azoimide N_3H

Preparation

(i) Hydrazine hydrochloride on treatment with sodium nitrite or nitrous acid gives hydrazoic acid.

$$NH_2NH_2.HCl + NaNO_2 \rightarrow NaCl + N_3H + 2H_2O$$

$$NH_2NH_2.HCl + HNO_2 \rightarrow HCl + N_3H + 2H_2O$$

Properties and reactions

It is a colourless liquid with a low boiling point of 37°C. It is very poisonous and has an irritating odour.

(i) It explodes on heating,

$$2N_3H \rightarrow H_2 + 2N_2$$

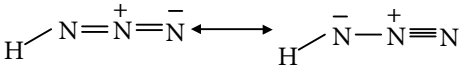
(ii) It behaves as a weak acid with a $K_a = 1.8 \times 10^{-5}$

(iii) Lead azide is prepared by precipitation reaction between sodium azide and lead nitrate.

$$2NaN_3 + Pb(NO_3)_2 \rightarrow Pb(N_3)_2 + 2NaNO_3$$

It is used as a detonator to set off high explosive charges.

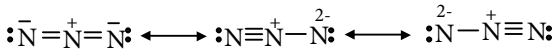
Sturcture of hydrazoic acid



All the three nitrogen atoms in N_3^- ion lie in a stright line with a bond length of 1.16 A° and the H – N – N bond angle being 110°, the H – Nbond length is 1.01 A°.

Structure of azide ion, N_3^-

Azide ion is linear with the N–N distance of 1.16 A° and may be represented as [N–N–N]$^-$. It may be represented as a resonance hybrid of the following structures.



Hydrazine, $NH_2 – NH_2$

It may be considered as a derivative of ammonia where one of its hydrogen atoms is replaced by $–NH_2$ group.

Preparation

It is made by Raschig process, which involves oxidation of ammonia by sodium hypochlorite in dilute aqueous solution in presence of glue or gelatin, which complexes any heavy metal ion as otherwise side reactions occur.

$$2NH_3 + NaOCl \rightarrow NH_2.NH_2 + NaCl + H_2O$$

Hydrazine forms a hydrate in presence of moisture and it is a fuming liquid.

Properties and reactions

It is soluble in water and alcohol.

(i) Decomposition: It decomposes on heating to give N_2 and NH_3.

$$3N_2H_4 \rightarrow N_2 + 4NH_3$$

(ii) It burns in air to give N_2.

$$N_2H_4 + O_2 \rightarrow N_2 + 2H_2O$$

(iii) Because of its basic nature. It forms two series of salts with acids like HCl and H_2SO_4.

$$N_2H_4 + HCl \rightarrow NH_2.NH_3Cl$$

$$N_2H_4 + 2HCl \rightarrow ClH_3N.NH_3Cl$$

(iv) It is also a strong reducing agent and reduces halogens to hydrohalic acids.

$$NH_2.NH_2 + 2Cl_2 \rightarrow 4HCl + N_2$$

(v) Silver platinum and gold salts are reduced to their metals. For example,

$$4AgNO_3 + NH_2NH_2 \rightarrow 4Ag + N_2 + 4HNO_3$$

$$PtCl_4 + NH_2NH_2 \rightarrow N_2 + 4HCl + Pt$$

(vi) Hydrazine reduces $KMnO_4$ to $MnSO_4$ in presence of dilute H_2SO_4. It is oxidized to N_2.

$$4KMnO_4 + 6H_2SO_4 + 5NH_2{-}NH_2 \rightarrow 2K_2SO_4 + 4MnSO_4 + 16H_2O + 5N_2\uparrow$$

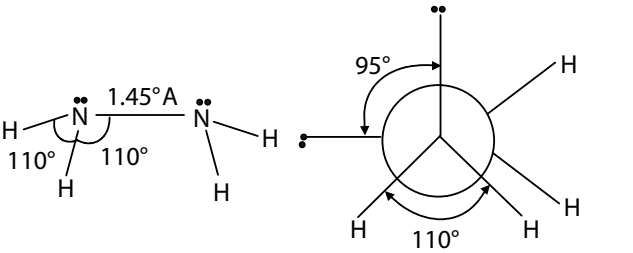
This reaction can be used for the titration of hydrazine and its salts.

Uses

Derivatives of hydrazine find application as a rocket fuel. Hydrazine is also used as an agricultural chemical.

Structure of hydrazine

Each nitrogen atom in hydrazine is tetrahedrally surrounded by one N atom, two H atoms and a lone pair of electrons. The N–N bond length is 1.45 A° and H–N–H bond is 110°. The nitrogen atom is sp^3 hybridised. The structure is unsymmetrical and N_2H_4 has a gauche. (non eclipsed) conformation.



Newman projection of N_2H_4

Oxides of nitrogen

Introduction

Nitrogen forms a number of oxides ranging in oxidation states from +1 to +5. They all exhibit pπ - pπ multiple bonding between nitrogen and oxygen atoms. The lower oxides are neutral while the higher ones are acidic. All are endothermic compounds.

The following table shows the various oxides and the oxidation number of nitrogen in these compounds.

Name	Formulae of oxide	Oxidation number of nitrogen in the compounds	State and colour of oxide	Magnetic behaviour
Nitrous oxide	N_2O	+1	Colourless gas	Diamagnetic
Nitric oxide	NO	+2	Colourless gas	Paramagnetic
Nitrogen trioxide (or nitrogen sesquioxide)	N_2O_3	+3	Pale blue solid dissociates into NO and NO_2 at –30°C.	Diamagnetic
Nitrogen dioxide, Nitrogen tetroxide	NO_2, N_2O_4	+4	NO_2 is red brown, poisonous gas. N_2O_4 is colourless	NO_2 is paramagnetic N_2O_4 is diamagnetic
Dinitrogen pentoxide	N_2O_5	+5	Colourless deliquescent solid	Diamagnetic

A brief account of the individual gases is given below:

Nitrous oxide, N_2O

Preparation

Nitrous oxide can be prepared heating ammonium nitrate

$$NH_4NO_3 \rightarrow N_2O + 2H_2O$$

Properties

It is a colourless gas with faint sweetish smell, quite stable unreactive and soluble in cold water but not in hot water. It is a neutral oxide and is not toxic. It is fairly soluble in water. It does not burn but is a supporter of combustion as it decomposes on heating to 600°C to give nitrogen and oxygen. A glowing splinter burns brightly in N_2O.

$$2N_2O \rightarrow 2N_2 + O_2$$

Uses

(i) It is used in the preparation of sodium azide and other azides.

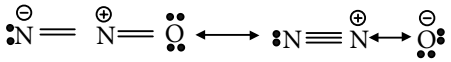
$$N_2O + 2NaNH_2 \rightarrow NaN_3 + NH_3 + NaOH$$

(ii) It also called 'laughing gas' and is used as an anaesthetic in dental operations.

(iii) It is widely used in making whipped ice cream.

Structure of N_2O

N_2O is a linear triatomic molecule and has sixteen valence electrons. The N – N and N – O bond lengths are 1.13 A° and 1.19 A° respectively. It is a resonance hybrid.



Nitric oxide, NO

Preparation

It is prepared in the laboratory by reduction of dilute (1 : 1) nitric acid with copper turnings.

$$3Cu + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 4H_2O + 2NO$$

Pure nitric oxide is formed by the addition of a concentrated solution of ferrous sulphate to a mixture of conc. H_2SO_4 and KNO_3.

$$2KNO_3 + 4H_2SO_4 + 6FeSO_4 \rightarrow 3Fe_2(SO_4)_3 + K_2SO_4 + 4H_2O + 2NO\uparrow$$

In an industrial scale, nitric oxide is prepared by the catalytic oxidation of ammonia with air (known as Ostwald's process) according to the reaction

$$4NH_3 + 5O_2 \xrightarrow[750^\circ C,\ 6\ atm]{Pt,\ gauze} 4NO + 6H_2O$$

Properties

It is a colourless gas and is sparingly soluble in water. Liquid NO is blue in colour and boils at –150°C. It is a neutral oxide. It is an odd electron molecule. The gas is paramagnetic and the liquid and solid are diamagnetic.

Reactions

(i) When mixed with oxygen it forms brown fumes of NO_2.

$$2NO + O_2 \rightarrow 2NO_2$$

(ii) It reacts with halogen like Cl_2, Br_2 to form nitrosyl halides.

$$2NO + Cl_2 \rightarrow 2NOCl$$

(iii) Nitric oxide is reduced to N_2 when passed over heated copper.

$$2Cu + 2NO \rightarrow 2CuO + N_2$$

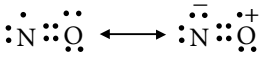
(iv) Nitric oxide forms co-ordination complexes with transition metals, which are known as nitrosyls. Two important nitrosyls are

(a) the complex formed in the brown ring test for nitrate ion. This complex ion may be represented as $[Fe(H_2O)_5NO]^{2+}$.

(b) sodium nitroprusside, $Na_2[Fe(CN)_5NO].2H_2O$.

Structure of NO

Nitric oxide is an odd electron molecule having eleven valence electrons. The gas is thus paramagnetic. The resonance structures of "NO" may be represented as



In the solid and liquid states, it is diamagnetic because it exists as a dimer, O=N – N=O, with the N – O bond length being 1.15 A°, which is intermediate between that of a double bond and a triple bond.

The molecular orbital diagram giving the distribution of 15 electrons in various molecular orbitals may be represented as

$$\sigma 1s^2,\ \sigma^* 1s^2\ \ \sigma 2s^2\ \ \sigma^* 2s^2\ \ \sigma 2p_z^2 \begin{cases} \pi 2p_x^2 \\ \pi 2p_y^2 \end{cases} \begin{cases} \pi^* 2p_x^1 \\ \pi^* 2p_y^0 \end{cases}$$ from

which the bond order is 2.5.

CONCEPT STRAND

Concept Strand 2

A colourless gas (A) combines with moist iron gives a colourless gas (B). Gas (A) gives a dark brown coloured complex with $FeSO_4$. Among gas (A), (B) and air, which is the supporter of combustion?

Solution

Gas (A) gives brown complex with $FeSO_4$.

$\therefore$ gas (A) is NO and complex formed is $\left[Fe(H_2O)_5NO\right]^{2+}$

NO with moist iron gives N_2O (B) and $Fe(OH)_2$.

$$2NO + Fe + H_2O \rightarrow N_2O + Fe(OH)_2$$

Among NO, N_2O and air, N_2O is a better supporter of combustion, because, N_2O on heating decomposes as follows.

$$2N_2O \rightarrow 2N_2 + O_2$$

So there is one in three parts oxygen while air contains one in five parts oxygen.

Nitrogen trioxide (dinitrogen sesquioxide) N_2O_3

Pure N_2O_3 exists at low temperature (< –30°C) as a blue liquid and as a pale blue solid with a melting point of –102°C. It is stable at low temperature. In gaseous state, it dissociates readily into nitric oxide and nitrogen dioxide.

$$N_2O_3 \rightleftharpoons NO + NO_2$$

N_2O_3 molecule is diamagnetic despite of its blue colour and is not an odd electron molecule.

Preparation

(i) It may be prepared by the condensation of equimolar amounts of NO and NO_2 below -30°C

$$NO + NO_2 \rightarrow N_2O_3$$

(ii) Another method of preparation involves the reaction of NO with required amount of O_2.

$$4NO + O_2 \rightarrow 2N_2O_3$$

Reactions

(i) It is an acidic oxide and forms nitrous acid on treatment with water. It is the anhydride of HNO_2.

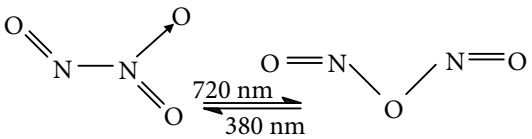
$$N_2O_3 + H_2O \rightarrow 2HNO_2$$

(ii) It reacts with alkalis to form nitrites.

$$N_2O_3 + 2NaOH \rightarrow 2NaNO_2 + H_2O$$

Structure of N_2O_3

The oxide exists in two different forms which may be interconverted by irradiation with light of appropriate wavelength or by the application of pressure.



asymmetrical form symmetrical form

The N – N bond length is 1.86 A° in the asymmetrical form, which is exceptionally long and so the N – N bond is weak compared to the N – N bond in hydrazine (1.45 A°).

Nitrogen dioxide, NO_2 and dinitrogen tetroxide, N_2O_4

Preparation

(i) It is prepared in the laboratory by heating lead nitrate.

$$2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$$

(ii) By the action of concentrated HNO_3 on copper, Ag, Pb etc.

$$Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O.$$

Properties

It is a brown gas.

Action of heat

(i) When brown NO_2 is cooled it condenses to a brown liquid, which has a boiling point of 21°C. It becomes pale in colour on further cooling (freezing point -11.2°C) and finally solidifies as a colourless solid.

This is due to the dimerisation of NO_2 to N_2O_4. Above 140°C, the colour of NO_2 becomes pale and its density decreases due to its dissociation to NO and O_2. The dissociation is complete at 620°C. Action of heat on nitrogen dioxide is shown below.

$$\underset{\text{colourless solid}}{N_2O_{4(s)}} \xrightarrow[-11.2^\circ C\ (m.p)]{260K} \underset{\text{colourless liquid}}{N_2O_{4(l)}} \xrightarrow[21^\circ C\ (b.p)]{294K}$$

$$\underset{\text{pale brown gas}}{N_2O_4} \xrightarrow[140^\circ C]{413K} \underset{\text{Brown gas}}{2NO_2} \xrightarrow[620^\circ C]{893K} \underset{\text{dissociation complete colourless gas}}{2NO + O_2}$$

Reaction of NO_2

(i) Hydrogen sulphide is oxidized to sulphur

$$NO_2 + H_2S \rightarrow NO + H_2O + S$$

(ii) On reaction with NO_2, finely divided copper gives cuprous oxide, which absorbs more of it to form 'nitro copper'.

$$2Cu + NO_2 \rightarrow Cu_2O + NO$$

(iii) Nitrogen dioxide reacts with alkalies to give nitrite and nitrate.

$$2KOH + 2NO_2 \rightarrow KNO_3 + KNO_2 + H_2O$$

Reaction of N_2O_4

(i) N_2O_4 reacts with water to give a mixture of nitrous and nitric acid and thus it is a mixed anhydride.

$$N_2O_4 + H_2O \rightarrow HNO_3 + HNO_2$$

(ii) Liquid N_2O_4 is a good solvent for preparing anhydrous metal nitrates.

$$ZnCl_2 + 2N_2O_4 \rightarrow Zn(NO_3)_2 + 2NOCl$$

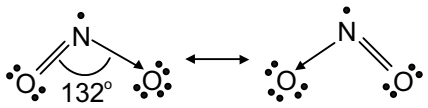
(iii) Liquid N_2O_4 undergoes self ionization according to

$$N_2O_4 \rightleftharpoons \underset{\text{Acid}}{NO^+} + \underset{\text{Base}}{NO_3^-}$$

Substances giving NO^+ ions in this solvent are acids and those giving NO_3^- ions are bases.

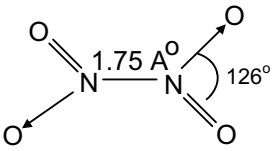
Structure of NO_2 and N_2O_4

NO_2 may be represented as



The molecule is bent with O – N – O angle of 132° and O – N bond length 1.2 A°. As seen from the structure given above, NO_2 is an odd electron molecule. It is thus paramagnetic and very reactive.

When NO_2 dimerises, electron pairing occurs and N_2O_4 formed is diamagnetic. Its structure may be given as



The molecule is planar. The N – N bond is weak due to the high value of bond length.

Nitrogen pentoxide (also known as dinitrogenpentoxide), N_2O_5

Preparation

(i) It is prepared by dehydrating concentrated nitric acid with P_2O_5 at low temperatures.

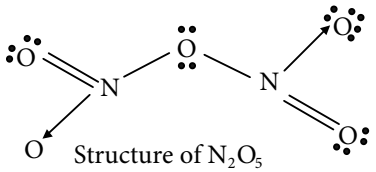
$$2HNO_3 + P_2O_5 \rightarrow 2HPO_3 + N_2O_5$$

(ii) Chlorine acts on solid silver nitrate to give N_2O_5 at 95°C

$$2Cl_2 + 4AgNO_3 \rightarrow 4AgCl + 2N_2O_5 + O_2$$

Properties

It is a deliquescent colourless solid and is very reactive. Solid N_2O_5 in its stable form exists in the ionic form $NO_2^+NO_3^-$ and is known as nitronium nitrate. N_2O_5 is covalent in solution and gas phase and has the structure



Structure of N_2O_5

Reactions

(i) It is the anhydride of nitric acid. It dissolves in water with a hissing sound to form HNO_3

$$N_2O_5 + H_2O \rightarrow 2HNO_3$$

(ii) It dissolves in alkali to form nitrates $N_2O_5 + 2NaOH \rightarrow 2NaNO_3 + H_2O$

(iii) It oxidizes iodine to iodine pentoxide.

$$I_2 + 5N_2O_5 \rightarrow I_2O_5 + 10NO_2$$

Oxyacids of nitrogen

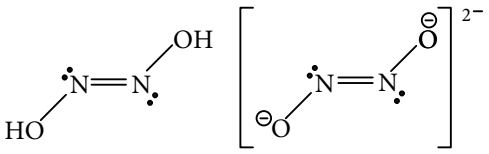
The following oxyacids, which are well known, will be considered in detail.

(i) Hyponitrous acid, $H_2N_2O_2$
(ii) Nitrous acid, HNO_2
(iii) Nitric acid, HNO_3

Structure of $H_2N_2O_2$ and $N_2O_2^{2-}$ ion

It has a non linear chain structure with N=N double bond.

The trans form of the acid is favoured (due to zero dipole moment).



Structures of $H_2N_2O_2$ and N_2O^{2-} ion

Nitrous acid, HNO_2

The free nitrous acid is not known in the liquid state but exists in the vapour phase. It is unstable except in dilute solution.

Preparation

(i) An aqueous solution of the acid, free of salts may be prepared by treating barium nitrite with sulphuric acid.

$$Ba(NO_2)_2 + H_2SO_4 \rightarrow BaSO_4 + 2HNO_2$$

(ii) Action of water on N_2O_3 gives nitrous acid

$$N_2O_3 + H_2O \rightarrow 2HNO_2$$

Properties

(i) In the gas phase, nitrous acid decomposes to NO and NO_2.

$$2HNO_2 \rightleftharpoons H_2O + NO + NO_2$$

$$(K_a = 4.5 \times 10^{-4})$$

(ii) Even in cold aqueous solution, HNO_2, undergoes disproportionation

$$3HNO_2 \rightarrow HNO_3 + H_2O + 2NO$$

(iii) Nitrous acid acts both as an oxidizing as well as a reducing agent.

HNO_2 as oxidizing agent: $2HNO_2 \rightarrow H_2O + 2NO + [O]$

HNO_2 as reducing agent: $HNO_2 + [O] \rightarrow HNO_3$.

(iv) Oxidizing action by HNO_2

(a) It oxidizes iodide ions to iodine in acid medium

$$2H^+ + 2I^- + 2HNO_2 \rightarrow I_2 + 2NO + 2H_2O$$

(b) It oxidises acidified ferrous sulphate to ferric sulphate and liberates NO.

$$2FeSO_4 + 2KNO_2 + 3H_2SO_4 \rightarrow Fe_2(SO_4)_3 + 2KHSO_4 + 2NO\uparrow + 2H_2O$$

(v) Reduing action by HNO_2

(a) $KMnO_4$ is reduced to manganous state by HNO_2 in presence of dil.H_2SO_4.

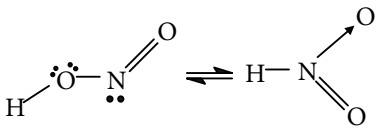
$$2KMnO_4 + 3H_2SO_4 + 5HNO_2 \rightarrow K_2SO_4 + 2MnSO_4 + 5HNO_3 + 3H_2O$$

(b) Acidified $K_2Cr_2O_7$ is reduced to green $Cr_2(SO_4)_3$.

$$K_2Cr_2O_7 + 4H_2SO_4 + 5HNO_2 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 3HNO_3 + 4H_2O$$

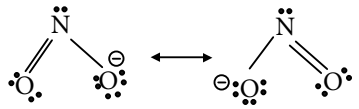
Structure of nitrous acid and nitrite ion

Nitrous acid is assumed to be a tautomeric mixture of the two forms



That is why, it forms two types of organic derivatives viz., nitrites (like methyl nitrite $CH_3 - ONO$) and nitro compounds (like nitro methane CH_3NO_2).

Nitrite ion has a bent structure with N atom at the centre



Resonance structure of $NO_2^{\ominus}$ ion

Nitric acid, HNO_3

Nitric acid is the most important oxyacid of nitrogen and has great commercial importance.

Manufacture

(i) Birkeland-Eyde process

This method involves the passing of an electric arc through atmospheric air (at a temperature of around 3500°C). At first nitric oxide is formed and on cooling it down to 500°C,

it combines with more oxygen to form NO_2. The NO_2 gas is passed into water to give a mixture of nitrous and nitric acid. The method is now obsolete.

$$N_2 + O_2 \rightarrow 2NO$$

$$2NO + O_2 \rightarrow 2NO_2$$

$$2NO_2 + H_2O \rightarrow HNO_2 + HNO_3$$

$$3HNO_2 \rightarrow HNO_3 + 2NO + H_2O$$

(ii) Ostwald's process

This process depends on the catalytic oxidation of NH_3 by air to NO in presence of platinum or rhodium catalyst at a pressure of 5 atm and a temperature of 850°C. This is followed by the oxidation of NO to NO_2 and conversion of NO_2 to HNO_3 by absorption in water in presence of air.

$$4NH_{3(g)} + 5O_{2(g)} \xrightarrow[5\text{ atm, }850^0\text{C}]{\text{Pt / Rh catalyst}} 4NO_{(g)} + 6H_2O_{(g)}$$

$$2NO_{(g)} + O_{2(g)} \rightarrow 2NO_{2(g)}$$

$$4NO_2 + 2H_2O + O_2 \rightarrow 4HNO_3$$

The method gives an acid concentration of about 60 % by weight. Distillation of the acid results in a constant boiling mixture containing 68% acid. This is the commercially available concentrated nitric acid. HNO_3, which is about 98% is obtained by dehydrating the 68% acid with conc. H_2SO_4. Fuming nitric acid is concentrated nitric acid (68% or 98%) containing dissolved NO_2.

Properties

It is a colourless liquid in the pure state with a boiling point of 84.1°C and freezing point of –41.5°C. It develops a yellow colour on exposure to light due to small amount of decomposition to NO_2 and O_2,

$$4HNO_3 \rightarrow 2H_2O + 4NO_2 + O_2$$

The pure acid undergoes self ionization giving nitronium ion (NO_2^+) and nitrate ion (NO_3^-) even at room temperature.

$$2HNO_3 \rightleftharpoons NO_2^+ + NO_3^- + H_2O$$

It is a strong acid and is completely dissociated in aqueous solution to give hydronium and nitrate ions.

$$HNO_3 + H_2O \rightarrow H_3O^+ + NO_3^-$$

It reacts with metallic carbonates, oxides and hydroxides to give the corresponding nitrates. It is a strong oxidizing agent, especially under hot and concentrated conditions.

$$2HNO_3 \rightarrow H_2O + 2NO_2 + [O]$$

Reactions

(i) Action on non-metals

(a) Sulphur is oxidized to H_2SO_4
$S + 6HNO_3 \rightarrow H_2SO_4 + 2H_2O + 6NO_2$

(b) Hot conc.HNO_3 oxidizes iodine to iodic acid.
$I_2 + 10HNO_3 \rightarrow 2HIO_3 + 4H_2O + 10NO_2$

(c) Phosphorus is oxidized to orthophosphoric acid.
$P_4 + 20HNO_3 \rightarrow 4H_3PO_4 + 4H_2O + 20NO_2$

(d) Carbon is oxidized to carbon dioxide.
$C + 4HNO_3 \rightarrow CO_2 + 2H_2O + 4NO_2$

(ii) Action on metals

Nitric acid reacts with most of the metals. The products, of such reactions depend on several factors such as concentration of acid, nature of the metal, temperature of the reaction etc. The various reactions of metals with nitric acid under different conditions are described below.

(a) Very dilute nitric acid reacts with metals like Mg, Mn liberating hydrogen.

$Mg + 2HNO_3 \rightarrow Mg(NO_3)_2 + H_2\uparrow$

$Mn + 2HNO_3 \rightarrow Mn(NO_3)_2 + H_2\uparrow$

(b) Weakly electropositive metals like Cu, Ag react with cold dilute nitric acid to give NO.

$3Cu + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 4H_2O + 2NO$

$3Ag + 4HNO_3 \rightarrow 3AgNO_3 + 2H_2O + NO\uparrow$

(c) Copper or silver reacts with hot conc.HNO_3 to give NO_2.

$Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2H_2O + 2NO_2$

$Ag + 2HNO_3 \rightarrow AgNO_3 + H_2O + NO_2$

(d) The reaction of mercury with dil.HNO_3 gives mercury (II) salts and nitric oxide

$3Hg + 8HNO_3 \rightarrow 3Hg(NO_3)_2 + 4H_2O + 2NO\uparrow$

When mercury is in excess, mercury (I) nitrate and NO are formed with dil.HNO_3.

$6Hg + 8HNO_3 \rightarrow 3Hg_2(NO_3)_2 + 4H_2O + 2NO\uparrow$

(e) With conc.HNO_3, mercury reacts to give mercury (II) salt with the liberation of NO_2.

$Hg + 4HNO_3 \rightarrow Hg(NO_3)_2 + 2H_2O + 2NO_2\uparrow$

(f) With very dil.HNO_3, iron reacts to form ferrous nitrate and ammonium nitrate.

$4Fe + 10HNO_3 \rightarrow 4Fe(NO_3)_2 + NH_4NO_3 + 3H_2O$

With dil.HNO_3, iron forms nitrous oxide and ferrous nitrate.

$4Fe + 10HNO_3 \rightarrow 4Fe(NO_3)_2 + N_2O + 5H_2O$

Moderately concentrated HNO_3 reacts with iron to form ferric nitrate and NO_2.

$Fe + 6HNO_3 \rightarrow Fe(NO_3)_3 + 3H_2O + 3NO_2$

Iron is rendered passive by concentrated nitric acid.

(g) More electropositive metals like Zn reduce very dil.HNO_3 (6%) to ammonia, which further reacts with acid to give NH_4NO_3.

$4Zn + 10HNO_3 \rightarrow 4Zn(NO_3)_2 + NH_4NO_3 + 3H_2O$

Zn reacts with 20% HNO_3 to form nitrous oxide

$4Zn + 10HNO_3 \rightarrow 4Zn(NO_3)_2 + N_2O + 5H_2O$

With conc.HNO_3, Zn reacts to give NO_2

$Zn + 4HNO_3 \rightarrow Zn(NO_3)_2 + 2H_2O + 2NO_2$

(h) Metals like Al, Cr, Fe, Ni are rendered passive by conc. HNO_3 due to a protective oxide film formation. Al is rendered passive even by dilute HNO_3.

(i) Noble metals like Pt, Au, Rh, and Ir are not attacked by HNO_3.

(iii) Action on some compounds

(a) An acidified solution of ferrous salt is oxidized to ferric salt and HNO_3 is reduced to NO.

$6FeSO_4 + 3H_2SO_4 + 2HNO_3 \rightarrow 3Fe_2(SO_4)_3 + 4H_2O + 2NO\uparrow$

This is the basis of the brown ring test used for detection of nitrates.

(b) SO_2 is oxidized to H_2SO_4

$SO_2 + 2HNO_3 \rightarrow H_2SO_4 + 2NO_2$

(c) H_2S is oxidized to sulphur

$H_2S + 2HNO_3 \rightarrow S\downarrow + 2NO_2\uparrow + 2H_2O$

(d) KI is oxidized to iodine.

$6KI + 8HNO_3 \rightarrow 6KNO_3 + 2NO + 4H_2O + 3I_2$

(iv) Nitration reaction

When concentrated nitric acid is mixed with concentrated sulphuric acid, the species, nitronium ion $\overset{+}{N}O_2$ is formed. Salts of nitronium ion such as $NO_2^+ClO_4^-$ are known. Nitronium ion is active species in the nitration of organic compounds. For example, benzene is nitrated to give nitrobenzene. Toluene is nitrated to give trinitrotoluene (TNT). Glycerol is nitrated to give trinitroglycerine and cellulose on nitration gives nitro cellulose (also known as gun cotton)

(v) Aqua regia

It is a mixture of 75% conc.HCl and 25% conc.HNO_3. Metals like gold, platinum which are insoluble in conc. HNO_3 dissolve in aqua regia. The ability of aqua regia to act as a solvent for these metals arises from the oxidising capacity of HNO_3 and the ability of chloride ions to form complexes. The reactions in the case of gold and platinum may be shown as follows

$HNO_3 + 3HCl \rightarrow 2[Cl] + 2H_2O + NOCl$

$NOCl \rightarrow NO\uparrow + [Cl]$

$Au + 3[Cl] \rightarrow AuCl_3$ } reactions in aqua regia

$Pt + 4[Cl] \rightarrow PtCl_4$

$AuCl_3$ and $PtCl_4$ exists as complexes $H[AuCl]_4$ and $H_2[PtCl_6]$

$Au + 3HCl + HNO_3 \rightarrow H[AuCl_4] + NO + 2H_2O$

$3Pt + 4HNO_3 + 18HCl \rightarrow 3H_2[PtCl_6] + 4NO + 8H_2O$

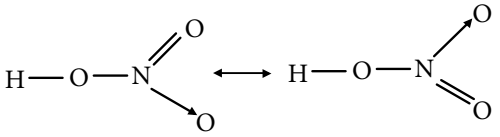
Uses of HNO_3

It finds a variety of uses such as

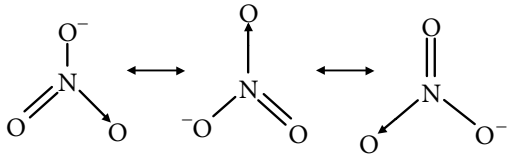
(i) in the manufacture of explosives such as TNT, gun cotton, nitroglycerin.
(ii) in the manufacture of rayon.
(iii) in the manufacture of fertilizers.
(iv) in the manufacture of batteries.
(v) in the manufacture of H_2SO_4 by lead chamber process.

Structure of HNO_3 and NO_3^- ion

HNO_3 may be considered to be a resonance hybrid of the following structures



NO_3^- ion can be represented as the resonance hybrid of the following structure



Ammonia, NH_3

Manufacture of ammonia

(i) Haber's process

On a large scale, ammonia is made from nitrogen obtained from air by a process known an Haber's process. In this process, a mixture of nitrogen and hydrogen in the ratio of 1 : 3 is passed over a catalyst of promoted iron (finely divided iron containing traces of molybdenum as promoter) at a temperature of about 450 - 550°C and a pressure of about 200–1000 atm. The resulting gaseous mixture is cooled to condense ammonia gas to liquid and the unreacted gases are recycled. The yield is 14% at 200 atm and 49% at 1000 atm.

$$N_2 + 3H_2 \rightleftharpoons 2NH_{3(g)}$$

It may be recalled that the equilibrium reaction is shifted in the direction of the product i.e., NH_3 at low temperatures and high pressure.

(ii) Cyanamide process

In cyanamide process, air is passed over calcium carbide heated to 800–1000°C when the nitrogen of air combines with CaC_2 to form calcium cyanamide. The method is obsolete now.

$$CaC_2 + N_2 \rightarrow CaCN_2 + C$$

Next, the calcium cyanamide is digested with steam at 180°C under a pressure of 3 – 4 atm to give out ammonia.

$$CaCN_2 + 3H_2O \rightarrow CaCO_3 + 2NH_3$$

Laboratory method of preparation of NH_3

In the laboratory, ammonia is prepared by heating ammonium chloride with slaked lime or caustic soda.

$$2NH_4Cl + Ca(OH)_2 \rightarrow CaCl_2 + 2NH_3 + 2H_2O$$

$$NH_4Cl + NaOH_{(aq)} \rightarrow NH_3 + NaCl + H_2O$$

This reaction forms the basis of a test for ammonium salts in qualitative analysis.

Test for ammonia

Ammonia can be detected by its characteristic pungent smell as well as by its effect on moist red litmus paper which turns blue in its presence. Nessler's reagent, K_2HgI_4 produces a reddish brown precipitate or colouration of HgO. $Hg(NH_2)I$ or iodide of Millon's base.

$$2K_2HgI_4 + NH_3 + 3KOH \rightarrow HgO.Hg(NH_2)I + 7KI + 2H_2O.$$

Properties of ammonia

Ammonia is a colourless gas with a strong pungent odour. It is extremely soluble in water with evolution of heat. One volume of water dissolves 1300 volumes of NH_3 at 0°C and 1 atm. On dissolution in water it forms small amounts of ammonium hydroxide which dissociates as a weak base. The reaction is

$$NH_3 + H_2O \rightleftharpoons NH_4OH \rightleftharpoons NH_4^+ + OH^-$$

It liquefies easily on cooling and compression. Liquid ammonia is colourless with a boiling point of –33°C and freezing point of –78°C.

Reactions of ammonia

(i) Ammonia burns in air or oxygen with a pale yellow flame to form N_2.

$$4NH_3 + 3O_2 \rightarrow 2N_2 + 6H_2O$$

(ii) It is oxidized by air to nitric oxide in presence of red hot platinum as catalyst.

$$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$$

(iii) Excess of ammonia reacts with Cl_2 or Br_2 to form the corresponding halide salt.

$$8NH_3 + 3Cl_2 \rightarrow 6NH_4Cl + N_2$$

(iv) When chlorine is in excess, an explosive yellow oil, nitrogen trichloride is formed.

$$NH_3 + 3Cl_2 \rightarrow NCl_3 + 3HCl$$

(v) Excess of ammonia reacts with alkali metals to form amides

$$2Na + 2NH_3 \rightarrow 2NaNH_2 + H_2$$

Amides are decomposed by water to release ammonia.

$$NaNH_2 + H_2O \rightarrow NaOH + NH_3$$

(vi) $KMnO_4$ oxidizes ammonia to nitrogen.

$$2KMnO_4 + 2NH_4OH \rightarrow 2MnO_2 + 2KOH + 4H_2O + N_2$$

(vii) It is also oxidized by hot cupric oxide to nitrogen.

$$2NH_3 + 3CuO \rightarrow 3Cu + 3H_2O + N_2$$

(viii) Ammonia is used for the manufacture of urea, an important fertilizer.

$$2NH_3 + CO_2 \xrightarrow[\text{high pressure}]{200^0\,C} \underset{\text{Ammonium carbamate}}{NH_2COONH_4} \rightarrow NH_2CONH_2 + H_2O$$

Urea hydrolyses in the soil to ammonium carbonate

$$NH_2CONH_2 + 2H_2O \rightarrow (NH_4)_2CO_3$$

(ix) Ammonia acts like a Lewis base by its capacity to donate its lone pair of electrons. Thus it can form

complexes with many cations like Ag^+, Cu^{2+}, Cd^{2+} etc. For example, the reaction with silver ions is

$$Ag^+ + 2NH_3 \rightarrow [Ag(NH_3)_2]^+$$

Liquid ammonia

Liquid ammonia boils at −33°C and freezes at −78°C. It undergoes self-ionization like water, which may be given as $2NH_3 \rightleftharpoons NH_4^+ + NH_2^-$

It is a good solvent for many salts. Substances that give NH_4^+ ions in this solvents are acids and those that give NH_2^- ions are bases (e.g., NH_4Cl and $NaNH_2$).

An interesting reaction involving liquid ammonia is that alkali and alkaline earth metals like Ca, Sr, Ba dissolve readily in this solvent and give highly electrolytically conducting solutions. The intense blue colour associated with dilute solutions of the above metals in liquid NH_3 is attributed to the presence of solvated electrons. The reaction is

$$Na \xrightarrow{\text{liquid amonia}} [Na(NH_3)_x]^+ + [e(NH_3)_y]^-$$

The metallic solutions in this solvent act as good reducing agents.

As the concentration increases, the blue colour change to bronze or copper coloured with metallic lustre. On standing colours fade until they disappear owing to formation of metal amides.

$$2Na + 2NH_3 \rightarrow NaNH_2 + H_2$$

CONCEPT STRAND

Concept Strand 3

Sodamide, $NaNH_2$, behaves as a base when dissolved in liquid ammonia. Explain

Solution

Liquid ammonia is a non-aqueous solvent. It undergoes self dissociation like water.

$$2NH_3 \rightleftharpoons NH_4^+ + NH_2^-$$

Salts which dissociates to NH_4^+ ion are acids and salts producing NH_2^- ion are bases when dissolved in liquid ammonia.

Sodamide dissociates in ammonia to produce NH_2^- ions. Therefore, it is a base.

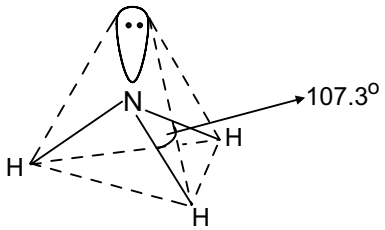
$$NaNH_2 \rightarrow Na^+ + NH_2^-$$

Ammonium salts behave as acids in ammonia because they produce NH_4^+ ions

$$NH_4Cl \rightarrow NH_4^+ + Cl^-$$

Structure of NH_3

The NH_3 molecule is trigonal pyramidal with the H – N – H bond angle equal to 107.3°. Nitrogen is in sp^3 hybridization.



Ammonium salts

Ammonium hydroxide being a weak base reacts with acids to form ammonium salts. Salts like NH_4Cl, NH_4NO_3 or $(NH_4)_2SO_4$ are acidic in solution due to hydrolysis.

Action of heat

Ammonium salts decompose readily on heating. When the anions are not oxidizing like Cl^-, SO_4^{2-}, CO_3^{2-}, the salts decompose with evolution of NH_3. For example,

$$NH_4Cl \xrightarrow{\text{heat}} NH_3 + HCl$$

When the anions are oxidising such as NO_3^-, NO_2^- heating results in the oxidation of NH_4^+ to N_2 or nitrous oxide.

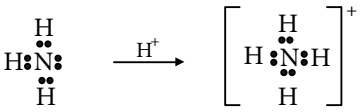
$$NH_4NO_3 \xrightarrow{\text{heat}} N_2O + 2H_2O$$

$$NH_4NO_2 \xrightarrow{\text{heat}} N_2 + 2H_2O$$

$$(NH_4)_2Cr_2O_7 \xrightarrow{\text{heat}} N_2 + Cr_2O_3 + 2H_2O$$

Structure of NH_4^+ ion

NH_4^+ ion is tetrahedral. It may be represented as being formed by donation of a lone pair of electrons from NH_3 to H^+ resulting in the formation of a co-ordinate covalent bond. Thereby a coordination number of four is achieved.



Uses of ammonium salts

NH_4Cl is used in 'dry cells' or Leclanche cells. It is also used as a flux in soldering metals. NH_4NO_3 mixed with $CaCO_3$ or $(NH_4)_2SO_4$ is used as fertilizer. $NH_4H_2PO_4$ and $(NH_4)_2HPO_4$ find application as fertilizers and in fire proofing wood, paper and textiles. NH_4ClO_4 is used as an oxidizing agent in solid rocket propellants.

Nitrogen halides

Nitrogen forms trihalides and with the exception of NF_3 halides are unstable. NCl_3 is explosive, and NBr_3 and NI_3 are known only as their adducts with ammonia i.e., $NBr_3.6NH_3$ and $NI_3.6NH_3$. NCl_3 is prepared by the combination of NH_3 and excess of Cl_2.

$$NH_3 + 3Cl_2 \rightarrow NCl_3 + 3HCl$$

NCl_3 is a dangereously explosive endothermic oil.

NCl_3 hydrolyses readily to give NH_4OH and HOCl

$$NCl_3 + 4H_2O \rightarrow NH_4OH + 3HOCl$$

NF_3 is a stable unreactive gas. It is prepared by the electrolysis of fused NH_4HF_2. It is not hydrolysed by water, dilute acids or alkali.

The trihalides are covalent and have a trigonal pyramidal structure.

Nitrogen does not form any pentahalides due to the absence of 'd' orbitals.

PHOSPHORUS (P)

Introduction

Phosphorus (P) belongs to group 15 of the periodic table, the other members of the group being nitrogen (N), arsenic (As), antimony (Sb) and bismuth (Bi). It is biologically an important element as it is essential for human life. It also takes part in metabolic processes in plants and animals.

Bones and teeth essentially contain calcium phosphate and nucleic acids such as DNA and RNA are made up of ester chains of phosphates. Phosphates are also important fertilizers.

Some important atomic properties of phosphorus are presented below:

Atomic number	Atomic weight	Electronic configuration	Atomic radius (A°)	Ionization energy (1st) (kJ mol^{-1})	Electro-negativity (Pauling scale)	Oxidation states
15	31.0	[Ne] $3s^2 3p^3$	1.10	1060	2.1	+3 and +5

Phosphorus compounds show co-ordination numbers of 5 or 6 in compounds like PCl_5 and in complex ion like $[PCl_6]^-$ due to the fact that phosphorus atoms possess vacant 'd' orbitals and thus exhibit sp^3d or sp^3d^2 hybridization. Phosphorus forms P^{3-} ions less readily compared to nitrogen although in a few cases like Ca_3P_2, it is observed. Phosphorus exists in three allotropic modifications, which will be subsequently discussed in detail. Phosphorus shows a limited degree for catenation in its compounds. Both cyclic and open chain structures involving several phosphorus atoms are possible in the elemental forms.

Occurrence

Phosphorus mainly occurs as phosphates and the following minerals contain largely calcium phosphate.

(i) Phosphorite, $Ca_3(PO_4)_2$
(ii) Chlorapatite, $3Ca_3(PO_4)_2.CaCl_2$
(iii) Fluorapatite, $3Ca_3(PO_4)_2.CaF_2$.

Apart from the above, bone ash is another important source of phosphorus in the form of $Ca_3(PO_4)_2$ to the extent of about 50%.

Manufacture/extraction

(i) From phosphate minerals

The mineral, containing calcium phosphate, is reduced by carbon in an electric furnace at a temperature of 1400-1500°C. Silica is added to liberate P_4O_{10} forming calcium silicate as slag. P_4O_{10} vapour is reduced by carbon to phosphorus which distills out. The vapour is passed through water to condense as crude yellow phosphorus. The crude product is purified by melting under water and treated with a 4% solution of acidified sodium dichromate, to oxidize any impurities. After leaving for a few hours, the chrome liquor is removed and the yellow phosphorus obtained is washed with hot water and filtered through canvas.

The reactions occurring are:

$$2Ca_3(PO_4)_2 + 6SiO_2 \rightarrow 6CaSiO_3\downarrow + P_4O_{10}$$

$$P_4O_{10} + 10C \rightarrow P_4 + 10CO$$

(ii) From bone ash

Bones contain about fifty percent calcium phosphate and the remaining part is gelatin, fat and other nitrogenous matter. After removing the extraneous matter in a suitable manner, the bones are burnt to bone ash. The ash is heated with conc. H_2SO_4, which results in the formation of insoluble calcium sulphate and phosphoric acid.

$$Ca_3(PO_4)_2 + 3H_2SO_4 \rightarrow 3CaSO_4 + 2H_3PO_4$$

The calcium sulphate is filtered off and the phosphoric acid solution is concentrated to a syrupy mass. It is then mixed with coke and heated in a muffle furnace. Metaphosphoric acid is formed first and is reduced by coke to phosphorus.

$$H_3PO_4 \rightarrow HPO_3 + H_2O$$

$$4HPO_3 + 10C \rightarrow 10CO\uparrow + P_4\uparrow + 2H_2O$$

The vapours of phosphorus and CO are passed through water so that phosphorus condenses in the yellow form. The impure phosphorus containing small amounts of dissolved carbon is melted under water and heated with chromic acid to oxidize the impurities. The pure phosphorus is cast in the form of sticks.

Uses

Phosphorus is mainly used in the manufacture of (i) matches (ii) phosphor bronze, an alloy of phosphorus, copper and tin, containing a variable proportion of phosphorus in the range of 0.3% to 2.5%. Radioactive phosphorous (P^{32}) is used in the treatment of leukemia.

Allotropes of phosphorus

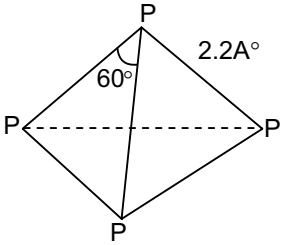
Several allotropic modifications of phosphorus such as yellow or white, red, scarlet, black and violet phosphorus have been reported but the most prominent of them, white, red and black forms are discussed below.

White or yellow phosphorus

White phosphorus is prepared by condensing phosphorus vapour under water. The yellow colour is believed to be due to small amounts of red phosphorus in it. It is a soft, waxy and reactive solid and has a melting point of 41°C. It exists as tetrahedral P_4 molecules and this structure is retained in the liquid and gaseous state. Above 800°C, P_4 begins to dissociate and the equilibrium $P_4 \rightleftharpoons 2P_2$ is established.

The dissociation is about 50% at 1200°C. White phosphorus is soluble in organic solvents like benzene and carbon disulphide. It reacts slowly with air at room temperature and gives out light, which is known as phosphorescence or chemiluminescence. It readily ignites in air at 35°C and hence it is stored under water. It melts at 44.1°C and boils at 280°C.

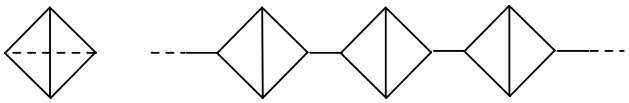
White phosphorus exists as P_4 molecules in liquid and vapour states. It has a tetrahedral structure with four P atoms at the corners of a tetrahedron as shown below with the P – P distance being 2.2 A°.



Structure of P_4 molecule

Red phosphorus

Red phosphorous is obtained by heating white phosphorus at about 250°C or at a low temperature in the presence of sunlight. Addition of a trace of iodine favours this conversion. This is a polymeric solid and is less reactive than the white phosphorus. It does not catch fire readily in air unless it is heated to about 400°C. It is insoluble in organic solvents but dissolves in alcoholic potash to give a red solution from which it can be precipitated by addition of acids. Red phosphorus melts at 592°C under pressure but sublimes at 416°C at 1 atm pressure. It has an extended chain structure.



White P4 red P (chain)

Black phosphorus

A crystalline form of black phosphorus is obtained by heating white phosphorus either under high pressure or, at 220 - 370°C for about eight days in presence of mercury as a catalyst. It has a layer structure (the adjacent layers are 3.68 A° apart), each P atom being bound to three neighbours. The closest P-P distance within each double layers is 2.2 A° with a P–P–P bond angle of 99°. This form is stable in air and can be ignited only with difficulty. It is an electrical conductor and it is insoluble in CS_2. Black phosphorus is thermodynamically the most stable form of phosphorus.

Reactions

The various forms of phosphorus show considerable difference in their reactivity. The white phosphorus is the most reactive form while the black variety is the least reactive.

(i) White phosphorus undergoes slow combustion to form P_2O_5 and heat of reaction is emitted as light as a green glow called phosphorescence.

(ii) White phosphorus catches fire on warming to 35°C in air forming phosphorus pentoxide

$$P_4 + 5O_2 \rightarrow P_4O_{10} \text{ (or } 2P_2O_5).$$

(iii) Phosphorus reacts with halogens forming tri and pentahalides.

$$P_4 + 6Cl_2 \rightarrow 4PCl_3$$

$$P_4 + 10Cl_2 \rightarrow 4PCl_5$$

With sulphur, phosphorus forms P_2S_3 and P_2S_5.

(iv) With metals like Na, Mg, Ca, Sr, phosphorus reacts to form metallic phosphides.

$$12Na + P_4 \rightarrow 4Na_3P$$

$$6Ca + P_4 \rightarrow 2Ca_3P_2$$

The phosphides are rapidly hydrolysed by water to give phosphine.

$$Na_3P + 3H_2O \rightarrow 3NaOH + PH_3$$

(v) Phosphorus acts as a strong reducing agent. It reduces copper sulphate to metallic copper.

$$P_4 + 10CuSO_4 + 16H_2O \rightarrow 10Cu + 4H_3PO_4 + 10H_2SO_4$$

(vi) Phosphorus is oxidized by conc.HNO_3 or conc.H_2SO_4 to orthophosphoric acid.

$$P_4 + 20HNO_3 \rightarrow 4H_3PO_4 + 20NO_2 + 4H_2O$$

$$P_4 + 10H_2SO_4 \rightarrow 4H_3PO_4 + 10SO_2 + 4H_2O$$

(vii) White phosphorus reacts with caustic soda to give phosphine. Red phosphorus does not undergo this reaction It is a disproportionation reaction.

$$P_4 + 3H_2O + 3NaOH \rightarrow 3NaH_2PO_2 + PH_3$$

Some compounds of phosphorus

Phosphorus trichloride, PCl_3

Preparation

By passing dry chlorine gas over heated phosphorus, PCl_3 is formed and distils over.

$$P_4 + 6Cl_2 \rightarrow 4PCl_3$$

Properties

PCl_3 is a colourless oily liquid with a pungent smell. It is covalent molecule.

Reactions

(i) It is readily hydrolysed by water to phosphorous acid.

$$PCl_3 + 3H_2O \rightarrow H_3PO_3 + 3HCl$$

(ii) PCl_3 reacts with alcohols to give alkyl chloride.

$$PCl_3 + 3C_2H_5OH \rightarrow H_3PO_3 + 3C_2H_5Cl$$

(iii) It is oxidized by oxygen or sulphur trioxide to form phosphorus oxychloride.

$$2PCl_3 + O_2 \rightarrow 2POCl_3$$

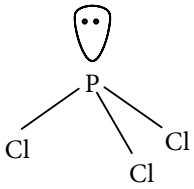
$$PCl_3 + SO_3 \rightarrow POCl_3 + SO_2$$

Uses

It is mainly used as a chlorinating agent (for example, conversion of alcohols to alkyl halides or organic acids to acid chlorides).

Structure of PCl_3

It has a trigonal pyramidal shape with the P atom being sp^3 hybridized.



Phosphorus pentachloride, PCl_5

Preparation

It is prepared by passing chlorine into PCl_3 cooled in freezing mixture.

$$PCl_3 + Cl_2 \rightarrow PCl_5$$

Properties

It is a yellowish white powder. It sublimes in the temperature range 100-160°C.

Reactions

(i) It fumes in moist air forming HCl and $POCl_3$ and on further reaction gives phosphoric acid.

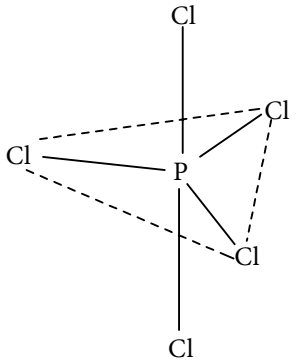
$$PCl_5 + H_2O \rightarrow POCl_3 + 2HCl$$
$$POCl_3 + 3H_2O \rightarrow H_3PO_4 + 3HCl$$

(ii) PCl_5 reacts with acids like H_2SO_4 or HNO_3 to give the chloro derivatives.

$$2PCl_5 + H_2SO_4 \rightarrow SO_2Cl_2 + 2POCl_3 + 2HCl$$
$$PCl_5 + HNO_3. \rightarrow NO_2Cl + POCl_3 + HCl$$

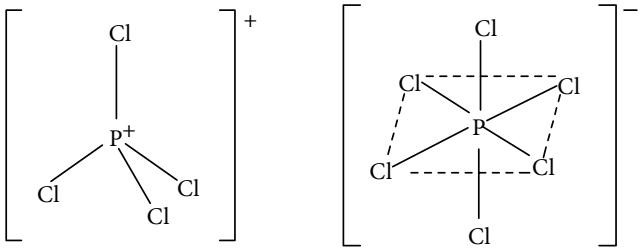
Structure

PCl_5 has a trigonal bipyramidal shape in vapour and liquid states and in solution (in non-polar solvents) as expected from VSEPR theory for a molecule having five electron pairs around the central atom. The P atom uses sp^3d hybridized orbitals for forming σ bonds with five chlorine atoms.



PCl_5 is a covalent molecule in the gaseous and liquid states but ionic in the solid state, existing as $[PCl_4]^+$ and $[PCl_6]^-$. The cation is tetrahedral but anion is octahedral. PBr_5 is ionic in the solid state, but PF_5 is covalent under the same conditions.

The tetrahedral and octahedral shapes of $[PCl_4]^+$ and $[PCl_6]^-$ ions are shown below.



Uses

PCl_5 is used as a chlorinating agent in organic chemistry.

CONCEPT STRAND

Concept Strand 4

Explain the properties of PCl_5 because of which it is used as a good chlorinating agent.

Solution

PCl_5 is a good chlorinating agent. The pentachloride of phosphous is not very stable because axial bonds are longer than the equatorial bonds. Hence PCl_5 decomposes to give PCl_3 and Cl_2.

$$PCl_5 \rightleftharpoons PCl_3 + Cl_2$$

The instability of PCl_5 makes it a good chlorinating agent.

Phosphorus oxychloride $POCl_3$ (Phosphoryl chloride),

It is prepared by the reaction of equimolar amount of water on PCl_5.

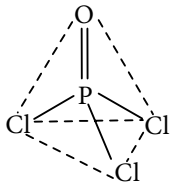
$$PCl_5 + H_2O \rightarrow POCl_3 + 2HCl$$

It is a colourless liquid and reacts with water to form phosphoric acid.

$$POCl_3 + 3H_2O \rightarrow H_3PO_4 + 3HCl$$

Structure

It has a tetrahedral structure as given below.



Oxides of phosphorus

Phosphorus forms several oxides, the most prominent of them being P_4O_6 and P_4O_{10}. Other oxides like PO_2, P_4O_7, P_4O_8, P_4O_9 are known but are not very common. The oxides P_4O_6 and P_4O_{10} will be discussed in detail.

Phosphorus trioxide, P_4O_6

It is a dimer of P_2O_3 and hence written as P_4O_6.

Preparation

(i) It is prepared by burning phosphorus in a limited quantity of air.

$$P_4 + 3O_2 \rightarrow P_4O_6$$

Properties

It is a colourless, crystalline solid with a garlic odour. It has a melting point of 23.8°C and boiling point of 178°C. It is poisonous and becomes waxy when impure. It dissolves in solvents like ether, benzene, chloroform, CS_2 and catches fire in contact with alcohol.

Reactions

(i) It burns in air to form P_4O_{10}.

$$P_4O_6 + 2O_2 \rightarrow P_4O_{10}$$

(ii) It hydrolyses on treatment with cold water forming phosphorous acid.

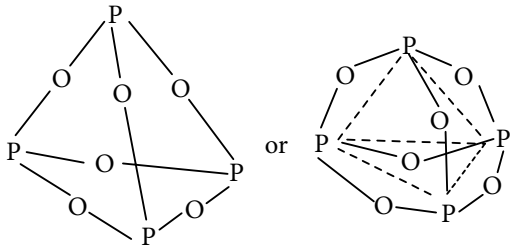
$$P_4O_6 + 6H_2O \rightarrow 4H_3PO_3$$

With hot water, phosphine and phosphoric acid are formed (disproportionation).

$$P_4O_6 + 6H_2O \rightarrow 3H_3PO_4 + PH_3$$

Structure of P_4O_6

Each P atom has five valence electrons of which three electrons form bonds with three oxygen atoms and the remaining two form a lone pair remaining outside a tetrahedral unit. So each phosphorous and each oxygen haven one and two lone pairs respectively. The structure may be represented as



Phosphorus pentoxide, P_4O_{10}

Phosphorus pentoxide is also dimeric.

Preparation

(i) It is made by burning phosphorus in excess of oxygen or air.

$$P_4 + 5O_2 \rightarrow P_4O_{10}$$

Properties

It is a white crystalline substance that sublimes at 360°C at 1 atm pressure. When heated under pressure it melts at 563°C and solidifies on cooling to a vitreous mass. It absorbs water from air and other substances and become sticky. It has strong affinity for water and hence used as a drying agent for gases.

Reactions

(i) Because of its strong dehydrating action, it can be used for preparing some acid anhydrides like acetic anhydride.

$$4CH_3COOH + P_4O_{10} \rightarrow 2(CH_3CO)_2O + 4HPO_3$$

i.e., $P_4O_{10} + 2H_2O \rightarrow 4HPO_3$ (metaphosphoric acid)

(ii) It is a powerful dehydrating or drying agent.

$$2CH_3CONH_2 + P_4O_{10} \rightarrow 2CH_3CN + 4HPO_3$$

$$2H_2SO_4 + P_4O_{10} \rightarrow 2SO_3 + 4HPO_3$$

$$4HNO_3 + P_4O_{10} \rightarrow 2N_2O_5 + 4HPO_3$$

(iii) It is an acidic oxide and it hydrolyses vigorously on treatment with water to give ortho phosphoric acid as the final product.

$$P_4O_{10} \xrightarrow{2H_2O} \underset{\text{metaphosphoric acid}}{4HPO_3} \xrightarrow{2H_2O}$$

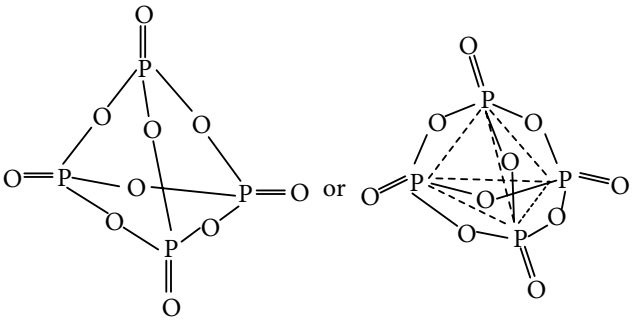
$$\underset{\text{Pyrophosphoric acid}}{2H_4P_2O_7} \xrightarrow{2H_2O} \underset{\text{orthophosphoric acid}}{4H_3PO_4}$$

(iv) It is reduced by carbon to red phosphorus.

$$P_4O_{10} + 10C \rightarrow P_4 + 10CO$$

Structure

In P_4O_{10}, the lone pairs on each of the 'P' atoms form a co-ordinate bond to an oxygen atom. The apical P–O bonds have appreciable double bond character through pπ-dπ bonding. The structure may be represented as



Oxyacids of phosphorus

Phosphorus forms many oxyacids all of which are based on a tetrahedral four co-ordinated phosphorus. These oxyacids contain at least one P = O, and one P – OH group and can be divided into two categories.

(i) Phosphorous series of acids in which P is in +3 oxidation state. These acids are good reducing agents.

(ii) Phosphoric series of acids in which P is in +5 oxidation state. These acids act as oxidising agents.

The structures of oxyacids of phosphorus can be easily arrived at by remembering that in all oxyacids phosphorus forms 5 covalent bonds.

The hydrogen atoms in P – OH groups are acidic and ionisable whereas the hydrogen atoms attached to phosphorus atom (as in phosphorous acid) are not ionisable and posses reducing properties. The various oxyacids are

(i) Hypophosphorous acid, H_3PO_2 (phosphiric acid)
(ii) Phosphorous acid, H_3PO_3 (phosphoric acid)
(iii) Hypophosphoric acid, $H_4P_2O_6$.
(iv) Phosphoric acid, H_3PO_4 (orthophosphoric acid)
(v) Pyrophosphoric acid, $H_4P_2O_7$
(vi) Metaphosphoric acid, HPO_3

A brief account of the various acids, salts of some of the acids and the structural aspects of acids are considered in the following pages.

Phosphorous acid, H_3PO_3

Introduction

This acid contains two acidic H atoms from two –OH groups and a H atom attached to P atom directly which is reducing in nature. The oxidation state of P in this molecule is +3. It is a dibasic acid, having successive dissociation constants of 1×10^{-2} and 2×10^{-7} respectively.

Preparation

(i) The pure acid is prepared by the action of water on PCl_3 or P_4O_6

$$PCl_3 + 3H_2O \rightarrow H_3PO_3 + 3HCl$$

$$P_4O_6 + 6H_2O \rightarrow 4H_3PO_3$$

On evaporation of H_3PO_3 solution, a syrupy liquid is formed which finally crystallizes out.

It is a deliquescent, colourless solid with a melting point of 70.2°C.

Reactions

(i) In alkaline solution, the acid and its salts are strong reducing agents. Copper and silver salts are reduced by the acid to the corresponding metals.

$$H_3PO_3 + CuSO_4 + H_2O \rightarrow H_3PO_4 + H_2SO_4 + Cu$$

$$H_3PO_3 + 2AgNO_3 + H_2O \rightarrow H_3PO_4 + 2HNO_3 + 2Ag$$

(ii) $HgCl_2$ is reduced to Hg_2Cl_2 and in boiling condition black mercury is also obtained.

$$H_3PO_3 + 2HgCl_2 + H_2O \rightarrow H_3PO_4 + Hg_2Cl_2 + 2HCl$$

(iii) SO_2 is reduced to sulphur by H_3PO_3

$$SO_2 + 2H_3PO_3 \rightarrow 2H_3PO_4 + S$$

(iv) When heated, it disproportionates into orthophosphoric acid and phosphine.

$$4H_3PO_3 \xrightarrow{200^\circ C} 3H_3PO_4 + PH_3$$

Salts of phosphorous acid

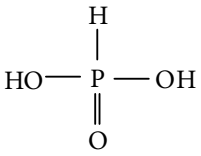
Two series of salts of this acid are known as:

(i) dihydrogen phosphites, e.g., NaH_2PO_3
(ii) monohydrogen phosphites, e.g., Na_2HPO_3

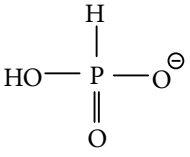
Phosphite salts (the salt of phosphorous acid) are strong reducing agents in basic solutions as they are readily oxidized to phosphates (PO_4^{3-} ions). For example, they can reduce $CuSO_4$ or $AgNO_3$ to the corresponding metals.

Structure of phosphorous acid

The structure of phosphorous acid may be given as



Structure of H_3PO_3



Structure of $H_2PO_2^-$

Hypophosphorous acid, H_3PO_2

It is a monobasic acid ($pK_a = 1.20$) as it contains only one replaceable hydrogen. The oxidation state of phosphorus in this molecule is +1.

Preparation

It is prepared by treatment of phosphorus with alkali. Phosphorus is boiled with baryta water, when insoluble barium hypophosphite is precipitated out.

$$2P_4 + 3Ba(OH)_2 + 6H_2O \rightarrow 3Ba(H_2PO_2)_2 + 2PH_3$$

The barium hypophosphite is next treated with calculated amount of sulphuric acid so that barium sulphate is precipitated and the acid is formed in solution.

$$Ba(H_2PO_2)_2 + H_2SO_4 \rightarrow 2H_3PO_2 + BaSO_4$$

The barium sulphate is filtered off and the solution is evaporated below 130°C to a syrupy liquid and can be crystallized to a white solid below 0°C. It forms colourless crystals with a melting point of 26.5°C.

The crystals are hygroscopic and readily soluble in water.

Reactions

Strong heating of the acid results in its disproportionation to phosphine and phosphoric acid.

$$2H_3PO_2 \rightarrow H_3PO_4 + PH_3$$

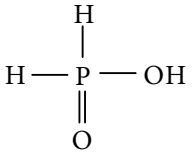
Hypophosphorous acid is a strong reducing agent. It reduces silver and gold salts to the corresponding metals.

$$H_3PO_2 + 4AgNO_3 + 2H_2O \rightarrow 4HNO_3 + H_3PO_4 + 4Ag$$

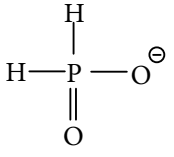
Salts of this acid are known as hypophosphites and sodium hypophosphite is used as a bleaching agent for wood and in paper manufacture.

Structure of hypophosphorous acid

It may be represented as



Structure of hypophosphorus acid



Structure of $H_2PO_2^-$ ion (tetrahedral)

Phosphoric acid, H_3PO_4

It is also known as orthophosphoric acid. This acid contains three replaceable hydrogen atoms and thus it is a tribasic acid.

The dissociation equlibria may be written as

$$H_3PO_4 \rightleftharpoons H_2PO_4^- + H^+ \quad K_1 = 7.5 \times 10^{-3}$$
$$H_2PO_4^- \rightleftharpoons HPO_4^{2-} + H^+ \quad K_2 = 6.2 \times 10^{-8}$$
$$HPO_4^{2-} \rightleftharpoons PO_4^{3-} + H^+ \quad K_3 = 1.0 \times 10^{-12}$$

Manufacture

(i) From phosphate rock

The phosphorite rock is treated with sulphuric acid. The calcium sulphate formed is hydrated to gypsum i.e.,

$CaSO_4.2H_2O$ and is filtered off. The fluoride ions present are removed as $Na_2[SiF_6]$. The aqueous acid, so obtained, is concentrated by evaporation.

$$Ca_3(PO_4)_2 + 3H_2SO_4 \rightarrow 3CaSO_4 + 2H_3PO_4$$

(ii) From bone ash

The bone ash is digested with dilute sulphuric acid. The filtrate from the calcium phosphate (main component of bone ash) containing phosphoric acid is concentrated by evaporation.

The reaction is

$$Ca_3(PO_4)_2 + 3H_2SO_4 + 6H_2O \rightarrow 2H_3PO_4 + 3CaSO_4.\ 2H_2O$$

Preparation

(i) Pure phosphoric acid is made by burning molten phosphorus in a furnace with air and steam. At first P_4O_{10} is formed by reaction between phosphorus and oxygen which is immediately hydrolysed to phosphoric acid

$$P_4 + 5O_2 \rightarrow P_4O_{10}$$
$$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$$

(ii) The acid is also obtained by action of conc.HNO_3 on phosphorus.

$$P + 5HNO_3 \rightarrow H_3PO_4 + H_2O + 5NO_2$$

Properties

In aqueous solution, H_3PO_4 is hydrogen bonded and this is responsible for the viscous and syrupy nature of the acid in concentrated solutions. It gives transparent colourless deliquescent crystals on cooling and the melting point of the crystals is 42°C. Phosphorous is in +5 oxidation state.

Reactions

(i) Phosphoric acid on heating loses water steadily

$$2H_3PO_4 \xrightarrow{220^0\,C} \underset{\text{pyrophosphoric acid}}{H_4P_2O_7} + H_2O$$

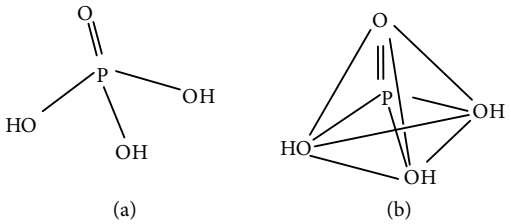
$$H_4P_2O_7 \xrightarrow{320^0\,C} \underset{\text{Metaphosphoric acid}}{2HPO_3} + H_2O$$

(ii) Phosphoric acid reacts with alcohols to form ester. For example,

$$O = P(OH)_3 + 3C_2H_5OH \rightarrow O = P(OC_2H_5)_3 + 3H_2O$$

Structure of phosphoric acid

It may be represented as



Phosphorus forms four sp^3 hybrid orbitals containing one electron each. They overlap with suitable orbitals of three oxygen atoms (of three –OH groups) and of the oxygen involved in double bond formation, to form σ bonds. The fifth electron in 3d orbitals overlaps with an unpaired electron in p orbital of oxygen atom to form dπ – pπ bond (thus explaining a dπ – pπ bond).

Salts of phosphoric acid

Being a tribasic acid, phosphoric acid gives three series of salts viz.,

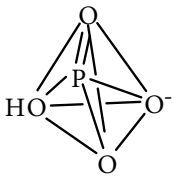
(i) dihydrogen phosphates such as sodium dihydrogen phosphate, NaH_2PO_4
(ii) monohydrogen phosphate such as Na_2HPO_4
(iii) normal phosphates such as Na_3PO_4

NaH_2PO_4 and Na_2HPO_4 salts are prepared by neutralization of phosphoric acid with Na_2CO_3. The NaH_2PO_4 is slightly acidic in solution due to the reaction

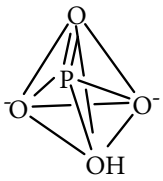
$$H_2PO_4^- + H_2O \rightarrow HPO_4^{2-} + H_3O^+$$

In the titration of a solution of phosphoric acid with NaOH, the first stage of ionization can be followed by titration with NaOH using methyl orange as indicator (pH = 4.7). The second stage of ionization can be followed by using thymolphthalein as indicator (pH = 9.6). The third stage cannot be followed by any indicator as the ionization is very small.

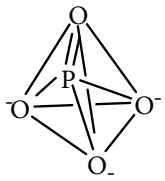
The structure of $H_2PO_4^-$, HPO_4^{2-} and PO_4^{3-} ions may be represented by



Structure of $H_2PO_4PO_4^-$ ion



Structure of HPO_4^{2-}



Structure of PO_4^{3-}

The acid strength of oxo acids of phosphorous follows the order $H_3PO_2 > H_3PO_3 > H_3PO_4$. It is attributed to the electron withdrawing effect of P = O bond. In H_3PO_2 the only P – OH group experience the effectfully. But in H_3PO_3 and H_3PO_4 the effect is distributed among more P – OH groups resulting in a progressive decrease in acidic nature.

Pyrophosphoric acid, $H_4P_2O_7$

Preparation

(i) By heating phosphoric acid for a sufficiently long time at 250– 260°C, pyrophosphoric acid is formed.

$$2H_3PO_4 \xrightarrow{200^\circ C} H_4P_2O_7 + H_2O$$

Properties

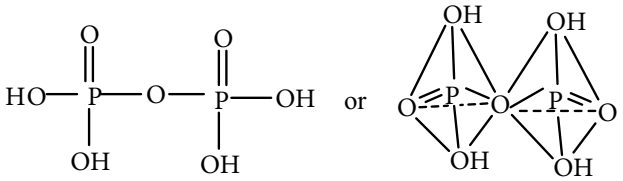
It is a syrupy liquid and crystallizes to a white crystalline solid with a melting point of 61°C Phosphorous is in +5 oxidation state. It is tetrabasic and its salts are known as pyrophosphates. Two sodium salts viz., $Na_4P_2O_7$ and $Na_2H_2P_2O_7$ have been isolated. Disodium dihydrogen pyrophosphate ($Na_2H_2P_2O_7$) mixed with $NaHCO_3$ is used as baking powder in bread making, to swell bread. The swelling occurs due to liberation of CO_2, when they are heated together.

$$Na_2H_2P_2O_7 + 2NaHCO_3 \rightarrow Na_4P_2O_7 + 2H_2O + 2CO_2$$

Calcium pyrophosphate $Ca_2P_2O_7$ is used in fluoride tooth paste as polishing agent.

Structure of pyrophosphoric acid

The structure may be represented as



Stucture of pyrophosphoric acid

Metaphosphoric acid, HPO_3

Preparation

(i) It may be prepared by heating orthophosphoric acid or pyrophosphoric acid at about 600K.

$$H_3PO_4 \rightarrow HPO_3 + H_2O$$
$$H_4P_2O_7 \rightarrow 2HPO_3 + H_2O$$

Properties

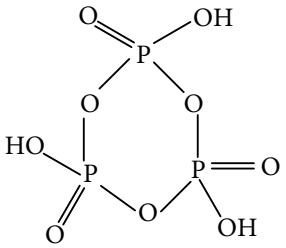
It is a colourless solid and form glassy crystals with phosphorus in +5 oxidation state. It exists as a polymer, $(HPO_3)_n$. They can be cyclic or linear. It reacts with water slowly to give phosphoric acid.

$$HPO_3 + H_2O \rightarrow H_3PO_4$$

The above reaction occurs more rapidly in boiling condition. The salts are known as metaphosphates. The salt, potassium metaphosphate is obtained on heating potassium dihydrogen phosphate.

$$KH_2PO_4 \rightarrow KPO_3 + H_2O$$

Neutral solutions of metaphosphates (like $NaPO_3$) gives gelatinous precipitate of $AgPO_3$ with silver nitrate. Salts of polymers of metaphosphoric acid, for example $(HPO_3)_4$, $(HPO_3)_6$ are known. The cyclic trimer acid has the structure.



Cyclotrimetaphosphoric acid

Sodium hexametaphosphate $(NaPO_3)_6$

It is prepared by dehydrating NaH_2PO_4 (or $Na_2H_2P_2O_7$) and fusing the mass by further heating. On rapid cooling of the fused mass, the salt $(NaPO_3)_6$ is obtained as a transparent mass. It is also called Graham's salt or calgon.

Properties

It is readily soluble in water. It forms soluble double salts containing calcium. Because of this property, insoluble calcium salts like calcium oxalate, calcium carbonate can be dissolved in excess sodium hexametaphosphate.

$$(NaPO_3)_6 \rightarrow Na_2[Na_4(PO_3)_6]$$

$$Na_2[Na_4(PO_3)_6] + 2CaCO_3 \rightarrow Na_2[Ca_2(PO_3)_6] + 2Na_2CO_3$$

Uses

It is useful in softening waters. It is also employed in removing crusts of calcium sulphate in boilers.

Hypophosphoric acid, $H_4P_2O_6$

Preparation

It is prepared by the hydrolysis of yellow phosphorus and oxidation of the intermediate product by air. The reaction is

$$P_4 + 4H_2O \rightarrow 2[H_4P_2O_2] \xrightarrow{4O_2} 2H_4P_2O_6$$

Properties

The solution of dihydrate intermediate formed in the above process loses water giving $H_4P_2O_6$ on evaporating in vacuum. It has a melting point 70°C. It forms colourless crystals soluble in water. It decomposes on heating into a mixture of phosphorus acid and metaphosphoric acid.

$$H_4P_2O_6 \rightarrow H_3PO_3 + HPO_3$$

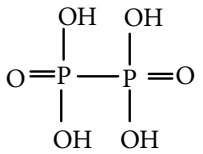
An aqueous solution of the acid, on boiling, undergoes slow hydrolysis to form phosphorous and phosphoric acids.

$$H_4P_2O_6 + H_2O \rightarrow H_3PO_3 + H_3PO_4$$

It is a tetrabasic acid having four replacable hydrogen atoms.

Structure of $H_4P_2O_6$

In this compound, there are no phosphorus–hydrogen bonds and hence is not a reducing agent with phosphorous in +4 oxidation state. It has a P – P bond. The structure of the acid may be represented as



Structure of $H_4P_2O_6$

Use of phosphates

A large part of phosphate production is used as fertilizer in industry in the form of superphosphate or ammonium phosphate. Sodium tripolyphosphate finds application in detergent industry. Trisodium phosphate is employed in water softening. Other phosphorus-based compounds include P_4S_3 which is used in the manufacture of matches. Organophosphorus compounds are employed as plasticisers and pesticides. Phosphorus acid is used for treatment of metals before electroplating.

Phosphine PH_3

Preparation

(i) Hydrolysis of metal phosphides like calcium or sodium phosphides gives phosphine. The reactions are

$$Ca_3P_2 + 6H_2O \rightarrow 3Ca(OH)_2 + 2PH_3\uparrow$$

$$Na_3P + 3H_2O \rightarrow 3NaOH + PH_3\uparrow$$

(ii) When white phosphorus is hydrolysed by heating with caustic potash, phosphine and potassium hypophosphite is formed

$$P_4 + 3KOH + 3H_2O \rightarrow 3KH_2PO_2 + PH_3$$

(iii) Pure phosphine can be prepared by warming phosphonium iodide with a solution of caustic potash. The reaction is

$$PH_4I + KOH \rightarrow KI + H_2O + PH_3\uparrow$$

CONCEPT STRAND

Concept Strand 5

Explain the properties of CaC_2 and Ca_3P_2 because of which they are used as Holmes's signal.

Solution

CaC_2 and Ca_3P_2 contained in a sealed tin can when dropped in water undergo hydrolysis.

$$CaC_2 + 2H_2O \rightarrow C_2H_2 + Ca(OH)_2$$

$$Ca_3P_2 + 6H_2O \rightarrow 2PH_3 + 3Ca(OH)_2$$

C_2H_2 and PH_3 catch fire. C_2H_2 produces a distinct brilliant flame visible over a long distance. PH_3 burns to give dense white fumes.

Thus Holme's signal helps in search and rescue at sea during day and night.

Properties

Phosphine is a colourless, highly toxic gas and has the smell of garlic or rotten fish. It is very reactive. Unlike NH_3, it is not very much soluble in water. It is more soluble in organic solvents like CS_2. It liquefies at –89°C and solidifies at –134°C. Phosphine is neutral to litmus.

Reactions

(i) The pure gas is stable in air but catches fire when heated above 100°C. But ordinary phospine containing diphosphine P_2H_4 as impurity spontaneously ignites to form white fumes of P_4O_{10}

$$4PH_3 + 8O_2 \rightarrow P_4O_{10} + 6H_2O$$

(ii) The gas decomposes into red phosphorus and hydrogen, when heated to 440°C.

$$2PH_3 \xrightarrow{440^0C} 2P + 3H_2$$

(iii) Phosphine is absorbed by $CuSO_4$ or $HgCl_2$ or $AgNO_3$ solutions, to give the corresponding phosphides.

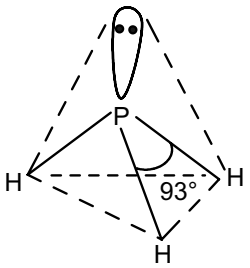
$$3CuSO_4 + 2PH_3 \rightarrow Cu_3P_2 + 3H_2SO_4$$

$$3HgCl_2 + 2PH_3 \rightarrow Hg_3P_2 + 6HCl$$

$$3AgNO_3 + PH_3 \rightarrow Ag_3P + 3HNO_3$$

Structure of Phosphine

Phosphine has a trigonal pyramidal structure with the central phosphorus atom in sp^3 hybridization. The H – P –H bond angle is 93°.

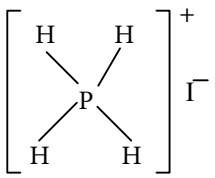


Salts of phosphine

Phosphonium salts such as $[PH_4^+]$ X^- where X is Cl, Br, I are formed when phosphine reacts with gaseous hydrogen halides. Phosphonium chloride and bromide, are however less stable compared to PH_4I.

$$PH_3 + HI \rightarrow PH_4I$$

Phosphonium ion, PH_4^+ is tetrahedral with the P – H distance of 1.414 A°. Phosphonium salts are similar to ammonium salts and the structure of PH_4I may be represented as



Structure of PH_4I

CONCEPT STRANDS

Concept Strand 6

Convert white phosphorus to red phosphorus through formation of phosphine.

Solution

Phosphine gas is evolved, when white phosphous is heated with alkali solution in inert atmosphere.

$$P_4 + 3NaOH + 3H_2O \xrightarrow[\text{heat}]{\text{coal gas}} 3NaH_2PO_2 + \underset{\text{phosphine}}{PH_3}$$

At 440°C, phosphine is decomposed into its elements i.e., red phosphous is formed.

$$4PH_3 \xrightarrow{440°C} \underset{\text{red phosphous}}{P_4} + 6H_{2(g)}$$

Concept Strand 7

Compare the covalent character of $P_2H_6^{2+}$, $P_2H_5^{+}$ and PH_4^{+}

Solution

The covalent character of the P–H bond depends on the formal charge distribution on each P–H bond. As the formal charge increases, the covalent character decreases.

$$\text{Formal charge distribution} = \frac{\text{total charge}}{\text{no. of P} - \text{H bonds}}$$

For each P–H bond in,

$$P_2H_6^{2+} = \frac{+2}{6} = 0.33$$

$$P_2H_5^{+} = \frac{+1}{5} = 0.2$$

$$PH_4^{+} = \frac{+1}{4} = 0.25$$

The formal charge distribution increases in the order $P_2H_5^{+} < PH_4^{+} < P_2H_6^{2+}$

Therefore, the covalent character decreases in the order $P_2H_5^{+} > PH_4^{+} > P_2H_6^{2+}$

OXYGEN (O)

Introduction

Oxygen (O) belongs to group 16 of the periodic table, the other members of the group being sulphur (S), selenium (Se), tellurium (Te) and polonium (Po). The first two elements of the group, oxygen and sulphur are typical non-metals whereas metallic properties begin to appear in selenium and tellurium, which are semimetals. Polonium is metallic and radioactive. Thus the metallic character increases with increasing atomic number. The first four members of the group together are referred as 'chalcogenides' or 'ore forming elements' because most ores occurs either as oxides or sulphides.

The general electronic configuration of the elements in this group is $ns^2\ np^4$ and hence they may attain a noble gas configuration by gaining two electrons to form doubly charged negative ions or by sharing two electrons to form two covalent bonds. Oxygen is an important element in that it reacts with almost all other elements and also it forms 21% by volume of the atmosphere. Some important atomic properties of oxygen are enumerated below:

Atomic number	Atomic weight	Electronic configuration	Atomic radius (A°)	1st ionization energy (kJ mol^{-1})	Electro–negativity (Pauling scale)	Oxidation states
8	16.0	[He] $2s^2 2p^4$	0.74	1314	3.5	–2, –1.

Oxygen also shows many differences from the rest of the members in its group. They may be ascribed to the following factors viz,.

(i) the small size of the oxygen atom
(ii) its high electronegativity, and
(iii) the absence of 'd' orbitals as evident from its electronic configuration

The important differences between oxygen and the other members are given below:

(i) Oxygen can use pπ orbitals form pπ-pπ double bonds. Although other members like selenium and tellurium can also form double bonds, they become weaker as the atomic number increases.
(ii) Oxygen forms strong H-bonds in compounds like water and H_2O_2 whereas other elements of the group do not form H-bonds (for example, H_2S).
(iii) Oxygen is a diatomic gas while others are solids.
(iv) The tendency for catenation decreases considerably as one goes down the group. However, sulphur shows greater tendency than oxygen to form chains and ring type structures. Considering the elements themselves, oxygen is diatomic while sulphur and selenium contain eight atoms in a puckered configuration.
(v) Oxygen shows a +2 oxidation state in very few compounds such as OF_2 but this is more common in other elements of the group.
(vi) Metallic oxides are ionic and contain O^{2-} ions due to large electronegativity differences between M and O whereas ions like S^{2-}, Se^{2-}, Te^{2-} are not common.
(vii) Oxygen shows a maximum covalency of two, due to the absence of d-orbitals whereas other elements in this group show a maximum covalency of six due to involvement of d-orbitals in bonding in their compounds.
(viii) Oxygen is paramagnetic in solid, liquid and gaseous states others are diamagnetic.

Occurrence/extraction of oxygen

(i) In an industrial scale, oxygen is produced by the fractional distillation of liquid air.
(ii) A commercial method of preparation of oxygen involves the electrolysis of water containing caustic soda as the electrolyte in an iron tank and employing nickel coated iron electrode.
(iii) On a laboratory scale, oxygen is prepared by the thermal decomposition of potassium chlorate using MnO_2 as catalyst.

$$2KClO_3 \xrightarrow[MnO_2 \text{ as catalyst}]{150^\circ C} 2KCl + 3O_2$$

(iv) It is also prepared by the decomposition of H_2O_2 with MnO_2 as catalyst.

$$2H_2O_2 \xrightarrow{MnO_2} 2H_2O + O_2$$

Properties

It is a colourless, odourless and tasteless gas. It occurs in three isotopic forms in nature O^{16}, O^{17}, and O^{18}. Liquid oxygen has a pale blue colour and boils at −182.5°C. It is appreciably soluble in water. It is transformed to a blue solid on further cooling (freezing point = −218.4°C). It is a supporter of combustion. Oxygen is paramagnetic in solid, liquid and gaseous states.

Uses of Oxygen

Oxygen is essential to life. It is the most important constituent in energy releasing reaction in plants and animals (known as respiration) given by

$$C_6H_{12}O_6 + 6O_2 \xrightarrow{\text{respiration}} 6CO_2 + 6H_2O + \text{energy}$$

In the human system, O_2 is transported to the various cells by the red pigment haemoglobin.

Oxygen is widely used in iron and steel industry especially in the manufacture of steel from pig iron. It is also used in the manufacture of H_2SO_4 and HNO_3. It finds application as a rocket fuel. It is also employed in the manufacture of ethylene oxide from ethylene.

Reactions

Oxygen forms compounds with all elements except some noble gases like He, Ne, Ar. It combines directly with all elements other than halogen and a few noble metals at ordinary or high temperature.

Oxides

Oxygen combines with metals and non-metals to form the corresponding oxides.

The oxides may be classified into several categories based on their acid-base properties as

(i) acidic oxides
(ii) basic oxides
(iii) amphoteric oxides
(iv) neutral oxides

Acidic oxides

When non-metals and some metals in high oxidation states combine with O_2, acidic oxides are formed. They dissolve in water to form oxy acids. Examples of this class of oxides include CO_2, SO_2, B_2O_3, SiO_2, N_2O_3, NO_2, N_2O_5, P_4O_{10}, SnO_2, V_2O_5, CrO_3, Mn_2O_7 etc.

For example,

$$CO_2 + H_2O \rightarrow H_2CO_3$$

The oxides are covalent and form acids on reaction with water.

In cases where an element exists in more than one oxidation state, the oxide in the higher oxidation state gives the most acidic solution. For example,

$$SO_2 + H_2O \rightarrow H_2SO_3$$

$$SO_3 + H_2O \rightarrow H_2SO_4$$

Acidic oxides react with bases to give salts.

$$SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$$

Basic oxides

Oxides formed by reaction of oxygen with metals are basic. Examples of this kind of oxides include Na_2O, CaO, MgO etc. Group I and group II metal oxides, which are more electropositive, are typical examples of this class. They are ionic and contain O^{2-} ions. These oxides react with water to give basic solutions called alkalies.

Some examples are

$$Na_2O + H_2O \rightarrow 2NaOH$$

$$MgO + H_2O \rightarrow Mg(OH)_2$$

Basic oxides react with acids to form salt and water. For example,

$$CaO + 2HCl \rightarrow CaCl_2 + H_2O$$

$$Na_2O + H_2SO_4 \rightarrow Na_2SO_4 + H_2O$$

When a metal exists in more than one oxidation state, the oxide in the lowest oxidation state is most basic and most ionic.

Amphoteric oxides

These are oxides of metals and they react with both acids and bases to from salts. Examples of this class are Al_2O_3, ZnO, BeO, Cr_2O_3, PbO, PbO_2 etc.

$$Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$$

$$Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$$

Neutral oxides

A few covalent oxides of non metals do not react with either acids or bases or water. Examples of such oxides are CO, NO, N_2O.

Oxides may also be classified on the basis of their structure as normal oxides, peroxides and suboxides.

Peroxides

These compounds contain more oxygen than that would be expected from the oxidation number of the metal with which they are combined. They may be ionic or covalent. Peroxides of group I and group II metals like Na_2O_2 and CaO_2 are ionic (contain O_2^{2-} ions) while others like per mono sulphuric acid (containing –O–O– group) are strong oxidizing agents and are hydrolysed by water to give H_2O_2. This is characteristic of a true peroxide. Some examples of such reactions are

$$Na_2O_2 + 2H_2O \rightarrow 2NaOH + H_2O_2$$

$$\underset{\substack{\text{peroxy mono}\\\text{sulphuric acid}}}{H_2SO_5} + H_2O \rightarrow H_2SO_4 + H_2O_2$$

In some cases, treatment of the peroxides with dilute acids liberate H_2O_2.

$$BaO_2 + H_2SO_4 \rightarrow BaSO_4 + H_2O_2$$

TiO_3 and CrO_5 are peroxides formed by transition metals.

Some oxides like PbO_2 and MnO_2 are not true peroxides, because they do not liberate H_2O_2 on treatment with dilute acid. For example,

$$2PbO_2 + 2H_2SO_4 \rightarrow 2PbSO_4 + 2H_2O + O_2$$

Superoxides

There is another class of oxide known as super oxides which contain even more oxygen (than even in case of peroxides), examples of such cases being LiO_2, NaO_2, KO_2 etc. They contain O_2^- ion, which contain an unpaired electron, and hence they are paramagnetic and coloured. They are even more stronger oxidizing agents than peroxides. They react with water or dilute acids to give H_2O_2 and O_2.

$$2KO_2 + 2H_2O \rightarrow 2KOH + H_2O_2 + O_2$$

Suboxides

Some oxides contain less oxygen than would be expected from the oxidation number of the metals. For example carbon suboxide, C_3O_2 (O = C = C= C = O), lead suboxide (Pb_2O), nitrous oxide (N_2O).

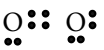
Mixed oxides

Some oxides are referred as mixed oxides which are a combination of two simple oxides. Some examples of such ox-

ides are Pb_3O_4 ($2PbO.PbO_2$), Fe_3O_4 ($FeO.Fe_2O_3$), Mn_3O_4 ($2MnO.MnO_2$).

Structure of dioxygen (O_2)

The electronic formula of oxygen on the basis of octet rule is



structure of oxygen
(using octet rule)

This does not explain the paramagnetic behaviour of oxygen, which suggests the presence of unpaired electrons.

The paramagnetic behaviour of oxygen is best explained on the basis of M.O. theory according to which the distribution of electrons in the various molecular orbitals is given as

$$\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \sigma 2p_z^2, \begin{cases} \pi 2p_x^2 \\ \pi 2p_y^2 \end{cases} \begin{cases} \pi^* 2p_x^1 \\ \pi^* 2p_y^1 \end{cases}$$

The antibonding orbitals $\pi^* 2p_x$ and $\pi^* 2p_y$ are occupied singly (unpaired with parallels spins) giving rise to paramagnetism.

CONCEPT STRAND

Concept Strand 8

A metal (M) forms an oxide M_2O_2. Give two reactions, which would help to identify whether it is a dioxide or a peroxide.

Solution

A true peroxide and a dioxide may be distinguished by the following tests.

(i) Action of dilute acid
If M_2O_2 is a peroxide, H_2O_2 should be evolved.

$$M_2O_2 + H_2SO_4 \rightarrow H_2O_2 + M_2SO_4$$

If it were a dioxide, no reaction would occur.

(ii) Action of concentrated H_2SO_4
A true peroxide does not react while a dioxide gives out oxygen gas.

$$2M_2O_2 + 2HSO_4 \rightarrow 2M_2SO_4 + 2H_2O + O_2$$

Ozone, O_3.

Ozone is the triatomic allotropic form of oxygen.

Preparation

It is commonly prepared by passing a silent electric discharge through pure dry oxygen in an apparatus, known as ozoniser. The Siemen's ozoniser, is an apparatus, to make ozone, As electric the discharge is applied via suitable electrical connection, the oxygen passing through the apparatus is converted to ozone

$$3O_2 \rightarrow 2O_3\ \Delta H = 143\ kJ\ mol^{-1}$$

Conversion of O_2 to O_3 is endothermic.

By this method, it is possible to get concentration of ozone upto 10%. Pure ozone can be obtained by fractional liquefaction of $O_2 - O_3$ mixtures. Pure ozone can also be prepared by the UV irradiation of oxygen.

Properties

Ozone is a light blue, unstable gas. It has the smell of rotten fish. It is a diamagnetic gas. Liquid ozone has a boiling point of −112°C and the pure liquid is highly explosive in nature. At −249.7°C, it forms violet black crystals. Ozone gas absorbs strongly in the UV region around 255 A°. Its colour is due to absorption in the red region (5570 to 6020 A°). It is heavier than air and is slightly soluble in water but highly soluble in turpentine oil, glacial acetic acid or CCl_4.

Reactions

Ozone is thermodynamically unstable with its decomposition to oxygen being associated with a decrease in Gibbs free energy.

$$2O_3 \rightarrow 3O_2 \qquad \Delta G = -163\ kJ\ mol^{-1}$$

The decomposition is also generally slow but is catalyzed by finely divided metals like Ag, Pt and Pd.

Ozone acts as a powerful oxidising agent because of its capacity to form atomic oxygen. Its oxidizing power is very nearly that of F_2.

$$O_3 \rightarrow O_2 + O$$

Some oxidation reactions of ozone are enumerated below

(i) It oxidizes sulphides of lead, copper, zinc or cadmium to corresponding sulphates.

$PbS + 4O_3 \rightarrow PbSO_4 + 4O_2$

$CuS + 4O_3 \rightarrow CuSO_4 + 4O_2$

(ii) It oxidizes NO_2 to N_2O_5

$2NO_2 + O_3 \rightarrow N_2O_5 + O_2$

(iii) Sulphur is oxidized to H_2SO_4. (All three oxygens of O_3 are used up)

$S + H_2O + O_3 \rightarrow H_2SO_4$

(iv) Hydrogen halides are oxidized to the corresponding halogen.

$2HCl + O_3 \rightarrow H_2O + Cl_2 + O_2$

(v) Potassium ferrocyanide solution is oxidized to ferricyanide according to the reaction

$2K_4[Fe(CN)_6] + H_2O + O_3 \rightarrow$

$2K_3[Fe(CN)_6] + 2KOH + O_2$

(vi) It reacts with KI to liberate iodine

$2KI + H_2O + O_3 \rightarrow 2KOH + O_2 + I_2$

This reaction can be used to estimate the amount of ozone in a gas mixture because the liberated iodine can be titrated against standard $Na_2S_2O_3$ solution under buffered conditions at pH of 9.0

(vii) SO_2 is oxidized to SO_3 (All three oxygens of O_3 are used up)

$3SO_2 + O_3 \rightarrow 3SO_3$

(viii) Stannous chloride is oxidized to stannic chloride in acid medium (All 3 oxygens are used)

$3SnCl_2 + 6HCl + O_3 \rightarrow 3SnCl_4 + 3H_2O$

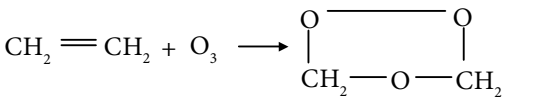
(ix) Ozone also acts as a reducing agent. It reduces H_2O_2 to water.

$H_2O_2 + O_3 \rightarrow H_2O + 2O_2\uparrow$

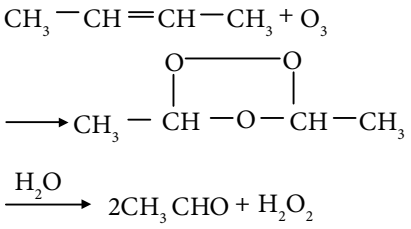
BaO_2 is reduced to BaO and oxygen is liberated.

$BaO_2 + O_3 \rightarrow BaO + 2O_2\uparrow$

(x) Ozone finds important application in organic chemistry as it adds on to double bonds forming compounds called ozonides. Ozonides on hydrolysis decompose into aldehydes and ketones.

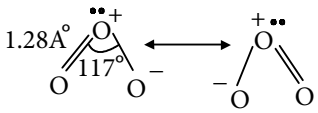


Ethylene ozonide



Structure of ozone

The structure of ozone is considered to be a resonance hybrid of the following contributing forms



The O – O bonds have considerable double bond character. The O – O bond distance is 1.28 A° which is in between that of a single bond (1.48 A° as in H_2O_2) and a double bond (1.21 A° as in O_2). The molecule is V-shaped with a bond angle of 116.8°.

Uses

1. Used as a germicide, disinfectant and sterilizer
2. For bleaching oils, ivory, starch etc.
3. As an oxidizing agent in the manufacture of $KMnO_4$

Hydrogen peroxide, H_2O_2

Preparation

(i) Sodium peroxide is added to an ice cold solution of 20% sulphuric acid.

$Na_2O_2 + H_2SO_4 \rightarrow Na_2SO_4 + H_2O_2$

The sodium sulphate thus formed may be removed by cooling as $Na_2SO_4.10H_2O$ thereby reducing the water content. H_2O_2 is formed as a 3% solution.

(ii) $BaO_2.8H_2O$ is decomposed by dilute sulphuric acid.

$BaO_2.8H_2O + H_2SO_4 \rightarrow BaSO_4 + 8H_2O + H_2O_2$

Anhydrous BaO_2 is not used for this purpose as a layer of $BaSO_4$ formed during the reaction hinders the formation of H_2O_2. In this reaction a higher concentration of H_2O_2 viz., 5% is obtained but the sample does not have good keeping properties due to the presence of impurities like bariumpersulphite. For this reason, phosphoric acid is used as an alternative to

H_2SO_4. This method has the advantage that any heavy metal impurities present are removed as insoluble phosphates.

$$3BaO_2 + 2H_3PO_4 \rightarrow Ba_3(PO_4)_2 + H_2O_2$$

(iii) Electrolytic method: The methods involves the electrolysis of 50% H_2SO_4 or $(NH_4)_2SO_4$–H_2SO_4 solution at a high current density of 1 amp dm^{-2} using Pt electrodes.

$$H_2SO_4 \rightleftharpoons H^+ + HSO_4^-$$

$$2HSO_4^- \rightleftharpoons H_2S_2O_8 + 2e^-$$

$H_2S_2O_8$ formed is withdrawn from near the anode and subjected to vaccum distillation. H_2O_2 is formed by hydrolysis.

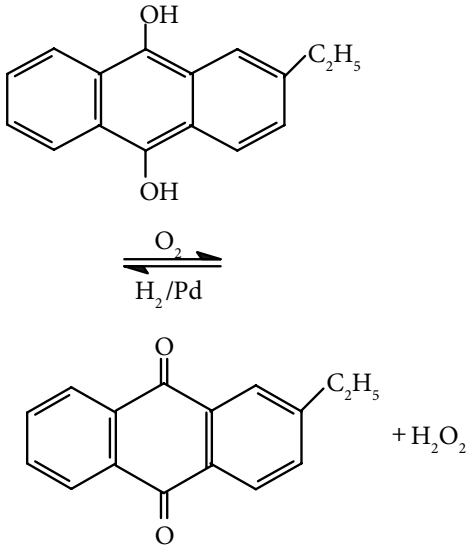
$$H_2S_2O_8 + H_2O \rightarrow H_2SO_5 + H_2SO_4$$

$$H_2SO_5 + H_2O \rightarrow H_2SO_4 + H_2O_2$$

The H_2O_2 formed is rapidly removed by distillation at high temperature and low pressure. Dilute solution of H_2O_2 obtained in this manner are further concentrated by vacuum distillation to about 32 wt% H_2O_2.

(iv) Manufacture of H_2O_2 on a large scale

2-Ethyl anthraquinol is dissolved in a mixture of organic solvents (an ester and a hydrocarbon) immiscible with water and air is bubbled through it which results in its oxidation to quinone and formation of H_2O_2. The anthraquinone is reduced back to anthraquinol with H_2 in presence of raney nickel or platinum or palladium at a moderate temperature.



H_2O_2 is extracted from the resulting solution with water and concentration of about 20% H_2O_2 is obtained in this manner.

(v) Expressing strength of H_2O_2 solution

The strength of H_2O_2 solution is expressed in terms of volume strength. For example, let x be the volume strength of a certain sample of H_2O_2. This implies that one mL of H_2O_2 gives x mL of O_2 at N.T.P.

$$2H_2O_2 \rightarrow 2H_2O + O_2$$

Properties of H_2O_2

Pure H_2O_2 is pale blue, syrupy liquid. It has a freezing point of –0.89°C and a boiling point of 152.1°C. It resembles water in many of its physical properties. It is even more H – bonded than water. It is denser than water with a density of 1.44 g cm^{-3} at 25°C. It is miscible with water in all proportions.

It is fairly stable in absence of sunlight in vessels of smooth surface. However, it decomposes readily in presence of traces of heavy metal ions, MnO_2, colloidal platinum rough glass surface etc into water and oxygen.

$$2H_2O_2 \rightleftharpoons 2H_2O + O_2$$

H_2O_2 is not stored in glass bottles, but in paraffin-wax water bottles to avoid its decomposition by alkali oxides present in the glass.

Reactions of H_2O_2

Hydrogen peroxide can act both as an oxidizing as well as a reducing agent. Its strong oxidizing action in acid and alkaline solutions is based on the equation.

$H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O$ (acid medium) $E° = 1.77V$

$HOO^- + H_2O + 2e \rightarrow 3OH^-$ (alkaline medium) $E° = 0.87V$

Its reducing action is based on the equation.

$H_2O_2 \rightarrow O_2 + 2H^+ + 2e$ $\quad E° = -0.68V$

(i) H_2O_2 as an oxidizing agent

(a) Acidiifed ferrous ions are oxidized to ferric ions according to the reaction.

$$2Fe^{2+} + H_2O_2 + 2H^+ \rightarrow 2Fe^{3+} + 2H_2O$$

(b) Iodide ions are oxidized to iodine in acidic solutions.

$$2I^- + H_2O_2 + 2H^+ \rightarrow I_2 + 2H_2O$$

(c) Arsenious acid is oxidized to arsenic acid.

$$H_3AsO_3 + H_2O_2 \rightarrow H_3AsO_4 + H_2O$$

(d) Sulphurous acid is oxidized to H_2SO_4.

$$H_2SO_3 + H_2O_2 \rightarrow H_2SO_4 + H_2O$$

(e) Lead sulphide is oxidized to lead sulphate.

$PbS + 4H_2O_2 \rightarrow PbSO_4 + 4H_2O$

(f) Ferrocyanide ion is oxidized to ferricyanide ion in acid solution

$2\left[Fe(CN)_6\right]^{4-} + H_2O_2 + 2H^+ \rightarrow$

$2[Fe(CN)_6]^{3-} + 2H_2O$

(g) Chromic acid in acid solution is oxidized to blue perchromic acid (CrO_5) which quickly decomposes to give chromic sulphate and oxygen.

$H_2Cr_2O_7 + 4H_2O \rightarrow 2CrO_5 + 5H_2O$

$2CrO_5 + 3H_2SO_4 \rightarrow Cr_2(SO_4)_3 + 3H_2O + \frac{7}{2}O_2$

(ii) H_2O_2 as a reducing agent

(a) Ferricyanide ions is reduced to ferrocyanide ion in alkaline medium

$2\left[Fe(CN)_6\right]^{3-} + H_2O_2 + 2OH^- \rightarrow 2[Fe(CN)_6]^{4-} + 2H_2O + O_2$

(b) Permanganate ion is reduced to Mn^{2+} state in acid medium

$2MnO_4^- + 5\,H_2O_2 + 6H^+ \rightarrow 2Mn^{2+} + 8H_2O + 5O_2$

In alkaline medium, MnO_2 is precipitated

$2MnO_4^- + H_2O_2 \rightarrow 2MnO_2 + 2OH^- + 2O_2$

(c) Chlorine is reduced to hydrochloric acid

$Cl_2 + H_2O_2 \rightarrow 2HCl + O_2$

(d) Ozone is reduced to oxygen

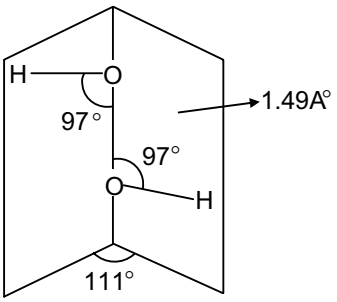
$H_2O_2 + O_3 \rightarrow H_2O + 2O_2\uparrow$

(e) Moist Ag_2O is reduced to metallic silver

$Ag_2O + H_2O_2 \rightarrow 2Ag + H_2O + O_2$

Structure of H_2O_2

H_2O_2 has a "skewed" structure with oxygen in sp^3 hybridization. It shows restricted rotation about the –O–O– bond possibly due to the repulsion between the OH groups and lone pairs. The O – O bond length is 1.49 A° and the O – H bond length is 0.97 A°. The H – O – O bond angle is 97°.



Structure of H_2O_2 (gas phase)

CONCEPT STRAND

Concept Strand 9

Calculate the amount of iodine that can be liberated from acidified KI solution by 10 mL of 40 volume H_2O_2 at STP.

Solution

The reaction of H_2O_2 with KI in acid will be

$H_2O_2 + 2H^+ + 2I^- \rightarrow I_2 + 2H_2O$

1 mole of H_2O_2 will liberate 1 mole of I_2.

Number of moles of H_2O_2

$2H_2O_2 \rightarrow 2H_2O + O_2$

2 moles of $H_2O_2 \rightarrow$ 22.4 L. of O_2 at STP.

$\therefore$ i.e., 40 volume (40 L) of $O_2 \rightarrow \frac{40}{22.4} \times 2 = 3.57$ moles

1 L contains 3.57 moles

10 mL i.e., 10×10^{-3} L contains $3.57 \times 10 \times 10^{-3} = 0.0357$ moles.

1 mole of H_2O_2 will liberate 1 mole of I_2

$\therefore$ 0.036 moles of H_2O_2 = 0.0357 moles of I_2

$= 0.0357 \times 254 = 9.07$ g.

or

$H_2O_2 \rightarrow H_2O + [O]$

$2KI + H_2SO_4 + [O] \rightarrow K_2SO_4 + H_2O + I_2$

1 mole of H_2O_2 = 0.5 mole of O_2 = 11.2 L of O_2 = 1 mole of I_2 = 254 g of I_2.

$\therefore$ Amount of iodine liberated $= \frac{254}{11200} \times 40 \times 10$

$= 9.07$ g.

SULPHUR (S)

Introduction

Sulphur (S) belongs to group 16 of the periodic table, the other members of the group being oxygen (O), selenium (Se), tellurium (Te) and polonium (Po). Sulphur is non-metallic and the non-metallic character decreases down the group. Physical properties like density, melting point, boiling point increase with increasing atomic number. Sulphur exists as a polymeric species forming S_8 molecules in puckered ring structure as shown later. Sulphur shows a greater tendency to form chains and ring type structures than other members of the group. Some examples are polysulphides, S_n^{2-}, sulphanes H–S_n–H and polysulphuric acids. The S–S bond is important in biological systems and in some proteins and enzymes. It forms an extensive range of compounds with nitrogen, not observed with other elements of the group. Sulphur forms many allotropes compared to other elements such as rhombic and monoclinic forms, which will be discussed later. Some important atomic properties of sulphur are given below:

Atomic number	Atomic weight	Electronic configuration	Atomic radius (A°)	Ionization energy (1st) (kJ mol^{-1})	Electro–negativity (Pauling scale)	Oxidation states
16	32.06	$[Ne]3s^2 3p^4$	1.03	1000	2.5	-2, +2, +4, +6 (+2 not common)

Extraction of sulphur

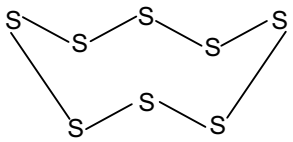
Sulphur is distributed in nature both in the free and combined state. Large deposits of sulphur are found in USA. Sulphur from these deposits is mined by a process known an Frasch process. It also occurs in the form of sulphides, in minerals like copper pyrites ($CuFeS_2$), iron pyrites (FeS_2), zinc blende (ZnS), cinnabar (HgS), and galena (or lead glance PbS). Another source of sulphur is in the form of sulphates in minerals such as gypsum ($CaSO_4.2H_2O$) and anhydrite ($CaSO_4$), Epsom salt ($MgSO_4.7H_2O$).

Frasch Process

In this method, three concentric pipes are sunk through a bore-hole until they meet the deposit of sulphur. Super heated steam at about 160°C is next passed down through the outer most pipe which results in the melting of sulphur. The next step consists in sending compressed air through the inner most pipe, which forces the molten sulphur up through the middle pipe. Sulphur in a high state of purity is obtained by this method.

Structure of sulphur

Sulphur exists as S_8 molecules at room temperature, and have a puckered ring structure as shown with a S–S bond distance of 2.12 A° and the S–S–S bond angle is 107°.



Structure of S_8 molecule

Effect of heat on sulphur

On heating, sulphur (S_8) melts to give a yellow transparent mobile liquid. As the temperature is raised, it becomes brown and at 160°C, the S_8 rings break and the diradicals formed polymerize to give long chains of up to a million atoms. The viscosity increases sharply as the temperature is raised beyond 160°C. The viscosity reaches a maximum at about 200°C and subsequently falls. At its boiling point of 444°C, the liquid sulphur again becomes a mobile dark red liquid. At boiling point vapour consists of S_8 rings with 1 – 2% S_2. At 600°C gas nearly consisits of S_2 molecules which are paramagnetic like O_2 and blue in colour.

Properties

Sulphur is a yellow solid at room temperature and exists in several allotropic modifications, which will be discussed subsequently. Sulphur is inflammable, burns in oxygen or air with a blue flame giving SO_2 and some amount of SO_3 as well.

$$S + O_2 \rightarrow SO_2$$

$$2SO_2 + O_2 \rightarrow 2SO_3$$

On heating sulphur with an oxidizing agent like potassium chlorate, an explosion occurs. Sulphur combines with most elements except gold and iridium at room temperature or when heated.

(i) Boiling sulphur acts on steam to form H_2S and SO_2.

$$3S + 2H_2O \rightarrow 2H_2S + SO_2$$

(ii) It combines with carbon to give CS_2 at red heat.

$$C + 2S \rightarrow CS_2$$

(iii) It combines with phosphorus and arsenic to give the corresponding suphides.

$$P_4 + 10S \rightarrow 2P_2S_5$$

$$2As + 3S \rightarrow As_2S_3$$

(iv) When hydrogen is passed into boiling sulphur, H_2S is formed. A similar reaction occurs with chlorine to give sulphur monochloride.

$$H_2 + S \rightarrow H_2S$$

$$2S + Cl_2 \rightarrow S_2Cl_2$$

(v) Sulphur is oxidized to SO_2 by conc.H_2SO_4.

$$S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$$

(vi) It also reacts with conc.HNO_3 to form sulphuric acid and liberates NO_2.

$$S + 6HNO_3 \rightarrow H_2SO_4 + 6NO_2 + 2H_2O$$

(vii) Potassium hydroxide reacts with sulphur to give potassium sulphide and potassium thiosulphate.

$$4S + 6KOH \rightarrow 2K_2S + K_2S_2O_3 + 3H_2O$$

Uses

Sulphur finds a variety of uses such as

(i) in the manufacture of H_2SO_4, CS_2, SO_2.
(ii) in the preparation of bisulphites.
(iii) in vulcanization of rubber.
(iv) in the preparation of matches, fireworks and gun powder.
(v) in agriculture as a fungicide.
(vi) in medicine for preparation of ointments.

Allotropes of sulphur

Preparation of various forms

(i) Rhombic sulphur (or octahedral sulphur or α-sulphur), is thermodynamically the most stable form at room temperature. This form occurs naturally as large yellow crystals in volcanic areas, with a melting point of 114.5°C. It is soluble in CS_2. Its crystal structure contains cyclic S_8 rings packed in a manner which leads to the formation of rhombic crystals. It is prepared by evaporating a solution of sulphur in carbondisulphide.

(ii) Monoclinic sulphur (or prismatic sulphur or β-sulphur), is stable above 95.5°C. It is prepared by melting rhombic sulphur in a dish and allowing it to cool until a crust is formed at the surface. By piercing the crust, the liquid, which has not solidified is poured out. When the crust is removed, needle-shaped yellow crystals of monoclinic sulphur become visible. Actually, the transition from α-sulphur to β-sulphur takes place at 95.5°C with the enthalpy of transition being about 0.4 kilo joule per gram atom at this temperature. This transition is slow and hence by rapid heating one can avoid it and reach the melting point of rhombic sulphur (or α-sulphur) i.e., 112.8°C. Monoclinic sulphur contains S_8 rings.

(iii) A third form of sulphur, known as γ-monoclinic sulphur can be prepared by chilling hot and concentrated solution of sulphur in solvents like carbon disulphide or ethanol. Like α-form and β-form, this form also has a puckered ring structure of S_8 ring. γ-sulphur slowly changes into α-and β-forms of sulphur.

(iv) A fourth form known as plastic sulphur (also referred as χ –sulphur) is prepared by pouring molten sulphur into ice water. It can be obtained in the form of long fibers by heating monoclinic sulphur in nitrogen at 300°C for five minutes and then quenching the same in a thin stream of ice water.

Properties

Rhombic sulphur

It forms transparent lemon yellow crystals, when sulphur dissolved in CS_2, is evaporated. It has a density of 2.07 g cm^{-3}. It is insoluble in water but slightly soluble in alcohol. It is fairly soluble in CS_2 and hot benzene. As mentioned earlier, it consists of S_8 rings with the S – S distance of 2.12 A°.

Monoclinic sulphur

This form melts at about 128°C and has a density of 1.97 g cm^{-3}. It is insoluble in water but soluble in CS_2. It occurs as S_8 rings both in liquid and vapour states.

γ–sulphur (γ–monoclinic sulphur)

It exists in the form of pale yellow needles and has a density of 2.19 g cm^{-3}. γ-sulphur changes into α-and β-forms.

Sulphur vapour

Sulphur vapour consists mostly of $S_{8\uparrow}$ rings in the temperature of 200°C but above 600°C, gaseous sulphur consists of

S_2 molecules. They are intense blue coloured and paramagnetic (similar to O_2).

Hydrogen sulphide, H_2S

Preparation

It is prepared in the laboratory in Kipp's apparatus by the action of dil.HCl or dil.H_2SO_4 on ferrous sulphide.

$$FeS + H_2SO_4 \rightarrow FeSO_4 + H_2S$$

A pure sample of H_2S may be obtained by the action of hot conc.HCl on antimony sulphide.

$$Sb_2S_3 + 6HCl \rightarrow 2SbCl_3 + 3H_2S\uparrow$$

A highly pure sample of H_2S can also be prepared by passing a mixture of hydrogen and sulphur over finely divided nickel at 450°C.

$$8H_2 + S_8 \xrightarrow[450^\circ C]{Ni} 8H_2S$$

Properties

It is a foul smelling (like that of rotten eggs) toxic gas. It is highly poisonous. Liquid H_2S boils at –60°C. The solid form has a melting point of –115°C. H_2S gas is soluble in water to the extent of 0.1 M at 1 atm at room temperature. A saturated solution of H_2S in water is used as reagent in qualitative analysis. However, the solution is not very stable and aerial oxidation slowly deposits sulphur from the solution. H_2S behaves like a weak dibasic acid in solution. The dissociation equilibria of H_2S in aqueous solution may be represented as

$$H_2S + H_2O \rightleftharpoons H_3O^+ + HS^- \quad K_1 = 1 \times 10^{-7}$$

$$HS^- + H_2O \rightleftharpoons H_3O^+ + S^{2-} \quad K_2 = 1.2 \times 10^{-15}$$

Since it is a dibasic acid, two series of salts are formed by H_2S, viz., the hydrosulphides and the normal sulphides.

$$H_2S + NaOH \rightarrow NaHS + H_2O$$

$$H_2S + 2NaOH \rightarrow Na_2S + H_2O$$

The alkali metal sulphides are extensively hydrolyzed in aqueous medium

$$Na_2S + 2H_2O \rightarrow 2NaOH + H_2S$$

As a result of the above reaction, the solution becomes alkaline.

Reactions of H_2S

(i) H_2S burns in air with a blue flame forming SO_2.

$$2H_2S + 3O_2 \rightarrow 2H_2O + 2SO_2$$

In restricted supply of oxygen, sulphur is formed due to incomplete combustion.

$$2H_2S + O_2 \rightarrow 2H_2O + 2S$$

(ii) Hydrogen sulphide is a good reducing agent.

(a) It reduces halogen to the corresponding hydrogen halides

$$H_2S + Cl_2 \rightarrow 2HCl + S$$

Similar reactions occur with bromine and iodine.

(b) It reduces acidified $KMnO_4$ to manganous salt.

$$2KMnO_4 + 3H_2SO_4 + 5H_2S \rightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 5S$$

(c) H_2S reduces acidified potassium dichromate to green chromic salt.

$$K_2Cr_2O_7 + 4H_2SO_4 + 3H_2S \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + 3S$$

(d) It reduces ferric ions to ferrous ions.

$$2FeCl_3 + H_2S \rightarrow 2FeCl_2 + 2HCl + S$$

(e) Nitric acid is reduced to NO_2.

$$2HNO_3 + H_2S \rightarrow 2NO_2 + 2H_2O + S$$

(f) SO_2 is reduced to sulphur.

$$SO_2 + 2H_2S \rightarrow 2H_2O + 3S$$

(iii) When H_2S is passed into a solution of $AgNO_3$, black Ag_2S is precipitated, which is soluble in hot dil.HNO_3.

$$2AgNO_3 + H_2S \rightarrow Ag_2S\downarrow + 2HNO_3$$

(iv) Reactions with Pb^{2+}

When H_2S is passed into a solution of lead acetate, black lead sulphide is precipitated. $(CH_3COO)_2Pb + H_2S \rightarrow PbS + 2CH_3COOH$

This is a test for H_2S.

Test for H_2S

H_2S may be identified by the blackening of a filter paper dipped in lead acetate solution on exposure to the gas. H_2S turns a solution of sodium nitroprusside in an alkaline medium to violet and the reaction is

$$Na_2S + Na_2[Fe(NO)(CN)_5] \rightarrow Na_4[Fe(NOS)(CN)_5]$$

Other hydrides

Straight chain polysulphanes of the type H_2S_2, H_2S_3, and H_2S_4 have been prepared. The structure of these hydrides are

H_2S_2	H – S – S –H	disulphane
H_2S_3	H – S – S – S –H	trisulphane
H_2S_4	H – S – S – S – S –H	tetrasulphane

H_2S_2 has a structure similar to that of H_2O_2. It is prepared by the addition of dil.H_2SO_4 to sodium persuphide

$$Na_2S_2 + H_2SO_4 \rightarrow Na_2SO_4 + H_2S_2$$

H_2S_2 is not very stable and decomposes to H_2S. The decomposition is catalysed by OH^- ions.

$$H_2S_2 \rightarrow H_2S + S$$

Oxides of sulphur

Although a number of oxides of sulphur like sulphur suboxide (S_2O), sulphur monoxide (SO), sulphur sesquioxide (S_2O_3), sulphur dioxide (SO_2), sulphur trioxide (SO_3), sulphur heptoxide (S_2O_7) and sulphur tetroxide (SO_4) are known, the most common and important oxides are sulphur dioxide, SO_2 and sulphur trioxide, SO_3. These oxides will be discussed in detail below.

Sulphur dioxide, SO_2

Preparation

(i) The simplest method of preparing SO_2 is by burning sulphur in air. The gas is also obtained as a byproduct during roasting of pyrites.

$$S + O_2 \rightarrow SO_2$$

(ii) The laboratory method of preparation of SO_2 is based on heating copper turnings or sulphur with conc. H_2SO_4 in conical flask .

$$Cu + 2H_2SO_4 \rightarrow CuSO_4 + 2H_2O + SO_2\uparrow$$

$$S + 2H_2SO_4 \rightarrow 2SO_2 + 2H_2O$$

(iii) A convenient laboratory preparation is by the action of dil.HCl in sodium sulphite or sodium bisulphate.

$$Na_2SO_3 + 2HCl \rightarrow 2NaCl + H_2O + SO_2$$

$$NaHSO_3 + HCl \rightarrow NaCl + H_2O + SO_2$$

Properties

Sulphur dioxide is a colourless gas with a suffocating smell. It can be condensed to a liquid which has a boiling point of −10.1°C. It is highly soluble in water. In concentrations greater than 5 ppm, it is hazardous to human beings and plants. SO_2 acts as a good bleaching agent as it forms nascent hydrogen on reaction with water while it is oxidized to H_2SO_4.

$$SO_2 + 2H_2O \rightarrow H_2SO_4 + 2H$$

Reactions of SO_2

(i) As mentioned above, SO_2 is highly soluble in water and forms H_2SO_3.

$$SO_2 + H_2O \rightleftharpoons H_2SO_3 \rightleftharpoons SO_3^{2-} + 2H^+$$

The solutions turns blue litmus red.

(ii) Potassium metal burns in SO_2 to form the sulphite and thiosulphate salts.

$$4K + 3SO_2 \rightarrow K_2SO_3 + K_2S_2O_3$$

(iii) Basic oxides like CaO react vigorously with the gas to form sulphites .

$$CaO + SO_2 \rightarrow CaSO_3$$

(iv) Sulphur dioxide reacts with alkali to give sulphites and bisulphites.

$$KOH + SO_2 + H_2O \rightarrow KHSO_3 + H_2O$$

$$2KOH + SO_2 + H_2O \rightarrow K_2SO_3 + 2H_2O$$

(v) It combines with Cl_2 in presence of sunlight to give sulphur chloride.

$$SO_2 + Cl_2 \rightarrow SO_2Cl_2$$

(vi) SO_2 is a good reducing agent. It reduces halogens to hydrohalic acids.

$$Br_2 + SO_2 + 2H_2O \rightarrow H_2SO_4 + 2HBr$$

The reaction with iodine is used for estimating sulphurous acid.

(vii) Fe^{3+} ions are reduced to Fe^{2+} by SO_2.

$$2Fe^{3+} + SO_2 + 2H_2O \rightarrow 2Fe^{2+} + 4H^+ + SO_4^{2-}$$

(viii) SO_2 reduces potassium dichromate in acid medium to green chromic salt.

$$K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + H_2O$$

The same reaction can be used as test to detect SO_2 by using a filter paper moistened with acidified $K_2Cr_2O_7$ (yellow) which turns green on exposure to the gas.

(ix) A solution of $KMnO_4$ is decolourised by SO_2 to give colourless manganous salt.

$$2KMnO_4 + 5SO_2 + 2H_2O \rightarrow K_2SO_4 + 2MnSO_4 + 2H_2SO_4$$

(x) SO_2 also acts as an oxidizing agent.
SO_2 oxidises H_2S to sulphur.

$$2H_2S + SO_2 \rightarrow 2H_2O + 3S$$

(xi) Carbon is oxidized to carbon monoxide by SO_2

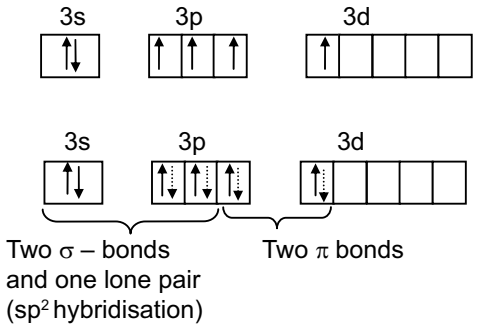
$$2C + SO_2 \rightarrow 2CO\uparrow + S\downarrow$$

Structure of SO_2

SO_2 is V- shaped molecule corresponding to a trigonal planar arrangement having two bonding electron pairs and

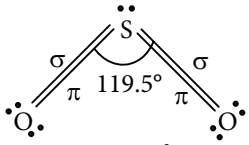
one lone pair. The central sulphur atom has to provide four unpaired electrons which is possible by excitation of its electrons as follows:

Outer electronic configuration of S in excited state.



SO_2 molecule is formed by supply of four electrons from two oxygen atoms as shown by dashed arrows.

Sulphur is assumed to form three sp^2 hybrid orbitals one of which is occupied by lone pair of electron. The two σ bonds between sulphur and oxygen formed by overlap of the remaining two sp^2 hybrid orbitals with a 'p' electron each from the two oxygen atom. One of the π-bonds in the molecule arises from pπ – pπ overlap and the other from pπ - dπ overlap. The structure may be represented as



Structure of SO_2

Uses of SO_2

It is used in the manufacture of H_2SO_4. It also finds application in the manufacture of sulphites which are used as bleaching agents.

Sulphur trioxide, SO_3

Preparation

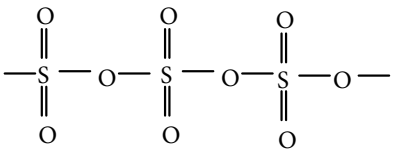
(i) When conc.H_2SO_4 is distilled with an excess of P_2O_5, SO_3 is formed.

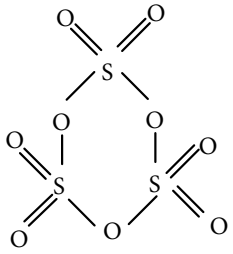
$H_2SO_4 + P_2O_5 \rightarrow 2HPO_3 + SO_3\uparrow$

(iii) On a large scale, SO_3 is manufactured by the contact process in which SO_2 reacts with O_2 in presence of a catalyst like V_2O_5 or platinized asbestos or spongy platinum. Usually it is not isolated but completely converted into H_2SO_4.

Properties

SO_3 is a solid at room temperature. It exists in three forms, known as α, β and γ forms. Both α and β varieties form white silky needles and are formed when SO_3 is kept for a long time or in the presence of a trace of water. β - SO_3 is dimeric (S_2O_6) and melts at 32.5°C. Above 50°C, it changes to α - form. The γ – form is formed when SO_3 vapour is condensed at –80°C or below. It is a cyclic trimer, $(SO_3)_3$ and has a melting point of 17°C. α – form is the most stable form, with a melting point 62.2°C. It has crosslinked chains. The structures of solid SO_3 in the cyclic and chain forms are shown below:


(a) chain form structure


(b) cyclic form

Reactions

(i) SO_3 reacts vigorously with water to form H_2SO_4 and it is a highly exothermic reaction. The evolved heat vapourizes water forming a thick mist of acid droplets.

$SO_3 + H_2O \rightarrow H_2SO_4$

(ii) SO_3 is a powerful oxidizing agent. It oxidizes HBr to bromine.

$6HBr + SO_3 \rightarrow 3Br_2 + 3H_2O + S$

(iii) It oxidizes phosphorus to its oxide, P_4O_{10}.

$3P_4 + 10SO_3 \rightarrow 3P_4O_{10} + 10S$

(v) Sulphur trioxide fumes in moist air giving rise to H_2SO_4 and oleum.

$SO_3 + H_2O \rightarrow H_2SO_4$

$2SO_3 + H_2O \rightarrow H_2S_2O_7$

(vi) It is a strong Lewis acid and readily reacts with basic oxides to form sulphates. For example, BaO reacts with SO_3 giving rise to $BaSO_4$.

$BaO + SO_3 \rightarrow Ba^{2+} + SO_4^{2-}$

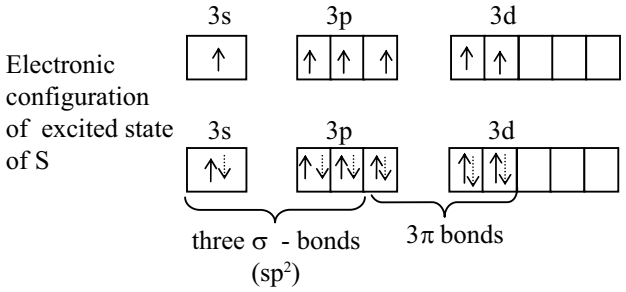
(vii) Reaction with HCl yields chlorosulphonic acid.

$SO_3 + HCl \rightarrow ClSO_3H$

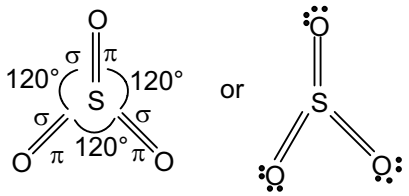
Structure of SO_3

In the gaseous state, SO_3 has a trigonal planar structure. To form the molecule, the central sulphur atom must provide six unpaired electrons by excitation to form six bonds.

The excited state electronic configuration of S atom may be shown as



The sulphur is assumed to form three sp^2 orbitals, which are used to form three σ bonds with a p-electron each from the three oxygen atoms. Of the three π bonds, one is formed by a pπ-pπ overlap and the other two are formed by dπ-pπ overlap. The O – S – O bond angles are 120° each. The structure may be represented as



Structure of S_3O

Uses of SO_3

It is mainly used for the manufacture of H_2SO_4. It is also used for sulphonating long chain alkyl benzenes because the sodium salts of the alkyl benzene sulphonates are good anionic surface active agents. These agents are useful as detergents. It is also used as a drying agent for gases.

Oxyacids of sulphur

Sulphur forms numerous oxyacids (also called oxoacids). Some of the oxyacids are not known in the free state but occurs in the form of their anions and salts. There are four main groups of oxyacids of sulphur viz., sulphurous acid group, sulphuric acid group, thionic acid group and peroxy acid group.

Sulphurous acid group

In this group, there are three acids,

(i) Sulphurous acid, H_2SO_3
(ii) Hyposulphurous acid (also known as dithionous acid), $H_2S_2O_4$ (the free acid is unknown)
(iii) Pyrosulphurous acid (also known as disulphurous acid), $H_2S_2O_5$ (the free acid is unknown)

Sulphuric acid group

In this group, there are three acids

(i) Sulphuric acid, H_2SO_4
(ii) Thiosulphuric acid, $H_2S_2O_3$
(iii) Pyrosulphuric acid, $H_2S_2O_7$

Thionic acid group

In this group, there are several acids

(i) Dithionic acid, $H_2S_2O_6$
(ii) Polythionic acid, $H_2S_{n+2}O_6$ (n = 1 – 12)

Peroxy acid group

In this group, there are two acids,

(i) Peroxy monosulphuric acid, H_2SO_5
(ii) Peroxy disulphuric acid, $H_2S_2O_8$

A detailed study of sulphurous acid, sulphuric acid and thiosulphuric acid is undertaken below:

Sulphurous acid, H_2SO_3

SO_2 is very much soluble in water and its aqueous solutions are referred to as sulphurous acid. SO_2 is generally known to be present as a hydrate in aqueous solution $SO_2.nH_2O$ ($n \approx 7$). However, it will be treated as H_2SO_3 for the present study. The first dissociation constant of the acid is 1.3×10^{-2} at 20°C. Sulphurous acid is a good reducing agent and all the reactions have been discussed under "reactions of SO_2".

Bisulphite and sulphite salts (salts of H_2SO_3)

Two series of salts, bisulphites containing HSO_3^- and sulphites containing SO_3^{2-} ions are known. The alkali metal sulphites and bisulphite are the most common among them.

Preparation of Na_2SO_3

SO_2 is passed into an aqueous solution of Na_2CO_3. At first, $NaHSO_3$ is formed which reacts further with Na_2CO_3 to give Na_2SO_3.

$$2Na_2CO_{3(aq)} + 2SO_2 \rightarrow 2Na_2SO_{3(aq)} + 2CO_2$$

Reactions

(i) Na_2SO_3 reacts with dil.HCl to give off SO_2.

$Na_2SO_3 + 2HCl \rightarrow 2NaCl + H_2O + SO_2$

(ii) Sulphites and bisulphites are strong reducing agents. They reduce iodine to HI.

$NaHSO_3 + I_2 + H_2O \rightarrow NaHSO_4 + 2HI$

Sulphites can be estimated by adding excess I_2 and after the reaction, the remaining I_2 is titrated with sodium thiosulphate to give sodium tetrathionate.

$2Na_2S_2O_3 + I_2 \rightarrow Na_2S_4O_6 + 2NaI$

(iii) When SO_2 is passed into a solution of Na_2SO_3, the salt $Na_2S_2O_5$ i.e., sodium disulphite is formed.

$Na_2SO_{3(aq)} + SO_2 \rightarrow Na_2S_2O_5$

(iv) H_2O_2 oxidizes Na_2SO_3 as well as other sulphites to Na_2SO_4.

$H_2O_2 + Na_2SO_3 \rightarrow Na_2SO_4 + H_2O$

(v) SO_3^{2-} is reduced to thiosulphate by sulphur.

$Na_2SO_3 + S \rightarrow Na_2S_2O_3$

Uses

Na_2SO_3 is mainly used as a bleaching agent for wood pulp in paper industry and also remove O_2 in boiler feed water.

Reactions of some sulphite salts

When a solution of sulphite is added to $BaCl_2$, white precipitate of $BaSO_3$ is formed.

$$SO_3^{2-} + Ba^{2+} \rightarrow BaSO_3$$

It dissolves in dil.HCl with the liberation of SO_2. $BaSO_3$ is insoluble in water.

$$BaSO_3 + 2HCl \rightarrow BaCl_2 + H_2O + SO_2$$

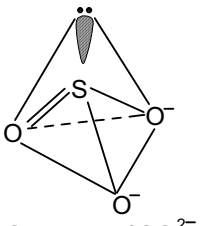
A similar reaction occurs with Ca and Sr salts. The precipitate of $BaSO_3$ on standing, is slowly oxidized to $BaSO_4$. The change can be hastened by warming with Br_2 water or H_2O_2.

$$2BaSO_3 + O_2 \rightarrow 2BaSO_4$$

$$BaSO_3 + Br_2 + H_2O \rightarrow BaSO_4 + 2HBr$$

Structure of SO_3^{2-} ion

The sulphite ion has a pyramidal structure i.e., a tetrahedral arrangement with three positions occupied by three oxygen atoms and a fourth one containing a lone pair of electrons. The central sulphur atom forms four sp^3 hybrid orbitals of which one contains a lone pair and the others contain an electron each. Three oxygen form three σ bonds with the three single electron in the three sp^3 hybrid orbitals. The electron of one of the oxygen atoms form π bond with the electron in 'd' orbital (dπ-pπ bond) of sulphur atom. The pyramidal structure of the sulphite ion is shown below:



Structure of SO_3^{2-}

Sulphuric acid, H_2SO_4

It is prepared on a large scale by two methods, lead chamber process and contact process.

(i) Lead chamber process

In this process the first step is the production of SO_2, which is done either by burning sulphur or iron pyrites in air.

$4FeS_2 + 11O_2 \rightarrow 2Fe_2O_3 + 8SO_2$

$S + O_2 \rightarrow SO_2$

Sometimes, SO_2 is generated from anhydrite or native $CaSO_4$. It is mixed with sand and coke and heated in a furnace, which results in a series of reaction forming SO_2.

$CaSO_4 + 2C \rightarrow CaS + 2CO_2$

$CaSO_4 + CaS + O_2 \rightarrow 2CaO + 2SO_2$

$2CaS + 4CaO + 6SiO_2 + 3O_2 \rightarrow 6CaSiO_3 + 2SO_2$

The gases formed contain some nitrogen from air and are known as burner gas. The burner gas is next passed through nitre pots contains a mixture of KNO_3 and H_2SO_4. The nitric acid so generated get mixed with the burner gases resulting in the following reactions.

$KNO_3 + H_2SO_4 \rightarrow KHSO_4 + HNO_3$

$2HNO_3 + 2SO_2 \rightarrow 2SO_3 + H_{\uparrow 2}O + NO + NO_2$

The gases are next passed through a denitrating tower and subsequently through a series of lead chambers (which are made of lead and supported in wood frames). Here, they come in contact with a spray of steam to form the chamber acid which is about 65% H_2SO_4 in concentration. The oxides of nitrogen catalyse the formation of H_2SO_4 from SO_2

through an intermediate formed during the course of reaction. The various reactions occurring may be represented as

$$SO_2 + NO_2 + H_2O \rightarrow H_2SO_4.NO$$

$$2H_2SO_4.NO + \frac{1}{2}O_2 \rightarrow$$

$$2HSO_4.NO(HO.SO_2.ONO) + H_2O$$

$$2HSO_4.NO + H_2O \rightarrow 2H_2SO_4 + N_2O_3$$

The N_2O_3 formed is absorbed by H_2SO_4 kept in another tower forming nitrosyl sulphuric acid again.

$$2H_2SO_4 + N_2O_3 \rightarrow 2HSO_4.NO + H_2O$$

The lead chamber process, which gives a concentration of 65% of H_2SO_4 is further concentrated by a cascade process, by which the concentration of the acid rises to 90%. Further concentration of the acid upto 98% is achieved by heating the acid in cast iron pans.

(ii) Contact process

In this method, SO_2 is oxidised to SO_3 under the influence of platinized asbestos as a catalyst. This catalyst has now been largely replaced by vanadium pentoxide, which is cheaper than platinized asbestos. The reaction is reversible and exothermic and may be represented as

$$2SO_2 + O_2 \rightleftharpoons 2SO_3 \quad \Delta H = -45.0 \text{ kcal}$$

According to Le-Chatelier's principle, high pressure and low temperature should favour the formation of SO_3. The optimum conditions for the formation of maximum amount of SO_3 in contact process were found to be 400 – 500°C and a pressure of 2 atm.

SO_3 is dissolved in 98% H_2SO_4, to form oleum.

$$H_2SO_4 + SO_4 \rightarrow H_2S_2O_7$$

Sulphuric acid of any desired concentration can be obtained from oleum by dilution with water.

$$H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$$

Properties

Pure H_2SO_4 is a colourless liquid, which is obtained from the commercial 98% H_2SO_4 by addition of SO_3 then adding water until the correct melting point is reached. The pure solidified acid has a melting point 10.5°C and it forms a viscous liquid having a specific gravity of 1.887 at 15°C. It is a strongly hydrogen bonded liquid. Concentrated sulphuric acid mixes with water in all proportions and it is a highly exothermic reaction. Like water, sulphuric acid undergoes extensive self ionization which may be represented as $2H_2SO_4 \rightleftharpoons H_3SO_4^+ + HSO_4^-$ and the value of the self ionization constant is 1.7×10^{-4} at 10°C.

In dilute aqueous solutions, H_2SO_4 is a strong acid. The dissociation in aqueous solution at 20°C may be represented as

$$H_2SO_4 + H_2O \rightarrow H_3O^+ + HSO_4^- \quad K_1 = 1 \times 10^3$$

$$HSO_4^- + H_2O \rightarrow H_3O^+ + SO_4^{2-} \quad K_2 = 1.3 \times 10^{-2}$$

Solutions containing bisulphite ions are acidic due to second reaction.

Reactions of H_2SO_4

(i) When metals like Zn, Fe, Mg are treated with dilute H_2SO_4, hydrogen gas is evolved.

$$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$$

(ii) The reaction of metals with conc. H_2SO_4, however, takes place in a different manner. With metals which have more negative reduction potential than hydrogen in the electrochemical series, the acid is reduced to SO_2.

For example, the reaction with zinc may be given as

$$Zn + 2H_2SO_4 \rightarrow ZnSO_4 + 2H_2O + SO_2$$

(iii) With metals, which are more electropositive than hydrogen in electrochemical series, like copper, silver, mercury although the end product is the same (liberation of SO_2). But the reaction is somewhat different.

For example, Cu first gets oxidised to CuO by the acid which then reacts with further acid to form $CuSO_4$. The overall reaction is

$$Cu + 2H_2SO_4 \xrightarrow{\Delta} CuSO_4 + 2H_2O + SO_2$$

Silver acts in a similar manner.

(iv) H_2SO_4 is a very good drying agent as it absorbs water readily. It is Therefore, used as a dehydrating agent for gases as well as other substances in a dessicator. This property is useful in organic nitration reactions, which occur with the formation of NO_2^+ ion from HNO_3.

$$HNO_3 + 2H_2SO_4 \rightarrow NO_2^+ + H_3O^+ + 2HSO_4^-$$

Organic compounds are charred by H_2SO_4. For example, sugar loses molecule of water and only carbon remains on treatment with conc.H_2SO_4. Oxalic acid is decomposed by the acid which loses a molecule of water.

$$C_{12}H_{22}O_{11} \xrightarrow{H_2SO_4} 12C + 11H_2O$$

$$(COOH)_2 \rightarrow CO_2 + CO + H_2O$$

(v) On heating potassium chlorate with conc.H_2SO_4, chlorine dioxide is evolved with explosion.

$$3KClO_3 + 3H_2SO_4 \rightarrow$$
$$3KHSO_4 + HClO_4 + 2ClO_2 + H_2O$$

(vi) Hot concentrated H_2SO_4 is a good oxidizing agent too. Sulphur and carbon are oxidized by H_2SO_4 to give SO_2 and CO_2 respectively.

$$S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$$
$$C + 2H_2SO_4 \rightarrow CO_2 + 2SO_2 + 2H_2O$$

Phosphorus is oxidized to phosphoric acid.

$$2P + 3H_2SO_4 \rightarrow 2H_3PO_3 + 3SO_2$$

(vii) Hydrobromic and hydroiodic acids are oxidized to the corresponding halogen.

$$2HBr + H_2SO_4 \rightarrow SO_2 + Br_2 + 2H_2O$$

(viii) Salts like sodium or potassium bromides or iodides are oxidized to bromine or iodine by conc.H_2SO_4

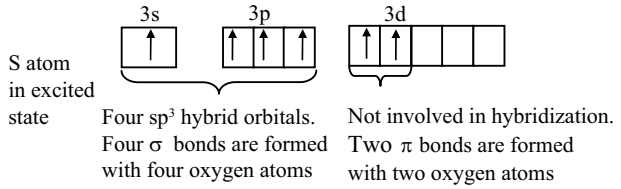
$$2KBr + 2H_2SO_4 \rightarrow K_2SO_4 + SO_2 + 2H_2O + Br_2$$
$$2KI + 2H_2SO_4 \rightarrow K_2SO_4 + SO_2 + 2H_2O + I_2.$$

Reactions of H_2SO_4 with metals

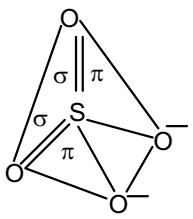
The acid forms two types of salts ie bisulphates and sulphates. Metals or metal oxides react with H_2SO_4 to give metal sulphates. Alkali metal sulphates are soluble in water but sulphates of alkaline earth metals are insoluble in water. $PbSO_4$ is only very sparingly soluble in water.

Structure of SO_4^{2-} ion

The sulphate ion has a tetrahedral structure with the central sulphur atom forming sp^3 hybrid orbitals as shown below.



The structure may be represented as



Tests for H_2SO_4

(i) When heated with copper, it evolves SO_2 which has a characteristic smell. The gas turns acidified $K_2Cr_2O_7$ solution green.

(ii) It forms a white precipitate (of $BaSO_4$) with $BaCl_2$ solution which is insoluble in conc.HCl and conc. HNO_3.

(iii) When treated with NaCl, it evolves HCl which forms white fumes with NH_4OH.

Thiosulphuric acid ($H_2S_2O_3$) and its salts

Thiosulphuric acid does not exist in the free state because the free acid decomposes into sulphur, H_2S, SO_2 and H_2SO_4. Several salts of the acid are, however, known which are stable and are called thiosulphates. Sodium thiosulphate, $Na_2S_2O_3.5H_2O$ is one of the most common thiosulphate known and is referred to as 'hypo'.

Preparation of $Na_2S_2O_3.5H_2O$

(i) By boiling sodiumsulphite Na_2SO_3 solution with sulphur, hypo can be prepared.

$$Na_2SO_3 + S \rightarrow Na_2S_2O_3$$

(ii) Oxidation of soluble sulphides or hydrosulphides in air results in the formation of $Na_2S_2O_3$.

$$2Na_2S + 2O_2 + H_2O \rightarrow Na_2S_2O_3 + 2NaOH$$

or $$2NaHS + 2O_2 \rightarrow Na_2S_2O_3 + H_2O$$

(iii) Pure sodium thiosulphate is prepared by allowing a solution of sodium hydrogen sulphide and sodium bisulphite to crystallise. The reaction is

$$2NaHS + 4NaHSO_3 \rightarrow 3Na_2S_2O_3 + 3H_2O$$

Properties of $Na_2S_2O_3$

It is a hydrated salt with five molecules of water. The crystals are colourless and hexagonal. The hydrate melts at 48°C. It is readily soluble in water.

Reactions of $Na_2S_2O_3$

(i) It reacts with dil.HCl to form SO_2 and S.

$$Na_2S_2O_3 + 2HCl \rightarrow 2NaCl + H_2O + SO_2 + S$$

(ii) Chlorine oxidizes sodium thiosulphate to sodium sulphate.

$$Na_2S_2O_3 + Cl_2 + H_2O \rightarrow Na_2SO_4 + 2HCl + S$$

Hypo can thus act as an antichlor agent to destroy excess chlorine on fabrics. It is also used to remove Cl_2

from heavily chlorinated water and there by reduce the chlorine smell.

(iii) $Na_2S_2O_3$ is oxidized by iodine to give sodium tetrathionate.

$$2Na_2S_2O_3 + I_2 \rightarrow Na_2S_4O_6 + 2NaI$$

This reaction is used for the estimation of iodine in volumetric analysis using starch as indicator.

(iv) When sodium thiosulphate solution is added to silver nitrate solution a white precipiate of silver thiosulphate is formed which quickly changes into yellow, red, brown and finally to black precipitate of silver sulphide (play of colours).

$$Na_2S_2O_3 + 2AgNO_3 \rightarrow Ag_2S_2O_3 + 2NaNO_3$$
$$Ag_2S_2O_3 + H_2O \rightarrow Ag_2S + H_2SO_4$$

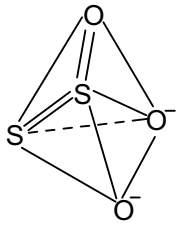
Uses of hypo

It has the property of dissolving silver halides forming the complex species $[Ag(S_2O_3)_3]^{5-}$. The dissolving power of hypo is applied in photography for fixing photographic films. After developing the film, it is put in a solution of hypo which forms a soluble complex with silver salts thus dissolving any unchanged silver salts in the emulsion.

$$Na_2S_2O_3 + 2AgBr \rightarrow Ag_2S_2O_3 \xrightarrow{3Na_2S_2O_3} 2Na_3[Ag(S_2O_3)_2]$$

Structure of $S_2O_3^{2-}$ ion

It has a tetrahedral geometry like SO_4^{2-} ion as shown below.



Structure of $S_2O_3^{2-}$

HALOGENS (F, CL, BR, I, AT)

Introduction

The halogens, fluorine (F), chlorine (Cl), bromine (Br), iodine (I) and astatine (At) belong to group 17 of the periodic table. They are typical non metals except astatine which is a metalloid. It is also radioactive. They derive the name halogen due to their salt forming tendency. All the elements have seven electrons in their valence shell with $ns^2\ np^5$ as the valence electronic configuration. Thus they have one "p" electron less than that of the noble gas, which follows it. They, Therefore, can attain the noble gas configuration either by acquiring an electron (through formation of an ionic bond) or by sharing an electron with another atom to form a covalent bond. The compounds formed by halogens with metals are ionic in nature and those formed with non-metals are covalent, some important atomic properties of halogens are listed below:

Element	At. no.	At. wt.	Electronic configuration	Covalent radius (A°)	Atomic radius (A°)	Ionization energy (1st) (kJ mol^{-1})	Electro-negativity	Oxidation states
Fluorine	9	19.00	$[He]2s^2 2p^5$	0.72	1.33	1680	4.0	−1
Chlorine	17	35.45	$[Ne]3s^2 3p^5$	0.99	1.81	1256	3.0	−1, +1, +3, +4, +5, +6, +7
Bromine	35	79.92	$[Ar]3d^{10} 4s^2 4p^5$	1.14	1.96	1142	2.8	−1, +1, +3, +4, +5, +6
Iodine	53	126.92	$[Kr]4d^{10} 5s^2 4p^5$	1.33	2.20	1008	2.5	−1, +1, +3, +5, +7
Astatine	85	210*	$[Xe]4f^{14} 5d^{10} 6s^2 6p^5$	---	---	---	---	---

(* mass number of the most stable isotope)

The halogens consist of diatomic molecules. The are coloured, fluorine is light yellow, chlorine is greenish yellow gas, bromine is reddish brown liquid and iodine is a dark grey solid with slight metallic lustre.

They exhibit close similarities in properties within the group although, fluorine, the first member of the group, differs in many aspects from the other members of the group. This is due to

(i) the small size of fluorine atom which results in a firm hold of its electrons.
(ii) the absence of d-orbitals which may be used in bonding
(iii) the high electronegativity of fluorine.

However, the properties of the succeeding pair of halogens, chlorine and bromine, are much more closer than the other pairs of elements in the group because of the closeness in the sizes of their ions.

Stability of X - X bond (X = halogen atom.)

In view of the small size of fluorine atom, one would expect the F – F bond length to be short and stable and correspondingly the bond dissociation energy to be highest for this molecule. But it has an abnormally low bond energy than either chlorine or bromine and makes it the weakest bond. This is because of the greater repulsion between the non-bonding electrons in F_2 and due to the non-existence of multiple bonding involving d-orbitals in F_2.

Halogen	Bond energy (kJ mol^{-1})	Bond length (A°)
F_2	126	1.43
Cl_2	210	1.99
Br_2	158	2.28
I_2	118	2.66

Ionization energies

The first ionisation energies decrease from fluorine to iodine. Since fluorine has a high first ionisation energy, it has no tendency to lose an electron to form F^+ species. The opposite is true in the case of iodine and I^+ species is well known in compound like ICl, ICN etc.

Oxidation states

Fluorine being the most electronegative element of the halogens, shows an oxidation state of –1. It doesn't show any positive oxidation state. The higher halogens, chlorine, bromine and iodine, have vacant d orbitals in their valence shells and Therefore, can exhibit higher positive oxidation state because of the possibility of electrons going into vacant d orbitals by excitation. The positive oxidation states arise for these halogens in their compounds with oxygen because they are less electronegative than oxygen (except fluorine). For example, chlorine shows oxidation states of +1, +3, +5, +7 in the compounds HClO, $HClO_4$ $HClO_3$ and $HClO_4$, respectively. In interhalogen compounds, the halogen with the lower electronegativity shows positive oxidation state, for example, iodine in ICl show a positive oxidation state of +1. Similarly, chlorine in ClF shows a positive oxidation sate of +1.

Halogens as oxidizing agents

All halogens have a tendency to take up electrons and thus act as strong oxidizing agents. Among them, fluorine is the strongest oxidizing agent which is reflected in its highly positive standard potential (E° = +2.85 V). The oxidizing power decreases from fluorine to iodine. Fluorine will oxidize other halide ions to halogens in solution or when dry.

$$F_2 + 2X^- \rightarrow 2F^- + X_2 \ (X = Cl, Br, I).$$

CONCEPT STRAND

Concept Strand 10

Give reasons for the following observations:

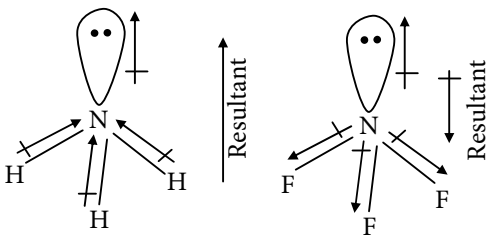
(i) Even though fluorine has a lower electron affinity than chlorine, the former surpasses latter in oxidising power. Explain.
(ii) NF_3 has a lesser dipole moment than NH_3.

Solution

(i) Because:
  (a) F_2 has a low dissociation energy than Cl_2 because of the repulsion between the lone pairs on the two F and the non existence of multiple bonding involving d – orbitals.
  (b) When elements combine with fluorine, the energies released are the highest due to small size of F.

(c) Lattice energy and hydration energy of fluorides are higher than those of chlorides.

(ii) In NH_3, the bond dipoles and lone pair dipoles are in the same direction and they augment each other. In NF_3, bond dipoles and lone pair dipoles are in opposite directions, so the resultant value is lesser.



Reactivity of halogens

Fluorine is the most chemically reactive of all halogens. It combines directly at ordinary or higher temperatures with all elements other than oxygen and some noble gases. It also attacks many compounds, particularly organic compounds, forming fluorides. Its high reactivity arises from

(i) high electronegativity
(ii) high oxidizing power
(iii) low dissociation energy of F – F bond
(iv) small size

The reactivity decreases from fluorine to iodine.

Variation in basic property of halogens

The basic character of halogens i.e., their tendency to form cations increases from fluorine to iodine, with iodine showing maximum basic character. This may be ascribed to its low electronegativity. Fluorine has no such tendency at all.

Fluorine

Occurrence and isolation

Fluorine is known to occur in the form of fluorspar or fluorite (CaF_2) for quite a long time. The most important minerals of fluorine are fluorspar (CaF_2), cryolite ($3NaF.AlF_3$) and fluorapatite ($CaF_2.3Ca_3(PO_4)_2$). The isolation of fluorine was first achieved by Moissan in 1886. For this purpose, CaF_2 was treated with conc.H_2SO_4 which results in the formation aqueous HF. The aqueous HF was distilled to get anhydrous liquid HF. It is then mixed with anhydrous potassium hydrogen fluoride, KHF_2 and electrolysed in an iridioplatinum vessel at –23°C to give hydrogen at cathode and fluorine at anode. The reactions are

$$CaF_2 + H_2SO_4 \rightarrow CaSO_4 + 2HF$$

$$HF + K[HF_2] \xrightarrow{\text{electrolysis}} H_2 + F_2 + KF$$

In a modification of the method, the electrolysis is carried by using fused KHF_2 and HF in a U-shaped copper vessel between steel cathode and carbon anode. The reactions are

$$KHF_2 \rightarrow KF + HF$$

$$HF \rightarrow H^+ + F^-$$

$$2H^+ + 2e^- \rightarrow H_2$$

$$2F^- \rightarrow F_2 + 2e^-$$

The manufacture of fluorine, in general, is associated with several practical difficulties, which are worthy of mention here.

(i) Anhydrous HF is a poor conductor of electricity due to its small ionization. For this reason, a mixture HF and KF (or NaF), which has a higher conductivity is used in electrolysis.

(ii) Water should be thoroughly excluded from the mixture as fluorine oxidizes water to O_2 and O_3. Therefore, an aqueous solution of Hydrofluoric acid cannot be electrolysed to get F_2

$$2H_2O + 2F_2 \rightarrow 4HF + O_2$$

$$3H_2O + 3F_2 \rightarrow 6HF + O_3$$

(iii) HF is highly corrosive and toxic both in gaseous and liquid states and hence must be handled carefully. Glass vessels must be avoided as HF can etch glass.

(iv) H_2 and F_2 liberated at cathode and anode must be separated by using a diaphragm cell as otherwise they will react violently causing explosions.

(v) Fluorine is very reactive and hence reaction vessels made of Cu/Ni alloy must be used during electrolysis.

(vi) Ordinary graphite electrodes cannot be used because they absorb F_2. Absorption of fluorine results in the expansion of graphite electrode and its subsequevent explosion. Therefore, copper impregnated special graphite electrodes are to be used for electrolysis.

Properties

Fluorine is a pale yellow gas. Liquid fluorine is also pale yellow.

Reactions

(i) It is a very reactive element. It reacts violently with hydrogen even in dark near zero Kelvin to form HF.

$$H_2 + F_2 \rightarrow 2HF$$

(ii) Oxygen and fluorine react under the influence of electric discharge at very low pressure and at liquid air temperature to give dioxygen difluoride, O_2F_2

$$O_{2(g)} + F_{2(g)} \rightarrow O_2F_2$$

(iii) When fluorine is passed into a dilute solution of NaOH, oxygen difluoride, OF_2 is formed, and when passed into conc.NaOH, O_2 is formed.

$$2F_2 + 2NaOH \rightarrow 2NaF + H_2O + OF_2$$

$$2F_2 + NaOH \rightarrow 4NaF + O_2 + 2H_2O$$

(iv) Fluorine reacts with ammonia to form N_2 and with H_2S to form SF_6 (H_2S burns).

$$2NH_3 + 3F_2 \rightarrow N_2 + 3H_2F_2$$

$$H_2S + 4F_2 \rightarrow SF_6 + H_2F_2$$

(v) Elements like S, Se, Te, P, Al, Si and B combine vigorously with fluorine. Also many metals react with fluorine forming fluorides.

$$2P + 3F_2 \rightarrow 2PF_3$$

$$2Al + 3F_2 \rightarrow 2AlF_3$$

$$Si + 2F_2 \rightarrow SiF_4$$

(vi) Fluorine reacts with chlorides displacing chlorine from them.

$$2NaCl + F_2 \rightarrow 2NaF + Cl_2$$

$$CCl_4 + 2F_2 \rightarrow CF_4 + 2Cl_2$$

(vii) Chlorine reacts with fluorine to form ClF_3.

$$3F_2 + Cl_2 \xrightarrow{200-300^\circ C} 2ClF_3$$

(viii) Fluorine oxidizes chlorates to perchlorates, iodates to periodates and bisulphates to peroxysulphates.

$$KClO_3 + F_2 + H_2O \rightarrow KClO_4 + H_2F_2$$

$$KIO_3 + F_2 + H_2O \rightarrow KIO_4 + H_2F_2$$

$$2NaHSO_4 + F_2 \rightarrow Na_2S_2O_8 + H_2F_2$$

Uses of fluorine

It is used in the separation of uranium isotopes. Especially, the isotope U^{235} is separated from other isotopes by forming UF_6 through gaseous diffusion process. It also finds application in the manufacture of fluorocarbons, which are chemically inert and non-inflammable. They find application as solvents, lubricants and chlorofluorocarbons are used as coolants in refrigeration and air conditioners. Teflon, which is polytetrafluroethylene is resistant to chemical attack and is used as an electrical insulator.

CONCEPT STRAND

Concept Strand 11

Explain the properties of fluoride ion because of which it reduces dental decay.

Solution

Traces of fluoride ion reduce dental decay. The F^- ion makes the enamel on the teeth much harder by converting hydroxyapatite [$3Ca(PO_4)_2.Ca(OH)_2$], the enamel on the surface of the teeth, into much harder fluorapatite [$3Ca_3(PO_4)_2.CaF_2$]. But higher quantities of fluoride above 2 ppm has a negative effect and causes yellowing of teeth and teeth become brittle (called mottling of teeth).

Compounds of fluorine

Since fluorine is more electronegative than oxygen, compounds of oxygen and fluorine are known as fluorides of oxygen. Two fluorides of oxygen are known

(i) oxygen difluoride, OF_2
(ii) dioxygen difluoride, O_2F_2.

Oxygen difluoride, OF_2

OF_2 is formed by passing F_2 into 0.2% NaOH

$$2F_2 + 2NaOH \rightarrow 2NaF + H_2O + OF_2$$

It is a pale yellow gas insoluble in water. It is a strong oxidizing agent.

(i) It oxidises KI to liberate iodine

$$OF_2 + 4KI + H_2O \rightarrow 2KF + 2KOH + 2I_2$$

(ii) It reacts with alkali to give fluoride ions and oxygen.

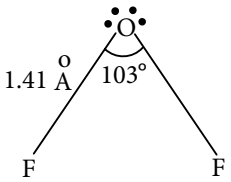
$$OF_2 + (2Na^+ + 2OH\uparrow) \rightarrow (2Na^+ + 2F^-) + H_2O + O_2$$

(iii) It reacts slowly with water but explodes with steam.

$$OF_2 + H_2O \rightarrow O_2 + 2HF$$

Structure of OF_2

It is a V-shaped molecule and the central oxygen atom is assumed to form sp^3 hybrid orbitals. The F – O – F angle is less than the tetrahedral angle and is 103°C. The O – F bond length is 1.41 A°.



Dioxygen difluoride, O_2F_2

O_2F_2 is formed by passing an electric are through a mixture of F_2 and O_2 cooled to about –185°C at very low pressure.

$$F_2 + O_2 \xrightarrow[\text{arc}]{\text{electric}} O_2F_2$$

It is yellow orange solid, melts at –163°C and decomposes into elements above –100°C. It is a strong fluorinating and oxidising agent. Its structure is similar to that of H_2O_2 with the O – O bond length of 1.22 A° (1.48 A° in H_2O_2) and O –F bond length of 1.58 A° (1.41 A° in OF_2).

Chlorine

Occurrence and isolation

Chlorine does not occur in the free state as it is a reactive element. In combination, it occurs mainly as common salt (NaCl), horn silver (AgCl), carnallite ($KCl.MgCl_2.6H_2O$). Sea water contains about 2.3% of NaCl.

Manufacture of chlorine

(i) Castner-Kellner Process

Chlorine is currently manufactured by the electrolysis of aqueous sodium chloride (brine) using Castner-Kellner process. Castner-Kellner cell consists of three compartments. The end compartments contain brine and the middle compartment contains very dilute NaOH solution. Carbon electrodes are used in the end compartments which act as anode. The cathode consists of a series of iron rods placed in the middle compartment. The bottom of the cell is covered with mercury. When current is passed through the cell, Cl^- ions travel to anode and are liberated as chlorine gas. The sodium ions get discharged at cathode and form sodium, which dissolves in mercury forming amalgam. The sodium amalgam reacts with water to liberate hydrogen. The reactions are

$$2Cl^- \rightarrow Cl_2 + 2e^- \text{ (anode)}$$

$$2Na^+ + 2e^- \rightarrow 2Na \text{ (cathode)}$$

$$2Na + 2H_2O \rightarrow 2NaOH + H_2$$

The overall reaction is

$$2NaCl + 2H_2O \rightarrow 2NaOH + H_2 + Cl_2$$

(ii) Deacon's process for manufacture of chlorine

This process was developed by Deacon and Hunter in 1868 and it involves the oxidation of HCl by the oxygen of air using cupric chloride as catalyst along with a promoter consisting of mixture of praseodymium and neodymium oxides. The reaction is carried out at 450°C and is given as

$$4HCl + O_2 \xrightarrow[450^\circ C]{CuCl_2} 2H_2O + 2Cl_2\uparrow$$

Preparation on a laboratory scale

(i) Hydrochloric acid is oxidised to chlorine using MnO_2 or $KMnO_4$ or $K_2Cr_2O_7$ or PbO_2

The reactions with the various oxidizing agents are given below

$$MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2\uparrow$$

$$2KMnO\uparrow_4 + 16HCl \rightarrow 2KCl + 2MnCl_2 + 8H_2O + 5Cl_2\uparrow$$

$$K_2Cr_2O_7 + 14HCl \rightarrow 2KCl + 2CrCl_3 + 7H_2O + 3Cl_2\uparrow$$

$$PbO_2 + 4HCl \rightarrow PbCl_2 + 2H_2O + Cl_2$$

The HCl in the above reaction can be generated by in situ heating NaCl with conc.H_2SO_4 and it can then be oxidised as above by MnO_2, $KMnO_4$ or PbO_2.

Properties

Chlorine is a greenish yellow toxic gas with a pungent suffocating smell. It liquefies under pressure at –3°C and liquid chlorine is yellowish in colour with a boiling point

of −34°C. At 0°C, crystals of chlorine hydrate, $Cl_2.8H_2O$ is formed. Chlorine is moderately soluble in water and reacts to form hypochlorous acid (HClO).

Reactions

(i) It reacts with hydrogen in presence of diffused sunlight forming HCl. Also H_2 can burn in Cl_2 to form HCl

$$H_2 + Cl_2 \rightarrow 2HCl$$

(ii) It oxidises H_2S releasing sulphur.

$$H_2S + Cl_2 \rightarrow 2HCl + S$$

(iii) In the presence of moisture, Cl_2 acts as an oxidizing agent due to the formation of HOCl along with HCl

$$H_2O + Cl_2 \rightarrow HCl + HOCl$$

HOCl decomposes to give nascent oxygen which accounts for the bleaching action of chlorine water.

$$HOCl \rightarrow HCl + [O]$$

(iv) It oxidizes ferrous salts to ferric salts.

$$2FeCl_2 + Cl_2 \rightarrow 2FeCl_3$$

(v) Sulphites are oxidized to sulphates by chlorine.

$$Na_2SO_3 + H_2O + Cl_2 \rightarrow Na_2SO_4 + 2HCl$$

(vi) Thiosulphates react with chlorine to form Na_2SO_4.

$$Na_2S_2O_3 + Cl_2 + H_2O \rightarrow Na_2SO_4 + 2HCl + S$$

(vii) Bromides and iodides react with chlorine to liberate bromine and iodine respectively.

$$2KBr + Cl_2 \rightarrow 2KCl + Br_2$$

$$2KI + Cl_2 \rightarrow 2KCl + I_2$$

(viii) SO_2 reacts with chlorine to form sulphuryl chloride.

$$SO_2 + Cl_2 \rightarrow SO_2Cl_2$$

(ix) Chlorine reacts with CO to form phosgene.

$$CO + Cl_2 \rightarrow COCl_2$$

(x) Chlorine adds to unsaturated hydrocarbons to form addition compounds.

$$CH_2 = CH_2 + Cl_2 \rightarrow CH_2Cl - CH_2Cl$$

(xi) Many metals and non-metals (except nitrogen, oxygen, carbon and inert gases) react with chlorine to form corresponding chlorides. Some typical reactions are given below

$$2Fe + 3Cl_2 \rightarrow 2FeCl_3$$

$$2Ag + Cl_2 \rightarrow 2AgCl$$

$$P_4 + 6Cl_2 \rightarrow 4PCl_3$$

$$2Sb + 3Cl_2 \rightarrow 2SbCl_3$$

(xii) With excess chlorine, NH_3 reacts to form nitrogen trichloride.

$$NH_3 + 3Cl_2 \rightarrow NCl_3 + 3HCl$$

NH_3 reacts with limited amounts of Cl_2 to form NH_4Cl and N_2

$$8NH_3 + 3Cl_2 \rightarrow 6NH_4Cl + N_2$$

(xii) Chlorine reacts with cold dilute alkali, to give chlorides and hypochlorites

$$2NaOH + Cl_2 \rightarrow NaCl + NaOCl + H_2O$$

If the same reaction is carried out with hot, concentrated alkali, the product formed is chlorate and chloride.

$$6NaOH + 3Cl_2 \rightarrow NaClO_3 + 5NaCl + 3H_2O$$

Uses

Chlorine is widely used in a variety of industries such as

(i) in the manufacture of vinyl chloride used for making PVC (poly vinyl chloride).
(ii) in the manufacture of 1,2-dichloroethane, an important raw material in making plastics.
(iii) in the manufacture of chemicals like benzene hexachloride, DDT, and for making chlorinated phenols.
(iv) in the manufacture of a large number of inorganic chemicals like bleaching powder, sodium hypochlorite (used in bleaching industry).
(v) for sterilizing drinking water.
(vi) in the manufacture of poisonous gases like phosgene ($COCl_2$), tear gas ($CCl_3.NO_2$) and mustard gas (Cl – C_2H_4 – S – C_2H_4 – Cl).

Bromine

Ocurrence and isolation

Bromine does not occur in the free state but occurs abundantly in the form of its compounds. The main sources of bromine are carnallite and sea water which contains about 65 ppm of bromine in the form of bromide salts of Ca, Mg and alkali metals.

Manufacture

(i) From Carnallite

On a large scale, bromine is made from the mother liquor left behind after the crystallization of chloride from the saturated solution of carnallite ($MgCl_2.KCl.6H_2O$). The mother liquor contains bromocarnallite, $MgBr_2.KBr.6H_2O$, which is used for the generation of bromine. By passage of chlorine through the hot mother liquor trickling down layers of earthenware balls, bromine is liberated and it is car-

ried up by steam into condensers where it collects in the liquid state. The reaction taking place is

$$MgBr_2 + Cl_2 \rightarrow MgCl_2 + Br_2$$

$$2KBr + Cl_2 \rightarrow Br_2 + 2KCl$$

Usually, small amount of chlorine and iodine are present as impurities in the bromine prepared as above. It is, Therefore, purified by distilling in the presence of KBr and ZnO to remove chlorine and iodine respectively.

(ii) From sea water

Some sources of sea water (in USA) contain a larger percentage of bromide which is profitably used for preparing bromine. Chlorine is passed through brine to liberate Br_2. The released bromine vapours are blown by a stream of air into a solution of Na_2CO_3. The bromine reacts forming NaBr and $NaBrO_3$. The solution is then acidified to release Bromine and distilled to collect pure bromine. The overall reaction is

$$3Na_2CO_3 + 3Br_2 \rightarrow 5NaBr + NaBrO_3 + 3CO_2$$

$$5NaBr + NaBrO_3 + 3H_2SO_4 \rightarrow 3Na_2SO_4 + 3Br_2 + 3H_2O$$

Properties

Bromine is a dense, reddish brown liquid at room temperature and has a pungent odour. Its vapours are highly irritating. It boils at 59°C and freezes to a brown solid at −7°C. It is moderately soluble in water giving a solution of about 0.4 M at 25°C. It is soluble in organic solvents like $CHCl_3$, CCl_4, CS_2 and benzene. A saturated solution of bromine in water, on cooling in freezing mixture, gives a crystalline hydrate with the composition $Br_2.8H_2O$. Bromine is a reactive halogen like chlorine.

Reactions

(i) It dissociates into bromine atoms when heated and the reaction is

$$Br_2 \rightleftharpoons 2Br$$

(ii) It reacts with many metals like Na, K, Zn, Al and with some non-metals like S, P, As, B to form the corresponding bromides

$$2P + 3Br_2 \rightarrow 2PBr_3$$

$$2As + 3Br_2 \rightarrow 2AsBr_3$$

$$2Na + Br_2 \rightarrow 2NaBr$$

$$2Al + 3Br_2 \rightarrow 2AlBr_3$$

(iii) It is a good oxidizing agent.

(a) It oxidizes SO_2 to H_2SO_4.

$$SO_2 + Br_2 + 2H_2O \rightarrow H_2SO_4 + 2HBr$$

(b) Sodium sulphite and sodium thiosulphate are oxidized to Na_2SO_4.

$$Na_2SO_3 + Br_2 + H_2O \rightarrow Na_2SO_4 + 2HBr$$

$$Na_2S_2O_3 + Br_2 + H_2O \rightarrow Na_2SO_4 + 2HBr + S$$

(iv) Br_2 reacts into cold dilute alkali to form bromide and hypobromite.

$$2NaOH + Br_2 \rightarrow NaOBr + NaBr + H_2O$$

With hot concentrated sodium hydroxide, the products are NaBr and $NaBrO_3$.

$$6NaOH + Br_2 \rightarrow NaBrO_3 + 5NaBr + 3H_2O$$

(v) Br_2 displaces I_2 from KI.

$$2KI + Br_2 \rightarrow 2KBr + I_2$$

Uses

Bromine is mainly used in making organic derivatives like methyl and ethyl bromides, which are used as pesticides and insecticides. It also finds application as a flame retardant. Compounds of bromine like AgBr are used in photography. 1,2-Dibromoethane is used as an additive to petrol to prevent the lead released from tetraethyl lead from collecting on the spark plug in the engine.

Iodine

Occurrence and isolation

Iodine is not a very common element occurring in nature. There are two sources of iodine. They are natural brines or seawater and Chile salt petre ($NaNO_3$) containing trace amount of $NaIO_3$ and $NaIO_4$.

From sea Water

Iodides are present only at low concentration (≈ 0.001%) in seawater, but they are absorbed and thereby get concentrated in seaweeds. Seaweeds are a source of iodine but this source is now replaced by others. The seaweeds are collected, dried and burnt into ash (also know as "kelp"). The ash is leached with water and the solution contains different salt such as chlorides, sulphates and some carbonates apart from bromides and iodides. All salts other than bromides and iodides are separated by fractional crystallization. The remaining liquor, which is rich in bromides and iodides, is treated with conc.H_2SO_4 and MnO_2 in iron retorts and heated.

The iodine vapours formed are condensed in earthenware receiver.

$$2NaI + 3H_2SO_4 + MnO_2 \rightarrow 2NaHSO_4 + MnSO_4 + 2H_2O + I_2$$

Pure iodine is obtained by sublimation.

From Chile salt petre ($NaNO_3$)

Chile salt petre contains trace amount of iodine in the form of sodium iodate ($NaIO_3$ about 0.5%) and sodium periodate ($NaIO_4$). It is dissolved in water and the major part of $NaNO_3$ is separated by crystallization. The mother liquor becomes concentrated in iodates. A part of it is reduced by sodium bisulphate to iodide and is added to the unreacted part, which results in liberation of iodine. It is filtered off and purified by sublimation.

$$2NaIO_3 + 6NaHSO_3 \rightarrow 2NaI + 6NaHSO_4$$

$$5NaI + NaIO_3 + 3H_2SO_4 \rightarrow 3Na_2SO_4 + 3H_2O + 3I_2$$

Properties

Iodine form dark violet (or black) and shiny crystals and has a slight metallic lustre. Its melting and boiling points are 114°C and 184°C respectively. At atmospheric pressure, it sublimes to give a violet vapour without melting. It is only slightly soluble in water (≈ 0.003 M), but is readily soluble in KI to give KI_3. It is readily soluble in organic solvents like CS_2, CCl_4 and benzene to give violet solutions.

Reactions

(i) Iodine reacts with hot alkali to give the iodide and iodate.

$$6NaOH + 5I_2 \rightarrow NaIO_3 + 5NaI + 3H_2O$$

In cold, iodide and hypoiodite formed,

$$2NaOH + I_2 \rightarrow NaI + NaOI + H_2O$$

(ii) Iodine reacts with chlorine to give iodine monochloride at first and finally iodine trichloride.

$$I_2 + Cl_2 \rightarrow 2ICl$$

$$ICl + Cl_2 \rightarrow ICl_3$$

(iii) H_2S reacts with I_2 to form HI is

$$H_2S + I_2 \rightarrow 2HI + S$$

(iv) I_2 reacts with $Na_2S_2O_3$ to give sodium tetrathionate

$$2Na_2S_2O_3 + I_2 \rightarrow Na_2S_4O_6 + 2NaI$$

This reaction is used in iodometric titration in presence of starch indicator.

(v) SO_2 is oxidized to H_2SO_4 by iodine.

$$SO_2 + I_2 + 2H_2O \rightarrow H_2SO_4 + 2HI$$

(vi) Iodine reacts with red phosphorus in presence of water to form phosphoric acid and HI.

$$P_4 + 10I_2 + 16H_2O \rightarrow 4H_3PO_4 + 20HI$$

(vii) Stannous chloride in acid medium is oxidised to stannic chloride.

$$SnCl_2 + 2HCl + I_2 \rightarrow SnCl_4 + 2HI$$

(viii) Iodine is used in determining the iodine value of unsaturated oils. The reaction is the addition of iodine atoms across double bond of the unsaturated compound.

$$RCH = CHR + I_2 \rightarrow RCHI - CHRI$$

(ix) The electropositive nature of iodine is exhibited by the formation of compounds such as iodine acetate $\left(I^{3+}(CH_3COO)_3^{3-}\right)$, iodine nitrate $\left(I^{3+}(NO_3)_3^{3-}\right)$, iodine percholorate $\left(I^{3+}(ClO_4)_3^{3-}\right)$ and iodine monochloride (ICl).

Uses

Iodine is used in the manufacture of iodoform and potassium iodide. It is used in medicine to prepare tincture iodine, which is used as an antiseptic. Small amounts of iodine are used in human diet as NaI to prevent goitre. Silver iodide finds application in photography.

Hydrohalic acids

Hydrogen fluoride, HF

Hydrogen and fluorine combine violently with explosion even in the dark to form HF, but this is not the generally adopted method of preparing HF.

Preparation

(i) On a large scale, HF is prepared by heating calcium fluoride (fluorspar) with conc.H_2SO_4. The heating is required because of the endothermic nature of the reaction. It is necessary to remove impurities like SiO_2 as they consume the HF produced forming SiF_4 and H_2SiF_6.

$$CaF_2 + H_2SO_4 \rightarrow CaSO_4 + 2HF$$

The HF vapours are condensed in a well cooled lead receiver and then fractionally distilled to give HF of about 99.7% purity.

(ii) Another method of obtaining anhydrous acid is to distill freshly crystallized KHF_2 (potassium hydrogen fluoride) or by passing dry H_2 over dry AgF in a copper retort at red heat.

$$2KHF_2 \rightarrow 2KF + 2HF$$

$$2AgF + H_2 \rightarrow 2HF + 2Ag.$$

The vapours are condensed in a receiver cooled in freezing mixture.

Properties

Hydrogen fluoride is a colourless, strongly fuming liquid. It is highly corrosive and toxic. It boils at 19.5°C and has a freezing point –84°C. It is generally stored in copper mond metal containers as it attacks glass. The liquid acid and the vapours at low temperatures consist of polymeric form $(HF)_x$ whereas at 90°C it exists as monomer.

In aqueous solution, HF is weak acid, the dissociation reaction being

$$HF \rightleftharpoons H^{\oplus} + F^{\ominus} \quad K_1 \approx 10^{-3}$$

Another equilibrium present in HF solution is

$$F^{\ominus} + HF \rightarrow HF_2^{\ominus} \quad \log K = 0.7$$

Because of the tendency to form hydrogen bond, metal fluorides are solvated by HF to give species, $HF_2^{\ominus}$. Potassium salts of the type KF.HF, KF.2HF, KF.3HF are known.

Reactions

(i) Dry HF does not attack metals except potassium under normal conditions. In presence of water, it dissolves metals forming fluorides liberating hydrogen.

(ii) It forms two series of salts viz., acid salts and normal salts. Thus it acts like a weak dibasic acid.

$$NaOH + H_2F_2 \rightarrow NaHF_2 + H_2O$$

$$NaHF_2 + NaOH \rightarrow 2NaF + H_2O$$

(iii) Carbonates react with H_2F_2 to liberate CO_2.

$$CaCO_3 + H_2F_2 \rightarrow CaF_2 + H_2O + CO_2$$

(iv) Oxides also react with H_2F_2 to form the corresponding fluorides.

$$Na_2O + H_2F_2 \rightarrow 2NaF + H_2O$$

(v) The reaction involved in etching of glass is the formation of SiF_4 and H_2SiF_6.

$$SiO_2 + 4HF \rightarrow SiF_4 + 2H_2O$$

$$SiF_4 + 4HF \rightarrow 2H_2SiF_6$$

(vi) One way of detection of fluoride is by heating it with conc.H_2SO_4. When a glass rod dipped in water is exposed to the evolving SiF_4, a white deposit is produced due to the formation of silicic acid.

$$SiO_2 + 4HF \rightarrow SiF_4 + 2H_2O$$

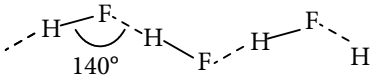
$$3SiF_4 + 3H_2O \rightarrow H_2SiO_3 + 2H_2SiF_6$$

Uses of HF

It is used for etching glass. Chloro fluorocarbon used as refrigeration fluid is made by the reaction of CCl_4 and HF. Anhydrous HF is used as catalyst for alkylation reaction in petroleum industry. Salts of HF like NaF are used as insecticides. CaF_2 is used as an additive in toothpastes to prevent the dental decay. UF_6 is used in the separation of Uranium isotopes.

Structure of HF

Vapour density and freezing point lowering of water by HF indicate the formula of hydrogen fluoride as H_2F_2 in aqueous solutions. It is represented as H_2F_2 in concentrated aqueous solutions. But in dilute aqueous solution, it consists mostly of HF molecules. Crystalline HF exists as associated species with zig-zag structure.



Hydrogen chloride, HCl

Preparation

(i) HCl is manufactured by what is known as the "salt cake" method. In this method, conc.H_2SO_4 is added to rock salt and the mixture is heated in two stages. The initial reaction at 150°C results in the formation of $NaHSO_4$, which forms a coating over the NaCl hindering further reaction. In the second stage, the reaction mixture is heated to a higher temperature of 550°C, which gives Na_2SO_4 and HCl. The reaction is carried in a salt cake muffle furnace.

$$NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl\uparrow$$

$$NaCl + NaHSO_4 \rightarrow Na_2SO_4 + HCl\uparrow$$

(ii) In the laboratory, HCl is prepared by the action of conc.H_2SO_4 on NH_4Cl. In this case, the reaction does not stop at half way stage because NH_4HSO_4 is water-soluble.

$$2NH_4Cl + H_2SO_4 \rightarrow 2(NH_4)_2SO_4 + 2HCl\uparrow$$

(iii) High purity HCl gas is however, prepared by buring hydrogen in an atmosphere of chlorine. The reaction

is highly exothermic (manufacture), and produces a white dazzling flame.

$$H_2 + Cl_2 \rightarrow 2HCl$$

Properties

HCl is a colourless, pungent smelling gas. It is highly soluble in water. A concentrated solution of HCl is about 12 N (about 38% by weight). It forms a constant boiling mixture with water and its composition is 35.4% by weight of HCl. Sometimes inferior grades of HCl are tinted yellow due to contamination by iron. It is a non-oxidising acid. It fumes in moist air. It is not a supporter of combustion and is not combustible too.

Reactions

(i) Gaseous HCl reacts with metals under heated conditions. However HCl solution attacks most metals in cold.

$$Fe + 2HCl \rightarrow FeCl_2 + H_2$$

(ii) Oxides, carbonates, sulphides, and sulphites react with HCl forming the corresponding chlorides.

$$MgO + 2HCl \rightarrow MgCl_2 + H_2O$$
$$ZnCO_3 + 2HCl \rightarrow ZnCl_2 + H_2O + CO_2\uparrow$$
$$FeS + 2HCl \rightarrow FeCl_2 + H_2S\uparrow$$
$$Na_2SO_3 + 2HCl \rightarrow 2NaCl + H_2O + SO_2\uparrow$$

(iii) HCl is oxidized to Cl_2 by atmospheric oxygen in presence of catalysts like copper salts.

$$4HCl + O_2 \rightarrow 2H_2O + 2Cl_2\uparrow$$

(iv) HCl reacts with fluorine to form HF.

$$F_2 + 2HCl \rightarrow 2HF + Cl_2\uparrow$$

(v) Metals such as Mg, Zn and Fe react with dilute HCl liberating hydrogen. Metals like tin and aluminium readily go into solution in warm concentrated acid. Copper metal forms cuprous chloride liberating H_2.

$$Mg + 2HCl \rightarrow MgCl_2 + H_2\uparrow$$
$$Zn + 2HCl \rightarrow ZnCl_2 + H_2\uparrow$$
$$2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2\uparrow$$
$$2Cu + 2HCl \rightarrow 2CuCl + H_2\uparrow$$

The Cu(I) chloride dissolves in excess conc.HCl to form a complex species.

$$CuCl + 2HCl \rightarrow H_2[CuCl_3]$$

(vi) An aqueous solution of HCl gives white precipitates of AgCl, $PbCl_2$ and Hg_2Cl_2 on treatment with silver, lead or mercurous salts respectively.

$$AgNO_3 + HCl \rightarrow AgCl\downarrow + HNO_3$$
$$Pb(NO_3)_2 + 2HCl \rightarrow PbCl_2\downarrow + 2HNO_3$$
$$Hg_2(NO_3)_2 + 2HCl \rightarrow Hg_2Cl_2\downarrow + 2HNO_3$$

Uses of HCl

It is used in removal of the oxide layer from metal surfaces, known as pickling of metal. It is also used for making metal chlorides, in dye stuff industry, in sugar industry, in the manufacture of chlorinated organic chemicals like vinyl chloride, chlorinated ethanes etc.

Hydrogen bromide, HBr

Preparation

(i) HBr is prepared by the reaction of concentrated phosphoric acid on a salt like KBr. Conc.H_2SO_4 cannot be used because it oxidizes HBr to bromine.

$$KBr + H_3PO_4 \rightarrow KH_2PO_4 + HBr$$

(ii) In the laboratory, HBr is prepared by the reduction of phosphorus with bromine in water.

$$P_4 + 6Br_2 \rightarrow 4PBr_3 \xrightarrow{+12H_2O} 12HBr + 2H_3PO_3$$

The vapours of HBr are passed through a U-tube containing phosphorus to free it from the vapours of bromine that may be carried along with HBr.

Properties

It is a colourless, pungent, fuming gas. In saturated solution, at room temperature it gives approximately a 5.5 molal solution. At low temperature and high pressure, it changes to liquid state which boils at −67°C and solidifies at −87°C. HBr forms a constant boiling mixture at 126°C at a composition of 48% HBr. The aqueous solution of the acid undergoes slight decomposition in sunlight to give bromine.

$$4HBr + O_2 \rightarrow 2Br_2 + 2H_2O$$

Reactions

(i) With silver, mercury(I) and lead salts, HBr gives sparingly soluble bromides.

$$AgNO_3 + HBr \rightarrow AgBr + HNO_3$$
$$Pb(NO_3)_2 + 2HBr \rightarrow PbBr_2 + 2HNO_3$$

(ii) Metals like zinc, tin and iron react with aqueous HBr to give the corresponding bromides and liberate hydrogen.

$$Zn + 2HBr \rightarrow ZnBr_2 + H_2\uparrow$$
$$Fe + 2HBr \rightarrow FeBr_2 + H_2\uparrow$$

(iii) HBr is oxidised by H_2O_2 or H_2SO_4 to bromine.

$$2HBr + H_2SO_4 \rightarrow 2H_2O + SO_2\uparrow + Br_2$$

$$2HBr + H_2O_2 \rightarrow Br_2 + 2H_2O$$

(iv) A bromide can be distinguished from chloride by the following tests.

(a) Br_2 vapours, which are dark red, are released when a bromide is heated with conc.H_2SO_4 and a little MnO_2, while chloride gives greenish yellow vapours.

(b) When Cl_2 is passed into a solution of bromide containing CCl_4 or $CHCl_3$, the bromine liberated turns the organic layer reddish brown, while a solution of chloride will remain colourless.

Hydrogen iodide, HI

The methods of preparation of HI are, in general, similar to those of HBr.

Preparation

(i) The reaction of a salt like NaI with concentrated phosphoric acid gives hydrogen iodide.

$$NaI + H_3PO_4 \rightarrow NaH_2PO_4 + HI$$

Conc.H_2SO_4 cannot be used in place of H_3PO_4, since it oxidises HI to I_2.

(ii) Another method involves the addition of water to a mixture of red phosphorus and iodine.

$$P_4 + 6I_2 \rightarrow 4PI_3$$

$$2PI_3 + 6H_2O \rightarrow 2H_3PO_3 + 6HI$$

Properties

HI is a colourless, pungent and irritating gas. It fumes in moist air. It is highly soluble in water. It forms a constant boiling mixture with water having a composition of 57% by weight of the acid. It can be liquefied (boiling point of –35.5°C) and solidified (melting point of –50.8°C).

Reactions

(i) HI decomposes into H_2 and I_2 on heating.

$$2HI \rightleftharpoons H_2 + I_2$$

(ii) Hydrogen iodide is a strong reducing agent. In most reactions, it gets oxidized to iodine.

(a) HI is oxidised by O_2 to iodine

$$4HI + O_2 \rightarrow 2H_2O + 2I_2$$

(b) H_2O_2 is reduced by HI to water

$$H_2O_2 + 2HI \rightarrow 2H_2O + I_2$$

(c) HNO_3 is reduced to NO by HI

$$2HNO_3 + 6HI \rightarrow 4H_2O + 2NO + I_2$$

(d) Ferric ions are reduced to Fe^{++} ions by HI.

$$Fe_2(SO_4)_3 + 2HI \rightarrow 2FeSO_4 + H_2SO_4 + I_2$$

(e) Copper (II) sulphate reacts with HI to form Cu (I) iodide.

$$2CuSO_4 + 4HI \rightarrow 2CuI + 2H_2SO_4 + I_2$$

HI finds some application as a reducing agent in organic reactions.

Some general considerations of hydrogen halides

(i) HF is a liquid at room temperature whereas HCl, HBr, HI are gases.

(ii) The boiling points increase from HCl to HI but HF has a much higher boiling point than others. This high boiling point of HF is due to the considerable hydrogen bonding in liquid HF. Such hydrogen bonding does not exist in HCl and HBr in the gas phase or liquid phase.

(iii) The hydrogen halides are covalent in gaseous state. In aqueous solutions, HCl, HBr, HI are completely dissociated but HF is only a weak acid.

Oxides of chlorine

Chlorine forms several oxides viz,

(i) dichlorine monoxide, Cl_2O
(ii) chlorine dioxide, ClO_2
(iii) dichlorine tetroxide, Cl_2O_4
(iv) dichlorine hexoxide, Cl_2O_6
(v) dichlorine heptoxide, Cl_2O_7

A detailed consideration of the preparation, properties of some of the oxides is presented below:

Dichlorine monoxide, Cl_2O

Preparation

It is prepared by treating freshly prepared yellow mercuric oxide with chlorine gas or with a solution of Cl_2 in CCl_4.

$$2HgO + 2Cl_2 \rightarrow HgCl_2.\ HgO + Cl_2O\uparrow$$

Properties

Cl_2O is yellowish brown gas at room temperature with a characteristic odour. It condenses to an orange liquid, which has a boiling point of 2°C.

Reactions

(i) It is readily soluble in water giving hypochlorous acid.

$$Cl_2O + H_2O \rightarrow 2HOCl$$

(ii) It dissolves in alkali forming hypochlorite.

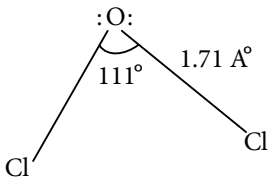
$$Cl_2O + 2NaOH \rightarrow 2NaOCl + H_2O$$

(iii) It is thermally unstable. When heated, it explodes giving Cl_2 and O_2.

$$2Cl_2O \rightarrow 2Cl_2 + O_2$$

Structure of Cl_2O

The oxygen uses sp^3 hybridized orbitals for bonding. It has a tetrahedral structure with two positions occupied by lone pairs. However Cl – O – Cl angle is more than the tetrahedral angle, being equal to 111°. This is because of steric crowding of the large halogen atoms. The O – Cl bond length is 1.71 A°. The structure may be represented as shown below



Chlorine dioxide, ClO_2

Preparation

(i) By the action of conc.H_2SO_4 on potassium chlorate, ClO_2 is liberated.

$$3KClO_3 + 3H_2SO_4 \rightarrow HClO_4 + 3KHSO_4 + H_2O + 2ClO_2\uparrow$$

(ii) Another method of preparation of ClO_2 involves the passage of dry chlorine gas over silver chlorate at 90°C. The resulting gaseous mixture is condensed in a freezing mixture to get ClO_2 gas condensed to a liquid.

$$2AgClO_3 + Cl_2 \rightarrow 2AgCl + O_2 + 2ClO_2\uparrow$$

Properties

It is a yellow gas and can be condensed to a red liquid with a boiling point of 11°C. It can be solidified at –79°C to orange red crystals. It is soluble in water and its aqueous solution is stable in dark but decomposes in light to give $HClO_2$ and $HClO_3$.

$$2ClO_2 + H_2O \rightarrow HClO_2 + HClO_3$$

Reactions

(i) It is dissolved in KOH to form potassium chlorite and potassium chlorate.

$$ClO_2 + 2KOH \rightarrow KClO_2 + KClO_3 + H_2O$$

(ii) ClO_2 is a powerful oxidizing and chlorinating agent. It reacts with ozone to form oxygen and Cl_2O_6.

$$2ClO_2 + 2O_3 \rightarrow Cl_2O_6 + 2O_2$$

(iii) It liberates iodine from acidified KI solution.

$$2ClO_2 + 10HI \rightarrow 2HCl + 4H_2O + 5I_2$$

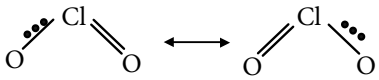
Uses

It is used for bleaching wood pulp and cellulose. It finds application as a chlorinating agent. Another important use of ClO_2 is for the manufacture of $NaClO_2$, which is also employed for bleaching textiles and paper.

$$2ClO_2 + 2NaOH + H_2O_2 \rightarrow 2NaClO_2 + 2H_2O + O_2$$

Structure of ClO_2

ClO_2 is paramagnetic and contains odd number of electrons (three electron bond). It is angular with the O – Cl – O angle being 118°. The O – Cl bond length is 1.47 A°.



Dichlorine Tetroxide, Cl_2O_4 ($ClO.ClO_3$)

This is also referred to as chlorine perchlorate ($Cl.ClO_4$).

Preparation

It is prepared by the reaction of cesium perchlorate with chlorofluoro sulfonic acid at –45°C.

$$CsClO_4 + ClSO_3F \rightarrow ClO.ClO_3 + Cs(SO_3)F$$

Properties

At room temperature, this gas is stable only for a short period of time and decomposes to O_2, Cl_2 and Cl_2O_6. It is also less stable than ClO_2.

$$3ClO.ClO_3 \rightarrow 2Cl_2 + Cl_2O_6 + 3O_2$$

Dichlorine hexoxide, Cl_2O_6

Preparation

It is prepared by the action of ozone and ClO_2 at 0°C.

$$2ClO_2 + 2O_3 \rightarrow Cl_2O_6 + 2O_2$$

Properties

It is a dark red oily liquid. It freezes to a yellow solid at –180°C. It is unstable and decomposes at its melting point into ClO_2 and O_2.

$$Cl_2O_6 \rightarrow 2ClO_2 + O_2$$

Both solid and liquid are diamagnetic and do not contain unpaired electrons. Its vapours are paramagnetic due to its partial dissociation to chlorine trioxide having an unpaired electron.

$$Cl_2O_6 \rightleftharpoons 2ClO_3$$

Reactions

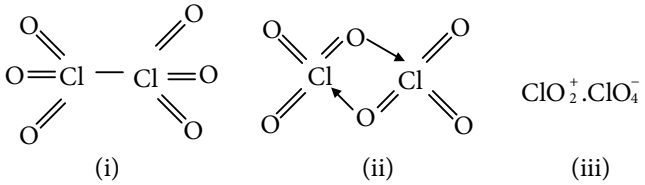
(i) It is hydrolysed by water to give chloric and perchloric acids.

$$Cl_2O_6 + H_2O \rightarrow HClO_3 + HClO_4$$

(ii) With NaOH, Cl_2O_6 gives sodium chlorate and perchlorate.

$$Cl_2O_6 + 2NaOH \rightarrow NaClO_3 + NaClO_4 + H_2O$$

Structure



Chlorine in Cl_2O_6 molecule appears to be sp^3 hybridized.

Dichlorine heptoxide, Cl_2O_7

Preparation

This compound is prepared by dehydration of perchloric acid with P_2O_5 or phosphoric acid at –10°C, followed by distillation at –35°C and a pressure of 1 mm of Hg.

$$2HClO_4 \xrightarrow[-10^0\,C]{P_2O_5} Cl_2O_7 + H_2O$$

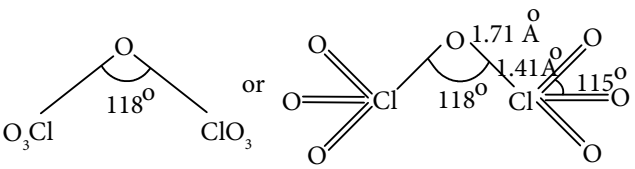
Properties

It is a colourless oily liquid with a boiling point of 82°C. It is the most stable of all chlorine oxides. However, it is sensitive to shocks and explodes on heating. It reacts with water forming perchloric acid.

$$Cl_2O_7 \underset{P_4O_{10}}{\overset{H_2O}{\rightleftharpoons}} 2HClO_4$$

Structure

It has a bent structure with a bond angle of 118° at the central oxygen. The structure may be given as



Oxyacids of chlorine

Chlorine forms four oxyacids (or oxoacids). They are,

(i) Hypochlorous acid (HOCl)
(ii) Chlorous acid ($HClO_2$)
(iii) Chloric acid ($HClO_3$)
(iv) Perchloric acid ($HClO_4$)

The oxidation states of chlorine in each of these acids are +1 (in HOCl), +3 (in $HClO_2$), +5 (in $HClO_3$) and +7 (in $HClO_4$).

In the case of other halogens, fluorine forms only HOF (hypofluorous acid), bromine and iodine form all acids other than HXO_2 (X = Br, I). Some general observations on these acids, which are of interest are

(i) The hydrogen of these acids is part of an –OH group and is not directly attached to the halogen atom.

(ii) In a given series of acids of a given halogen, say chlorine atom, the acidic nature and thermal stability increases with the increase in the oxidation state of halogen (ie, $HOCl < HClO_2 < HClO_3 < HClO_4$).

(iii) Considering different halogens in the same oxidation state, the acidic character and thermal stability of series of oxyacids decrease with increasing atomic weight.

Hypochlorous acid (chloric(I) acid), HOCl

Preparation

(i) It is known only in solution. HOCl is prepared by passing chlorine into a freshly prepared and well agitated suspension of mercuric oxide in water.

$$2HgO + 2Cl_2 + H_2O \rightarrow HgO.HgCl_2 + 2HOCl$$

The liquid is then distilled over to obtain a dilute solution of the acid (about 25%).

(ii) Another method of preparation consists of passing CO_2 through a suspension of bleaching powder in water.

$$2CaCl(OCl) + H_2O + CO_2 \rightarrow CaCl_2 + CaCO_3 + 2HOCl$$

(iii) Hypochlorous acid is also formed in aqueous solutions of chlorine. In such a solution, not only solvated Cl_2 molecules are present but also HCl and HOCl, due to rapid disproportionation reaction as shown below.

$$Cl_{2(g)} + H_2O \rightarrow Cl_{2(aq)}$$

$$Cl_{2(aq)} \rightleftharpoons HCl + HOCl$$

Properties

The acid is known only in aqueous solutions. It is a weak acid with a dissociation constant of 3.4×10^{-8} at 20°C and is not very stable. The acid decomposes under heat and light to give Cl_2 and O_2.

$$2HOCl \rightarrow 2HCl + O_2$$

$$HCl + HOCl \rightarrow H_2O + Cl_2$$

Reactions

(i) HOCl is a powerful oxidizing agent as well as a bleaching agent because it liberates nascent oxygen.

$$HOCl \rightarrow HCl + (O)$$

(ii) Hypochlorous acid can be distinguished from chlorine water by its action on mercury. Mercury reacts with HOCl to give light brown precipitate of basic mercuric chloride, which dissolves in HCl.

$$Hg + HOCl \rightarrow Hg(OH)Cl$$

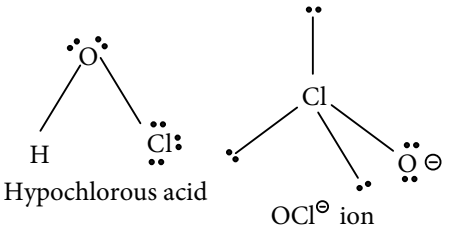
$$Hg(OH)Cl + HCl \rightarrow HgCl_2 + H_2O$$

Chlorine water, on the other hand, reacts with mercury to give white mercurous chloride, which is insoluble in HCl.

$$2Hg + Cl_2 \rightarrow Hg_2Cl_2$$

Structure of HOCl

The HOCl is considered to have a distorted tetrahedral structure. The sp^3 hybrid orbitals of the halogen atoms form fairly weak σ bonds since 's' and 'p' orbitals have appreciable energy difference. The ion OCl^- is stable due to pπ – dπ bonding between filled 2p orbitals of oxygen atom with vacant 'd' orbitals of halogen atom. The structures of HOCl and OCl^- ion may be given as



Salts of hypochlorous acid

Among the salts of hypochlorous acid, sodium hypochlorite is an important one and its preparation and properties are briefly discussed below.

Preparation

When Cl_2 is passed into a cold, dilute solution of NaOH or Na_2CO_3, NaOCl is formed.

$$2NaOH + Cl_2 \rightarrow NaCl + NaOCl + H_2O$$

On a commercial scale, NaOCl is prepared by electrolysis of cold brine under effective stirring conditions. At cathode, hydrogen gas is liberated. At anode, the liberated chlorine reacts with OH^- ions to give Cl^- and OCl^- ions.

$$2Cl^- \rightarrow Cl_2 + 2e^-$$

$$Cl_2 + 2OH^- \rightarrow Cl^- + OCl^- + H_2O$$

Reactions

(i) A solution of NaOCl decomposes to NaCl and $NaClO_3$ on heating above 80°C.

$$3NaOCl \xrightarrow{\Delta} NaClO_3 + 2NaCl$$

(ii) Being a good oxidizing agent, NaOCl oxidises HCl to Cl_2.

$$2HCl + NaOCl \rightarrow NaCl + H_2O + Cl_2$$

(iv) Potassium iodide is oxidized to iodine by NaOCl.

$$NaOCl + 2KI + 2HCl \rightarrow NaCl + 2KCl + H_2O + I_2$$

(v) It oxidizes $Cr(OH)_3$ to chromate in alkaline solutions.

$$3NaOCl + 2Cr(OH)_3 + 4NaOH \rightarrow 2Na_2CrO_4 + 3NaCl + 5H_2O$$

Uses

It is used for bleaching cotton fabrics. It also finds application as a disinfectant and a sterilizing agent.

Chlorous acid (chloric (III) acid), $HClO_2$

Preparation

This acid is known only in solution. It is prepared by treating barium chlorite with H_2SO_4 and filtering off the $BaSO_4$ formed.

$$Ba(ClO_2)_2 + H_2SO_4 \rightarrow BaSO_4\downarrow + 2HClO_2$$

Properties

The freshly prepared solution is colourless but it soon decomposes to ClO_2 which makes the solution yellow.

$$5HClO_2 \rightarrow 4ClO_2 + 4HCl + 2H_2O$$

It is a moderately weak acid with $K_a = 10^{-2}$. Chlorous acid undergoes auto oxidation to give HOCl and $HClO_3$.

$$2HClO_2 \rightarrow HOCl + HClO_3$$

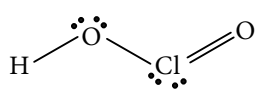
Alkali chlorites are excellent bleaching agents. In alkaline medium, chlorites are relatively stable but are decomposed by acids.

When heated they give chlorates and chlorides.

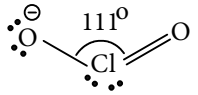
$$4HClO_2 \rightarrow 2ClO_2 + ClO_3^- + Cl^- + 2H^+ + H_2O$$

Structure of chlorous acid and chlorite ion

Chlorite ion has an angular shape which results from the sp^3 hybridization of chlorine atom. In chlorite ion the O – Cl – O bond angle is 111° and (due to resonance) both the O – Cl bond lengths are 1.64 A°



Structure of chlorous acid



Structure of chlorite ion

Chloric acid (chloric(V) acid), $HClO_3$

Chloric acid is prepared by heating barium chlorate with H_2SO_4 and filtering off the $BaSO_4$ formed. After barium sulphate is filtered off, the filtrate is evaporated in vacuum dessicator over conc.H_2SO_4 till about 40 percent concentration is reached.

$$Ba(ClO_3)_2 + H_2SO_4 \rightarrow BaSO_4\downarrow + 2HClO_3$$

Another method is based on the reaction of hexafluorosilicic acid (H_2SiF_6) with potassium chlorate.

$$H_2SiF_6 + 2KClO_3 \rightarrow 2HClO_3 + K_2SiF_6\downarrow$$

Properties

Chloric acid is a colourless liquid with a pungent smell. It is stable in the dark but gradually decomposes in sunlight.

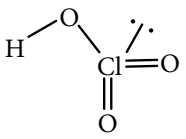
$$3HClO_3 \rightarrow HClO_4 + H_2O + Cl_2 + 2O_2$$

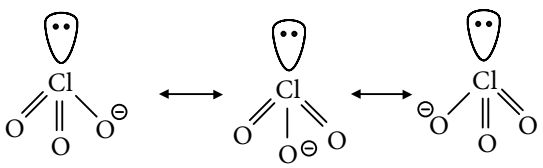
$HClO_3$ decomposes violently when heated to dryness.

$$4HClO_3 \rightarrow 4ClO_2 + 2H_2O + O_2$$

Structure of chloric acid and chlorate ion

Chlorine is sp^3 hybridized in chloric acid and chlorate ion. ClO_3^- ion has pyramidal shape and is the resonance hybride of the following structures.





Iodic and Bromic acids

Among the halic acids, iodic acid, HIO_3 is stable and it exists as a white solid. It is obtained by oxidising I_2 with conc.HNO_3 or ozone.

$$I_2 + 10HNO_3 \rightarrow 2HIO_3 + 10NO_2 + 4H_2O$$

$$I_2 + H_2O + 5O_3 \rightarrow 2HIO_3 + 5O_2$$

Bromic acid, $HBrO_3$ may be prepared in a similar manner to that of $HClO_3$.

All halic acids are strong acids and are powerful oxidizing agents.

Salts of chloric acid (chlorates)

Preparation

Two methods are available for the preparation of chlorates. The preparation of $NaClO_3$ will be considered as an example

(i) By passing chlorine into a hot solution of NaOH at 80°C, $NaClO_3$ is obtained.

$6NaOH + 3Cl_2 \rightarrow NaClO_3 + 5NaCl + 3H_2O$

(ii) By electrolysis of a vigorously stirred and hot solution of chloride, $NaClO_3$ is formed.

$2Cl^- \rightarrow Cl_2 + 2e^-$

$2H_2O + 2e^- \rightarrow 2OH^- + H_2$

$6NaOH + 3Cl_2 \rightarrow NaClO_3 + 5NaCl + H_2O$

Sodium chlorate is used widely as a weed killer. Potassium chlorate is prepared in a similar manner.

Properties and reactions

As a typical example, the properties of $KClO_3$ is considered. It is a white crystalline solid. It is sparingly soluble in cold water but the solubility increases with temperature.

(i) It decomposes on heating in the temperature range of 400 – 500°C to liberate oxygen. MnO_2 acts as a catalyst for this reaction.

$2KClO_3 \rightarrow 2KCl + 3O_2$

(ii) On heating with hydrochloric acid, $KClO_3$ forms ClO_2.

$2KClO_3 + 4HCl \rightarrow 2KCl + 2ClO_2 + Cl_2 + 2H_2O$

(iii) $KClO_3$ reacts with conc.H_2SO_4 with the liberation of ClO_2.

$3KClO_3 + 3H_2SO_4 \rightarrow 3KHSO_4 + HClO_4 + H_2O + 2ClO_2$

(iv) SO_2 reduces $KClO_3$ to KCl

$KClO_3 + 3SO_2 + 3H_2O \rightarrow KCl + 3H_2SO_4$

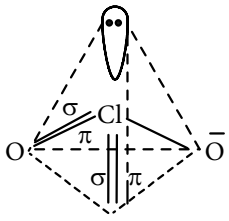
(v) Iodine in presence of HNO_3 evolves chlorine from $KClO_3$.

$2KClO_3 + I_2 \rightarrow 2KIO_3 + Cl_2$

Solid potassium chlorate yields an explosive mixture with sulphur or phosphorus and even with charcoal. It is used in the manufacture of matches and in firework materials.

Structure of chlorate ion

The central chlorine atom is sp^3 hybridized and one of the four hybrid orbitals is occupied by a lone pair. Three oxygen atoms form three σ bonds with the single electron in the three other sp^3 hybrid orbitals of chlorine atom. In addition, two pπ-dπ bonds are formed between the single electrons in the d-orbitals of chlorine and p-orbitals of oxygen. The ion thus has two double bonds and one single bond. It has a trigonal pyramidal shape. The structure may be represented as



Structure of ClO_3^- ion

Perchloric acid (chloric(VII) acid), $HClO_4$

Preparation

(i) When $NaClO_4$ is warmed with conc.HCl, $HClO_4$ is formed.

$NaClO_4 + HCl \rightarrow NaCl + HClO_4$

(ii) Ammonium perchlorate, on treatment with dil.HNO_3, gives $HClO_4$

$NH_4ClO_4 + HNO_3 \rightarrow NH_4NO_3 + HClO_4$

(iii) Chloric acid, on heating, decomposes to give $HClO_4$

$3HClO_3 \rightarrow HClO_4 + H_2O + Cl_2 + 2O_2$

Properties

$HClO_4$ is a colourless, mobile hygroscopic liquid. Commercially it is available in concentration of 70-72% by weight. It forms a constant boiling mixture with water with the concentration of the acid being around 72.5 wt%. It is one of

the strongest acids known. It is a powerful oxidizing agent and explodes in contact with organic matter like wood, pulp and grease.

Metals like iron and zinc react with dilute solutions of the acid forming the corresponding perchlorates.

$$Zn + 2HClO_4 \rightarrow Zn(ClO_4)_2 + H_2$$

The oxidizing action of the acid is due to the liberation of O_2 on heating the acid.

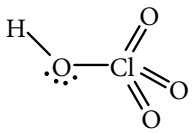
$$4HClO_4 \rightarrow 2Cl_2 + 2H_2O + 7O_2$$

When dehydrated with P_4O_{10} it yields Cl_2O_7.

$$4HClO_4 + P_4O_{10} \rightarrow 4HPO_3 + 2Cl_2O_7$$

Structure of $HClO_4$

Cl is sp^3 hybridized.



Uses

It finds application as an oxidant in analytical chemistry. It is used for destroying any organic matter in qualitative inorganic analysis.

Salts of perchloric acid

Potassium perchlorate, $KClO_4$

(i) By heating $KClO_3$, it first fuses, but the fused mass on further heating is converted to $KClO_4$.

$$4KClO_3 \rightarrow 3KClO_4 + KCl$$

The resulting mixture is dissolved in water and the $KClO_4$ is separated because it is only sparingly soluble in water.

(ii) It is also made by the anodic oxidation of $NaClO_3$ to $NaClO_4$. The sodium perchlorate is heated with KCl solution to form sparingly soluble $KClO_4$. It is filtered and dried. $KClO_4$ is used in making fireworks and flames.

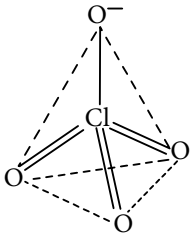
$$NaClO_3 + H_2O \xrightarrow{\text{at anode}} NaClO_4 + 2H^+ + 2Cl^-$$

$$NaClO_4 + KCl \rightarrow KClO_4 + NaCl$$

Other perchlorate of importance are $Mg(ClO_4)_2$ and NH_4ClO_4. Magnesium perchlorate is used as a desiccant and as electrolyte in dry batteries. NH_4ClO_4 finds application in rockets.

Structure of ClO_4^- ion

The ion has a tetrahedral shape with chlorine atom having sp^3 hybrid orbitals. The structure of this ion may be represented as shown below.



Structure of ClO_4^- ion

It is a resonance hybrid of four structures. This accounts for the fact that all Cl – O bond lengths in ClO_4^- are exactly the same.

Bleaching powder, Ca(OCl)Cl

Preparation

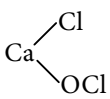
It is prepared by the action of chlorine on moderately dry slaked lime at temperature below 40°C. A small amount of water (about 4%) is actually necessary for the reaction. The gas is absorbed to the extent of about 36%.

$$Ca(OH)_2 + Cl_2 \rightarrow Ca(OCl)Cl + H_2O$$

The bleaching powder is a white powdery material with an irritating smell. It possesses both disinfecting and bleaching properties. It has an approximate composition of $Ca(OCl)_2.CaCl_2.Ca(OH)_2.2H_2O$.

Properties

Bleaching powder is an unstable compound and smells strongly of chlorine. It is soluble in water but a clear solution is never formed due to impurities. It decomposes gradually in presence of moist air. The products of decomposition are $Ca(OH)_2$ and Cl_2. Generally the amount of available chlorine in bleaching powder is of the order of 38%. $CaOCl_2$ is considered as a mixed salt of hydrochloric acid and hypochlorous acid is represented by



Reactions

(i) It is easily decomposed by dilute acids like HCl, H_2SO_4 and CO_2. Cl_2 is liberated.

$$Ca(OCl)Cl + 2HCl \rightarrow CaCl_2 + H_2O + Cl_2$$

$$Ca(OCl)Cl + CO_2 \rightarrow CaCO_3 + Cl_2$$

The bleaching action of bleaching powder is due to the chlorine liberated during the above reaction.
The amount of chlorine obtained from a sample of bleaching powder by treatment with excess of dilute acids is called available chlorine, which is about 33-38% in commercial samples.

(ii) Bleaching powder reacts with an acidified solution of KI liberating iodine

$$Ca(OCl)Cl + 2KI + 2HCl \rightarrow CaCl_2 + 2KCl + H_2O + I_2$$

This reaction forms the basis for the estimation of available chlorine in bleaching powder and thereby determine its effectiveness as a bleaching agent and as a disinfectant.

(iii) On long standing, bleaching powder decomposes to give calcium chlorate and $CaCl_2$.

$$6Ca(OCl)Cl \rightarrow Ca(ClO_3)_2 + 5CaCl_2$$

(iv) In solution, it decomposes into calcium hypochlorite and $CaCl_2$.

$$2Ca(OCl)Cl \rightarrow Ca(OCl)_2 + CaCl_2$$

(vii) When bleaching powder reacts with NH_4OH, N_2 is liberated.

$$3Ca(OCl)Cl + 2NH_4OH \rightarrow 3CaCl_2 + 5H_2O + N_2$$

(viii) Acetone reacts with bleaching powder to form chloroform .

$$3Ca(OCl)Cl + 2CH_3COCH_3 \rightarrow$$

$$2CHCl_3 + (CH_3COO)_2Ca + 2Ca(OH)_2$$

(ix) Bleaching powder is decomposed catalytically by CoO or $CoCl_2$ in aqueous solution to liberate oxygen.

$$2CaOCl_2 \xrightarrow{CoO} 2CaCl_2 + O_2$$

Uses

Bleaching powder is mainly used as a bleaching agent for paper and textiles. It is also used as disinfectant and germicide.

XENON FLUORIDES

Introduction

Xenon (rare gas with atomic number 54) forms three binary compounds with fluorine, namely, xenon difluoride (XeF_2), xenon tetrafluoride (XeF_4) and xenon hexafluoride (XeF_6). The oxidation states of xenon in these compounds are +2, +4 and +6 respectively. A brief account of their preparation and properties are given below:

Xenon difluorides, XeF_2

Preparation

(i) When a mixture of xenon and fluorine in the ratio of 2 : 1 is heated at 400°C in a sealed nickel vessel, XeF_2 is formed.

$$Xe + F_2 \rightarrow XeF_2.$$

(ii) Another method of preparing XeF_2 is by irradiating equimolar mixture of xenon and fluorine with light from a high pressure mercury lamp.

$$Xe + F_2 \xrightarrow{\text{mercury lamp}} XeF_2$$

(iii) Dioxygen difluoride reacts with Xe at about 118°C to give XeF_2.

$$Xe + O_2F_2 \rightarrow XeF_2 + O_2.$$

Properties

It is a white crystalline solid having a melting point of 129°C. It can be sublimed at room temperature. It acts as a good fluorinating as well as oxidizing agent.

Reactions

(i) XeF_2 can fluorinate compounds like ethylene, propylene and benzene.

$$2XeF_2 + 2CH_2 = CH_2 \rightarrow CH_2F - CH_2F + CH_3CHF_2 + 2Xe$$

$$C_6H_6 + XeF_2 \rightarrow C_6H_5F + HF + Xe$$

(ii) Inorganic compounds like NO and SO_3 can also be fluorinated. For example,

$$2NO + XeF_2 \rightarrow 2NOF + Xe$$

$$2SO_3 + XeF_2 \rightarrow S_2O_6F_2 + Xe$$

(iii) XeF_2 oxidizes a large number of compounds and it itself gets reduced to Xe. For example, H_2 is oxidized to HF, I_2 to IF, Cl^- ions to Cl_2 etc.

$$H_2 + XeF_2 \rightarrow 2HF + Xe$$

$$I_2 + XeF_2 \xrightarrow{BF_3} 2IF + Xe$$

$$2Cl^- + XeF_2 \rightarrow Cl_2 + 2F^- + Xe$$

(iv) XeF_2 is soluble in water or in acid solutions but it undergoes slow hydrolysis.

$$2XeF_2 + 2H_2O \rightarrow 4HF + O_2 + 2Xe$$

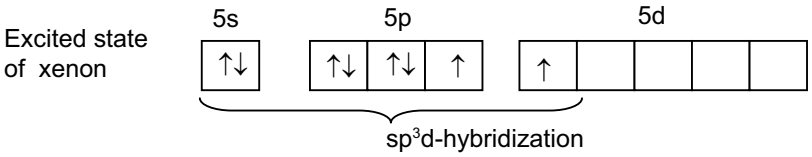
The hydrolysis is faster in alkaline solutions.

(v) XeF_2 forms addition compounds or adducts with pentafluorides of P, As and Sb. In these reactions, XeF_2 acts as a Lewis base by donating a fluoride ion and the pentafluoride acts as a Lewis acid by accepting fluoride ion. For example,

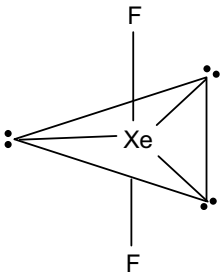
$$XeF_2 + PF_5 \rightarrow XeF_2.PF_5 \text{ or } XeF^+ PF_6^-$$

Structure of XeF_2

XeF_2 is a linear molecule with sp^3d hybridization of Xe and both the Xe – F distance being 2 A°. The structure may be explained in simple manner on the basis of the promotion of an electron from 5p level of Xe to 5d level as shown below:



The two unpaired electrons form bonds with two fluorine atoms giving XeF_2. There are five electrons pairs of which two are bond pairs and three are lone pairs. The three lone pairs occupy equitorial positions and the two bond pairs occupy apex positions as shown in the diagram below:



Structure of XeF_2

Xenon tetrafluoride, XeF_4

Preparation

When xenon and fluorine in the ratio 1 : 5 are heated at 400°C in a nickel vessel under a pressure of 6 atm for few hours, XeF_4 is formed.

$$Xe + 2F_2 \rightarrow XeF_4$$

Properties

It is a white solid having a melting point of 117°C. It can be sublimed at room temperature and stored indefinitely in nickel container. It is a good fluorinating as well as a good oxidizing agent.

Reactions

(i) Hydrogen is oxidised to HF

$$XeF_4 + 2H_2 \rightarrow 4HF + Xe$$

(iv) A vigorous reaction occurs with water resulting in the formation of highly explosive XeO_3.

$$6XeF_4 + 12H_2O \rightarrow 2XeO_3 + 24HF + 3O_2 + 4Xe$$

(vi) The fluorinating capacity of XeF_4 is revealed by the following reactions.

(a) Pt metal reacts with XeF_4 to give PtF_4

$$Pt + XeF_4 \rightarrow PtF_4 + Xe$$

(b) Benzene, ethylene, NO, NO_2 are fluorinated by XeF_4 to give the corresponding fluoride.

$$XeF_4 + 2C_6H_6 \rightarrow 2C_6H_5F + 2HF + Xe$$

$$XeF_4 + 2C_2H_4 \rightarrow FCH_2 - CH_2F + CH_3CHF_2 + Xe$$

$$4NO + XeF_4 \rightarrow 4NOF + Xe$$

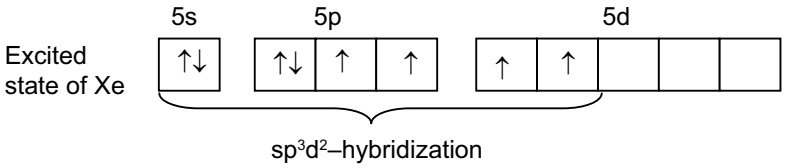
$$4NO_2 + XeF_4 \rightarrow \underset{\text{Nitryl fluoride}}{4NO_2F} + Xe$$

(vii) XeF_4 also forms addition compounds with pentafluorides of P, As, Sb and transition elements to form various adducts.

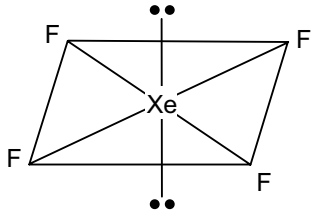
$$XeF_4 + SbF_5 \rightarrow XeF_4.SbF_5 \text{ or } [XeF_3]^+ [SbF_6]^-$$

Structure of XeF_4

XeF_4 has a square planar structure with the Xe–F distances of 1.95 A°. On the basis of valence bond theory, it may be explained by considering the excitation of two electrons from the 5p level of Xe to 5d level as shown below:



The four unpaired electrons form four bonds with four fluorine atoms. There are now six electron pairs, of which four are bonded to four fluorine atoms in equatorial positions and the remaining two lone pairs are in the vertical (or apical) position. The structure is shown as



Structure of XeF_4

Xenon hexafluoride, XeF_6

Preparation

Xenon and fluorine are mixed in the ratio of 1 : 20 and heated in a sealed nickel vessel at 400°C at 5 atm pressure. XeF_6 is formed, with a yield of about 95%.

$$Xe + 3F_2 \rightarrow XeF_6$$

Properties

It is a white solid having a melting point of 49.6°C. It can be sublimed at room temperature. XeF_6 is very susceptible to hydrolysis and hence even traces of water must be eliminated during storage. It acts as a strong oxidising as well as a fluorinating agent.

Reactions

(i) It oxidises hydrogen to HF

$$XeF_6 + 3H_2 \rightarrow Xe + 6HF$$

(ii) It reacts violently with water forming XeO_3

$$XeF_6 + 3H_2O \rightarrow XeO_3 + 6HF$$

The same reaction takes place slowly in the presence of moisture in air. XeO_3 is a white hygroscopic solid and explosive in nature.

(iii) In presence of small amounts of water, partial hydrolysis takes place resulting in the formation xenon oxofluorides, $XeOF_4$ and XeO_2F_2 . $XeOF_4$ is formed when XeF_6 reacts with silica or glass.

$$XeF_6 + H_2O \rightarrow XeOF_4 + 2HF$$

$$2XeF_6 + SiO_2 \rightarrow XeOF_4 + SiF_4$$

$$XeF_6 + 2H_2O \rightarrow XeO_2F_2 + 4HF$$

(iv) It oxidises ammonia to nitrogen.

$$XeF_6 + 8NH_3 \rightarrow 6NH_4F + N_2 + Xe$$

(vi) XeF_6 forms addition compounds with compounds like AsF_5, PF_5, BF_3. It functions as a fluoride ion donor.

$$XeF_6 + AsF_5 \rightarrow XeF_6.AsF_5 \text{ or } [XeF_5]^+ [AsF_6]^-$$

$$XeF_6 + BF_3 \rightarrow XeF_6.BF_3 \text{ or } [XeF_5]^+ [BF_4]^-$$

(vii) XeF_6 can also act as a fluoride ion acceptor.

$$XeF_6 + RbF \rightarrow Rb^+ [XeF_7]^-$$

(ix) In strongly basic solution, XeF_6 is converted to hydrogen xenate ion, $HXeO_4^-$, which then disproportionates to perxenate ion XeO_6^{4-} and Xe.

$$XeF_6 + 3H_2O \rightarrow XeO_3 + 6HF$$

$$XeO_3 + OH^- \rightarrow HXeO_4^-$$

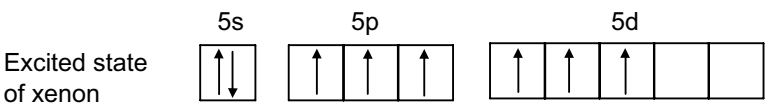
$$2HXeO_4^- + 2OH^- \rightarrow XeO_6^{4-} + Xe + O_2 + 2H_2O$$

(x) XeF_6 attacks glass and silica and the product of such a reaction is SiF_4.

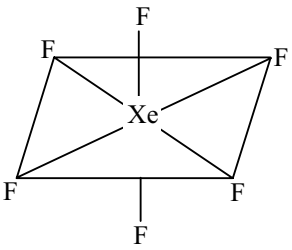
$$2XeF_6 + 3SiO_2 \rightarrow 3SiF_4 + 2XeO_3$$

Structure of XeF_6

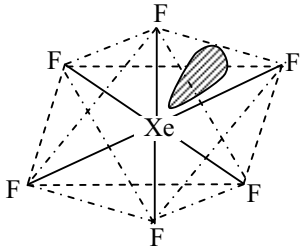
XeF_6 is considered to have the structure of a distorted octahedron which may be explained as follows: The six electrons in the 5p level of Xe are assumed to be promoted to the 5d level to give the following electronic distribution in the excited state of Xe on the basis of valance bond theory.



Six fluorine atoms form bonds with six unpaired electrons forming XeF_6. There are thus seven electron pairs, one of them being a lone pair, distributed in seven orbitals. One would expect an sp^3d^3–hybridisation leading to a pentagonal bipyramidal structure but the hybridisation theory fails in this case. The structure may be described as a distorted octahedron. The lone pair points through one of the faces of the octahedron. The structure may be designated as



Structure of XeF_6
(lone pair not shown)



Distorted Octahedron

CONCEPT STRAND

Concept Strand 12

What happens when XeF_4 is hydrolysed by water? Write the equation for the reaction.

Solution

A disproportionation reaction takes place.

$$6XeF_4 + 12H_2O \rightarrow 4Xe + 2XeO_3 + 24HF + 3O_2$$

COMMERCIALLY AVAILABLE FERTILIZERS

Introduction

Fertilizers are chemical compounds added to improve the fertility of the soil by making up any deficiency of the essential elements like nitrogen, phosphorous and potassium. When they are added in appropriate quantity, they enhance the yield of the crops. Fertilizers are generally compounds containing nitrogen, phosphorous and potassium. Fertilizers are classified into five categories.

(i) Nitrogenous fertilizers
(ii) Phosphatic or phosphorus containing fertilizers
(iii) Potassium containing fertilizers
(iv) NP fertilizers containing mainly the two elements nitrogen and phosphorus
(v) NPK type fertilizers which contain all the three essential elements viz nitrogen, phosphorous and potassium.

A brief account of the important fertilizers in the various classes is presented, below.

Nitrogenous fertilizers

These fertilizers are commonly used in the combined form as ammonium nitrate (NH_4NO_3), ammonium sulphate ($(NH_4)_2SO_4$) or as urea (NH_2CONH_2). They are essential for plant growth since nitrogen is an important constituent of amino acid and proteins.

$(NH_4)_2SO_4$

The important gas required in the manufacture of $(NH_4)_2SO_4$, as employed at the Sindri fertilizers factory, is NH_3. The sources for obtaining nitrogen and hydrogen gases used to produce NH_3 are air and steam respectively. They are passed over red hot coke.

The reactions occurring are

red hot coke + air $\rightarrow CO + N_2$

red hot coke + steam $\rightarrow CO + H_2$

The carbon monoxide gas in the above mixture is oxidized to CO_2 by passing over hot ferric and chromic oxide mixed catalyst and dissolved in water under pressure and removed from the system.

Nitrogen and hydrogen, are now mixed in the ratio of 1 : 3 and passed over Fe_2O_3 catalyst in presence of a promoter (Al_2O_3 and K_2O) at a temperature of 450-500°C under a pressure of 150-200 atm. This process is known an Haber's process. The ammonia formed is passed through a suspension of $CaSO_4$ in water through which a stream of CO_2 also is passed.

The reaction leading to the formation of $(NH_4)_2SO_4$ is

$$2NH_3 + H_2O + CO_2 + CaSO_4 \rightarrow (NH_4)_2SO_4 + CaCO_3\downarrow$$

It may be mentioned that there are other ways of obtaining NH_3 required in the above reaction but will not be dealt with here. The precipitate $CaCO_3$ is filtered off and the ammonium sulphate solution is concentrated by evaporation under vacuum.

A second method of obtaining $(NH_4)_2SO_4$ on a large scale is by the absorption of synthetic ammonia in sulphuric acid which gives $(NH_4)_2SO_4$.

$$2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$$

A third method of manufacture of $(NH_4)_2SO_4$ is by making use of the ammoniacal liquor obtained as a byproduct in the manufacture of coke. The ammoniacal liquor contains some free ammonia and some bound ammonia in the form of volatile and fixed ammonium salts. The volatile salts are decomposed by boiling so that they give off NH_3. The bound ammonia in the form of salts like $(NH_4)_2SO_4$ and NH_4Cl are decomposed by lime water, thereby releasing ammonia through the following reactions.

$$(NH_4)_2SO_4 + Ca(OH)_2 \rightarrow CaSO_4 + 2H_2O + 2NH_3$$

$$2NH_4Cl + Ca(OH)_2 \rightarrow CaCl_2 + 2H_2O + 2NH_3$$

The ammonia so formed is passed into a tank containing 60% H_2SO_4 to give $(NH_4)_2SO_4$.

$$2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$$

$(NH_4)_2SO_4$ as fertilizer

It contains about 25% NH_3 and when applied to the soil, the $(NH_4)_2SO_4$ is converted into nitrate by nitrifying bacteria in the soil. The nitrates are easily assimilated by plants.

Other ammonium compounds like ammonium dihydrogen phosphate, $NH_4H_2PO_4$ and diammonium hydrogen phosphate $(NH_4)_2HPO_4$ also find considerable use as fertilizers.

Urea, NH_2CONH_2

Urea is manufactured by the reaction of ammonia with CO_2 in the temperature range of 180 – 200°C under high pressure. Ammonium carbamate is formed as an intermediate which decomposes to urea and water.

$$2NH_3 + CO_2 \rightarrow NH_2COONH_4 \rightarrow (NH_2)_2CO + H_2O$$

Properties

It is highly soluble in water and Therefore, absorbed by the soil rapidly. It has a high nitrogen content of about 47% urea reacts with water in the soil slowly and forms $(NH_4)_2CO_3$.

$$(NH_2)_2CO + 2H_2O \rightarrow (NH_4)_2\,CO_3$$

Urea has the advantage that it does not change the pH of the soil and it can also be used for a variety of crops

Ammonium nitrate, NH_4NO_3

Ammonium nitrate is manufactured by the neutralization of ammonia with nitric acid. Ammonia required for the purpose is synthesized by Haber's process described earlier (under the manufacture of $(NH_4)_2SO_4$). Ammonia obtained by this method is mixed with air (in the ratio of 1 : 10 by volume) and is allowed to pass over heated platinum gauge kept at a temperature of 800 – 900°C. The ammonia gets oxidized to 'NO'.

$$4NH_3 + 5O_2 \rightleftharpoons 4NO + 6H_2O$$

The 'NO' formed is further oxidized by air to NO_2 which on passing into water gives HNO_3. $2NO + O_2 \rightarrow 2NO_2$.

$$3NO_2 + H_2O \rightarrow 2HNO_3 + NO$$

The nitric acid so formed is neutralized with NH_3 around a temperature of 70°C to form NH_4NO_3.

$$NH_3 + HNO_3 \rightarrow NH_4NO_3$$

It is obtained as liquor and contains about 85% NH_4NO_3. This liquor is concentrated by heating under vacuum.

It finds wide application as a nitrogenous fertilizers. It is deliquescent and is used in the soil in combination with $(NH_4)_2CO_3$ or $CaCO_3$ to avoid explosive decomposition. It may be noted that NH_4NO_3 alone decomposes with great violence at temperature above 300°C liberating N_2 and O_2.

Calcium ammonium nitrate (CAN), $Ca(NO_3)_2.NH_4NO_3$ (CAN)

It is also referred as nitro limestone. The manufacture of this fertilizers is based on two basic constituents ammonia and nitric acid. Ammonia synthesised by Haber's process is converted partially to nitric acid by Ostwald's process. When they neutralise each other, ammonium nitrate is formed. A concentrated solution of NH_4NO_3 containing

some HNO_3 is stirred with finely powdered limestone. Calcium nitrate is formed which combines with NH_4NO_3 to form pellets of calcium ammonium nitrate (CAN).

$$CaCO_3 + 2HNO_3 \rightarrow Ca(NO_3)_2 + H_2O + CO_2$$

$$Ca(NO_3)_2 + NH_4NO_3 \rightarrow Ca(NO_3)_2.NH_4NO_3 \text{ (CAN)}$$

Since NH_4NO_3 is an explosive material, limestone acts as a filler to prevent it from exploding. It is hygroscopic and to prevent the pellets of CAN becoming wet, they are coated with a layer of sodium silicate (soap stone).

CAN contains about 20% nitrogen and it is assimilated by plants easily because of its higher solubility in water.

Phosphatic fertilizers

Phosphorus containing compounds are also used as an important fertilizers essential for the growth of the roots of plants. They are used in the form of calcium sulphur phosphate and calcium triple superphosphate.

Calcium super phosphate, $Ca(H_2PO_4).2(CaSO_4.2H_2O)$

The starting material for the manufacture of calcium super phosphate (composition $Ca(H_2PO_4)_2.2(CaSO_4.2H_2O)$ is phosphate rock containing fluorapatite, $Ca_3(PO_4)_2.CaF_2$. It is treated with calculated amount of H_2SO_4. Since phosphate rock usually contains $CaCO_3$ and CaF_2, their reactions with sulphuric acid give calcium sulphate.

$$CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + CO_2 + H_2O$$

$$CaF_2 + H_2SO_4 \rightarrow CaSO_4 + 2HF$$

The liberated HF reacts with silica to form SiF_4 which then reacts with water to give hydrofluosilicic acid. The reactions are

$$SiO_2 + 4HF \rightarrow SiF_4 + 2H_2O$$

$$3SiF_4 + 2H_2O \rightarrow SiO_2 + 2H_2SiF_6$$

Having eliminated CaF_2 in the above manner, the remaining part of phosphate rock (i.e.,) $Ca_3(PO_4)_2$ reacts with H_2SO_4 according to

$$Ca_3(PO_4)_2 + 2H_2SO_4 + 4H_2O \rightarrow \underset{\text{Calcium superphosphate}}{Ca(H_2PO_4)_2} + 2(CaSO_4.2H_2O)$$

Calcium superphosphate, also called superphosphate of lime, contains about 16 – 20% P_2O_5. The acid phosphate part of the fertiliser is more soluble and gradually dissolves in the soil water.

Calcium triple superphosphate, $Ca(H_2PO_4)_2$

In the preparation of triple superphosphate, phosphate rock is heated with phosphoric acid rather than sulphuric acid to avoid the formation of $CaSO_4$. The reaction is

$$[Ca_3(PO_4)_2.CaF_2] + 6H_3PO_4 \rightarrow \underset{\text{triple superphosphate}}{4Ca(H_2PO_4)_2} + 2HF$$

Mixed fertilizers (NPK fertilizers)

Mixed fertilizers contain nitrogenous, phosphatic and potash fertilisers. As nitrogenous fertilisers, $(NH_4)_2SO_4$, NH_4NO_3 or CAN can be used. Calcium super phosphate acts as a source of phosphorous while potassium chloride, or potassium sulphate is used as a supplier of potassium.

CONCEPT STRAND

Concept Strand 13

Compounds, which contained deuterium, were found to be stable. Suggest a synthesis for the following compounds, taking deuterium oxide as the only source of deuterium.

(i) D_3BO_3
(ii) PD_3
(iii) $[Ag(ND_3)_2]^+$

Solution

(i) H_3BO_3 is prepared by dissolving boron in HNO_3.
Therefore, DNO_3 is prepared by treating NO_2 with D_2O

$$2NO_2 + D_2O \rightarrow DNO_3 + DNO_2$$

$$B + 3DNO_3 \rightarrow D_3BO_3 + 3NO_2$$

(ii) PH_3 is obtaining by treating phosphorous with aqueous alkali.

$P_4 + 3NaOH + 3H_2O \rightarrow 2NaHPO_2 + PH_3$.

Heavy water is treated with sodium when sodium deutroxide is obtained.

$2Na + 2D_2O \rightarrow 2NaOD + D_2$

$P_4 + 3NaOD + 3D_2O \rightarrow 3NaDPO_2 + PD_3$

(iii) The complex $[Ag(ND_3)_2]^+$ is formed on treating a silver salt with deuterated ammonia.

To get $[Ag(ND_3)_2]^+$, the Ag salt should be treated with deutro ammonia.

So deuterium is prepared by reducing deuterium oxide with zinc and treating with nitrogen.

$Zn + D_2O \xrightarrow{1000^0 C} ZnO + D_2$

[or electrolysis of D_2O using LiCl electrolyte]

$N_2 + 3D_2 \xrightarrow{catalyst} 2ND_3$ Heavy ammonia (deutro-ammonia)

$Ag^+ + 2ND_3 \rightarrow [Ag(ND_3)_2]^+$ (stable deutro ammonia complex)

SUMMARY

	p–block elements
Boron	$Na_2B_4O_2 10H_2O$–Borax
	$Na_2B_4O_7.4H_2O$–Kernite
	Preparation
	Reduction of B_2O_3 using Na or Mg– decomposition of BI_3 or B_2H_6
	–Reactions with O_2, N_2, Cl_2, HNO_3, NaOH and NH_3
	B_2O_3, B_2H_6, Borohydrides structure of diborane–Borazole
	Boric acids – Borax
	Borax bead test.
Aluminum compounds	Alumina–preparation and properties
	Aluminium chloride–structure of Al_2Cl_6
	Alums–Preparation and uses
Carbon	Allotropes of carbon–diamond and graphite–Structure and uses
	Carbon monoxide–preparation, properties and structure.
	Carbondioxide–preparation, properties–detection–photosynthesis
	Green house effect–Preparation , properties and structure of
	Carbon suboxide, C_3O_2
	Carbonic acid–Ka_1, Ka_2
	Structure of CO_3^{2-} and HCO_3^-
Silicon	Occurrence, preparation reactions and Uses of silicon,
	Silicon dioxide–different forms of silica, uses of silica
	Preparation and properties of silanes SiH_4, Silicon halides SiF_4 and $SiCl_4$
	Silicones, sodium silicate, silcon carbide–uses of SiC
	Silicates–ortho, pyro, cyclic, chain, sheet and network silicates and their examples. Silicones and their uses.

Nitrogen	General discussion of group 15 elements. Occurrence of nitrogen—difference of Nitrogen from other elements of the group.
	Preparation uses and reactions of N_2
	Nitrogen cycle
	Preparation, properties, structure and uses of
	Hydrazoic acid–N_3H
	Hydrazine–N_3H_4
	Nitrous oxide–N_2O
	Nitric oxide–NO
	Nitrogen trioxide–N_2O_3
	Nitrogen dioxide–NO_2 and N_2O_4
	Nitrogenpentoxide–N_2O_5
	Hyponitrous acid–$H_2N_2O_2$ or HNO
	Nitrous acid–HNO_2
	Nitric acid–HNO_3
	Birkeland–Eyde and Ostwald's process
	Of HNO_3 manufacture
	Ammonia–NH_3
	Haber process
	Test of NH_4^+
	Liquid ammonia as a solvent
	Ammonium salts
Phosphorous	Occurrence–Phosphorite $Ca_3(PO_4)_2$
	Fluorapatite and Chlorapatite–Extraction of phosphorous–preparation and structure of allotropes of phosphorous,–Reaction and use of Phosphorous
	Preparation, properties, reactions, structure and use of phosphorous trichloride–PCl_3
	Phosphorous Oxychloride–$POCl_3$
	or phosphoryl chloride
	Phosphorus trioxide–P_4O_6
	Phosphorus pentoxide–P_4O_{10}
	Phosphorous acid–H_3PO_3
	Hypophosphorous acid–H_3PO_2
	Phosphoric acid–H_3PO_4
	Pyrophosphoric acid–$H_4P_2O_7$
	Metaphosphoric acid–HPO_3
	Sodium hexametaphosphate–$(NaPO_3)_6$
	Hypophosphoric acid–$H_4P_2O_6$
	Phosphine–PH_3

Oxygen	General discussion of group 16
	Occurrence, extraction, properties, reactions and uses of oxygen.
	Oxides–acidic, basic , amphoteric and neutral oxides with examples
	Peroxides, superoxides, suboxides and mixed oxides, structure of O_2
	Preparation, properties, reactions, structure and uses of Ozone–O_3
	Hydrogen peroxide – H_2O_2
Sulphur	General discussion–Extraction of sulphur–Frasch Process– Structure of sulphur–Effect of heat on sulphur–Properties, reactions and uses of sulphur allotropes of sulphur–Rhombic, Monoclinic and plastic sulphur
	Preparation, properties, reactions, structure and uses of
	Hydrogen sulphide–H_2S
	Polysulphides such as H_2S_2
	Sulphur dioxide–SO_2
	Sulphur trioxide–SO_3
	Sulphurous acid–H_2SO_3
	Na_2SO_3, SO_3^{3-} ion
	Sulphuric acid–H_2SO_4
	Lead chamber process and contact process of manufacture of sulphuric acid
	Tests for H_2SO_4
	Thiosulphuric acid $H_2S_2O_3$
	Sodium thiosulphate $Na_2S_2O_3.5H_2O$
Halogens	General discussion of group 17
	Preparation, properties, reactions, uses and structure of fluorine–F_2
	Oxygen difluoride–OF_2
	Dioxygen difluoride–O_2F_2
	Chlorine–Cl_2
	Castner–Kellner and Deacons process of chlorine manufacture
	Bromine–Br_2
	Manufacture of Bromine from Carnallite and Sea water
	Iodine–I_2
	Isolation of iodine from sea water and chile saltpeter
	Hydrogen fluoride and Hydrofluoric acid–Hydrogen chloride and hydrochloric acid–Hydrogen bromide, Hydrogen iodide
	chlorine monoxide–Cl_2O
	chlorine dioxide–ClO_2

Halogens	chlorine tetroxide–Cl_2O_4 Hypochlorous acid–HOCl Chlorous acid–$HClO_2$ Chloric acid–$HClO_3$ Chorates, structure of ClO_3^- Perchloric acid–$HClO_4$. Potassium perchlorate–$KClO_4$ Bleaching powder–Ca(OCl)Cl
Xenon fluorides	Preparation, properties, reactions uses and structure of Xenon difluoride–XeF_2 Xenon tetrafluoride–XeF_4 Xenon hexafluoride–XeF_6
Commercially available fertilizers	Manufacture, application and merits of Ammonium sulphate $(NH_4)_2SO_4$ Urea $(NH_2)_2CO$ Ammonium Nitrate NH_4NO_3 Calcium Ammonium Nitrate (CAN) – $Ca(NO_3)_2.NH_4NO_3$ Calcium superphosphate $Ca(H_2PO_4).2(CaSO_4.2H_2O)$ Calcium triple superphosphate–$Ca(H_2PO_4)_2$ Mixed fertilizers or NPK fertilizers.

TOPIC GRIP

Subjective Questions

1. Give any two reagents that can be used to distinguish between ammonia and phosphine.
2. About 0.2 moles of an alkene used for an ozonolysis experiment gave 5 fragments per molecule. If the same amount of ozone was passed through moist iodine, what would be the weight of the acid formed?
3. Sodium carbonate is treated with carbonic acid and orthoboric acid. Which would react faster and why?
4. A colourless gas with a foul smell is produced on reduction of BCl_3. Give a method of preparation of a good reducing agent in organic chemistry from this gas produced.
5. Anhydrous $AlCl_3$ is an ionic solid at low temperatures. This changes at higher temperatures to give a non ionic solid containing the dimer Al_2Cl_6. This transition leads to large decrease in density. Explain.
6. Give reasons for the fact that unlike $SiCl_4$, CCl_4
   (i) does not undergo hydrolysis
   (ii) does not form complexes
7. Give any two characteristic properties of the compound formed on dissolving SO_2 gas in water. Give the structure of the compound so formed.
8. On treating a halogen with alkali a gas is evolved. Identify the halogen and substantiate.
9. Why is it difficult to liquify noble gases? How does this change as atomic weight increases?
10. Even though the lifting power of helium is only 92% that of hydrogen, helium is preferred over hydrogen for filling weather balloons. Explain.

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

11. The specific heat ratio for all noble gases will be
    (a) 1.66 (b) 1.33 (c) $\cong 1$ (d) 1.52
12. Unlike other halogens fluorine does not form higher oxyacids because
    (a) fluorine has a small size
    (b) fluorine does not have vacant d orbital
    (c) fluorine is highly electronegative
    (d) fluorine shows hydrogen bonding
13. Metals like copper, silver etc are oxidised by sulphuric acid. The gas liberated during this reaction
    (a) dissolves in pyragallol (b) turns lime water milky
    (c) turns lead acetate paper black (d) there is no gas liberated
14. A mixture of NaOH and aluminium can be used to rectify clogged drains because
    (a) hydrogen is liberated during the mixing
    (b) aluminium forms a protective oxide layer

(c) formation of aluminium oxide is highly exothermic
(d) aluminium dissolves in sodium hydroxide

15. Which of the following an thermal decomposition yields a basic as well as an acidic oxide ?
(a) Na_2CO_3 (b) $CaCO_3$
(c) $NaNO_3$ (d) NH_4NO_3



Assertion–Reason Type Questions

Directions: **Each question contains Statement -1 and Statement -2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**
(a) Statement -1 is True, Statement -2 is True; Statement -2 is a correct explanation for Statement -1
(b) Statement -1 is True, Statement -2 is True; Statement -2 is NOT a correct explanation for Statement -1
(c) Statement -1 is True, Statement -2 is False
(d) Statement -1 is False, Statement -2 is True

16. Statement 1
The compound $(SiH_3)_3N$ is more basic than $(CH_3)_3N$.

and

Statement 2
$(SiH_3)_3N$ contains. $p\pi - d\pi$ back bonds.

17. Statement 1
With the exception of the first element, the hydrides of group 16 elements are foul smelling, poisonous gases.

and

Statement 2
On moving down the group, covalent character of these hydrides increases.

18. Statement 1
In the preparation of CO_2 from marble, HCl is preferred over H_2SO_4.

and

Statement 2
$CaCl_2$ is water soluble while $CaSO_4$ is not.

19. Statement 1
B–Cl distance in BCl_3 is shorter than the theoretical B – Cl single bond distance.

and

Statement 2
BCl_3 does not dimerise like $AlCl_3$.

20. Statement 1
Nitrogen is replaced by helium in oxygen cylinders used by deep sea divers.

and

Statement 2
Even though helium and nitrogen are inert in nature, helium is much more unreactive than nitrogen.

Linked Comprehension Type Questions

Directions: **This section contains 2 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

Hydrogen sulphide occurs in volcanic gases and in several spring waters. It is obtained in the laboratory by the action of dilute sulphuric acid on ferrous sulphide.

$$FeS + H_2SO_4 \rightarrow FeSO_4 + H_2S$$

It is a colourless gas with unpleasant odour. It is slightly heavier than air and moderately soluble in cold water. Solublility decreases in hot water.

21. The gas released on combustion of H_2S, when it burns with a blue flame is
(a) H_2 (b) SO_2 (c) SO_3 (d) SO_2 and SO_3

22. H_2S used in qualitative analysis in the lab, may be prepared by the hydrolysis of
(a) thioacetamide (b) thiols (c) ZnS (d) CuS

23. Which of the following oxidizing agents will not convert H_2S into sulphur
(a) chlorine (b) $FeCl_3$ (c) ozone (d) dichromate

Passage II

For the halogens, four series of oxoacids are known. The structures of their anions are based on a tetrahedron . The sp^3 hybrid orbitals used for bonding form only weak σ bonds. The ions are stabilized by strong pπ – dπ bonding . Many of the oxoacids are known only in solution or as their salts.

24. The oxoacid known only for chlorine is
(a) HOX (b) HXO_2 (c) HXO_3 (d) HXO_4

25. The σ orbitals formed in these oxyacids are weak because
(a) the halogens take positive oxidation state
(b) oxygen has smaller size than halogens
(c) among halogens only fluorine is more electronegative than oxygen
(d) s and p levels differ in energy appreciably

26. The pi bonds formed in these oxyacids are always between
(a) filled 2p orbital of oxygen and vacant d orbital of halogen
(b) filled 2p orbital of halogen and vacant 2p orbital of oxygen
(c) vacant 2p orbital of oxygen and filled d orbital of halogen
(d) pi bonds differ based on the halogen

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers of which ONE OR MORE answers will be correct.**

27. The compounds that would liberate oxygen on reaction with ClO_2 is/are
(a) ozone (b) alkaline H_2O_2 (c) HI (d) aluminium

28. Carbon monoxide is a stable gas. But a SiO analog for silicon does not exist because
(a) silicon is larger than carbon
(b) silicon contains vacant d orbitals
(c) carbon monoxide is resonance stabilized
(d) silicon is less electronegative than carbon

29. When diborane is heated with ammonia under different conditions, the products that can be formed are
(a) Borax (b) borazole
(c) boron nitride (d) diammoniate of diborane



Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

30.

Column I	**Column II**
(a) F_2	(p) Exists as gas at STP
(b) Cl_2	(q) Liberates O_2 with alkali
(c) Br_2	(r) Bleaching agent
(d) I_2	(s) oxidised by ozone

IIT ASSIGNMENT EXERCISE

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

31. The correct order of reactivity of boron halides with ammonia is
(a) $BF_3 > BCl_3 > BBr_3 > BI_3$ (b) $BCl_3 > BF_3 > BBr_3 > BI_3$
(c) $BCl_3 > BBr_3 > BI_3 > BF_3$ (d) $BI_3 > BBr_3 > BCl_3 > BF_3$

32. The gas formed on treating BF_3 gas with sodium hydride at 180°c is
(a) Borax (b) HBF_4 (c) B_2H_6 (d) $NaBH_6$

33. The number of hydrogens that can be methylated in a molecule of diborane is
(a) 6 (b) 2 (c) 4 (d) 1

34. The ion that will readily disproportionate is
(a) B^{3+} (b) Al^{3+} (c) Ga^{+} (d) Te^{+}

35. Nitric acid can be stored in aluminium vessel because
(a) Aluminium is inert (b) Aluminium forms a protective oxide coating
(c) Aluminium is brittle below its melting point (d) Aluminium is a good reducing agent

36. Aluminium dissolves in excess alkali due to the formation of
(a) $Na_3 AlO_3$ (b) $Al(OH)_3$
(c) Al_2O_3 (d) $Al(OH)_3$ and $Na_3 AlO_3$

37. Among the following, the one that is not an alum is
(a) $(NH_4)_2SO_4\ Al_2\ (SO_4)_3.\ 24H_2O$ (b) $FeSO_4.\ Al_2(SO_4)_3.\ 24H_2O$
(c) $K_2SO_4.\ Fe_2(SO_4)_3.\ 24H_2O$ (d) $(NH_4)_2\ SO_4.\ Cr_2(SO_4)_3.\ 24H_2O$

38. The compound which shows acidic property among the following is
(a) $Be(OH)_2$ (b) $Al(OH)_3$ (c) $B(OH)_3$ (d) $Ga(OH)_3$

39. The pair of elements having the closest atomic radii value is
(a) B and Al (b) Al and Ga (c) Ga and In (d) Ga and Tl

40. An example of an interstitial carbide among the following is
(a) WC (b) CaC_2 (c) Al_4C_3 (d) SiC

41. Tin on reaction with NaOH liberates a gas. The same gas is also liberated by the reaction of tin with
(a) H_2SO_4 (b) Conc.HNO_3 (c) dil.HNO_3 (d) HCl

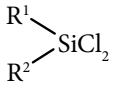
42. In group 14, on going down the group, the electropositive character
(a) increases (b) decreases
(c) does not show clear trend (d) decrease first, increases after Ge

43. A good reducing agent among the following is
(a) $PbCl_4$ (b) $SnCl_4$ (c) $PbCl_2$ (d) $SnCl_2$

44. In the presence of sunlight, carbon monoxide combines with chlorine to form
(a) carbon tetrachloride (b) phosgene (c) phosphine (d) carbonic acid

45. Carbon sub oxide has
(a) linear structure
(b) sp^2 hybridised carbon
(c) no multiple bonds
(d) least stability among oxides of carbon

46. Amorphous silicon dissolves readily in
(a) Con HCl (b) Con H_2SO_4 (c) KOH (d) both (a) and (b)

47. A dimer alone is formed on hydrolysis of
(a) R_3SiCl (b) R_2SiCl_2 (c) $R\,SiCl_3$ (d) 

48. The units present in pyrosilicate is
(a) SiO_4^{4-} (b) $Si_2O_7^{6-}$ (c) $Si_3O_9^{6-}$ (d) $Si_6O_{18}^{12-}$

49. Nitrogen can be obtained by the action of nitrous acid on
(a) urea
(b) ammonium dichromate
(c) metal azide
(d) ammonium chloride

50. The halide that will not undergo hydrolysis is
(a) NCl_3 (b) NF_3 (c) PCl_3 (d) PF_3

51. BiH_3 is a stronger reducing agent than NH_3 because of
(a) large size of bismuth
(b) absence of vacant d orbitals in nitrogen
(c) lower bond angle in bismuth
(d) lower electronegativity of Bi

52. The water insoluble oxide among the following is
(a) N_2O_5 (b) P_2O_5 (c) As_2O_5 (d) Sb_2O_5

53. When NH_4Cl is added to 1.48g of calcium hydroxide, the volume of gas liberated at STP is
(a) 4.5 L (b) 0.9 L (c) 9 L (d) 0.45 L

54. The compound that would turn acidified dichromate paper green is
(a) HNO_2 (b) HNO_3 (c) N_2O_5 (d) $Hg(NO_3)_2$

55. When nitric acid comes in contact with phosphine, it burns. The gas liberated is
(a) NH_3 (b) P_2O_5 (c) NO_2 (d) O_2

56. The yellow colour of commercial sample of HNO_3 is due to
(a) +5 oxidation state of nitrogen
(b) oxidation of atmospheric nitrogen
(c) presence of impurities of copper
(d) decomposition of HNO_3

57. The hybridization of phosphorus in polymeric species of PO_3^- is
(a) sp^3d (b) sp^3 (c) sp^2 (d) sp

58. When a metal halide is treated with sulphuric acid, a gas is released, which turns acidified dichromate paper green. The metal halide is
(a) NaBr (b) KI (c) NaCl (d) either (a) or (b)

59. The reagent formed on boiling sodium hydroxide with sulphur is
(a) hypo (b) sodium sulphite (c) sodium bisulphate (d) colloidal sulphur

60. The most acidic hydride among the group 16 elements is
(a) H_2O (b) H_2S (c) H_2Se (d) H_2Te

61. Hydrogen halide and hydrogen peroxide on oxidation with ozone give (respectively)
(a) oxyacid and hydrogen (b) halogen and oxygen (c) halogen and hydrogen (d) oxyacid and water

62. Ozone is a powerful oxidizing agent because
(a) its decomposition is exothermic
(b) both bonds have equal bond strength
(c) it forms atomic oxygen
(d) ozone is diamagnetic

63. Atomicity of rhombic sulphur is
(a) 2 (b) 4 (c) 6 (d) 8

64. Oxygen is bonded by peroxo linkages in
(a) H_2SO_5 (b) H_2SO_3 (c) $H_2S_2O_2$ (d) $H_2S_2O_6$

65. Bond angle in H_2S is smaller than that in H_2O because
(a) sulphur is a larger atom than oxygen
(b) sulphur is less electronegative than oxygen
(c) H_2S has back bonding due to vacant 3d orbital
(d) there is no possibility of hydrogen bonds in H_2S

66. H_2S gas should not be dried using con. H_2SO_4 because
(a) con. H_2SO_4 is not a good drying agent
(b) H_2S is a reducing agent
(c) reaction of H_2SO_4 with water is exothermic
(d) H_2S does not absorb moisture at all

67. Among halogens, iodine alone has some metallic character like
(a) formation of IBr
(b) high density in liquid state
(c) physical state of iodine is solid
(d) iodine has highest melting point in the group

68. For the reaction, HX (hydrated) $\rightarrow HX_{(g)}$ (anhydrous), the enthalpy of dehydration is highest when (X) is
(a) Fluorine (b) Astatine (c) Iodine (d) Chlorine

69. The halogen that cannot displace any halogen from its halide,
(a) combines even with noble gases
(b) decomposes water
(c) is sparingly soluble in water
(d) reacts readily with hydrogen

70. The hydrogen halide with highest dipole moment is
(a) HF (b) HCl (c) HBr (d) HI

71. Hydrofluoric acid cannot be stored in glass bottles because
(a) Gaseous HF is extremely toxic
(b) HF undergoes auto ionization
(c) HF enclosed in bottle develops high pressure
(d) HF dissolves silicates

72. The oxide which has a chlorine-chlorine bond is
(a) chlorine monoxide (b) chlorine dioxide (c) Dichlorine hexoxide (d) Dichlorine heptoxide

73. Even though $PbCl_4$ exists, PbI_4 does not because
(a) iodine is much larger than chlorine
(b) iodine has low electronegativity
(c) iodine is a solid and chlorine is a gas
(d) iodine is a weak oxidizing agent

74. Tetrafluorides of carbon, sulphur and xenon have
(a) same dipole moment
(b) same number of lone pairs
(c) different hybridization
(d) regular geometry

75. The gas evolved when bleaching powder is treated with moist carbon dioxide
(a) chlorine (b) carbon monoxide (c) oxygen (d) water vapour

76. In their reaction with fluorine or oxygen, noble gases behave as
(a) reducing agent (b) oxidizing agent (c) lewis acid (d) no specific agent

77. The property that is highest in value for noble gases is
(a) electronaffinity (b) ionization energy (c) electronegativity (d) solubility in water

78. The true statement about noble gases is
(a) They do not burn and do not help in burning
(b) They are colourless gases with pungent smell
(c) They are good electrical insulators
(d) Ionization energy does not show a proper trend

79. The monoatomic nature of noble gases can be easily proved using
(a) dipole moment values
(b) bond order values
(c) ionisation energy values
(d) spectra of noble gas

80. The product which is not formed during hydrolysis of XeF_2 is
(a) Xe (b) $XeOF_2$ (c) O_2 (d) HF

81. Diborane on reaction with oxygen gives a boron compound A. 'A' can also be obtained by strongly heating
(a) H_3BO_3 (b) BCl_3 (c) $NaBH_4$ (d) $B_3N_3H_6$

82. Gallium melts at 30° C which is extremely low in the group. This low melting point is due to
(a) intervening d orbitals
(b) small size of gallium atom
(c) its discrete diatomic structure
(d) presence of impurities in its crystal lattice

83. A non metal in group 13 reacts with halogen at high temperature to give a compound, which gives two acids on hydrolysis. The weaker of the two acids is
(a) HCl (b) H_3BO_3 (c) BCl_3 (d) $Al(OH)_3$

84. CO_2 is a gas while SiO_2 is a solid. When they are treated with aluminium, the products formed are respectively
(a) carbon, no reaction
(b) aluminium carbide, no reaction
(c) aluminium carbide, aluminium silicide
(d) carbon, silicon

85. Aluminium chloride is a lewis acid but its fluoride is not because
(a) of difference in hybridization of Al
(b) fluorine is more electronegative than chlorine
(c) fluorine forms ionic AlF_3 unlike chlorine
(d) AlF_3 has much lower molecular weight than $AlCl_3$

86. Silicon forms organo silicon polymers, while carbon, which belongs to the same group does not form such polymers. This is because
(a) silicon has higher atomic size than carbon.
(b) silicon can hold three hydroxyl groups.
(c) polymerisation does not occur in carbon.
(d) catenation is preferred in carbon.

87. Bond order in graphite is
(a) 1 (b) $1\frac{1}{3}$ (c) $2\frac{3}{4}$ (d) 4

88. The bond distance (C – C) and bond angle in diamond are same as those of
(a) ethene (b) benzene (c) ethane (d) cyclopentane

89. The silicate that is water soluble is
(a) Li_2SiO_3 (b) $MgSiO_3$ (c) $ZrSiO_3$ (d) $Zn_4Si_2O_7(OH)_2$

90. Silicon tetrachloride can be converted to crystalline silicon
(a) by treating $SiCl_4$ with Zn
(b) by passing $SiCl_4$ vapours into molten aluminium
(c) by reducing with coke
(d) by passing it through stream of hydrogen at high temperature

91. The combination of reagents that can be used to quantitatively estimate carbon monoxide is
(a) Iodine and thiosulphate
(b) KI and thiosulphate
(c) KI and peroxomonosulphuric acid
(d) I_2O_5 and thiosulphate

92. $SnCl_4$ is partially soluble in water. But this can be completely dissolved on addition of
(a) excess water (b) HCl (c) any mineral acid (d) aqua regia

93. The combination of compounds that would not liberate nitrogen on heating is
(a) hydrazoic acid and lithium
(b) oxidation of NH_3 with calcium hypochlorite
(c) oxidation of NH_3 with oxygen on platinum
(d) oxidation of ammonia with bromine water

94. The common characteristic seen in all oxides and oxyacids of nitrogen is
(a) diamagnetic property
(b) acidic nature
(c) symmetric structure
(d) presence of pπ -pπ multiple bonding between N and O

95. The reagent that would reduce NO_3^- to ammonia is
(a) Zinc (b) Zn in NaOH (c) ferrous sulphate (d) Pt

96. Trioxides of Arsenic and antimony can be prepared by burning the metal in air which is not possible for phosphorus because
(a) phosphorus is a non metal
(b) higher oxides of As and Sb are oxidising
(c) As and Sb have less tendency to form higher oxides
(d) As cannot form the tetrahedral structure

97. In solid state, PCl_5 exists as PCl_4^+ and PCl_6^-
(a) In this phosphorus undergoes change of hybridization
(b) This is true for both PBr_5 and PI_5
(c) This is true only for PBr_5 and PI_5 does not exist
(d) PBr_5 and PI_5 exist as PX_4^+ and X^-

98. Phosphorus has the least oxidation state in
(a) metaphosphoric acid
(b) hypophosphorous acid
(c) orthophosphoric acid
(d) phosphorous acid

99. The most volatile hydride of the VI group element is
(a) H_2Te
(b) H_2Se
(c) H_2S
(d) There is no regular trend

100. The sulphur compound formed by passing chlorine gas in CS_2, has the structure similar to
(a) methane (b) sulphur dioxide (c) PCl_5 (d) hydrogen peroxide

101. Ice melts to water with a sudden decrease in volume at 0°C. This is because
(a) water is H–bonded while ice is not
(b) Ice has greater number of H–bonds than water
(c) Ice has an open cage structure formed by H–bonds. On melting the cage structure is disrupted due to breaking of some of the H–bonds
(d) Ice and water have a negative coefficient of linear expansion

102. Under normal conditions, oxygen is a gas while sulphur is a solid. This difference is due to
(a) Oxygen forms pπ – pπ bonds while sulphur does not.
(b) absence of vacant d orbitals in oxygen
(c) high electronegativity of oxygen
(d) presence of unpaired electrons in oxygen

103. When sulphuric acid reacts with sugar, it behaves as a
(a) acid (Bronsted) (b) Lewis acid (c) oxidizing agent (d) dehydrating agent

104. The basis of commercial extraction of bromine from sea water is
(a) chlorine is a better oxidizing agent than bromine
(b) chlorine is more water soluble than bromine
(c) lower boiling of bromine compared to salt solution
(d) fractional crystallization of bromine at its melting point

105. The reaction which is true for all halogens is
(a) formation of pentahalide with phosphorus
(b) reaction with sulphur to form S_2X_2
(c) oxidize H_2S gas to sulphur
(d) oxidize ammonia to nitrogen

106. Bleaching powder liberates Cl_2 on acidification. The type of change is
(a) Elimination
(b) Reduction cum oxidation
(c) Oxidation only
(d) Reduction only

107. A halogen is found to readily and completely decompose water. The corresponding halide can be qualitatively tested using
(a) acidified dichromate (b) chlorine water (c) calcium chloride (d) lead acetate

108. When a compound of the type KXO_3 (where X is a halogen) is decomposed on heating, there is no evolution of gases, then (X) is
(a) iodine (b) bromine (c) chlorine (d) fluorine

109. Xenon difluoride is a linear molecule. The number of electrons present in the 5d orbitals of an excited xenon atom before bonding according to VBT is,
(a) 0 (b) 1 (c) 2 (d) 3

110. The noble gas that does not solidify on cooling even to absolute zero, is
(a) Radon (b) xenon (c) Helium (d) Neon



Assertion–Reason Type Questions

Directions: **Each question contains Statement -1 and Statement -2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**
(a) Statement -1 is True, Statement -2 is True; Statement -2 is a correct explanation for Statement -1
(b) Statement -1 is True, Statement -2 is True; Statement -2 is NOT a correct explanation for Statement -1
(c) Statement -1 is True, Statement -2 is False
(d) Statement -1 is False, Statement -2 is True

111. **Statement 1**
The most acidic hydride among the Group 16 elements is the least polar among them.

and

Statement 2
Acid strength is directly proportional to dipole moment.

112. **Statement 1**
There are 16 lone pairs of electrons totally in a molecule of phosphorus trioxide.

and

Statement 2
All the phosphorus and oxygen atoms have 8 electrons each in their valence shell.

113. Statement 1
In the presence of a bisulphite, H_2S cannot be used as a reagent for qualitative analysis.

and

Statement 2
Bisulphite decomposes to give SO_3^{2-} and SO_2.



Linked Comprehension Type Questions

Directions: **This section contains 1 paragraph. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

CO_2 is used as a fire extinguisher-The use of a common fire extinguisher is based on the production of CO_2. A soda acid fire extinguisher contains a concentrated solution of sodium bicarbonate and a bottle containing con H_2SO_4. When apparatus is turned upside down, the acid flows out of the bottle, reacts with bicarbonate and CO_2 is released

114. The advantage on replacing con H_2SO_4 with aluminium sulphate in a soda acid fire extinguisher is that,
(a) storage problems can be overcome
(b) Con H_2SO_4 is also oxidizing in nature
(c) $Al_2(SO_4)_3$ leads to formation of $Al(OH)_3$
(d) $Al_2(SO_4)_3$ can be stored in powder form

115. In a laboratory, fire is caused due to large spills of potassium. In this situation
(a) CO_2 can be used to extinguish the fire as it is not a supporter of combustion
(b) CO_2 can be used to extinguish the fire as it is heavier than air
(c) CO_2 cannot be used as to extinguish the fire as potassium continue burning in it
(d) CO_2 cannot be used to extinguish the fire as it will react with potassium

116. The compound that can be used to absorb carbondioxide is
(a) Ethanolamine (b) $CaCO_3$ (c) $NaHCO_3$ (d) Ethanol



Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers of which ONE OR MORE answers will be correct.**

117. An oxide M_2O_2 is a true peroxide. Which of the following statements is/are true for M_2O_2.
(a) It liberates oxygen with con H_2SO_4
(b) It does not react with both concentrated or dilute H_2SO_4
(c) As the size 'M' increases M_2O_2 becomes more stable
(d) The metal sulphate is formed with dil H_2SO_4

118. Among the following Group III elements, the ones that are not affected by alkali is/are
(a) Boron (b) Gallium (c) Indium (d) Thallium

119. The shape of NO_3^- and CO_3^{2-} are trigonal planar. While going down their group, we have tetrahedral PO_4^{3-} and SiO_4^{4-} because
(a) size increases down the group
(b) both carbon and nitrogen have attained stable configuration
(c) both carbon and nitrogen have restriction of eight electrons in the outer most shell
(d) carbon and nitrogen are more electronegative than Si and phosphorus respectively

Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

120.

Column I	**Column** II
(a) N_2O_3	(p) gives $NaNO_2$ with NaOH
(b) N_2O_4	(q) neutral oxide
(c) N_2O	(r) presence of N–N Bond
(d) NO	(s) odd electron molecule

ADDITIONAL PRACTICE EXERCISE

Additional Practice Exercise Questions

121. Explain the relative Lewis acid character of trihalides of boron.

122. When a nonmetal in group III is treated with conc.H_2SO_4, a gas is released. What happens on passing the gas through $CuCl_2$ solution?

123. When aluminium sulphate is heated strongly, a white solid (A) is formed along with a gas (B). When the solid (A) is treated with sulphur monochloride and chlorine gas, a white solid (C) and a gas (D) are formed. Explain the reactions with equations. Give a method to prepare gas (B) from gas (D).

124. A compound (A), was heated with concentrated sulphuric acid, when violet vapours were evolved which turned starch paper blue. What would happen on passing the vapours through ammonia?

125. Compare the strengths of the following pairs of acids: H_2CO_3 and H_3BO_3.

126. Write a short note on action of concentreated sulphuric acid on alkali metal halides.

127. Explain the stability of potassium hydrogen halides.

128. Bleaching powder loses its bleaching action when kept in an open bottle for a long time, why?

129. H_2S acts only as a reducing agent whereas, SO_2 can act as both as a reducing agent and an oxidizing agent. Explain.

130. Complete the reactions

$CO + Cl_2 \rightarrow (A) \xrightarrow{H_2O} (B) + (C)$

$(A) + (E) \rightarrow (B) + (D)$

$(C) + (E) \xrightarrow[\text{pressure}]{180^\circ C} (D) + H_2O$

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

131. A metal which forms its trihalide and having great affinity for oxygen, is dissolved in NaOH. It can be reprecipitated back

(a) by passing CO_2 gas
(b) by heating with Na_2CO_3
(c) by crystallizing
(d) by adding excess NaOH

132. An element 'M' of group 13 forms an ion $[MF_6]^{3-}$. The true statement about the metal is

(a) Its hydroxide is acidic
(b) It forms amphoteric oxides
(c) It does not form electron deficient hydrides
(d) It has high melting point

133. On passing chlorine over heated mixture of coke, B_2O_3 and Al_2O_3, the compounds formed are

(a) both solids
(b) a fuming liquid and a solid respectively
(c) both gaseous substances which solidify only on cooling
(d) a gas and fuming liquid respectively

134. A compound 'X' on heating, loses water, swells and finally gives a transparent glassy solid. We can say,
(a) Its aqueous solution is slightly acidic
(b) X cannot be used in a buffer solution
(c) the transparent bead is an oxysalt
(d) 'X' exists as a decahydrate

135. One mole of orthoboric acid reacts with one mole of sodium hydroxide, then
(a) one mole of sodium orthoborate is formed.
(b) reaction proceeds to completion
(c) the structure of boric acid changes from trigonal planar to tetrahedral
(d) 0.5 moles of sodium orthoborate is formed.

136. Alums are double salts of the formula M(I) M'(III)$(SO_4)_2$.12H_2O. In this formula, M cannot be lithium because
(a) Lithium has no vacant d orbitals
(b) Lithium can form only covalent compounds
(c) Lithium has small size
(d) Lithium compounds have low water solubility

137. The gas liberated when anhydrous aluminium sulphate is heated above 800°C is
(a) O_2 (b) SO_2 (c) SO_3 (d) N_2O

138. Among the following halides, the most stable halide is
(a) CF_4 (b) $SiCl_4$ (c) $SnBr_4$ (d) CI_4

139. Electrical conductivity of graphite is confined only to one direction because
(a) electron moves only within a layer
(b) carbon shows its π bonds with three neighbours
(c) graphite has both covalent bonds and van der Waals forces
(d) of slippery nature of graphite

140. A gas is liberated at room temperature on heating formic acid with con.H_2SO_4. The gas can be detected using a metal chloride. During the detection
(a) a coloured complex is formed
(b) a soluble, colourless complex is form
(c) the metal ion is reduced to the metal
(d) the metal ion is oxidised to a coloured higher oxidation state

141. The compound that will liberate CO_2 on hydrolysis is
(a) phosgene (b) calcium carbide (c) calcium carbonate (d) metal carbonyls

142. Dry ice is molecular solid while silica is an atomic solid. The reason for the difference is,
(a) Dry ice contains pπ – dπ bonds
(b) Silica contains no pπ – pπ bonds
(c) Bonds in dry ice are stronger than those in silica
(d) Silica has high lattice energy

143. A cyclic silicate contains three silicate units. The number of cations of M^{2+} required to form the formula unit is
(a) 1.5 (b) 2 (c) 3 (c) 6

144. The product formed, on passing methyl chloride over a copper silicon alloy at 300°C, is hydrolysed. The compound formed.
(a) can be classified as organometallics
(b) contain a Si = O bond
(c) always form three dimensional structures
(d) are always dimers

145. Among the halide of nitrogen, the strongest Lewis base is
(a) NF_3 (b) NCl_3 (c) NBr_3 (d) NI_3

146. About 5 g each of P_4 and oxygen are heated, such that both phosphorus and oxygen are used up. The weight of P_4O_6 formed is
(a) 8.8 g (b) 3.3 g (c) 5.8 g (d) 8.47 g

147. The properties of hydrides of group 15 elements which shows an increasing trend down the group is
(a) thermal stability (b) Bond angle (c) K_b values (d) reducing character

148. Considering the molecules NCl, NCl^+ and NCl^-. The wrong statement about their magnetic moment is
(a) all are paramagnetic species
(b) both NCl^+ and NCl^- will have same magnetic moment
(c) NCl will have the least magnetic moment
(d) the number of paired electrons in NCl^+ and NCl are same

149. Hydrazine is obtained by partial oxidation of ammonia by sodium hypochlorite. This is done in the presence of gelatin
(a) to avoid formation of nitrogen
(b) to avoid further oxidation to oxides of nitrogen
(c) to avoid explosive reaction
(d) to increase activity of hypochlorite

150. Super phosphate of lime is prepared by heating
(a) calcium phosphate with phosphoric acid (b) calcium phosphate with sulphuric acid
(c) calcium hydroxide with phosphorus pentoxide (d) calcium hydroxide with orthophosphoric acid

151. On heating ortho phosphoric acid above 500°C, the product formed is
(a) pyrophosphoric acid (b) meta phosphoric acid (c) phosphorus pentoxide (d) phosphine

152. A 10 cm^3 solution of H_2O_2 liberated 1.015 g of iodine from an acidified solution of KI. The molarity of H_2O_2 is
(a) 0.1 (b) 0.3 (c) 0.2 (d) 0.4

153. Oxygen is a colourless gas. It condenses to a pale blue liquid or solid at low temperatures. In which state does oxygen show paramagnetic property?
(a) In gaseous state only. (b) In solid state only.
(c) In liquid and solid states. (d) In gaseous, liquid and solid states.

154. The conditions under which sulphur may become paramagnetic and blue coloured like oxygen is
(a) On heating sulphur to 600°C it consists of S_2 molecules
(b) sulphur on heating forms diradicals and polymerise
(c) On cooling sulphur to low temperatures
(d) On extracting sulphur from organic solvents like toluene

155. SiO_4^{4-} units polymerise leading to formation of silicates. This type of polymerization is not seen in SO_4^{2-} units because
(a) sulphur is more electronegative than silicon (b) sulphur is a smaller atom than silicon
(c) sulphur and oxygen belong to same group (d) SiO_4^{4-} has a lower charge than SO_4^{2-}

156. The only acid in which sulphur is in –2 oxidation state also is
(a) thiosulphuric acid (b) dithionic acid
(c) peroxomonosulphuric acid (d) sulphurous acid

157. The false statement about oxides of group 16 elements is
(a) Both SO_2 and SeO_2 are water soluble
(b) SO_2, SeO_2, TeO_2 are all acidic oxides
(c) SO_2, SeO_2, TeO_2 are soluble in alkali hydroxide forming salts
(d) TeO_2 is water insoluble

158. Dichloride of sulphur has sp^3 hybridised sulphur with two lone pairs. The Cl – S – Cl bond angle is
(a) greater than water (b) less than oxygen fluoride
(c) greater than $TeBr_2$ (d) greater than CCl_4

159. With respect to solubility of halogens
(a) Halogens are more soluble in water than in organic solvents
(b) Solubility in water increases from Cl to I
(c) Fluorine does not dissolve in water
(d) Halogens exist in different molecular forms in organic solvent and in gas phase.

160. A dilute solution of HF and HI are exposed to air at room temperature. After some time
(a) Both HF and HI turn yellow
(b) Both are oxidized but there is no colour change
(c) only HI is oxidized but there is no colour change
(d) only HI acquires a brown colour

161. The most thermally stable acid among the following is
(a) HClO (b) HBrO (c) $HClO_3$ (d) $HClO_4$

162. Compared to the others in group 17, there is a greater tendency for
(a) fluorine to form $[AlF_6]^{3-}$ ion
(b) fluorine to stabilize lower valencies
(c) hydrogen fluoride to behave as a strong acid
(d) iodine to react with water and air

163. If 22.8 g of hydrated periodic acid is heated to about 200°C, the weight of product formed is
(a) 21.0 g (b) 19.2 g (c) 33.4 g (d) 16.7 g

164. A compound 'X' on reaction with fluorine gives out an element (present in the compound X) as one of the products. X may be
(a) C_6H_6 (b) $KClO_3$ (c) NH_3 (d) H_2S

165. Which of the following salts cannot be oxidised using acidified dichromate?
(a) HF (b) HCl and HF (c) HBr and HI (d) HI

166. During the formation of clathrate compounds
(a) all the cavities in the crystal lattice are filled
(b) Ar, Kr, Xe only form clatharate compounds
(c) Once trapped, the gas molecules cannot be completely removed from the clatharate compound
(d) radioactive Kr and Xe cannot be stored

167. Xenon forms compounds with fluorine, oxygen etc. This is not seen in helium or neon because
(a) He and Ne have low atomic weight
(b) fluorine has much smaller size than Xe
(c) Xe has lower ionization energy than He or Ne
(d) He and Ne cannot expand their octet beyond 8

168. The fluoride of xenon that undergoes slow hydrolysis with water is
(a) XeF_2 (b) XeF_4
(c) XeF_6 (d) All hydrolyse at almost same rate

169. Among the following geometries, the one with same number of lone pairs of electrons as in water is
(a) XeF_4 (b) XeF_6 (c) XeO_3 (d) XeO_2F_2

170. Xenon forms compounds with fluorine, but it is not true for bromine because
(a) fluorine is smaller than bromine (b) fluorine is more electronegative than bromine
(c) Br^- ions can be easily oxidised compared to F^- (d) bromine does not allow back bonding

Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement -1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement -1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

171. Statement 1
Aluminium chloride exists as a dimer in benzene but the dimer breaks in water.

and

Statement 2
Water has high dielectric constant.

172. Statement 1
Borax swells up on heating.

and

Statement 2
Anhydrous borax on strong heating forms a transparent glassy mass.

173. Statement 1
Silanes are more susceptible to nucleophilic attack compared to alkanes.

and

Statement 2
Silicon is a larger atom than carbon and is less electronegative than hydrogen.

174. Statement 1
In a silicate structure, two SiO_4^{2-} units can never have two O^{2-} ions in common.

and

Statement 2
In silicates, silicon is four coordinate and is surrounded by four oxygen atoms at the corner of a tetrahedra.

175. Statement 1
NaN_3 can be used to inflate the air bags used as safety devices in cars.

and

Statement 2
N_3^- ion has a bent structure.

176. Statement 1
If urea is added to con.HNO_3, It becomes a strong oxidizing agent.

and

Statement 2
Oxidizing action of HNO_3 is due to presence of NO_2.

177. Statement 1
Inspite of different type of bonds between sulphur and the two oxygen atoms in SO_2 molecule, both the S – O bond lengths are equal.

and

Statement 2
Sulphur is sp^2 hybridized in SO_2 and bond angle is less than expected 120°.

178. Statement 1
In aqueous solution Cl_2 is a stronger oxidizing agent than F_2 due to higher electron affinity.

and

Statement 2
F – F bond length in F_2 is shorter than Cl – Cl bond length in Cl_2.

179. Statement 1
ICl_2^- has a linear geometry but unsymmetric arrangement of lone pairs.

and

Statement 2
ICl_2^- has three lone pairs of electrons.

180. Statement 1
Radon forms many compounds similar to those of xenon.

and

Statement 2
Radon has lower ionization energy than xenon.

Linked Comprehension Type Questions

Directions: **This section contains 3 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

Manufacture of Nitric acid
The ostwald process for preparation of nitric acid depends on the catalytic oxidation of ammonia to NO, followed by oxidation of NO to NO_2 and conversion of NO with water to HNO_3. This gives HNO_3 of concentration 60% by weight. Distillation gives 68% acid only. Therefore, 98% HNO_3 is obtained by dehydrating with con.H_2SO_4.

181. The overall reaction for formation of nitric acid from ammonia can be written as
(a) $NH_3 + 2O_2 \rightarrow HNO_3 + H_2O$ (b) $4NH_3 + 5O_2 + 2HNO_2 \rightarrow 4HNO_3 + H_2O + 2NO$
(c) $NH_3 + 3NO_2 + H_2O \rightarrow HNO_3 + 3NO + 2H_2$ (d) $2NO + O_2 \rightleftharpoons 2NO_2 \xrightarrow{H_2O} 2HNO_3$

182. Distillation of HNO_3 does not give 98% HNO_3 but only 68% HNO_3 because
(a) HNO_3 decomposes at its boiling point (b) HNO_3 decomposes on heating
(c) HNO_3 and H_2O form a constant boiling mixture (d) HNO_3 reacts with H_2O on heating

183. The ion formed when HNO_3 is treated with con.H_2SO_4 is
(a) NO_2^+ (b) NO_2^- (c) NO^+ (d) NO_3^-

Passage II

Compounds of group 14 elements form covalent hydrides of the type MH_4. The ease of formation of hydrides decreases on moving down the group. The number of compounds formed with hydrogen by each element differs greatly. Carbon forms vast number of chain and ring compounds. Silicon forms limited number of saturated hydrides called silanes .

184. Among CH_4 and SiH_4, the one that is more thermally stable is
(a) CH_4, due to its low reactivity
(b) CH_4, due to higher electronegativity of carbon
(c) SiH_4, due to large size of silicon
(d) SiH_4, due to presence of vacant d orbitals

185. The hydride that can be hydrolysed even by neutral water is
(a) CH_4 (b) SiH_4 (c) GeH_4 (d) SnH_4

186. Germanium hydrides can be prepared by treating the chloride or oxide of germanium with
(a) sodium amalgam (b) NH_3 (c) $LiAlH_4$ (d) dil acids

Passage III

$A_{(g)} + B_{(g)} \xrightarrow{\text{water}}$ yellow turbid solution.
A turns lime water milky and both A and B are sulphur compounds which turn acidified dichromate paper green.

187. The compound A is
(a) water insoluble
(b) used as a non aqueous solvent
(c) a gas and the similar compound on going down the group are also gases
(d) easily oxidized to a dibasic organic acid

188. The hybridization of the central atom in A and B are
(a) sp^2, sp^3
(b) sp^3, dsp^2
(c) both are sp^3 hybridized
(d) Both are dsp^2 hybridized

189. The product causing turbidity is
(a) oxidation product of A
(b) oxidation product of B
(c) addition product of A and B
(d) a salt formed due to neutralization

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers of which ONE OR MORE answers will be correct.**

190. Among group 13 elements, the unusual properties shown by gallium is/are
(a) It has the longest liquid range of any known substance
(b) It has discrete diatomic molecular structure
(c) Ga liquid expands on forming solid
(d) Gallium dissolves in aqueous NaOH

191. Graphite the allotrope of carbon
(a) is crystalline in nature
(b) is less stable than diamond under standard conditions
(c) carbon is at a lesser distance from carbon in it own layer than carbon in the superposed layer
(d) is less dense than diamond.

192. The nonexistent compound(s) is/are
(a) PbI_4 (b) $Tl\,I_3$ (c) BH_3 (d) NF_5

193. The acid among the following which are both dibasic and reducing in nature are
(a) orthophosphorus acid (b) hypophosphoric acid (c) hypophosphorous acid (d) pyrophosphorousacid

194. Sulphur dioxide can be qualitatively or quantitatively determined using
(a) starch iodate paper (b) starch iodide paper (c) oxidation to H_2SO_4 (d) $K_2[HgCl_4]$

195. Consider the potassium salts of $HClO_4$ and HBF_4. The common characteristics of these two compounds is
(a) presence of π bonds (b) similar structure
(c) water solubility (d) tendency of anions to form complexes

196. On long standing, bleaching powder decomposes. The decomposition products formed are
(a) $Ca(ClO_3)_2$ (b) $Ca(OCl)_2$ (c) O_2 (d) $CaCl_2$

197. The product(s) of hydrolysis of Xenon tetrafluoride in excess of water is/are
(a) F_2 (b) Xe (c) $XeOF_2$ (d) XeO_3

Matrix-Match Type Questions Questions

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

198. Match the lettered products in each equation in the **Column I** with the properties given in **Column II**

Column I	**Column II**
(a) $Na[BH_4] + I_2 \rightarrow [A] + \text{salt} + H_2$	(p) Lewis acid
(b) $CaF_2 + B_2O_3 + H_2SO_4 \xrightarrow[-H_2O]{} [B] + \text{salt}$	(q) e^- deficient compound
(c) $Ca_2B_6O_{11} + SO_2 + H_2O \rightarrow [C] + \text{calcium salt}$	(r) planar molecule
(d) $Al_2O_3 + 3C + 3Cl_2 \rightarrow [D] + \text{gas}$	(s) intermolecular hydrogen bonding

199.

Column I	**Column II**
(a) Carbon	(p) Stable tetrahalides
(b) Silicon	(q) Gives hydrogen with NaOH
(c) Tin	(r) Acidic dioxide
(d) Lead	(s) Unstable hydrides

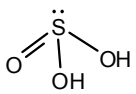
200.

Column I	**Column II**
(a) BO_3^{3-}	(p) Planar
(b) ClO_3^-	(q) sp^3 hybridized
(c) SO_3^{2-}	(r) Presence of lone pair
(d) NO_3^-	(s) Dative bonds

SOLUTIONS

ANSWER KEYS

Topic Grip

1. $AgNO_3$ and $CuSO_4$
2. Wt. of HIO_3 formed = 56.32 g
3. Carbonic acid is stronger and reacts faster
4. The gas is B_2H_6
   $B_2H_6 + 2NaH \rightarrow 2Na[BH_4]$
5. At low temperatures $AlCl_3$ exists as a close packed lattice of Cl^- with Al^{3+} occupying octahedral voids. On heating Al_2Cl_6 species are formed and the volume of the solid greatly.
6. Si contain available 3d orbitals and its co-ordination number can increase to 6
7. H_2SO_3 is a dibasic acid. It is a reducing agent 
8. The halogen is fluorine & gas evolved in O_2
9. Weak van der Waals force between molecules
10. Helium in noninflamable while H_2 is inflammable.

11. (a) **12.** (b) **13.** (b)
14. (a) **15.** (b) **16.** (d)
17. (b) **18.** (a) **19.** (b)
20. (b) **21.** (b) **22.** (a)
23. (c) **24.** (b) **25.** (d)
26. (a)
27. (a), (b)
28. (a), (c), (d)
29. (b), (c), (d)
30. (a) → (p), (q)
(b) → (p), (r)
(c) → (r), (s)
(d) → (s)

IIT Asignment Exercise

31. (d) **32.** (c) **33.** (c)
34. (c) **35.** (b) **36.** (a)
37. (b) **38.** (c) **39.** (b)
40. (a) **41.** (d) **42.** (a)
43. (d) **44.** (b) **45.** (a)
46. (c) **47.** (a) **48.** (b)
49. (a) **50.** (b) **51.** (a)
52. (d) **53.** (b) **54.** (a)
55. (c) **56.** (d) **57.** (b)
58. (d) **59.** (a) **60.** (d)
61. (b) **62.** (c) **63.** (d)
64. (a) **65.** (b) **66.** (b)
67. (a) **68.** (a) **69.** (c)
70. (a) **71.** (d) **72.** (c)
73. (d) **74** c) **75.** (a)
76. (a) **77.** (b) **78.** (a)
79. (b) **80.** (b) **81.** (a)
82. (c) **83.** (b) **84.** (d)
85. (c) **86.** (b) **87.** (b)
88. (c) **89.** (a) **90.** (b)
91. (d) **92.** (b) **93.** (c)
94. (d) **95.** (b) **96.** (c)
97. (a) **98.** (b) **99.** (c)
100. (d) **101.** (c) **102.** (a)
103. (d) **104.** (a) **105.** (c)
106. (b) **107.** (c) **108.** (a)
109. (b) **110.** (c) **111.** (c)
112. (b) **113.** (a) **114.** (c)
115. (d)
116. (a)
117. (c), (d)
118. (c), (d)
119. (a), (b), (c)
120. (a) → (p), (r)
(b) → (p), (r)
(c) → (q), (r)
(d) → (q), (s)

Additional Practice Exercise

121. $BI_3 > BBr_3 > BCl_3 > BF_3$
122. It is boron, the gas in SO_2
$2CuCl_2 + SO_2 + 2H_2O \rightarrow 2HCl + H_2SO_4 + Cu_2Cl_2$
123. $Al_2(SO_4)_3 \rightarrow \underset{(A)}{Al_2O_3} + \underset{(B)}{3SO_3}$
$4Al_2O_3 + 3S_2Cl_2 + 9Cl_2 \rightarrow \underset{(C)}{8AlCl_3} + \underset{(D)}{6SO_2}$
$2SO_2 + O_2 \rightarrow 2SO_3$
124. A is an iodide
$3I_2 + 2NH_3 \rightarrow NI_3.NH_3 + 3HI$
125. H_2CO_3 is more acidic than H_3BO_3
126. Refer SM
127. KHF_2 is formed by HF. No similar compound by other HX.
128. $CaOCl_2$ converts to $CaCl_2$ and $Ca(OCl_3)_2$
129. Sulphur is in lowest o.s.in H_2S while it is in intermediate oxidation state in SO_2
130. $CO + Cl_2 \rightarrow \underset{(A)}{COCl_2}$
$COCl_2 + H_2O \rightarrow \underset{(B)}{2HCl} + \underset{(C)}{CO_2}$
$CO_2 + \underset{(E)}{NH_3} \rightarrow \underset{(D)}{(NH_2)_2CO} + 2HCl$

131. (a) **132.** (b) **133.** (b)
134. (d) **135.** (c) **136.** (c)
137. (c) **138.** (a) **139.** (a)
140. (c) **141.** (a) **142.** (b)
143. (c) **144.** (a) **145.** (d)
146. (d) **147.** (d) **148.** (d)
149. (a) **150.** (b) **151.** (b)
152. (d) **153.** (d) **154.** (a)
155. (b) **156.** (a) **157.** (b)
158. (c) **159.** (c) **160.** (d)
161. (d) **162.** (a) **163.** (d)

164. (c) **165.** (a) **166.** (b)
167. (c) **168.** (a) **169.** (a)
170. (b) **171.** (a) **172.** (b)
173. (a) **174.** (b) **175.** (c)
176. (d) **177.** (b) **178.** (d)
179. (d) **180.** (d) **181.** (a)
182. (c) **183.** (a) **184.** (c)
185. (b) **186.** (c) **187.** (b)
188. (a)
189. (b)
190. (a), (b), (c)
191. (a), (c), (d)
192. (a), (c), (d)
193. (a), (d)
194. (a), (c), (d)
195. (b), (c), (d)
196. (a), (d)
197. (b), (d)
198. (a) → (p), (q)
(b) → (p), (r)
(c) → (p), (q), (r), (s)
(d) → (p), (q)
199. (a) → (p), (r)
(b) → (p), (q), (r)
(c) → (p), (q), (s)
(d) → (q), (s)
200. (a) → (p)
(b) → (q), (r)
(c) → (q), (r)
(d) → (p), (s)

HINTS AND EXPLANATIONS

Topic Grip

1. The two reagents are silver nitrate and copper sulphate

$AgNO_3$: NH_3 forms a soluble complex $Ag(NH_3)_2NO_3$ PH_3 forms a black precipitate of silver phosphide Ag_3P

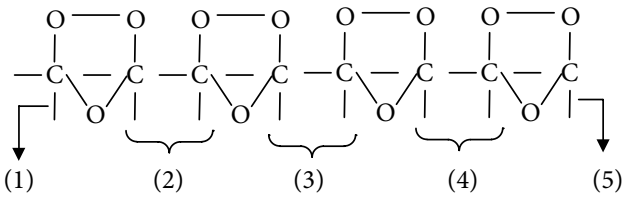
$3AgNO_3 + PH_3 \rightarrow Ag_3P + 3HNO_3$

$CuSO_4$: NH_3 forms a blue coloured soluble complex $Cu(NH_3)_4SO_4$

PH_3 forms a black precipitate of copper phosphide Cu_3P_2.

$3CuSO_4 + 2PH_3 \rightarrow Cu_3P_2 + 3H_2O$

2. If the alkene gives 5 products, it must contain 4 double bonds.



$\therefore$ 1 mole of the alkene requires 4 moles of ozone.

$\therefore$ 0.2 moles of alkene requires 0.8 moles of ozone.

According to the reaction

$I_2 + H_2O + 5O_3 \rightarrow 2HIO_3 + 5O_2$.

5 moles of ozone gives 2 moles of HIO_3.

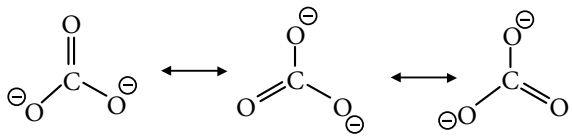
$\therefore$ 0.8 moles of ozone gives $\frac{0.8}{5} \times 2 = 0.32$ moles of HIO_3

Mol. Wt. of HIO_3 = 1 + 127 + 48 = 176

Wt. of HIO_3 obtained = 176 × 0.32 = 56.32 g.

3. Sodium carbonate is a weak base.

Among carbonic acid and orthoboric acid, carbonic acid is a stronger acid hence it will react faster. Carbonic acid is a stronger acid because the carbonate ion formed is stabilized by resonance.



4. Compound of boron, which is a foul smelling colourless gas is diborane.

A compound of boron which is a good reducing agent is sodium borohydride.

Diborane can be prepared by reducing BCl_3 with $LiAlH_4$

$4BCl_3 + 3LiAlH_4 \rightarrow 2B_2H_6 + 3AlCl_3 + 3LiCl$

Sodium borohydride can be prepared by action of sodium hydride on B_2H_6

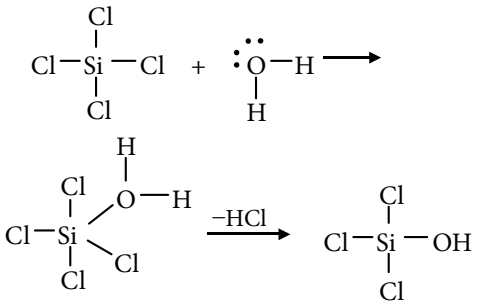
$B_2H_6 + 2NaH \rightarrow 2Na[BH_4]$

5. When aluminium chloride is crystallized as ionic solid, with close packed Cl^- ions with Al^{3+} in octahedral holes which have considerable electrostatic force of attraction. This will lead to a decrease in interionic distance in the lattice of the crystal.

On thermal decomposition, it forms a neutral Al_2Cl_6 leading to a greater distance in lattice structure. So density decreases considerably.

6. This behaviour of CCl_4 is because carbon does not have any d orbitals in it outer shell.

(i) Hydrolysis of $SiCl_4$ takes place by formation of five coordinated intermediate $SiCl_4(H_2O)$ which then dissociates with elimination of HCl giving $SiCl_3OH$. Due to non availability of d orbitals for bond formation in carbon, the five coordinated intermediate is not formed hence hydrolysis does not take place.



(ii) As $SiCl_4$ can increase its coordination from four to six due to availability of d orbitals, it forms complexes of the type $[SiF_6]^{2-}$, $[SiCl_6]^{2-}$ etc. This is not possible in the case of carbon due to non availability of d orbitals.

7. $H_2O + SO_2 \rightleftharpoons \underset{\text{sulphurous acid}}{H_2SO_3}$

SO_2 dissolves in water as mainly $SO_{2(aq)}$ and forming traces of H_2SO_3

Properties of H_2SO_3

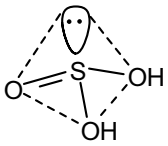
(a) diprotic, fairly strong acid

$H_2SO_3 \rightarrow H^+ + HSO_3^-$

$HSO_3^- \rightarrow H^+ + SO_3^{2-}$

(b) Strong reducing agent.

It gets oxidized to H_2SO_4 even by atmospheric oxygen. Structure.



Sulphur is sp^3 hybridized three sigma bonds and one lone pair

Pi bond is formed by $p\pi - d\pi$ overlap from oxygen and sulphur respectively.

8. The halogen is fluorine

Fluorine is the only halogen that liberates gases oxygen or oxygen difluoride on reaction with NaOH.

$\underset{\text{(cold \& dilute)}}{2NaOH} + 2F_2 \rightarrow 2NaF + OF_2 + H_2O$

$\underset{\text{(conc.)}}{4NaOH} + 2F_2 \rightarrow 4NaF + O_2\uparrow + H_2O$

with other halogens, no gas is liberated

$\underset{\text{cold \&dilute}}{2NaOH} + X_2 \rightarrow NaX + NaXO + h_2O$

$\underset{\text{conc}}{6NaOH} + 3X_2 \rightarrow 5NaX + NaXO_3 + 3H_2O$

9. It is difficult to liquify noble gases because the van der Waal's forces or intermolecular forces existing between these atoms are extremely weak.

As the atomic weight increases, atomic volume increases the van der Waal's forces increase, hence liquification is easier. So on moving from helium to xenon liquification becomes easier.

10. Inspite of a lower lifting power, helium is preferred over hydrogen because helium is inert, unreactive and non inflammable. Due to these safety aspects its disadvantage is over looked.

11. All noble gases are monoatomic in nature.

For monoatomic gases $\frac{C_P}{C_V} = 1.66$

12. Fluorine forms HOF, but higher oxy acids are not possible due to the absence of vacant d–orbitals.

13. $Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$

Initially CuO is formed which further reacts with H_2SO_4 to give $CuSO_4$. SO_2 turns limewater milky.

$Ca(OH)_2 + SO_2 \rightarrow CaSO_3\downarrow + H_2O$

14. $2Al + 2NaOH + 2H_2O \rightarrow 2NaAlO_2 + 3H_2\uparrow$

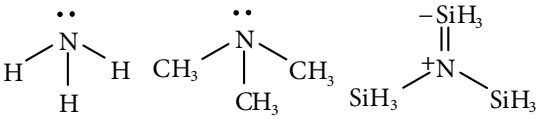
$2Al + 6NaOH \xrightarrow{\text{fused}} 2Na_3AlO_3 + 3H_2\uparrow$

Hydrogen gas produced during the process build up pressure and the pressure of the gas helps to open up clogged drains.

15. $CO_2 + 4Na \rightarrow 2Na_2O + C$

Since carbon is less electropositive, the metal will have more affinity for oxygen. Hence the reaction will continue. So CO_2 cannot be used to extinguish fire caused by burning active metals like Na, Mg etc.

16.



Due to presence of vacant d orbitals, the lone pair forms a $p\pi - d\pi$ bond between nitrogen and silicon. Due to non-availability of lone pair $N(SiH_3)$ is less basic than $N(CH_3)_3$ where lone pair is available on nitrogen.

17. All hydrides of group 16 elements except water are gases. This is due to presence of hydrogen bonding in water due to high eletronegativity of oxygen.

Covalent character increases down the group, due to increase in size of anion (Fajan's rule). Covalent character does not markedly change the state of hydrides. It is only due to hydrogen bonding.

18. $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$

$CaCO_3 + H_2SO_4 \rightarrow CaSO_4\downarrow + H_2O + CO_2$

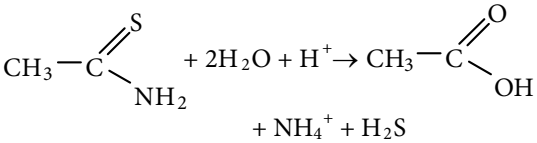
$CaSO_4$ being insoluble forms a layer on $CaCO_3$ and prevents further reaction.

19. B – Cl distance is shorter because of back bonding leading to a double bond formation. Since B – Cl bond has partial double bond character bond length is shorter. Due to small size of boron when compared to Al, back bonding is possible and so it does not dimerise.

20. Nitrogen is replaced by helium because helium is less water soluble than nitrogen so it does not cause diver's

bends as seriously as nitrogen. Nitrogen is inert due to half filled stable configuration while helium has completely filled stable configuration.

21. $2H_2S + 3O_2 \rightarrow 2H_2O + SO_2\uparrow$
in limited supply of oxygen sulphur is formed.

22. 
$+ 2H_2O + H^+ \rightarrow$

$+ NH_4^+ + H_2S$

23. $H_2S + 2O_3 \rightarrow H_2SO_4 + O_2$
$H_2S + Cl_2 \rightarrow 2HCl + S\downarrow$
$2FeCl_3 + H_2S \rightarrow 2FeCl_2 + 2HCl + S\downarrow$
$3H_2S + Cr_2O_7^{2-} + 8H^+ \rightarrow 3S\downarrow + 2Cr^{3+} + 7H_2O$

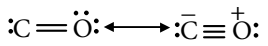
24. In the series HXO_2only $HClO_2$ is known.

25. Due to appreciable energy difference between s and p orbitals, there is uneven mixing during the formation of sp^3 hybrid orbital leading to a weak bond overlap.

26. $H - O - Cl \underset{\sigma}{\overset{\pi}{\rightleftharpoons}} O$

27. $2ClO_2 + 2O_3 \rightarrow Cl_2O_6 + 2O_2$
$2ClO_2 + 2NaOH + H_2O_2 \rightarrow 2NaClO_2 + O_2 + 2H_2O$
$10HI + 2ClO_2 \rightarrow 2HCl + 4H_2O + 5I_2$
$Al + 3(ClO_2) \rightarrow Al(ClO_2)_3$

28. Due to small size and high electronegativity of carbon, it can easily form $p\pi - p\pi$ multiple bonds. Thus forming C ≡ O stable carbon monoxide. It is resonance stabilized.



This type of π overlap is not seen in silicon.

29. B_2H_6 (Ratio 1:2) $+ 2NH_3 \xrightarrow{-120°C} B_2H_6.2NH_3$ diammoniate of diborane

$2(B_2H_6.2NH_3) \xrightarrow{200°C} 2B_3N_3H_6 + 12H_2$ Borazole

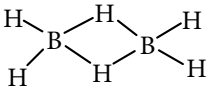
$B_2H_6 + xNH_3$ (excess) $\xrightarrow{\text{high temp.}}$ $(BN)_x$ boron nitride

30. (a) → (p), (q)
(b) → (p) (r)
(c) → (r), (s)
(d) → (s)
Fluorine and chlorine as gases at STP.
Br_2 is a liquid while I_2 is a solid.
F_2 liberates oxygen with alkali while other halogens form NaOX and NaCl or $NaXO_3$ and NaCl.
Cl_2 and Br_2 are good bleaching agent. Cl_2 being a better bleaching agent than Br_2.
I_2 and to some extent Br_2 are oxidized by ozone to iodates and bromates.

IIT Asignment Exercise

31. The greater the Lewis acid character, the faster the reaction with ammonia.
The Lewis acid character is in the order $BI_3 > BBr_3 > BCl_3 > BF_3$ this is due to greater back bonding due to small size of fluorine back bonding decreases on going down the group due to increase in size.

32. $2BF_3 + 6NaH \xrightarrow{180°c} B_2H_6(g) + 6NaF$

33. 
Only the terminal hydrogens can be methylated.

34. $3Ga^+ \rightarrow Ga^{3+} + 2Ga$

35. Dilute and con.HNO_3 render Al passive. This is due to surface oxidation and formation of a thin oxide film which prevents further reaction

36. Aluminium on reaction with alkali forms gelatinous white precipitate of $Al(OH)_3$. This is soluble in excess NaOH
$2Al + 6NaOH \rightarrow 2Na_3AlO_3 + 3H_2$

37. Metal should be in +I and +3 oxidation state

38. H_3BO_3 is a monobasic weak acid, Lewis acid

39. Gallium is below aluminium in the group but 3d electrons are filled up on going down due to poor shielding of d electrons, increase in size is not appreciable.

40. WC, carbon occupies the interstitial lattice spaces in tungsten

41. $Sn + 2\ NaOH \rightarrow Na_2SnO_2 + H_2$
$Sn + 2\ HCl \rightarrow SnCl_2 + H_2$

42. On going down any group, electropositive character increases i.e., metallic nature increases

43. $PbCl_2$ and S_nCl_4 are stable
$PbCl_4$ is a good oxidizing agent $Pb^{4+} \xrightarrow{2e^-} Pb^{2+}$

$SnCl_2$ is a good reducing agent

$Sn^{2+} \rightarrow Sn^{4+} + 2e^-$

44. $CO + Cl_2 \xrightarrow{\text{sunlight}} \underset{\text{Phosgene}}{COCl_2}$

45. O = C = C = C = O

Carbon is sp hybridized. Oxides like C_2O, C_2O_3 are still less stable.

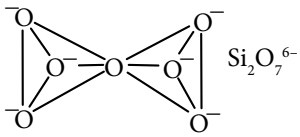
46. Silicon dissolves only in a mixture of HNO_3 and HCl or aqueous KOH

$Si + 2KOH + H_2O \rightarrow K_2SiO_3 + 2H_2$

47. $R_3SiCl + H_2O \rightarrow R_3Si - OH + HCl$

$R_3Si - OH + HOSi - R_3 \rightarrow R_3 - Si - O - SiR_3$

48. Pyrosilicates contain $2SiO_4^{4-}$ tetrahedral units



49. $NH_2CONH_2 + 2HNO_2 \rightarrow 2N_2\uparrow + CO_2 + 3H_2O$

50. NF_3 does not undergo hydrolysis because both nitrogen and fluorine does not have vacant d orbitals

51. As the size of the central atom increases, the strength of the metal hydrogen bond decreases.

It is easier to remove the hydrogen for a larger atom so BrH_3 is more reducing than NH_3.

52. Acidic nature of the oxide decreases down the group. N_2O_5, P_2O_5, As_2O_5 are water soluble while Sb_2O_5 and Bi_2O_5 are water insoluble but soluble in alkali.

53. The reaction is

$2NH_4Cl + \underset{74g}{Ca(OH)_2} \rightarrow 2NH_3 + CaCl_2 + 2H_2O$

$1.48g = \underset{\text{moles}}{0.02} \quad \underset{\text{moles}}{0.04}$

At STP 1 mole = 22.4 L

0.04 moles = 22.4×0.04

$= 0.896$ L

$\cong 0.9$ L

54. In HNO_2 N is in +3 oxidation state, so it can be oxidized to +5 state.

Nitrogen in (b), (c), (d) are in +5 state

$\underset{\text{orange}}{K_2Cr_2O_7} + 4H_2SO_4 + 3HNO_2 \rightarrow$

$K_2SO_4 + \underset{\text{green}}{Cr_2(SO_4)_3} + 4H_2O$

55. $2PH_3 + 16HNO_3 \rightarrow P_2O_5 + 16NO_2 + 11H_2O$

HNO_3 oxidizes PH_3

P_2O_5 is a solid.

56. On decomposition of HNO_3 in the presence of sunlight, NO_2 is formed which imparts the yellow colour

$4HNO_3 \xrightarrow{\text{Sunlight}} 4NO_2 + 2H_2O + O_2$

57. Phosphorus in PO_3^- is sp^3 hybridized.

58. Br^- or I^- on reaction with H_2SO_4 liberate HBr and HI respectively which are reducing

$NaBr + H_2SO_4 \rightarrow NaHSO_4 + HBr$

$2HBr + H_2SO_4 \rightarrow 2H_2O + SO_2 + Br_2$

SO_2 being reducing turns dichromate paper green.

59. Sodium thiosulphate $Na_2S_2O_3 \cdot 5H_2O$

$6NaOH + 4S \rightarrow Na_2S_2O_3 + 2Na_2S + 3H_2O$

60. As the atomic size increases, the metal hydrogen bond strength decreases, hence H^+ can easily dissociate

61. $2HCl + O_3 \rightarrow H_2O + Cl_2 + O_2$

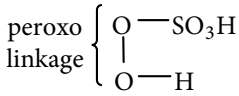
$H_2O_2 + O_3 \rightarrow H_2O + 2O_2$

62. $O_3 \rightarrow O_2 + O$

Due to formation of atomic oxygen, it is oxidizing

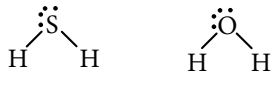
63. Sulphur forms 8 membered puckered ring structure.

64. H_2SO_5



Peroxy monosulphuric acid or caro's acid

65. Due to low electronegativity of sulphur, bonded pair is farther away than in H_2O, so the $b_p - b_p$ repulsion is small, hence bond angle is small.



66. Since H_2S is a reducing agent, sulphur in −2 oxidation state, it will easily get oxidized in the presence of H_2SO_4

67. Iodine forms I^+, formation of positive ion and slight metallic lustre are characteristic of metallic nature.

68. Due to strong hydrogen bonding in fluorine, its value will be the highest.

69. The halogen is iodine.
Iodine does not react with noble gases, water reacts with hydrogen in the presence of catalyst on heating.

70. HF
Since electronegativity decreases down the group, dipole moment also decreases down the group dipole moment order HF > HCl > HBr > HI

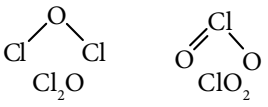
71. HF attacks silicates as well as silica and dissolves them forming soluble fluorosilicates

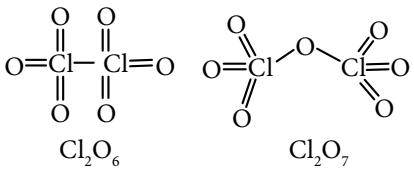
$SiO_2 + 4HF \rightarrow SiF_4 + 2H_2O$

$SiF_4 + 2HF \rightarrow H_2SiF_6$

Soluble fluorosilicate

72.





73. I_2 is a weak oxidizing agent ie I^- is strongly reducing I^- reduces Pb^{4+} to Pb^{2+} hence it does not exist.

74. CF_4 – Sp^3, tetrahedral, no lp, $\mu = 0$

SF_4 – Sp^3d, see saw , 1 lp $\mu \neq 0$

XeF_4 – sp^3d^2 square planar, 2 lp $\mu = 0$

75. $CaOCl_2 + CO_2 \rightarrow CaCO_3 + Cl_2$
Amount of chlorine released is called available chlorine.

76. $Xe + F_2 \rightarrow XeF_2$
F is reduced to F^- due to high oxidizing nature of fluorine.

77. As it is very difficult to remove an electron, they have high ionization energy.

78. 'a' is true
Noble gases are odourless, fairly high electrical insulators and I.E are high and decreases down the group.

79. Bond order = 0 so there are no bonds hence monoatomic.
cp/cv value also show that they are monoatomic.

80. $2XeF_2 + 2H_2O \rightarrow 2Xe + 4HF + O_2$

81. $B_2H_6 + 3O_2 \rightarrow B_2O_3 + 3H_2O$

$H_3BO_3 \xrightarrow[-H_2O]{100^\circ} 4HBO_2 \xrightarrow{160^\circ} H_2B_4O_7$

$\xrightarrow{\Delta} 2B_2O_3 + H_2O$

82. Gallium has an unusual structure.
It tends towards a discrete diatomic molecules rather than a metallic structure. This accounts for the incredibly low melting point.

83. $B + Cl_2 \xrightarrow[\text{Temp}]{\text{high}} BCI_3$

$BCl_3 + 3H_2O \rightarrow 3HCl + H_3BO_3$

H_3BO_3 is a weak acid.

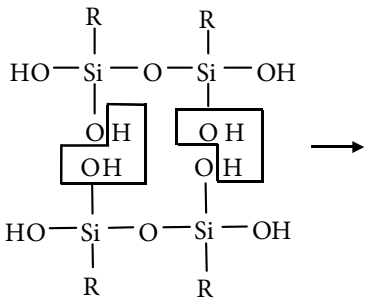
84. Aluminium is a strong reducing agent so it also reduces oxides of non metals.

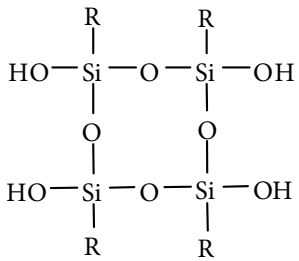
$3CO_2 + 4Al \rightarrow 2Al_2O_3 + 3C$

$3SiO_2 + 4Al \rightarrow 2Al_2O_3 + 3Si$

85. Fluoride of aluminium is ionic, so aluminium is not electron deficient.
$AlCl_3$ is covalent with sp^3 hybridized aluminium.

86. Unlike carbon, which cannot hold more than one hydroxyl group, the silicon atom can hold three hydroxyl groups $RSi(OH)_3$. It is this property of silicon that makes formation of organo silicon possible.





87. In graphite, the bond order of carbon is $1\frac{1}{3}$ $\left(\text{or}\frac{4}{3}\right)$

88. In diamond carbon bond distance = 1.54A° and bond angle is the tetrahedral angle as in C_2H_6.

89. Silicates of alkali metals are water soluble.

90. $3SiCl_4 + 4Al \rightarrow 3Si + 4AlCl_3$.
Dissolved silicon in molten aluminium crystallizes out an cooling.

91. $I_2O_5 + 5CO \rightarrow 5CO_2 + I_2$
Iodine liberated is estimated using thiosulphate.

92. $SnCl_4$ would dissolve in solutions containing appropriate halogen acid.

$$Sn(OH)_4 \underset{H_2O}{\overset{HCl}{\rightleftharpoons}} SnCl_4 \underset{H_2O}{\overset{HCl}{\rightleftharpoons}} [SnCl_6]^{2-}$$

93. $4NH_3 + 5O_2 \xrightarrow{\text{ptgauze}} 4NO + 6H_2O$
a – $6N_3H + 4Li \rightarrow 4LiN_3 + 2NH_3 + 2N_2 \uparrow$
b – $4NH_3 + 3Ca(OCl)_2 \rightarrow 3CaCl_2 + 6H_2O + 2N_2 \uparrow$
d – $8NH_3 + 3Br_2 \rightarrow 6NH_4Br + N_2 \uparrow$

94. All oxides and oxyacid have a N = O NO_2 is paramagnetic, N_2O is neutral oxide N_2O is not symmetric, N_2O_3 also has one asymmetric structure.

95. Reduction of nitrates generally gives NO or NO_2 but in alkaline solution with metals, ammonia is formed e.g., Devarda's alloy (Cu/Al/Zn), Al or Zn Cu/Al/Zn + NaOH → H.
$NO_3^- + 9H \rightarrow NH_3 + 3H_2O$

96. If phosphorus is burnt in air it would form P_4O_{10} so P_4O_6 cannot be prepared by this method since As and Sb do not have a high tendency to form, the pentoxide, they stop with formation of trioxide on burning.

97. PCl_5 – sp^3d hybridized while
PCl_4^+ – sp^3 and PCl_6^- is sp^3d^2
PI_5 does not exist and PBr_5 exists as PBr_4^+ and Br^-

98.

(a)	Meta phosphoric acid	HPO_3	–	+5
(b)	Hypo phosphorous acid	H_3PO_2	–	+1
(c)	Orthophosphoric acid	H_3PO_4	–	+5
(d)	Phosphorous acid	H_3PO_3	–	+3

99. Volatile nature decreases down the group. Since H_2O has hydrogen bonds, it is not volatile So H_2S is the most volatile hydride.
$H_2O < H_2Te < H_2Se < H_2S$ most volatile

100. $Cs_2 + 3Cl_2 \rightarrow CCl_4 + S_2Cl_2$
S_2Cl_2 has open book type structure like hydrogen peroxide.

101. Ice has an open cage like structure due to extensive hydrogen bonding. On melting some of the H–bonds break and the molecules get folded together resulting in a reduction in volume.

102. Due to small size of oxygen atom, it forms double bonds ie pπ pπ bonds between two oxygen atoms and Therefore, it exists as discrete diatomic molecule. While sulphur exists as puckered 8-membered rings.

103. When sulphuric acid reacts with sugar it, removes water and gives carbon. As it removes water in this reaction it is a dehydrating agent.

104. Chlorine is passed through salt water containing Br^-. Cl_2 displaces bromine from its solution as chlorine is more oxidizing.

105. (c) is true for all halogens
(a) is not true for iodine. PI_5 does not exist
(b) is true only for chlorine and bromine
(d) is not true for iodine

106.

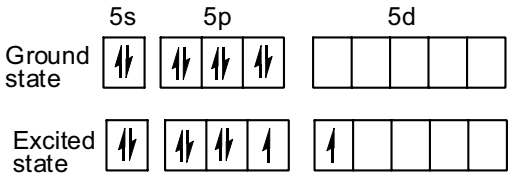
$$H_2SO_4 + Ca\begin{matrix} \nearrow Cl\ (-1) \\ \searrow OCl\ (+1) \end{matrix} \xrightarrow{H_2O} Cl^\circ_2 + CaSO_4 + H_2O$$

107. Fluorine completely decomposes water to O_2 and O_3 fluoride can be qualitatively estimated using $CaCl_2$

$$NaF + CaCl_2 \rightarrow \underset{\text{whiteppt}}{CaF_2} + 2NaCl$$

108. Fluorine does not form KXO_3.
When (X) is Br or Cl, the reaction is
$2KXO_3 \rightarrow 2KX + 3O_2$
When (X) is I, the reaction is
$4KIO_3 \rightarrow 3KIO_4 + KI$

109. Xe is sp^3d hybridized in XeF_2.



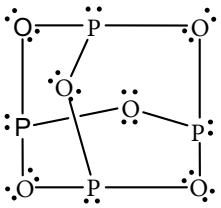
110. Helium has lowest known boiling point On cooling it only forms helium liquid. It solidifies only under high pressure of 25 atm.

111. Hydrides of group 16 are H_2O, H_2S, H_2Se, H_2Te. On going down the group, atomic size of the chalcogen increases. Hence H – M bond strength decreases. Moreover M^- formed is more stable as size increases So acid strength order is $H_2O < H_2S < H_2Se < H_2Te$ most acidic.

On going down the group electronegativity decreases hence dipole moment decreases so polarity decreases down the group.

$$\underset{\text{high } \mu}{H_2O} > H_2S > H_2Se > H_2Te$$

112. P_4O_6



Phosphorus atoms lie in the corners of a tetrahedron. Oxygen atoms lie in between the phosphorus atoms and are covalently bonded to two phosphorus atoms.

Two lone pairs on oxygen($6 \times 2 = 12$)

One lone pair on phosphorus($4 \times 1 = 4$)

Total of 16 lone pairs.

113. Bisulphite decomposes and forms sulphite along with liberation of SO_2 gas.

$$2NaHSO_3 \rightarrow Na_2SO_3 + H_2O + SO_2$$

In the presence of SO_2 gas, H_2S is oxidized to colloidal sulphur.

$$2H_2S + SO_2 \rightarrow 2H_2O + 3S$$

Overall Rxn . $2NaHSO_3 + 2H_2S \rightarrow Na_2SO_3 + 3S + 3H_2O$

As all S^{2-} ions are oxidized in the presence of HSO_3^- H_2S cannot be used as a reagent

114. On reaction with $Al_2(SO_4)_3$, bicarbonate produces CO_2 gas along with white foam of $Al(OH)_3$ formation of $Al(OH)_3$ is advantageous as it deposits on burning articles and cuts off their air supply.

115. Even though CO_2 is not combustible and is not a supporter of combustion, when used for metals like Na, K etc, it reacts with these active metals, hence the reaction continues and the fire is not put out.

$$4K + CO_2 \rightarrow 2K_2O + C$$

116. $2HOCH_2CH_2NH_2 + CO_2 + H_2O \underset{100 - 150°C}{\overset{30 - 60°C}{\rightleftharpoons}} (HOCH_2CH_2NH_3)_2CO_3$

This is called Girbotol process

CO_2 can be recovered by absorbing it in either aqueous Na_2CO_3 or ethanolamine.

117. Peroxide reacts with dil acid to give H_2O_2 but does not liberate oxygen even with con H_2SO_4

$M_2O_2 + H_2SO_4 \rightarrow M_2SO_4 + H_2O_2$

As size of 'M' increases, M_2O_2 becomes more stable because a large cation is stabilised by a large anion.

118. Indium and Thallium are not affected by alkali

Boron $\quad 2B + 6NaOH \rightarrow 2Na_3BO_3 + 3H_2$

Al and Ga – $\quad 2M + 2NaOH + 2H_2O \rightarrow 2NaMO_2 + 3H_2$

119. The difference in shape of the oxyacids is because of

(a) small size of C and N so they cannot accommodate more oxygen atoms.

(b) as they cannot utilize the d orbitals they cannot expand their Octet.

120. (a) – (p), (r)

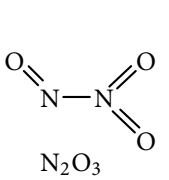
(b) – (p), (r)

(c) – (q), (r)

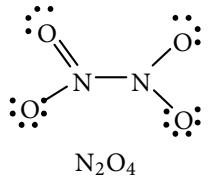
(d) – (q), (s)

$N_2O_3 + 2NaOH \rightarrow 2NaNO_2 + H_2O$

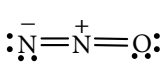
$N_2O_4 + 2NaOH \rightarrow NaNO_2 + NaNO_3 + H_2O$



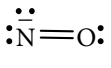
N_2O_3



N_2O_4



N_2O



NO

NO is an odd electron molecule with II electrons and is a neutral oxide.

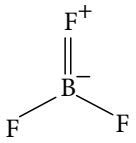
Additional Practice Exercise

121. The Lewis acid character is in the order $BI_3 > BBr_3 > BCl_3 > BF_3$, though the order should have been the

reverse on the basis of electronegativity of the halogen. This anomalous behaviour can be explained on the basis of relative tendency of the halogen atom to back donate its unutilized electrons to the vacant 'p' orbitals of boron atom. As a result of back bonding of electrons the deficiency of electrons on the boron atom is reduced.

The tendency for the formation of back bonding ($p\pi$-$p\pi$ bond) is maximum in BF_3 and decreases rapidly from BF_3 to BI_3 because efficiency of overlapping decreases as the size of the halogen atom increases.

$\therefore$ BF_3 is a weaker Lewis acid.



122. Nonmetal in the group III is boron while all others are metals.

$$2B + 3H_2SO_4 \rightarrow 2H_3BO_3 + 3SO_2\uparrow.$$

SO_2 gas released is passed through $CuCl_2$. Sulphur dioxide reduces $CuCl_2$ (which is blue) to Cu_2Cl_2 (which is white precipitate).

$$\underset{\text{blue cupric chloride}}{2CuCl_2} + SO_2 + 2H_2O \rightarrow 2HCl + H_2SO_4 + \underset{\text{white ppt}}{Cu_2Cl_2}\downarrow$$

123. Aluminium sulphate on heating gives alumina and sulphur trioxide.

$$Al_2(SO_4)_3 \xrightarrow{\text{strong heat}} \underset{\substack{\text{Alumina}\\(A)}}{Al_2O_3} + \underset{\substack{\text{sulphur trioxide}\\(B)}}{3SO_3}$$

Al_2O_3 is treated with S_2Cl_2 and Cl_2

$$4Al_2O_3 + 3S_2Cl_2 + 9Cl_2 \rightarrow \underset{\substack{\text{aluminium chloride}\\(C)}}{8AlCl_3} + \underset{\substack{\text{sulphur dioxide}\\(D)}}{6SO_2}$$

SO_2 is oxidised to SO_3 by contact process and it is an exothermic reaction favoured by low temperature and high pressure.V_2O_5 is used as a catalyst.

$$2SO_2 + O_2 \rightarrow 2SO_3$$

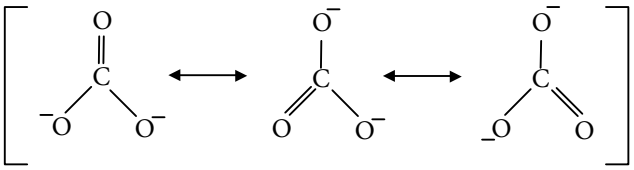
124. Since violet vapours turn starch paper blue, they are iodine vapours. So compound (A) is an iodide. Iodine on reaction with ammonia forms nitrogen triiodide, which is a black explosive powder.

$$3I_2 + 2NH_3 \rightarrow NI_3.NH_3 + 3HI$$

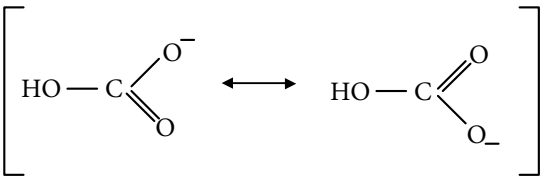
$$8NI_3.NH_3 \rightarrow 5N_2 + 9I_2 + 6NH_4I$$

125. Both H_2CO_3 and H_3BO_3 are weak acids.

In the case of H_2CO_3, the anion formed is CO_3^{2-}. This is stabilized by resonance.



HCO_3^- ion is also resonance stabilized.



This type of stabilization of the anion by resonance does not occur in $[B(OH)_4]^-$ from H_3BO_3.

$$B(OH)_3 + H_2O \rightarrow H^+ + [B(OH)_4]^-$$

Therefore, H_2CO_3 is more acidic than H_3BO_3.

126. Sodium salt of F^-, Cl^-, Br^-, I^- on reaction with H_2SO_4

NaCl reacts with H_2SO_4 to give sodium sulphate and hydrogen chloride

$$2NaCl + H_2SO_4 \rightarrow Na_2SO_4 + 2HCl$$

Both NaBr and NaI evolve hydrogen halides which are oxidized to the halogen being strong reducing agents and reduce H_2SO_4 to SO_2, S and H_2S.

$$NaBr + H_2SO_4 \rightarrow NaHSO_4 + HBr$$

$$2HBr + H_2SO_4 \rightarrow SO_2 + Br_2 + 2H_2O.$$

$$NaI + H_2SO_4 \rightarrow NaHSO_4 + HI$$

$$2HI + H_2SO_4 \rightarrow SO_2 + I_2 + 2H_2O.$$

$$6HI + H_2SO_4 \rightarrow 4H_2O + S + I_2$$

$$8HI + H_2SO_4 \rightarrow 4H_2O + H_2S + 4I_2$$

127. Among the halogens, only potassium hydrogen fluoride exists whereas $KHCl_2$, $KHBr_2$, KHI_2 do not exist.

H_2F_2 exists as dimeric molecule due to hydrogen boding. Hence it gives two series of salts KHF_2 as [K^+ and F^-…H–F] and KF[K^+ and F^-] whereas in HCl and HBr, hydrogen bonding is not possible.

128. Bleaching action of bleaching powder is due to the release of Cl_2 at the time of application. Chlorine is

lost by bleaching powder on long standing in open air by two processes.

(a) $CaOCl_2 + CO_2(\text{from air}) \rightarrow CaCO_3 + Cl_2$

(b) $6CaOCl_2 \rightarrow 5CaCl_2 + Ca(ClO_3)_2$

129. On the basis of the electronic configuration of sulphur ($3s^2\ 3p^4$), the maximum and minimum oxidation states it can exhibit are +6 and −2 respectively. In H_2S, sulphur is in −2 oxidation state. It can only increase the oxidation state and not decrease. So H_2S can act only as a reducing agent. In SO_2, sulphur is in +4 oxidation state, which can be increased as well as decreased. Thus SO_2 can act as both reducing and oxidizing agents.

130. $CO + Cl_2 \rightarrow \underset{(A)}{COCl_2}$

(A) is phosgene or carbonyl chloride

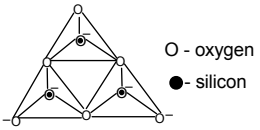
$$COCl_2 \xrightarrow{H_2O} \underset{(B)}{2HCl} + \underset{(C)}{CO_2}$$

$COCl_2 + (E) \rightarrow 2HCl + (D)$ and

$CO_2 + (E) \rightarrow (D) + H_2O$

Comparing the two equations (E) should be ammonia.

$$COCl_2 + 2NH_3 \longrightarrow \underset{\underset{\underset{\

143. 

It is a $Si_3O_9^{6-}$ unit. So the silicate salt is $M_3Si_3O_9$

144. Compound formed is a silicone

$$2CH_3Cl + Si \xrightarrow[300°C]{Cu} (CH_3)_2SiCl_2 \xrightarrow{H_2O}$$

$$(CH_3)_2Si(OH)_2$$

These condense with elimination of water.

Depending on number of R group they can be three dimensional or dimer. They do not have an Si = 0. Organometallic compounds are compounds in which there is a direct covalent bond from carbon to a metal or to a les electro negative atom like Si, B, Ge, Sb, As.

145. Due to low electronegativity of iodine among the halogens, NI_3 is the strongest Lewis base due to the availability of lone pair on nitrogen.

146.

	P_4	$3O_2$		P_4O_6
	$\frac{5}{124}$	$\frac{5}{32}$		
Initial	0.04	0.15		
After	0	0.03		0.04 moles
	P_4O_6 +		$2O_2$	$\rightarrow P_4O_{10}$
Initial	0.04		0.03	
After	0.04 – 0.0015 = 0.0385		0	0.0015

Weight of $P_4O_6 = 0.0385 \times 220 = 8.47$ g

147. Reducing character increases down the group as size increases down the group, so the M – H bond strength decreases.

148. Going by MOT

NCl has two unpaired electrons

NCl^+ and NCl^- have one unpaired electron each.

149. $2NH_3 + NaOCl \rightarrow N_2H_4 + NaCl + H_2O$

NH_2Cl is formed as an intermediate

Glue or gelatin is added to inhibit the side reaction.

$2NH_2Cl + N_2H_4 \rightarrow N_2 + 2NH_4Cl$

and thereby avoid loss of N_2H_4

150. $Ca_3(PO_4)_2 + 2H_2SO_4 \xrightarrow{\Delta}$

$$\underbrace{Ca(H_2PO_4)_2 + 2CaSO_4.2H_2O}_{\text{super phosphate of lime}}$$

option (a) gives triple superphosphate.

151. $\underset{\text{ortho phosphoric acid}}{2H_3PO_4} \xrightarrow[-H_2O]{250°C} \underset{\text{pyrophosphoric acid}}{H_4P_2O_7} \xrightarrow[-H_2O]{500°C} \underset{\text{Metaphosphoric acid}}{2HPO_3}$

152. $H_2O_2 + 2I^- + 2H^+ \rightarrow 2H_2O + I_2$

(2×17) (2×127)

254 g of I_2 is liberated from 34 g of H_2O_2

1.015 g of I_2 is liberated from $\frac{34}{254} \times 1.015$

$= 0.136$ g of H_2O_2.

$$\text{Molarity} = \frac{\text{no. of moles}}{\text{volume in litres}}$$

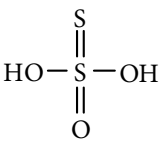
$$= \frac{0.136/34}{10/1000} = \frac{0.004}{0.01} = 0.4 \text{ M}$$

153. Oxygen shows paramagnetic property in all the three states. Its ground state electronic configuration has two unpaired electrons, thereby showing paramagnetic property.

154. On heating to 600°C, sulphur exists as S_2 molecules. These are paramagnetic and blue coloured and presumably has similar bonding like oxygen.

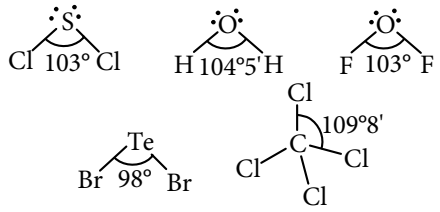
155. Moving across the period, atomic size decreases. Due to smaller atomic size, i.e., presence of smaller d orbitals overlap leading to $p\pi - d\pi$ double bonds is easier in sulphur and polymerisation does not occur to a large extent. Smaller the size like Cl there is no polymerization at all. While in SiO_4 as the π bonds are weaker, polymerization occurs to a larger extent forming Si – O – Si linkages.

156.



Sulphur is in both +6 oxidation state as well as –2 oxidation state.

157. SO_2, SeO_2 – are acidic and water soluble and also dissolve in alkali forming sulphates and selenates. TeO_2 is amphoteric so they dissolve in NaOH. But are water insoluble.

158. 

Tellurium is very less electronegative than sulphur, the bond pair is closer to the halogen. So due to repellsion the lone pair can easily compress the bond pair.

159. Statement (a), (b) and (d) are false.
I_2 dissolves in KI solution as KI_3

160. I^- is reducing in nature rather than F^-. So on exposure to air I^- gets oxidised to I_2 which does not occur in F^-.

$$4HI + O_2 \rightarrow 2I_2 + 2H_2O$$

161. As oxidation state of Cl increases, covalent character of Cl – O bond increases, so thermal stability increases. Between HClO and HBrO , is less stable due to low electronegativity.

162. Fluorine forms complexes of the type $[AlF_6]^{3-}$ due to its small size.

163. Para periodic acid is hydrated periodic acid.
$HIO_4.\ 2H_2O$ i.e., H_5IO_6

$$\underset{\substack{2\times228\ g\\ 22.8\ g}}{2H_5IO_6} \xrightarrow[-4H_2O]{100^\circ C} \underset{\text{periodic acid}}{2HIO_4} \xrightarrow{200^\circ C}$$

$$\underset{\substack{\text{iodine}\\ \text{pentoxide}\\ 334\ g\\ \frac{334}{2\times228}\times228\\ =16.7\ g}}{I_2O_5} + O_2 + H_2O$$

164. $3NH_3 + 3F_2 \rightarrow N_2 + 6HF$

$$C_6H_6 + 9F_2 \xrightarrow[265^\circ C]{\text{Cu gauze}} C_6F_{12} + 6HF;$$

$$KClO_3 + F_2 + H_2O \rightarrow KClO_4 + H_2F_2$$

$$H_2S + 4F_2 \rightarrow SF_6 + 2HF$$

165. HF is stable and not oxidized even by strongly oxidizing agents.

166. Only Ar, Kr, Xe form clathanate compounds. It is not possible for He and Ne due to small size.
(a), (c), (d) are false.

167. Xe has low ionization energy. Due to lower ionization energy it is able to form compounds.

168. XeF_2 undergoes slow hydrolysis, while XeF_4 and XeF_6 are hydrolyzed violently.

169. XeF_4 has 2 lone pairs – square planar geometry while (b), (c), (d) have one lone pair each.

170. Xe has lower ionization energy.
The element which is highly electronegative would react.
Fluorine being highly electronegative forms compounds.

171. Benzene is a nonpolar solvent. Al_2Cl_6 exists as a dimer in benzene.

In water, which is a polar solvent with high dielectric constant, the dimer breaks due to stabilization by water molecules forming $[Al(H_2O)_6]^{3+}$ with high heat of hydration.

172. $\underset{\text{Borax}}{Na_2B_4O_7.10H_2O} \xrightarrow[-10H_2O]{\text{heat}} Na_2B_4O_7 \longrightarrow$

$$\underbrace{2NaBO_2 + B_2O_3}_{\text{glassy mass}}$$

Due to the presence of 10 molecules of water, on heating it releases the water molecules and Therefore, swells up.
After this, on heating further it forms a glassy bead.

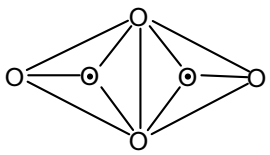
173. Silanes are more susceptible to nucleophilic attack compared to alkenes because

(i) Silicon is less electronegative than hydrogen, which is not so for carbon. [Si – 1.92; H – 2.30; C – 2.54]

(ii) Si is a larger atom than carbon, so it provides greater surface area for attack by nucleophile.

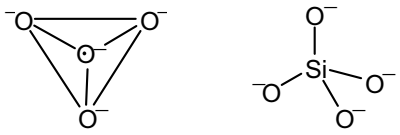
(iii) Silicon has vacant low lying d orbitals which accept electron pairs from nucleophiles.

174. Both statements are true
Two SiO_4^{4-} units cannot have two oxygen atoms in common or an edge of a tetrahedra in common.

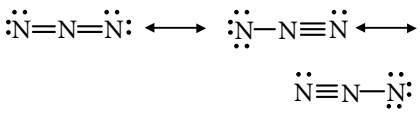


Such a sharing is not possible because it would be unstable owing to close approach of the O^{2-} ions that

it would bring about. In silicates silicon is surrounded tetrahedrally by four oxygen atoms.



175. N_3^- ion has a linear structure



NaN_3 is used to inflate air bags, because it undergoes thermal decomposition to liberate Nitrogen gas

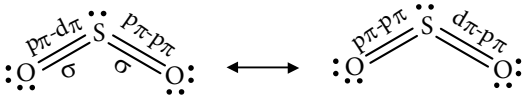
$2NaN_3 \rightarrow 3N_2 + 2Na$

176. $2HNO_3 \rightarrow 2NO_2 + H_2O + O$

Oxidizing action of HNO_3 is more due to NO_2 rather than oxygen released.

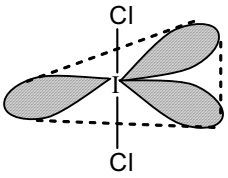
If urea is added, it will react with NO_2 and oxidizing action of HNO_3 will be reduced.

177. Sulphur is sp^2 hybridized. Bond angle is reduced due to lone pair-bond pair repulsion and is 119°5'. One sigma bond is formed each between s and two oxygen atom. π bond formed is pπ – pπ bond and the other is a pπ–dπ bond. So the two S – O bonds are different but the bond lengths are equal because of resonance.



178. Statement (2) is correct. F – F bond length shortest due to smaller atomic size. Inspite of higher electron affinity of Cl_2 compared to F_2, F_2 is a stronger oxidizing agent because F_2 has a small size so it has high heat of hydration.

179. ICl_2^- has linear geometry and symmetric arrangement. It has symmetric arrangement of the three lone pairs around the central iodine atoms and the net bond pair lone pair repulsion is the same along the axial direction.



180. Radon has lower ionization energy than Xenon. Even though radon is expected to form many compounds similar to Xenon, it is not so because radon is radioactive with isotopes of short half lives.

181. $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$

$2NO + O_2 \rightleftharpoons 2NO_2$

$2NO_2 + H_2O \rightarrow HNO_3 + HNO_2$

$2HNO_2 \rightarrow H_2O + NO_2 + NO$

$3NO_2 + H_2O \rightarrow 2HNO_3 + NO$

$NH_3 + 2O_2 \rightarrow HNO_3 + H_2O$

182. HNO_3 and H_2O form a constant boiling mixture so they distill out as an azeotropic mixture with concentration of 68% HNO_3.

183. Nitronium ion NO_2^+ is formed

In the presence of strong acid, HNO_3 behaves as a base $HO^-.NO_2^+$.

184. On going down the group due to larger sizes M – H bond is longer and weaker hence thermal stability reduces.

185. $SiH_4 + 3H_2O \rightarrow H_2SiO_3 + 4H_2$

$Si_2H_6 + 6H_2O \rightarrow 2H_2SiO_3 + 7H_2$

Stability of hydrides towards water follows the order. Only SiH_4 is hydrolysed by water.

$$CH_4 > GeH_4 > SnH_4 > SiH_4$$

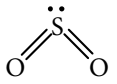
Si has the smallest empty d – orbitals among Si, Ge, Sn. Therefore, it can be easily coordinated by water molecules which is the first step in hydrolysis.

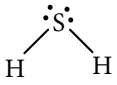
186. Si, Ge, Sn can be prepared by treating their halide or oxide with strong reducing agents

$$GeCl_4 + LiAlH_4 \xrightarrow{\text{dry ether}} GeH_4 + LiCl + AlCl_3$$

187. A is SO_2 gas. It is water soluble. It forms discrete v shaped molecule and is a gas. On going down the group, SeO_2, TeO_2 etc form infinite chains and are solids, SO_2 can be easily oxidized to H_2SO_4 dibasic mineral acid. Liquid SO_2 is a self dissociating non aqueous solvent.

$$2SO_2 \rightleftharpoons SO^{2+} + SO_3^{2-}$$

188. A is SO_2  sp^2 hybridized sulphur

B is H_2S  sp^3 hybridized

189. $\underset{A}{SO_2} + \underset{B}{2H_2S} \xrightarrow{\text{aqueous}} 3S + 2H_2O$

turbidity is due to formation of colloidal sulphur

SO_2 is reduced

H_2S is oxidized

190. d is true for B and Al also

Ga melts at 30°C and boils at 2070°C

It forms discrete diatomic structure rather than a metallic structure.

191. Graphite is the most stable crystalline allotrope of carbon. In the layer carbon is covalently bonded and the different layers are held by van der Waals forces. So the inter layer distance is greater.

Due to the open structure of graphite, it is less dense than diamond.

Graphite = 2.22 g/cm^3; Diamond 3.5 g/cm^3.

192. PbI_4 does not exist because Pb^{4+} is highly oxidizing and I^- is strongly reducing which results in the conversion to PbI_2

TlI_3 exists as $Tl^+ I_3^-$

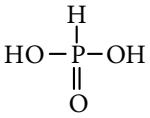
BH_3 (Borine or Borane) does not exist as it is electron deficient.

It dimerises to B_2H_6 and polymerizes to higher borone

NF_5 does not exist as N contains no d-orbitals. One s and three p- orbital in the outer most shell of N can form a maximum of 4 single covalent bonds.

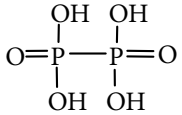
$\therefore$ only NF_3 is formed.

193. Phosphorus acids are reducing due to presence of P – H bond.

Ortho phosphorus acid H_3PO_3 

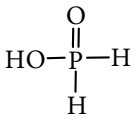
dibasic, reducing

Hypo phosphoric acid $H_4P_2O_6$



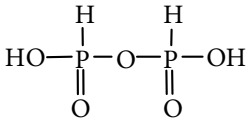
tetrabasic, not reducing

Hypo phosphorous acid H_3PO_2



Monobasic, reducing

Pyro phosphorous acid $H_4P_2O_5$



dibasic, reducing

194. $2KIO_3 + 5SO_2 + 4H_2O \rightarrow I_2 + KHSO_4 + 3H_2SO_4$

Starch iodate paper turns blue.

SO_2 on oxidation with H_2O_2 gives H_2SO_4 which can be estimated.

$$SO_2 + H_2O_2 \rightarrow H_2SO_4$$

$K_2[HgCl_4]$ is a mercury complex which is estimated colorimetrically with pararosaniline dye

$$K_2[HgCl_4] + 2SO_2 + 2H_2O \rightarrow K_2[Hg(SO_3)_2] + 4HCl$$

195. Both have tetrahedral structure.

$KClO_4$ and KBF_4 are sparingly soluble in water they do not have a tendency to form complex ions in aqueous solution.

196. On long standing, bleaching powder decomposes as

$$6CaOCl_2 \rightarrow Ca(ClO_3)_2 + 5CaCl_2$$

It loses oxygen when it decomposes in presence of $CoCl_2$ and forms $CaCl_2$ in solution.

197. $6XeF_4 + \underset{\substack{\text{excess}\\\text{water}}}{12H_2O} \rightarrow 4Xe + 2XeO_3 + 24HF + 3O_2$

$$2XeF_4 + 3H_2O \rightarrow Xe + XeO_3 + 6HF + F_2$$

198. (a) $\rightarrow$ (p), (q)

(b) $\rightarrow$ (p), (r)

(c) $\rightarrow$ (p), (q), (r), (s)

(d) $\rightarrow$ (p), (q)

$2Na[BH_4^-] + I_2 \rightarrow B_2H_6 + 2NaI + H_2$

A is B_2H_6. It has 3C – 2e bonds reacts with ammonia accepting its lone pair so it is a weak Lewis acid.

$3CaF_2 + B_2O_3 + 3H_2SO_4$

$\rightarrow 2BF_3 + 3CaSO_4 + 3H_2O$

BF_3 is a Lewis acid and a planar molecule. Boron is sp^2 hybridized.

$$Co_2B_6O_{11} + 4SO_2 + 11H_2O \rightarrow 2Ca(HSO_3)_2 + 6H_3BO_3$$

H_3BO_3 is a OH^- acceptor, hence a Lewis acid. Due to the presence of covalent O – H bond, it has inter molecular hydrogen bonds. B is sp^2 hybridized and molecule is planar and electron deficient.

$$2Al_2O_3 + 3C + 3Cl_2 \rightarrow 2AlCl_3 + 3CO\uparrow$$

$AlCl_3$ is a electron acceptor and is an electron deficient species.

It exists as a dimer where aluminium is tetrahedrally bonded to 4 chlorine atoms.

199. (a) $\rightarrow$ (p), (r)
(b) $\rightarrow$ (p), (q), (r)
(c) $\rightarrow$ (p), (q), (s)
(d) $\rightarrow$ (q), (s)

Carbon, silicon and tin form stable tetrahalides while lead does not form tetrahalides due to inert pair effect.

Si, Sn and lead dissolve in NaOH and liberate hydrogen gas.

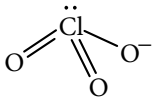
SiO_2 and CO_2 are acidic. While PbO_2 and SnO_2 are basic. Basic character increases down the group.

Carbon and silicon form stable hydrides.

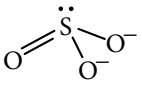
200. (a) $\rightarrow$ (p)
(b) $\rightarrow$ (q), (r)
(c) $\rightarrow$ (q), (r)
(d) $\rightarrow$ (p), (s)

In BO_3^{3-}, boron is sp^2 hybridized and has a planar trigonal structure.

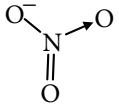
In ClO_3^-, Cl is sp^3 hybridized, with one lone pair of electrons, has a pyramidal structure with $2p\pi-d\pi$ double bonds.



In SO_3^{2-}, sulphur is sp^3 hybridized . It has one lone pair of electrons, has a pyramidal structure.



NO_3^- has sp^2 hybridized nitrogen. It has two covalent bonds and one dative bond and has a trigonal planar structure.



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CHAPTER

3

ORES AND MINERALS, EXTRACTIVE METALLURGY, TRANSITION ELEMENTS

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INTRODUCTION

Metals are an important group of substances in which each atom is surrounded by a large number of other atoms of the same kind. The bonding is three dimensional and usually strong. The substance is characterized by high electrical conductivity which decreases with increase of temperature.

There are not enough valence shell orbitals or electrons for each metal atom to form two centre two electron bonds to each of its neighbours. Hence there must be multicentre orbitals. The electrical conductivity which is due to electronic mobility proves that the multicentre orbitals extend over the whole crystal.

Metallic bond

Metals are made up of positive ions packed together. Electrons originating from the neutral metal atoms hold the positive metal ions together. The high electrical and thermal conductivities of metals are because of the mobility of these electrons through the lattice.

Bond length

If the valence electrons in a metal are spread over a large number of bonds, each bond should be longer and hence weaker. For example, the alkali metals exist as diatomic molecules in the gaseous state and the interatomic distances in the metal crystal are longer than in the diatomic molecule. The total bonding energy is greater in the metal crystal than in the M_2 molecule as there are many more of the bonds in the metal crystal.

Theories of bonding in metals

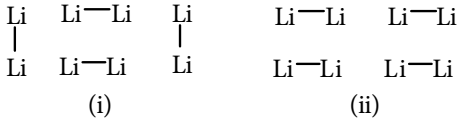
Different theories are proposed to account for the behaviour of metals. The important ones are

(i) Free electron theory

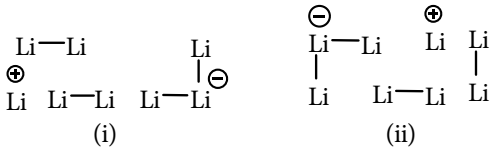
It was suggested that metals comprise a lattice of positive ions, embedded in a gas of free valency electrons which could move in the interstices. This model explains the free movement of electrons. Cohesion results from electrostatic attraction between the positive ions and the electron cloud.

(ii) Valence bond theory

Bonding in metals can be explained by taking a simple metal such as lithium. A lithium atom has one electron in its outer shell which may be shared with one of its neighbours forming a normal two electron bond. The atom could be bonded to any of its other eight neighbours in the lattice and hence many different arrangements are possible as represented below.



A lithium atom may ionize to give Li^- and Li^+. Then it can form many structures of the type represented above.

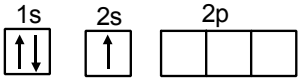


The true structure is a mixture of all the many possible bonding forms.

The presence of ions could explain the electrical conduction, but this theory does not explain the conduction of heat in solids.

(iii) Molecular orbital theory or band theory

Electronic configuration of lithium atom is



The bonding occurs using the 2s atomic orbitals. There are three empty 2p orbitals in the valency shell of each lithium atom. Presence of empty atomic orbitals is essential for metallic properties. That is why nitrogen, oxygen, fluorine, neon, etc., are non-metals as they all lack empty atomic orbitals.

Empty atomic orbitals may be used to form additional bonds in two different ways

1. Empty atomic orbitals may accept lone pair of electrons from other atoms forming coordinate bonds.
2. Each atom may share its few electrons with several of its neighbours and obtain a share in their electrons, which results in the formation of cluster compounds.

Consider the combination of two lithium atoms to give Li_2. Ignoring the inner electrons, the 2s atomic orbitals on each of the two Li atoms combine to form two molecular orbitals. The bonding molecular orbital is occupied by two electrons and the antibonding molecular orbital is empty.

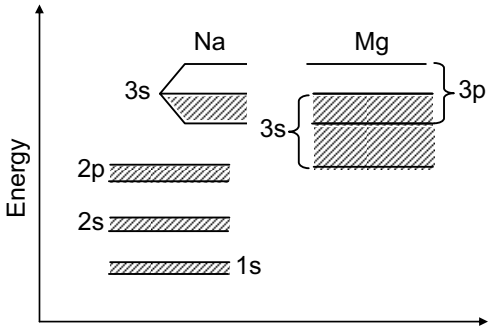
When three Li atoms combine to give Li_3, three atomic orbitals combine to form three molecular orbitals — one bonding, one non–bonding and one anti–bonding. There are three valence electrons from three Li atoms. Two of them occupy the bonding molecular orbital and the third one goes to the non–bonding molecular orbital which lies between bonding and anti–bonding molecular orbitals.

In Li_4–, the four 2s atomic orbitals would form four molecular orbitals — two bonding and two anti-bonding. Valence electrons occupy the bonding orbitals only and the anti-bonding orbitals are empty.

In a metal, lattice there are an extremely large number of metal atoms. The 2s atomic orbitals on these atoms form twice the number of molecular orbitals; half of them are bonding and the other half anti–bonding. All these molecular orbitals together form an energy band that is only half – filled.

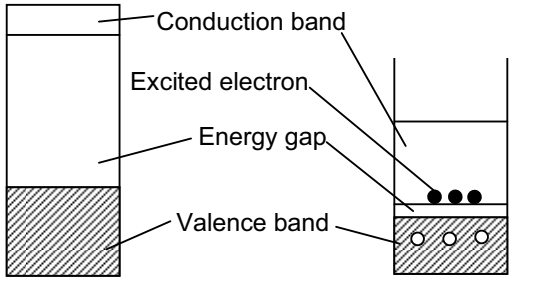
In different metals, different types of atomic orbitals can overlap to form different energy bands. The arrangement of electrons in different energy bands determines the characteristics of a metal. The energy bands formed from different atomic orbitals may overlap or be separated from each other. The highest occupied energy band is called the valence band while the lowest unoccupied energy band is called conduction band. The energy gap between the top of the valence band and the bottom of the conduction band is called the energy gap. In the case of metals the valence band may be half–filled or may be overlapping with the conduction band. This makes it possible for the electrons to move into vacant bands, making metals good conductors.

In the case of insulators, the energy gap is very large and in semi conductors the gap is very small and increase in temperature gives enough thermal energy to some of the electrons in valence band to move into conduction band. Sodium and magnesium are good conductors. In sodium the valence band is only half filled. In magnesium, the valence band overlaps with conduction band.



Formation of energy band in Na and Mg (representation)

Energies of filled and empty bands in an insulator and a semiconductor in which a few electrons have been promoted to the conduction band are shown as



ORES AND MINERALS

Metallurgy is the science and art of winning and refining metals, and preparing them for use. The process of extraction of a metal from the ore is called winning of the metal. Refining is the purification of the metal that has been extracted from the ore.

Metals occur in nature either in the free state or in the combined state. When the metal is found free it is said to occur as native. Gold, platinum silver, copper and mercury are the metals that are found in native state. All the other metals exist as their compounds. These metal compounds are called minerals. The minerals from which the metal can be extracted economically are referred to as ores.

Some of the common minerals and their composition are given below:

Table 3.1 Name and composition of certain minerals

Metal	Name	composition	Metal	Name	composition
Magnesium	Magnesite	$MgCO_3$	Iron	Haematite	Fe_2O_3
	Carnallite	$KCl.MgCl_2.6H_2O$		Magnetite	Fe_3O_4
	Epsom salt	$MgSO_4.7H_2O$		Limonite	$Fe_2O_3.3H_2O$
Aluminum	Bauxite	$Al_2O_3.2H_2O$		Siderite	$FeCO_3$
	Cryolite	Na_3AlF_6	Copper	Cuprite	Cu_2O
Tin	Cassiterite (tin stone)	SnO_2		Copper pyrites	$CuFeS_2$
				Malachite	$CuCO_3.Cu(OH)_2$
Lead	Galena	PbS	Silver	Argentite	Ag_2S
	Anglesite	$PbSO_4$		Horn silver	AgCl
	Cerrusite	$PbCO_3$	Zinc	Zinc blende	ZnS
				Calamine	$ZnCO_3$
				Zincite	ZnO

EXTRACTIVE METALLURGY

An ore is a complex mixture of a mineral containing the metal and gangue, which consists of sand, clay and other earthly impurities. The various steps in the extraction, of a metal from an ore are

(i) Concentration of the ore
(ii) Heat treatment of concentrated ore–Calcination and Roasting
(iii) Reduction to the free metal
(iv) Refining and purification of the metal.

In addition to extraction, metallurgy involves production of alloys, namely metallic solution composed of two or more elements.

(i) Concentration of the ore or ore dressing

Ores are concentrated by separating the mineral from gangue or matrix. The separation method makes use of the density difference in gravity separation, magnetic property difference in electromagnetic separation and wettability difference in froth floatation.

Froth floatation method

The metal sulphide ores are concentrated by froth floatation. This process makes use of the difference in the ability of water and oil to wet surfaces of the mineral and gangue. For example chalcopyrite ($CuFeS_2$) is suspended in water to which collectors such as pine oil and froth stabilizers such as cresols and aniline are added. Air is blown through it. The gangue becomes moistened by polar water molecules and sinks to the bottom. The metal sulphide particles become wet by oil and are carried to the surface by froth formed which is skimmed off, at the top of the tank.

Leaching

Another method of concentration is by chemical treatment or leaching. In the Baeyer's process, the Al_2O_3 in bauxite ore is separated from Fe_2O_3 impurities by treating the ore with hot aqueous NaOH. The amphoteric Al_2O_3 dissolves as $Al(OH)_4^-$, while Fe_2O_3 does not. Fe_2O_3 is filtered off. The filtrate containing $Al(OH)_4^-$ is neutralized by passing CO_2 and seeded with freshly precipitated $Al(OH)_3$ which induces precipitation of $Al(OH)_3$. $Al(OH)_3$ is filtered, dried and calcined to get pure alumina.

$$Al_2O_{3(s)} + 2(OH^-) + 3H_2O_{(\ell)} \rightarrow 2Al(OH)_4^-$$

$$Al(OH)_4^- + CO_2 \rightarrow Al(OH)_3 + HCO_3^-$$

$$2Al(OH)_3 \rightarrow Al_2O_3 + 3H_2O$$

(ii) Heat treatment of concentrated ore– Calcination and Roasting

Calcination

Calcination is used to remove volatile impurities associated with concentrated ore by heating in limited supply of air. It is usually adopted for carbonate ores and hydrated oxide ores.

$$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$$

Roasting

Roasting is heating of the ore below its melting point in a free supply of air to oxidize it.

Roasting is generally employed to convert sulphide ores to oxides as in the case of ZnS. The sulphur dioxide produced is used to manufacture H_2SO_4.

$$2ZnS + 3O_2 \xrightarrow{\text{heat}} 2ZnO + 2SO_2$$

(iii) Winning of metal

The concentrated ore is reduced to the free metal by a suitable method.

Extraction of metal from metal compound involves reduction. A suitable reducing agent is necessary to bring about reduction. Metals are electropositive, hence they are themselves good reducing agents. In order to reduce a metal compound to metal a reducing agent stronger than the metal itself is needed. The reducing power of a metal can be ascertained with the help of its standard reduction potential. It is known that a reducing agent having lesser reduction potential than the given metal can bring about the reduction of the metal salt. This is applicable only in aqueous medium since enthalpy of hydration plays an important role in standard electrode potentials.

The method used for obtaining metal from concentrated ore usually depends on the activity of the metal as indicated by its standard reduction potential. The most active metals have the most negative standard reduction potentials. These are not easily reduced. The least active metals with positive standard reduction potentials are easily reduced. The following gives the method employed in reduction of different ores.

Table 3.2 Method of extracting metal

Metal	Method
Cu, Hg, Pb	Roasting of metal sulphide (self – reduction)
Fe, Zn, Sn	Reduction of metal oxide with carbon
Al	Electrolysis of molten Al_2O_3 in cryolite
Li, Na, Mg	Electrolysis of molten metal chloride
Au, Ag	Reduction of cyanide complex with Zn

Two important metals Au and Pt are noble metals and are usually found in nature in the native state.

Electrolytic reduction

Electron is the strongest reducing agent. Any ionic substance may be electrolysed and reduction will occur at the cathode. Electrolysis can be performed in

(i) aqueous solution
(ii) other solvents
(iii) fused melts

The electrolysis is done in aqueous solution if the products do not react with water. Copper and zinc may be obtained by electrolysis in aqueous solution, which is a cheap method.

Self reduction

Sulphide ore of copper may be roasted in air until a part is converted into oxide. This is further heated in absence of air to get the metal. This is known as self reduction method.

$$3Cu_2S \xrightarrow[-2SO_2]{\text{roasted in air}} 2Cu_2O + Cu_2S \xrightarrow[\text{absence of air}]{\text{heated in}} 6Cu + SO_2$$

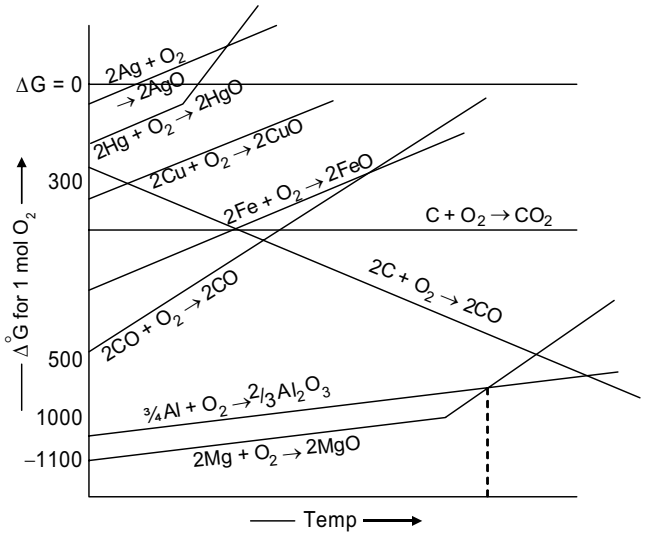
Here S^{2-} ion reduces Cu^{2+} to Cu. S^{2-} is oxidized to S^{4-} in SO_2

Smelting is the process of removing gangue from concentrated ore as molten slag and obtaining the purified ore or the metal in molten form by fusing with suitable flux and other required materials.

Copper pyrites is smelted to get Cu_2S (matte) and magnetite is smelted to get iron in the molten state.

Reduction of metal compounds under anhydrous condition require some other parameter. Free energy is a parameter which can be used under such conditions. Usually concentrated ores are converted to the oxides and the oxides are reduced to metals. The free energy change for the formation of oxide can be considered for assessing the ease of reducing the oxide. Lesser the free energy of formation of the oxide, greater is its stability and greater is the difficulty to reduce it to metal. The ΔG° of formation of oxide depends on temperature. The plot of ΔG° of formation for oxide per mole of oxygen consumed vs temperature is

known as Ellingham diagram. Ellingham diagram for various metals plotted together is used for deciding the reducibility of an oxide compared to others.



An element corresponding to lower line in the diagram reduces a higher metal oxide to metal. This diagram explains the theory behind alumino–thermy, carbon and CO reduction, magnesium reduction, etc.

Hydrometallurgy

In hydrometallurgy the mineral containing metal or metal compound is leached with a suitable reagent to form a soluble metal compound. The metal is recovered from the solution by adding a more electropositive metal. Copper from low grade copper, silver from argentite, gold from auriferous quartz etc are obtained by this method.

(iv) Refining and purification of metals

The metals obtained by different extraction procedures have to be purified. The methods adopted depends on the nature of metal to be purified and the impurity that is to be removed. Some of the refining methods for various elements are listed below:

Table 3.3 Refining methods

Method of refining	Elements
Liquation	Sn
Distillation	Hg, Zn
Zone refining	Ge, Si
Cupellation	Ag
Poling	Cu
Electrolytic refining	Cu, Al, Zn, Pb, Ag
Vapour phase refining:	
(i) Mond's process	Ni
(ii) van Arkel process	Ti, Zr

CONCEPT STRANDS

Concept Strand 1

Fluorine is obtained by electrolysis of KHF_2 dissolved in anhydrous HF. Why?

Solution

Fluorine reacts violently with water and hence the electrolysis of KHF_2 is carried out in anhydrous HF. Elements reacting with water are extracted from the fused melts of their ionic salts. Aluminium is obtained by electrolysis of fused mixture of Al_2O_3 and cryolyte, Na_3AlF_6.

Na and Cl_2 are obtained by electrolysis of fused NaCl. $CaCl_2$ is added to reduce the melting point of the fused melt from 803°C to 505°C.

Concept Strand 2

What are the factors to be considered to choose a particular extraction procedure?

Solution

(i) Unreactive elements may exist in the free state in nature and no specific procedure need to be used.
(ii) The thermal stability of compound should be analysed.
(iii) The nature of compound, ionic or not and stability of the element in water must be checked.
(iv) If the sulphide ore is present, it may be roasted in air to get oxide and then heated in the absence of air.

(v) If it does not form a carbide, carbon reduction method may be employed.
(vi) If other methods are unsuitable, electrolysis may be used for ionic substances. If elements are stable in water, electrolysis in aqueous solution is cheaper.

Concept Strand 3

How is Cu used for electrical wiring refined?

Solution

Acidified (H_2SO_4) copper sulphate solution is electrolysed using thick sheets of impure copper as anode and thin sheets of pure copper as cathode. Copper at the anode is oxidised to Cu^{2+} and the dissolved Cu^{2+} ions of the aqueous solution are reduced to copper and deposited on the cathode as pure (99.99%) copper. This method of purification is called *electrorefining*.

PRINCIPLES AND REACTIONS INVOLVED IN EXTRACTION OF CERTAIN METALS

Carbon reduction method

(i) Extraction of iron

The iron ore, haematite (Fe_2O_3), is reduced by carbon monoxide in a blast furnace. A charge of iron ore, coke and limestone ($CaCO_3$) is introduced at the top of the furnace and a blast of hot air is sent in at the bottom. Coke burns yielding CO at a temperature of about 2000°C.

$$C + O_2 \rightarrow CO_2 \qquad \Delta H = -393.5 \text{ kJ}$$

$$C + CO_2 \rightarrow 2CO \qquad \Delta H = +172.5 \text{ kJ}$$

As the charge descends and CO rises, a complex series of high temperature reactions occurs in different regions of the furnace as given below:

At 300 – 600°C

$$3Fe_2O_3 + CO \rightarrow 2Fe_3O_4 + CO_2$$

$$Fe_3O_4 + CO \rightarrow 3FeO + CO_2$$

At 700 – 1000°C

$$C + CO_2 \rightarrow 2CO$$

$$FeO + CO \rightarrow CO_2 + Fe(\text{spongy ion})$$

$$CaCO_3 \rightarrow CaO + CO_2$$

At 1000 – 1200°C

Molten slag forms and impure Fe melts.

$$CaO + SiO_2 \rightarrow CaSiO_3$$

At 1500 – 1800°C

$$C + O_2 \rightarrow CO_2$$

$$C + CO_2 \rightarrow 2CO$$

The overall reaction is the reduction of Fe_2O_3 to Fe metal (m.p. = 1538°C), which is obtained as an impure liquid at the bottom of the furnace.

$$Fe_2O_{3(s)} + 3CO_{(g)} \rightarrow 2Fe_{(\ell)} + 3CO_{2(g)}$$

Limestone is added to remove the gangue from the iron ore. At the high temperature of the furnace, the limestone decomposes yielding lime (CaO), which reacts with SiO_2 and other acidic oxides present in the gangue. The product referred to as slag is a molten material consisting mainly of calcium silicate.

$$CaCO_{3(s)} \rightarrow CaO_{(s)} + CO_{2(g)}$$

$$CaO_{(s)} + SiO_{2(s)} \rightarrow \underset{(\text{slag})}{CaSiO_{3(\ell)}}$$

The slag floats on surface of Fe since it is immiscible with molten iron and has less density relative to Fe. Iron and slag are removed separately from the bottom through separate taps. The iron formed is called cast iron or pig iron. This contains 4% elemental carbon and other impurities like Si, P, S and Mn. Fe is purified by the basic oxygen process. Molten Fe from blast furnace is exposed to a jet of pure oxygen for 20 minutes in a furnace lined with CaO.

The impurities in Fe are oxidized and acidic oxides formed react with CaO to yield a molten slag, which can be poured off. Now the required amount of carbon is added in the form of Spiegeleisen an alloy of Mn, Fe and C. If required, other alloying metals are also added.

Wrought iron is the purest form of Fe containing the lowest percentage of carbon (0.12 to 0.25%) and impurities S, P, Si and Mn to the extent of 0.3%. Steel contains 0.15 to 1.5% carbon. It is classified as mild steel (0.15 – 0.5% carbon) and hard steel (0.6 to 1.5% carbon).

CONCEPT STRAND

Concept Strand 4

Write the reactions involving the impurities of P and Mn with oxygen.

Solution

$$P_{4(\ell)} + 5O_{2(g)} \rightarrow P_4O_{10(\ell)}$$

$$6CaO_{(s)} + P_4O_{10(\ell)} \rightarrow \underset{(slag)}{2Ca_3(PO_4)_{2(\ell)}}$$

$$2Mn_{(s)} + O_{2(g)} \rightarrow 2MnO_{(s)}$$

$$MnO_{(s)} + SiO_{2(g)} \rightarrow \underset{(sa\,lg)}{MnSiO_{3(\ell)}}$$

(ii) Extraction of tin

The main ore of tin is cassiterite or tin stone, SnO_2. The ore, which is heavy is first dressed by levigation and roasted in moderate heat to avoid clotting of sulphides present. The arsenic present, is removed as volatile As_4O_6. Sulphur burns off as SO_2, and Cu and Fe get converted to corresponding sulphates.

The tungstates of iron and manganese present in the roasted ore are removed by dropping it on a travelling belt over electromagnetic rollers. The tungstates collect as a heap near the magnets and SnO_2 collects as a heap away. The roasted ore is moistened and left for a few days and then agitated in a tank with water. Copper and iron sulphates dissolve and tin and iron oxides settle down due to density factor. The lighter ferric oxide is separated by washing.

The concentrated tin oxide is called black tin and contains 70% tin as oxide. This is then mixed with anthracite coal and heated in a reverberatory furnace. A little lime or fluorspar is added in case silica is present. The oxide is reduced to metal.

$$SnO_2 + 2C \rightarrow Sn + 2CO$$

Liquation process is used to refine tin. The crude mass called pig tin is heated in another reverberatory furnace to a regulated temperature. Pure tin fuses readily and the first molten portion is collected in a cast iron vessel. The metal is then subjected to poling in a refining kettle. The molten metal is stirred with logs of green wood. The impurities collect around the wood and are removed. It is further purified by electrolysis using impure tin as anode and pure tin as cathode. The electrolyte is hydrofluorosilicic acid (H_2SiF_6) containing tin sulphate acidified with sulphuric acid.

Self reduction method

(i) Extraction of copper

The important ores of copper are chalcopyrites or copper pyrites, $CuFeS_2$, malachite, $CuCO_3.Cu(OH)_2$, chalcocite or copper glance, Cu_2S and cuprite, Cu_2O.

The copper pyrites ore is dressed by froth floatation and roasted in a current of air on the hearth of a reverberatory furnace. The sulphur present is oxidized and removed as SO_2. As and Sb present are removed as volatile oxides. The pyrites is converted to a mixture of Cu_2S and FeS. A small part of the sulphides are further oxidized to FeO and Cu_2O.

$$2CuFeS_2 + O_2 \rightarrow Cu_2S + 2FeS + SO_2$$

$$2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$$

$$2FeS + 3O_2 \rightarrow 2FeO + 2SO_2$$

The roasted ore is transferred to a mini blast furnace and heated with coke and sand in presence of excess air. FeS is almost completely converted into FeO, which forms slag and is removed.

$$2FeS + 3O_2 \rightarrow 2FeO + 2SO_2$$

$$FeO + SiO_2 \rightarrow FeSiO_3 \text{ (slag)}$$

A molten mixture of Cu_2S with a small amount of FeS called matte is obtained. The molten matte collects at the bottom of the furnace. Copper metal is then recovered by bessemerisation. The reactions taking place during bessemerisation are the following. Part of the Cu_2S is oxidised to Cu_2O. FeS is converted to FeO and is removed as slag ($FeSiO_3$).

$$2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$$

$$Cu_2O + FeS \rightarrow Cu_2S + FeO$$

$$FeO + SiO_2 \rightarrow \underset{(slag)}{FeSiO_3}$$

$$2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$$

The metal obtained is called blister copper. It is 98% pure. The crude copper which contains impurities like Ni, Zn, Pb, Sn, etc. is purified by electrolytic refining, in which the impure copper acts as anode and pure copper in plate form is the cathode and electrolyte used is acidified (H_2SO_4) $CuSO_4$ for refining. The crude copper becomes pure after refining (99.99%).

(ii) Extraction of lead

The main ore of lead is galena, PbS. The sulphide ore can be converted to lead by one of the following two processes.

(i) Air reduction process
(ii) Carbon reduction process

In the air reduction process the concentrated galena is heated in a reverberatory furnace at moderate temperature in excess supply of air. Lead sulphide is partly oxidized to oxide and partly to sulphate.

$$2PbS + 3O_2 \rightarrow 2PbO + 2SO_2$$

$$PbS + 2O_2 \rightarrow PbSO_4$$

Air supply is reduced, more galena added and again the temperature increased. Both PbO and $PbSO_4$ formed react with PbS to form metallic lead.

$$PbSO_4 + PbS \rightarrow 2Pb + 2SO_2$$

$$2PbO + PbS \rightarrow 3Pb + SO_2$$

Molten lead is drawn off and purified. The silver is removed by desilverisation. Desilverisation is done by Pattinson's process or by Parke's process.

CONCEPT STRAND

Concept Strand 5

Describe Pattinson's process of desilverization.

Solution

It involves fractional crystallization of silver and lead from an alloy involving them. The fused alloy is gradually cooled when lead deposits first. It is removed. After a series of lead depositions, a mixture of Pb and Ag begin to separate at the eutectic point (303°C). The alloy rich in silver is cupelled and we get refined Ag. The Pattinson's process uses the fact that alloy of Pb and Ag containing less than 2.25% of silver has a lower melting point than pure lead. Pure lead has a higher density. The first portions of Pb recovered contain little Ag. The desilverized Pb is refined by electrolysis. The electrolyte is $PbSiF_6$ containing 8–12% of H_2SiF_6. The anode is crude lead and cathode is pure lead. A small amount of gelatin is added to the fluorosilicate for smooth deposition of lead.

Electrolytic reduction method

(i) Extraction of magnesium

Important ores are magnesite, $MgCO_3$, dolomite, $MgCO_3.CaCO_3$ and carnallite, $KCl.MgCl_2.6H_2O$.

Mg can be prepared by

(i) Electrolysis method
(ii) Thermal reduction method

Here we discuss the electrolysis of fused $MgCl_2$. Fused carnallite, $MgCl_2.KCl.6H_2O$ is made anhydrous before electrolysis. Initially four water molecules are removed. Further removal of water does not produce anhydrous $MgCl_2$ but gives MgO, which cannot be melted and electrolysed.

$$MgCl_2.2H_2O \rightarrow MgO + 2HCl + H_2O$$

Therefore, the final stages of dehydration are carried out in dry HCl gas at 350°C, when anhydrous $MgCl_2$ is obtained.

The electrolysis is done in iron crucibles. The bath temperature is 700°C. The anodes of carbon or graphite are lead covered and have a porcelain hood dipping into electrolyte. The chlorine evolved is removed through pipes in hoods and used in the magnesium chloride plant. The

iron crucible containing the electrolyte is the cathode. Magnesium collects at the cathode in the form of drops, which coalesce and rise to the surface and is protected by hood from attack by Cl_2. Air in the apparatus is displaced by an inert gas like coal gas or hydrogen to prevent the oxidation of liberated magnesium by the oxygen of air. The metal separated from electrolyte is 99.9% pure. It is refined by remelting in a flux of anhydrous $MgCl_2$ and NaCl.

(ii) Extraction of aluminium

The main ore of aluminium is bauxite, $Al_2O_3.2H_2O$. Bauxite is purified to get pure alumina by one of the two processes.

1. Baeyer's process

Bauxite is roasted to oxidize FeO to Fe_2O_3 and digested with concentrated NaOH. Bauxite dissolves as sodium aluminate.

$$Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2Na[Al(OH)_4]$$

Insoluble Fe_2O_3, TiO_2 and SiO_2 are removed by filtration. Part of SiO_2 dissolves as Na_2SiO_3. Solution is diluted, partially neutralized by passing CO_2, and seeded with freshly precipitated $Al(OH)_3$. $Al(OH)_3$ is precipitated and Na_2SiO_3 remains in solution. $Al(OH)_3$ is filtered, dried and strongly heated to get pure alumina.

$$Na[Al(OH)_4] \rightarrow NaOH + Al(OH)_3$$

$$2NaOH + CO_2 \rightarrow Na_2CO_3 + H_2O$$

$$2Al(OH)_3 \rightarrow Al_2O_3 + 3H_2O$$

2. Hall's process

At first, bauxite is concentrated by fusing with sodium carbonate. Al_2O_3 dissolves in Na_2CO_3 to form sodium metaaluminate, $NaAlO_2$. The fused mass is extracted with water. The solution is heated to 50–60°C in a current of CO_2. Then aluminium hydroxide is precipitated. $Al(OH)_3$ is filtered off and washed. Then it is calcined at 1500°C when we get Al_2O_3.

$$Al_2O_3.2H_2O + Na_2CO_3 \rightarrow 2NaAlO_2 + 2H_2O + CO_2$$

$$2NaAlO_2 + 3H_2O + CO_2 \rightarrow 2Al(OH)_3 + Na_2CO_3$$

$$2Al(OH)_3 \rightarrow Al_2O_3 + 3H_2O$$

Electrolysis of fused alumina to get aluminium

Now Al_2O_3 is first electrolytically reduced by Hall–Heroult process by dissolving Al_2O_3 in fused cryolite, $AlF_3.3NaF$. Electrolysis is done in an iron tank lined with carbon. Carbon lining round the furnace acts as the cathode. The anodes are composed of carbon rods. The temperature is maintained at 950° – 1000°C. The oxygen liberated at anode react with carbon of the anodes to form CO and CO_2 which escape. Aluminium liberated at cathode gets deposited in the base of the bath. Aluminium of 99.8% purity is obtained.

Electrolytic refining is then done using Hoop's electrolytic process which consists of an iron cell containing three fused layers.

(i) the bottom anode layer consists of molten impure aluminium containing Cu, Si and Fe as impurities.
(ii) the middle layer of molten cryolite and BaF_2 which acts as the electrolyte
(iii) the top cathode layer of molten pure Al obtained from the reduction process.

The aluminium obtained from this refining cell is very ductile and has high resistance to corrosion.

Cyanide process

Cyanide process is employed for the extraction of gold and silver

(i) Extraction of silver

Ores are argentite or silver glance, Ag_2S, horn silver, AgCl and native silver. The Mac–Arthur–Forrest cyanide process can be used for extraction of Ag (and Au). The sulphide ore is first concentrated by froth floatation process. The sulphide comes to the top and silicious impurities settle at the bottom. The concentrated ore is ground to a fine powder and treated with 0.4–0.6% NaCN solution and the entire mass is agitated by passing air. Silver complexes with NaCN to form $Na[Ag(CN)_2]$. Air supplied serves to remove Na_2S from the equilibrium mixture and reaction proceeds in forward direction.

$$Ag_2S + 4NaCN \rightleftharpoons 2Na[Ag(CN)_2] + Na_2S$$

$$4Na_2S + 5O_2 + 2H_2O \rightarrow 2Na_2SO_4 + 4NaOH + 2S$$

The soluble sodium argentocyanide is removed by filtration. When zinc powder is added to a clear solution of $Na[Ag(CN)_2]$, silver is precipitated as a dark amorphous mass.

$$2Na[Ag(CN)_2] + Zn \rightarrow 2Ag + Na_2[Zn(CN)_4]$$

The precipitated silver is removed by filtration, dried and fused with KNO_3 or borax when a bright compact mass is obtained. Silver is refined electrolytically from a solution of $AgNO_3$ containing about 10% HNO_3. The impure silver blocks acts as anode and a sheet of pure silver act as cathode.

(ii) Extraction of gold

Rocks containing traces of gold are crushed and treated with a 0.1 – 0.2% solution of NaCN and aerated. Gold dissolves in NaCN forming sodium aurocyanide complex.

$$4Au + 8NaCN + 2H_2O + O_2 \rightarrow 4Na[Au(CN)_2] + 4NaOH$$

Gold is recovered from the solution either by adding zinc or by electrolysis:

$$2Na(Au(CN)_2] + Zn \rightarrow 2NaCN + Zn(CN)_2 + 2Au$$

The precipitated gold is treated with dil.H_2SO_4 to remove the excess of zinc and changed into compact mass by fusion. The gold obtained contain other base metals as impurities. These are removed by electrolytic refining.

In electrolytic refining, 4% gold chloride solution acidified with 10% hydrochloric acid is used as the electrolyte. Impure gold is made the anode and cathode is a plate of pure gold.

TRANSITION ELEMENTS–3d SERIES

Introduction to transition elements

The transition elements are located in the central region of the periodic table between the s–block and p–block elements. *These elements are also called d–block elements since the penultimate d–orbitals in the atoms of these elements are progressively filled with valence electrons.* Each d–subshell has five orbitals (d_{xy}, d_{xz}, d_{yz}, $d_{x^2-y^2}$ and d_{z^2}) and can accommodate a maximum of 10 electrons. There are four transition series and each series consists of 10 elements. The elements in the four series are

3d–series : $_{21}Sc$, $_{22}Ti$, $_{23}V$, $_{24}Cr$, $_{25}Mn$, $_{26}Fe$, $_{27}Co$, $_{28}Ni$, $_{29}Cu$, $_{30}Zn$

4d–series : $_{39}Y$, $_{40}Zr$, $_{41}Nb$, $_{42}Mo$, $_{43}Tc$, $_{44}Ru$, $_{45}Rh$, $_{46}Pd$, $_{47}Ag$, $_{48}Cd$

5d–series: $_{57}La$, $_{72}Hf$, $_{73}Ta$, $_{74}W$, $_{75}Re$, $_{76}Os$, $_{77}Ir$, $_{78}Pt$, $_{79}Au$, $_{80}Hg$

6d–series : $_{89}Ac$, $_{104}Rf$, $_{105}Ha$, $_{106}Sg$, $_{107}Bh$, $_{108}Hs$, $_{109}Mt$, $_{110}Uun$, $_{111}Uuu$, $_{112}Uub$

The 14 elements between La and Hf are called lanthanides. Similarly 14 elements following $_{89}Ac$ are called actinides. Lanthanides and actinides are called f–block elements.

In this chapter, a discussion of the characteristic properties of the 3d–series of transition elements will be presented.

The role of transition elements

The transition elements have played a crucial role in the development of civilization. Iron is the main constituent of steel, which is produced all over the world as an important building and structural material. Titanium is a major component in modern supersonic jet aircrafts.

The transition elements and their compounds are important catalysts in

(i) automobile catalytic converters
(ii) industrial synthesis of H_2SO_4, HNO_3, NH_3, CH_3OH and petrochemicals

They play a vital role in living systems. Iron present in haemoglobin transports oxygen from lungs to cells in other parts of the body. Cobalt is a component of vitamin B_{12}. Ni, Cu and Zn are important constituents of many enzymes.

CHARACTERISTICS OF 3d–SERIES TRANSITION ELEMENTS

Electronic configuration

The general electronic configuration is $(n-1)d^{1-10}\ ns^{0-2}$ outside a noble gas core

In the first transition series, the 3d orbitals are being filled with electrons. The names of elements with symbols, electronic configuration, melting point, boiling point, density, oxidation states and atomic radius are presented in the table 3.4.

Table 3.4 Electronic configuration and some other properties and 3d transition metals

	Scandium	Titanium	Vanadium	Chromium	Manganese
Symbol	Sc	Ti	V	Cr	Mn
Atomic number	21	22	23	24	25
Electronic configuration	$3d^1\,4s^2$	$3d^2\,4s^2$	$3d^3\,4s^2$	$3d^5\,4s^1$	$3d^5\,4s^2$
Density (g cm^{-3})	2.99	4.51	6.0	7.15	7.3
Melting point (°C)	1541	1668	1910	1907	1246
Boiling point (°C)	2836	3287	3407	2671	2061
Atomic radius (pm)	162	147	134	128	127
Oxidation states	3	2, 3, 4	2, 3, 4, 5	1, 2, 3, 4, 5, 6	2, 3, 4, 5, 6, 7

	Iron	Cobalt	Nickel	Copper	Zinc
Symbol	Fe	Co	Ni	Cu	Zn
Atomic number	26	27	28	29	30
Electronic configuration	$3d^6\,4s^2$	$3d^7\,4s^2$	$3d^8\,4s^2$	$3d^{10}\,4s^1$	$3d^{10}\,4s^2$
Density (g cm^{-3})	7.87	8.86	8.9	8.96	7.14
Melting point (°C)	1538	1495	1455	1085	420
Boiling point (°C)	2861	2927	2913	2562	907
Atomic radius (pm)	126	125	124	128	134
Oxidation states	2, 3, 4, 6	2, 3, 4, 5	2, 3, 4	1, 2	2

From table 3.4, it is seen that the filling of 3d subshell proceeds according to Hund's rule with two exceptions. In the case of Cr, the configuration is $3d^5 4s^1$ and for Cu, it is $3d^{10}\,4s^1$ due to extra stability of half filled and completely filled orbitals.

Variable oxidation states

Unlike the s and p block elements, the transition elements can exist in a large number of different oxidation states because the (n–1) d electrons have an energy comparable to that of ns electrons. Therefore, d electrons also can be ionized one by one to form positive ions. Valency increases by units of one in transition elements in contrast to the representative elements where it increases in units of two.

In $FeCl_3$, oxidation state of Fe is +3 and in $FeSO_4$, the oxidation state is +2. Chromium can exist in oxidation states +1 to +6. Mn exists in oxidation states +2 to +7. All the d–block elements have a minimum of two oxidation states except zinc and cadmium. Maximum oxidation states increases from Sc (+3) to Mn (+7) because the outer electrons increase from 3 to 7.

After Mn, oxidation states decrease because as we go from left to right in the series, the energy of the d orbitals progressively decreases and from iron onwards, it becomes more and more difficult to involve all the valence electrons in bonding. Therefore, the maximum oxidation states decreases from Mn (+7) to Zn (+2). Highest oxidation state is +8 for Os and Ru in their tetroxides.

Metallic character

All transition elements are typical metals. They conduct heat and electricity. They are hard, strong, malleable, ductile and show metallic lustre.

Complex formation

The most important property of a transition metal is the tendency to form complexes with Lewis bases, namely groups which can donate an electron pair. These are referred to as ligands. Ligands can be neutral molecules like CO, NH_3 or ions like Cl^-, SO_4^{2-}, NO^+.

Maximum number of complexes are formed by cobalt. Unlike the p–block elements, transition elements easily form complexes because

(a) They have small atomic/ionic radii.
(b) They form highly charged ions.

(c) They posses vacant low energy orbitals to accept electron pairs from ligands.

More details on complexes will be presented in the coordination chemistry section (next chapter).

Atomic and ionic sizes

The covalent radii of the elements decrease from left to right across a period in the transition series. At the end of the series there is a slight increase in size from Ni to Zn. As one moves from left to right, protons get added to the nucleus and there are extra orbital electrons. These electrons shield the nucleus. But the shielding is poor and this leads to a higher effective nuclear charge (Z_{eff}). The poor shielding leads to reduction in size. The transition elements are smaller in size compared to group I and group II elements in the same horizontal period. A few irregularities in the general trend are due to crystal field effects. Atomic and ionic radii increase from first (3d) series to the second (4d) series of transition elements. But the 4d and 5d series have almost the same radii due to lanthanide contraction.

Density

The increased nuclear charge and poor shielding of d electron leads to low atomic volumes. This leads to high densities in transition metals. Osmium and iridium have highest densities in the range 22 – 23 g cm^{-3}. Densities increase from left to right and from top to bottom.

Melting and boiling points

Transition metals generally melt at a temperature greater than 1000°C. Exceptions are La, Ag, Zn, Cd and Hg. Zn, Cd and Hg have very low melting points because they have a completed d–subshell and the metallic bonding in these metals is due to only two s–electrons in the outermost shell.

Metal reactivity

Many of the transition metals are reactive because of their electropositive character and liberate hydrogen from mineral acids. Some of the higher series of elements like Ru, Rh, Pd, Os, Ir, Pt and Au are noble due to their large positive standard electrode potentials. This in turn is due to their high enthalpy of sublimation, high ionisation energy and low enthalpy of solvation.

Ionization potentials

The first ionization potentials of transition elements lie in between those of s and p block elements. They are less electropositive than group I and group II metals. Depending on conditions, they form ionic and covalent bonds. When a metal exists in two oxidation states, the lower oxidation state is predominantly ionic (Example– +1, +2). Higher oxidation states are predominantly covalent (e.g., +4 to +8). First ionization energy increases from Sc to Co and then decreases to Cu and increases to Zn. The variations can be explained on the basis of decrease in size and special stability of half–filled and completely filled subshells.

Colour

NaCl, NaBr and NaI are ionic, but all of them are colourless. The silver compound AgI is yellow coloured. Na_2CO_3 has no colour, but Ag_2CO_3 is coloured. Na_2O is colourless. But Ag_2O and Ag_2S are black. Thus the ionic and covalent compounds of many transition metals are coloured. When light is absorbed by a material, the transmitted light is coloured displaying the colour complimentary to that absorbed. Origin of colour of transition metal compounds is due to three mechanisms.

(a) d–d transition

Electronic energy changes are responsible for absorption and we can get electronic spectra. In the case of transition metal ions, one of the reasons for coloured compounds is *the promotion of electron from one d–level to another d–level. Such transitions are called d–d transitions.*

In gaseous transition metal ions d orbitals are degenerate. But when these ions are surrounded by anions or ligands, d orbitals split into two or more energy levels. By absorbing visible or ultraviolet light, electrons are excited from one level to a higher level. This absorption brings about colour to transition metal compounds.

The colour of transition metal complexes depends on energy difference between two levels. This is decided by nature of ligand and type of complex formed.

If we consider the three nickel complexes, $[Ni(NH_3)_6]^{2+}$, $[Ni(H_2O)_6]^{2+}$ and $[Ni(NO_2)_6]^{4-}$, their colours are blue, green and reddish brown respectively. The shape of the complex and number of ligands attached to the metal (coordination number of metal) also play a role in the colour shown by the complex.

(b) Polarization

If a cation polarizes an anion, the compound becomes coloured. The sizes of the anion and cation are important for colour of compounds by polarisation. Smaller cation has high polarizing power and larger anion has high polarizability (Fajans rule) and hence AgI and CdS are both yellow.

(c) Charge–transfer spectra

Some complexes involving ions like MnO_4^-, VO_2^+ and CrO_4^{2-} are coloured. For V(+5), Cr(+6) and Mn (+7), there are no d–electrons. So d–d transition is not possible. Still the above mentioned ions are coloured. This is due to charge transfer. The purple colour of MnO_4^- arises because an electron is momentarily transferred from oxygen to the metal, thus changing O^{2-} to O^-. The oxidation state of Mn reduces from +7 to +6. For a charge transfer, the two energy levels must be close to each other. The s and p–block elements do not have partially filled d–shells and the energy required to promote s or p electrons is very large and hence compounds of s and p block elements are generally not coloured. The transition may occur in UV light.

Magnetic properties

When a substance is introduced into a magnetic field of strength H, the intensity of the magnetic field in the substance may be higher or lower than H. If it is greater than H, we call the substance paramagnetic. Magnetic lines of force travel more easily through a paramagnetic material than through vacuum. *A paramagnetic material moves from a weaker to stronger part of the field. In other words, it is attracted to a magnetic field.* The presence of unpaired electron in an atom or compound leads to paramagnetism. *Diamagnetic substances are repelled from a magnetic field. They contain only paired electrons and they are repelled very weakly from a magnetic field.*

There is another class of compounds called ferromagnetic and Fe, Co and Ni belong to this category. Ferromagnetic materials *are special cases of paramagnetic materials in which the magnetic moments on individual atoms are aligned and point in the same direction under the influence of an external magnetic field.* Such materials can become permanent magnets. *If we pair moments of adjacent atoms which point in opposite directions, we get antiferromagnetism.* Simple salts of Fe^{2+} and Mn^{2+} are antiferromagnetic.

Ferromagnetism and antiferromagnetism are related to orientation and hence will disappear in the solution state.

Spin–only magnetic moments

Many compounds of transition metals are paramagnetic and the number of unpaired electrons can be calculated using magnetic moment measurements. The magnetic moments can be obtained from magnetic susceptibility values, which are measured using a Gouy balance. If the magnetic moment of a compound arises only due to the spin of unpaired electrons, then μ_S, the spin–only magnetic moment in Bohr Magnetons, is given by the formula $\mu_S = \sqrt{4S(S+1)}$ where S is the total spin quantum number of the ion and is equal to ns, where $s = \frac{1}{2}$, the spin quantum number of an electron and n is the number of unpaired electrons in the ion. So $\mu_s = \sqrt{n(n+2)}$ BM.

Table 3.5 gives the spin–only magnetic moments for different number of unpaired electrons in units of BM (Bohr Magneton).

Number of unpaired electrons	Magnetic moment (BM)	Total spin value(S)
1	1.73	$\frac{1}{2}$
2	2.83	1
3	3.87	$\frac{3}{2}$
4	4.90	2
5	5.92	$\frac{5}{2}$

The spin–only magnetic moment will be different for different transition metal complexes depending on the number of unpaired electrons. It is possible to find the oxidation state of the central metal ion using the magnetic moment data. The actual or observed magnetic moments may be slightly modified by contribution from orbital magnetic moment (μ_L).

Catalytic properties

Many transition metals and compounds have catalytic properties. The catalytic activity is due to

(i) variable valency
(ii) ability to form complexes with reactants
(iii) residual valencies on the surface of metals

Some of these compounds along with their use are listed below:

$TiCl_3$	Used as Ziegler – Natta catalyst in production of polythene.
V_2O_5	Used in manufacture of H_2SO_4 for converting SO_2 to SO_3
MnO_2	Used as catalyst for producing oxygen from $KClO_3$ or H_2O_2.
Fe	Used in preparation of NH_3
Fe/Cl_2 Fe/Br_2	Used in halogenation of benzene
$FeSO_4$ and H_2O_2	Fenton's reagent in oxidizing alcohols to aldehydes.
ZnO/Cr_2O_3	Preparation of CH_3OH from CO and H_2
Ni	Raney nickel used in reduction of organic compounds.
Pd / $BaSO_4$ in quinoline	Lindlar's catalyst used in reduction of alkynes to alkenes.

Interstitial compounds

Transition elements form interstitial compounds with small atoms like H, B, C, N etc. They are non-stoichiometric in composition.

Example $TiH_{1.7}$, $VH_{0.56}$

(i) They have higher melting points than the pure metals.
(ii) They are very hard. Some approach diamond in hardness.
(iii) They retain metallic conductivity.
(iv) They are chemically inert.

CONCEPT STRANDS

Concept Strand 6

How many unpaired electrons are there in chromium and copper? Explain.

Solution

Chromium has six unpaired electrons due to the configuration $3d^5 4s^1$. This configuration is preferred over $3d^4 4s^2$ since half filled configuration leads to more stability. Copper has one unpaired electron due to the configuration $3d^{10} 4s^1$. This configuration is preferred over $3d^9 4s^2$ since the d–shell becomes completely filled.

Concept Strand 7

Co exists in oxidation states +2, +3, +4 and +5, but zinc exists in +2 state only. Why?

Solution

The electronic configuration of zinc is $3d^{10} 4s^2$. By loss of two electrons, zinc acquires the stable d^{10} configuration. Hence it does not exist in any oxidation state other than +2. The presence of incomplete d–shell permits Co to have different oxidation states.

Concept Strand 8

The melting points of Zn, Cd and Hg are low compared to the other transition elements. Explain.

Solution

Zn, Cd and Hg have their d–shells complete. Therefore, d–electrons do not participate in metallic bonding. Hence their melting points are low.

Concept Strand 9

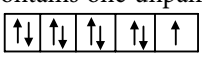
Find the oxidation state of the metal in each of the following.

(i) $[RbCl_6]^{3-}$ (ii) $Ni(CO)_4$
(iii) $[Ni(NH_3)_6]^{2+}$ (iv) $CuCl_2$

Does $CuCl_2$ contain unpaired electron?

Solution

(i) +3 (ii) Zero
(iii) +2 (iv) +2

$CuCl_2$ contains one unpaired electron since the configuration is 

Concept Strand 10

(i) Arrange the 3d series elements in the order of increasing Z_{eff}.
(ii) Which M^{2+} is the strongest reducing agent?
(iii) Between VO_4^{3-} and FeO_4^{3-}, which is a stronger oxidizing agent?

Solution

(i) Sc < Ti < V < Cr < Mn < Fe < Co < Ni < Cu < Zn
(ii) Ti^{2+} ($3d^2\,4s^0$) is the strongest reducing agent and Zn^{2+} ($3d^{10}\,4s^0$) is the weakest reducing agent. (Sc^{2+} with $3d^1\,4s^0$ is so unstable that it is nonexistent)
(iii) FeO_4^{3-} is a stronger oxidizing agent than VO_4^{3-}.

Concept Strand 11

Copper has the completely filled configuration d^{10} in its +1 state, but still Cu^{2+} compounds are more stable than Cu^{+} compounds. Explain.

Solution

Cu^{+} undergoes disproportionation as $2Cu^{+} \rightleftharpoons Cu + Cu^{2+}$. The equilibrium constant for this disproportionation reaction is very high (equal to 1.6×10^6). Hence reaction proceeds towards the right direction leading to Cu and Cu^{2+}. With respect to Cu(I), Cu(II) is further stabilized by greater solvation energy and lattice energy.

Concept Strand 12

Zn and Ti are transition metals, but still $ZnSO_4$ and TiO_2 are white. Explain.

Solution

Zn^{2+} has d^{10} configuration and Ti^{4+} has d^0 configuration. So in both ions, d–d transition is not possible. Hence the compounds are white.

Concept Strand 13

Using the formula $\mu_S = \sqrt{4S(S+1)}$, derive a formula to calculate μ_S from the number of unpaired electrons 'n' in an ion.

Solution

Since one electron has a spin of $\frac{1}{2}$, n unpaired electrons will have a spin of $\frac{n}{2}$.

$$S = \frac{n}{2}$$

Hence,

$$\mu_S = \sqrt{\frac{4n}{2}\left(\frac{n}{2}+1\right)} = \sqrt{\frac{2n}{2}(n+2)} = \sqrt{n(n+2)}\ \text{BM}$$

Concept Strand 14

Use the values of spin–only magnetic moments given below to find the oxidation state of the transition metal ion.

(i)	Titanium	1.7 BM
(ii)	Vanadium	2.8 BM
(iii)	Iron	5.0 BM
(iv)	Manganese	5.8 BM
(v)	Nickel	2.9 BM

Solution

(i) Ti has the electronic configuration of d^2s^2. Spin–only magnetic moment of 1.7 BM corresponds to one unpaired electron. One d–electron remains by the removal of three electrons. This corresponds to 1.7 BM. Hence oxidation state of Ti is +3.
(ii) V has the electronic configuration of d^3s^2. Spin–only magnetic moment of 2.8 BM corresponds to two unpaired electrons. Removal of three electrons leads to d^2 configuration. Thus the oxidation state of V is +3.
(iii) Fe has the electronic configuration of d^6s^2. Spin–only magnetic moment of 5 BM corresponds to four unpaired electrons. If Fe has d^6 configuration in its ion, then we have four unpaired electrons. Hence Fe is in +2 oxidation state.
(iv) Mn has the electronic configuration of d^5s^2. Spin–only magnetic moment of 5.8 BM corresponds to five unpaired electrons. Hence Mn is in +2 oxidation state.
(v) Ni has the electronic configuration of d^8s^2. Spin–only magnetic moment of 2.9 BM corresponds to two unpaired electrons and the configuration of the ion is d^8. Thus Ni is in +2 oxidation state

Concept Strand 15

Describe briefly the method of determining magnetic moment using a Gouy balance.

Solution

The volume susceptibility of a sample is measured by weighing it in and out of a magnetic field. The molar susceptibility χ_m is then calculated. The value of χ_m is related to μ and μ_B by $\frac{\mu}{\mu_B} = 797.5\sqrt{\chi_m T}$. Knowing μ and μ_B, we can find the number of unpaired electrons using $\mu = \sqrt{n(n+2)}\ \mu_B$. The value we calculate is called spin only magnetic moment. The unpaired electron can give rise to a magnetic field because of its spin and orbital angular momentum. The general equation for $\mu_{(S+L)}$ is $\mu_{S+L} = \sqrt{4S(S+1) + L(L+1)}\ \mu_B$, where S is the total spin quantum number and L is the resultant of the orbital angular momentum quantum numbers of all electrons in the molecule.

In many compounds of the first transition series, the orbital contribution is quenched by the electric fields of the surrounding atoms. As an approximation we ignore this contribution and use the spin–only values.

Concept Strand 16

The compound $CuSO_4.5H_2O$ at 293K has the spin–only magnetic moment equal to 1.93 BM. Find χ_m, the molar magnetic susceptibility including diamagnetic correction.

Solution

$$\frac{\mu}{\mu_B} = 1.93 = 797.5\sqrt{\chi_m \times 293}$$

$$\chi_m \times 293 = \left(\frac{1.93}{797.5}\right)^2$$

$$\chi_m = \left(\frac{1.93}{797.5}\right)^2 \times \frac{1}{293}$$

$$= 2 \times 10^{-8}\ m^3\ mol^{-1}$$

The various ions and molecules in a compound have a diamagnetic contribution to magnetic moment. The value calculated from Gouy balance study includes these diamagnetic contributions also. But the diamagnetic contributions are so negligible that they may be overlooked except in very accurate measurements.

SUMMARY

Extractive metallurgy	
Metals	Three dimensional bonding–Conductivity decreases with increase of temperature
Metallic bond	Enhances electrical conductivity
Bond length	Longer in metallic crystal than in gaseous diatomic molecules
Theories of Bonding in metals	Free electron theory, Valence bond theory, Molecular orbital or Band theory
Metallurgy	The science and art of winning metals from minerals
Mineral	Naturally occurring metal or metal compound
Ore	Mineral from which metal can be extracted economically
Some important minerals and their composition	Minerals of Magnesium, Aluminium, Tin, lead, Iron, Copper, Silver and Zinc
Steps in metallurgy	Concentration of ore–Heat treatment–reduction to metal–refining

Ore concentration	Gravity separation – magnetic separation– froth floatation – leaching
Heat treatment	Calcination Roasting
Winning of metal	Electrolytic reduction– self reduction– carbon or carbon monoxide reduction– cyanide process
Ellingham diagram	Variation of standard free energy change for formation of metal oxide with temperature (represented graphically)
Hydrometallurgy	Getting the metal from the solution obtained by leaching the mineral by applying suitable method
Refining of metal	Different methods of purifying the impure metal obtained: Liquation – Zone refining – cupellation – poling – electrolytic refining – vapour phase refining
Extraction of metals	
Iron	By carbon–carbon monoxide reduction of haematite in blast furnace.
Tin	From tin stone by reduction with carbon $SnO_2 + 2C \rightarrow Sn + 2CO$
Copper	From $CuFeS_2$ Roasted – matte – self reduction $2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$
Lead	From galena PbS – Roasted to $PbSO_4$– $PbSO_4 + PbS \rightarrow 2Pb + 2SO_2$
Magnesium	From carnallite ($KCl.MgCl_2.6H_2O$) by electrolysis of fused $MgCl_2$
Aluminium	From Bauxite – $Al_2O_3.2H_2O$ purified by Baeyers process – Alumina dissolved in molten cryolite–electrolysed using carbon electrodes
Silver	Leaching Argentite (Ag_2S) using NaCN in presence of air and treating the solution with zinc
Gold	Aerating powdered auriferous rock with dil NaCN and extracting the soluble gold complex–gold recovered by treatment with metallic zinc
Transition elements	
Transition elements	d – block elements–group 3 to group 12 metals
Characteristic proportion of transition elements	Variable oxidation number–metallic character–complex formation– reactivity–colour in aqueous medium–magnetic properties–catalytic properties–alloy formation, etc.

TOPIC GRIP

Subjective Questions

1. With example and reason, give the set of metals that exist in combined state and native state comment on their standard electrode potentials.

2. What is the principle behind Leaching? Explain with example.

3. Give reasons for using zinc as a suitable solvent for extracting silver from lead.(Parke's Process).

4. Give Reasons
(i) Copper formed by auto reduction of Cu_2S is blister copper
(ii) Anodes should be repeatedly replaced during aluminium extraction
(iii) Gold is soluble in aqua regia, but silver is not

5. 6.87kg of Mn_3O_4 was heated with 2.16kg of Aluminium. Comment on this reaction and calculate the amount of oxide formed as product.

6. All transition metals of 3d series except Zn exhibit a minimum of two oxidation states.
Explain why Zn exhibits only +2 state. Does Zn form complexes?

7. Give reason.
(i) When Cu^{2+} is treated with H_2S, insoluble CuS is precipitated. But this method is not used to estimate Cu gravimetrically.
(ii) $K_2Cr_2O_7$ is used as primary standard but not $Na_2Cr_2O_7$.
(iii) NaI is not coloured but AgI is coloured.
(iv) $ZnSO_4$ is white but $KMnO_4$ is pink.

8. $KMnO_4$ can be prepared by fusing together equimolar amounts of MnO_2, KOH and KNO_3, collecting the dark green solidified melt, which is K_2MnO_4, crushing it and treating it with dil.H_2SO_4. On adding dil.H_2SO_4 a solution of $KMnO_4$ and a precipitate of MnO_2 are formed. Write the relevant equations for the process.

9. Give the electronic configuration of the metal in
(i) FeO_4^{2-} (ii) $Cr(OH)_3$ (iii) VCl_5 (iv) $[Ni(H_2O)_6]^{2+}$
Give their calculated magnetic moment.

10. With reference to atomic size of transition metals, explain the observation that covalent radii decreases from left to right across a row in the transition series, but near the end size increases.

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

11. The process of hydraulic washing for the concentration of ores is based on the
(a) difference in specific gravity of ore and gangue particles
(b) electrical conductivity of ore particles
(c) partial solubility
(d) difference in melting points of ore and gangue particles

12. Siderite ore is decomposed in excess of air. the solid product formed is
(a) Fe_3O_4 (b) PbO (c) Fe_2O_3 (d) FeS_2

13. The process that produces 99.99% pure aluminium is
(a) Halls process (b) Hall Haroult process (c) Hoops Process (d) Serpecks Process

14. The most acidic oxide among the following is
(a) V_2O_5 (b) VO (c) V_2O_3 (d) VO_2

15. Ti^{3+} has same spin only magnetic moment as that of
(a) Mn^{2+} (b) VF_5 (c) VBr_4 (d) VI_3



Assertion–Reason Type Questions

Directions: **Each question contains Statement–1 and Statement–2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**
(a) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1
(b) Statement–1 is True, Statement–2 is True; Statement–2 is NOT a correct explanation for Statement–1
(c) Statement–1 is True, Statement–2 is False
(d) Statement–1 is False, Statement–2 is True

16. Statement 1
Roasting of sphalerite and cinnabar give the metal oxide and metal respectively.

and

Statement 2
During roasting, the ore is heated strongly in the presence of air above its melting point.

17. Statement 1
Impure titanium, when subjected to van – Arkel method for purification, produces a stable volatile compound.

and

Statement 2
The compound is formed at a lower temperature than its decomposition temperature.

18. Statement 1
Decrease in atomic radius from sodium to chlorine is greater than that from scandium to copper.

and

Statement 2
The increase in number of electrons on moving from sodium to chlorine is greater than from Sc to Cu.

19. Statement 1
In a transition element, the total magnetic moment of a substance is the resultant of magnetic moment of all the individual electrons.

and

Statement 2
Each electron behaves as a micromagnet having a certain value of magnetic moment.

20. Statement 1
It is ordinarily not possible for Lanthanum to exhibit oxidation state greater than +3.

and

Statement 2
There are no electrons in the 4f orbital of Lanthanum.

Linked Comprehension Type Questions

Directions: **This section contains 2 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

The oxide of metal (A) is used in converting SO_2 to SO_3 in contact process. The penta halide of (A) cannot give d–d spectrum. When metal (A) reacts with HF, (B) is formed The tetrachloride of (A) dissociates to trichloride and chlorine even at room temperature. Other tetrahalides and trihalides undergoes disproportionation. The pentafluoride forms complex with F^- ion.

21. Why is it not possible for A in +5 oxidation states to give rise to a spectrum by d–d transition?
(a) A^{5+} has d^o configuration (b) A^{5+} has d^{10} configuration
(c) A^{5+} is unstable in solution (d) A^{5+} disproportionates

22. The geometry of ACl_4 is same as that of
(a) $[Ni(CN)_4]^{2-}$ (b) $[NiCl_4]^{2-}$ (c) XeF_4 (d) $Fe(CO)_5$

23. ACl_4 is paramagnetic like
(a) $[Ni(Cl)_4]^{2-}$ (b) $[Co(NH_3)_6]^{3+}$ (c) $[Ni(CN)_4]^{2-}$ (d) $[Co(en)_3]^{3+}$

Passage II

The process of froth flotation is based on the principle of preferential wetting of the solid surface by liquid. Metallic sulphides for example are wetted by certain oils.

The finely divided ore is put in water to which a small quantity of oil has been added. On violent shaking, froth is formed and the ore particles are carried away by the froth while the gangue particles remain in aqueous medium.

24. The ore that is concentrated by froth flotation is
(a) Copper glance (b) Barytes (c) Magnesite (d) Cassiterite

25. The compound used as collectors, in order to take all the ore particles into the froth is
(a) Ethyl acetate (b) Potassium ethyl tartarate
(c) Ethyl xanthate (d) sodium Tartarate

26. Galena is concentrated by froth flotation but it contains ZnS impurity. ZnS can be separated using
(a) $CuSO_4$ (b) NaCN (c) Ethyl xanthate (d) Sodium Tartarate

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers of which ONE OR MORE answers will be correct.**

27. The ore(s) that can be concentrated by magnetic separation method are/is
(a) Chromite ore (b) copper pyrites (c) Ilmenite (d) Cinnabar

28. During extraction of Aluminium Bauxite ore can be leached using
(a) $CaSiO_3$ (b) $NaOH_{(aq)}$ (c) $Na_2CO_{3(aq)}$ (d) CO_2

29. The metals that can be reduced from their sulphides without any external reducing agent is (are)

(a) Copper (b) lead (c) Silver (d) Gold



Matrix–Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

30. Column I has refining methods and Column II contains metals

Column I	**Column II**
(a) Electro Refining	(p) Copper
(b) Poling	(q) Lead
(c) Vapour Phase Refining	(r) Nickel
(d) Liquation	(s) Tin

IIT ASSIGNMENT EXERCISE

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

31. If bauxite ore is calcined, the entity that will be removed as a volatile impurity is
(a) CO_2 (b) CO (c) SO_2 (d) H_2O

32. The process that converts a sulphide ore into the oxide is
(a) Calcination (b) Roasting (c) Smelting (d) both b and c

33. The choice of flux used to form the slag depends mainly on
(a) type of slag to be formed (b) density of slag required
(c) type of impurity in the ore (d) process employed for metal extraction

34. Amalgamation method of extraction of metal is possible because
(a) of low melting point of mercury (b) low reactivity of mercury
(c) amalgamated metal is less reactive (d) mercury is readily available

35. Froth flotation is a concentration method employed for sulphide ores. This method is not applied to the sulphide ore
(a) Galena (b) Cinnabar (c) Argentite (d) Sphalerite

36. The Van Arkel method for purification of zirconium is based on
(a) Electrolysis (b) Carbon reduction
(c) Hydro metallurgy (d) Thermal decomposition

37. Copper pyrites contains another metal sulphide as impurity. The flux used to remove this impurity is
(a) SiO_2 (b) $MgCO_3$ (c) Lime (d) Haematite

38. Among the following an example of thermite process is
(a) $2\ Fe + Al_2O_3 \rightarrow 2Al + Fe_2O_3$ (b) $3MnO_2 + 4\ Al \rightarrow 2Al_2O_3 + 3Mn$
(c) $2Al + 6H_2SO_4 \rightarrow Al_2O_3(SO_4)_3 + 6H_2O + 3SO_2$ (d) $Al_2O_3 + 3C + 3Cl_2 \rightarrow 2AlCl_3 + 3CO$

39. The charge in a blast furnace contains limestone during the extraction of iron. This limestone is added as
(a) slag (b) reducing agent (c) flux (d) oxidizing agent

40. Which of the following pairs contain more than 2% carbon
(a) pig iron and cast iron (b) pig iron and steel
(c) cast iron and wrought iron (d) steel and wrought iron

41. Gallium is finally refined after extraction by
(a) Cupellation (b) Distillation (c) Vapour phase refining (d) Electrolytic refining

42. The reducing agent that actually reduces iron ore in the blast furnace is
(a) Carbon (b) Coke (c) CO (d) CO_2

43. In the Hoops process for refining of Aluminum, the impure Al, pure Al and electrolyte form different separated layers because they are separated
(a) by a perforated steel sheet (b) by carbon sheet which is anode
(c) by density difference between them (d) by keeping them in different containers

44. The true statement regarding roasting of zinc blende ore is
(a) Both zinc oxide and zinc sulphate are formed during the process
(b) Impurities like arsenic and antimony cannot be removed
(c) Roasting is done to remove organic impurities
(d) Chemical composition of ore does not change

45. The chemical composition of matte is
(a) $FeS + SiO_2$ (b) $Cu_2O + FeS$ (c) $Cu_2O + Cu_2S$ (d) $Cu_2S + FeS$

46 Which of the following is used as a depressant in separation of ZnS?
(a) NaCN (b) Al_2O_3 (c) $Na[Ag(CN)_2]$ (d) Na_2S

47. Carbon reduction method is not useful for extraction of
(a) Fe (b) Pb (c) W (d) Sn

48. The ore that exists as the dihydrate is
(a) Chromite ore (b) Bauxite (c) Galena (d) Cassiterite

49. A haematite sample contains phosphorus impurity. The slag formed due to this impurity in the blast furnace is chemically
(a) $CaSiO_3.P_2O_5$ (b) $Ca(HPO_4)$ (c) $MgSiO_3.P_2O_5$ (d) $Ca_3(PO_4)_2$

50. Copper pyrites contains both Fe and Cu. During extraction of copper from copper pyrites, Fe is removed
(a) during concentration by froth flotation (b) by adding silica
(c) as volatile impurity during roasting (d) by distillation

51. The electrolyte used for the electrolytic refining of tin is
(a) tinsulphate acidified with H_2SO_4
(b) molten tinsulphate
(c) hydrofluorosilicic acid containing $SnSO_4$ acidified with H_2SO_4
(d) aqueous stannous chloride acidified with HCl

52. During roasting of leadsulphide ore, the formation of $PbSO_4$, and $PbSiO_3$ are prevented by the addition of
(a) Lime (b) Silica (c) Magnesium oxide (d) lead

53. In metallurgy formation of slag, using a flux is commonly adopted because
(a) slag floats on molten metal (b) flux required is either acidic or basic
(c) slag is used for other purposes (d) infusible impurity is converted to fusible material

54. The chemical reaction behind the poling process is
(a) oxidation (b) reduction (c) neutralisation (d) pyrolysis

55. The complex formed when argentite is treated with aqueous sodium cyanide is
(a) $Na[Ag(CN)_2]$ (b) $Na_2[Ag(CN)_2]$ (c) $Na_2[Ag(CN)_4]$ (d) $Na[Ag(CN)_4]$

56. Mercury exists as a liquid at room temperature due to
(a) high lattice enthalpy (b) d^{10} configuration and large size of atoms
(c) positive electrode potential (d) small size of mercury

57. When transition metal elements form interstitial compounds with carbon or hydrogen, their
(a) hardness increases (b) electrode potentials increase
(c) Electron gain enthalpy decreases (d) ionization energy increases

58. In comparison to the other transition metals, cadmium melts at a very low temperature because of
(a) absence of van der Waals forces.
(b) high ionisation energy.

(c) completely filled d orbital
(d) highest atomic weight in the second transition series

59. The highest magnetic moment is possessed by
(a) V^{3+} (b) Cr^{3+} (c) Cr^{2+} (d) Ni^{2+}

60. Which of the following is coloured in aqueous medium?
(a) Sc^{3+} (b) Ti^{4+} (c) Zn^{2+} (d) Ni^{2+}

61. Which among the following is an amphoteric oxide?
(a) Na_2O (b) CrO (c) Cr_2O_3 (d) MgO

62. The element with highest electrical conductivity among the following is
(a) Copper (b) Silver (c) Zinc (d) Gold

63. The metallic character and covalent character of the transition elements are due to
(a) ns electrons
(b) ns and (n–1)d electrons respectively
(c) ns and nd electrons respectively
(d) (n–1)d electrons

64. The number of electrons in the d shell of Cr^{3+} is equal to
(a) no of electrons in d shell of Cr atom
(b) no of s electrons in lithium
(c) no of d electrons in V^{3+} ion
(d) no of p electrons in phosphorus

65. Generally sulphides of metals are black or coloured except Zinc which is white. This is because
(a) size of S^{2-} and Zn^{2+} are comparable
(b) charges on anion and cation are equal and opposite
(c) Zinc has completely filled d orbitals
(d) cation is much smaller than anion

66. The highest oxidation state among transition elements is seen in
(a) Fe (b) Mn (c) Os (d) Re

67. Transition elements tend to form alloys very easily because
(a) similar metallic radii
(b) they have incomplete d orbitals
(c) of similar reactivity of elements
(d) of low heat of sublimation

68. $[Ti(H_2O)_6]Cl_3$ (A) is violet while $TiCl_4$(B) is colourless. This is because
(a) (A) is a complex while (B) is not
(b) In (B) Ti cation is smaller than that in A
(c) Oxygen present in H_2O is more electronegative than Cl
(d) There are no d electrons in Ti cation of B

69. Among the following species Fe^{2+}, Mn^{+} and Cr, the pair with the same magnetic moment is
(a) Mn^{+} and Fe^{2+} (b) Mn^{+} and Cr (c) Fe^{2+} and Cr (d) Fe^{2+}, Mn^{+}, Cr

70. Zn, Cd and Hg do not show the characteristic property of transition elements because
(a) they have completely filled d orbital
(b) they have high ionization energy
(c) they have a large atomic volume
(d) they form colourless compounds

71. A transition element of 3d series attains noble gas configuration in its +3 oxidation state. The element is
(a) Scandium (b) Titanium (c) Vanadium (d) Zinc

72. Transition elements are used as catalysts in many chemical reactions because
(a) They are highly electropositive
(b) They have paramagnetic property
(c) They show variable oxidation state
(d) They are of small size

73. For the 3d series, as the oxidation state of the element increases, the ionic radius
(a) decreases (b) increases
(c) does not change significantly (d) there is no particular trend

74. On moving from scandium to titanium, an increase in enthalpy of fusion and enthalpy of vapourization are noticed. This is due to
(a) slight increase in atomic radius (b) increase in nuclear charge
(c) increase in ionisation energy (d) increase in number of d electrons

75. The maximum oxidation state shown by elements with z = 21 to z = 25 is equal to the number of
(a) ns electrons (b) ns and nd electrons
(c) ns and (n – 1)d electrons (d) (n – 1) s and nd electrons

76. All lanthanides are
(a) capable of forming coloured ions (b) strong reducing agents
(c) capable of forming halides of the type MX_4 (d) good conductors of electricity

77. Actinides have higher tendency to form complex compared to lanthanides because
(a) they have high charge density (b) poor shielding of 5f electrons compared to 4f
(c) they show variable oxidation state (d) they have larger atomic weights

78. Considering La^{3+}, Lu^{3+} and Gd^{3+} the order of decreasing ionic radius is
(a) $La^{3+} > Lu^{3+} > Gd^{3+}$ (b) $Lu^{3+} > Gd^{3+} > La^{3+}$ (c) $La^{3+} \cong Gd^{3+} > Lu^{3+}$ (d) $La^{3+} > Gd^{3+} > Lu^{3+}$

79. Unlike other lanthanides, cerium also shows +4 oxidation state other than +3 state because
(a) of poor shielding of 4f electrons (b) cerium has small size due to lanthanide contraction
(c) in +4 state it attains xenon gas configuration (d) in +4 state it attains half filled configuration

80. Lanthanides and Actinides are similar with respect to
(a) decrease in ionic radii with increase in atomic number
(b) radioactivity
(c) variable oxidation state
(d) formation of oxo – ions.

81. Bauxite is converted into anhydrous alumina by
(a) Leaching (b) Calcination (c) Roasting (d) Oredressing

82. The metal that can be obtained by electrolysis of aqueous solution of sulphate is
(a) Cu (b) Pb (c) Al (d) Mg

83. Which of the following sulphides on roasting forms the corresponding metal and SO_2?
(a) ZnS (b) MoS_2 (c) HgS (d) MgS

84. Consider a metal M, having E° values –1.0V. Such a metal's oxide can be reduced to give the metal by
(a) using very mild condition (b) using very drastic reducing condition
(c) using another metal of more negative E° (d) using another metal of less negative E°

85. Silver residues from laboratories are not leached but converted to Ag_2O and heated to get silver. Silver residue present as AgCl can be converted to Ag_2O using
(a) Na_2CO_3 (b) $KMnO_4$ (c) NaCl (d) Fe_2O_3

86. A sudden change in slope seen when free energy change for formation of oxide is plotted graphically against temperature is due to
(a) free energy increases with temperature (b) change of state from solid metal to liquid metal
(c) cutting across the $\Delta G = 0$ line (d) decomposition of oxide at that temperature

87. Which of the following halides of (V) does not undergo disproportionation?.
(a) VF_4 (b) VF_5 (c) VBr_4 (d) VI_3

88. The true statement about metal oxide based on the Ellingham diagram is
(a) Change in entropy for formation of metal oxide is positive
(b) No line crosses the $\Delta G = 0$ line in the Ellingham diagram
(c) MgO can never be reduced using carbon
(d) All metal oxides can be thermally decomposed to give the metal, theoretically

89. The gas released on calcination of malachite are
(a) CO_2 and H_2O (b) CO_2 only (c) SO_2 and H_2O (d) SO_2

90. The process that should be used to concentrate a bauxite ore that contains silica a impurity ie., white Bauxite is
(a) Baeyer's Process (b) Hoops Process (c) Serpecks Process (d) Halls Process

91. The least stable Cu(II) halide is
(a) CuF_2 (b) $CuCl_2$ (c) $CuBr_2$ (d) CuI_2

92. In the McArthur Cyanide process for leaching Argentite, it is important to pass a current of oxygen because
(a) oxygen acts as a promoter
(b) oxygen is an electronegative element
(c) the reaction is other wise reversible
(d) it does not allow the formation of any other complex other than Ag

93. Silicon is purified by zone refining. In the process, the melt formed will contain,
(a) same concentration of silicon as the solid from which it is formed
(b) lesser percentage of silicon than the solid from which it is formed
(c) slightly greater percentage of silicon than the solid from which it is formed
(d) at least about 99% silicon

94. On going down the group ionisation energy decreases. This trend is not seen among transition elements due to
(a) small size of the metals (b) high heat of sublimation
(c) diffused nature of 4f orbital (d) poor shielding of d electrons

95. In their highest oxidation states, transition metals are usually combined with
(a) N and O (b) Cl and O (c) F and O (d) F and I

96. Liquid mercury at room temperature has appreciable vapour pressure. The atomicity of these vapours may be compared to those of
(a) halogens (b) iodine only (c) bromine only (d) noble gases

97. The second ionisation energy of which of the following elements is expected to be abnormally high
(a) Ti (b) Cr (c) Ni (d) Zn

98. Transition elements form non stoichiometric compound. This property is possible because transition elements
(a) show variable oxidation state (b) have small size
(c) have similar property (d) are generally less reactive

99. The statement that shows that cobalt is a transition element is
(a) cobalt does not react with cold water
(b) the second ionisation energy of cobalt is greater than 1st I.E
(c) hydrated $CoCl_2$ changes colour when converted to $CoCl_3$
(d) ionic radii of Co(III) is almost half that of its atomic radii

100. The densities of the 5d series transition elements are much higher than 4d or 3d. This can be explained by
(a) lanthanide contraction (b) high heat of sublimation
(c) increase in atomic weight (d) formation of clusters

101. Electrons present in the p block elements do not produce colour like d block elements because
(a) p electrons are not diffused like d electrons (b) transitions in p electrons does not fall in visible region
(c) all the p orbitals are of same shape (d) there are only three p orbitals

102. Identify the pair of atoms/ion which do not have almost same atomic radii
(a) Fe, Ni (b) Zr, Hf (c) Nb, Ta (d) Cr, Mo

103. CuCl is insoluble in water but is readily soluble in aqueous NH_3. This is because
(a) aqueous ammonia is more polar than water
(b) aquo complex of Cu(I) is unstable while its amino complex is stable
(c) Ammonia removes Cl^- as NH_4Cl
(d) CuCl is the salt of a weak base and very strong acid

104. The transition metal for which only +3 oxidation state is known is
(a) Scandium (b) Cobalt (c) Molybdenum (d) Tungsten

105. A transition metal M in 3d series is used as catalyst for hydrogenation of oil. The magnetic moment of M in its chloride is
(a) 3.8 BM (b) 5.8 BM (c) 2.83 BM (d) 1.73 BM

106. It is very difficult to separate the lanthanides by general techniques. The factor that allows separation is
(a) They have similar size and charge
(b) They have different ionization energy
(c) Though very similar the ionic radii are different
(d) They have slight difference in reactivity of metal

107. In the periodic table, the lanthanides and actinides are placed
(a) separately in Group 3 and Group 4 (b) both in Group 3
(c) in separate box and not in any group (d) both in group 2

108. If hydroxides of lanthanides are expressed as $Ln(OH)_3$, the correct order of basic strength is
(a) $Ca(OH)_2 > Al(OH)_3 > Ln(OH)_3$ (b) $Ln(OH)_3 > Ca(OH)_2 > Al(OH)_3$
(c) $Ln(OH)_3 \cong Ca(OH)_2 > Al(OH)_3$ (d) $Ca(OH)_2 > Ln(OH)_3 > Al(OH)_3$

109. The reducing property of lanthanides
(a) decrease from lanthanum to lutetium (b) increase from lanthanum to lutetium
(c) increases upto gadolinium and remains constant (d) does not show a proper trend

110. Among the following, which configuration corresponds to the ion that will not be coloured?
(a) $[54Xe]4f^{12}$ (b) $[54Xe]4f^{13}$ (c) $[54Xe]4f^{3}$ (d) $[54Xe]4f^{7}$



Assertion–Reason Type Questions

Directions: **Each question contains Statement–1 and Statement–2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1
(b) Statement–1 is True, Statement–2 is True; Statement–2 is NOT a correct explanation for Statement–1

(c) Statement–1 is True, Statement–2 is False
(d) Statement–1 is False, Statement–2 is True

111. Statement 1
There is always a good agreement between experimental and calculated magnetic moments of first row transition elements.

and

Statement 2
The unpaired electron gives rise to a magnetic field because of spin as well as orbital angular momentum.

112. Statement 1
In the blast furnace, during extraction of iron, iron melts at a much lower temperature than its melting point.

and

Statement 2
Iron produced in the blast furnace is impure in nature.

113. Statement 1
Copper, Silver and Gold in the +1 oxidation State are known to be stable in solid state as well as solution.

and

Statement 2
In the +1 state, they attain completely filled d^{10} stable configuration.

Linked Comprehension Type Questions

Directions: **This section contains 1 paragraph. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

The ore containing Fe and Cr is heated with carbon in an electric furnace to form an alloy (A). The ore on treatment with NaOH and oxygen at 1100°C gives (B), Fe_2O_3 and H_2O. (B) reacts with soluble lead salts to give yellow precipitate. When (B) is acidified, we get (C). (C) is reduced with carbon to give (D), Na_2CO_3 and CO. (D) by thermite reduction with (E) gives metal with ground state valance electronic configuration d^5s^1.

114. The alloy (A) formed is
(a) bronze (b) brass (c) silicomanganese (d) ferrochrome

115. The yellow precipitate formed with lead acetate is
(a) PbI_2 (b) $Pb(CH_3COO)_2$ (c) $PbCrO_4$ (d) $PbCl_4$

116. The metal Cr is rendered passive in
(a) HNO_3 (b) HCl (c) CH_3COOH (d) H_3PO_3

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers of which ONE OR MORE answers will be correct.**

117. Cassiterite is concentrated by
(a) Magnetic separation (b) Gravity separation (c) Froth flotation (d) Leaching

118. Even though hydrogen is a good reducing agent, it is rarely used to reduce metal oxides because
(a) hydrogen has high rate of diffusion
(b) The reaction becomes reversible for most metal oxides at higher temperature
(c) hydrogen reacts with metal forming hydrides
(d) hydrogen may react explosively with oxygen

119. Due to lanthanide contraction, the 4d and 5d series transition elements have almost same size. Owing to this the other properties that may be similar are
(a) solvation energy
(b) ionization energy
(c) stable higher oxidation state
(d) ionic radii of cation of the same charge



Matrix–Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

120.

	Column I		**Column II**
(a)	diamagnetic	(p)	$ScCl_3$
(b)	colourless	(q)	CuCl
(c)	shows variable oxidation state	(r)	$KMnO_4$
(d)	coloured due to charge transfer spectra	(s)	$CrCl_3$

ADDITIONAL PRACTICE EXERCISE

Subjective Questions

121. Even though Ti^{2+} and Ti^{3+} ions exist in aqueous solution Ti^{+4} hydrated ions do not exist explain.

122. Perovskite and Ilmenite are minerals with composition $CaTiO_3$ and $FeTiO_3$ respectively. Give one method for preparing calcium titanate in the laboratory.

123. Eventhough all the transition metals of the first transition series except copper have high negative value of standard electrode potentials, they are not good reducing agents as alkali metals or alkaline earth metals. Explain why?

124. Briefly explain the cause of Lanthanide contraction.

125. Explain why Yttrium though not a member of lanthanide series or even of the same period is regarded for all practical purposes as a member of lanthanide series.

126. Plutonium –238 is used as a lightweight power source in cardiac pacemakers. why?

127. For actinides it is considered that for higher oxidation states co–valent bonding must certainly be involved unlike the lanthanides. Why?

128. The observed paramagnetic moment of $Co^{2+}_{(aq)}$ is 4.3 – 5.2 BM which is more than the calculated value 3.87 BM. Why?

129. Why Titanium metal is not prepared by electrolysis of molten salt or carbon reduction of oxide?

130. $[Re_2Cl_8]^{2-}$ which is isostructural with $[Mo_2Cl_8]^{4-}$ is made up of two square planar $ReCl_4$ units linked by very short Re– Re bond. This bond is interpreted to be a quadruple bond. Briefly discuss the structure of $[Re_2Cl_8]^{2-}$

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

131. Copper is produced from copper pyrites by self reduction process, the element that gets reduced and oxidised are respectively

(a) copper and oxygen
(b) oxygen and copper
(c) copper and sulphur
(d) sulphur and oxygen

132. Zone refining cannot be applied when

(a) impurity is an oxide of the metal
(b) molecular size of impurity and metal are similar
(c) the metal oxide formed is unstable
(d) the metal can be refined electrolytically

133. Zinc sulphide is heated in air at a low temperature and the product(s) is dissolved in H_2SO_4. Zn now exists as compound A. Zn can be extracted from A by

(a) carbon reduction
(b) hydrometallurgy
(c) reduction with carbon monoxide
(d) electrolytic process

134. From its ore cadmium is obtained as $CdSO_4$ which contains $ZnSO_4$. Cd can be extracted from this mixture by

(a) electrolysis
(b) addition of zinc powder
(c) leaching with Na_2CO_3
(d) calcination followed by electrolysis

135. During electrolytic reduction of $MgCl_2$, 30% NaCl is added to the electrolyte. This is done to
(a) slightly decrease conductivity for controlled deposition
(b) prevent any reaction between Mg and Cl_2
(c) decrease the fusion temperature of anhydrous $MgCl_2$
(d) to prevent formation of MgO

136. During extraction of iron, it is preferable to oxidize FeO to Fe_2O_3 because
(a) Fe_2O_3 is more reactive with carbon
(b) FeO may act as flux for SiO_2
(c) reaction of FeO with carbon does not go to completion
(d) carbon monoxide does not react with FeO

137. An ore contains PbS, ZnS and FeS_2. During separation of these three by froth flotation, the role of $CuSO_4$ is
(a) to make ZnS wettable by froth
(b) to depress the floatation of FeS_2
(c) to react with excess NaCN
(d) to stabilize ZnS in froth

138. During electrolytic extraction of magnesium, the anode and cathode should be separated using a porcelain tube because
(a) oxygen is liberated at the anode
(b) products formed at anode and cathode react with each other
(c) magnesium formed at cathode may get oxidized
(d) in all such processes, anode and cathode are separated to avoid explosion

139. An ore is subjected to roasting. On doing so, a gas A is liberated and the metal is obtained in free state. If gas turns acidified dichromate paper green, the ore is
(a) cassiterite
(b) copper glance
(c) malachite
(d) sphalerite

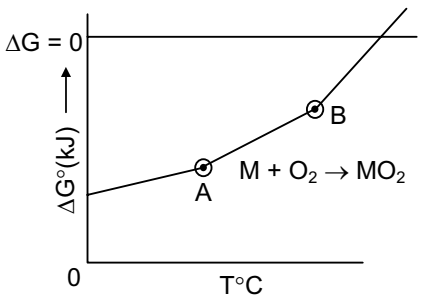
140. Iron oxide can be reduced to iron with the help of
(a) Aluminium
(b) carbondioxide
(c) silver
(d) self reduction

141. An ore was concentrated by froth flotation followed by roasting. If the compound constituting the ore is pale white coloured, the ore can be reduced using
(a) hydrogen
(b) carbon
(c) self reduction
(d) heat

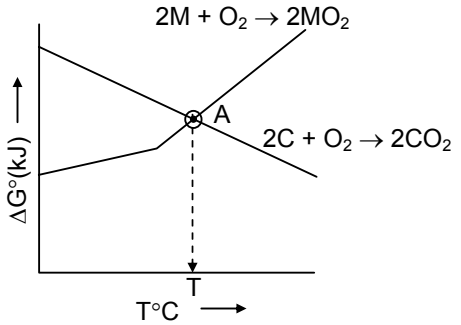
142. Among the following, the ore which on roasting gets converted to the metal rather than the metal oxide is
(a) ZnS
(b) FeS
(c) HgS
(d) $Al_2O_3.2H_2O$

143. Consider the reaction $M^{n+}_{(aq)} + ne^- \rightarrow M_{(s)}$. If E° for this reaction has most negative values, the metal can be reduced by
(a) roasting
(b) smelting
(c) electrolytic reduction
(d) heating with another metal

144. The Ellingham diagram of a certain metal M is represented in the figure. The points A and B respectively correspond to
(a) oxidation, reduction of metal
(b) beginning and end of reduction process
(c) minimum and maximum value of ΔG
(d) melting point and boiling point of metal



145. From the following graph we can say that



(a) At A, ΔG for MO + C → M + CO is zero
(b) carbon can reduce MO below T°C
(c) carbon can reduce MO at all temperatures
(d) carbon is converted to carbide after reducing the metal oxide

146. Ore $\xrightarrow{\text{concentration}}$ aqueous solution of the ore $\xrightarrow{\text{electolysis}}$ metal. The metal may be
(a) Magnesium (b) zinc (c) sodium (d) aluminium

147. The slag obtained after extraction of tin is worked up separately by smelting with coke. From this statement we can say that
(a) slag is more dense than molten metal
(b) the tin ore was not concentrated properly
(c) a considerable amount of tin is present in the slag
(d) tin forms a complex and remains in the slag

148. If phosphorus is present as an impurity, the slag formed due to this is
(a) $Fe_3(PO_4)_2$ (b) P_2O_5 (c) $P_2O_5.SiO_2$ (d) $Ca_3(PO_4)_2$

149. AgI is yellow coloured because
(a) of d–d transition
(b) of charge transfer spectrum
(c) Ag^+ polarizes iodide ion
(d) AgI is ionic

150. The compound/ion, whose colour does not arise due to charge transfer is
(a) $[Fe(NO)(H_2O)_5]^{2+}$ (b) $KMnO_4$ (c) Ag_2CO_3 (d) CrO_4^{2-}

151. The third row transition elements are least reactive when compared to other series of transition elements, because of
(a) high heat of sublimation
(b) high ionization energy
(c) low heat of hydration
(d) All the above

152. The only true statement among the following is
(a) Zn form complexes with great ease
(b) AgI is yellow due to d–d transition
(c) Zn^{2+} solution is less acidic than Cu^{2+} solution for same concentration
(d) Zn exhibits variable valency

153. Exposed mercury surfaces should always be covered with toluene. This is done because
(a) Mercury has appreciable vapour pressure at room temperature
(b) Mercury in gaseous state is monoatomic
(c) Mercury is very reactive
(d) Mercury forms both monovalent and divalent compounds

154. A few compounds from the main group elements are isostructural with the transition metal compounds. An example of this is
(a) SO_3 and CrO_3 (b) VCl_5 and PCl_5 (c) CrO_4^{2-} and SO_4^{2-} (d) VF_4 and $SiCl_4$

155. Cobalt chloride paper is used to test for water. In the presence of water the blue colour turns pink. The change in oxidation state of cobalt is
(a) +1 (b) –1 (c) +2 (d) 0

156. Even though both copper and zinc have d^{10} configuration (completely filled d orbitals), copper behaves as a transition metal while zinc does not. This is because
(a) the increase in size of zinc compared to copper
(b) most common oxidation state exhibited by Cu^{2+} contain incomplete 'd' subshell
(c) cubic close packed structure in the case of copper
(d) high heat of sublimation

157. Both group 11 metals as well as the group 1 metal have similar outer electron configuration and they exhibit similarities in properties. Which among the following property is not similar for both
(a) electrical conductivity (b) enthalpy of sublimation
(c) formal stoichiometry of compounds (d) lustre due to oscillation of electron

158. Among the following the pair of ions with same magnetic moment is
(a) Ca and Sc^+ (b) Sc^{3+} and Cu^+ (c) Mn^{4+} and Co^{3+} (d) Fe^{2+} and Ni^{2+}

159. With reference to the oxides of transition elements the wrong statement is
(a) Along a period, acidic nature of oxide in a given oxidation state increases
(b) As oxidation state increases, covalent character of the oxide increases for a given transition element
(c) for a given element, as oxidation state of the metal in oxide increases the acidic character decreases
(d) the acidic oxide of transition elements dissolve in water to form oxoacids

160. The explanation given for pale yellow colour of VO_2^+ ion will be same as that given for
(a) Ti^{4+} (b) Cu^+ (c) $[Cr(H_2O)_6]^{2+}$ (d) CrO_4^{2-}

161. The measurement that confirms that Hg^+ exists as a dimer is
(a) magnetic moment (b) colorimetry (c) reactivity (d) dipole moment

162. A transition element exhibits two valency states x and y. If compound in x state is ionic and y state is covalent we can say that
(a) $y - x = 2$ (b) $x > y$ (c) $x < y$ (d) $x \geq y$

163. Even though both potassium and copper have $4s^1$ configuration in the ground state, the ionization energy of copper is
(a) less than potassium when moved across the period
(b) less than potassium as it attains argon gas configuration
(c) higher than potassium due to poor shielding effect of d electrons
(d) higher than potassium due to larger size of potassium ion

164. The ion that disproportionates in water is
(a) Cu^{2+} (b) Ag^+ (c) Au^+ (d) Au^{3+}

165. The common characteristic shown by d–block transition elements is
(a) paramagnetic character (b) variable oxidation state
(c) coloured compounds (d) one oxidation state equal to group number

166. Among the following the ion that is not coloured is
(a) VO_2^+ (b) $[ScF_6]^{3-}$ (c) $TiCl_3$ (d) $FeCl_3.6H_2O$

167. Lanthanide contraction causes
(a) similar size of Fe, Co, Ni
(b) small increase in size towards the end in the transition series
(c) similar size of elements in the same group belonging to second and third transition series elements
(d) almost the same size among the lanthanides

168. The colour of the lanthanide compounds does not change with change in ligand bonded to the element because
(a) lanthanides have very small size compared to the ligand
(b) the f electrons have very poor shielding effect
(c) spin orbit coupling should be considered for lanthanides
(d) f orbitals are deep inside the atom

169. Which among the following lanthanides posses low enthalpy of atomization compared to other lanthanide elements?
(a) Eu and Yb (b) Gd and Tb (c) Ce and Lu (d) Pr and Nd

170. Among the following, the property which is similar for both lanthanides and actinides is
(a) tendency to form complex
(b) reaction with water
(c) density
(d) tendency to form oxoanions



Assertion–Reason Type Questions

Directions: **Each question contains Statement–1 and Statement–2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**
(a) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1
(b) Statement–1 is True, Statement–2 is True; Statement–2 is NOT a correct explanation for Statement–1
(c) Statement–1 is True, Statement–2 is False
(d) Statement–1 is False, Statement–2 is True

171. **Statement 1**
For the change occurring during roasting of sphalerite, the entropy change is negative.

and

Statement 2
Two different compounds of zinc are formed during roasting.

172. **Statement 1**
Extraction of gold which occurs in native form, can be done by forming an amalgam.

and

Statement 2
Due to completely filled d orbitals mercury has low melting point.

173. **Statement 1**
Magnesium oxide can be reduced by carbon but it should be followed by rapid cooling.

and

Statement 2
All metals can be reduced by carbon at a temperature characteristic of the metal.

174. Statement 1
For extraction of copper, malachite is calcined but chalcocite is roasted.

and

Statement 2
Both roasting and calcination convert all the ores into their oxides.

175. Statement 1
Zn^{2+} has a smaller ionic radii than Cu^{2+}.

and

Statement 2
The ionic radii of ions in the same oxidation state decreases with decrease in atomic number for transition elements.

176. Statement 1
Zn liberates hydrogen gas with HCl but this is not seen with HNO_3.

and

Statement 2
HNO_3 is an oxidizing acid.

177. Statement 1
Iron and osmium are in the same group, but show a maximum oxidation sate of +6 and +8 respectively.

and

Statement 2
For transition elements the maximum oxidation state is equal to number of (n – 1)d and ns electrons.

178. Statement 1
As the oxidation state of metal ion increases, its oxide becomes more acidic.

and

Statement 2
The oxide becomes less ionic with increase in oxidation state.

179. Statement 1
The group 12 elements are more reactive than the copper group.

and

Statement 2
Group 12 metals are very soft compared to the copper group.

180. Statement 1
The chemical properties of lanthanoids have very close similarity with each other.

and

Statement 2
The 4f electrons are buried deep inside and do not take part in chemical bonding.

Linked Comprehension Type Questions

Directions: **This section contains 3 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

Metal 'M' is extracted from its ore, which is a double salt. It is a salt of M and of potassium. The ore is concentrated and water of hydration present in the double salt is removed. The ore is then reduced to get the metal.

The metal so produced is used to make Grignard reagent.

181. The ore of the metal in the passage is

(a) carnallite (b) magnesite (c) spodumene (d) epsomite

182. The best method employed to remove the water of crystallization from the ore is

(a) calcination (b) roasting

(c) heating in a atmosphere of HCl gas (d) heating followed by rapid cooling in inert atmosphere

183. The compound of metal obtained from the ore is reduced to the metal by

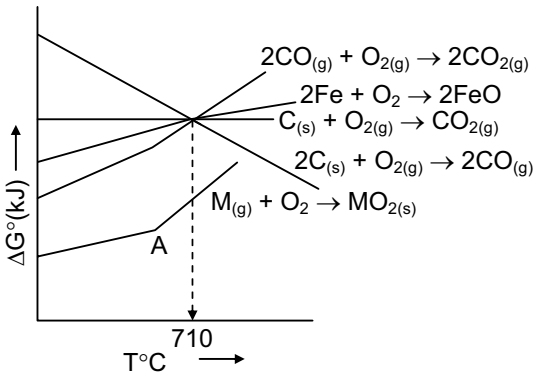
(a) smelting (b) heating with carbon and silica

(c) hydrogen reduction (d) electrolysis of fused compound

Passage II

Ellingham diagrams are the plots of variation of ΔG with increase in temperature for the oxidation of some common metals to the metal oxide.

Thermodynamically, the oxidation reaction would be feasible and oxide formed would be stable. If the free energy change ΔG of the oxidation reaction is negative.



184. The point A for the metal M represents its

(a) melting point (b) sublimation point

(c) temperature of oxidation (d) reduction of metal oxide

185. Identify the true statement among the following.

(a) Fe can reduce the metal oxide MO_2 to M

(b) Carbon can reduce both FeO and MO_2 below 710°C

(c) Metal M can reduce FeO to Fe

(d) The diagram holds even if 2 moles of O_2 is used during the reaction

186. Considering the oxidation of Fe with respect to those of carbon, we can say
(a) below 710°C reduction of FeO with C will give CO
(b) Above 710°C only, CO can reduce FeO
(c) carbon can reduce FeO at all temperature
(d) below 710°C the formation of CO_2 is not favourable

Passage III

The d block or transition elements are those elements which have partly filled d orbitals. The general electronic configuration is $(n-1)d^{1-10}\,ns^{1-2}$. The last element zinc has maximum permissible electrons in the d orbital. Based on the definition of d block elements, it excludes zinc from d block because zinc or any of it common ions do not have completely filled d orbitals. Zn, Cd and Hg fall in this group 12.

187. It is preferred to substitute zinc compounds for lead compounds in white paints because
(a) zinc compounds are not coloured
(b) zinc shows only one oxidation state
(c) zinc does not have any magnetic property
(d) zinc does not have properties of transition element

188. The yellow colour of hot zinc oxide is due to
(a) d – d spectra
(b) charge transfer
(c) completely filled d orbital
(d) defects in solid structure

189. Zn and Mg have ns^2 configuration. They are similar to each other in
(a) formation of divalent, basic oxide
(b) formation of covalent compounds
(c) formation of hydrated sulphates
(d) tendency to form complexes



Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers of which ONE OR MORE answers will be correct.**

190. In the electrolytic reduction of pure alumina to produce aluminium, the true statement(s) is/are
(a) the current passed helps to heat the electrolyte
(b) the aluminium formed sinks to the bottom
(c) addition of cryolite increases electrical conductivity
(d) A part of aluminium vapourizes because it is low melting than alumina

191. When the gangue particles are mixed with flux, slag is formed. The slag
(a) is partially soluble in the metal
(b) protects the molten metal
(c) is fusible in nature
(d) is generally acidic

192. In metallurgy, an ore may be dissolved in water and the element may be displaced from solution using another cheap element. An example for this is
(a) solution of cadmium + zinc metal
(b) sea water containing bromine + Cl_2
(c) solution of Fe + Cu metal
(d) solution of Zn + Ag metal

193. In which of the following processes, calcium silicate slag is formed. Extraction of
(a) tin from cassiterite
(b) copper from copper pyrite
(c) Zinc from zinc blende
(d) iron from heamatite

194. Potassium and silver differ from each other with respect to
(a) ionic size
(b) thermal stability of carbonates
(c) thermal decomposition of nitrates at low temperature
(d) solubility of chlorides

195. In which of the following aspects WO_3 and CrO_3, differ?
(a) oxidizing property (b) melting points (c) acidic nature (d) solubility in NaOH

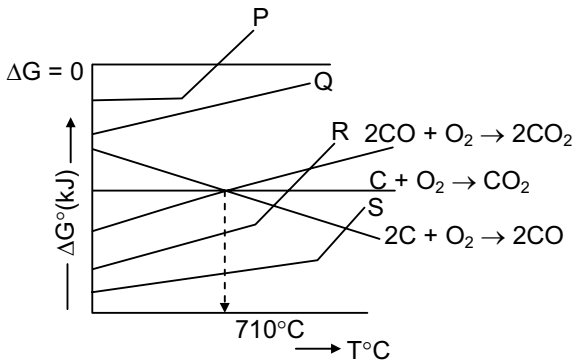
196. There is a large decrease in melting point on moving from
(a) scandium to vanadium (b) Manganese to iron
(c) chromium to manganese (d) copper to zinc

197. Among the following, the metal(s) that liberates hydrogen from dilute acid is/are
(a) Mn (b) Cr (c) Fe (d) Cu

Matrix–Match Type Questions

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

198. Based on the Ellingham diagram for formation of oxide. Match the matrix given.



	Column I		**Column II**
(a)	Reduced on heating below 710°C	(p)	P
(b)	Reduced by carbon below 710°C	(q)	Q
(c)	Reduced by metal S	(r)	R
(d)	Reduced by CO only above 710°C	(s)	S

199.

	Column I		**Column II**
(a)	Radioactive	(p)	Pm^{3+}
(b)	Forms complex	(q)	Ac^{3+}
(c)	Shows only one oxidation state	(r)	Th^{4+}
(d)	Colourless in aqueous medium	(s)	Lu^{3+}

200.

	Column I **Property of oxide**		**Column II** **Metal oxide**
(a)	Amphoteric oxide	(p)	ZnO
(b)	Coloured oxide	(q)	Cr_2O_3
(c)	Oxide dissolves in HCl forming complex	(r)	Mn_2O_3
(d)	Prepared by thermal decomposition of a compound	(s)	Fe_2O_3

SOLUTIONS

ANSWERS KEYS

Topic Grip

1. Cu, Ag, Au, Pt least reactive
2. Dissolving the mineral a suitable reagent which do not dissolve the impurities.
3. Molten lead and zinc are immiscible and there is greater solubility for Ag in molten Zn
4. (i) due to escaping SO_2 Copper formed has blistered appearance
   (ii) O_2 liberated oxidise carbon electrodes
   (iii) Ag forms insoluble AgCl
5. 4.08 kg Al_2O_3
6. Zn^{2+} has $3d^{10}$ configuration, it is stable. It does not exhibit other oxidation states. Zn(II) complexes are typically either tetrahedral or octahedral.
7. (i) In addition to CuS, oxide of Cu is also produced.
   (ii) $Na_2Cr_2O_7$ is hygroscopic.
   (iii) AgI is coloured because of polarisation.
   (iv) In $ZnSO_4$ there is no d–d transition. Colour of $KMnO_4$ is due to charge transfer
8. $3MnO_2 + 4KOH + 2KNO_3 \rightarrow 3K_2MnO_4 + 2NO + 2H_2O$
   $2MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$
9. $Fe_3O_4^{2-}$: iron in +6 state = 2.84 BM
   $Cr(OH)_3$ = 3.87 BM
   VCl_5 = 0 BM
   $[Ni(H_2O)_6]^{2+}$ = 2.84 BM
10. Refer SM
11. (a) **12.** (c) **13.** (c)
14. (a) **15.** (c) **16.** (c)
17. (a) **18.** (c) **19.** (a)
20. (b) **21.** (a) **22.** (b)
23. (a) **24.** (a) **25.** (c)
26. (b)
27. (a), (c)
28. (b)
29. a, b
30. (a) →(p), (q), (r), (s)
    (b) →(p), (s)
    (c) → (r)
    (d) →(q), (s)

IIT Assignment Exercise

31. (d) **32.** (b) **33.** (c)
34. (a) **35.** (c) **36.** (d)
37. (a) **38.** (b) **39.** (c)
40. (a) **41.** (d) **42.** (c)
43. (c) **44.** (a) **45.** (d)
46. (a) **47.** (c) **48.** (b)
49. (d) **50.** (b) **51.** (c)
52. (a) **53.** (d) **54.** (b)
55. (a) **56.** (b) **57.** (a)
58. (c) **59.** (c) **60.** (d)
61. (c) **62.** (b) **63.** (b)
64. (b) **65.** (c) **66.** (c)
67. (a) **68.** (d) **69.** (b)
70. (a) **71.** (a) **72.** (c)
73. (a) **74** (d) **75.** (c)
76. (b) **77.** (a) **78.** (d)
79. (c) **80.** (a) **81.** (b)
82. (a) **83.** (c) **84.** (c)
85. (a) **86.** (b) **87.** (b)
88. (d) **89.** (a) **90.** (c)
91. (d) **92.** (c) **93.** (b)
94. (c) **95.** (c) **96.** (d)
97. (b) **98.** (a) **99.** (c)
100. (a) **101.** (b) **102.** (d)
103. (b) **104.** (a) **105.** (c)
106. (c) **107.** (b) **108.** (d)
109. (a) **110.** (d) **111.** (b)
112. (a) **113.** (d) **114.** (d)
115. (c)
116. (a)
117. (a), (b)
118. (b), (c), (d)
119. (a), (b), (c), (d)
120. (a) →(p), (q), (r)
    (b)→(p), (q)
    (c) →(r), (s)
    (d) → (r)

Additional Practice Exercise

121. Because of their high charges – Salts are hydrolysed
122. Fusing. $CaCO_3$ and TiO_2 together
123. high IE, low heat of hydration, high heat of vapourisation
124. Refer SM
125. similarity in size
126. α emitter– $t_{1/2}$ 86 yrs
127. 5f orbitals extent farther
128. orbital angular momentum contribution
129. m.p high and reacts with air readily
130. There is a quadruple bond between Re atoms consisting of one σ, 2π and one δ bonds
131. (c) **132.** (b) **133.** (d)
134. (b) **135.** (c) **136.** (b)
137. (a) **138.** (b) **139.** (b)
140. (a) **141.** (b) **142.** (c)
143. (c) **144.** (d) **145.** (a)
146. (b) **147.** (c) **148.** (d)
149. (c) **150.** (c) **151.** (d)
152. (c) **153.** (a) **154.** (c)
155. (d) **156.** (b) **157.** (b)
158. (b) **159.** (c) **160.** (d)
161. (a) **162.** (c) **163.** (c)
164. (c) **165.** (a) **166.** (b)
167. (c) **168.** (d) **169.** (a)

170. (b) **171.** (b) **172.** (a)
173. (b) **174.** (c) **175.** (d)
176. (a) **177.** (c) **178.** (a)
179. (b) **180.** (b) **181.** (a)
182. (c) **183.** (d) **184.** (a)
185. (c) **186.** (b) **187.** (a)
188. (d)
189. (c)
190. (a), (b), (c)
191. (b), (c)
192. (a), (b)
193. (a), (d)
194. (b), (d)
195. (a), (b)
196. (c), (d)
197. (a), (c)
198. (a) → (p)
(b)→(p), (q)
(c) →(p), (q), (r)
(d) →(p), (q), (r)
199. (a) →(p), (q), (r)
(b)→(q), (r)
(c) →(p), (q), (s)
(d) →(q), (r), (s)
200. (a) →(p), (q), (r)
(b)→(q), (r), (s)
(c) →(p), (q), (r)
(d) →(p), (q), (s)

HINTS AND EXPLANATIONS

Topic Grip

1. Metals which have little or no affinity for oxygen and which resist the attack of water and other chemical reagents occur in free state or native state. These are called noble metals.

e.g., Gold, Silver, Platinum etc. They occur as lumps of pure metal called nuggets

They have a positive standard reduction potentials i.e., their ions are easily reduced to the metal

Metals which have a tendency to be converted to their ions and thereby forming compounds occur in nature in combined state as their oxides, sulphides etc. These metals are called active metals and occur as compounds called minerals e.g., Na, Fe, Cu etc.

They have a negative standard reduction potential and so are oxidized easily to their ions or their ions do not have any tendency to be reduced to the metallic state.

2. Leaching is a method of concentrating the ore. The principle behind the process is to treat powdered ore with a suitable reagent such that only the ore particles completely dissolve in the reagent leaving behind the impurities. The impurities can be filtered off.

Example Bauxite ore is treated with NaOH solution. Only Al_2O_3 dissolves in NaOH forming soluble sodium meta aluminate while the impurities remain and can be filtered off.

3. Zinc is a suitable solvent for extracting silver from lead because

(a) molten zinc and lead are immiscible.

(b) silver does not react with lead.

(c) alloy of zinc and silver melts at higher temperature than lead.

(d) molten zinc floats over molten lead.

(e) due to low boiling point of zinc, it is easily to distill out silver.

4. (i) copper obtained by self reduction of Cu_2S contains SO_2 gas. The copper obtained is called blister copper due to the blistered appearance produced by SO_2

$$2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$$

$$2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$$

(ii) Aluminum is extracted by electrolytic reduction of Al_2O_3

Graphite lined steel tank is the cathode and graphite rods are the anode.

During the process, Oxygen is liberated at the anode which reacts with the carbon rod giving CO_2 gas. Hence the anode rods have to be replaced regularly.

(iii) Gold dissolves in aquaregia due to formation of chloroauric acid.

$$3HNO_3 + 11HCl + 2Au \rightarrow 2\,HAuCl_4 + 6H_2O + 3NOCl$$

Whereas, silver remains insoluble due to the formation of insoluble AgCl on reaction with aqua regia.

5. The reaction is aluminothermite process and it is a highly exothermic process. Aluminum being highly electropositive, it reduces the metal oxide to the metal when heated with it

$$\underset{\substack{3\times[(3\times55)+(16\times4)]\\687\text{ g}\\6.87\text{ kg}}}{3Mn_3O_4} + \underset{\substack{[8\times27]\\216\text{ g}\\2.16\text{ kg}}}{8\,Al} \rightarrow \underset{\substack{4[(2\times27)+(3\times16)]\\408\text{ g}\\4.08\text{ kg}}}{4\,Al_2O_3} + 9Mn$$

The oxide formed is Al_2O_3 and the amount of Al_2O_3 formed is 4.08kg

6. Zn has $3d^{10}4s^2$ configuration. By losing two electrons, it acquires d^{10} configuration, which is very stable. Hence it does not exhibit other oxidation states. Zn(II) complexes are typically either tetrahedral (Example $[ZnCl_4]^{2-}$) or octahedral (Example $[Zn(H_2O)_6]^{2+}$)

7. (i) Heating of CuS will lead to its decomposition. Hence quantitative estimation by CuS is not possible.

(ii) $Na_2Cr_2O_7$ is hygroscopic. So it cannot be used as a primary standard.

(iii) AgI is coloured because the small Ag^+ polarizes the large iodide ion.

(iv) $ZnSO_4$ is a Zn^{2+} salt and has $3d^{10}$ configuration. There is no possibility of d–d transition. Since $KMnO_4$ contains Mn^{7+} which has a d^0 configuration, no d–d transition is possible. The colour is due to charge transfer by which electrons are excited from oxygen to Mn^{7+}.

8. The reaction taking place on fusion is

$3MnO_2 + 4KOH + 2KNO_3 \rightarrow 3K_2MnO_4 + 2NO + 2H_2O$

K_2MnO_4 undergoes disproportionation in dil.H_2SO_4

$3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

9. (i) FeO_4^{2-}

Fe is in the +6 state

[Ar] $3d^2$

$\mu = \sqrt{2(2+2)} = \sqrt{8} = 2.84$ BM

(ii) $Cr(OH)_3$

Cr is in +3 state

[Ar] $3d^3$

$\mu = \sqrt{3(3+2)} = \sqrt{15} = 3.87$ BM

(iii) VCl_5

Vanadium is in +5 state

[Ar]$3d^0$

$\mu = 0$

(iv) $[Ni(H_2O)_6]^{2+}$

Nickel is in +2 state

[Ar] $3d^8$

$\mu = \sqrt{2(2+2)} = \sqrt{8} = 2.84$ BM

10. The increase in size towards the end of series can be explained by considering the increased force of repulsion on addition of electrons. The repulsive forces among the electrons exceeds the attractive forces from the nucleus. So there is an increase in the covalent radius towards the end of the series.

11. Due to difference in density between the ore particles and impurity particles, this separation is possible.

12. Fe_2O_3

Siderite ore is $FeCO_3$

$4\,FeCO_3 + O_2 \rightarrow 2\,Fe_2O_3 + 4\,CO_2$

On roasting ores are converted to their oxides.

13. Hoops process is electrolytic refining of aluminium.

14. The more the number of oxygen atoms bonded to the metal, greater is the acidity of the oxide.

15. Ti^{3+} has configuration d^1.

Oxidation state of V in VBr_4 is +4. It is also a d^1 ion.

16. Ore is heated below its melting point during roasting. Sphalerite or Zinc blende is ZnS

Cinnabar is HgS.

On roasting ZnS gives ZnO while HgS gives Hg. This is because the electrode reduction potential of Hg is less than zero.

So HgO formed gets converted to Hg on heating.

17. Van Arkel method of refining Titanium involves formation of stable, volatile T_iI_4 which decomposes on heating.

$$T_1 + 2I_2 \xrightarrow{150^\circ} TiI_4 \xrightarrow{1800^\circ C} \underset{\text{pure}}{Ti} + 2I_{2(g)}$$

18. Number of electrons added is less for Na_{11}^{23} to Cl_{17}^{35}. The decrease in size is greater because of the low shielding power of the incoming d electrons into the penultimate 3d shell on moving from scandium to copper.

19. Just like motion of electric charges creates a magnetic field, the motion of an electron, due to the presence of a negative charge on it creates a magnetic field. So it behaves as a micro magnet and the magnetic moment is dependent on the number of individual electrons.

20. Electronic configuration of La is [Xe] $5d^1\,6s^2$ there are no electrons in 4f orbital.

La^{3+}[Xe]

La in the 3+ state attains noble gas Xenon configuration so it is very difficult to remove an electron from La^{3+}, so it is not possible to attain an oxidation state greater than 3+.

I V_2O_5 from $V \rightarrow (A)$

VX_5 will have V in d^o configuration.

So no d–d transition is possible.

$V + 4HF \rightarrow VF_4 + 2H_2$

$2VCl_4 \rightarrow 2VCl_3 + Cl_2$,

$2VF_4 \rightarrow VF_5 + VF_3$

$2VF_3 \rightarrow VF_4 + VF_2$

$VF_5 + F^- \rightarrow [VF_6]^-$

21. A(V) has d^0 configuration.

22. It is tetrahedral similar to $[NiCl_4]^{2-}$.

23. VCl_4 and $[NiCl_4]^{2-}$ are paramagnetic compounds. The magnetic moment is 2.8 BM in $[NiCl_4]^{2-}$ and 1.8 BM in VCl_4.

24. Sulphide ores are concentrated by froth flotation copper glance is a sulphide ore.

25. Ethyl Xanthate and potassium ethyl xanthate are used as collecters. They attach themselves to the mineral ore

and make them water repellent and hence mineral ore comes with the froth.

26. NaCN is here called depressant. On adding NaCN impurity ZnS in galena forms a complex and dissolves. So PbS is exclusively wetted by froth in the presence of ZnS.

27. Ore that have magnetic properties are concentrated using magnetic separator. Chromite ore – $FeCr_2O_4$, Ilmenite $FeTiO_3$.

28. Aqueous NaOH on reaction with bauxite ore form soluble sodium aluminate. While the impurities remain and are filtered out.

29. Sulphides of Cu and Pb can be reduced to their metals by auto reduction or self reduction without the use of any external reducing agent.

Example $2PbS + 3O_2 \xrightarrow{\text{Roasting}} 2PbO + 2SO_2$

$$2PbO + PbS \xrightarrow[\Delta]{\text{absence of air}} \underset{\text{self reduction}}{3Pb} + SO_2 \uparrow$$

30. (a) → (p), (q), (r), (s)
(b) → (p), (s)
(c) → (r)
(d) → (q), (s)

Electro refining is done for most metals Poling is done when the oxide of the metal is present as impurity. This is seen in Cu and Sn. Nickel is refined by Mond's process.

Liquation is applicable to metals with low melting point.

IIT Assignment Exercise

31. Bauxite ore is $Al_2O_3.2H_2O$
On calcination hydrated oxide ores are converted to anhydrous form

$$Al_2O_3.2H_2O \xrightarrow{\text{calcination}} Al_2O_3 + 2H_2O$$

32. Roasting is a process of heating the ore in excess of air below its melting point.

33. The choice of flux depends on the type of impurity present.
Acidic impurity is treated with basic flux while basic impurity is treated with acidic flux.

34. When metal comes in contact with mercury, it immediately forms the amalgam.
It is very convenient to get back the metal simply through distillation due to low melting point of mercury.

35. Argentite is Ag_2S
Silver is extracted by leaching

36. $Zr + 2I_2 \rightarrow ZrI_4 \xrightarrow{\Delta} \underset{\text{pure}}{Zr} + 2I_2$

37. Copper pyrites is $CuFeS_2$
On roasting FeO is formed an acidic flux is Therefore, required.

$$\underset{\text{impurity}}{FeO} + \underset{\text{flux}}{SiO_2} \rightarrow \underset{\text{slag}}{FeSiO_3}$$

38. In aluminothermite process, metallic aluminium is oxidized and the metal oxide is reduced to give a large amount of energy.

39. Limestone is added as flux to remove the acidic gangue particles as slag.

40. Iron obtained from the blast furnace is called pig iron. It contain about 4% carbon.
Cast iron is obtained by melting pig iron with scrap iron and coke in blast furnace. It contain about 3% carbon.

41. Ga like Al is refined electolytically.

42. The charge contains coke while adding in the blast furnace. But in the zone of reduction, CO reduces Fe_2O_3.

43. Impure Al forms the lower layer, the electrolyte forms the middle layer and pure aluminium the top layer. They remain separate due to density difference.

44. $ZnS + O_2 \rightarrow ZnO + SO_2$
$ZnS + 2O_2 \rightarrow ZnSO_4$

45. A molten mixture of Cu_2S and with small amount of FeS is called matte.

46. NaCN acts as depressent for ZnS but does not prevent PbS from the formation of the froth.

47. W, since W form the carbide WC.

48. Bauxite $Al_2O_3.2H_2O$

49. flux added in blast furnace is $CaCO_3$ i e, CaO

$P_4 + SO_2 \rightarrow P_4O_{10}$;
$P_4O_{10} + 6CaO \rightarrow 2Ca_3(PO_4)_2$

50. FeS in the ore is converted to FeO during roasting. FeO is removed as slag $FeSiO_3$.

51. (c)

52. Any $PbSiO_3$ formed, lime reacts to form slag and PbO

$$PbSiO_3 + CaO \rightarrow PbO + \underset{\text{slag}}{CaSiO_3}$$

53. The infusible impurity present in the ore is converted into a fusible slag and hence can be easily removed from the metal.

54. Huge logs of green wood are used to reduce the metal oxide to the metal.

55. MacArthur forest cyanide process.

$$Ag_2S + 4NaCN \rightleftharpoons 2Na[Ag(CN)_2] + Na_2S$$

Sodium argento cyanide

56. Mercury is the only metal which is a liquid at room temperature. This is due to completed d^{10} configuration metallic bonding is due to two s electrons only. Bonding further weakens as size is large.

57. Hardness increases, example steel

58. Due to d^{10} configuration, there is no covalent bonding leading to high melting point in Zn, Cd, Hg.

59. $V^{3+} - d^2$; $Cr^{3+} - d^3$, $Cr^{2+} - d^4$, $Ni^{2+} - d^8$ (two unpaired electrons)

Spin only magnetic moment = $\sqrt{n(n+2)}$. More the number of unpaired electrons, greater the spin only magnetic moment. It is Cr^{2+} with four unpaired electrons.

60. Sc^{3+} and Ti^{4+} have d° configuration Zn^{2+} has d^{10} configuration. Ni^{2+} has d^8 configuration. So it has d–d transition, hence coloured.

61. Cr_2O_3 is amphoteric.

62. (b)

63. The valence shell electrons that is ns electrons are responsible for the metallic character covalent bonding occurs between (n – 1)d electrons.

64. Cr – [Ar] $3d^5\ 4s^1$ Cr^{3+} [Ar] $3d^3$

s electrons in Li – $1s^2\ 2s^1$ – 3 electrons

d electrons in V^{3+}. [Ar] $3d^2$ – 2 electrons

p electrons in P – [Ne] $3s^2\ 3p^3$ – 3 + 6 = 9 electrons

65. Due to completely filled d orbital, there is no absorption in visible region leaving most zinc salts colourless.

66. Osmium and Ruthenium show a maximum oxidation state of +8.

67. Alloys are formed when two or more metals form a homogenous solid solution due to similar atomic radii one element can easily mix with the other to form solid solution.

68. In A – Ti is Ti^{3+} i.e. $3d^1$ configuration but in B. Ti is Ti^{4+} i.e. 3d° configuration hence 'B' is colourless as there is no d – d transition.

69. Cr – (24) $3d^5\ 4s^1$ $\mu = \sqrt{6(6+2)} = 6.9$ BM

Mn^+ 24 $3d^5\ 4s^1$ $\mu = \sqrt{6(6+2)} = 6.9$ BM

Fe^{2+} 24 $3d^6\ 4s^\circ$ $\mu = \sqrt{4(4+2)} = 4.89$ BM

70. Because of completely filled d orbitals, they do not show the characteristic property of transition elements.

71. Sc. [Ar] $3d^1\ 4s^2$

Sc^{3+} [Ar]

72. As they show variable oxidation state, they form unstable intermediate compounds of lower activation energy, thereby increasing rate of reaction.

73. As oxidation state increases, ionic radius decreases as the number of e^- decreases and nuclear charge remains same

74. On addition of d electrons, there is larger contribution to the covalent bonding between each atom, hence the enthalpy values increases

75. The maximum oxidation state is equal to the total number of electrons in the 3d and 4s orbitals. i.e., (n –1) d and ns orbitals.

76. They have fairly low ionization potential so are good reducing agents La^{3+}, Gd^{3+}, Lu^{3+} form colourless ion. They form halides of the type MX_3 They are not good conductors of heat and electricity.

77. Due to small size and high charge, the tendency to form complex is greater.

78. Due to lanthanide contraction size of lanthanides decreases from lanthanum to lutetium.

79. Ce – [Xe] $4f^1\ 5d^1\ 6s^2$

Ce^{3+} – [xe] $4f^1$

Ce^{4+} – [Xe]

As it attains [Xe] configuration it shows +4 state

80. Both Lanthanide contraction and actinide contraction are known.

81. $\underset{\text{Bauxite}}{Al_2O_3.2H_2O} \xrightarrow[\Delta]{-H_2O} \underset{\text{Anhydrous Alumina}}{Al_2O_3}$

This is done by calcination

82. Cu and Zn

83. $HgS + O_2 \rightarrow Hg + SO_2$

84. Positive to slightly negative or 0 to –0.5VE° requires mild conditions –0.5 to –1.5V requires another metal of more negative E°. E° values greater than –1.5V requires drastic conditions like electrolytic reduction.

85. $2AgCl + Na_2CO_3 \rightarrow Ag_2CO_3 + 2NaCl$

$$Ag_2CO_3 \xrightarrow{\Delta} Ag_2O + CO_2$$
$$\downarrow \Delta$$
$$2Ag + \tfrac{1}{2}O_2$$

86. $M_{(s)} + O_{2(g)} \rightarrow MO_{2\,(s)}$ ΔS = –ve

When the solid metal changes to the liquid metal at its melting point, ΔS becomes more negative so there is a sudden change in slope.

87. +5 is the highest oxidation state of vanadium.

88. Change in entropy is negative

$M_{(s)} + O_{2(g)} \rightarrow MO_{2(s)}$

Depending on the temperature the lines cross $\Delta G = 0$ line

Since $C \rightarrow CO$ line slopes downwards and

$Mg \rightarrow MgO$ slopes upwards, it is possible to reduce MgO with carbon but at a very high temp

All oxides can be theoretically decomposed to the metal because as they slope upward at a certain temperature they cross $\Delta G = 0$ line. But the corresponding temperature may be too high for practical purposes.

89. $CuCO_3.Cu(OH)_2 \xrightarrow{\Delta} 2CuO + CO_2 + H_2O$.

At that temperature H_2O is also a gas so option (a) is the right one

90. Serpeck's process

Powdered bauxite is mixed with coke and nitrogen AIN formed is converted to $Al(OH)_3$ with water. This on heating gives Al_2O_3

$Al_2O_3.2H_2O + 3C + N_2 \rightarrow 2AlN + 3CO\uparrow + 2H_2O\uparrow$

$SiO_2 + 2C \rightarrow Si\uparrow + 2CO\uparrow$

$AlN + 3H_2O \rightarrow Al(OH)_3 + NH_3$

$2Al(OH)_3 \xrightarrow{\Delta} Al_2O_3 + 3H_2O$

91. CuI_2 is the least stable I^- reduce Cu^{2+} to Cu^+

92. $Ag_2S + 4NaCN \rightleftharpoons 2Na[Ag(CN)_2] + Na_2S$ this reaction is reversible

On passing O_2, Na_2S is converted to Na_2SO_4 making reaction irreversible

93. In zone refining, the impure metal melts and only the pure metal crystallizes. So the melt will contain greater amount of impurities.

94. Due to diffused nature of f orbitals they have poor shielding effect. So on moving from 4d to 5d series, there is no increase in size but nuclear charge increase so ionisation energy increases contrary to other groups.

95. The highest oxidation state in a compound arises by a combination of transition metal with F or O. CrF_5, $KMnO_4$, $K_2Fe_2O_4$ are powerful oxidizing agents.

96. Mercury vapour are mono atomic.

97. On removing one electron, it attains stable half filled configuration. So the second ionization energy is expected to be high. ‘

98. As they show variable oxidation state, in a compound the same metal shows two different oxidation states.

99. a, b, d may be true for other elements as well but statement (c) shows that compounds of cobalt are coloured and vary with oxidation state. This is true for transition elements.

100. Due to lanthanide contraction the atomic size of 5d elements are very small almost comparable with 4d. But atomic weight increases as expected. Due to decrease in atomic volume, densities are very high.

101. Transition in the s and p electrons falls in the uv region so colour is not visible to naked eye.

102. Cr is in 3d and Mo is in 4d

Fe, Co, Ni have almost same radii

4d and 5d elements in a given group have same radii.

103. Since the aquo complex of Cu(I) is soluble CuCl is insoluble. But in aqueous NH_3 a stable amino complex is formed and hence it is soluble.

104. Sc configuration $3d^1\ 4s^2$

Sc^{3+} – [Ar]

105. M is Nickel

Nickel forms $NiCl_2$. ie Ni^{2+}

Electronic conf. –



Unpaired electrons = 2

$$\mu = \sqrt{2(2+2)} = 2.83\,BM$$

106. Due to lanthanide contraction, there is a slight difference in size of their trivalent ion. This leads to slight difference in properties like solubility, complex formation, hydration etc.

The metals have almost identical chemical properties

107. All lanthanide elements are placed in group 3 along with lanthanum and actinides are placed along with Actinium. Both in Group3.

108. Lanthanide hydroxide are less basic than $Ca(OH)_2$ but more basic than amphoteric $Al(OH)_3$

109. Ionisation energy values are low and they increase on moving from La to Lu due to increase in effective nuclear charge

110. $[54Xe]4f^7$ configuration does not show colour. It is Gd^{3+}.

111. Statement–2 is true. In the transition elements, the orbital contribution is quenched by the surrounding atoms/legends

So the magnetic moment calculated using the formula based on number of unpaired electrons holds.

112. During reduction of iron in the blast furnace, iron oxide is reduced to iron by CO. The iron formed contains impurities, so it melts at a much lower temperature of 1150°C to 1250°C even though melting point of pure iron is 1536°C

113. Configuration of Cu – $[Ar]\ 3d^{10}\ 4s^1$

Ag – $[Cr]\ 4d^{10}\ 5s^1$

Au – $[Xe]\ 5d^{10}\ 6s^1$

So in the +1 state, they will attain d^{10} configuration

But in the +1 state, only Ag is stable

Both Cu and Au disproportionate in solution

$$2Cu^+ \rightleftharpoons Cu^{2+} + Cu$$

$$2Au^+ \rightleftharpoons Au^{3+} + 2Au$$

II $$2FeCr_2O_4 + 8NaOH + \frac{7}{2}O_2 \xrightarrow{1100^0C} \underset{(B)}{4Na_2CrO_4} + Fe_2O_3 + 4H_2O$$

$$\underset{(B)}{Na_2CrO_4} + Pb^{2+} \rightarrow \underset{(yellow)}{PbCrO_4} + 2Na^+$$

Na_2CrO_4 on acidification gives $Na_2Cr_2O_7$

$$Na_2CrO_4 + 2H^+ \rightleftharpoons Na_2Cr_2O_7 + 2Na^+ + H_2O$$

$$\underset{(C)}{Na_2Cr_2O_7} + 2C \rightarrow \underset{(D)}{Cr_2O_3} + Na_2CO_3 + CO$$

$$Cr_2O_3 + \underset{(E)}{2Al} \rightarrow 2Cr + Al_2O_3$$

114. Ferrochrome (Cr = 65% and Fe = 35%) is obtained when chromite is reduced with carbon in an electric furnace.

115. $PbCrO_4$

116. HNO_3

117. Cassiterite or tin stone – SnO_2 is first concentrated by Gravity separation to remove the heavier gangue particles.

It is then passed through magnetic separator to remove wolframite ore impurity.

118. Hydrogen used for reducing reacts with the metal formed.

At the high temperature of furnace it may react with oxygen.

119. Due to lanthanide contraction all the properties mentioned are more or less same for 4d and 5d series transition elements.

120. (a) → (p), (q), (r)

(b) → (p), (q)

(c) → (q), (r), (s)

(d) → (r)

Sc^{3+} exhibits only +3 oxidation state.

It has d° configuration, so it is colourless and diamagnetic.

Cu^+ exhibits +1 and +2 states.

Cu^+ has d^{10} configuration so it is diamagnetic and colourless.

Mn^{7+} shows variable oxidation state from +1 to +7 It has d° configuration, hence diamagnetic and is coloured due to charge transfer Cr^{3+} shows variable oxidation state from +1 to +6. It has d^3 configuration.

So it is paramagnetic and is coloured due to d –d transition.

Additional Practice Exercise

121. Ti^{+4}, Zr^{+4} and Hf^{+4} cannot exist in solution because of their high charges. Their soluble derivatives are hydrolyzed.

Eg $TiCl_4$ form $TiCl_4.2H_2O$ by co–ordinating water, which then hydrolyzes and the hydroxide is formed. The hydroxide polymerizes and finally forms $TiO_2.2H_2O$.

122. $CaTiO_3$ can be prepared by fusing equimolar amounts of $CaCO_3$ and TiO_2

$$CaCO_3 + TiO_2 \xrightarrow{\Delta} CaTiO_3 + CO_2$$

123. The less reducing nature is due to high heat of vapourisation, high ionization potential and low heat of hydration of their ions.

124. As the nuclear charge increase by one unit one electron is added to 4f level. The shielding of one electron is 4f level by another one in the same level is very poor. Due to negligible amount of mutual shielding between 4f electrons the increase in nuclear charge experienced on the valence shell becomes increased and the size of atom decrease progressively.

125. The radii of Y^{3+} and Er^{3+} are almost the same (88 pm). This similarity along with equality in ionic charge accounts for the occurance of 'Y' with heavier lanthanides. The chemical properties, crystal structure and solubility of Yttrium compounds are similar to that of heavier lanthanide elements. Hence 'Y' cannot be easily separated from them. So for all practical purposes 'Y' is considered to be a member of lanthanide series. This is a consequence of lanthanide contraction.

126. It is an α emitter requiring very little shielding, since they are easily stopped by surrounding atoms and their kinetic energy is converted to heat. The heat so produced is used to produce electricity by means of thermopiles. Hence it is used as a power source in cardiac pacemakers and satellites . Other actinide isotopes used for this purpose are Am – 241 ($T_{1/2}$ 458 y) Cm–242 ($T_{1/2}$ 162 days)

$T_{1/2}$ of Pu–238 is 86 y.

The power obtained from an isotope is inversely proportional to its $T_{1/2}$.

127. It may be due to 5f orbitals extent farther from the nucleus and are involved in bonding

Or because the energy difference between 5f, 6d, 7p and 7s orbitals are within range of chemical bonding and hence appropriate valency states for co–valently bonded species are easily attained.

128. This is because of an appreciable orbital angular momentum contribution towards magnetic moment. Co^{2+} has $t_{2g}^5 eg^2$ configuration.

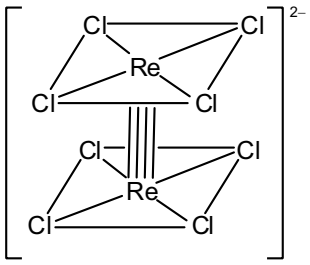
Configurations $t_{2g}^1 eg°$, $t_{2g}^2 eg°$, $t_{2g}^4 eg^2$ and $t_{2g}^5 eg^2$ have orbital contribution in octahedral complexes. This is because if an orbital can be converted to another degenerate orbital by rotation there will be an orbital angular moment and hence an orbital contribution towards magnetic moment.

$d_{x^2-y^2}$ to d_{z^2} rotation is impossible. Between t_{2g} rotation is possible only when they are not half filled or completely filled.

129. Melting point of Titanium is high and it react readily with air. O_2, N_2 and H_2 at elevated temperature. It form carbides hence can not be reduced by carbon or CO and further TiO_2 is very stable.

Hence Kroll process or Van Arkel method using volatile $TiCl_4$ or TiI_4 is employed for its production.

130. The structure of the ion is



The square planar $ReCl_4$ units use s, p_x, p_y and $d_{x^2-y^2}$ for bonding ie., the hybridization is dsp^2. In Re – Re bonding σ bond formed have contribution from p_z and d_{z^2} orbitals which lie along the bonding axis. With eclipsed conformations the d_{xz} and d_{yz} orbitals on Re– atoms from π bonds, by sideways overlap. The d_{xy} orbitals on each Re atom which lie in the two $ReCl_4$ planes overlap each other to form δ bond.

131
$$\overset{+2\ -2}{CuS} \xrightarrow{air} \begin{matrix}\overset{+2\ -2}{CuO} \\ + \\ \underset{+2\ -2}{CuS}\end{matrix} \xrightarrow[\text{without air}]{\text{roast}} \underset{0}{Cu} + \underset{+4\ -2}{SO_2}$$

132. The principle of zone refining is based on crystallization of the metal alone from the melt. So if the impurity is of the same size, it might also crystallize.

133. $ZnS + O_2 \rightarrow ZnO + SO_2$

$ZnS + 2O_2 \rightarrow ZnSO_4$

ZnO and $ZnSO_4$ are dissolved in H_2SO_4.

So Zn is in the form of $ZnSO_4$.

From $ZnSO_4$, Zn be extracted by electrolytic process.

134. Cadmium is below zinc in the electrochemical series. Therefore, on adding zinc powder into the solution, zinc will replace cadmium from its salt leaving a precipitate of cadmium.

$$Zn_{(s)} + Cd_{(aq)}^{2+} \rightarrow Zn_{(aq)}^{2+} + Cd_{(s)}$$

135. 30% NaCl is added to $MgCl_2$ inorder to decrease the fusion temperature of anhydrous $MgCl_2$ and to increase the conductivity of $MgCl_2$.

136. FeO present is preferably converted to Fe_2O_3. As FeO is basic, it may react with SiO_2 and be lost as slag. Fe_2O_3 is acidic.

137. $CuSO_4$ is a activator

During separation NaCN is added to depress floatation of ZnS and FeS_2 and remove PbS exclusively. After all PbS is removed, $CuSO_4$ is now added, to activate the floating character of ZnS. After which ZnS is exclusively removed leaving behind FeS_2.

138. At anode $2Cl^- \rightarrow Cl_2 + 2e^-$

At cathode $Mg^{2+} + 2e^- \rightarrow Mg$

If the electrodes are not separated there would be reaction between magnesium and chlorine to give back $MgCl_2$.

139. The ore is subjected to roasting and it gives out SO_2 gas [gas turns $K_2Cr_2O_7$ green]. So it is a sulphide ore. Since metal is given out, it undergoes self reduction.

This happens for copper and lead

Since copper glance is Cu_2S, it should be (b).

140. Aluminium line being below Fe line in the Ellingham diagram can be used to reduce iron oxide.

141. The ore is concentrated by froth floatation and is roasted. So the ore is a sulphide ore. Generally sulphides are not white except zinc compounds. Hence the ore is ZnS. ZnS is reduced using carbon–smelting.

142. Since the ΔG value for formation of oxide of Hg is close to O, HgO can be converted to Hg even on heating. So on roasting Hg is formed.

$$HgS + O_2 \xrightarrow[\text{Roasting}]{\Delta} Hg + SO_2$$

143. $M \rightarrow M^{n+} + ne^-$ E is highly positive

i.e., for formation of oxide ΔG is negative. So the metal compound cannot be reduced easily and requires very drastic methods like electrolysis.

144. $M_{(s)} + O_{2(g)} \rightarrow MO_{2(s)}$

ΔS is –ve, so ΔG is +ve so the ΔG vs T line curves upward. When $M_{(s)} \rightarrow M_{(l)}$ at its melting point upward slope increases.

When $M_{(l)} \rightarrow M_{(g)}$ at its boiling point, the upward slope further increases.

145. At point A, both M and C have equal affinity for oxygen. ΔG is equal and opposite for he two reactions. MO can be reduced by carbon only above T°C.

146. a, c, d cannot be reduced from their aqueous solution. Only the molten salt is electrolyzed.

147. Tin stone SnO_2 is reduced with coke by smelting.

If the slag formed is to be again treated with carbon then it means that slag contains considerable amount of tin.

Generally the slag retains 20 – 40% of tin.

148. If a slag is formed due to phosphorous then a basic flux must be used. So the slag may be used.

$$\underset{\text{impurity}}{P_2O_5} + 3\underset{\text{flux}}{CaO} \longrightarrow \underset{\text{slag}}{Ca_3(PO_4)_2}$$

149. The colour of any compound ion arises by (a) d–d transition (b) polarization of anion (c) charge transfer. In AgI, Ag^+ has d^{10} configuration, so it cannot undergo d–d transition; charge transfer is also not observed. The small Ag^+ polarizes large I^- to give the yellow colour.

150. Ag_2CO_3 is yellow coloured due to charge polarization of CO_3^- ion by small Ag^+ cation.

151. All the three factors contribute to the low reactivity of third row transition elements.

152. Zn^{2+} is less acidic than Cu^{2+} is correct.

153. Mercury is a liquid at room temperature and Therefore, has appreciable vapour pressure. As mercury is poisonous in nature, it surface should be covered with substances like toluene to prevent vapourization.

154. Both CrO_4^{2-} and SO_4^{2-} have tetrahedral structure.

Other example is $SiCl_4$ and $TiCl_4$. As the element is in the highest oxidation state both s and d electrons participate in the bonding. So the property depends

on size and valency. So the main group element with same oxidation state are seen to be isostructural.

155. There is no change in oxidation state. The tetrahedral blue complex changes into pink in the presence of water.

$$\underset{\text{blue}}{\left[Co(H_2O)_4\right]^{2+}} + 2H_2O \rightleftharpoons \underset{\text{pink}}{\left[Co(H_2O)_6\right]^{2+}}$$

156. The most common oxidation state exhibited by copper is +2 which has d^9 configuration while the common oxidation state of zinc has d^{10} configuration.

157. a, c, d are similar for both group 11 and 1 but due to the presence of d electrons, the group 11 elements are involved in metallic bonding. So they have higher melting points and enthalpies of sublimation.

158. Sc^{3+} and Cu^{+} both do not have unpaired electrons.

159. For a given transition metal oxide, as the oxidation state increases, covalent character increases and acidic character increases.

160. VO_2^{+} has vanadium in 5+ state i.e., d° configuration colour is due to charge transfer like in the case of CrO_4^{2-} or MnO_4^{-}.

161. For Hg^{+}, there is one unpaired electron. So it should be paramagnetic. But Hg^{+} is diamagnetic, this is possible only if it dimerises.

162. Generally lower valency state, the compound is ionic as oxidation state increases size decreases so charge/size ration increases so it becomes more polarizing hence covalent.

163. Even though both have $4s^1$ configuration, due to the low shielding effect of d electrons, effective nuclear charge is greater, hence the ionization energy is higher.
i.e., K – 4.34 eV, Cu – 7.726 eV

164. $3Au^{+} \rightarrow 2Au + Au^{3+}$
Cu^{+} is also unstable in aqueous.

165. b, c and d are not exhibited by all d–block elements

166. $[ScF_6]^{3-}$ scandium is in +3 oxidation state and has d° configuration. So it is not coloured.
VO_2^{+} is pale yellow coloured due to charge transfer.

167. Due to the decrease in size of the lanthanides due to poor shielding of f orbitals, the 5d series also small in size. So the increase in size on going down the group i.e., From 4d to 5d is not seen.

168. Colour appears due to f – f transition. The f orbitals lay deep inside the atom as they are the anti penultimate orbital. So the ligands approaching the atom do not affect the f electron.

169. The low enthalpies of europium and Ytterbium are probably due to involvement of 6s electrons only in the formation of bonds in these elements because of stability of nonbonding configurations of $4f^7$ and $4f^{14}$.

170. Both actinides and lanthanides are reactive metals and react with hot water and tarnish in air forming oxide coating.

171. Sphalerite or zinc blende is ZnS.
On roasting ZnO and $ZnSO_4$ are formed .
For both reactions, entropy change is negative.

$$2ZnS_{(s)} + 3O_{2(g)} \rightarrow 2ZnO_{(s)} + 2SO_{2(g)} \quad \Delta S = -ve$$

$$ZnS_{(s)} + 2O_{2(g)} \rightarrow ZnSO_{4(s)} \quad \Delta S = -ve$$

172. Amalgamation process is used for extraction of gold which occurs in native state. The finely crushed auriferous quartz is brought into contact with mercury which takes up the metal forming an amalgam.
From the amalgam, the metal can be easily recovered by distillation of the amalgam. This distillation is possible due to the good difference in melting point between gold and mercuty.

173. In the Ellingham diagram, all the metal metal oxide lines slope upwards while the C – CO line slopes downwards. So at a certain temperature the lines would meet and the metal oxide can be reduced with carbon.
MgO can be reduced with carbon, but this happens at 2000°C. Which is higher than the melting point of magnesium, hence it has to be cooled rapidly.

174. Generally sulphide ors are roasted and carbonates, hydrated ores are calcined. Here malachite is $CuCO_3$. $Cu(OH)_2$ hence calcined while chalcocite is Cu_2S which is roasted .
Calcination and roasting convert the ores into oxides generally.
But during roasting sulphide ore can also be converted into water soluble sulphates. EgCuS converted to $CuSO_4$, ZnS converted to $ZnSO_4$.

175. Zn^{2+} is larger than Cu^{2+}. Towards the end of the series due to electron electron repulsion, there is a slight increase in size.
But generally on moving across the period in the case of transition element, there is decrease in size due to poor shielding effect of d electrons.

176. Zn on reaction with HCl liberates hydrogen gas

$$Zn + 2HCl \rightarrow ZnCl_2 + H_2$$

But because HNO_3 is an oxidizing acid, hydrogen is not liberated and it is reduced

$$3Zn + 8HNO_3 \rightarrow 3Zn(NO_3)_2 + 2NO + 4H_2O$$

$$Zn + \underset{\text{conc and hot}}{4HNO_3} \rightarrow Zn(NO_3)_2 + 2NO_2 + 2H_2O$$

177. Maximum oxidation state = (n – 1)d + ns electron is true only up to d^5 configuration generally. In iron the configuration is $3d^64s^2$. Only the unpaired d electrons take part in bonding. So it shows maximum of +6 state.

Osmium and ruthenium show +8 oxidation state as on going down the group higher oxidation state is stabilized. This is due to increase in size on going down the group.

178. As the oxidation state increases, the charge increases and ionic size decreases. So the charge/size ratio in the cation increases thereby it becomes more polarizing. So as charge increases, covalent character increases. So the oxides becomes more acidic.

179. Group 12 ie zinc group are more reactive than copper group because of there low melting point. Due to completely filled d orbials, there are no covalent bonds, hence they have heat of sublimation leading to higher reactivity.

The group 12 metals are soft due to completely filled d orbitals, hence there is no covalent bonding.

180. The 4f orbital is the antepenultimate shell and so it does not take part in bonding. Since the electronic configuration of the last and penultimate orbitals remains the same for all the lanthanoid elements, they show similar chemical properties.

181. Carnallite is $MgCl_2.KCl.6H_2O$.

182. On heating, the magnesium may react with the water of crystallization to give $Mg(OH)_2$ and finally MgO. So avoid this it should be heated in an atmosphere of HCl.

183. Generally alkali and alkaline earth metals are reduced by electrolysis of the fused metal with addition of some impurity.

184. At point A, there is a change in the slope towards upwards ie., entropy change has become more negative. This happens when metal melts and turns into a liquid.

185. The metal–metal oxide line which the below can be used to reduce the metal oxide whose line is above Ellinghams diagrams are drawn for reactions involving oxidation with one mole of oxygen.

186. Formation of (i) CO_2 is favourable below 710°C
(ii) CO is favourable above 710°C

187. Since no compound of zinc is coloured, any reaction occurring there is no colouring of painted surface or blackening of painted surface as seen in lead.

188. Colour of hot ZnO is yellow and it is white when cold. This colouration is due to defects in the solid structure.

189. Both their hydrates are isomorphous . They form divalent oxides but ZnO is amphoteric. Due to small size and higher nuclear charge of zinc, its compounds are more covalent and has greater tendency to form complex

190. Current passed for electrolysis also helps to heat the electrolyte.

Aluminium being heavier than electrolyte, sinks to the bottom and is tapped from a hole at the bottom.

Addition of cryolite increase conductivity and decreases melting point.

Even though alumimium has lower mpt, it does not vapourise as mpt of alumina is reduced.

191. The slag formed is insoluble in metal and is less dense so it floats over the metal. Since it forms a layer on the metal it protects the metal from atmosphere. The gangue is infusible, it is made into fusible slag.

192. Any element may be displaced from solution by another element which is higher in the electrochemical series. So only (a) and (b) are possible.

$Zn + Cd^{2+} \rightarrow Zn^{2+} + Cd$

$Cl_2 + 2Br^- \rightarrow Br_2 + 2Cl^-$

193. During extraction of Sn and Fe $CaSiO_3$ is formed.

During extraction of Cu, $FeSiO_3$ slag is formed and for zinc no slag is formed.

194. K^+ and Ag^+ has almost same ionic size133 pm and 126pm respectively.

K_2CO_3 is thermally stable while Ag_2CO_3 decomposes to the oxide.

Both nitrates decompose to the nitrite at lower temp.

$AgNO_3$ decomposes to the oxide at red heat .

AgCl is water insoluble while KCl is water soluble.

195. On going down the group higher oxidation state is more stable hence CrO_3 is oxidizing while WO_3 is not. CrO_3 melts at 197°C while WO_3 melts at 1473°C. Both the oxides are acidic and dissolve in NaOH.

196. Presence of unpaired electrons in the d orbital leads to higher interatomic forces leading to higher melting point.

On moving from Cu to Zn, due to d^{10} configuration of Zn it has very low melting point.

Manganese inspite of the d^5 configuration shows a low melting point probably because of its complex structure.

197. Chromium forms a protective oxide layer, so it does not liberate hydrogen from acid.

Copper has a positive reduction potential so it does not liberate H_2 from acid.

198. (a) → (p)
(b) → (p), (q)
(c) → (p), (q), (r)
(d) → (p), (q), (r)

From the Ellingham diagram–the metal that is below can reduce the one that is above when the metal–metal oxide line crosses the $\Delta G = 0$ line at that temperature the metal can be reduced thermally.

199 (a) → (p), (q), (r)
(b) → (q), (r)
(c) → (p), (s)
(d) → (q), (r) (s)

Pm^{3+} and Lu^{3+} are lanthanides

Ac^{3+} and Th^{4+} are actinides

All actinides and promethium among lanthanides are radioactive.

Actinides form complexes easily and show variable oxidation state. Lanthanides do not form stable complexes easily.

Ac^{3+}, Th^{4+} have f° configuration

Lu^{3+} has f^{14} configuration

So all the three do not have ff transition so are not coloured.

200. (a) → (p), (q), (r)
(b) → (q), (r), (s)
(c) → (p), (q), (r)
(d) → (p), (q) (s)

As oxidation state increases, metal oxide changes from basic to amphoteric to acidic.

Zn compounds are generally white. The other oxides are coloured due to d–d transition. Basic and amphoteric oxides generally dissolve in non oxidizing acids like HCl to form hexa aquo ions of the type $[M(H_2O)_6]^{n+}$.

Certain oxides are prepared by decomposition

$$\underset{\text{calamine}}{ZnCO_3} \xrightarrow{\Delta} ZnO + CO_2$$

$$(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + 4H_2O$$

$$Fe_2(SO_4)_3 \xrightarrow{\Delta} Fe_2O_3 + 3SO_3\uparrow$$

$$2FeSO_4 \xrightarrow{\Delta} Fe_2O_3 + SO_2 + SO_3\uparrow$$

CHAPTER

4

COORDINATION COMPOUNDS, PREPARATION AND PROPERTIES OF SOME METAL COMPOUNDS

CHAPTER OUTLINE

Preview

STUDY MATERIAL

Coordination Compounds

Werner's Theory
- Concept Strands (1-3)

Some Common Terms used in Co-Ordination Compounds

Writing Formula of Mononuclear Coordination Entities

Nomenclature of Coordination Compounds
- Concept Strands (4-5)

Isomerism in Coordination Compounds
- Concept Strands (6-8)

Bonding in Coordination Compounds
- Concept Strands (9-10)

Limitation of Valence Bond Theory

Crystal Field Theory
- Concept Strand (11)

Crystal Field Splitting in Tetrahedral Coordination Entities

Limitations of Crystal Field Theory

Stability of Coordination Compounds

Applications of Complexes

Biological Importance of Complexes

Preparation and Properties of Some Metal Compounds
- Concept Strand (12-20)

TOPIC GRIP
- Subjective Questions (10)
- Straight Objective Type Questions (5)
- Assertion–Reason Type Questions (5)
- Linked Comprehension Type Questions (6)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

IIT ASSIGNMENT EXERCISE
- Straight Objective Type Questions (80)
- Assertion–Reason Type Questions (3)
- Linked Comprehension Type Questions (3)
- Multiple Correct Objective Type Questions (3)
- Matrix-Match Type Question (1)

ADDITIONAL PRACTICE EXERCISE
- Subjective Questions (10)
- Straight Objective Type Questions (40)
- Assertion–Reason Type Questions (10)
- Linked Comprehension Type Questions (9)
- Multiple Correct Objective Type Questions (8)
- Matrix-Match Type Questions (3)

COORDINATION COMPOUNDS

When two or more simple stable compounds are combined in stoichiometric ratio the products obtained are called molecular compounds or addition compounds. Addition compounds are of two different types, viz., double salts and complexes.

(i) Double salts

Double salts are those addition compounds which retain their identity in solid state but undergo complete ionization in polar solvents.

Example Carnallite $KCl.MgCl_2.6H_2O$
Mohr's salt $FeSO_4(NH_4)_2SO_4.6H_2O$
Potash alum $KAl(SO_4)_2.12H_2O$

When one mole each of KCl and $MgCl_2$ are dissolved in water and evaporated, crystals of $KCl.MgCl_2.6H_2O$ are obtained. But when these crystals are dissolved in water they again produce K^+, Mg^{2+} and Cl^- ions i.e., the double salts exist only in solid state. The properties of double salts are the same as those of the individual compounds from which they are formed.

(ii) Complexes or coordination compounds

Complexes are addition compounds which do not undergo complete ionization in aqueous or polar solvents. They retain their identity both in solid state and in solution.

Example Potassium ferricyanide $K_3[Fe(CN)_6]$
Schwitzers reagent $[Cu(NH_3)_4](OH)_2$

Complexes consists of a definite number of ionic or nonionic species bonded to the central atom through coordinate bonds. Hence they are also known as co-ordination compounds.

When excess KCN is added to an aqueous $Fe(CN)_3$ solution and evaporated, crystals of addition compound $K_3[Fe(CN)_6]$ are formed. When these crystals are dissolved in water, K^+ ions and $[Fe(CN)_6]^{3-}$ ions are formed. The ion $[Fe(CN)_6]^{3-}$ retains its identity both in solid state and in solution. The properties of complexes are different from those of the compounds from which they are formed.

WERNER'S THEORY

The bonding in coordination compounds was first explained by Werner in his theory in 1893. Inspite of the absence of modern instrumental techniques like X-ray diffraction, Werner was able to explain the structure and bonding in transition metal complexes. The important postulates of Werner's theory are

(i) Central metal atom in the complex shows two types of valencies namely primary valency and a secondary valency.
(ii) Primary valencies are ionisable and are satisfied by negative ions.
(iii) Secondary valencies are non-ionisable and are satisfied by neutral molecules or negative ions or even positive ions.
(iv) The ions or groups bonded by secondary valencies are arranged around the central atom in a definite geometric pattern characteristic of the number of groups.

Primary valency

These are non-directional. Generally the central metal atom exists as a positive ion. *The primary valency refers to the number of positive charges on the central metal atom.* Primary valency applies to simple salts also. Consider $FeCl_2$ (Fe^{2+} and $2Cl^-$). The primary valency is 2 (two ionic bonds). If we consider the complex $[Co(NH_3)_6]Br_3$, there is a positive ion $[Co(NH_3)_6]^{3+}$ and $3Br^-$. The primary valency is 3. In $[CoBr(NH_3)_5]Br_2$ also, primary valency is 3.

In modern terms, primary valency is the oxidation number of the central atom.

Secondary valency

These are directional. *The number of secondary valencies equals the number of donor atoms of ligands coordinated to the metal atom. This is presently denoted as coordination number.*

Geometry of complex

Ligands may be negative ions or neutral molecules or positive ions. The ligands are arranged symmetrically around the central atom to form a polyhedron.

In $[Co(NH_3)_6]Br_3$, three Br^- ions satisfy the primary valencies and six NH_3 groups satisfy the secondary valencies. For this complex, the shape is octahedral since six ligands surround the metal ion.

In addition to six-coordinate complexes, there are complexes with coordination numbers 2, 3, 4, 5, 7, 8 etc. Complexes of the type $[Ag(NH_3)_2]^+$, $[Au(CN)_2]^-$ and $[Hg(CN)_2]$, which have d^{10} configuration belong to coordination number 2 are linear. The complex $(Ni(CO)_4)$ is tetrahedral and $[PtCl_4]^{2-}$ square planar. The most common coordination numbers are 4 and 6.

Werner also postulated that for co-ordination compounds of transition metals octahedral, square planar and tetrahedral geometries are more common.

CONCEPT STRANDS

Concept Strand 1

Find the primary valencies of the metal in the following complexes. Also indicate the number of moles of AgCl precipitated, when $AgNO_3$ solution is added to one mole of each of them.

(i) $[Co(NH_3)_6]Cl_3$
(ii) $[CoCl(NH_3)_5]Cl_2$
(iii) $[CoCl_2(NH_3)_4]Cl$

Solution

(i) Primary valency is 3 and 3 moles of AgCl get precipitated.
(ii) Primary valency is 3 and 2 moles of AgCl get precipitated.
(iii) Primary valency is 3 and 1 mole of AgCl gets precipitated.

Concept Strand 2

Cis-platin is an anti-cancer drug with the name cis-diamminedichloroplatinum(II). Write the formula of the complex. If there is another geometrical isomer of this complex, predict its geometry.

Solution

The complex is $[PtCl_2(NH_3)_2]$. Since the coordination number is 4, shape can be tetrahedral or square planar. Only square planar complex exists in cis and trans forms. Hence the complex is square planar.

Concept Strand 3

The freezing point of a 2 molal aqueous solution of an electrolyte is –11.16°C. Which of the following complexes at the same molal concentration will have the same freezing point?

(K_f of water is 1.86 K kg mol^{-1})

(i) $[Co(NH_3)_6]Br_3$
(ii) $[CoBr(NH_3)_5]Br_2$
(iii) $[CoBr_2(NH_3)_4]Br$

Solution

$$\Delta T = iK_f m$$

$$i = \frac{\Delta T}{K_f m} = \frac{11.16}{1.86 \times 2} = 3$$

The complex (ii) has three particles and this corresponds to a depression of freezing point of 11.16° and the new freezing point will be –11.16°C

$$\Delta T_f = 3 \times 1.86 \times 2 = 11.16$$

The electrolyte and $[CoBr(NH_3)_5]Br_2$ will have the same depression of freezing point.

SOME COMMON TERMS USED

(i) Coordination entity

It consists of a central metal atom or metal ion bonded by a fixed number of ions or molecules. The coordination entity can be anion, cation or neutral. Example, $[Fe(CN)_6]^{3-}$, $[Co(NH_3)_6]^{3+}$, $[Ni(CO)_4]$.

(ii) Central atom or ion

It is the central atom or ion in coordination entity to which a definite number of atoms or ions are coordinated and are arranged around in definite geometrical pattern. The central atom or ion in a coordination entity always acts as Lewis acid. It may be a metal or a non metal.

Example, Fe^{2+} in $[Fe(CN)_6]^{4-}$,B^{3+} in $[BF_4]^-$

(iii) Ligands

These are ions or molecules in a coordination entity that are coordinated to the central atom or ion. They may be anions, cations or neutral simple molecules like NH_3, H_2O or bigger molecules such as ethylene diamine or macro molecules such as proteins. All ligands are Lewis bases.

(iv) Denticity of ligands

It is the number of donor atoms present in a ligand. Monodentate ligands possess only one donor atom per ligand.

Example, Cl^-, H_2O, etc.

Didentate ligands have two donor atoms per ligand

Example, $H_2NCH_2CH_2NH_2$ i.e., ethylene diamine, oxalate $C_2O_4^{2-}$

Polydentate ligands have several donor atoms per ligand

Example, EDTA have denticity 6.

Ambidientate ligands are monodentate ligands which can coordinate through two different donor atoms in different complexes.

Example, NO_2^-, SCN^-

Flexidentate ligands are those which can exhibit more than one denticity. Some polydentate ligands are also flexidentate.

(v) Coordination number

It is the number of ligand donor atoms coordinated to a central atom in a coordination entity. Coordination number is the secondary valency in Werner's theory.

(vi) Coordination sphere

It is the central metal surrounded by ligands in definite geometrical pattern; represented within square brackets. The charge on the coordination sphere may be positive, negative or even zero. The ions associated with the coordination sphere are called counter ions. i.e., the coordination sphere along with counter ion form the coordination compound or complex.

(vii) Coordination polyhedron

It is the spatial arrangement in which the ligands are arranged around the central metal atom or ion. It can be linear, square planar, tetrahedral, octahedral etc. It depends on the co-ordination number.

(viii) Oxidation number of the central metal

It is the charge on the central metal atom when all the ligands along with their co-ordinated electron pairs were removed from the co-ordination entity.

(ix) Homoleptic and Heteroleptic complexes

Complexes containing only one type of ligands are said to be homoleptic and complexes in which metal is bound to more than one type of ligands are heteroleptic.

WRITING FORMULA OF MONONUCLEAR COORDINATION ENTITIES

The following rules are used for writing the formula of coordination entities.

(i) The central atom is listed first.
(ii) The ligands are arranged after the central metal in alphabetical order. The charge on the ligands are not considered for priority.

(iii) In the case of abbreviation of ligands, such as in the case of polydentate ligands the first letter of abbreviation is considered for priority.
(iv) The formula of the coordination entity is enclosed within square brackets when the ligands are polyatomic or abbreviations are used for ligands. Their formulae are enclosed in parenthesis. Example, $[Co(CN)_6]^{3-}$, $[Cr(en)_3]^{3+}$
(v) Within the co-ordination sphere there should be no space between ligands and metal.
(vi) The charge on the co-ordination entity is written as a right superscript with the number before the sign, when the co-ordination entity without counter ion is represented.
(vii) The charge on cations should be balanced by charge on anions.

NOMENCLATURE OF COORDINATION COMPOUNDS

The transition metals form complex compounds of different kinds. The (ligands) attached to the central metal may be neutral, negative ions, positive ions, organic groups and so on. Due to the occurrence of a variety of coordination compounds, we can follow a systematic nomenclature for these compounds.

The IUPAC nomenclature is followed and the rules of naming coordination compounds are given below.

(i) If a complex has positive and negative ions, the positive ion is named first followed by the negative ion.
(ii) The ligands are named in alphabetical order regardless of their charge followed by the metal.
(iii) Names of anionic ligands end is –O
Example, Cl^- chlorido, Br^- – bromido, OH^- – hydroxo
names of neutral and cationic ligands are the same, except aqua for H_2O, ammine for NH_3, carbonyl for CO and nitrosyl for NO.
(iv) If there are several ligands of the same kind, we use prefixes like di, tri, tetra, penta and hexa to show the number of such ligands. If the ligand includes a number as in the case of dipyridyl or ethylenediamine, we use the prefixes bis, tris, tetrakis instead of di, tri, tetra and name of ligand is placed in brackets.
(v) Roman numerals in parentheses are used to show the oxidation state of central metal as indicated below.
Titanium (III)
(vi) Complex positive ions and neutral molecules have no specific ending. But negative complex ions end in 'ate'; $[ZnCl_4]^{2-}$ ends in zincate.
(vii) *Sometimes a ligand may be attached through different atoms. They are ambidentate ligands.* For example, in the complex M–NO_2, nitrogen is attached to the metal. But in M–ONO, oxygen is attached to the metal. The first one named is Nitrito – N and the second one Nitrito–O.
(viii) Water or solvent molecules may be present as solvent of crystallization. In such cases, these follow the name and are preceeded by the number of groups of the solvent molecule. For example, $AlK(SO_4)_2.12H_2O$ is called aluminium potassium sulphate dodecahydrate.
(ix) Name of a complex ion or molecule is written as one word.

CONCEPT STRANDS

Concept Strand 4

Give the IUPAC names of the following complexes.

(i) $[Co(NH_3)_6]Cl_3$
(ii) $[Zn(NCS)_4]^{2-}$
(iii) $[Cd(SCN)_4]^{2-}$
(iv) $Li[AlH_4]$
(v) $Na_2[ZnCl_4]$
(vi) $Na_3[Ag(S_2O_3)_2]$
(vii) $[Cr(en)_3]Cl_3$

Solution

(i) This is a complex involving the chloride anion and is named as hexaamminecobalt(III) chloride.
(ii) Tetrathiocyanato–N–zincate(II) ion
(iii) Tetrathiocyanato–S-cadmiate(II)
(iv) Lithium tetrahydridoaluminate(III) ion
(v) Sodium tetrachloridozincate(II)
(vi) Sodium di(thiosulphato)argentate(I)
(vii) Tris(ethylenediamine)chromium(III) chloride.

Concept Strand 5

Find the formulae of the following complexes:
(i) Tetraamminedichloridoplatinum(IV) tetrachloridoplatinate(II)
(ii) Dicyanoaurate(I) ion
(iii) Potassium tetracyanonickelate(II)

Solution

(i) $[Pt(NH_3)_4Cl_2][PtCl_4]$
(ii) $[Au(CN)_2]^-$
(iii) $K_2[Ni(CN)_4]$

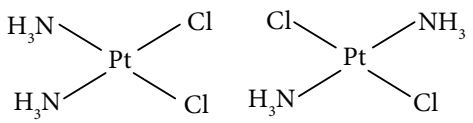
ISOMERISM IN COORDINATION COMPOUNDS

Stereo isomerism

Stereo isomerism is of two different types viz., geometrical and optical

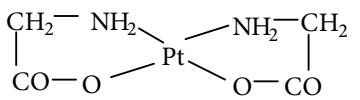
(i) Geometrical isomerism

In complexes like $[Pt(NH_3)_2Cl_2]$, similar ligands (NH_3 and Cl^-) can be adjacent or opposite to each other. This leads to cis-trans isomerism. Cis-trans isomers are also examples for diastereomers.

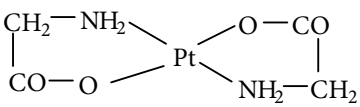


cis trans

The trans complex has zero dipole moment, but the cis compound has a non-zero dipole moment. The amino acid, glycine, complexes with Pt^{2+} to give cis-trans isomers as given below:

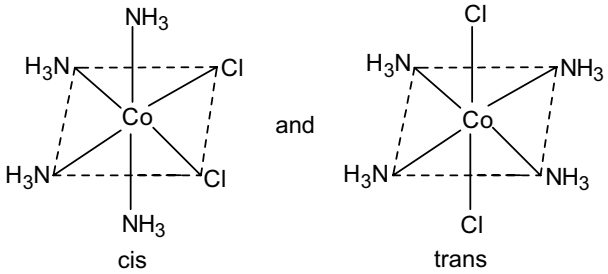


cis



trans

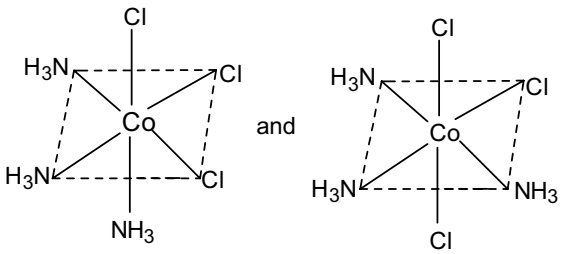
Octahedral complexes of the type Ma_4b_2 exhibit cis-trans isomerism Example, $[Co(NH_3)_4Cl_2]^+$



cis trans

Octahedral complexes of the type Ma_3b_3 exhibit facial-meridional isomerism

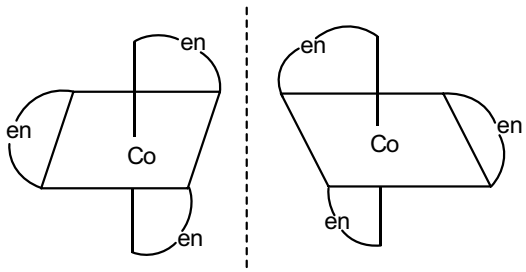
Example, $[Co(NH_3)_3Cl_3]$



facial (fac) meridional (mer)

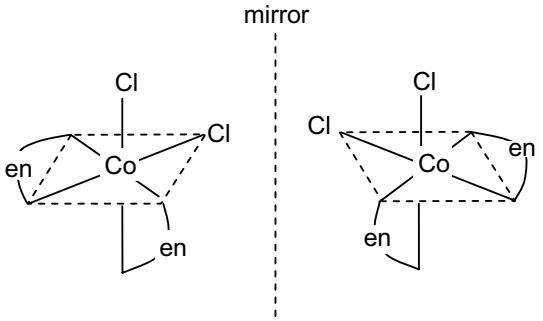
(ii) Optical isomerism

Optical isomerism is exhibited by octahedral complexes of the type $M(aa)_3$ where (aa) is a bidentate ligand.

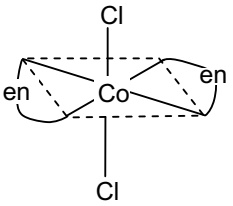


d & ℓ forms of the complex

Octahedral complexes having chelating ligands like ethylenediamine also exhibit optical activity. Example $[Co(en)_2Cl_2]^+$, $[Co(en)_3]^{3+}$



d and l forms of cis-dichlorobis (ethylenediamine) cobalt (III) ion



Trans-dichlorobis (ethylenediamine) cobalt (III) ion is optically inactive (meso form)

For example, $[Co(en)_2Cl_2]^+$ has the cis form giving rise to d and ℓ isomers. The trans isomer does not show optical activity. The cis and trans compounds are shown above.

CONCEPT STRANDS

Concept Strand 6

Which of the following compounds exhibit cis-trans isomerism?

(i) $Ni(CO)_4$
(ii) $[CrCl_2(NH_3)_4]^+$
(iii) $[CoCl_2(NH_3)_4]^+$
(iv) $[Co(NO_2)_3(NH_3)_3]$
(v) $[PtBrCl(NH_3)_2]$

Solution

(ii), (iii), (iv) and (v) exhibit geometrical isomerism. In (iv) which has the general formula Ma_3b_3, all three similar groups may be adjacent or two out of three may be trans in the octahedral geometry (facial – meridional).

$[Ni(CO)_4]$ is tetrahedral

Concept Strand 7

Which of the following coordination compounds exhibits optical activity? Which of them are dissymmetric?

(i) $[Co(en)_3]^{3+}$
(ii) $[CrCl_2(H_2O)_4]^+$
(iii) $[CrCl_2(en)(NH_3)_2]^+$

Solution

(i) and (iii) are optically active and dissymmetric. A compound is optically active when it is disymmetric or chiral.

Structural isomerism

(i) Hydrate isomerism

Hydrate isomerism arises when the number of molecules of water inside the complex positive ion is different.

$[Cr(H_2O)_6]Cl_3$ – violet

$[CrCl(H_2O)_5]Cl_2.H_2O$ – green

$[CrCl_2(H_2O)_4]Cl.2H_2O$ – dark green

(ii) Ionization isomerism

Ionization isomerism belong to the category of constitutional isomerism. This type of isomerism is due to exchange of anionic ligands between inside and outside of the coordination sphere. $[Co(NH_3)_5Br]SO_4$ and $[Co(NH_3)_5SO_4]Br$ are examples of ionization isomers. In the first case, SO_4^{2-} is outside the coordinated positive ion. In the second case, Br^- is outside the coordination sphere. The first complex gives precipitate with $BaCl_2$ but the second does not. The

second compound gives a precipitate with $AgNO_3$ but the first does not. The first complex is red-violet and the second complex is red.

(iii) Linkage isomerism

Linkage isomerism belong to the category of constitutional isomerism. They arise when a ligand can bond to a metal through either of the two different donor atoms. For example, the nitrite ion NO_2^- forms two pentamminecobalt(III) complexes. The yellow nitrito–N complex contains the Co–N bond and has the formula $[Co(NH_3)_5NO_2]^{2+}$. The second complex is red and it is a nitrito–O complex. Such ligands are called *ambidentate ligands*.

(iv) Coordination isomerism

If the complex is ionic and both positive ion and negative ion are complex ions, isomerism can arise by the interchange of ligands one by one between the anion and cation. Consider $[Co(NH_3)_6][Cr(CN)_6]$ and $[Cr(NH_3)_6][Co(CN)_6]$. The NH_3 and CN^- ligands interchange between cation and anion one by one producing a series of coordination isomers.

CONCEPT STRAND

Concept Strand 8

If one molal solution of $[Cr(H_2O)_6]Cl_3$, $[CrCl(H_2O)_5]Cl_2.H_2O$ and $[CrCl_2(H_2O)_4]Cl.2H_2O$ are prepared in water, assuming complete dissociation, calculate the new freezing points of water in the three cases.

Solution

$[Cr(H_2O)_6]Cl_3$ has 4 particles or ions.

Hence $\Delta T_f = 4 \times 1.86 \times 1 = 7.44^oC$

New freezing point is -7.44^oC

For $[Cr(H_2O)_5Cl]Cl_2.H_2O$, the total number of ions is 3. Hence the new freezing point will be -5.58^oC. In the third complex, there are two ions and the new freezing point will be -3.72^oC.

BONDING IN COORDINATION COMPOUNDS

Three theories of bonding in transition metal complexes have been developed after Werner's theory. These are

(i) Valence bond theory (VBT)
(ii) Crystal field theory (CFT)
(iii) Molecular orbital theory (MOT)

Molecular orbital theory allows covalent and ionic contributions. The quantitative calculation involved are lengthy and subjects of symmetry and group theory have to be used to discuss this theory. Hence the discussion in this study material is restricted to V.B. theory and crystal field theory.

Valence bond theory

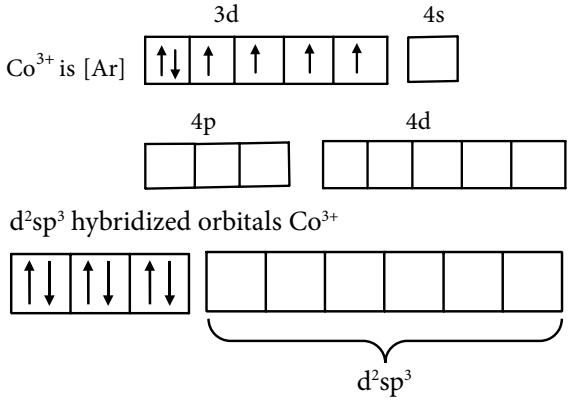
Linus Pauling developed the V.B. theory. According to this theory,

(i) Coordination compounds contain complex ions, in which ligands form coordinate bonds with the metal.
(ii) The ligands must have a lone pair of electrons and the metal must have empty orbitals of suitable energy available for bonding.
(iii) When the required number of empty orbitals are not available on the central atom, they are made available by pairing the electrons in the (n – 1)d orbitals or by promoting the (n – 1)d electrons to any of the outer orbitals or by using outer nd orbitals themselves.
(iv) The empty orbitals used for bonding are hybridized and hybridization increases the strength of the bonds and gives the complex its shape.

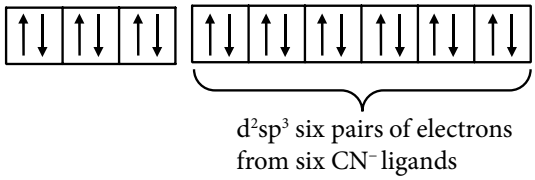
The theory explains which orbital will be used in bonding and then predicts the shape, stability and magnetic properties of the complex.

Let us now consider, how V.B. theory is able to describe the bonding in transition metal complexes. It is possible to predict the geometry of the complex from the knowledge of its magnetic moment.

Let us now consider the complex diamagnetic $[Co(CN)_6]^{3-}$ formed between Co^{3+} and CN^- Co^{3+} is



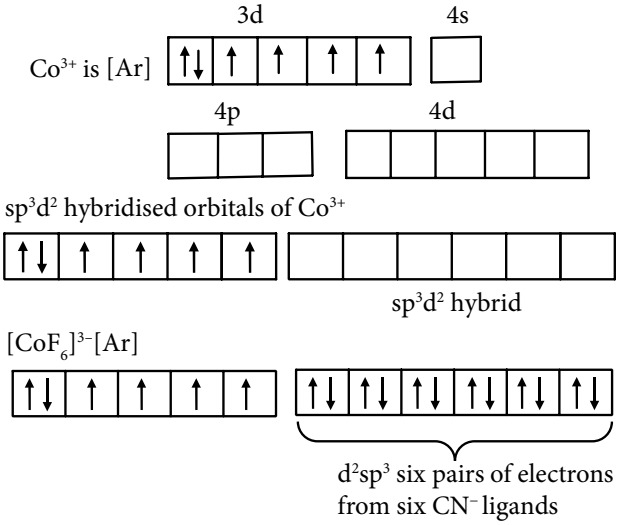
The complex has the orbital diagram,
$[Co(CN)_6]^{3-}$ low spin complex.



The hybridization is d^2sp^3 and the complex is octahedral and diamagnetic with no unpaired electrons.

In $[Ni(CN)_4]^{2-}$, Ni^{2+} is dsp^2 hybridized and the complex is square planar.

Let us consider a complex formed between cobalt and fluorine. The formula of the complex is $[CoF_6]^{3-}$ and it is paramagnetic. Co has the electronic configuration $[Ar]3d^74s^2$. Co^{3+} has the electronic configuration [Ar] $3d^6$. The electrons are arranged as follows.



The paramagnetic octahedral complex $[CoF_6]^{3-}$ uses outer 4d orbitals for sp^3d^2 hybridization. It is therefore called outer orbital or high spin or spin free complex.

As a result there are four unpaired electrons. The spin-only magnetic moment will be close to 4.9 BM. It will be octahedral. It has been found to be so.

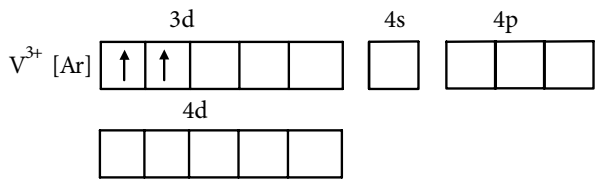
CONCEPT STRANDS

Concept Strand 9

Write the electronic configuration of vanadium and give the V.B. description for the complex $[V(NH_3)_6]^{3+}$. What will be the spin only magnetic moment of the complex?

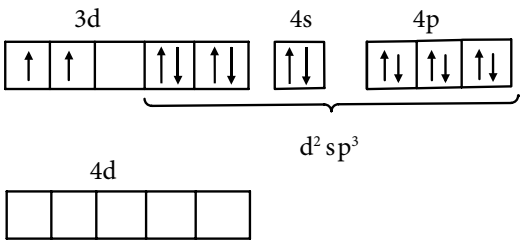
Solution

V has the configuration $[Ar]3d^34s^2$. V^{3+} has the configuration $[Ar]3d^2$. The orbital diagram is



Since many 3d orbitals are vacant and d^2sp^3 hybrids are lower in energy than sp^3d^2 hybrids, the hybridization used is d^2sp^3. The complex has the orbital diagram,

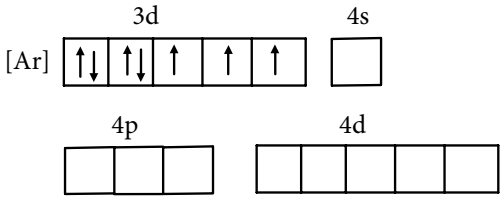
$[V(NH_3)_6]^{3+}$ [Ar]



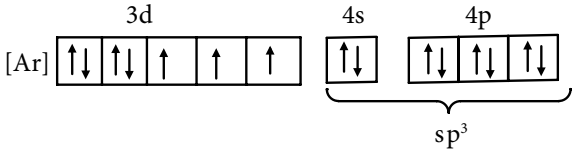
There are two unpaired electrons and spin-only magnetic moment is 2.9 BM.

Let us now consider 4-coordinate complexes. The 4-coordinate complex can be tetrahedral or square planar. Let us consider a tetrahedral complex $[CoCl_4]^{2-}$.

Co^{2+} has the orbital diagram,



For the complex $[CoCl_4]^{2-}$, the diagram is



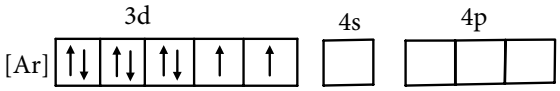
There are three unpaired electrons and spin-only magnetic moment is 3.9 BM.

Concept Strand 10

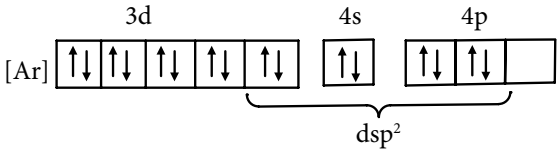
Use the dsp^2 hybridization for a square planar complex of nickel with CN^- and discuss the orbital diagram of the complex and obtain its spin only magnetic moment.

Solution

Ni has the electronic configuration $[Ar]3d^84s^2$ and Ni^{2+} has the electronic configuration $[Ar]3d^8$. The orbital diagram of Ni^{2+} is



In the complex $[Ni(CN)_4]^{2-}$, the diagram will be



There are no unpaired electrons. Spin-only magnetic moment is zero. It is diamagnetic.

Depending on whether a ligand is weak field ligand or strong field ligand, the complex is divided into two categories, namely, high spin complex and low spin complex respectively. Alternatively, these are called outer orbital and inner orbital complexes respectively.

LIMITATION OF VALENCE BOND THEORY

The demerits of valence bond theory are the following

(i) It involves a number of assumptions.
(ii) It does not explain the colour of co-ordination compounds.
(iii) The magnetic properties of co-ordination compounds are not accounted for quantitatively.
(iv) Kinetic and thermodynamic stabilities of complexes are not considered.
(v) It does not distinguish between weak field and strong field ligands.
(vi) In complexes with CN–4, V.B theory cannot predict whether it would be tetrahedral or square planar.

CRYSTAL FIELD THEORY

Valence bond theory considers the overlap of orbitals of central metal and ligands in forming the complex. In crystal field theory the metal-ligand bond is considered to be purely ionic in nature. These ionic bonds are considered to be arising due to electrostatic interaction between metal ion and ligand.

The anionic ligands are considered to be point charges and neutral ligands are considered to be point dipoles. All

the five d orbitals on the isolated gaseous metal atom (ion) are considered to be degenerate. They remain degenerate in a spherically symmetrical field of negative charges. But in the field of negative charge produced by ligands the degeneracy of the d orbitals is lifted and they split into groups of orbitals. The splitting pattern depends on the nature of the crystal field.

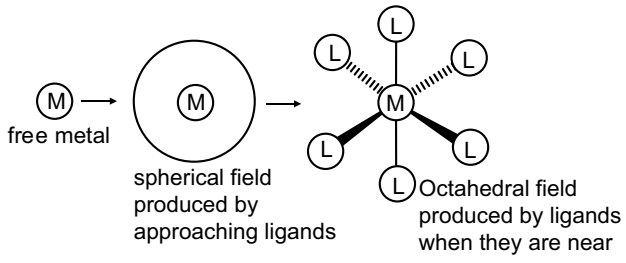
The five d orbitals are of two different types. The d_{z^2} and $d_{x^2-y^2}$ have their lobes directed along the axes and are known as e_g orbitals and the three orbitals d_{xy}, d_{xz} and d_{yz} have their lobes directed between the axes and are known as t_{2g} orbitals. In the absence of external field or in a spherically symmetrical negative field both e_g and t_{2g} orbitals have the same energy.

Crystal field splitting in octahedral field

Consider the case of six ligands forming a complex. The ligands can be considered to be positioned symmetrically along the axes with the metal present at the origin. The ligands may be considered to be producing a spherical field when they are at a distance which repel all the d-orbitals to the same extent. This result in the increase in an overall energy of the d orbitals. As the ligands moves closer, the spherical field gets converted to octahedral field which splits the d-orbitals into two groups. As the ligands are approaching along the axes they repel the e_g orbitals which are directed along the axes more strongly than the t_{2g} orbitals which are directed in between the axes. The d orbitals then get split into two sets with e_g orbitals at a higher energy and t_{2g} orbitals at a lower energy than the unsplit d-orbitals . This splitting does not change the bary centre or centre of gravity of the orbitals. To maintain constant bary centre it is necessary for the two e_g orbitals to be repelled more than the t_{2g} orbitals.

The extent to which e_g and t_{2g} orbitals are separated in an octahedral complex is denoted by Δ_0 the crystal field splitting energy.

Hence it is observed that the energy of e_g orbitals increase by 0.6 Δ_0 and that of t_{2g} orbitals decrease by $0.4\Delta_0$. This is represented as



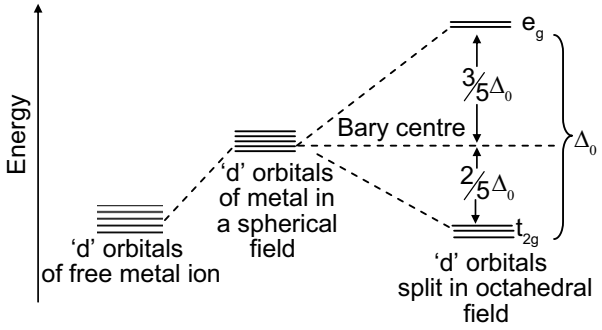
The energy level diagram of d-orbitals in gaseous ion and the splitting in octahedral field are shown in next page.

Spectrochemical series

Ligands can be arranged in a series in the order of ligand field strength. This arrangement is known as spectrochemical series. A spectrochemical series of common ligands is given below.

$$I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS < edta < NH_3 < en < CN^- < CO$$

It is an experimentally determined series based on absorption of light by complexes with different ligands.



The crystal field splitting energy Δ_0 depends on the field produced by the ligands and charge on the metal ion. Some ligands are able to produce strong fields where the splitting will be large and some ligands produce only weak fields and splitting will be small.

For a metal in an octahedral field if 1, 2 or 3 electrons are there in the d orbitals they will be present in the t_{2g} orbitals. But when there are 4 electrons, the fourth one may occupy the t_{2g} orbitals if the energy for pairing with an already existing electron in t_{2g} level is less than Δ_0 and will occupy e_g level if pairing energy is more compared to Δ_0. In other words Δ_0 is more in a strong field and there will be pairing. Thus the d^4 configuration is different in weak field and strong field. Similar is the case for d^5, d^6 and d^7 configuration.

Calculation of crystal field stabilization energy

Allot +0.6 Δ_0 for every electron in t_{2g} orbitals and –0.4 Δ_0 for every electron in t_{2g} orbitals and get the sum. The value obtained is called crystal field stabilization energy (CFSE).

i.e., CFSE = $\Sigma 0.6\ \Delta_0\ nt_{2g} + \Sigma -0.4\ \Delta_0\ ne_g$ where, nt_{2g} and ne_g are electrons present in t_{2g} and e_g orbitals.

Table gives the electronic arrangements in the complex ion in the weak field and strong field environments. It can be seen that the high spin and low spin distinction exists only for d^4, d^5, d^6 and d^7 systems.

d-electron configuration	Weak field case		Spin only magnetic moment (BM)	Strong field case		Spin only magnetic moment (BM)
	t_{2g}	e_g		t_{2g}	e_g	
d^1	↑ _ _	– –	1.73	↑ _ _	– –	1.73
d^2	↑ ↑ _	– –	2.83	↑ ↑ _	– –	2.83
d^3	↑ ↑ ↑	– –	3.87	↑ ↑ ↑	– –	3.87
d^4	↑ ↑ ↑	↑ _	4.9	↑↓ ↑ ↑	– –	2.83
d^5	↑ ↑ ↑	↑ ↑	5.92	↑↓ ↑↓ ↑	– –	1.73
d^6	↑↓ ↑ ↑	↑ ↑	4.9	↑↓ ↑↓ ↑↓	– –	0.0
d^7	↑↓ ↑↓ ↑	↑ ↑	3.87	↑↓ ↑↓ ↑↓	↑ _	1.73
d^8	↑↓ ↑↓ ↑↓	↑ ↑	2.83	↑↓ ↑↓ ↑↓	↑ ↑	2.83
d^9	↑↓ ↑↓ ↑↓	↑↓ ↑	1.73	↑↓ ↑↓ ↑↓	↑↓ ↑	1.73
d^{10}	↑↓ ↑↓ ↑↓	↑↓ ↑↓	0.0	↑↓ ↑↓ ↑↓	↑↓ ↑↓	0.0

CONCEPT STRAND

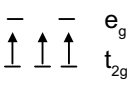
Concept Strand 11

Draw the crystal field orbital energy diagrams and predict the spin only magnetic moment for the following complexes.

(i) $[Cr(en)_3]^{3+}$
(ii) $[Mn(CN)_6]^{3-}$
(iii) $[Co(H_2O)_6]^{2+}$

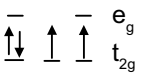
Solution

(i) Cr^{3+} has configuration $[Ar]3d^3$ and when complexed with ethylenediamine in octahedral fashion, the orbital diagram is



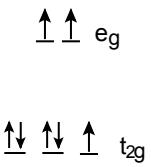
There are three unpaired electrons and spin only magnetic moment is 3.87 BM.

(ii) The manganese complex has Mn in +3 oxidation sate. The configuration is $3d^4$. This will be a low spin complex because CN^- is a strong field ligand and hence the orbital diagram is



There are two unpaired electrons. It is a low spin complex with spin only magnetic moment is 2.83 BM.

(iii) The cobalt complex has the d^7 configuration, it will be a high spin complex since H_2O is a weak field ligand. The orbital diagram is



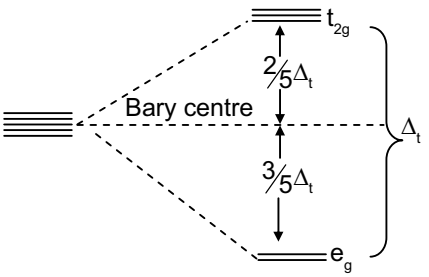
The three unpaired electrons give rise to spin only magnetic moment of 3.87 BM.

CRYSTAL FIELD SPLITTING IN TETRAHEDRAL COORDINATION ENTITIES

In tetrahedral complexes the ligands do not approach any of the metal d orbitals directly but they are closer to the t_{2g} orbitals. Hence their energy is raised while e_g orbitals are stabilized relative to the bary centre. The energy of t_{2g} are raised by 0.4 Δt and that of the e_g orbitals are lowered by 0.6 Δt. The crystal field splitting is smaller than that in octahedral field as there are only two third ligands as in octahedral complex and they have a less direct effect on the d orbitals.

For the same metal ion, same ligands and same metal-ligand distance. $\Delta_t = \frac{4}{9}\Delta_0$. Hence the orbital splitting energy in tetrahedral complexes are generally not large enough to force electrons to pair and so low spin configurations have never been observed. Under conditions favouring strong crystal field geometries other than tetrahedral are preferred.

Since the crystal field splitting is small for tetrahedral complexes, there is only high spin arrangement whether it is a weak field ligand or strong field ligand.



Tetrahedral complexes are favoured when

(i) Ligands are large and bulky and cause crowding in octahedral complex.

(ii) Weak field ligands are present.

(iii) Configuration of metal ion is d^0, d^1, d^5, d^6 and d^{10}. $[NiCl_4]^{2-}$, $Ni(CO)_4$, $[CoCl_4]^{2-}$ and $[VCl_4]^-$ are some tetrahedral complexes.

LIMITATIONS OF CRYSTAL FIELD THEORY

Though the theory successfully explains the formation, structure, colour and magnetic properties of complexes it has the following limitations.

(i) It does not take into account the covalent character of bonding between metal and ligands.

(ii) Ligands are considered to be point charges which in reality are not.

(iii) Anion ligands should have highest splitting effect but neutral ligands such as ethylene diamine, ammonia etc are found to be more strong compared to anionic ligands such as halogens.

STABILITY OF COORDINATION COMPOUNDS

The stability of a species is a measure of the extend to which the species will form from or will be transformed into another species under certain conditions when the system has reached equilibrium.

The stability of a complex is measured in terms of the equilibrium constant for the formation of the complex from the metal ion (M) and ligands (L).

$$M + nL \rightleftharpoons ML_n \qquad K = \frac{[ML_n]}{[M][L]^n}$$

K is called the overall formation constant or the stability constant of the complex. (Here M and L may or may not have charges which have been omitted for simplicity). K is a measure of the stability of a complex.

Consider the formation of $[Cu(NH_3)_4]^{2+}$ and $[Cu(CN)_4]^{2-}$

$$Cu^{2+} + 4NH_3 \rightleftharpoons [Cu(NH_3)_4]^{2+} \quad K = 4.5 \times 10^{11}$$

$$Cu^{2+} + 4CN^- \rightleftharpoons [Cu(CN)_4]^{2-} \quad K = 2 \times 10^{27}$$

Since K for the second reaction is much (10^{16} times) greater than that of the first reaction $[Cu(CN)_4]^{2-}$ is stable than $[Cu(NH_3)_4]^{2+}$. We can also say that $CN^{\ominus}$ is a stronger ligand than NH_3.

APPLICATIONS OF COMPLEXES

(i) Applications in qualitative analysis

In the confirmatory test for chloride, it is precipitated as a white precipitate of AgCl, which dissolves readily in ammonium hydroxide as a soluble complex.

$$\underset{\text{ppt}}{AgCl} + 2NH_4OH \rightarrow \underset{\text{Soluble}}{[Ag(NH_3)_2]Cl} + 2H_2O$$

Cu^{2+} ion produces an intense blue solution when excess of NH_4OH is added to it.

$$CuSO_4 + 2NH_4OH \rightarrow \underset{\text{pale blue ppt}}{Cu(OH)_2}\downarrow + (NH_4)_2SO_4$$

$$Cu(OH)_2 + 4NH_4OH \rightarrow \underset{\text{Intense blue solution}}{[Cu(NH_3)_4]}(OH)_2 + 4H_2O$$

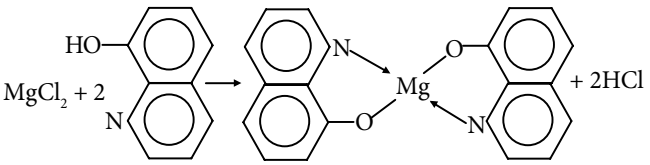
Fe^{2+} and Fe^{3+} develop intense blue precipitates with $K_3[Fe(CN)]_6$ and $K_4[Fe(CN)_6]$ respectively. Ni^{2+} is detected and estimated as a rose–red precipitate formed with dimethylglyoxime.

Anions and cations are detected by 'spot tests' in which a drop of a solution of the ion is mixed with a reagent, which is usually a complexing agent or a dye, which develops a characteristic colours even in ppm or ppb concentrations.

Thus Cu^{2+} ion develops a green colouration with cupron (benzoin-α-oxime) (concentration limit is 1 in 10^5 parts). Fe^{3+} develops a blood red colour, with ammonium thiocyanate (concentration limit is 1 in 10^5).

(ii) Applications in quantitative analysis

Ni^{2+} is estimated gravimetrically as dimethylglyoximate precipitate as described earlier. Mg^{2+} and Al^{3+} can be estimated as their oxinates gravimetrically.



oxine (8-hydroxy quinoline) magnesium oxinate (yellow ppt.)

Ethylenediaminetetraacetate (EDTA) is used to estimate Zn^{2+}, Mg^{2+}, Ca^{2+}, Ni^{2+} etc volumetrically or more precisely complexometrically.

$$Mg^{2+} + [EDTA]^{4-} \rightarrow [Mg(EDTA)]^{2-}$$

(iii) Applications in metallurgy

Ag and Au can be extracted from their sulphide ores by Mac Arthur-Forrest cyanide process by complexing with NaCN and reprecipitating the metal with Zn dust.

$$Ag_2S \xrightarrow[O_2]{NaCN} Na[Ag(CN)_2] \xrightarrow{Zn} Ag$$

In Mond's process, Ni is purified by converting Ni into $Ni(CO)_4$, a volatile liquid, distilling it and decomposing it into pure nickel by heating.

$$Ni \xrightarrow[350K]{4CO} Ni(CO)_4 \xrightarrow[450K]{} Ni + 4CO$$

Lanthanides and actinides are separated by ion-exchange chromatography where they are eluted by complexing agents like citric acid–ammonium citrate buffer.

(iv) Catalysis

An increasing number of complexes are used as catalyst.

	Catalyst	Process
1.	Ziegler-Natta catalyst $TiCl_4 + (C_2H_5)_3Al$	Polymerisation of alkenes at room temperature and pressure. $nCH_2 = CH_2 \rightarrow -(CH_2 - CH_2) - n$
2.	Wilkinson's catalyst $[Rh\ Cl\ (PPh_3)_3]$	Hydrogenation of alkenes.
3.	$[Rh\ H(CO)\ (PPh_3)_3]$	Hydroformylation of alkenes. $RCH = CH_2 + H_2 + CO \rightarrow RCH_2CH_2CHO$
4.	$[Pd\ Cl_4]^{2-}$	Oxidation of alkenes to aldehydes $C_2H_4 + \frac{1}{2}O_2 \rightarrow CH_3CHO$

(v) Photography

Photographic image is fixed by washing the developed film with sodium thiosulphate solution to remove unreacted AgBr by complexation.

$$AgBr + 2Na_2S_2O_3 \rightarrow Na_3[Ag\ (S_2O_3)_2] + NaBr$$

(vi) Medicine

Toxic amounts of copper, iron and radioactive metals can be reduced by removing them from the body as soluble complexes with chelating ligands like D-pencillamine, disferroxime–B and EDTA respectively. C is platin is used to treat cancer. Mercurochrome is used as an antiseptic. Salvarsan was used in the treatment of syphilis. Some organoarsenic compounds are used as carcinostatic drugs.

BIOLOGICAL IMPORTANCE OF COMPLEXES

Complexes play an important role in biological systems. Chlorophyll contains a magnesium ion and hemoglobin contains an Fe^{2+} ion coordinated at the centre of a porphyrin ring system. They assist in photosynthesis and respiration respectively. Vitamin B_{12}, cyanocobalamine, is the antipernicious anaemia factor in human body. Here a cobalt ion is coordinated at the centre of a corrin ring system. Nitrogen fixation takes place in bacteria and in blue green algae with the help of 'nitrogenases', which contain Fe and Mo.

PREPARATION AND PROPERTIES OF SOME METAL COMPOUNDS

(i) Chlorides and oxides of tin and lead

Tin forms compounds in oxidation states +2 and +4. The +2 state is less stable than +4 state and the former is a powerful reducing agent. But for Pb, +2 state is more stable than +4 state and therefore, the latter is an oxidising agent. This is due *to 'inert pair effect' i.e., a pair of s electrons in the outermost shell does not participate in bonding, and they remain paired.* The energy required to unpair the s electrons exceeds the energy released during bond formation. This effect is also displayed by Tl. This explains why Sn^{2+} is a good reducing agent and Pb^{2+} is more stable than Pb^{4+}.

(A) CHLORIDES OF TIN

The important chlorides of tin are stannous chloride $SnCl_2$ and stannic chloride $SnCl_4$.

$SnCl_2$: Tin dissolves in conc. HCl to give $SnCl_2$. Sn and HCl can be used to reduce nitro compounds to amines. Sn and $HgCl_2$ (solid) when distilled together produce anhydrous $SnCl_2$. $SnCl_2$ undergoes hydrolysis in water to give basic chloride.

$$Sn + HgCl_2 \rightarrow SnCl_2 + Hg$$

$$SnCl_2 + H_2O \rightarrow Sn(OH)Cl + HCl$$

$SnCl_2$ can be used as a reducing agent.

(i) $2HgCl_2 + SnCl_2 \rightarrow \underset{\text{white}}{Hg_2Cl_2} + SnCl_4$

$$Hg_2Cl_2 + SnCl_2 \rightarrow \underset{\text{grey}}{2Hg} + SnCl_4$$

(ii) $2FeCl_3 + SnCl_2 \rightarrow SnCl_4 + 2FeCl_2$

(iii) $SnCl_2 + I_2 + 2HCl \rightarrow SnCl_4 + 2HI$

It reduces cupric salts to cuprous salts and is used as a mordant in dyeing.

$SnCl_4$: Passage of chlorine over heated tin foil kept in a retort leads to $SnCl_4$. The stannic chloride formed volatilizes and can be condensed to a colourless liquid in a water cooled receiver, protected from atmospheric moisture by $CaCl_2$ guard tube.

The second method is to distill tin with excess $HgCl_2$. $SnCl_4$ reacts with HCl to give $H_2[SnCl_6]$. This contains H^+ ions and $SnCl_6^{2-}$ complex ion.

(B) OXIDES OF TIN

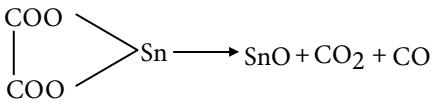
There are two oxides of tin: SnO and SnO_2.

SnO: SnO is obtained by heating $Sn(OH)_2$, which in turn is obtained by addition of NaOH to $SnCl_2$.

$$SnCl_2 + 2NaOH \rightarrow Sn(OH)_2 + 2NaCl$$

$$Sn(OH)_2 \rightarrow SnO + H_2O$$

It is also obtained by heating $SnCl_2$ with Na_2CO_3 or by heating stannous oxalate in the absence of air.



SnO is an amphoteric oxide. It burns in air with incandescence forming SnO_2.

SnO_2: It occurs as the ore cassiterite (or tin stone) in nature. When tin is burnt in air, SnO_2 is formed. $SnCl_4$ when treated with NH_4OH solution gives a precipitate $Sn(OH)_4$, which on washing and drying produces SnO_2. Stannic oxide finds use as a polishing powder and in glass and pottery manufacture.

(C) OXIDES OF LEAD

PbO (Litharge): PbS (galena) or Pb can be heated in air to a high temperature to give PbO. It is lemon yellow to reddish yellow in colour and becomes brownish on heating. Litharge reacts with alkali to form plumbites.

$$PbO + 2NaOH \rightarrow Na_2PbO_2 + H_2O$$

Pb_3O_4 (Red lead): It is obtained by heating PbO in air in a reverberatory furnace at 400°C. It decomposes at 470°C to give PbO and O_2.

$$PbO \xrightarrow[400^\circ C]{O_2} 2Pb_3O_4 \underset{400\,^\circ C}{\overset{470\,^\circ C}{\rightleftharpoons}} 6PbO + O_2$$

It oxidizes conc.HCl to Cl_2 and gives O_2 with hot conc. H_2SO_4

$$Pb_3O_4 + 8HCl \rightarrow 3PbCl_2 + 4H_2O + Cl_2$$

$$2Pb_3O_4 + 6H_2SO_4 \rightarrow 6PbSO_4 + 6H_2O + O_2$$

Pb_3O_4 can be considered as a mixed oxide $PbO_2.2PbO$. It is used as a protective paint for iron and steel. It is used in match industry.

PbO_2 (Lead dioxide): It is a dark reddish brown powder and is obtained by heating red lead with conc. HNO_3 and filtering out the residue PbO_2

$$Pb_3O_4 + 4HNO_3 \rightarrow PbO_2 + 2Pb(NO_3)_2 + 2H_2O$$

It can be conveniently prepared by the action of bleaching powder on $(CH_3COO)_2Pb$.

$$(CH_3COO)_2Pb + Ca(OH)_2 \rightarrow Pb(OH)_2 + (CH_3COO)_2Ca$$

$$Pb(OH)_2 + CaOCl_2 \rightarrow PbO_2 + CaCl_2 + H_2O$$

PbO_2 is a powerful oxidising agent. It liberates Cl_2 from conc.HCl on heating.

$$PbO_2 + 4HCl \rightarrow PbCl_2 + 2H_2O + Cl_2$$

It dissolves in cold conc.HCl forming $PbCl_4$.

$$PbO_2 + 4HCl \rightarrow PbCl_4 + 2H_2O$$

(D) CHLORIDES OF LEAD

$PbCl_2$: When dil.HCl is added to a lead salt solution, $PbCl_2$ is precipitated.

$$Pb(NO_3)_2 + 2HCl \rightarrow PbCl_2 + 2HNO_3$$

$PbCl_4$: Prepared by dissolving PbO_2 in cold conc.HCl. It is a yellow oil, stable below 0°C.

Above 50°C, $PbCl_4$ converts to $PbCl_2$ and Cl_2.

CONCEPT STRAND

Concept Strand 12

How is sodium stannite, Na_2SnO_2 prepared? What is its use?

Solution

$SnCl_2$ reacts with excess of NaOH in the absence of air. A white precipitate $Sn(OH)_2$ dissolves in excess of NaOH forming Na_2SnO_2 or $Na_2[Sn(OH)_4]$

$$SnCl_2 + 2NaOH \rightarrow Sn(OH)_2 + 2NaCl$$

$$2NaOH + Sn(OH)_2 \rightarrow Na_2SnO_2 + 2H_2O$$

Sodium stannite reduces bismuth salts to black precipitate of Bi (test for Bi^{3+} salts). It is used in the manufacture of glass, paints and varnishes. When mixed with glycerine, it is employed as glass-metal cement.

(ii) Oxides chlorides and sulphate of Fe^{2+}, Cu^{2+} and Zn^{2+}

(A) OXIDES, CHLORIDES AND SULPHATES OF IRON

Iron(II) salts are pale green in colour when hydrated and contain $[Fe(H_2O)_6]^{2+}$ ion. $FeSO_4.7H_2O$ (green vitriol) is an example. They are colourless in the anhydrous state.

FeO: FeO is a non-stoichiometric solid and is written as $Fe_{0.95}O$. It is formed as a black powder when ferrous oxalate is heated in a vacuum and then it is quenched to prevent disproportionation.

$$Fe(C_2O_4) \rightarrow FeO + CO_2 + CO$$

$$4Fe^{II}O \rightarrow \underset{(Fe^{II}Fe_2^{III}O_4)}{Fe_3O_4} + Fe^{(0)}$$

FeO dissolves in acids and is completely basic.

Fe_2O_3: Fe_2O_3 occurs as haematite. It is prepared by heating ferric carbonate, nitrate or sulphate in air.

$$4Fe(NO_3)_3 \xrightarrow{\Delta} 2Fe_2O_3 + 12NO_2 + 3O_2$$

Fe_3O_4: Ignition of Fe_2O_3 at 1400°C gives Fe_3O_4. It is also non-stoichiometric. Fe_3O_4 is basic.

CHLORIDES OF IRON

$FeCl_2$ and $FeCl_3$: Fe dissolves in hydrochloric acid in the absence of air and from these solutions, the hydrated dihalides, $FeF_2.8H_2O$ (white), $FeCl_2.4H_2O$ (pale green) and $FeBr_2.4H_2O$ (pale green) are obtained. $FeCl_2.4H_2O$ has octahedral structure with the chlorines occupying trans positions.

Anhydrous FeF_2 and $FeCl_2$ are obtained by heating the metal with gaseous HF or HCl. If metal reacts with halogens, we get $FeCl_3$ and FeF_3. $FeBr_2$ and FeI_2 are obtained by heating Fe with Br_2 or I_2. Fe(II) halides react with gaseous NH_3 to form $[Fe(NH_3)_6]^{2+}$ complex. $FeCl_3$ is coloured due to charge transfer spectra. Anhydrous $FeCl_3$ is obtained by passing Cl_2 over red hot iron. It is a shining black crystalline solid.

Ferrous sulphate ($FeSO_4.7H_2O$): $FeSO_4.7H_2O$ is called green vitriol. It is a byproduct in Kipp's apparatus in the preparation of H_2S. FeS_2 (pyrites), on oxidation in water and air gives ferrous sulphate.

$$2FeS_2 + 7O_2 + 2H_2O \rightarrow 2FeSO_4 + 2H_2SO_4$$

Scrap iron reacts with H_2SO_4 to give $FeSO_4$.

$$Fe + H_2SO_4 \rightarrow FeSO_4 + H_2$$

When $FeSO_4$ is heated, we get Fe_2O_3, a red variety is used as a pigment (venetian red).

$$2FeSO_4 \rightarrow Fe_2O_3 + SO_2 + SO_3$$

Mohr's salt is a double salt of $FeSO_4$ with $(NH_4)_2SO_4$ and has the formula, $FeSO_4(NH_4)_2SO_4.6H_2O$. Solutions of Mohr salt are not oxidised by a atmospheric oxygen like $FeSO_4$. Therefore, former is preferred to latter to make volumetric ferrous salt solutions.

The other isomorphous crystals of $FeSO_4.7H_{\uparrow 2}O$ are $MgSO_4.7H_2O$ (epsom) and $ZnSO_4.7H_2O$ (white vitriol).

Ferric sulphate : Oxidation of $FeSO_4$ by HNO_3 in presence of H_2SO_4 gives to $Fe_2(SO_4)_3$.

$$6FeSO_4 + 3H_2SO_4 + 2HNO_3 \rightarrow 3Fe_2(SO_4)_3 + 2NO + 4H_2O$$

(B) CHLORIDES AND SULPHATE OF COPPER

Cu(I) halides: Cu(I) compounds are generally colourless since they have d^{10} configuration and no d-d transition is

possible. Exceptions are Cu_2O (yellow), Cu_2CO_3 (yellow) and CuBr (brown). Colour arises due to charge transfer in these compounds.

Cu(II) halides: $CuCl_2$ is prepared by burning Cu in excess of Cl_2. It is dark coloured in anhydrous state and is greenish blue as $CuCl_2.2H_2O$. CuF_2 is obtained by dissolving CuO in HF.

CuO: Cu_2O is heated in air to give CuO. Malachite, $CuCO_3.Cu(OH)_2$, on heating gives CuO.

$$CuCO_3.Cu(OH)_2 \rightarrow 2CuO + H_2O + CO_2$$

Copper sulphate: $CuSO_4.5H_2O$ is called blue vitriol. Cu scrapings are dissolved in hot dil.H_2SO_4 in presence of air.

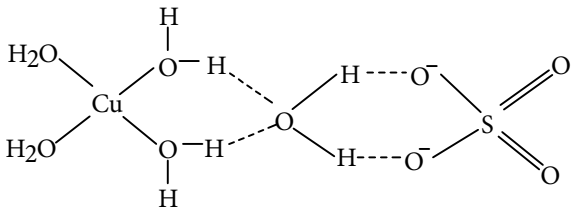
$$2Cu + 2H_2SO_4 + O_2 \rightarrow 2CuSO_4 + 2H_2O$$

Copper sulphate has several hydrates.

$$\underset{\text{(blue)}}{CuSO_4.5H_2O} \xrightarrow{\text{expose to air}} \underset{\text{(pale blue)}}{CuSO_4.3H_2O} \xrightarrow{100^\circ C}$$

$$\underset{\text{(bluish-white)}}{CuSO_4.H_2O} \xrightarrow{250^\circ C} \underset{\text{(white)}}{CuSO_4}$$

The structure of $CuSO_4.5H_2O$ is



The four molecules of H_2O are coordinated to Cu^{2+} and the fifth one is attached to anion and to the two water molecules through hydrogen bonding.

(C) OXIDES, CHLORIDE AND SULPHATE OF ZINC

ZnO: $ZnSO_4$ is reacted with Na_2CO_3 and the $ZnCO_3$ precipitated is ignited to give ZnO.

$$ZnSO_4 + Na_2CO_3 \rightarrow ZnCO_3 + Na_2SO_4$$

$$ZnCO_3 \rightarrow ZnO + CO_2$$

ZnO is used in the processing of rubber and it shortens the time for vulcanization to occur. It is used as a white pigment in paint. ZnO is amphoteric. In alkali, it forms zincates. ZnO is covalent and sublimes on heating. ZnO is white when cold but becomes yellow on heating. This is used to identify Zn^{2+} salts in qualitative analysis by heating salts and noting their colours when hot and cold.

$ZnCl_2$: The chloride of zinc is partly covalent. ZnF_2 is not much soluble in water due to its high lattice energy and its inability to form complexes in solution. $ZnCl_2$, $ZnBr_2$ and ZnI_2 are hygroscopic. The chloride form complexes like $[ZnCl_4(H_2O)_2]^{2+}$, which are easily soluble in water.

Zinc sulphate, $ZnSO_4.7H_2O$ (white vitriol): Zinc blende is taken and the ZnS in it is oxidized by roasting. Zn reacts with H_2SO_4 to give $ZnSO_4$. $ZnSO_4$ forms different hydrates depending on temperature.

At 800°C, $ZnSO_4$ decomposes to give ZnO.

$$2ZnSO_4 \rightarrow 2ZnO + 2SO_2 + O_2$$

A solution of $ZnSO_4$ is used as eye lotion. It is used in zinc plating baths.

CONCEPT STRANDS

Concept Strand 13

What are the structures of $FeCl_3.6H_2O$ and $[FeCl_4]^-$?

Solution

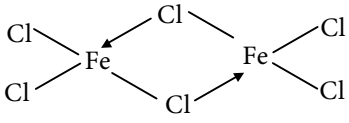
$FeCl_3.6H_2O$ is octahedral and occurs in trans form, $[FeCl_2(H_2O)_4]Cl.2H_2O$. In conc.HCl, it forms the tetrahedral $[FeCl_4]^-$ ion.

Concept Strand 14

Vapour density of anhydrous $FeCl_3$ at 400°C gives a molecular weight of 324.7 for ferric chloride. What is the structure of ferric chloride at this temperature?

Solution

Molecular weight of 324.7 shows that it has dimeric structure with halogen bridges. Fe is tetrahedrally surrounded by chlorine atoms.



Concept Strand 15

Since Cu(I) takes up the d^{10} configuration, we expect the Cu(I) compound to be stable. But it is actually found that Cu(II) compounds are more stable. Why?

Solution

Cu(I) compounds undergo disproportionation and equilibrium constant for disproportionation,

$$2Cu^{+} \rightarrow Cu^{2+} + Cu \text{ , is } 1.6 \times 10^{6}.$$

Hence we find Cu^{2+} compounds to be more stable. Cu^{2+} compounds are further stablized over Cu^{+} compounds by greater lattice energy and solvation energy.

Concept Strand 16

Dilute solution of copper(II) chloride is blue. But concentrated solution is green. Explain.

Solution

In dilute solution, it exists as $[Cu(H_2O)_4]^{2+}$, which is blue. In concentrated solutions, it forms a mixture of complex cation $[Cu(H_2O)_4]^{2+}$ and $[CuCl_4]^{2-}$ anion. This is green. There is an equilibrium given by $2[Cu(H_2O)_4]Cl_2 \rightleftharpoons [Cu(H_2O)_4]^{2+} + [CuCl_4]^{2-} + 4H_2O$

Concept Strand 17

How is Cu^{2+} estimated volumetrically?

Solution

Addition of KI solution to Cu^{2+} salt solutions in the neutral or weakly acidic medium results in the liberation of iodine. I_2 formed is estimated volumetrically using $Na_2S_2O_3$ and starch indicator.

$$2Cu^{2+} + 4I^{-} \rightarrow Cu_2I_2 + I_2$$

$$2Na_2S_2O_3 + I_2 \rightarrow Na_2S_4O_6 + 2NaI$$

Thus Cu^{2+} is estimated by adding excess KI to neutral or slightly acidified solution of Cu^{2+} and titrating I_2 liberated with $Na_2S_2O_3$.

Fehling's test: Cu^{2+} is reduced to Cu_2O by mild reducing agents, which is the basis of Fehling's test for reducing sugars. Fehling's-A and Fehling's-B are mixed together, sugar is added and warmed. Deep blue colour changes to red precipitate of Cu_2O. Other reducing agents like aldehydes answer this test.

(iii) Compounds of Ag

OXIDE, NITRATE AND THIOSULPHATE OF SILVER

Ag_2O: $AgNO_3$ solution on treating with KOH solution to give Ag_2O.

$$2AgNO_3 + 2KOH \rightarrow Ag_2O + H_2O + 2KNO_3$$

On heating to 300°C, it decomposes to Ag and O_2.

$$2Ag_2O \rightarrow 4Ag + O_2$$

Ag_2O is basic. One of the important uses of Ag_2O is it that is used in moist form (moist Ag_2O) to convert quaternary ammonium salts to quaternary hydroxides in Hoffmann elimination procedure to obtain alkenes.

$AgNO_3$: $AgNO_3$ is called lunar caustic and is obtained by dissolving Ag metal in warm dilute nitric acid.

$$3Ag + 4HNO_3 \rightarrow 3AgNO_3 + 2H_2O + NO$$

Silver nitrate decomposes to Ag on heating,

$$2AgNO_3 \xrightarrow{723\,K} 2AgNO_2 + O_2 \xrightarrow{980\,K} 2Ag + 2NO_2 + O_2$$

Ammoniacal silver nitrate is used to identify terminal alkynes. When it is added to terminal alkynes, a white precipitate is formed. The presence of halides, Cl^{-}, Br^{-} and I^{-} is also identified by treating these with $AgNO_3$. White, pale yellow and yellow precipitates indicate the presence of Cl^{-}, Br^{-} and I^{-} respectively.

$Ag_2S_2O_3$: When dilute $AgNO_3$ solution is gradually added to $Na_2S_2O_3$ solution which is in excess, we get white precipitate of $Ag_2S_2O_3$. The precipitate is washed and dissolved in NH_4OH and reprecipitated by HNO_3 and quickly dried. $Ag_2S_2O_3$ is soluble in excess of $Na_2S_2O_3$ solution and forms a complex.

$$Ag_2S_2O_3 + 2Na_2S_2O_3 \rightarrow 2Na_3[Ag(S_2O_3)_2]$$

On standing, $Ag_2S_2O_3$ undergoes decomposition and on hydrolyzis, to black Ag_2S. Colour changes rapidly from white to yellow, red, brown and finally black (play of colours).

$$\underset{\text{white}}{Ag_2S_2O_3} + H_2O \rightarrow \underset{\text{black}}{Ag_2S} + H_2S$$

(iv) Preparation and properties of $KMnO_4$

PREPARATION

(a) MnO_2 is oxidized by KNO_3 or atmospheric oxygen in the presence of KOH or K_2CO_3 to give K_2MnO_4

$$2MnO_2 + 4KOH + O_2 \rightarrow 2K_2MnO_4 + 2H_2O$$

$$2MnO_2 + 2K_2CO_3 + O_2 \rightarrow 2K_2MnO_4 + 2CO_2$$

K_2MnO_4 formed undergoes disproportionation by H_2SO_4 or by CO_2 in water, to form $KMnO_4$ and MnO_2.

$$3K_2MnO_4 + 2H_2SO_4 \rightarrow 2K_2SO_4 + 2KMnO_4 + MnO_2 + 2H_2O$$

$$3K_2MnO_4 + 2H_2O + 4CO_2 \rightarrow 2KMnO_4 + MnO_2 + 4KHCO_3$$

(b) Electrolytic oxidation between iron electrodes separated by a diaphragm is also used.

$$2K_2MnO_4 + H_2O + [O] \rightarrow 2KMnO_4 + 2KOH$$

$KMnO_4$ AS AN OXIDIZING AGENT

$KMnO_4$ solutions are used in volumetric analysis. Solutions of $KMnO_4$ decompose slowly in neutral but faster in acid medium. Decomposition is catalysed by sunlight. Hence $KMnO_4$ solutions are stored in dark bottle and are standardized frequently.

$KMnO_4$ is a strong oxidizing agent in acidic medium.

$$2KMnO_4 + 3H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5(O)$$

By ion-electron method, we have

$$MnO_4^- + 8H^+ + 5\bar{e} \rightarrow Mn^{2+} + 4H_2O$$

Since Mn^{7+} is converted to Mn^{2+}, the equivalent weight of Mn in such reaction is $\frac{\text{molecular weight}}{5}$ which is equal to 31.6 g.

In acidic medium, it oxidizes H_2S to S, SO_3 to H_2SO_4, $C_2O_4^{2-}$ to CO_2 or Fe^{2+} to Fe^{3+}.

$$2MnO_4^- + 16H^+ + 5S^{2-} \rightarrow 2Mn^{2+} + 8H_2O + 5S$$

$$2MnO_4^- + 16H^+ + 5C_2O_4^{2-} \rightarrow 2Mn^{2+} + 8H_2O + 10CO_2$$

$$MnO_4^- + 46H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$$

In neutral medium, the changes in oxidation state remains the same as in alkaline medium. In alkaline medium, $KMnO_4$ is reduced to brown precipitate of MnO_2 through the formation of a transient green intermediate, K_2MnO_4.

$$2KMnO_4 + 2KOH \rightarrow 2K_2MnO_4 + H_2O + (O)$$

$$2K_2MnO_4 + 2H_2O \rightarrow 2MnO_2 + 4KOH + 2(O)$$

The overall reaction is

$$2KMnO_4 + H_2O \rightarrow 2MnO_2 + 2KOH + 3(O)$$

By ion electron method we have

$$MnO_4^- + 2H_2O + 3e \rightarrow MnO_2 + 4OH^-$$

The equivalent weight of $KMnO_4$ in neutral and alkaline media is $\frac{158}{3} = 52.67$ g

$KMnO_4$ is used in oxidation of many organic compounds.

In qualitative analysis, the presence of Mn^{2+} is confirmed by treating the solution with sodium bismuthate, when purple colour of MnO_4^- is formed.

Uses of $KMnO_4$: It is used in volumetric analysis to estimate reducing agents like ferrous salts, oxalic acid and H_2O_2. Alkaline $KMnO_4$, is used in organic chemistry for oxidation of unsaturated compounds. The colour of $KMnO_4$ is due to charge transfer spectrum.

CONCEPT STRANDS

Concept Strand 18

Write ion-electron equations for oxidation of (i) I^- and (ii) H_2O_2 with $KMnO_4$ in acid medium?

Solution

(i) $2MnO_4^- + 16H^+ + 10I^- \rightarrow 2Mn^{2+} + 8H_2O + 5I_2$

(ii) $2MnO_4^- + 6H^+ + 5H_2O_2 \rightarrow 2Mn^{2+} + 8H_2O + 5O_2$

Concept Strand 19

Write the reactions of $KMnO_4$ with (i) NH_3 and (ii) KI in alkaline condition.

Solution

(i) $2KMnO_4 + H_2O \rightarrow 2KOH + 2MnO_2 + 3(O)$

$2NH_3 + 3(O) \rightarrow N_2 + 3H_2O$

$2KMnO_4 + 2NH_3 \rightarrow 2KOH + 2MnO_2 + N_2 + 2H_2O$

(ii) $2KMnO_4 + H_2O + KI \rightarrow 2MnO_2 + 2KOH + KIO_3$

(v) Preparation and properties of $K_2Cr_2O_7$

Chromite ore, $FeO.Cr_2O_3$ is roasted with Na_2CO_3 with free access to air in a reverberatory furnace. Chromite ore is oxidised, according to the equation,

$$\left[2(FeO.Cr_2O_3) + (O) \rightarrow Fe_2O_3 + 2Cr_2O_3\right] \times 2$$

$$[2Na_2CO_3 + Cr_2O_3 + 3(O) \rightarrow 2Na_2CrO_4 + 2CO_2] \times 4$$

$$4(FeO.Cr_2O_3) + 8Na_2CO_3 + 7O_2 \rightarrow 2Fe_2O_3 + 8Na_2CrO_4 + 8CO_2$$

Sodium chromate is then extracted with water and acidified with sulphuric acid to obtain sodium dichromate.

$$2Na_2CrO_4 + H_2SO_4 \rightarrow Na_2Cr_2O_7 + Na_2SO_4 + H_2O$$

The solution of Na_2CrO_7 is concentrated and treated with calculated amounts of KCl, when $K_2Cr_2O_7$ being less soluble crystallies out.

$$Na_2Cr_2O_7 + 2KCl \rightarrow 2NaCl + K_2Cr_2O_7$$

$K_2Cr_2O_7$ is preferred to $Na_2Cr_2O_7$ in titration because sodium dichromate is hydroscopic and is not suitable as a primary standard to prepare standard solutions.

$K_2Cr_2O_7$ AS OXIDIZING AGENT

The ion electron equation in acid medium is

$$Cr_2O_7^{2-} + 14H^+ + 6\bar{e} \rightarrow 2Cr^{3+} + 7H_2O$$

Chromate and dichromates are interconvertible in aqueous solution by changing pH.

In acid medium, $2CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{2-} + H_2O$

In alkaline medium $Cr_2O_7^{2-} + 2OH^- \rightarrow CrO_4^{2-} + H_2O$

Uses of $K_2Cr_2O_7$: It is used in volumetric analysis as a primary standard. A mixture of $K_2Cr_2O_7$ and conc.H_2SO_4, known as chromic acid mixture is used as cleansing agent for glassware.

CONCEPT STRAND

Concept Strand 20

Write the ion electron equations for reactions of $K_2Cr_2O_7$ with (i) KI and (ii) Fe^{2+}.

Solution

(i) $Cr_2O_7^{2-} + 14H^+ + 6I^- \rightarrow 2Cr^{3+} + 7H_2O + 3I_2$

(ii) $Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$

SUMMARY

Co-ordination compounds	
Double salts	Completely ionized in solution
Complexes	Do not undergo complete ionization to ions from which they are formed
Werners theory	Primary valency
	Secondary valency
	Geometry of complex

Common terms used in co-ordination chemistry	Co-ordination entity–central atom–ligands–denticity of ligands–co-ordination number, co-ordination sphere–co-ordination polyhedron Oxidation number of central atom Homoleptic and Heteroleptic complexes
Witting formulae of mononuclear co-ordination entities	Central atom–ligands in alphabetical order
Nomenclature of complexes	IUPAC rules
Isomerism in co-ordination compounds	Stereo isomerism–geometrical and optical–structural isomerism – hydrate, ionization, linkage, co-ordination isomerisms
Bonding in co-ordination compounds	Valence bond and crystal field theory
Valence band theory	Different postulates of the theory–limitations
Crystal field theory	Crystal field splitting in octahedral field–spectochemical series–crystal field stabilization energy Crystal field splitting in tetrahedral complexes–limitations of crystal field theory
Stability of co-ordination compounds	Stability constant
Application of complexes	Application in qualitative analysis, in quantitative analysis, in metallurgy, catalysis, Photography, medicine
Preparation and properties of some metal compounds	
Chlorides and oxides of tin and lead	stannous chloride, stannic chloride, stannous oxide, stannic oxide Litharage, red lead, lead dioxide $PbCl_2$, $PbCl_4$
Oxides, chlorides and sulphates of Fe^{2+}, Cu^{2+} and Zn^{2+}	FeO, Fe_2O_3, Fe_3O_4,. $FeCl_2$, $FeCl_3$ $FeSO_4.7H_2O$, Cu(I) compounds, Cu(II) compounds $CuSO_4.5H_2O$, ZnO, $ZnCl_2$, $ZnSO_4.7H_2O$

TOPIC GRIP

Subjective Questions

1. Give the electronic configurations of Co^{3+} and Co^{2+}. Give the names of $[Co(NH_3)_6]Cl_2$ and $[Co(NH_3)_6]Cl_3$. Using crystal field energy level diagram arrive at the values of their spin only magnetic moments.

2. Find the effective atomic number of the metal in the following complexes and indicate their geometry.
(i) $Ni(CO)_4$ (ii) $Fe(CO)_5$ (iii) $[Zn(NH_3)_4]^{2+}$
(iv) $[Fe(CN)_6]^{4-}$ (v) $[NiCl_4]^{2-}$

3. Classify the following pairs of isomers into specific category of isomerism and explain.
(i) $[CoCl_3(NH_3)_6]$ and $[Co(NH_3)_6][CoCl_6]$
(ii) Two forms of $[PtCl_2(NH_3)_2]$
(iii) $[Co(ONO)(NH_3)_5]Cl_2$ and $[Co(NO_2)(NH_3)_5]Cl_2$

4. The M-C bond in a metal carbonyl consists of a σ-bond and a π-bond. They are said to be synergic or complementary to each other. Explain.

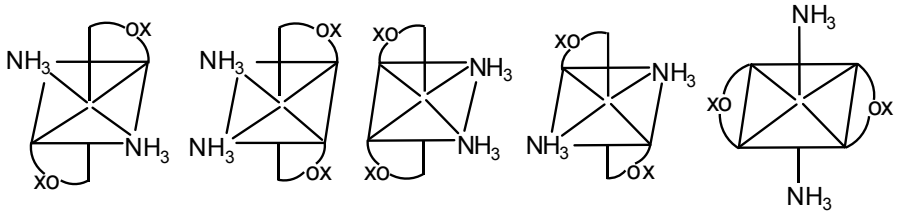
5. A monomeric complex of cobalt gives the following results on analysis.

Cl^-	NH_3	SO_4^{2-}	H_2O
12.81%	24.77%	34.65%	6.53%

The compound is diamagnetic. Find the molecular formula of the complex if the molecular weight is equal to 276.5 g. Find the formula of the complex and give its IUPAC name. Explain why it is diamagnetic. Give the name of a complex of Co^{3+}, which shows paramagnetism.

6. Give the IUPAC names of the following complexes. Draw their crystal field orbital energy diagram, calculate the spin only magnetic moment in Bohr magneton and give the geometry.
(i) $[Cr(en)_3]Cl_3$ (ii) $(NH_4)_3[Mn(CN)_6]$ (iii) $[Co(H_2O)_6]SO_4$

7. From the following set of compounds, identify the
(i) geometric isomers (ii) optical isomers (iii) identical structures



8. An oxide (A) of a transition metal with configuration d^5s^2 is prepared by using a pink coloured compound (B) in alkaline medium. The oxide reacts with HCl to give a gas. This oxide with hot conc.H_2SO_4 gives out O_2. The oxide (A) reacts with KNO_2 in NaOH to produce an intermediate compound which by electrolytic oxidation gives pink coloured compound (B). (B) oxidizes $CH_3CH{=}CH_2$ in hot alkaline condition to CH_3COOH and CO_2. Find (A) and (B) and write all reactions involved and balance the equation.

9. (i) Blue vitriol is heated to 720°C and the substance obtained is heated with carbon to a higher temperature. What is the product formed? (gaseous products if any are removed).
(ii) Give balanced equation for reactions of $CuSO_4$ and KCN. If excess CN^- is added what happens?

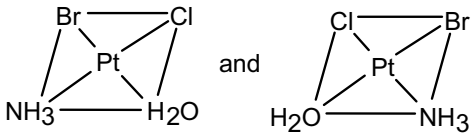
10. Write the balanced equation for the following reaction.
(i) Cu with dilute HNO_3
(ii) Cu with conc.HNO_3
(iii) Cu^{2+} with KI followed by addition of $Na_2S_2O_3$

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

11. The complex $[CoCl_2(NH_3)_4]$ shows
(a) geometric isomerism only
(b) optical isomerism only
(c) both geometric and optical isomerism
(d) ionization isomerism which is optically active

12. Consider the two complexes



These are
(a) geometric isomers (b) optical isomers (c) linkage isomers (d) not isomers

13. The oxidation number and coordination number of the central metal atom in $[Ag(CN)_2]^-$ is
(a) +1, 2 (b) −1, 2 (c) +2, 4 (d) +1, 4

14. Among the following complexes, the one that has highest molar conductivity is
(a) $[Co(H_2O)_3(NO_2)_3]$
(b) $[Cr(NH_3)_5CN]_2[Co(CN)_6]$
(c) $Na[Co(NH_3)_2Cl_4]$
(d) $[NH_4][Ni(H_2O)_3Cl_3]$

15. In which of the following, the solution would not turn green?
(a) $Cr_2O_7^{2-} + H_2O_2 \xrightarrow{\text{acid medium}}$
(b) $Cr_2O_7^{2-} + H_2S \xrightarrow{\text{acid medium}}$
(c) $Cr^{2+} + H_2O_2 \xrightarrow{\text{acid medium}}$
(d) $CrO_4^{2-} + H_3O^+ \xrightarrow{\text{acid medium}}$

Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

16. Statement 1
Rust stains on clothes can be removed using oxalic acid.

and

Statement 2
Rust stains are oxidized and hence removed easily from fabric.

17. **Statement 1**
Both $V(CO)_6$ and $Cr(CO)_6$ are diamagnetic species.

and

Statement 2
Generally in metal carbonyls, the metal exhibits zero oxidation state.

18. **Statement 1**
For the complex containing same ligand, colour changes with change in metal ion.

and

Statement 2
Colour depends on crystal field splitting energy of metal ion.

19. **Statement 1**
Square planar complexes containing 4 different ligands [Mabcd] type only are optically active.

and

Statement 2
Square planar complexes of the type [Mabcd] have three geometric isomers.

20. **Statement 1**
Copper on reaction with concentrated sulphuric acid liberates sulphur dioxide gas.

and

Statement 2
Non oxidizing acids dissolve copper in the presence of air forming salt and water.

Linked Comprehension Type Questions

Directions: **This section contains 2 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

Potassium dichromate is a powerful oxidizing agent. It is a useful reagent in organic chemistry as well as in volumetric analysis. It is formed as orange red crystals soluble in water. It decomposes on heating and liberates oxygen gas.

21. The number of moles of oxygen gas liberated per mole of $K_2Cr_2O_7$ on heating is
(a) 1 (b) 3 (c) 0.4 (d) 0.75

22. $K_2Cr_2O_7$ is treated with aqueous KOH, the product formed
(a) has a lesser oxidation state
(b) has a higher oxidation state
(c) has the same colour
(d) can be easily converted back to dichromate by treatment with dil acid

23. Which among the following formed on passing H_2S gas through acidified solution of dichromate, solution is
(a) SO_2 gas (b) SO_3 gas (c) colloidal sulphur (d) H_2SO_4

Passage II

A metal 'M' is refined due to the unique properties of its complex 'A' with carbon monoxide. Metal M forms a square planar complex 'B' with 4 molecules of ammonia and an octahedral complex 'C' with 4 molecules of ammonia and 2 molecules of water.

24. The formula of complex A is
(a) $[M_2(CO)_9]$ (b) $[M(CO)_5]$ (c) $[M_2(CO)_{10}]$ (d) $[M(CO)_4]$

25. The complex that is paramagnetic is
(a) A (b) B
(c) C (d) All three are diamagnetic

26. The complex which exhibits geometric isomerism is
(a) A (b) B (c) C (d) Both B and C



Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers out of which ONE OR MORE answers will be correct.**

27. Among the following the process that uses the formation of a complex is
(a) vapour phase refining of Nickel
(b) separation of lead and zinc during concentration of their sulphide ores
(c) estimation of Cu^{2+} by gravimetric analysis
(d) oxidation of oxalic acid by $KMnO_4$

28. The complexes $[CoCl(NH_3)_5]Cl_2$ and $[Co(NH_3)_6]Cl_3$ can be distinguished by
(a) Magnetic moment (b) depression in freezing point measurements
(c) $AgNO_3$ test (d) reaction with MnO_2 in acidic medium

29. Among the following compounds that will give Prussian blue on reaction with $K_4[Fe(CN)_6]$ is/are
(a) $FeBr_3$ (b) $FeCl_3$ (c) $FeSO_4$ (d) FeF_3



Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

30. Column I contains a complex and column II contains the types of isomerism it can show

Column I	**Column II**
(a) $[Co(NH_3)_5NO_3]SO_4$	(p) Ionization isomerism
(b) $[Cr(H_2O)_5NO_2]Cl_2$	(q) solvate isomerism
(c) $[Co(en)_2NO_2Cl]SCN$	(r) linkage isomerism
(d) $[Co(NH_3)_4(CH_3C_6H_4NH_2)Cl]Cl_2$	(s) ligand isomerism

IIT ASSIGNMENT EXERCISE

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

31. The oxidation number of central metal atom in the complex $Na_4[Ni(CN)_6]$ is
(a) +2 (b) +1 (c) 0 (d) +3

32. The purple complex formed when sodium nitroprusside reacts with sulphide ions is
(a) $[Fe(CN)_5NO]^{3-}$ (b) $[Fe(CN)_5NOS]^{3-}$ (c) $[Fe(CN)_5NO]^{2-}$ (d) $[Fe(CN)_5NOS]^{4-}$

33. The presence of nitrates is confirmed by the formation of brown ring. The ligand present in the brown ring complex formed from NO_3^- is
(a) NO_2^+ (b) NO (c) NO_3^- (d) NO^+

34. The coordination number and oxidation number of the central metal atom in the complex $[Cr(H_2O)_5CO_3]ClO_4$ is
(a) 6, 2 (b) 6, 3 (c) 7, 3 (d) 7, 2

35. The complex $[Ni(en)_3]Cl_2$ [A] is more stable than $[Ni(NH_3)_6]Br_2$ [B] because
(a) A uses 3d orbitals while B uses 4d orbitals for hybridization
(b) A shows geometric isomerism while B does not
(c) A contains bidendate ligands while B has monodendate ligands
(d) The anion in complex B is larger than that in complex A

36. The complex showing maximum ionic conductivity is
(a) $[Cu(NH_3)_4]Cl_2$ (b) $Na_3[Cu(CN)_4]$ (c) $K[Ag(CN)_2]$ (d) $[Ni(CO)_4]$

37. Geometric isomerism is not exhibited by
(a) $[Cr(en)_3]^{3+}$ (b) $[Cr(en)_2Cl_2]^+$ (c) $[CrCl_3(H_2O)_3]$ (d) $[Cr\,Cl_2(H_2O)_4]^+$

38. The oxidation number and coordination number of central metal atom in Na[Co(EDTA)] is
(a) +2, 6 (b) +2, 4 (c) +3, 4 (d) +3, 6

39. The hybridization of platinum in the complex $[Pt(NH_3)_2Cl_2]$ is
(a) d^2sp^3 (b) sp^3d^2 (c) dsp^2 (d) sp^3

40. The number of ions formed on dissolving $[Cr(NH_3)_5(NO_2)]_3\,[Co(NO_2)_6]_2$ is
(a) 3 (b) 4 (c) 5 (d) 6

41. The nomenclature of $[Co(NH_3)_5(ONO)](NO_3)_2$ is
(a) pentaammine nitrite cobalt (II) nitrate
(b) pentaamminenitrito-O-cobalt (III) nitrate
(c) pentaammine nitro cobalt (III) dinitrate
(d) penta ammine-O-nitrito cobalt (III) dinitrate

42. Among the following a complex that does not exist is
(a) $[CuI_4]^{2-}$ (b) $[Cu(CN)_4]^{3-}$ (c) $[CuCl_4]^{2-}$ (d) $[Cu(H_2O)_4]^{2+}$

43. A metal complex of the type $[M(aa)_3]$ would show
(a) linkage isomerism
(b) geometric isomerism
(c) coordination isomerism
(d) optical isomerism

44. The complex ion present in a molecule of copper sulphate pentahydrate is
(a) $[Cu(H_2O)_4]^{2+}$ (b) $[Cu(H_2O)_3SO_4]$ (c) $[Cu(H_2O)_2SO_4]$ (d) $[Cu(H_2O)_5]^{2+}$

45. Pyridine is classified as a
(a) monodendate ligand (b) bidendate ligand (c) hexadendate ligand (d) chelating ligand

46. Linkage isomerism is generally seen when there are
(a) bidendate ligands (b) ambidendate ligands
(c) water of hydration (d) two coordination spheres

47. The complex formed between one mole of a metal and three moles of a bidendate ligand would show geometric isomerism if the bidendate ligand is
(a) oxalate (b) glycinate (c) ethylene diamine (d) o-phenanthroline

48. Among the following complexes, the one that shows either optical or geometric isomerism, but not both is
(a) $[Pt(py)_2(NH_3)_2Cl_2]^{2+}$ (b) $[Cu(NH_3)_4]^{2+}$ (c) $[Cr(en)_2Cl_2]^{+}$ (d) $[Cr(en)_3]^{3+}$

49. Total number of stereo isomers possible for the complex $[Cr(C_2O_4)_2(H_2O)_2]$ is
(a) 0 (b) 2 (c) 3 (d) 4

50. Complex with molecular formula $[Mn(H_2O)_4Cl_2]^{+}$ exists in two different colours the different coloured complexes are
(a) geometrical isomers (b) optical isomers (c) ionization isomers (d) solvate isomers

51. Among the following a complex that can exhibit meridional form is
(a) $[NiCl_2Br_2]$ (b) $Co(NH_3)_3(NO_3)_3]$ (c) $[Cr(en)_2Cl_2]^{+}$ (d) $[Cr(en)_3]^{3+}$

52. The complex $[Co(H_2O)_5NO_2]SO_4$ can exhibit
(a) Ionization isomerism (b) Hydrate isomerism
(c) Geometric isomerism (d) Both (a) and (b)

53. Geometric isomerism is not seen in tetrahedral complexes because
(a) it is a three dimensional structure
(b) all 4 positions are equivalent in space
(c) it is possible only when four different ligands are present
(d) ligands have sterical strain at the corners

54. The complexes $[Cr(NH_3)_6][Co(CN)_6]$ and $[Co(NH_3)_6][Cr(CN)_6]$ are
(a) same compounds (b) coordination isomers
(c) solvate isomers (d) linkage isomers

55. For the complex $[Fe(CN)_6]^{4-}$ and $[Fe(CN)_6]^{3-}$, the true statement is
(a) Both are diamagnetic (b) Both are paramagnetic
(c) $Fe(CN)_6^{3-}$ only is paramagnetic (d) $Fe(CN)_6^{4-}$ only is paramagnetic

56. The number of geometric isomers possible for $[NiCl_2Br_2]^{2-}$ is
(a) 0 (b) 2 (c) 3 (d) 4

57. According to valence bond theory, $[FeF_6]^{3-}$ will have a magnetic moment of
(a) 0 BM (b) 2.83 BM (c) 5.85 BM (d) 3.73 BM

58. The crystal field stabilization energy for zinc complexes $[Zn(NH_3)_4]^{2+}$ is
(a) −4 Dq (b) +4 Dq (c) −8 Dq (d) 0 Dq

59. Among the following, the wrong statement regarding assumptions of crystal field theory is
(a) All the d orbitals are affected by ligand filled
(b) Metal orbitals and ligand orbitals interact with each other.
(c) Electrostatic forces operate between ligand and metal ion,
(d) Complex formation destroys the degeneracy of metal orbitals

60. For crystal field splitting , the CN^- is a strong field ligand. This is because of
(a) π bonding from metal to ligand
(b) presence of a triple bond in CN
(c) difference in electronegativity between carbon and nitrogen
(d) availability of unpaired electrons which can pair up

61. SnI_4 is orange coloured because of
(a) d-d transition
(b) polarization of iodide by Sn^{4+}
(c) charge transfer
(d) paramagnetism

62. The compound used in volumetric analysis as a primary standard is
(a) $FeSO_4$ (b) $Fe_2(SO_4)_3$ (c) $FeCl_2$ (d) $FeSO_4.(NH_4)_2SO_4.6H_2O$

63. Potassium dichromate should be preferred over sodium dichromate for use as primary standard in volumetric analysis because
(a) Sodium has smaller size than potassium
(b) $Na_2Cr_2O_7$ is less water soluble than $K_2Cr_2O_7$
(c) $Na_2Cr_2O_7$ is hygroscopic while $K_2Cr_2O_7$ is not
(d) KOH is more basic than NaOH

64. Monuments made of copper, turn green on exposure to atmosphere due to
(a) reaction with sulphides in the atmosphere
(b) deposits of CuS oxidized to $CuSO_4$
(c) oxidation of impurities present in copper
(d) formation of basic copper carbonate

65. Dichromate ion changes to a yellow colour at pH = x and reverts back to orange colour at pH = y. The values of x and y respectively can be
(a) x = 9 y = 11 (b) x = y = 7 (c) x = 8, y = 6 (d) x = 6, y = 4

66. Potassium dichromate is prepared from sodium dichromate by addition of KCl. This method makes use of
(a) low reactivity of NaCl formed
(b) low solubility of $K_2Cr_2O_7$
(c) high solubility of NaCl formed
(d) hygroscopic nature of $Na_2Cr_2O_7$

67. The equivalent weight of $KMnO_4$ in acidic, alkaline and neutral solutions. Where it is converted to Mn^{2+}, MnO_4^- and MnO_2 respectively are [M.wt = 158 g/mol]
(a) 31.6, 52.7, 52.7 (b) 31.6, 158.7, 52.7 (c) 158, 52.7, 31.6 (d) 52.7, 31.6, 31.6

68. $KMnO_4$ solution should always be stored in dark bottles because
(a) the solution is purple coloured
(b) the solution is oxidizing in nature
(c) the solution has a low boiling point
(d) the decomposition of $KMnO_4$ is catalysed by sunlight

69. MnO_4^{2-} is treated with moist carbon dioxide. In this reaction MnO_4^{2-} undergoes
(a) Reduction (b) oxidation (c) disproportionation (d) decomposition

70. The compound of silver that will rapidly absorb carbondioxide is
(a) $AgNO_3$ (b) Ag_2O (c) $Ag_2S_2O_3$ (d) AgCl

71. Silver nitrate gives a yellow coloured precipitate with
(a) NaBr (b) Na_3PO_4 (c) Na_2S (d) Na_2CrO_4

72. When a silver salt is treated with excess sodium thiosulphate, the product formed is
(a) white precipitate of $Ag_2S_2O_3$
(b) black precipitate of Ag_2S
(c) white $Ag_2S_2O_3$ changing to red, brown and finally black
(d) a soluble complex

73. The compound formed on heating basic zinc carbonate is
(a) $ZnCO_3$ (b) ZnO (c) $Zn(OH)_2$ (d) ZnO and $ZnCO_3$

74. The gas released on heating $ZnSO_4$ above 800°C is/are
(a) SO_2 (b) SO_3 (c) O_2 (d) SO_2 and O_2

75. A metal chloride on reaction with H_2S gives a white precipitate. The metal chloride on reaction with NaOH gives
(a) a soluble hydroxide (b) a hydroxide which decomposes to the oxide
(c) a soluble complex (d) an insoluble complex

76. $CuCl_2$ can be converted to Cu_2Cl_2 on reaction with Zn dust. A gas that can be used instead of Zn is
(a) HCl (b) SO_2 (c) O_3 (d) SO_3

77. When air is blown through scrap copper dipped in dil H_2SO_4, the compound formed is
(a) Cu_2S (b) CuO (c) CuS (d) $CuSO_4$

78. When a green solution of ferrous sulphate is exposed to air, it turns brown due to
(a) formation of an oxide layer on the surface (b) formation of $Fe_2(SO_4)_3$
(c) formation of basic ferric sulphate (d) formation of FeS_2

79. An aqueous solution of $FeCl_3$ can be stabilized by adding
(a) $FeCl_2$ (b) Cl_2 gas (c) dil HCl (d) excess water

80. Lead dioxide on reaction with HCl liberates chlorine gas because
(a) Pb^{4+} is less stable than Pb^{2+} due to inert pair effect
(b) HCl is a good reducing agent
(c) $PbCl_2$ formed remains insoluble hence further reaction does not occur
(d) $PbCl_4$ formed is unstable

81. $CoCl_2$ is treated with sodium nitrite when a cobalt complex is formed. The oxidation number of cobalt in this complex is
(a) +1 (b) +2 (c) +3 (d) 0

82. The reaction $[CoCl_4(H_2O)_2]^{2-}_{(aq)} + EDTA^{4-} \rightarrow [Co(EDTA)]^{2-}_{(aq)} + 4Cl^-_{(aq)} + 2H_2O_{(\ell)}$ is expected to
(a) occur only at high temperature
(b) be feasible because the complex formed is more stable
(c) proceed in backward direction due to higher charge on Co
(d) to occur only under drastic conditions

83. The nomenclature of the complex $Na_3[Fe(NO_2)_5CO]$ is
(a) sodium carbonyl pentanitritoferrate (II)
(b) sodium carbonyl pentanitro ferrate (III)
(c) sodium pentanitro carbonyl ferrate (III)
(d) sodium carbonyl pentanitro ferrate (II)

84. $HgCl_2$ dissolves in excess KI because
(a) of formation of a complex (b) of large size of I^-
(c) Hg^{2+} gets reduced to Hg_2^{2+} (d) I^- is a good reducing agent

85. Among the following metal ions, one that will form the least stable complex with CN^- ligand is
(a) Na^+ (b) Cu^{2+} (c) Cs^+ (d) K^+

86. Iron deficiency causes yellowing of leaves. This is prevalent in alkaline soil and is treated by adding Fe(III) as an EDTA complex. This because
(a) EDTA reduces oxidation state of Fe^{3+} to Fe^{2+} (b) in alkaline soil Fe^{3+} decomposes
(c) only a complex is soluble in alkaline solution (d) Fe^{3+} in alkaline system dissolves as Fe^{2+}

87. The oxidation state of nitrogen in the complex $[Co(NH_3)_5Cl]Br_2$ is
(a) −3 (b) +1 (c) +3 (d) −1

88. A metal in the 3d series forms a tetrahedral complex with Cl^- ligands and has magnetic moment of 5.86 BM. If the complex ion is neutralized by two moles of ammonium ion per mole, the metal is
(a) Cr (b) Fe (c) Mn (d) Co

89. Consider a complex formed by Co with 5 moles of NH_3 and 3 moles of Cl^-. When all the ammonia molecules are coordinated to Co, the number of moles of AgCl precipitated on treating the complex with $AgNO_3$ is
(a) 1 (b) 2 (c) 3 (d) none

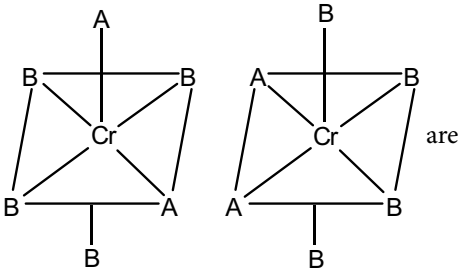
90. Which among the following species is colourless?
(a) $[Cu(NH_3)_4](OH)_2$ (b) $[Zn(H_2O)_4]SO_4$ (c) $[Fe(H_2O)_5NO]SO_4$ (d) $[Co(NH_3)_4Cl_2]Cl$

91. Fe^{3+} in $FeCl_3$ is identified by adding SCN^- to its aqueous solution. The blood red colour is due to the complex
(a) $[Fe(H_2O)_5SCN]^{2+}$ (b) $[Fe\ Cl_3(SCN)]^{2-}$ (c) $[Fe(H_2O)_4SCN\ Cl]^+$ (d) $[Fe(H_2O)_3SCN]^{2+}$

92. The ligand that would show cis and trans forms when it forms a square planar complex with Pt is
(a) ethylene diamine (b) glycine anion (c) oxalate (d) dipyridyl

93. The complex $[Pt\ (H_2O)_2(NO_2)_2Br_2]$ exists in two different forms having two different colours. This is possible because it exhibits
(a) geometric isomerism (b) ionization isomerism
(c) coordination isomerism (d) hydrate isomerism

94. Consider the complex $[Cr\ A_2B_4]$ which are represented as



are

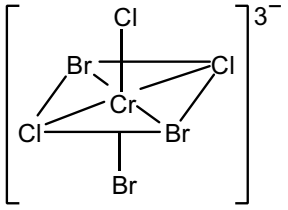
(a) same compounds (b) cis and trans form
(c) trans and cis form (d) both are trans forms of same isomer

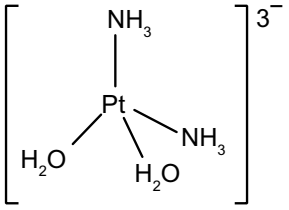
95. Which among the following is true regarding the complex $[Co(NH_3)_3Cl_3]$
(a) it is colourless
(b) it exhibits optical isomerism
(c) cis-variety is optically active
(d) it has two geometrical isomers facial and meridional

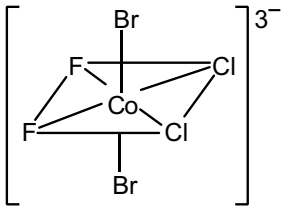
96. Consider the complex of the type $CuCl_2.nH_2O$ where n is a positive integer. The complex with highest conductivity formed from this will
(a) show optical activity
(b) show geometric isomerism
(c) give white precipitate with $AgNO_3$
(d) contain both neutral and negative ligands

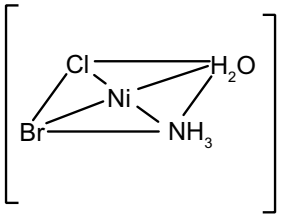
97. The geometrical isomers possible for the complex ion $[Co(NH_3)_2(H_2O)_2Cl_2]^+$ is
(a) 0 (b) 2 (c) 5 (d) 8

98. Which among the following in optically active

(a) 

(b) 

(c) 

(d) 

99. $KMnO_4$ solution cannot be used to estimate
(a) Iron (b) Nitrites (c) Iodide (d) Propene

100. During chromyl chloride test, if the yellow solution obtained with NaOH is neutralized with H_2SO_4 and a few drops of excess is added. The colour of solution will now be
(a) orange (b) colourless (c) yellow (d) white

101. Manganese dioxide can be estimated using
(a) HCl, dichromate
(b) Oxalic acid, permanganate
(c) HCl, thiosulphate
(d) H_2SO_4, permanganate

102. The colour change observed when ferric chloride is treated with $SnCl_2$ is
(a) green to yellow (b) Fenton's reagent (c) yellow to green (d) yellow to brown

103. Among the following reagents, the one that is prepared from copper sulphate is
(a) Tollens Reagent (b) Baeyer's solution (c) Benedicts Solution (d) Jone's Reagent

104. The acid (s) given out on treating $AgNO_3$ solution with PH_3 is
(a) HNO_2 and H_3PO_4 (b) HNO_2 and H_3PO_3 (c) HNO_3 and H_3PO_4 (d) HNO_3 and H_3PO_3

105. Metallic copper will dissolve in
(a) dil. HCl (b) aqueous $FeCl_3$ (c) aqueous NaOH (d) aqueous $ZnSO_4$

106. The compound that will not turn dichromate solution green is
(a) H_2O_2 (b) SO_2 (c) $FeCl_2$ (d) NaI

107. Which of the following statements is wrong?
(a) PbI_4 does not exist.
(b) SnI_4 has orange colour due to charge-transfer spectrum.
(c) $(NH_4)_2Cr_2O_7$ on decomposition gives N_2O.
(d) The equivalent weight of $KMnO_4$ in alkaline medium is 52.67.

108. The wrong statement about Fe and its compounds is
(a) $Fe(OH)_3$ dissolves in acid but not in excess base.
(b) $Fe(OH)_3$ has a very low solubility product value of the order of 10^{-39}.
(c) when Fe reacts with HCl in absence of air, it gives $[Fe(H_2O)_6]^{3+}$; but in presence of air it gives $[Fe(H_2O)_6]^{2+}$.
(d) when Fe reacts with HCl in absence of air, we have $[Fe(H_2O)_6]^{2+}$; but in presence of air, Fe(II) oxidizes to give $[Fe(H_2O)_6]^{3+}$.

109. $KMnO_4$ is mixed with con.H_2SO_4 accidentally instead of dil H_2SO_4 for a certain titration in acidic medium. The outcome would be
(a) The titre values be less than expected as H_2SO_4 is oxidizing.
(b) Titration cannot be done as con. H_2SO_4 will char the other substance to be estimated.
(c) The titre value would be greater than expected due to reaction of $KMnO_4$.
(d) An explosion would occur.

110. The compound that can liberate chlorine gas from con.HCl is
(a) PbI_2 (b) PbO_2 (c) SnO (d) SnI_4

Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

111. Statement 1
Ligands like CO, CN^- etc stabilize the metal complex.

and

Statement 2
CO, CN^- are strong field ligands and increase the crystal field stabilization energy.

112. Statement 1
The freezing point depression of aqueous solution of $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_5Cl]Cl_2$. H_2O are different.

and

Statement 2
Even though both the complexes have same empirical formula, $[Cr(H_2O)_6]Cl_3$ does not show geometric isomerism while $[Cr(H_2O)_5Cl]$. H_2O has a geometric isomer.

113. Statement 1
All the halides of iron are prepared by direct reaction of the metal with halogen except FeI_3.

and

Statement 2
Electron affinity of halogens is in the order Cl > Br > F > I.

Linked Comprehension Type Questions

Directions: **This section contains 1 paragraph. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

KCl is an important compound used in the preparation of salt bridge to make EMF measurements. The structure of KCl has been studied by X-ray. The structure predicted by X-rays for KCl is incorrect. KOH in ethanol is used for dehydrohalogenation

of alkyl halides. Compound (A) is another alkali metal chloride, whose structure is fcc from X-ray studies. (B) and (C) are other halides of the alkali metal in (A). (A), (B), (C) are colourless but (D) is another metal halide (MX) which is yellow coloured. MX is not soluble in water but the nitrate, fluoride and perchlorate of (M) are soluble. The metal (M) forms two coordinate complexes. (M') is another metal which has d^{10} configuration. The metal (M') gives sulphide precipitate in group IV cation analysis. Potassium in combination with a metal (E) with d^5s^2 configuration along with oxygen forms another important compound (F) used in volumetric analysis.

114. The colour of (F) is due to
(a) d-d transition (b) charge transfer (c) polarization of anion (d) its paramagnetism

115. Which of the statements about metal (M') is wrong?
(a) (M') has high melting point
(b) (M') forms compounds only in +2 oxidation state
(c) (M') rarely behaves as a transition metal
(d) Salts of (M') are white

116. The metal (E) in +2 oxidation state, forms a complex with cyanide ion having octahedral geometry which has the spin only magnetic moment equal to
(a) 2.83 BM (b) 3.87 BM (c) 4.9 BM (d) 1.73 BM

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers out of which ONE OR MORE answers will be correct.**

117. In a complexometric titration, the advantage of using a polydendate ligand as against a monodendate ligand is (are)
(a) complex formed is stable and reaction proceeds to completion
(b) error is minimized
(c) complex formed are strongly coloured
(d) complex formed is always high spin

118. Among the following the complexes which are paramagnetic in nature are
(a) $[Cu(NH_3)_4]SO_4$ (b) $[Co(NH_3)_6]Cl_3$ (c) $[Cr(CN)_6]^{3-}$ (d) $[Ni(CN)_4]^{2-}$

119. The process that would produce zinc oxide is (are)
(a) heating basic zinc carbonate
(b) heating zinc nitrate
(c) treating zinc chloride with bicarbonate
(d) heating zinc in steam to red heat

Matrix-Match Type Question

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

120.

Column I	**Column II**
(a) $ZnCl_2.2H_2O$	(p) forms oxide of lower oxidation state
(b) $K_2Cr_2O_7$	(q) liberates a gas
(c) $KMnO_4$	(r) forms oxy salt
(d) $CuSO_4.5H_2O$	(s) changes colour

ADDITIONAL PRACTICE EXERCISE

Subjective Questions

121. Calculate EAN for $V(CO)_6$, $Mn_2(CO)_{10}$ and $Fe_3(CO)_{12}$.

122. The nitroprusside ion $[Fe(CN)_5NO]^{2-}$ is diamagnetic. Briefly discuss the oxidation state of iron and nature of the ligand 'NO'.

123. Chromium exhibits similarity to sulphur. Give three examples in which this aspect is illustrated.

124. The heploxide of manganeseMn_2O_7 is volatile and explosive. It is an acidic oxide producting MnO_4^- ions in aqueous solution. Its salt $AgMnO_4$ is sparingly soluble. Identify another oxide behaving similarly.

125. $K_2Cr_2O_7$ is fused with Borie acid and the product formed is heated with water. A green product is obtained. What is the name of the product formed? Give its chemical composition and use.

126. Chromyl chloride in the acid chloride of chromic acid. Justify.

127. $KMnO_4$ can be prepared from K_2MnO_4 by dissolving in water and passing CO_2 through the solution. How CO_2 facilitate the oxidation process?

128. When KNO_2 is added to $Co(NO_3)_2$ solution acidified with acetic acid a yellow precipitate of Fischer's salt also known as cobalt yellow or Indian yellow is formed. Give the chemical composition of the compound. In what name this compound is usually known?

129. How does aqueous $CuSO_4$ react with (a) KI (b) KCN in excess.

130. What happen when KI is added in excess to an aqueous solution of $HgCl_2$. Give the formula of the compound formed and its IUPAC name. Which reagent is prepared by making use of their reaction?

Straight Objective Type Questions

Directions: **This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

131. For the hypothetical complex $[Co(gly)(DMG)_2]$ the oxidation number and coordination number of cobalt are
(a) 6, 3 (b) 6, 2 (c) 4, 3 (d) 4, 2

132. Consider the complex $[PtCl_2(NH_3)_2[PtCl_4]$, the oxidation state of platinum in the cation and anion are
(a) +4, +4 (b) +2, +4 (c) +2, +2 (d) +4, +2

133. A certain metal complex containing weak field ligand has a magnetic moment of 5.92 BM. The metal in the same oxidation state, with a strong field ligand will have a magnetic moment
(a) zero (b) 1.73 BM (c) 2.8 BM (d) 4.9 BM

134. The organomercury compounds are unaffected by air or water. This stability may be due to
(a) linear structure of the compound
(b) low polarity of Hg-C bond
(c) very high bond energy
(d) same size of Hg and C

135. Arrange the following in increasing order of stability
(A) - metal-alkyl compounds
(B) - metal-alkynyl compounds
(C) - metal-aryl compounds
(a) B < A < C (b) C < B < A (c) A < C < B (d) A < B < C

136. In the octahedral complexes, the configurations having the same non-zero spin only magnetic moment in weak field and in strong field are
(a) d^1, d^2, d^3, d^8, d^9 and d^{10} (b) d^4 and d^5 (c) $d^6$and d^7 (d) d^5 and d^7

137. When CuCN (insoluble in water) is shaken with aqueous KCN, the freezing point of aqueous solution will
(a) increase (b) decrease (c) remain unchanged (d) be zero Kelvin

138. A complex of cobalt contains 34.65% of SO_4^{2-}, 12.81% of Cl^-, 24.77% of NH_3 and remaining the metal. The empirical formula of the complex is same as molecular formula. The complex is diamagnetic. Water of hydration may be present. It gives precipitate with $AgNO_3$. The complex is
(a) $[Co(SO_4)(NH_3)_4(H_2O)]Cl$ (b) $[CoCl(NH_3)_4(H_2O)]SO_4$
(c) $[CoCl(NH_3)_5]SO_4$ (d) $[Co(SO_4)(NH_3)_5]Cl$

139. Even though scandium is a transition metal, it does not form alkene complex because
(a) it has a small size (b) Sc^{3+} has inert gas configuration
(c) d^1 configuration does not allow back bonding (d) alkene complex is of very large size

140. The absorption spectrum of a complex shows a single band at 240 kJ mol^{-1}. The wavelength of the d – d transition is approximately
(a) 500nm (b) 830 nm (c) 500μm (d) 830Å

141. Consider a complex formed with a lanthanide ion as the central metal atom. The colour of such a complex
(a) changes with any change in ligand
(b) becomes intense for a strong field ligand
(c) becomes colorless for a weak field ligand
(d) remains a constant irrespective of the ligand

142. For a metal complex, the ligand that does not particularly favour a low valent state or zero valent of the metal is
(a) CN^- (b) CO (c) EDTA (d) dipyridyl

143. When pink coloured hydrated $CoCl_2.6H_2O$ is heated it losses two molecules of H_2O and the complex so formed is blue coloured. The change incolour is due to
(a) change in oxidation state (b) change in hybridization of Co^{2+}
(c) change in the type of ligand (d) increase in number of ligands

144. The moity whose presence will prevent the precipitation of barium as its sulphate is
(a) EDTA (b) Cl^- (c) NH_3 (d) CN^-

145. The reaction $Cu^{2+} + 4NH_3 \rightarrow [Cu(NH_3)_4]^{2+}$ may be considered as
(a) an auto redox reaction (b) oxidation of copper ions
(c) acid base neutralisation (d) reduction of ammonia

146. The ligand in a complex that can show ligand isomerism is
(a) aniline (b) diaminopropane (c) ethylene diamine (d) dipyridyl

147. A square planar complex $[MA_4]$ adds two molecules of water to give $[MA_4(H_2O)_2]$, the new compound is
(a) a trans isomer (b) a cis isomer
(c) would not show cis trans isomerism (d) about 90% cis and 10% transmixture

148. A coordination compound of empirical formula.$[PtBr_2(H_2O)_2]$ does not give water with con H_2SO_4, nor any precipitate with $AgNO_3$. It gives two ions in aqueous solution. The complex exhibits
(a) ionization isomerism
(b) geometric isomerism
(c) coordination isomerism
(d) linkage isomerism

149. The total number of stereoisomers possible for the complex $[CrCl_2(ox)_2]^{3-}$ is same as total number of isomers possible for
(a) $[Pt(NH_3)_2Cl_2]$
(b) square planar [M abcd]
(c) $[Co(NH_3)_5Br]NO_2$
(d) $[Cr(H_2O)_6]Cl_3$

150. Consider the complex $[Cr(H_2O)_6]Cl_3$. The hydrate isomer of this complex, which shows facial form
(a) gives 1 mole of AgCl with $AgNO_3$
(b) does not exist
(c) has an optically inactive trans form
(d) shows reduction in weight on leaving it over con.H_2SO_4

151. Consider a complex $Cr(NH_3)_4Cl_2I$. Two complexes A and B can be formed such that they give an ammonia soluble and ammonia insoluble precipitates respectively with $AgNO_3$ we can say that
(a) Both A and B have same hybridization and magnetic moment
(b) A and B have same hybridization and colours
(c) A and B have different hybridization but same magnetic moment
(d) A is low spin while B is a high spin complex

152. Both geometric as well as optical isomerism is possible in
(a) $[Cu(NH_3)_2Cl_2]$ (b) $[Cr(en)_3]Cl_3$ (c) $[Co(en)_2(OH)_2]Cl$ (d) $[Mn(en)_2Cl\ Br]$

153. The complex that will show facial and meridional isomerism is
(a) $[Co(NH_3)_2Cl_4]^-$ (b) $[Co(en)_3]^{3+}$ (c) $[Co(dien)_3]^{3+}$ (d) $[Co(EDTA)]^-$

154. A complex formed with a metal M associated with 6 different monodendate ligands will form
(a) 15 enantiomers
(b) 30 enantiomers
(c) 24 enantiomers including mirror images
(d) 30 diasteriomers

155. Consider the complex $[Zn(H_2O)(NH_3)BrCl]$. The complex
(a) has three different geometric isomers
(b) is chiral
(c) cannot be optically active due to plane of symmetry
(d) has three geometric isomers but of which one is optically active

156. The tetrahedral complex of zinc bisbenzoylacetanato zinc II, exists as two optically active forms, but this is not true for bis acetylacetanato zinc(II) because
(a) presence of aromatic C_6H_5 group favours optical activity
(b) tetrahedral complexes show optical activity only for large ions
(c) complex with ligand (acac)cannot be resolved due to small energy difference
(d) ligand (acac) shows symmetry

157. Identify the complex which can exists in two isomeric forms
(a) $[Pd(gly)_2]$ (b) $[Pt(en)_2]^{2+}$ (c) $[Ni(en)Cl_2]$ (d) $[ZnCl_2(NH_3)_2]$

158. chromic oxide on reaction with calculated amounts of ammonia gives a compound which on heating liberates a gas. The gas is
(a) oxygen (b) nitrogen (c) ammonia (d) hydrogen

159. A reagent A reacts with $CuSO_4$ as well as $ZnSO_4$ giving a reddish brown precipitate and white precipitate respectively. The reagent A on heating will liberate
(a) O_2 (b) CO
(c) CO_2 (d) N_2

160. The compound that will cause a white turbidity on reaction with $KMnO_4$ is
(a) Na_2SO_4 (b) SO_2 gas
(c) Na_2S (d) $Na_2S_2O_3$

161. On passing ozone through a solution obtained by treating pyrolusite ore with KOH in the presence of air, the solution would turn
(a) colourless (b) purple
(c) green (d) black turbid

162. The precipitate formed on treating aqueous $CuSO_4$ with KI is
(a) CuI (b) CuI_2
(c) $Cu(OH)_3$ (d) $[CuI_4]SO_4$

163. When an alkaline solution of potassium chromate is treated with H_2O_2, K_3CrO_8 is formed. The oxidation state of chromium changes by 1. Then K_3CrO_8 is
(a) a diamagnetic species (b) a colour species
(c) a tetraperoxo species (d) a diperoxo species

164. After the chromyl chloride test, if lead acetate was replaced by $AgNO_3$ for addition into the alkaline solution, the result would be
(a) no conspicuous reaction (b) a yellow solution
(c) a brick red precipitate (d) a white precipitate

165. $SnCl_2$ on reaction with a compound A, gives a white ppt B. B with excess $SnCl_2$ turns grey $SnCl_2$ is
(a) Oxidized and then reduced (b) Reduced and oxidized
(c) Reduced both the times (d) Oxidized both the times

166. An aqueous solution of ferric chloride can be reduced to ferrous chloride using tin because
(a) Sn^{2+} salts are reducing in nature (b) Tin is more stable in +2 state
(c) Tin is more stable in +4 state (d) solution of $FeCl_3$ is acidic

167. The gases liberated on treating red lead with HCl and H_2SO_4 are respectively
(a) Cl_2 and O_2 (b) $Cl_2$and SO_2
(c) H_2 and O_2 (d) H_2 and Cl_2

168. The pigment Rinmann's green is
(a) cobalt and zinc chloride (b) cobalt zincate
(c) Nickel boride and manganese (d) Manganese zincate

169. Copper sulphate dissolves in excess thiosulphate. During this process
(a) solution turns acidic
(b) oxidation state of copper does not change
(c) the colour of solution is discharged
(d) boiling of solution, makes product more stable

170. Consider the structures of chromate ion and dichromate ion. The true statement is
(a) The two chromiums in chromate, form a bond to form dichromate.
(b) Three Cr–O bonds in chromate are of equal length.
(c) Chromate is planar, while dichromate is non planar.
(d) In dichromate, Cr–O bond length in the tetrahedral are shorter than the Cr–O bond linking the two tetrahedra.

Assertion–Reason Type Questions

Directions: **Each question contains Statement-1 and Statement-2 and has the following choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False
(d) Statement-1 is False, Statement-2 is True

171. Statement 1
Comparing $Co(CO)_4$ and $V(CO)_6$, $Co(CO)_4$ dimerises and becomes stable while $V(CO)_6$ does not.

and

Statement 2
$V(CO)_6$ obeys EAN rule while $Co(CO)_4$ does not.

172. Statement 1
An aqueous solution of $CrCl_3$ and the complex $[Cr(H_2O)_6]Cl_3$ will have different colours.

and

Statement 2
Both H_2O and Cl^- are weak field ligands.

173. Statement 1
For formation of a complex, NF_3 is a weaker ligand than $N(CH_3)_3$.

and

Statement 2
NF_3 has a pyramidal structure, while $N(CH_3)_3$ is planar.

174. Statement 1
In the presence of EDTA, in alkaline solution, Ba^{2+} will not form any precipitate with SO_4^{2-} ions.

and

Statement 2
Ba^{2+} forms a chelated complex with EDTA of the type $[Ba(EDTA)]^{2-}$.

175. Statement 1
The complex $[Co(SCN)_4]^{2-}$ and its linkage isomer will have the same magnetic moment.

and

Statement 2
Linkage isomerism occurs when the ligand contains more than one atom which could donate an electron pair.

176. Statement 1
A square planar complex of the type $[MA_2(aa)]$ can exist only in the cis form and cannot exist as the trans isomer.

and

Statement 2
Square planar complex $[MA_2(aa)]$ can show cis trans isomerism if bidentate ligand (aa) is replaced by bindentate ligand (ab).

177. Statement 1
Ferrous sulphate on exposure to air turns brown.

and

Statement 2
Fe^{2+} has a d^6 configuration, and doesnot have unpaired electrons when bonded with strong field ligands.

178. Statement 1
Disproportionation of Cu^+ ions in aqueous is controlled by addition of Cl^- ions.

and

Statement 2
Cu^+ dissolves in excess Cl^- ions due to formation of complex.

179. Zn^{2+} ions are treated with a base to give a precipitate X.

Statement 1
X dissolves in both HCl as well as NaOH.

and

Statement 2
X is amphoteric and forms a tetrahedral complex.

180. Statement 1
The compound formed by passing ozone through silver is a good oxidizing agent.

and

Statement 2
Silver is stable only in the +1 state.



Linked Comprehension Type Questions

Directions: **This section contains 3 paragraphs. Based upon the paragraph, 3 multiple choice questions have to be answered. Each question has 4 choices (a), (b), (c) and (d), out of which ONLY ONE is correct.**

Passage I

Cuprous Iodide is obtained as a white crystalline powder by the action of potassium iodide on cupric sulphate solution. The unstable cupric iodide first formed immediately decomposes to yield cuprous iodide precipitate with liberation of iodine.

$$2CuSO_4 + 4KI \rightarrow 2CuI_2 + 2K_2SO_4$$

$$2CuI_2 \rightarrow 2CuI + I_2$$

The formation of cuprous iodide plus liberation of iodine serve as the basis for the iodometric estimation of cupric salts.

181. Even through Cu^{2+} is a stable oxidation state and $CuCl_2$ exists, CuI_2 decomposes immediately on formation because
(a) $CuCl_2$ exists as a dihydrate
(b) Iodine has a larger size than chlorine
(c) formation of a stable complex is possible in $CuCl_2$
(d) iodide ion is reducing in nature compared to Cl^-

182. Among the following compounds, the one that would stop the liberation of iodine is
(a) $FeSO_4$ (b) Ozone (c) atmospheric air (d) Chromium trioxide

183. In the given reaction, the colour of the final products formed apart from violet iodine is
(a) light blue (b) intense blue (c) white (d) reddish brown

Passage II

Chromium metal completely reacts with a molar H_2SO_4 solution to give a blue coloured solution A. When KCN is added to A a solution B is formed.

The formation of A occurs with evolution of hydrogen gas. A compound similar to A is formed on reaction with HCl but not with HNO_3

184. The coordination number and oxidation number of A is
(a) 6, 3 (b) 6, 2 (c) 4, 3 (d) 'A' is not a complex

185. Comparing A and B we can say that
(a) A is square pyramidal and B is octahedral.
(b) 'A' has greater magnetic moment than 'B'.
(c) 'B' shows geometric isomerism white A does not.
(d) A is a salt and B is a complex.

186. Formation of A occurs with H_2SO_4 and HCl but not with HNO_3. This is because
(a) HNO_3 is an oxidizing acid.
(b) Chromium is inert for reaction with HNO_3.
(c) due to slow reaction of HNO_3, Chromium reacts with atmospheric air.
(d) of higher ionistion energy of nitrogen.

Passage III

Nickel is consistently bivalent in its compounds. Complexes of nickel are blue, red or yellow. Nickel forms a tetraammine complex (A) which on reaction with con HCl forms a complex in which two of the NH_3 ligands are replaced by Cl^- ligands. The complex so formed exists in two forms (B) and (C).

187. The wrong statement about complex (B) and (C) is
(a) Both (B) and (C) will have same geometry
(b) Both (B) and (C) will have same magnetic moment
(c) (B) and (C) will react in different manner with ethylene diamine
(d) (B) and (C) can show geometric isomerism but not optical isomerism

188. During preparation of complex (A), the medium is changed from alkaline to acidic. Then
(a) the hydrated complex $[Ni(H_2O)_4]^+$ is formed
(b) the ammonia complex will not be formed
(c) The oxidation state of copper increases
(d) copper is reduced hence no complex is formed

189. For the octahedral complex of nickel $[Ni(NH_3)_4Cl_2]NO_2$, there is no possibility of
(a) ligand isomerism (b) linkage isomerism (c) Ionization isomerism (d) optical isomerism

Multiple Correct Objective Type Questions

Directions: **Each question in this section has four suggested answers out of which ONE OR MORE answers will be correct.**

190. Consider a complex $[Cu(NH_3)_5NO_2]$ Br. The types of isomerism it can show are
(a) Linkage isomerism (b) geometrical isomerism (c) ligand isomerism (d) ionization isomerism

191. The complexes which may hace zero crystal field stabilization energy are
(a) $[Hg(SCN)_4]^{2-}$ (b) $[CdCl_5]^{3-}$ (c) $[Cu(CN)_4]^{3-}$ (d) $[Fe(CN)^6]^{4-}$

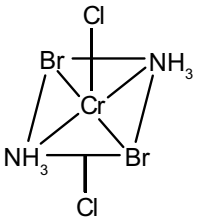
192. In an octahedral complex $[Ma_6]$ where all the ligands are weak field, the magnetic moment is 5.92 BM. If all the ligands are replaced by strong field ligands, the magnetic moment cannot be

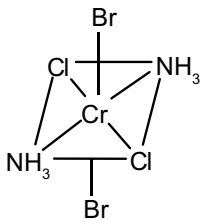
(a) 1.73 (b) 2.83 (c) 5.92 (d) 0

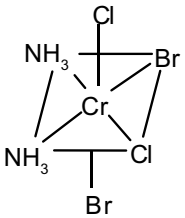
193. Among the following, the complexes that will not show any change in magnetic moment, on replacing a strong field ligand by weak field ligand or vice versa is/are

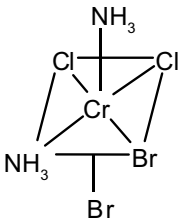
(a) $[Cr(C_2O_4)_3]^{3-}$ (b) $[NiCl_4]^{2-}$ (c) $[TiCl_6]^{2-}$ (d) $[MnF_6]^{4-}$

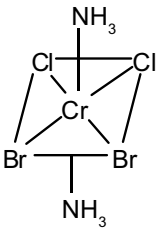
194. Identify the geometric isomer of the complex



(a) 

(b) 

(c) 

(d) 

195. The sulphate that liberates two different gases on heating is/are

(a) $FeSO_4.7H_2O$ (b) $CuSO_4.5H_2O$ (c) $FeSO_4$ (d) $AgNO_3$

196. Copper sulphide dissolves in sodium cyanide, in this reaction cyanide ions act as

(a) solvating agents (b) fluxating agents (c) reducing agent (d) complexing agents

197. Both $KMnO_4$ and $K_2Cr_2O_7$ are used in redox titrations . The advantage of using dichromate over permanganate is/are

(a) due to intense colour $K_2Cr_2O_7$ can be used as self indicator
(b) $k_2Cr_2O_7$ does not affect glass ware
(c) $k_2Cr_2O_7$ is a primary standared
(d) $k_2Cr_2O_7$ can be used in HCl medium

Matrix-Match Type Questions

Directions: **Match the elements of Column I to elements of Column II. There can be single or multiple matches.**

198. Column I has pair of isomers and Column II has methods of identifying

Column I	**Column II**
(a) cis and trans $[Pt(NH_3)_4Cl_2]$	(p) Dipole Moment
(b) $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_5Cl]Cl_2$	(q) NMR spectroscopy
(c) $[Co(NH_3)_5NO_3]SO_4$ and $[Co(NH_3)_5SO_4]NO_3$	(r) Freezing Point Depression
(d) two forms of $[Pd(gly)_2]$	(s) Conductivity

199.

	Column I		**Column II**
(a)	Optical isomerism	(p)	$[Co(H_2O)_2(NH_3)_2Cl_2]^+$
(b)	Cis and trans isomerism	(q)	$[Cr(ox)_3]^{3-}$
(c)	paramagnetic	(r)	$[NiCl_2(en)]$
(d)	metal ion in +2 state	(s)	$[CuBrCl(NH_3)_2]$

200.

	Column I		**Column II**
(a)	$AgNO_3$	(p)	NaOH
(b)	ZnO	(q)	NH_4OH
(c)	CuCl	(r)	HCl
(d)	$SnCl_2$	(s)	$Na_2S_2O_3$

SOLUTIONS

ANSWER KEYS

Topic Grip

1. Co^{3+} is a d^6 ion
$[Co(NH_3)_6]^{3+}$ is a low spin complex
$\mu = 0$

2. (i) tetrahedral – 36
(ii) trigonal bipyramidal – 36
(iii) tetrahedral – 36
(iv) octahedral – 36
(v) tetrahedral –34

3. (i) Polymerisation isomer
(ii) Cis – trans isomers
(iii) linkage isomers

4. Metal –carbonyl π back bonding strengthens metal carbon bond

5. Empirical formula $[Co(NH_3)_4(H_2O)(SO_4)Cl]$ complex is diamagnetic.
$[CoF_6]^{3-}$

6. (i) B is ethylene diamine chromium (III) chloride – octahedral $\mu = 3.87$ BM
(ii) Ammonium hexacyano manganate (III) octahedral: $\mu = 2.89$ BM
(iii) hexa aqua cobalt (II) sulphate:octahedral $\mu = 3.87$ BM

7. Geometrical isomers(B), C and E
Optical isomers B and C
Identical (A), D and E

8. The pink solution in MnO_4^-
A is $nMnO_2$
B is MnO_4^-

9. $CuSO_4.5H_2O \rightarrow CuSO_4$
$CuSO_4 \rightarrow CuO + SO_3$
$CuO + C \rightarrow Cu + CO$
$2CuSO_4 + 4KCN \rightarrow 2CuCN + 2K_2SO + C_2N_2$
$CuCN + 3KCN \rightarrow K_3(Cu(CN)_4]$

10. (i) $Cu(NO_3)_2$ and NO are formed
(ii) $Cu(NO_3)_2$ and NO_2 are formed
(iii) Cu_2I_2 and I_2 are formed

11. (c) **12.** (a) **13.** (a)
14. (b) **15.** (d) **16.** (c)
17. (d) **18.** (a) **19.** (d)
20. (b) **21.** (d) **22.** (d)
23. (c) **24.** (d) **25.** (c)
26. (c)
27. (a), (b)
28. (b), (c)
29. (a), (b)
30. (a) →(p), (q)
(b) →(p), (q), (r)
(c) →(p), (r)
(d) →(q), (s)

IIT Assingment Exercise

31. (a) **32.** (d) **33.** (d)
34. (b) **35.** (c) **36.** (b)
37. (a) **38.** (d) **39.** (c)
40. (c) **41.** (b) **42.** (a)
43. (d) **44.** (a) **45.** (a)
46. (b) **47.** (b) **48.** (d)
49. (c) **50.** (a) **51.** (b)
52. (d) **53.** (b) **54.** (b)
55. (c) **56.** (a) **57.** (c)
58. (d) **59.** (b) **60.** (a)
61. (b) **62.** (d) **63.** (c)
64. (d) **65.** (c) **66.** (b)
67. (b) **68.** (d) **69.** (c)
70. (b) **71.** (b) **72.** (d)
73. (b) **74** (d) **75.** (c)
76. (b) **77.** (d) **78.** (c)
79. (c) **80.** (a) **81.** (c)
82. (b) **83.** (d) **84.** (a)
85. (c) **86.** (c) **87.** (a)
88. (c) **89.** (b) **90.** (b)
91. (a) **92.** (b) **93.** (a)
94. (a) **95.** (d) **96.** (c)
97. (c) **98.** (c) **99.** (d)
100. (a) **101.** (b) **102.** (c)
103. (c) **104.** (d) **105.** (b)
106. **(a)** **107.** (c) **108.** (c)
109. (d) **110.** (b) **111.** (b)
112. (c) **113.** (c) **114.** (c)
115. (a)
116. (b)
117. (a), (b)
118. **(a)**, (c)
119. (a), (b), (d)
120. (a) → (r)
(b) →(p), (q), (r), (s)
(c) →(p), (q), (r), (s)
(d) →(q), (s)

Additional Practice Exercise

121. 36, 36, 36
122. Fe^{2+} and NO^+
123. CrO_3, SO_3 acidic CrO_2Cl_2 and SO_2Cl_2 similar
124. Cl_2O_7
125. $Cr_2O_3.2H_2O$ guignets green
126. $CrO_2(OH)_2$ is chromic acid CrO_2Cl_2 is chromylchloride
127. disproportionation of MnO_4^{2-} facilitated
128. $K_3[Co(NO_2)_6]$ potassium cobaltinitrate
129. with KI - I_2 and CuI with KCN - $K_3[Cu(CN)_4]$
130. K_2HgI_4 – Nesslerss reagent
131. (b) **132.** (d) **133.** (b)
134. (b) **135.** (d) **136.** (a)
137. (a) **138.** (a) **139.** (c)
140. (a) **141.** (d) **142.** (c)
143. (b) **144.** (a) **145.** (c)
146. (b) **147.** (a) **148.** (c)
149. (b) **150.** (d) **151.** (a)
152. (c) **153.** (c) **154.** (b)
155. (b) **156.** (d) **157.** (a)
158. (b) **159.** (d) **160.** (c)

161. (b) **162.** (a) **163.** (c)
164. (c) **165.** (d) **166.** (a)
167. (a) **168.** (b) **169.** (c)
170. (d) **171.** (c) **172.** (b)
173. (c) **174.** (a) **175.** (d)
176. (c) **177.** (b) **178.** (b)
179. (a) **180.** (a) **181.** (d)
182. (a) **183.** (c) **184.** (b)
185. (b) **186.** (a) **187.** (c)
188. (b)
189. (a)

190. (a), (d)
191. (a), (b), (c)
192. (b), (c), (d)
193. (a), (c)
194. (b), (c)
195. (b), (c), (d)
196. (c), (d)
197. (c), (d)
198. (a) → (p), (s)
(b) → (r), (s)
(c) → (s)
(d) → (p)

199. (a) → (p), (q)
(b) → (p), (s)
(c) → (q), (s)
(d) → (r), (s)
200. (a) → (p), (q), (r), (s)
(b) → (p), (q), (r)
(c) → (q), (s)
(d) → (p)

HINTS AND EXPLANATIONS

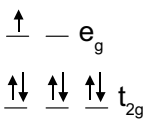
Topic Grip

1. Electronic configuration of Co – $3d^7 4s^2$.

$\therefore$ Co^{3+} – $3d^6$ and Co^{2+} – $3d^7$.

NH_3 is a strong field ligand and a low spin complex is formed. $[Co(NH_3)_6]Cl_2$ is hexaamminecobalt(II) chloride.

$d^7 - (t_{2g})^6 (e_g)^1$



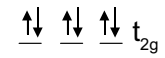
n = 1

$\therefore$ $\mu = \sqrt{n(n+2)} = 1.73$ BM.

$[Co(NH_3)_6]Cl_3$ is

hexaamminecobalt(III) chloride.

$d^6 - (t_{2g})^6 e_g^0$



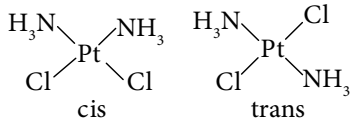
No unpaired electrons. Hence $\mu = 0$

2.

	Complex	Geometry	EAN
(i)	$Ni(CO)_4$	tetrahedral	28 – 0 + 8 = 36
(ii)	$Fe(CO)_5$	trigonal bipyramidal	26 – 0 + 10 = 36
(iii)	$[Zn(NH_3)_4]^{2+}$	tetrahedral	30 – 2 + 8 = 36
(iv)	$[Fe(CN)_6]^{4-}$	octahedral	26 – 2 + 12 = 36
(v)	$[NiCl_4]^{2-}$	tetrahedral	28 – 2 + 8 = 34

3. (i) $[CoCl_3(NH_3)_3]$ and $[Co(NH_3)_6][CoCl_6]$ are polymerization isomers. The latter is the dimer of the former.

(ii) $[PtCl_2(NH_3)_2]$ can exist as cis and trans isomers as given below.



This can also exhibit polymerization isomerism.

(iii) In $[Co(ONO)(NH_3)_5]Cl_2$, the metal is linked to oxygen of O – N – O and in the other complex $[CoNO_2(NH_3)_5]Cl_2$, the metal is linked to nitrogen. So they are linkage isomers.

4. In the carbonyl compounds, a carbon to metal σ bond is formed first, transferring electronic charge to metal. Apart from this, the nonbonding d orbitals of the metal, overlap with the antibonding orbitals of CO forming a π-bond. During this back bonding, electronic charge is transferred from the filled nonbonding orbitals of the metal to the antibonding orbitals of the ligand CO. σ-bond increases the electron density on metal making the back bonding stronger. π-bond increases the electron density on carbon making the σ bond stronger. Hence the two bonds are complimentary to each other or synergic. Back bonding increases the bond order of M-C bond and decreases that of C-O bond.

5.

constituent	%	Dividing with mol mass	Dividing with smallest quotient
Co	21.24	$\frac{21.24}{59} = 0.36$	1
NH_3	24.77	$\frac{24.77}{17} = 1.46$	4
Cl^-	12.81	$\frac{12.81}{35.5} = 0.36$	1
SO_4^{2-}	34.65	$\frac{34.65}{96} = 0.36$	1
H_2O	6.53	$\frac{6.53}{18} = 0.36$	1

The empirical formula is $Co(NH_3)_4(H_2O)(SO_4)Cl$

Empirical formula weight

$= 59 + 17 \times 4 + 18 + 96 + 35.5$

$= 276.5$ g. So the complex could be

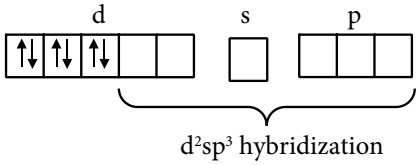
(i) $[CoCl(NH_3)_4(H_2O)]SO_4$

Tetraammineaquachlorocobalt(III) sulphate

or

(ii) $[Co(SO_4)(NH_3)_4(H_2O)]Cl$ Tetraammineaquasul phatocobalt(III) chloride.

$Co^{3+} \rightarrow d^6$. Since (NH_3) is a strong field ligand and complex is low spin we have the arrangement.



d^2sp^3 hybridization

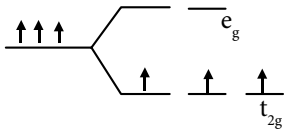
$[CoF_6]^{3-}$ is paramagnetic as it contains four unpaired electrons, sp^3d^2 hybridization.

6.

(i) $[Cr(en)_3]Cl_3$ Tris (ethylenediamine) chromium (III) chloride

CFSE Cr – $3d^5\ 4s^1$

Cr^{3+} – $3d^3$



$\mu = \sqrt{n(n+2)} = \sqrt{3(3+2)} = 3.87$ BM

Geometry – octahedral

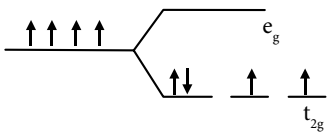
(ii) $(NH_4)_3[Mn(CN)_6]$ Ammonium hexacyano manganate (III)

Mn – $3d^5\ 4s^2$

Mn^{3+} – $3d^4$

CN^- is a strong field ligand so electron pairing occurs against hunds rule. So the complex is low spin.

CFSE



Unpaired electrons = 2

$\mu = \sqrt{2(2+2)} = 2.89$ BM

Geometry – Octahedral

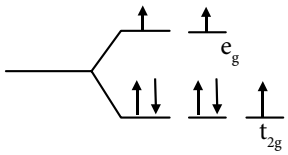
(iii) $[Co(H_2O)_6]SO_4$

Hexa aqua cobalt (II) sulphate

Co – $3s^7\ 4s^2$

Co^{2+} – $3d^7$

H_2O is a weak field ligand. So electron pairing does not occur. It is a high spin complex.

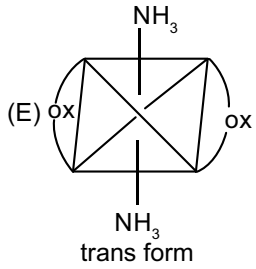


Unpaired electrons = 3

$\mu = \sqrt{3(3+2)} = 3.87$ BM

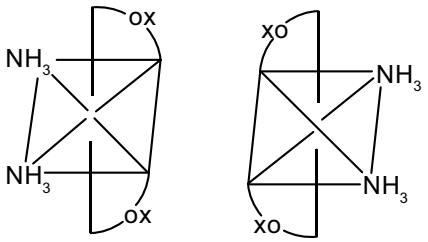
Geometry – Octahedral

7. (a) Geometric isomers



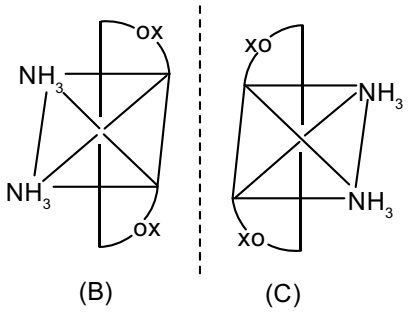
A, D and E are trans forms

(B) and (C) are cis forms



(b) Optical isomers

(B) and (C) are non super imposable mirror images.



(c) Identical structures

(A), (D), (E) are all identical structure.

All of them are the trans isomers.

8. $Mn \rightarrow d^5s^2$ (A)$\rightarrow MnO_2$

$\underset{(B)}{MnO_4^-} + 2H_2O + 3e^- \rightarrow \underset{(A)}{MnO_2} + 4OH^-$

$MnO_2 + 4HCl \rightarrow Mn^{2+} + 2Cl^- + Cl_2 + 2H_2O$

$2MnO_2 + 2H_2SO_4 \rightarrow 2MnSO_4 + O_2 + 2H_2O$

$MnO_2 + 2KNO_2 \rightarrow K_2MnO_4 + 2NO$

$K_2MnO_4 + 2H_2O \rightarrow \underset{(B)}{KMnO_4} + KOH + H_2$

9. (i) Blue vitriol – $CuSO_4.5H_2O$

$CuSO_4.5H_2O \xrightarrow{100^\circ C} CuSO_4.H_2O \xrightarrow{230^\circ C}$

$CuSO_4 \xrightarrow{720^\circ C} CuO + SO_3$.

$CuO + C \xrightarrow{\Delta} Cu + CO$

Product formed is copper metal.

(ii) $2Cu^{2+} + 4CN^- \rightarrow 2CuCN + (CN)_2$

$CuCN + 3CN^- \rightarrow [Cu(CN)_4]^{3-}$

CuCN is a precipitate, which dissolves in excess CN^- to form a complex which is tetrahedral.

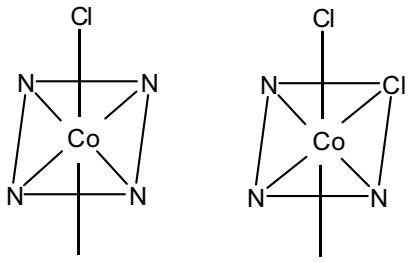
10. (i) $3Cu + 8HNO_3 \rightarrow 2NO + 3Cu(NO_3)_2 + 4H_2O$

(ii) $Cu + 4HNO_3 \rightarrow 2NO_2 + Cu(NO_3)_2 + 2H_2O$

(iii) $2Cu^{2+} + 4I^- \rightarrow 2CuI + I_2$

$2Na_2S_2O_3 + I_2 \rightarrow Na_2S_4O_6 + 2NaI$

11.



trans form optically inactive

cis form optically active

12. They are not isomers

Optical isomerism is not seen in square planar geometry.

Note: Correct option is (d)

13. Central metal atom is Ag

Exists as Ag^+ and is surrounded by $2CN^-$ ligands.

14. Greater the number of charges on the ions, greater the molar conductivity.

(a) no ions, no charge

(b) 4 negative and 4 positive charges

(c) one positive charge and one negative charge

(d) one positive charge and one negative charge

15. (a) $Cr_2O_7^{2-} + 3H_2O_2 + 8H_3O^+ \rightarrow 2Cr^{3+} + 3O_2 + 15H_2O$

(b) $Cr_2O_7^{2-} + 3H_2S + 8H^+ \rightarrow 3S + 2Cr^{3+} + 7H_2O$

(c) $2Cr^{2+} + H_2O_2 + 2H_3O^+ \rightarrow 2Cr^{3+} + 4H_2O$

(d) $2CrO_4^{2-} + 2H_3O^+ \rightarrow Cr_2O_7^{2-} + 3H_2O$

Formation of Cr^{3+} gives green colour.

16. Oxalic acid removes rust stains from fabric through complex formation.

Iron coordinates easily through oxygen and forms complex with oxygen containing ligands.

Rust ($Fe_2O_3.xH_2O$) is reduced to ferrous and this forms a soluble complex with oxalic acid and hence removed form clothes.

$2Fe^{3+} + H_2C_2O_4 \rightarrow 2Fe^{2+} + 2H^+ + 2CO_2$

$2Fe^{2+} + 4H_2C_2O_4 \rightarrow 2Fe(C_2O_4)_2 + 4H^+$

$2Fe(C_2O_4)_2 + 2H_2C_2O_4 \rightarrow 2[Fe(C_2O_4)_3]^{4-} + 4H^+$

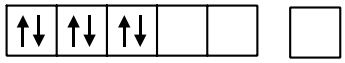
17. In metal carbonyls, metal exhibits zero or low oxidation state to allow back bonding i.e., Flow of electrons from metal to ligand. In $V(CO)_6$ and $Cr(CO)_6$, $V(CO)_6$ is paramagnetic while $Cr(CO)_6$ is diamagnetic.

In this carbonyl, both metals have zero oxidation state and have sp^3d^2 hybridizaition.

Carbonyl being strong ligand, electron pairs against hund's rule.

$Cr(CO)_6$

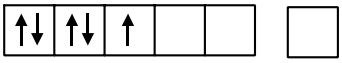
$Cr - 3d^5\ 4s^1$

In $Cr(CO)_6$ 

No unpaired electrons hence diamagnetic.

$V(CO)_6$

$V - 3d^3\ 4s^2$

In $V(CO)_6$ 

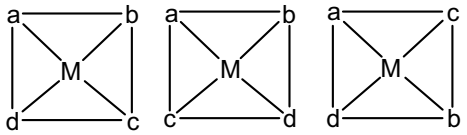
Presence of one unpaired electron hence paramagnetic

$\mu = 1.73$ BM

18. The colour of complex depends on the CFSE which in turn depends on the metal ion, its oxidation state and the type of ligands.

19. Square planar complexes have a plane of symmetry because metal and all the four ligands lie in the same plane.

Square planar complexes of the type [Mabcd] show only three geometric forms.



All other combinations will be a repetition of one of these forms.

20. Dilute HCl, dilute H_2SO_4 are non oxidizing acid. They do not react with copper in the absence of air. In the presence of air they dissolve Cu.

$$Cu + \underset{dil}{H_2SO_4} + \frac{1}{2}O_2 \longrightarrow CuSO_4 + H_2O$$

Concentrated sulphuric acid is oxidizing. It oxidizes Cu and itself gets reduced liberating SO_2 gas

$$Cu + \underset{con.}{2H_2SO_4} \longrightarrow CuSO_4 + 2H_2O + SO_2\uparrow$$

21. $4K_2Cr_2O_7 \longrightarrow 4K_2CrO_4 + 2Cr_2O_3 + 3O_2$

22. $\underset{orange}{K_2Cr_2O_7} + 2KOH \longrightarrow \underset{yellow}{K_2CrO_4} + H_2O$

Both are in 6+ oxidation state.

The reaction is reversible

$$Cr_2O_7^{2-} + H_2O \rightleftharpoons 2CrO_4^{2-} + 2H^+$$

23. H_2S is oxidized to sulphur

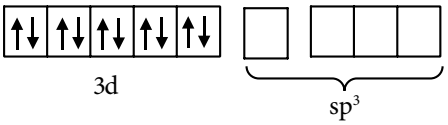
$$Cr_2O_7^{2-} + 3H_2S + 8H^+ \rightarrow 2Cr^{3+} + 3S + 7H_2O$$

24. Nickel unlike other metals forms a volatile complex with carbon monoxide. Hence it is used in refining by vapour phase refining. The complex is $[Ni(CO)_4]$.

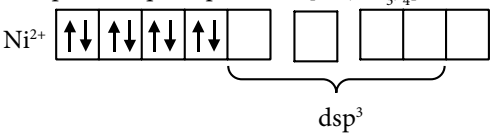
25. $Ni(CO)_4$ tetrahedral geometry

Ni is in zero oxidation state.

Ni in $Ni(CO)_4$



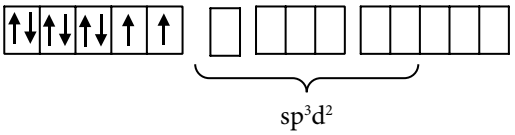
Complex B, square plannar $[Ni(Nh_3)_4]^{2+}$



$\mu = 0$

Complex C octahedral $[Ni(NH_3)_4(H_2O)_2]^{2+}$

Ni^{2+}

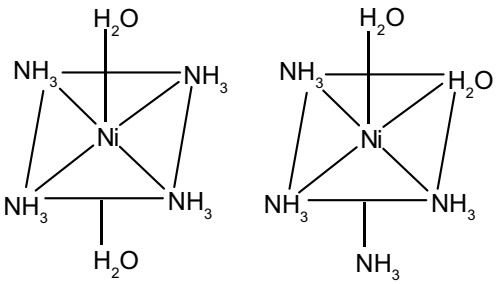


$$\mu = \sqrt{n(n+2)} = \sqrt{2(2+2)} = 2.83 \text{ BM}$$

26. A does not exhibit geometric isomerism because it is tetrahedral.

In B all four ligands are same. So it will not show geometric isomerism.

(c) is $[Ni(NH_3)_4(H_2O)_2]$ will show cis and trans forms



27. (a) Mond's process proceeds through formation of $[Ni(CO)_4]$

(b) Froth flotation. Zn is deactivated in the presence of lead by addition of NaCN. The complex $Na_2[Zn(CN)_4]$ is formed.

28. Due to difference in number of ions formed, there will be a difference in the freezing point measurement.

$AgNO_3$ test—one mole of each complex on reaction with excess $AgNO_3$ will give two moles and three moles of AgCl respectively.

Since Co is in the same oxidation state and number of NH_3 ligands differ by one only there will be no difference in magnetic moment.

MnO_2 test is not quantitative hence number of Cl^- ions cannot be identified.

29. Prussian blue is formed by reaction of Fe^{3+} salts with $K_4[Fe(CN)_6]$

$$Fe^{3+} + [Fe(CN)_6]^{4-} \longrightarrow \underset{\text{Prussian blue}}{Fe[Fe(CN)_6]}$$

FeF_3 does not give any test for Fe^{3+} due to very low ionization.

30. (a) → (p), (q)
(b) → (p), (q), (r)
(c) → (p), (r)
(d) → (q), (s)

$$[Co(NH_3)_5NO_3]SO_4 - \underset{\text{Ionisation I}}{[Co(NH_3)_5SO_4]NO_3}\ ,$$

$$\underset{\text{ammoniate I}}{[Co(NH_3)_4NO_3SO_4].NH_3}$$

$$[Cr(H_2O)_5NO_2]Cl_2 - \underset{(P)}{[Co(H_2O)_5Cl_2]NO_2}\ ,$$

$$\underset{(R)}{[Co(H_2O)_5ONO]Cl_2}\ ,$$

$$\underset{(Q)}{[Co(H_2O)_4NO_2Cl]Cl.H_2O}$$

$$[Co(en)_2NO_2Cl]SCN - \underset{(P)}{[Co(en)_2NO_2SCN]Cl}$$

$$\underset{(R)}{[Co(en)_2ONO,Cl]SCN}$$

$$[Co(NH_3)_4(CH_3C_6H_4NH_2)Cl]Cl_2$$

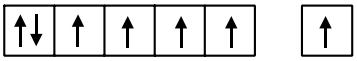
$$\underset{(Q)}{[Co(NH_3)_3(CH_3C_6H_4NH_2)Cl]Cl.NH_3}$$

$CH_3C_6H_4NH_2$ can exist as ortho meta or para forms.

IIT Assignment Exercise

31. The complex is $[Ni(CN)_6]^{4-}$
Since CN has −1 charge, Ni is in the +2 oxidation state.

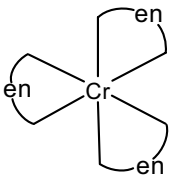
32. $Na_2S + Na_2[Fe(CN)_5NO] \rightarrow Na_4[Fe(CN)_5NOS]$

33. The brown ring complex formed is $[Fe(H_2O)_5NO]SO_4$ here Fe is in +1 state.
ie., Ar 
five unpaired electrons.

34. ClO_4^- has −1 charge
So complex ion is $[Cr(H_2O)_5CO_3]^+$
H_2O is neutral and CO_3^{2-} has −2 charge
So charge on Cr is +3
It is coordinated to $5H_2O$ and one CO_3^{2-}
Hence coordination number is 6.

35. Since complex A contains bidendate ligand, it forms a chelate structure. This increases the stability of the ligand.

36. $Na_3[Cu(CN)_4]$ gives 4 ions per mole which is maximum among the four

37. 

en is $NH_2 - CH_2 - CH_2 - NH_2$
Since it is symmetric it will not show geometric isomerism.

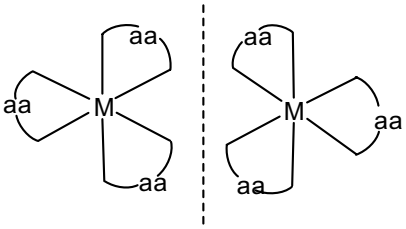
38. EDTA is a hexadendate ligand with
charge = −4
So complex is $[Co(EDTA)]^-$
So oxidation number = +3
Coordination number = +6

39. The complex is square planar. So the hybridization is dsp^2.

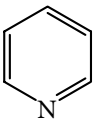
40. There are three cations and two anions.

41. (b)

42. In $[CuI_4]^{2-}$, I^- is a reducing ion. So it would reduce Cu^{2+} to Cu^+ if making such a complex is attempted. So $[CuI_4]^{2-}$ does not exist.

43. 

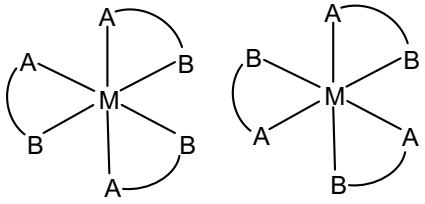
44. The structure of copper sulphate pentahydrate is $[Cu(H_2O)_4]SO_4.H_2O$

45. 

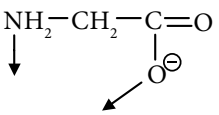
Pyridine coordinates through the nitrogen.

46. Linkage isomerism is seen when the ligand can coordinate to the metal through one or more centres. This is possible in an ambidendate ligand.

47.



When the bidendate ligand is unsymmetric, the complex would show geometric isomerism. Glycinato is

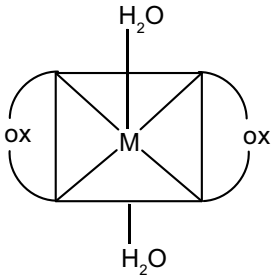


Symmetric bidendate ligands would have similar atoms, so they will not show geometric isomerism.

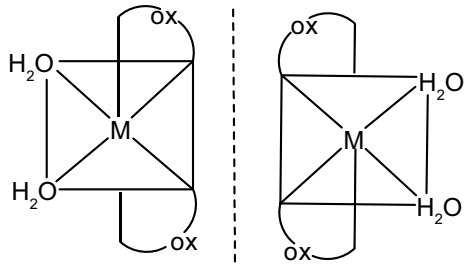
48. $[Cr(en)_3]^{3+}$ shows optical isomerism but does not show geometric isomerism.

(a), (c) show both geometric as well as optical isomerism (b) does not show both stereoisomerism.

49. It forms one optically active cis isomer, so there are two d and l isomer and an optically inactive trans isomer.



trans form
optically inactive



cis form
optically active

50. The complex exhibits geometric isomerism and has different colours for its cis and trans forms.

51. Complex of the type Ma_3b_3 exhibit meridional and facial isomers.

52. $[Co(H_2O)_5NO_2]SO_4$ and $[Co(H_2O)_5SO_4]NO_2$– ionization isomers

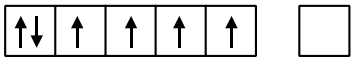
$[Co(H_2O)_5NO_2]SO_4$ and

$[Co(H_2O)_4NO_2SO_4].H_2O$ Hydrate isomer

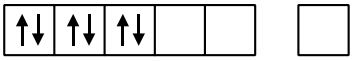
Complex of the type MA_5B do not show geometric isomerism.

53. All four positions are equivalent and are at the same bond angles. So all four ligands are adjacent to each other, so there can be no geometric isomerism.

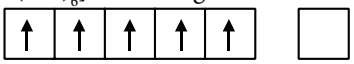
54. There is a change in the ligands with respect to the metal atom.

55. Fe^{2+} 

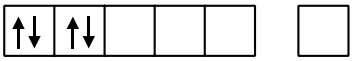
CN^- is a strong ligand so electron pairs giving



So $[Fe(CN)_6]^{4-}$ is diamagnetic

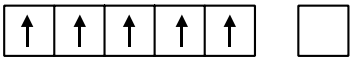
Fe^{3+} 

In $[Fe(CN)_6]^{3-}$



So it is paramagnetic with $\mu = 1.73$ BM

56. $[NiCl_2Br_2]^{2-}$ has a tetrahedral structure. So it will not show geometric isomerism.

57. Fe^{3+} 

As F^- is weak ligand electron pairing does not take place. So there are five unpaired electrons.

$$\mu = \sqrt{5(5+2)} = \sqrt{35} = 5.85 \text{ BM}$$

58. Zn has d^{10} configuration. So CFSE will always be O.

59. There is no interaction between metal orbitals and ligand orbitals.

Ligands are treated as point charges.

60. In the metal. CN^- complex formation, the filled t_{2g} orbitals of metal donates electrons to the vacant orbitals of the ligands.

This increases the CFSE thereby making CN^- a strong field ligand.

61. SnI_4 is coloured due to polarization.

62. $FeSO_4.(NH_4)_2SO_4.6H_2O$.
Mohr's salt is less readily oxidized in solid state.

63. $Na_2Cr_2O_7$ is hygroscopic in nature. So during the process of weighing the substance and making a standard solution error would creep in.

64. On prolonged exposure to oxygen, carbondioxide and moist air, copper forms basic copper carbonate.
$2Cu + O_2 + CO_2 + H_2O \rightarrow Cu_2(OH)_2CO_3$

65. In acid medium
$2CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{2-} + H_2O$
In alkaline medium
$Cr_2O_7^{2-} + 2OH^- \rightarrow CrO_4^{2-} + H_2O$

66. Preparation of $K_2Cr_2O_7$
$Na_2Cr_2O_7 + 2KCl \rightarrow 2NaCl + K_2C_2O_7$
Due to low solubility of $K_2Cr_2O_7$. It crystallizes out.

67. In acidic medium
$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$

$\therefore$ Eq. wt. $= \frac{158}{5} = 31.6$

In alkaline medium
$2KMnO_4 + 2KOH \rightarrow 2K_2MnO_4 + H_2O + (O)$

Eq. wt $= \frac{158}{1} = 158$

In neutral medium
$2KMnO_4 + H_2O \rightarrow 2MnO_2 + 2KOH + 3(O)$

Eq.wt $= \frac{158}{3} = 52.67$ g

68. As $KMnO_4$ decomposes in the presence of sunlight, it is stored in dark bottles to avoid decomposition.

69. MnO_4^{2-} disproportionates to MnO_4^- and MnO_2
$3K_2MnO_4 + 2H_2O + 4CO_2 \rightarrow$
$2KMnO_4 + MnO_2 + 4KHCO_3$

70. Ag_2O is basic. So it instantly absorbs CO_2 gas to form Ag_2CO_3.

71.

AgBr	–	pale yellow ppt
Ag_3PO_4	–	yellow ppt
Ag_2S	–	black
Ag_2CrO_4	–	brick red

72. $Ag_2S_2O_3$ formed is soluble in excess $Na_2S_2O_3$
$2Ag^+ + Na_2S_2O_3 \rightarrow Ag_2S_2O_3 + 2Na^+$
$Ag_2S_2O_3 + 3Na_2S_2O_3 \rightarrow$
$2Na_3[Ag(S_2O_3)_2]$
Sodium argento thiosulphate

73. Heating basic zinc carbonate is a preparatory procedure for pure ZnO.
$ZnCO_3.3Zn(OH)_2 \xrightarrow{\Delta} 4ZnO + 3H_2O + CO_2$

74. $ZnSO_4 \xrightarrow{800°C} ZnO + SO_3$
SO_3 decomposes to SO_2 and O_2

75. Only ZnS is white coloured
So the metal chloride is $ZnCl_2$
$ZnCl_2$ dissolves in NaOH forming a complex
$ZnCl_2 + 2NaOH \rightarrow Zn(OH)_2 + 2NaCl$
$Zn(OH)_2 + 2NaOH \rightarrow Na_2ZnO_2 + 2H_2O$

76. The function of Zn is reducing agent.
Among the options a reducing gas is SO_2
$2CuCl_2 + Zn \rightarrow Cu_2Cl_2 + ZnCl_2$
$2CuCl_2 + 2H_2O + SO_2 \rightarrow$
$Cu_2Cl_2 + 2HCl + H_2SO_4$

77. This is a commercial method of preparation of copper sulphate.
From the solution so formed $CuSO_4$ is crystallized out.
$Cu + H_2SO_4 + \frac{1}{2}O_2 \rightarrow CuSO_4 + H_2O$

78. On exposure to atmosphere $FeSO_4$ forms basic ferric sulphate.
$4FeSO_4 + 2H_2O + O_2 \rightarrow 4Fe(OH)SO_4$

79. Aqueous solution of $FeCl_3$ is acidic due to hydrolysis.
$FeCl_3 + 3H_2O \rightleftharpoons Fe(OH)_3 + 3HCl$
Addition of HCl will prevent this hydrolysis

80. Pb^{4+} is less stable due to inert pair effect. So PbO_2 is good oxidizing agent so Cl^- is oxidized to Cl_2
$PbO_2 + 4HCl \rightarrow PbCl_2 + Cl_2 + 2H_2O$

81. +3
When sodium nitrite in acetic acid is added to Co^{2+}, the nitrite ion oxidize Co^{2+} to Co^{3+} and then cobalt-nitrite complex is formed

$Co^{2+} \rightarrow Co^{3+} + e^-$

$NO_2^- + 2H^+ + e^- \rightarrow NO + H_2O$

$Co^{3+} + 6NO_2^- \rightarrow \underset{\text{hexanitrito-N cobaltate ion}}{\left[Co(NO_2)_6\right]^{3-}}$

82. $[Co(EDTA)]^{2-}$ is a more stable complex due to formation of chelates. So the reaction is expected to be feasible.

83. CO and $(NO_2)_5$ have to be written in alphabetical order and oxidation number of Iron is II. The correct name is sodium carbonylpentanitrito-N-ferrate(II).

84. $4KI + HgCl_2 \rightarrow \underset{\text{complex formation}}{K_2HgI_4} + 2KCl$

85. For a given ligand, the metal with highest charge by size ratio will form the most stable complex Cs^+ has large size, hence least charge/size ratio.

86. Fe^{3+} in alkaline solution gets converted to insoluble $Fe(OH)_3$, so it cannot be absorbed by the plants. So it is added as soluble Fe(III)–EDTA complex which is water soluble.

87. Oxidation state of nitrogen is same as that of nitrogen in ammonia.

88. From description complex is $[NH_4]_2\,[MCl_4]$
so metal is in +2 oxidation state.
Only manganese has 5 unpaired electrons in +2 state.

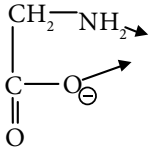
89. Complex would be $[Co(NH_3)_5Cl]Cl_2$
$2Cl^-$ can be ionized

90. Zn in $[Zn(H_2O)_4]\,SO_4$ has +2 oxidation state with d^{10} s^0 configuration. As there is no unpaired electron it is colourless.

91. $SCN^- + FeCl_3 + H_2O \rightarrow$

$\underset{\text{Blood red colour complex}}{\left[Fe(H_2O)_5SCN\right]}Cl_2 + Cl^-$

92. Cis trans isomerism is possible only when the two atoms attached are different. This is true for glycinato.



93. The different geometrical isomers differ in their colour.

94. They both are the cis form of the same compound.

95. $[Co(NH_3)_3Cl_3]$ exhibits fac-mer isomerism.

96. The complex with highest conductivity is $[Co(H_2O)_6]Cl_2$.
This would give most number of ions in solution $[Co(H_2O)_6]^{2+}$ and 2Cl
a, b, d do not tally with this complex.

97. There are five geometrical isomers for the complex.

98. The ion is chiral and exhibits optical isomerism.

99. $KMnO_4$ can only identity unsaturation (i.e., the double bond/triple bond can be detected). The reaction is not quantitative. Secondary reactions also occur.

100. On adding NaOH, CrO_2Cl_2 is converted to Na_2CrO_4. CrO_4^{2-} is converted to $Cr_2O_7^{2-}$ in acidic solution.

101. MnO_2 dissolves in oxalic acid liberating CO_2 due to oxidation. It is dissolved in excess and the excess acid is titrated against $KMnO_4$.

$MnO_2 + H_2C_2O_4 + H_2SO_4 \rightarrow$
$MnSO_4 + 2H_2O + 2CO_2\uparrow$

$2MnO_4^- + 5(COOH)_2 + 6H^+ \rightarrow$
$2Mn^{2+} + 10CO_2 + 8H_2O$

102. $\underset{\text{yellow}}{2FeCl_3} + SnCl_2 \rightarrow 2\underset{\text{green}}{FeCl_2} + SnCl_4$

$FeCl_3$ acts as an oxidizing agents and is converted to $FeCl_2$

Sn^{4+} is move stable than Sn^{2+}

103. Benedicts solution $CuSO_4$ and citrate ion
i.e., Cu^{2+} ion complexed to citrate ions

Note: Correct keys are (b) and (c)

104. $6AgNO_3 + PH_3 + 3H_2O \rightarrow Ag + 6HNO_3 + H_3PO_3$
PH_3 reduces $AgNO_3$ to metallic Ag

105. Metallic copper dissolves in aqueous $FeCl_3$. $FeCl_3$ is reduced to $FeCl_2$.

106. H_2O_2 forms chromic peroxide which is blue violet coloured

$Cr_2O_7^{2-} + 2H^+ + 4H_2O_2 \rightarrow 2CrO(O_2)_2 + 5H_2O$

b, c, d are oxidized by dichromate thereby forming green Cr^{3+}

107. $(NH_4)_2Cr_2O_7$ decomposes to give nitrogen.

108. $[Fe(H_2O)_6]^{2+}$ in aqueous medium can be readily oxidised even on exposure to air to Fe^{3+} ion.

$$2Fe^{2+}_{(aq)} + \frac{1}{2}O_2 + 2H^+_{(aq)} \rightarrow 2Fe^{3+}_{(aq)} + H_2O$$

109. dil H_2SO_4 does not react with $KMnO_4$ while conc H_2SO_4 reacts with $KMnO_4$ to form Mn_2O_7 which is explosive in nature

$$2KMnO_4 + H_2SO_4 \rightarrow K_2SO_4 + Mn_2O_7 + H_2O$$

Mn_2O_7 highly explosive

$$2Mn_2O_7 \rightarrow 4MnO_2 + 3\,O_2$$

110. Con HCl is oxidized to Cl_2. The best oxidizing agent among the options is PbO_2

Pb^{2+} is more stable than Pb^{4+} due to inert pair effect. So the reaction

$$Pb^{4+} + 2e^- \rightarrow Pb^{2+} \text{ occurs easily}$$

111. Both I and II happen due to back bonding. The empty π^* orbitals of CO and CN^- with the correct symmetry overlap with the t_{2g} orbitals of the metal. This leads to stable bonds and higher CFSE

112. The number of ions given out are different, so the depression in freezing point changes

$[Cr(H_2O)_6]Cl_3$ – 4 ions

$[Cr(H_2O)_5\,Cl]Cl_2$ – 3 ions

Both the complexes will not show geometric isomerism.

113. FeF_3, $FeCl_3$, $FeBr_3$ are prepared by direct combination of Iron with halogen

FeI_3 does not exist due to high reducing power of I^-

The electron affinity order is Cl > F > Br > I

The electron affinity in electron volts is respectively. 3.6, 3.4, 3.36 and 3.07 Electron affinity decreases down the group, but due to small size of fluorine and repelsion between electrons, fluorine has lower E.A than chlorine.

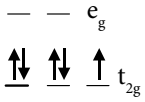
(A) – NaCl (B) – NaBr, (C) – NaI, (D) AgI

M = Ag, M' = Zn, (E) = Mn, F = $KMnO_4$

114. $KMnO_4$ is coloured due to the charge transfer.
Note: Correct option is (b)

115. M' has a high melting point is a wrong statement.

116. $[Mn(CN)_6)]^{4-}$ is a low spin complex in which Mn^{2+} is in d^5 configuration.



There is one unpaired electron with μ = 1.73 BM

Note: Correct key is (d)

117. The complex formed if hexadendate then reaction goes to completion in 1 : 1 ratio

Due to chelation, complex is stable

As such ligands are organic, water solubility is minimum. Due to large groups, molecular weight is high. So error is minimized

(c), (d) are seen in monodendate ligands and are not advantageous either.

118. (a), (c)

119. $ZnCO_3.\,3Zn(OH)_2 \xrightarrow{\text{heat}}$

$4ZnO + 3H_2O + CO_2$

$2Zn(NO_3)_2 \xrightarrow{\Delta} 2\,ZnO + 4NO_2 + O_2$

$ZnCl_2 + 2NaHCO_3 \longrightarrow ZnCO_3 + 2NaCl$

$H_2O + CO_2$

$Zn + \underset{\text{steam}}{H_2O} \xrightarrow{\Delta} ZnO + H_2$

120. (a) → (r)
(b) → (p), (q), (r), (s)
(c) → (p), (q), (r), (s)
(d) → (q), (s)

$ZnCl_2 . 2H_2O \xrightarrow{\Delta} Zn(OH)Cl + HCl$

(zinc hydroxyl chloride)

$2Zn(OH)Cl \xrightarrow{\Delta} Zn_2OCl_2 + H_2O$

zinc oxychloride

$K_2Cr_2O_7 \longrightarrow 2Cr_2O_3 + K_2CrO_4 + O_2$

$2KMnO_4 \longrightarrow K_2MnO_4 + MnO_2 + O_2$

$CuSO_4.\,5H_2O \longrightarrow CuSO_4 \longrightarrow CuO + SO_3$

Additional Practice Exercise

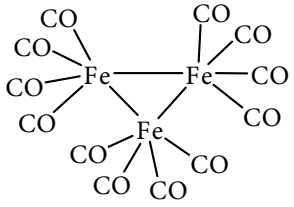
121. EAN for $V(CO)_6 = 23 + (6 \times 2) = 35$

EAN for $Mn_2(CO)_{10}$

There is one Mn–Mn bond in the carbonyl

$$\text{Hence EAN} = \frac{(2 \times 25)+(10 \times 2)+2}{2} = \frac{72}{2} = 36$$

The structure of $Fe(CO)_{12}$



EAN for $Fe_3(CO)_{12}$

There are three Fe – Fe bond

$$\text{EAN} = \frac{(26 \times 3)+(12 \times 2)+(3 \times 2)}{3}$$

$$= \frac{108}{3} = 36$$

122. This is an example of a complex containing the ligand NO^+. The paramagnetic ligand NO once forming the co-ordinate bond transfer its odd electron to the central Fe^{3+} ion containing an unpaired electron and form a pair. This reduces the O.N of Fe to +2 and NO becomes NO^+. Since there is no lone electron on either NO^+ or Fe^{2+} the species is diamagnetic and the over all charge on the co-ordination entity is –2.

123. (i) CrO_3 and SO_3 are both acidic and react with water to form chromic acid H_2CrO_4 and sulphuric acid H_2SO_4.

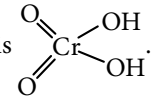
(ii) Chromates and sulphates are isomorphous. $BaSO_4$ and $BaCrO_4$, $PbSO_4$ and $PbCrO_4$ are insoluble in water.

(iii) CrO_2Cl_2 and SO_2Cl_2 both react with water and give H_2CrO_4 and H_2SO_4 respectively.

124. Cl_2O_7 behave similarly. It is an explosive acidic oxide. It dissolve in water to form $HClO_4$. Perchlorates are isomophous with permanganate. $AgClO_4$ is sparingly soluble.

125. The green product is guignet's green. It is chemically $Cr_2O_3.2H_2O$ used as a pigment.

126. Chromyl chloride is CrO_2Cl_2 and chromic acid is H_2CrO_4.

Structure of chromic acid is 

. When both the 'OH' groups of chromic acid are replaced by chlorine, chromyl chloride is formed. Hence it the acid chloride of chromic acid. Further, hydrolysis of chromyl chloride produce chromic acid and HCl

127. In aqueous medium MnO_4^{2-} disproportionate to MnO_4^- and MnO_2 producing OH^- ions CO_2 keeps the medium acidic and shift the equilibrium to the products side.

$$3K_2MnO_4 + 2H_2O + 4CO_2 \rightarrow$$

$$2KMnO_4 + MnO_2 + 4KHCO_3$$

128. It is $K_3[Co(NO_2)_6]$ known as potassium cobaltnitrite. The HNO_2 formed oxidizes Co^{2+} to Co^{3+} getting itself reduced to NO. NO_2^- ligands forms the complex.

129. (a) With KI, $CuSO_4$ liberate I_2 and CuI is precipitated

$$2CuSO_4 + 4KI \rightarrow 2CuI + 2K_2SO_4 + I_2$$

(b) With excess KCN, $CuSO_4$ forms the complex $K_3[Cu(CN)_4]$

Like I^-, CN^- reduce Cu^{2+} to Cu^+ and form CuCN.

$$2CuSO_4 + 4KCN \rightarrow 2CuCN + 2K_2SO_4 + (CN)_2$$

CuCN form complex with excess KCN

$$CuCN + 3KCN \rightarrow K_3[Cu(CN)_4]$$

130. When KI is added to $HgCl_2$, HgI_2 is formd which dissolve in excess of KI to form the complex $K_2[HgI_4]$

$$HgCl_2 + 2KI \rightarrow HgI_2 + 2KCl$$

$$HgI_2 + 2KI \rightarrow K_2[HgI_4]$$

Its IUPAC name is Potassium tetra iodomercurate.

This reaction is made use in preparing Nessler's reagent.

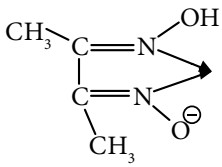
131. [Co (gly) $(DMG)_2$]

[Co(gly)$(DMG]_2$]

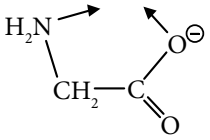
gly and DMG are bidendate ligands

so coordination number = 6

Charge on DMG is negative (–1) on forming complex by releasing an H^+.



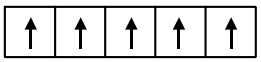
gly has one negative charge so that cobalt is in +3 charge



Note: Correct option is 3, 6

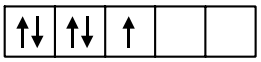
132. The cation and anion are $[PtCl_2(NH_3)_2]^{2+}$ $[PCl_4]^{2-}$
Therefore the oxidation state are +4 and +2 respectively

133. If $\mu = 5.9$ Bm, the number of unpaired electrons is five. Since oxidation state is same, number of unpaired e^- is same on using a strong field ligand raising would take place.



$\mu = \sqrt{5(5+2)} = 5.92$

weak field



Strong field

$\mu = \sqrt{1(1+2)} = 1.73$ βm

134. Due to the low polarity of the Hg–C bond and low affinity of mercury for oxygen, they are not attacked by air or water. But they are photochemically and thermally unstable.

135. Alkynyl compounds act as π acceptor and σ donor. So they are highly stable. Transfer of charge from filled dπ orbitals of metal to empty π anti-bonding molecular orbitals of aryl ligand, makes it more stable than metal-alkynyl bond. Metal alkyl complex has M–C σ bond, hence least stable.

136. d^1, d^2, d^3, d^8, d^9, d^{10} have same spin only magnetic moment whether complex is high spin or low spin.

137. The freezing point increases when KCN is added. The number of ions decreases. This leads to an increase in freezing point.

$$CuCN + 3KCN \rightarrow K_3[Cu(CN)_4]$$

138. The percentage indicate the ratio of moles of Co : NH_3 : Cl : SO_4 in the complex as 1 : 4 : 1 : 1. The correct formula of complex structure is $[CoSO_4(NH_3)_4(H_2O)]Cl$.

139. Due to d^1 configuration, they are not sufficiently populated with electrons, therefore there is no back bonding between the metal and the alkene. Hence the complex is not stable and the complex is not formed

140. $E = 240$ KJ mol^{-1}

$$= \frac{240}{6.022\times10^{23}} \text{ kJ molecule}^{-1}$$

$$E = \frac{hc}{\lambda}$$

$$\frac{240}{6.022\times10^{23}} = \frac{6.63\times10^{-34}\times3\times10^{8}}{\lambda}$$

$$\lambda = 496.6\text{nm} \cong 500\text{nm}$$

141. The colour of a lanthanide complex depends on the f – f transition.
The 4f orbitals are the anti penultimate shell, so they are well shielded by the 5s and 5p orbitals. So the f-electrons are unaffected by complex formation and colour does not change with the ligands.

142. Low oxidation states particularly occur when metal bonds with π bonding ligands

Back bonding is seen in $C\overline{N}$, CO and dipyridyl

This is not seen in EDTA.

143. $[Co(H_2O)_6]Cl_2 \overset{\Delta}{\rightleftharpoons} [Co(H_2O)_4]Cl_2 + 2H_2O$ number of H_2O molecules change from six to four hybridization changes from sp^3d^2 to sp^3 this leads to change in CFSE leading to colour change.

144. EDTA will form a chelated complex. So Ba cannot be precipitated as sulphate in its presence.

145. Cu^{2+} acts as a lewis acid which accepts electrons into its empty orbital from a lewis base i.e., NH_3 the ligand. So its an acid base reaction.

146. A ligand that can itself show isomerism exhibits ligand isomerism.
diaminopropane can show two isomeric forms 1, 2 diamino propane and 1, 3 diaminopropane.

147. As it is a square planar complex initially attack of H_2O is from above or below.
So the complex will be a trans complex.

148. Since both the ligands H_2O and Br are within the coordination sphere and the complex gives two ions in solutions, the complex should be
$[Pt(H_2O)_4]$ $[Pt\ Br_4]$.

149. $[Cr\ Cl_2(Ox)_2]$ shows cis and trans isomerism
The cis isomer is optically active
[M abcd] can also be represented in three forms

a – shows cis trans isomerism – two isomers

c – ionization and linkage isomerism – total 4 isomers

d – hydrate isomers – total 4 isomers

150. hydrate isomer of $[Cr(H_2O)_6]Cl_3$ are

$[Cr(H_2O)_5Cl]Cl_2.H_2O$,

$[Cr(H_2O)_4Cl_2]Cl.2H_2O$, $[Cr(H_2O)_3Cl_3].3H_2O$

Facial and meridional isomerism is shown by

Complexes of the type $[MA_3B_3]$

i.e., $[Cr(H_2O)_3Cl_3].3H_2O$

This does not have ionisable CI^- and trans form is optically active.

It loses the three molecules of water of hydration with con.H_2SO_4.

151. Complex A contains ionizable CI^-. AgCl dissolves in ammonia. So A is

$[Cr(NH_3)_4ClI]Cl$

Complex B has ionizable I^- AgI does not dissolve in ammonia

Complex B is $[Cr(NH_3)_4Cl_2]I$

Ligand strength of halogens are comparable so both will form high spin complexes

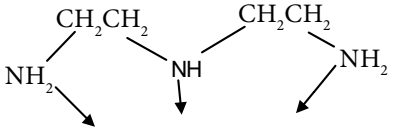
So both will be d^2sp^3 hybridization and

$\mu = \sqrt{3(3+2)} = 3.89$

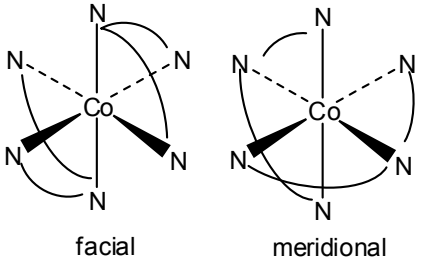
152. (a) and (b) are complexes with coordination 4. So they will not show both types together. $[Cr(cn)_3]Cl_3$ shows only optical isomerism.

$[co(em)_2(OH_2]^+$ shows geometric isomerism of which the cis form is optically active.

153. (dien–di ethylene triammine is a tridendate ligand



It shows facial and meridional isomerism

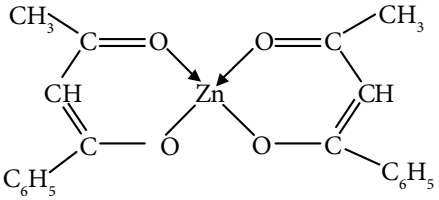


154. For a complex of the type [M abcdef] there are 15 diastereomers each of which will have a mirror image therefore there will be 30 enantiomers.

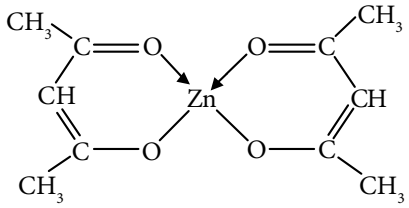
155. Zn complexes are tetrahedral.

So it will not show geometric isomerism due to the inherent symmetry of the tetrahedron, due to presence of four different groups, it will be chiral.

156. With bisbenzoylacetanato ligand due to asymmetry non super imposable mirror image can be formed. But with acetylacetanato mirror images are super imposable.



can form non super imposable mirror image hence optically active.



157. Only $[Pd(gly)_2]$ will show cis trans isomerism because glycinato is an unsymmetric bidendate ligand

(b) (en) is symmetric

(c) in order to form trans en cannot be stretched across the trans position

(d) it is a tetrahedral complex

158. $2CrO_3 + 2NH_3 + H_2O \rightarrow \underset{\text{ammonium dichromate}}{(NH_4)_2Cr_2O_7}$

$(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2\uparrow + Cr_2O_3 + 4H_2O$

159. (Reagent A is $K_4[Fe(CN)_6]$

$2CuSO_4 + K_4[Fe(CN)_6] \rightarrow$

$\underset{\text{Reddish Brown}}{Cu_2[Fe(CN)_6]\downarrow} + 2K_2SO_4$

$K_4[Fe(CN)_6] + ZnSO_4 \rightarrow Zn_2[Fe(CN)_6] + 2K_2SO_4$

$K_4[Fe(CN)_6] \xrightarrow{\Delta} 4KCN + FeC_2 + N_2$

160. $KMnO_4$ is an oxidising agent. White turbidity is caused by colloidal sulphur

Among the options on S^{2-} can be oxidized to S. all others have to be reduced to S

$$2MnO_4^- + 16H^+ + 3S^{2-} \rightarrow 2Mn^{2+} + 5S + 8H_2O$$

161. $\underset{\text{pyrolusite}}{2MnO_2} + 4KOH + O_2 \rightarrow \underset{\text{green}}{2K_2MnO_4} + 2H_2O$

$$K_2MnO_4 + H_2O + O_3 \rightarrow \underset{\text{purple}}{2KMnO_4} + 2KOH + O_2$$

162. I^- is a good reducing agent

It reduces Cu^{2+} to Cu^+

$$2CuSO_4 + 4KI \rightarrow 2K_2SO_4 + 2CuI + I_2.$$

163. The oxidation state of Cr in K_2CrO_4 is +6

so as oxidation state changes by 1, Cr should be in +5 state

so it is a peroxo complex. So it is both coloured and paramagnetic

satisfy +5 oxidation state it should have four peroxo linkages

K_3CrO_8 exists as $K_3[Cr(O_2)_4]$

$$2K_2CrO_4 + 7H_2O_2 + 2KOH \rightarrow 2K_3CrO_8 + 8H_2O$$

164. The solution obtained contains K_2CrO_4.

K_2CrO_4 with $AgNO_3$ gives brick red precipitate of Ag_2CrO_4 – silver chromate

165. Reagent A is $HgCl_2$

$$SnCl_2 + 2HgCl_2 \rightarrow \underset{\text{white ppt(B)}}{Hg_2Cl_2} + SnCl_4$$

$$SnCl_2 + Hg_2Cl_2 \rightarrow \underset{\text{grey}}{2Hg} + SnCl_4$$

In both cases $SnCl_2$ is converted to $SnCl_4$ Sn^{2+} oxidized to Sn^{4+}

166. The aqueous solution of $FeCl_3$ is acidic in nature

Sn liberates hydrogen from acidic solution which reduces Fe^{3+} to Fe^{2+}.

167. $Pb_3O_4 + 8\,HCl \rightarrow 3PbCl_2 + 4\,H_2O + Cl_2$

$2Pb_3O_4 + 6H_2SO_4 \rightarrow 6PbSO_4 + 6H_2O + O_2$

168. Rinmann's green is cobalt zincate – $CoZnO_2$

(ZnO.CoO) prepared by heating zinc with cobalt nitrate.

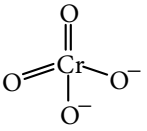
169. $CuSO_4 + Na_2S_2O_3 \rightarrow Na_2SO_4 + CuS_2O_3$

$2CuS_2O_3 + 2Na_2S_2O_3 \rightarrow 2NaCuS_2O_3 + Na_2S_4O_6$

On boiling, Cu_2S is precipitated

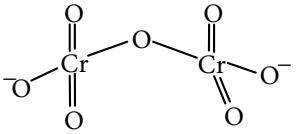
$2NaCuS_2O_3 + H_2O \rightarrow Cu_2S + Na_2S_2O_3 + H_2SO_4$

170.



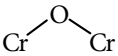
chromate

All bonds are equal



dichromate

Two chromate tetrahedral CrO_4^{2-} units having common 'O' atom Cr–O bond in tetrahedral has partial double bond character and hence bond length is small (1.62Å).



Cr bond is longer (1.78Å)

171. Both $V(CO)_6$ and $Co(CO)_4$ do not obey EAN rule

$V(CO)_6 - 23 + (2 \times 6) = 35e^-$

$Co(Co)_4 - 27 + (2 \times 4) = 35e^-$

$Co(CO)_4$ alone dimerises because it can increase its coordination number from four to six.

While $V(CO)_6$ can show maximum coordination number of 6, hence cannot dimerises.

172. Aqueous solution of $CrCl_3$ would contain other complexes like $[Cr(H_2O)_5Cl]^{2+}$, $[Cr(H_2O)_4Cl_2]$, $[Cr(H_2O)_3Cl_3]$ etc. These are green in colour while the complex $[Cr(H_2O)_6]\,Cl_3$ is violet coloured.

173. In NF_3. fluorine is more electronegative, pulls the shared pair of electrons towards itself, so lone pairs are not available for bonding. In $N(CH_3)_3$, lone pairs are more available than NF_3

Both NF_3 and $N(CH_3)_3$ have pyramidal structure.

174. In the presence of EDTA, Ba^{2+} forms a very stable complex due to chelation. Charge on EDTA is –4, so the complex formed would be $[Ba(EDTA]^{2\pm}$.

Due to the formation of this stable complex, reaction of Ba^{2+} is not seen, so white precipitate of $BaSO_4$ is not formed.

175. The linkage isomer of $[Co(SCN)_4]^{2-}$ is $[Co(NCS)_4]^{2-}$ SCN is a weak ligand, therefore there is no pairing

of electrons. So it is a tetrahedral complex with 3 unpaired electrons i.e.,

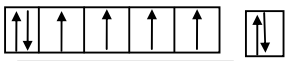
μ = 3.87 βM while NCS being a strong ligand, forms square planar complex due to pairing of electrons so number of unpaired electrons is 1, so μ = 1.73 βM.

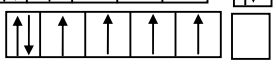
176. Complex can show only cis form because in the trans complex, the bidendate ligand has to span across.

For the same reason [M A_2 (ab)] will not show trans form.

177. Fe SO_4 on exposure to air gets oxidized to Fe^{3+}

$$4FeSO_4 + 2H_2O + O_2 \rightarrow \underset{\text{Brown}}{4Fe(OH)SO_4}$$

Fe – [Ar] 

Fe^{2+}[Ar] 

In the presence of strong ligand, pairing would lead to no unpaired electrons

178. Cu^+ ion is stabilized by formation of insoluble CuCl in the presence of Cl^- ions

$2Cu^+ + 2Cl^- \rightarrow 2CuCl$

So disproportion reaction is reversed

But in excess Cl^-, it forms a complex

$CuCl + Cl^- \rightarrow [CuCl_2]^-$

179. $Zn^{2+} + 2OH^- \rightarrow \underset{\text{ppt}}{Zn(OH)2} \downarrow$

$Zn(OH)_2 + 2HCl \rightarrow ZnCl_2 + 2H_2O$

Aqueous $ZnCl_2$ exists as $[Zn(H_2O)_4]Cl_2$

$Zn(OH)_2 + 2NaOH \rightarrow [Zn(OH)_4]^{2-} + 2Na^+$

Both are water soluble and have tetrahedral structure

180. The compound formed on passing ozone over silver is AgO.

Here Ag is in the +2 oxidation state which is not stable.

Therefore it moves to the +1 state thereby acting as an oxidizing agent.

181. Due to reducing nature of I^-, CuI_2 formed immediately decomposes to CuI where Cu^{2+} is reduced to Cu^+ and iodide is oxidized to iodine.

182. Addition of a reducing agent like $FeSO_4$, SO_2 etc would prevent formation of I_2.

Among the options only (a) is reducing

$$2CuSO_4 + 2FeSO_4 + 2KI \rightarrow$$
$$2CuI\downarrow + Fe_2(SO_4)_3 + K_2SO_4$$

183. By the equation CuI and K_2SO_4 are formed K_2SO_4 is colourless. In CuI, Cu^+ has a d^{10} configuration. Hence it is also white

184. A is $[Cr(H_2O)_6]SO_4$

C.NO = 6

O.NO = 2

185. A is $[Cr(H_2O)_6]SO_4$

Has four unpaired electrons

$$\mu = \sqrt{4+(4+2)} = 4.89\ \beta M$$

While B is $K_4[Cr(CN)_6]$

As CN^- is a strong ligand, electron pairing occurs against Hund's rule. So there are only two unpaired electrons. So $\mu = \sqrt{2(2+2)} = 2.89\beta M$

186. As HNO_3 is an oxidizing acid, chromium is rendered passive.

187. B and C are geometric isomers

Ethylene diamine can react with the cis isomer to give a complex [$Ni(NH_3)_2$ en] but will not react with the trans isomer as it cannot bond across the diagonal.

188. In acidic medium NH_3 would be converted to NH_4^+. So it will not behave like a ligand so no ammonia complex would e formed.

189. It would show ionization isomerism

$[Ni(NH_3)_4\ ClNO_2]Cl$ this would show linkage isomerism.

Complex of type $[MA_4B_2]$ Cis form shows optical isomerism.

190. $[Cu(NH_3)_5\ NO_2]Br$ – $[Cu(NH_3)_5\ ONO]Br$ linkage

$[Cu(NH_3)_5NO_2]Br$ – $[Cu(NH_3)_5\ Br]\ NO_2$ Ionization

191. In all the three central metal ion has a d^{10} configuration

So CFSE will be zero

192. If μ = 5.92BM with weak field ligand, the value of μ will be 1.73 BM in strong field ligand.

193. (a) $[Cr(C_2O_4)_3]^{3-}$ has 3 electrons hence t^3_{2g} irrespective of ligand, pairing will not occur.

(b) $[NiCl_4]^{2-}$ is a tetrahedral complex. Since CFSE is small, pairing takes place after all the orbitals are singly filled-But hybridization changes with strong field ligand.

(c) $[TiCl_6]^{2-}$ has a d° configuration so μ is always zero.

(d) $[MnF_6]^{4-}$ weak field will have 5 unpaired electrons strong field will have 1 unpaired electrons.

194. (a) is the same as the given compound
(b) and (c) are cis isomers in all respects
(d) is also a geometrical isomer of the complex

195. $2[FeSO_4.7H_2O] \xrightarrow{\Delta} Fe_2O_3 + H_2SO_4 + SO_2 + 13H_2O$

$2FeSO_4 \xrightarrow{\Delta} Fe_2O_3 + SO_2 + SO_3$

$CuSO_4 . 5H_2O \xrightarrow{100°C} CuSO_4 . H_2O$

$\xrightarrow{230°C} CuSO_4 \xrightarrow{720°} CuO + SO_3 \rightarrow$

$SO_2 + O_2$

$2AgNO_3 \xrightarrow{\text{red heat}} 2Ag + 2NO_2 + O_2$

196. $2Cu^{2+} + 2CN^- \rightarrow 2Cu^+ + C_2N_2$

$2Cu^+ + 2CN^- \rightarrow 2CuCN$

$CuCN + 3CN^- \rightarrow \underset{\text{soluble}}{[Cu(CN)_4]^{3-}} -$

CuS dissolves in NaCN due to formation of $Cu(CN)_4]^{3-}$

Where, Cu is in +1 oxidation state. Cu^{2+} is reduced before complexation

197. c, d

Both $K_2Cr_2O_7$ and $KMnO_4$ can be used as self indicators Both do not affect glass ware

Only $K_2Cr_2O_7$ is a primary standard because $KMnO_4$ cannot be obtained in pure state, it is always contaminated with MnO_2.

Permanganate liberates Cl_2 gas from HCl. While $K_2Cr_2O_7$ has no effect on HCl. Only H_2SO_4 should be used make the medium acidic for $KMnO_4$ titrations.

198. (a) → (p), (q)
(b) → (r), (s), (p), (q)
(c) → (s), (p)
(d) → (p)

A – Isomeric complexes containing hydrogen atoms in different chemical environment produce different H'NMR signals.

A and D – Trans isomer has low or zero dipole moment while cis isomer has higher dipole moment.

As glycinate is an unsymmetric bidendate ligand it shows cis trans forms.

B – gives out different number of ions hence can be studied by both conductivity as well as freezing point measurements.

C – gives ions with different charges so they can be studied by molar conductivity measurements.

199. (a)→ (p), (q)
(b) → (p), (s)
(c)→ (q), (s)
(d) → (r), (s)

P – Co^{3+} ion, it shows 6 geometric isomers out of two are optically active

It is d^6 system, NH_3 is a strong ligand. So diamagnetic

Q – shows only optical isomerism.

Cr is in +3 state and is d^3 system. Hence paramagnetic.

R – square planar complex of Ni^{2+} can show only cis form.

S– Square planar complex of Cu^{2+} has one unpaired electron hence paramagnetic shows cis trans isomerism.

200. (a) → (p), (q), (r), (s)
(b) → (p), (q), (r)
(c) → (s), (q)
(d) → (p)

$2AgNO_3 + 2NH_4OH \rightarrow Ag_2O + 2NH_4NO_3 + H_2O$

$Ag_2O + 2NH_4NO_3 + 2NH_4OH \rightarrow$

$\underset{\text{soluble complex}}{2\left[Ag(NH_3)_2\right]NO_3} + 3H_2O$

$2AgNO_3 + Na_2S_2O_3 \rightarrow 2NaNO_3 + Ag_2S_2O_3$

$3Na_2S_2O_3 + Ag_2S_2O_3 \rightarrow \underset{\text{soluble complex}}{2Na_3\left[Ag(S_2O_3)_2\right]}$

$SnCl_2 + 2NaOH \rightarrow Sn(OH)_2 + 2NaCl$

$Sn(OH)_2 + 2NaOH \rightarrow \underset{\text{soluble}}{Na_2SnO_2} + 2H_2O$